

# Status of the Rougheye and Blackspotted Rockfishes stock off the U.S. West Coast in 2025

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## 1 Executive Summary

### 1.1 Stock

This assessment reports the status of the Rougheye (*Sebastes aleutianus*) and Blackspotted (*Sebastes melanostictus*) Rockfishes that reside in the waters off California, Oregon, and Washington from the U.S.- Canadian border in the north to the U.S.-Mexico border in the south.

These two species are difficult to differentiate visually in the catch, thus they are commonly reported and treated as one management complex. Despite recent advances in species identification of Rougheye/Blackspotted Rockfishes as genetically distinct species, there is still little ability to reliably separate historical fishery data in order to differentiate these two species into two stocks. Therefore, this assessment maintains a species complex approach, though given absolute presence off the U.S. West Coast, this may be considered more of a Rougheye than Blackspotted Rockfish stock assessment. While we treat these species as one assessed stock complex, we recognize and are mindful of the above species distinctions as we conduct our analyses.

### 1.2 Catches

Rougheye/Blackspotted Rockfishes are not targeted by a specific fishery, but are desirable and marketable, thus are typically retained when caught.

The historical reconstruction of landings for Rougheye/Blackspotted Rockfishes suggests that non-trawl (largely hook-and-line gear) fisheries have caught Rougheye/Blackspotted Rockfishes since the turn of the 20th century. The bottom trawl fishery developed in the 1930s and 1940s, and trawl landings for Rougheye/Blackspotted Rockfishes first peaked in the 1960's and 1970s, when the foreign trawl fleet was targeting Pacific Ocean perch. The declaration of the Exclusive Economic Zone resulted in the buildup of a domestic fleet and landings increased rapidly into the late 1980's and early 1990's. Subsequently, landings declined in the late 1990's, with catches under 250 metric tons in the last two decades. The contribution of mid-water trawl catches gradually grew over the past 15 years, and now they represent majority of the trawl removals.

Since Rougheye/Blackspotted Rockfishes are a desirable market species, discarding has been low historically. However, management restrictions (e.g., trip limits) have resulted in increased discarding since early 2000s. Trawl rationalization was introduced in 2011, and since then very little discarding of Rougheye/Blackspotted Rockfishes has occurred.

Rougheye/Blackspotted Rockfishes also has long been bycaught in the fishery for the coastal population of Pacific hake, which is almost exclusively conducted with mid-water trawl.

Time series of landings are shown Figure [i](#), with recent landings detailed in Table [i](#).

Table i: Recent landings by fleet, total landings summed across fleets, and the total dead catch including discards.

| Year | Trawl | Trawl discard | Non-trawl | Non-trawl discard | Midwater trawl | At-sea-hake | Total Landings | Total Dead (mt) |
|------|-------|---------------|-----------|-------------------|----------------|-------------|----------------|-----------------|
| 2015 | 30.67 | 0.01          | 46.56     | 13.79             | 19.26          | 21.80       | 132.09         | 132.09          |
| 2016 | 30.79 | 0.11          | 60.27     | 12.61             | 15.53          | 29.63       | 148.95         | 148.95          |
| 2017 | 21.93 | 0.00          | 59.03     | 34.42             | 2.48           | 38.15       | 156.02         | 156.02          |
| 2018 | 16.49 | 0.00          | 46.67     | 14.55             | 2.58           | 161.24      | 241.52         | 241.52          |
| 2019 | 22.06 | 0.04          | 38.75     | 31.13             | 9.25           | 125.37      | 226.59         | 226.59          |
| 2020 | 9.86  | 0.03          | 24.35     | 1.03              | 28.92          | 41.88       | 106.07         | 106.07          |
| 2021 | 10.33 | 0.01          | 21.06     | 2.36              | 21.39          | 37.62       | 92.76          | 92.76           |
| 2022 | 11.54 | 0.02          | 19.06     | 2.72              | 18.63          | 65.46       | 117.43         | 117.43          |
| 2023 | 13.29 | 0.48          | 18.67     | 0.48              | 26.22          | 38.50       | 97.63          | 97.63           |
| 2024 | 9.97  | 0.12          | 9.90      | 0.48              | 69.15          | 29.32       | 118.94         | 118.94          |

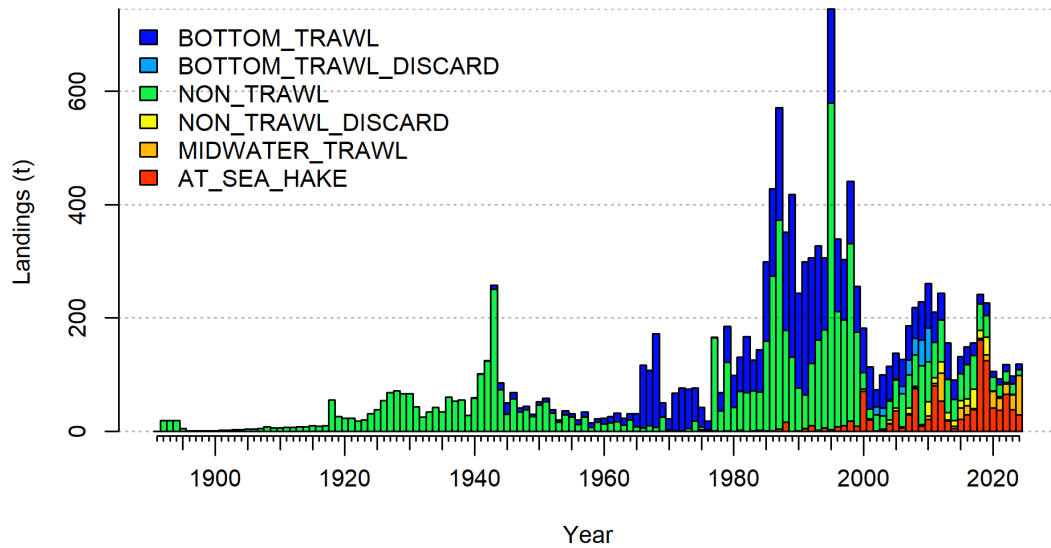


Figure i: Landings in metric tons (mt) by year for each fleet.

### 1.3 Data and Assessments

The only previous benchmark stock assessment for Rougheye/Blackspotted Rockfishes for the U.S. West Coast area was done in 2013 ([Hicks, Wetzel, and Harms 2013](#)). It was conducted with Stock Synthesis statistical catch-at-age modelling framework ([Methot and Wetzel 2013](#)).

This 2025 assessment also uses Stock Synthesis, version 3.30.23.1. The modeling period begins in 1892, and the stock prior to that is assumed to be in an unfished equilibrium condition.

Rougheye/Blackspotted Rockfishes fishery-dependent data in this assessment are divided among six fleets, treating discard catches separately from the retained fisheries. Following 2013 assessment, it maintains the at-sea-hake fishery as its own fleet, and adds a mid-water fishery that has emerged in the last decade. This stock assessment adds 10+ years of additional length data, and several more years of age data (included as conditioned on length data). The same four surveys (the West Coast Groundfish Bottom Trawl Survey (WCGBTS), AFSC/NWFSC Triennial Shelf Survey, AFSC Slope Survey and NWFSC Slope Survey) as used in the last stock assessment are used here, with an extension to 2024 to the the WCGBTS. The index standardization of all survey data uses the newer approach of applying spatiotemporal generalized linear mixed models.

This is a sex-specific model. Females and males have separate growth curves and sex-specific weight-at-length parameters. Growth is assumed to follow the von Bertalanffy growth model, and the assessment explicitly estimates all parameters describing somatic growth. The natural mortality for females is estimated in the assessment and natural mortality for males is fixed at the value generated from meta-analytical study. Externally estimated life history parameters, including those defining the length-weight relationship, female fecundity and maturity schedule were revised for this assessment to incorporate new information. Recruitment dynamics are assumed to follow the Beverton-Holt stock-recruit function, and recruitment deviations are estimated. Stock-recruitment steepness is fixed at the value generated from meta-analytical study. The base model estimates parameters for selectivity based on length data, and estimated selectivity curves are a mix of dome-shaped (for bottom trawl gears) and logistic (for mid-water trawl).

#### 1.4 Stock Output and Dynamics

The model estimates that the stock complex currently is in a healthy state, well above management target (Figure ii, Figure iii). Approximate confidence intervals based on the asymptotic variance estimates show that the uncertainty in the estimated spawning output is high. Estimates of spawning biomass in most recent decade are shown in Table ii.

Fraction unfished shows slight decline between 1940s and the 1960s, also also gradual decline since early-1980s, which correspond to catch history.

Table ii: Estimated recent trend in spawning output and the fraction unfished and the 95 percent confidence intervals.

| Year | Spawning output | Lower Interval (mt) | Upper Interval (mt) | Fraction Unfished | Lower Interval | Upper Interval |
|------|-----------------|---------------------|---------------------|-------------------|----------------|----------------|
| 2015 | 4,980,900       | -2,394,288          | 12,356,088          | 0.882             | 0.648          | 1.115          |
| 2016 | 4,969,060       | -2,407,049          | 12,345,169          | 0.880             | 0.644          | 1.116          |
| 2017 | 4,954,890       | -2,421,278          | 12,331,058          | 0.877             | 0.638          | 1.116          |
| 2018 | 4,940,000       | -2,435,893          | 12,315,893          | 0.875             | 0.633          | 1.116          |

|      |           |            |            |       |       |       |
|------|-----------|------------|------------|-------|-------|-------|
| 2019 | 4,914,900 | -2,461,287 | 12,291,087 | 0.870 | 0.623 | 1.117 |
| 2020 | 4,893,960 | -2,484,403 | 12,272,323 | 0.867 | 0.615 | 1.118 |
| 2021 | 4,891,660 | -2,492,289 | 12,275,609 | 0.866 | 0.613 | 1.119 |
| 2022 | 4,895,290 | -2,498,792 | 12,289,372 | 0.867 | 0.613 | 1.121 |
| 2023 | 4,900,660 | -2,508,945 | 12,310,265 | 0.868 | 0.612 | 1.123 |
| 2024 | 4,913,840 | -2,516,952 | 12,344,632 | 0.870 | 0.614 | 1.126 |
| 2025 | 4,929,140 | -2,527,955 | 12,386,235 | 0.873 | 0.615 | 1.130 |

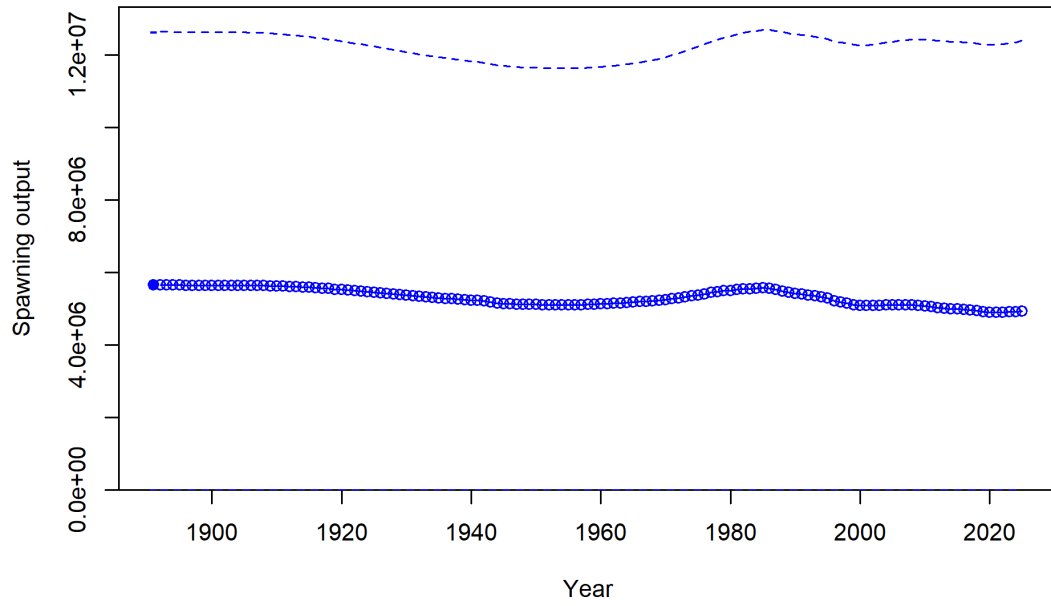


Figure ii: Estimated time series of spawning output (millions of eggs) for the base model.

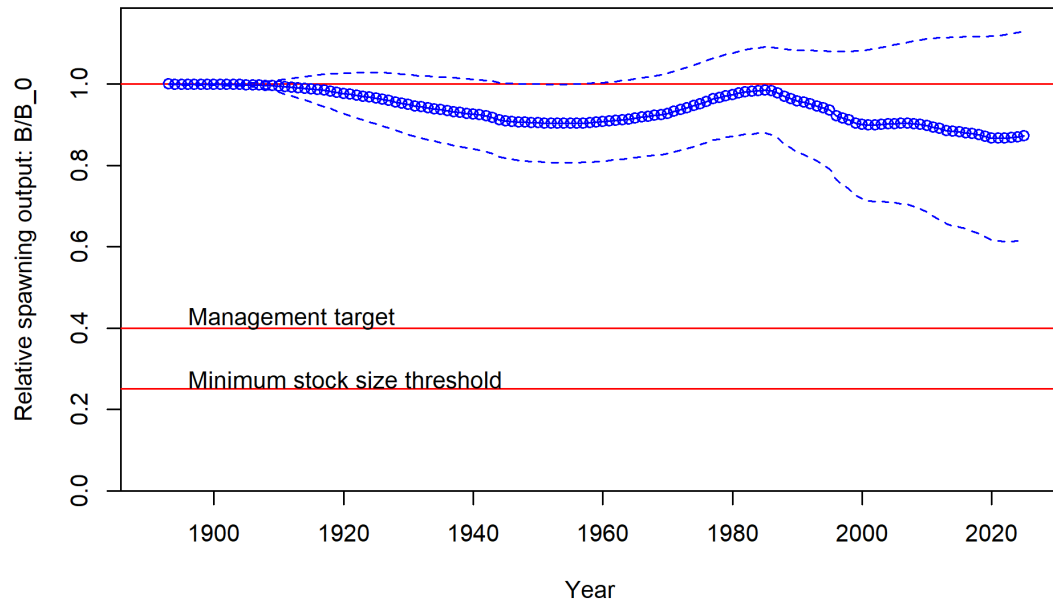


Figure iii: Estimated time series of fraction of unfished spawning output for the base model.

### 1.5 Recruitment

Recruitment dynamics (Table iii, Figure iv) are assumed to follow Beverton-Holt stock-recruit function and the steepness parameter was fixed at the value of 0.72, which is the mean of steepness prior probability distribution, derived from meta-analysis of rockfish stocks. The level of virgin recruitment ( $R_0$ ) is estimated to inform the magnitude of the initial stock size. Annual recruitment is treated as stochastic. “Main” recruitment deviations were estimated for modeled years between 1892 and 2023, with forecast recruitment period starting in 2024 (Figure v).

Table iii: Estimated recent trend in recruitment (1,000s) and recruitment deviations and the 95 percent confidence intervals.

| Year | Recruit-<br>ment<br>(1,000s) | Lower<br>Interval<br>(1,000s) | Upper<br>Interval<br>(1,000s) | Recruit-<br>ment<br>Deviations | Lower<br>Interval | Upper<br>Interval |
|------|------------------------------|-------------------------------|-------------------------------|--------------------------------|-------------------|-------------------|
| 2015 | 659                          | 173                           | 2,515                         | -0.445                         | -1.271            | 0.382             |
| 2016 | 870                          | 228                           | 3,328                         | -0.172                         | -1.012            | 0.668             |
| 2017 | 2,172                        | 584                           | 8,071                         | 0.738                          | -0.009            | 1.484             |
| 2018 | 1,341                        | 354                           | 5,087                         | 0.251                          | -0.565            | 1.066             |
| 2019 | 1,197                        | 317                           | 4,517                         | 0.132                          | -0.678            | 0.943             |
| 2020 | 846                          | 221                           | 3,242                         | -0.220                         | -1.083            | 0.643             |
| 2021 | 923                          | 236                           | 3,608                         | -0.138                         | -1.050            | 0.774             |
| 2022 | 1,017                        | 254                           | 4,067                         | -0.046                         | -1.008            | 0.916             |
| 2023 | 1,067                        | 266                           | 4,286                         | -0.003                         | -0.982            | 0.975             |
| 2024 | 1,076                        | 268                           | 4,323                         | 0.000                          | -0.980            | 0.980             |
| 2025 | 1,076                        | 268                           | 4,324                         | 0.000                          | -0.980            | 0.980             |

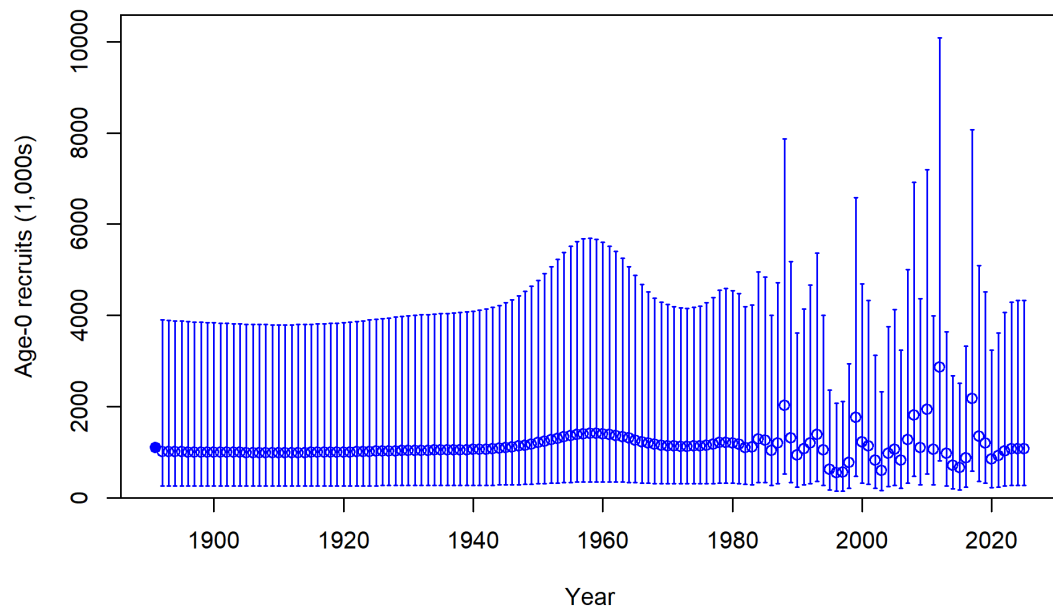


Figure iv: Estimated time series of age-0 recruits for the base model.



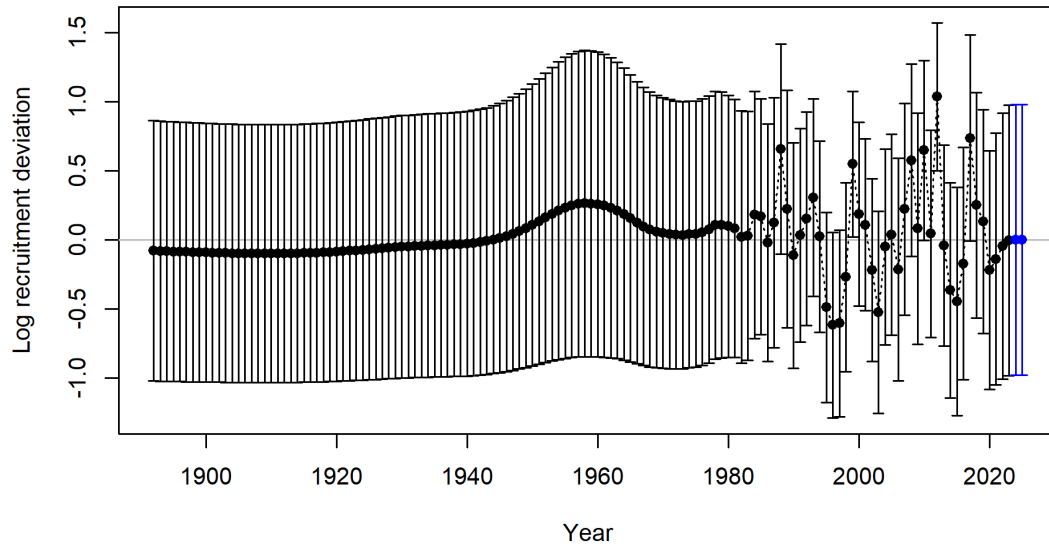


Figure v: Estimated time series of recruitment deviations for the base model.

### 1.6 Exploitation Status

Two measures of exploitation are fishing intensity and exploitation rate. Fishing intensity is defined here as  $1 - \text{SPR}$ , where SPR (Spawning Potential Ratio) is the equilibrium spawning output at a given combination of  $F$  and selectivity relative to spawning output at unfished equilibrium. Using the units of  $1 - \text{SPR}$  means that more intense fishing is associated with a higher value. The value of  $1 - \text{SPR}$  in the absence of fishing is 0 and the maximum is 1.0 if all spawning fish are being killed before spawning. The Pacific Fishery Management Council (PFMC) has chosen an SPR target of 0.5 for Rougheye/Blackspotted Rockfishes so harvest which leads to SPR below 0.5, or fishing intensity ( $1 - \text{SPR}$ ) greater than 0.5 would be overfishing. Exploitation rate is defined as the catch relative to age 26+ biomass. This metric is included because interpretation is simple, but it is not used as a basis for management.

Exploitation rates were below the management target of a fishing intensity that leads to a SPR of 0.5 throughout most of the time series except for 1995, when the catch peaked at 744 metric tons (Table iv, Figure vi).

Table iv: Estimated recent trend in fishing intensity 1-SPR, where SPR is the spawning potential ratio, and the exploitation rate, along with the 95 percent confidence intervals for both quantities.

| Year | 1-SPR | Lower Interval (SPR) | Upper Interval (SPR) | Exploitation Rate | Lower Interval (Rate) | Upper Interval (Rate) |
|------|-------|----------------------|----------------------|-------------------|-----------------------|-----------------------|
| 2015 | 0.114 | -0.038               | 0.266                | 0.004             | -0.002                | 0.011                 |
| 2016 | 0.127 | -0.041               | 0.296                | 0.005             | -0.002                | 0.013                 |
| 2017 | 0.133 | -0.042               | 0.307                | 0.005             | -0.003                | 0.013                 |
| 2018 | 0.188 | -0.047               | 0.423                | 0.008             | -0.004                | 0.021                 |
| 2019 | 0.180 | -0.048               | 0.408                | 0.008             | -0.004                | 0.019                 |
| 2020 | 0.090 | -0.034               | 0.213                | 0.004             | -0.002                | 0.009                 |
| 2021 | 0.079 | -0.031               | 0.190                | 0.003             | -0.002                | 0.008                 |
| 2022 | 0.098 | -0.036               | 0.233                | 0.004             | -0.002                | 0.010                 |
| 2023 | 0.083 | -0.032               | 0.199                | 0.003             | -0.002                | 0.009                 |
| 2024 | 0.098 | -0.036               | 0.233                | 0.004             | -0.002                | 0.011                 |

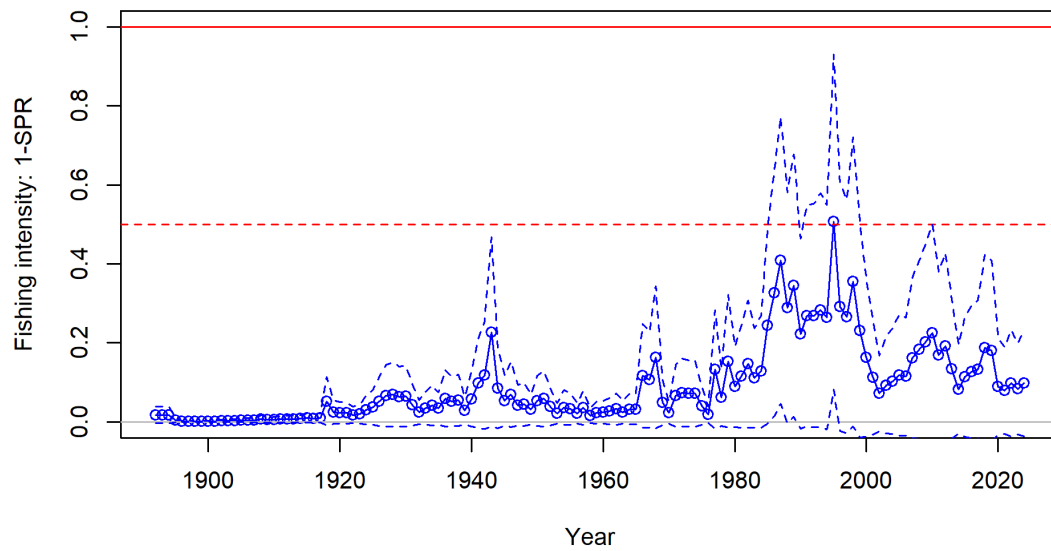


Figure vi: Estimated time series of the fishing intensity ( $1 - \text{SPR}$ ), where SPR is the spawning potential ratio, with approximate 95% asymptotic intervals. The horizontal line at 0.5 corresponds to  $\text{SPR} = 0.5$ , the management reference point. The horizontal line at 1.0 corresponds to  $\text{SPR} = 0$  (all spawning fish removed from the population).

## 1.7 Ecosystem Consideration

Rockfishes are an important component of the California Current ecosystem along the U.S. West Coast, with its many dozens of species filling various niches in both soft and hard bottom habitats from the nearshore to the continental slope. Rougheye/Blackspotted Rockfishes are one of the larger species of rockfishes and occupy shelf areas when they are young and move into deeper slope waters with age. As they age, they tend to become more solitary, but may form aggregations during the spawning season. Due to a paucity of life-history data for Rougheye/Blackspotted Rockfishes, most ecosystem considerations are implied from the understanding of rockfishes in general.

Recruitment is one mechanism by which the ecosystem may directly impact the population dynamics of Rougheye/Blackspotted Rockfishes. The specific pathways through which environmental conditions exert influence on Rougheye/Blackspotted Rockfishes dynamics are unclear, however, changes in water temperature and currents, distribution of prey and predators, and the amount and timing of upwelling are all possible linkages. Changes in the environment may also result in changes in age-at-maturity, fecundity, growth, and survival which can affect how the status of the stock and its susceptibility to fishing are determined. Unfortunately, there are no data for Rougheye/Blackspotted Rockfishes that provide insights into these effects.

## 1.8 Reference Points

Estimates of the current state of the population, as well as reference points based on 1) a target unfished spawning output of 40%, 2) a spawning potential ratio of 0.5, and 3) the model estimate of maximum sustainable yield, are all listed in Table v. Equilibrium yield curve for the base case model is shown in Figure viii.

Unfished spawning stock output for Rougheye/Blackspotted Rockfishes is estimated to be 5,647,660 million eggs (95% confidence interval: ,<1–12,581,189 million eggs). The management biomass target for Rougheye/Blackspotted Rockfishes is defined as 40% of the unfished spawning output ( 40%), which is estimated by the model to be 2,259,070 million eggs (95% confidence interval: <1–5,032,478 million eggs), which corresponds, in a theoretical equilibrium state, to an exploitation rate (catch / age 26+ biomass) of 0.048 (Table v, Figure viii). This harvest rate provides an equilibrium yield of 553 mt at 40% (95% confidence interval: 0–1,224 mt). Catch limits are determined by an SPR = 50% reference point which is associated with equilibrium exploitation rate of 0.040. The model estimate of maximum sustainable yield (MSY) is 592 mt (95% confidence interval: 0–1,311 mt). The estimated spawning stock output at MSY is 1,497,160 million eggs (95% confidence interval: 0–3,337,633 million eggs). The exploitation rate corresponding to the estimated FMSY proxy of SPR = 34% is 0.085.

Table v: Summary of reference points and management quantities, including estimates of the 95 percent confidence intervals. SO is spawning output, SPR is the spawning potential ratio, and MSY is maximum sustainable yield.

| Reference Point                                | Estimate    | Lower Interval | Upper Interval |
|------------------------------------------------|-------------|----------------|----------------|
| Unfished Spawning output                       | 5,647,680.0 | -1,285,908.2   | 12,581,268.2   |
| Unfished Age 26+ Biomass (mt)                  | 33,631      | -7,664         | 74,925         |
| Unfished Recruitment (R0)                      | 1,092       | -237           | 2,420          |
| 2025 Spawning output                           | 4,929,140   | -2,527,955     | 12,386,235     |
| 2025 Fraction Unfished                         | 0.873       | 0.615          | 1.130          |
| Reference Points Based SO40%                   | —           | —              | —              |
| Proxy Spawning output SO40%                    | 2,259,070   | -514,357       | 5,032,497      |
| SPR Resulting in SO40%                         | 0.458       | 0.458          | 0.458          |
| Exploitation Rate Resulting in SO40%           | 0.048       | 0.045          | 0.051          |
| Yield with SPR Based On SO40% (mt)             | 553         | -119           | 1,224          |
| Reference Points Based on SPR Proxy for MSY    | —           | —              | —              |
| Proxy Spawning output (SPR50)                  | 2,519,730   | -573,720       | 5,613,180      |
| SPR50                                          | 0.500       | —              | —              |
| Exploitation Rate Corresponding to SPR50       | 0.040       | 0.038          | 0.042          |
| Yield with SPR50 at SO SPR (mt)                | 526         | -113           | 1,165          |
| Reference Points Based on Estimated MSY Values | —           | —              | —              |
| Spawning output at MSY (SO MSY)                | 1,497,170   | -343,318       | 3,337,658      |
| SPR MSY                                        | 0.337       | 0.334          | 0.339          |
| Exploitation Rate Corresponding to SPR MSY     | 0.085       | 0.079          | 0.090          |
| MSY (mt)                                       | 592         | -127           | 1,311          |

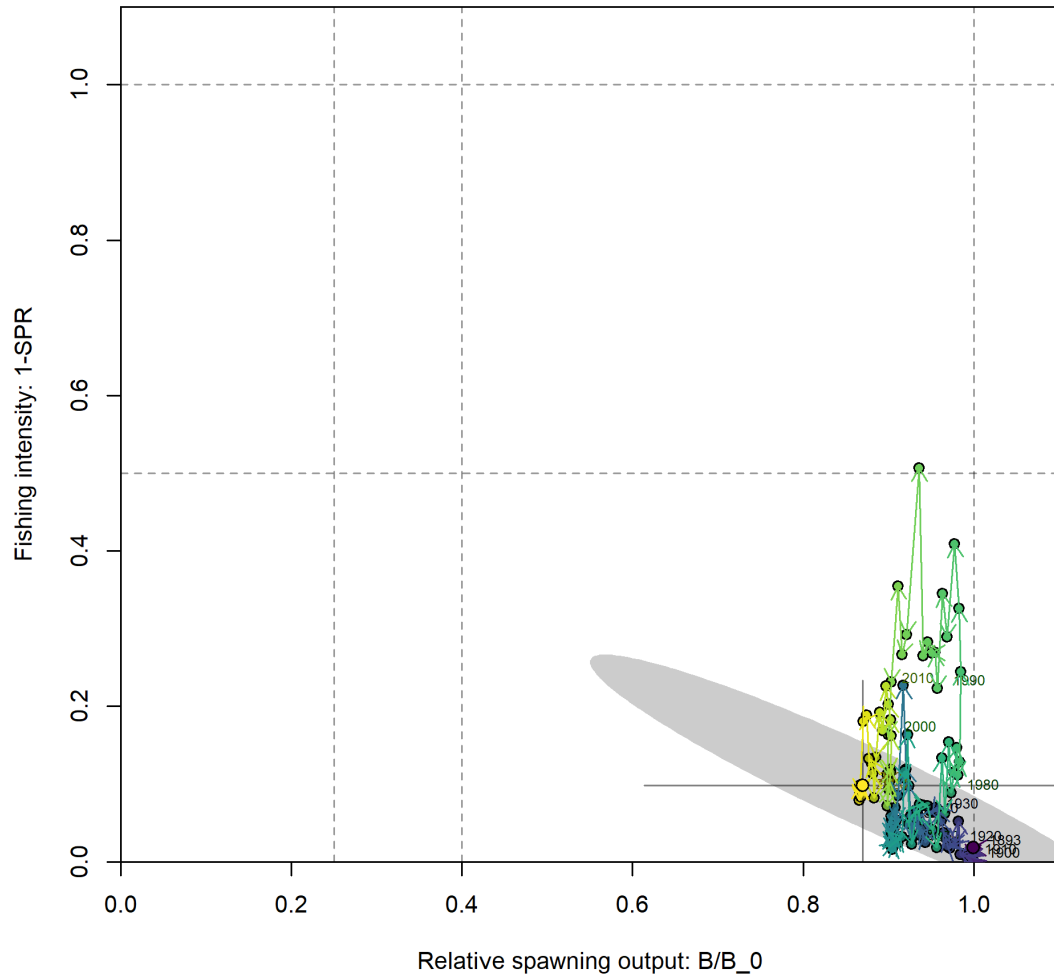


Figure vii: Phase plot of fishing intensity versus fraction unfished.

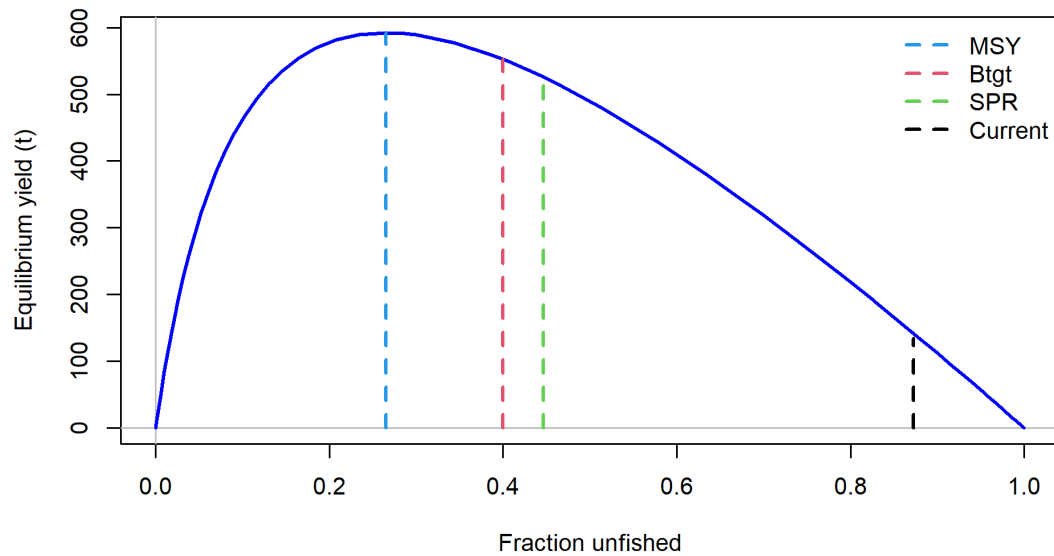


Figure viii: Equilibrium yield curve for the base case model. Values are based on the most recent fishery selectivities and retention curves and with steepness fixed at 0.72.

### 1.9 Management Performance

In the last ten years total dead catches of Rougheye/Blackspotted Rockfishes have been below the annual catch limit. The last ten years total dead catches for Rougheye/Blackspotted Rockfishes against the overfishing limits (OFLs), the acceptable biological catches (ABCs), the annual catch limits (ACLs) are shown in Table vi.

Table vi: Recent trend in the overfishing limits (OFL), the acceptable biological catches (ABCs), the annual catch limits (ACLs), and the total dead catch (landings + discards) all in metric tons (mt). Specifications are combined across north and south of 40°10'.

| Year | OFL (mt) | ABC (mt) | ACL (mt) | Catch (mt) |
|------|----------|----------|----------|------------|
| 2015 | 206      | 188.1    | 188.1    | 132.1      |
| 2016 | 211      | 192.7    | 192.7    | 149.0      |
| 2017 | 215      | 196.3    | 196.3    | 156.0      |
| 2018 | 219      | 199.9    | 199.9    | 241.5      |
| 2019 | 222      | 202.7    | 202.7    | 226.6      |
| 2020 | 224      | 204.5    | 204.5    | 106.1      |
| 2021 | 237      | 195.8    | 195.8    | 92.8       |
| 2022 | 238      | 194.7    | 194.7    | 117.4      |

|      |     |       |       |       |
|------|-----|-------|-------|-------|
| 2023 | 238 | 192.8 | 192.8 | 97.6  |
| 2024 | 238 | 194.7 | 194.7 | 118.9 |

### 1.10 Unresolved problems and major uncertainties

Rougheye/Blackspotted Rockfishes is relatively lightly exploited stock. The maximum catch amount of 744 metric tons was observed in 1995, and the average removals over the entire time period of exploitation is slightly over 100 metric tons. For comparison, Widow Rockfish catches for in the recent years were reaching over 12,000 metric tons, and in the early 1980s they were exceeding 28,000 metric tons. Most likely, catches of Rougheye/Blackspotted Rockfishes were not substantial because the complex is associated with steep slopes and boulder habitats, difficult to access with bottom trawl gears. Plus, it primarily occurs in the northern part of the assessed area (off Oregon and Washington), and is only rarely taken by fisheries in California. Given this relatively light exploitation, the scale of Rougheye/Blackspotted Rockfishes stock is uncertain, as we did not observe large contrasts in the stock dynamics. Such contrasts (“two-way trips”) provide information about carrying capacity, the intrinsic growth rate as well as scale of the population.

Through extensive explorations, we found that stock scale is sensitive to assumptions about shape of the selectivity curves, especially for the bottom trawl fleet. In the previous assessment all fleets (except for Triennial Survey) were assumed asymptotic. In this assessment, we provide flexibility for the bottom trawl and non-trawl fleets and bottom trawl surveys to estimate dome-shaped selectivity curves. Changes in selectivity assumptions allow to substantially improve fits to length and age composition data in fisheries and surveys, but also result in a substantial increase of stock scale.

Stock scale is also sensitive to assumptions about natural mortality. Rougheye/Blackspotted Rockfishes are one of the longest lived species of rockfish on the West Coast, with maximum ages of 169 years in this assessment. Individual as old as 205 years are reported in literature. It is, therefore, likely that natural mortality for Rougheye/Blackspotted Rockfishes is lower than for other rockfish species. With length and age data available only for years after 1994, there is limited ability to monitor the long-term changes of aging cohorts. Therefore, estimates of natural mortality are uncertain. This assessment attempts to capture uncertainty in this parameter and integrate it into the derived biomass estimates by estimating natural mortality for females, while fixing the male natural mortality at the value from meta-analytical study.

### 1.11 Scientific Uncertainty

The model estimated uncertainty ( $\sigma$ ) around the 2025 spawning output for is 0.6836465. The uncertainty around the OFL is 0.6713586. These values underestimate the overall uncertainty as they do not incorporate the model structural uncertainty and do not account for any time-varying dynamics other than recruitment. The estimated uncertainty values

are higher than the Category 1 default of 0.5, so all projections will use the estimated ( $\sigma$ ).

#### 1.12 Harvest Projections and Decision Tables

Projections of the overfishing limit, acceptable biological catch, and annual catch limit, all based on a  $P^*$  of 0.45 and a log-space standard deviation of the overfishing limit ( $\sigma$ ) of 0.68 (as estimated in this model) are included in Table [vii](#). Assumed catches for 2025 and 2026 for this projection were provided by the PFMC Groundfish Management Team, and catches from 2027 onward assume full attainment of the acceptable biological catch.

No decision table needed in draft assessments undergoing review.



Table vii: Potential OFLs (mt), ABCs (mt), ACLs (mt), the buffer between the OFL and ABC, estimated spawning output, and fraction unfished with adopted OFLs and ACLs and assumed catch for the first two years of the projection period.

| Year | Adopted<br>OFL (mt) | Adopted<br>ACL (mt) | Assumed<br>Catch<br>(mt) | OFL (mt) | Buffer | ABC (mt) | ACL (mt) | Spawning<br>output | Fraction<br>Unfished |
|------|---------------------|---------------------|--------------------------|----------|--------|----------|----------|--------------------|----------------------|
| 2025 | —                   | —                   | 141                      | —        | —      | —        | —        | 4,929,140.000      | 0.873                |
| 2026 | —                   | —                   | 174                      | —        | —      | —        | —        | 4,945,630.000      | 0.876                |
| 2027 | —                   | —                   | —                        | 980      | 0.913  | 894      | 894      | 4,960,900.000      | 0.878                |
| 2028 | —                   | —                   | —                        | 968      | 0.908  | 878      | 878      | 4,888,090.000      | 0.866                |
| 2029 | —                   | —                   | —                        | 955      | 0.902  | 862      | 862      | 4,818,790.000      | 0.853                |
| 2030 | —                   | —                   | —                        | 944      | 0.896  | 846      | 846      | 4,752,560.000      | 0.842                |
| 2031 | —                   | —                   | —                        | 932      | 0.890  | 830      | 830      | 4,688,990.000      | 0.830                |
| 2032 | —                   | —                   | —                        | 921      | 0.885  | 815      | 815      | 4,627,620.000      | 0.819                |
| 2033 | —                   | —                   | —                        | 909      | 0.879  | 799      | 799      | 4,567,870.000      | 0.809                |
| 2034 | —                   | —                   | —                        | 898      | 0.874  | 784      | 784      | 4,509,380.000      | 0.798                |
| 2035 | —                   | —                   | —                        | 886      | 0.868  | 769      | 769      | 4,451,710.000      | 0.788                |
| 2036 | —                   | —                   | —                        | 874      | 0.863  | 755      | 755      | 4,394,830.000      | 0.778                |

### 1.13 Research and Data Needs

There are many areas of research that would benefit the understanding and assessment of Rougheye/Blackspotted Rockfishes. Below, we identify several of them that we consider particularly important.

- **Understanding the stock structure and biology of Rougheye and Blackspotted Rockfishes:** This assessment reports the status of Rougheye/Blackspotted Rockfishes as a pooled complex because it is extremely difficult to separate the catches of each species even in recent data, and attempting to do so would greatly increase the uncertainty in the predictions. Because little is known about the respective biology and catch histories of the two species, it is unclear whether managing them as a complex may place one species at disproportionate risk of overfishing relative to the other. Additional research that will provide insight into the distribution, life history, biological characteristics, and catch and discard profiles of the two species is recommended. Such an endeavor would like require the efforts of at sea observers in all fleets, biologists aboard fishery-independent surveys, and port samplers along the entire West Coast requiring broad, inter-agency collaboration.
- **Understanding of coastwide stock structure, connectivity, and distribution:** This is a stock assessment for Rougheye/Blackspotted Rockfishes off of the west coast of the U.S. and does not consider data from British Columbia or Alaska. Further investigating and comparing the data and predictions from British Columbia and Alaska to determine if there are similarities with the U.S. West Coast observations would help to define the connectivity between rougheye rockfish north of the U.S.-Canada border.
- **Natural mortality:** Uncertainty in natural mortality translates into uncertain estimates of status and sustainable fishing levels for Rougheye/Blackspotted Rockfishes. Assessment model was able to estimate female natural mortality, consistent with maximum age based meta-analytical prior, but male natural mortality was fixed in the model. The collection of additional age data and further research of the life-history of Rougheye/Blackspotted Rockfishes may improve our understanding of Rougheye/Blackspotted Rockfishes natural mortality.
- **Historical landings and discards, including investigation of fisheries selectivity:** The substantial progress has been made in reconstructing historical landings of rockfishes on the U.S. West Coast, including those for Rougheye/Blackspotted Rockfishes. This assessment highlighted the importance of understanding of fishery selectivity assumptions associated with removals and how fishery selectivity changes throughout the years. Further understanding of this area would help reduce uncertainty in estimated scale of the stock.

## 1 Introduction

Rougheye (*Sebastes aleutianus*) and Blackspotted (*Sebastes melanostictus*) rockfishes are two species that form one management complex.

Rougheye rockfish are a long-lived rockfish named after a series of 2-10 spines along the lower rim of their eyes. They have also been called blackthroat or blacktip rockfish (Love, Yoklavich, and Thorsteinson 2002; Love 2011). Blackspotted rockfish are distributed in similar locations as rougheye rockfish and it is very difficult to visually distinguish the two species. These two species may hybridize on occasion (Love 2011).

It has only been from recent genetic studies that these two separate species have been identified (A. J. Gharrett et al. 2006; S. L. Hawkins et al. 2005) and have had phenotypic characteristics useful for identifying the species in the field identified (A. J. Gharrett et al. 2006; Orr and Hawkins 2008). Before then, data are available for one species called Rougheye Rockfish which included Rougheye Rockfish and Blackspotted Rockfish. Due to the difficulty in distinguishing these two species and the lack of historical separation of the species in all of the data, this assessment combines any data for Blackspotted Rockfish with Rougheye Rockfish into Rougheye/Blackspotted Rockfishes and provides management advice for the two species combined. These species are also closely related to Shortraker Rockfish (*Sebastes borealis*) and are sometimes difficult to distinguish from Shortraker Rockfish without looking at the gill rakers.

### 1.1 Stock Structure

There are at least two questions to think about when considering stock structure for Rougheye/Blackspotted Rockfishes when doing a stock assessment.

- Since Rougheye and Blackspotted Rockfishes are two different species, can they be separated as two stocks and conduct separate assessments?

Rougheye rockfish were first described in 1811 as *Perca variabilis* by German zoologist Peter Simon Pallas (Jordan and Evermann 1898), and assigned to various taxa at least 15 times since (Love, Yoklavich, and Thorsteinson 2002). Some descriptions noted both light and dark color morphs, which, along with possible confusion with several morphologically similar co-occurring species (e.g., *Sebastes borealis* and *Sebastes melanostomus*) have contributed to the persistent ambiguity in formal descriptions of Rougheye Rockfish (Orr and Hawkins 2008). The first genetic studies conducted in the late 1960s and early 1970s (Tsuyuki et al. 1968; Tsuyuki and Westrheim 1970) observed diversity suggestive of two genetic types within specimens identified as Rougheye Rockfish. Allozyme studies

conducted over the next two decades (Seeb 1986; S. Hawkins, Heifetz, and Pohl 1997; S. L. Hawkins et al. 2005) provided additional evidence suggesting two separate genetic types within field-identified Rougheye Rockfish. Genetic variation between the two types, supported by both nuclear and mitochondrial DNA, was determined to be sufficiently conclusive to separate two species: “Type I” and “Type II” Rougheye Rockfish (Anthony J. Gharrett et al. 2005). Meristic and morphometric comparisons of the two species suggested certain characters, such as gill raker counts and length, snout length, anal base length, and pectoral fin base, were significantly different, and in combination could reliably, though not definitively, distinguish between the species (A. J. Gharrett et al. 2006). The two separate species were formally re-described by Orr and Hawkins (2008) with the Type II group retaining *Sebastes aleutianus* and the common name Rougheye Rockfish. Blackspotted Rockfish was proposed as the common name for the Type I group along with the scientific name of *Sebastes melanostictus*, re-establishing nomenclature from one of the species complex’s earlier descriptions (Matsubara 1934).

These two species remain difficult to consistently differentiate visually in the catch, thus are still commonly reported and treated as a species complex. Otolith morphometrics (e.g., shape, size, weight) have shown some promise in possibly identifying these species in Alaskan waters (97.3% Blackspotted and 86.2% of Rougheye rockfishes were accurately identified) and possibly using older otoliths to break out historical information by species (Harris, Hutchinson, and Wildes 2019). Frey et al. (in prep.) provided insight into the ability of the Northwest Fisheries Science Center West Coast Groundfish Bottom Trawl Survey biologists to identify the two species, with 90% of genetically identified Rougheye rockfish being correctly identified in the field. When mis-identifications occurred, it was usually a Blackspotted rockfish being mis-identified as a Rougheye rockfish. There were few mis-identifications when a fish was identified as a Blackspotted rockfish. While this is promising for potential future species-specific data coming from the survey, it does not alleviate the historical problem of separating fishery data into the two species. Frey et al. (in prep.) therefore also considered whether ecological factors like depth or latitude could help separate samples by species. They found that both species occur within the range of this assessment’s considered areas (California to Washington), and heavily spatially overlap. Interestingly, there seem to be relative hot spots for these species where one species is more common than the other, and in general, Rougheye Rockfish seems to be more common than Blackspotted Rockfish (however, Blackspotted Rockfish may be the more common of the two in parts of Alaska; Anthony J. Gharrett et al. (2005); S. L. Hawkins et al. (2005); Orr and Hawkins (2008)).

Despite recent advances in species identification of Rougheye/Blackspotted Rockfishes as genetically distinct species, there is little ability to separate current or historical fishery data reliably in order to separate these two species into two stocks. Therefore, this assessment maintains a species complex approach, though given absolute presence off the U.S. West Coast, this may be considered more of a Rougheye than Blackspotted Rockfish stock assessment. We also note that throughout the range of these stocks, all current assessments to this point have maintained a species complex approach. While we treat

these species as one assessed stock complex, we recognize and are mindful of the above species distinctions as we conduct our analyses.

- Both species range into Canada and Alaska – are they one stock?

While genetics studies have focused mostly on identification of the two species, little is known about the population structure of either species. This assessment and the 2013 assessment (Hicks, Wetzel, and Harms 2013) represent the most southerly range of these species. Comparing the absolute abundance of the 2013 assessment to the most current estimates of the Alaskan stocks, the absolute number in this southerly range is much smaller than in the Gulf of Alaska (GOA), but higher than in the Bering Sea/Aleutian Island (BSAI) stock (Figure 2). The two smaller stocks have similar trend of decline and stabilization, whereas the higher biomass GOA stock looks to have not dropped at all over the time period considered (Figure 3). We assume here that the west coast stocks of Rougheye/Blackspotted Rockfishes are distinct management units from those in Alaska.

## 1.2 Distribution

Rougheye/Blackspotted Rockfishes range from northern California up to and throughout Alaska and into Japan (Anthony J. Gharrett et al. 2005; S. L. Hawkins et al. 2005; Orr and Hawkins 2008). Both are long-lived (>100 years), with Rougheye Rockfish having the distinction of the oldest ever aged *Sebastes* species at 205 years old. They both greatly overlap in latitude and depth (shallower than 100 m to at least 439 m), and are generally considered slope rockfish, with an ontogenetic shift from shallower to deeper, and adults commonly found at 360 m (around 200 fathoms). Rougheye seems to be proportionally more abundant when survey samples are genetically identified, and Blackspotted Rockfish tend to be found, on average, deeper than Rougheye Rockfish (S. L. Hawkins et al. 2005; Orr and Hawkins 2008).

Rougheye/Blackspotted Rockfishes are often associated with structure, such as hard, rocky bottoms and steep habitats. They are rarely found on the deep flats. They can be found alone or in aggregations (Love, Yoklavich, and Thorsteinson 2002), with aggregations often differentiated by age. Younger fish may school and are often found in shallower waters on the shelf, juveniles and subadults can be found together, and larger fish may form larger aggregations in the Pacific Northwest during the autumn and winter. These two species may also hybridize on occasion (Love 2011). These species are closely related to Shortraker Rockfish (*S. borealis*) and are sometimes difficult to distinguish from Shortraker Rockfish without looking at the gill rakers. One major distinguishing feature of Rougheye Rockfish are the 2–10 spines along the lower rim of their eyes, hence the common name “rougheye”.

### 1.3 A Map Showing the Scope of the Assessment

This assessment treats the U.S. Rougheye/Blackspotted Rockfishes resource from the Mexican border to the Canadian border as a single coastwide stock (Figure 1). The U.S.-Canadian border is the northern boundary for the assessed stock, although the basis for this choice is due to political and current management needs rather than the population dynamics. The assessment excludes consideration of the Puget Sound and Salish Sea.

### 1.4 Life History

Like all *Sebastes* species, Rougheye/Blackspotted Rockfishes give birth to live young. Larvae released has been documented between February and June and extrusion lengths are between 4.5-5.3 mm (Love, Yoklavich, and Thorsteinson 2002). Dick et al. (2017) showed that rockfishes exhibit a non-proportional relationship of fecundity to weight, with larger individuals producing more eggs than expected only by weight. Although Neither Rougheye or Blackspotted Rockfishes have a species- or subfamily-specific estimate for this relationship, this stock assessment uses the unobserved Genus *Sebastes* values to inform fecundity to weight relationship for Rougheye/Blackspotted Rockfishes.

A wide range of prey items make up the diet of Rougheye/Blackspotted Rockfishes. Crangid and pandalid shrimps make up the majority of their diets, and larger individuals, greater than 30 cm, feeding upon other fishes (Love 2011). They are also known to feed upon gammarid amphipods; mysids, crabs, polychaetes, and octopuses (Love, Yoklavich, and Thorsteinson 2002; Love 2011).

### 1.5 Ecosystem Considerations

### 1.6 Historical and Current Fishery Information

Rougheye/Blackspotted Rockfishes are not targeted by a specific fishery, but are desirable and marketable, thus are typically retained when captured. They are often caught in bottom trawl, mid-water trawl, and longline fisheries. Small numbers have been observed in pot, shrimp, and recreational fisheries.

Longline catches of Rougheye/Blackspotted Rockfishes are present from the turn of the century and continue in recent years, targeting sablefish and halibut.

After many attempts to start trawl fisheries off the west coast of the United States in the late 1800's, the availability of the otter trawl and the diesel engine in the mid-1920's

helped the trawl fisheries expand (Douglas and Division 1998). The trawl fisheries became established during World War II when demand increased for bottomfish. A mink food fishery also developed during World War II (Jones and Harry 1961), and post-war catches for rockfishes, including Rougheye/Blackspotted Rockfishes, increased (Niska 1976). Between mid-1960s and mid-1970s, foreign trawl fleets from the former Soviet Union, Japan, Poland, Bulgaria and East Germany targeted aggregations of Pacific ocean perch in the Northeast Pacific Ocean, in the waters off the U.S. West Coast (Love, Yoklavich, and Thorsteinson 2002), until the EEZ was implemented in 1977 (J. B. Rogers 2003).

Also, large-scale harvesting of Pacific hake in the United States began in late-1960s, when factory trawlers from the Soviet Union and other countries began targeting this stock. After the 200-mile U.S. Rougheye/Blackspotted Rockfishes is commonly caught in this fishery. Exclusive Economic Zone was declared in 1977, a Joint-Venture fishery was initiated between United States trawlers and Soviet factory trawlers acting as mother-ships (larger, slower ships for fish processing and storage while at sea). By 1989 the U.S. fleet capacity had grown to a level sufficient to harvest the entire quota, and no further foreign fishing was allowed.

Since 1977, landings of rockfish were higher until management restrictions were implemented in 2000. Rougheye/Blackspotted Rockfishes inhabit deeper water as adults, which were fished less often historically. More detailed information of the fisheries by state is given in Section 2.1, where the reconstructed catches are discussed. The catches by state in fleets as well as for the Pacific whiting at-sea fleet are shown in Figure 5.

## 1.7 Summary of Management History and Management Performance

Rougheye/Blackspotted Rockfishes has been a small component of groundfish fisheries and catches of Rougheye/Blackspotted Rockfishes have been governed by restrictions on assemblages of species, of which these species are a member. However, the distribution of fishing effort in areas where Rougheye/Blackspotted Rockfishes might be encountered has also been affected by catch restrictions on co-occurring, rebuilding species, as well as associated area closures instituted to promote rebuilding. The first imposed landings limits on a coastwide *Sebastes* complex (Rougheye/Blackspotted Rockfishes being one of the 50 rockfishes in the complex) were instituted in 1983.

This complex was divided in to two management areas north and south of 43°00' N in 1994. Ongoing concern that shelf and slope rockfishes may be undergoing overfishing led the attempt by J. S. Rogers et al. (1996) to describe the status of most rockfishes in the *Sebastes* complex. Rougheye/Blackspotted Rockfishes information content was low. To estimated exploitation rates of Rougheye Rockfish J. S. Rogers et al. (1996) assumed fishing mortality being to equal to natural mortality and used AFSC/NWFSC Triennial

Shelf Survey to calculate an average biomass. The analysis found that the stock was undergoing very high exploitation rates in both management areas.

The dividing line between the northern and southern management areas was shifted to 40°10' N latitude in 1999 and the *Sebastes* complex was subsequently divided into nearshore, shelf, and slope complexes in 2000. Rougheye Rockfish has been managed under trip limits for the minor slope rockfish complex in both north and south management areas since this time.

Table 1 summarizes major management changes since 2000. Some important changes include the implementation of Rockfish Conservation Areas (RCA's) in 2002, the beginning of trawl rationalization in 2011, and the lifting of the RCAs beginning in 2020 with the removal of the trawl RCA in Oregon and California and loosening restrictions in the non-trawl RCAs in 2023 and 2024.

Though managed as part of a complex, OFL contributions for Rougheye/Blackspotted Rockfishes were calculated using DB-SRA in 2010 for the 2011-2012 management cycle. This led to the observation that recent catches had frequently exceeded the OFL contribution estimated using data-poor, catch-only methods provided a strong indication that a more thorough evaluation of Rougheye/Blackspotted Rockfishes stock status and sustainable harvest levels be undertaken, using all available data. A full assessment of Rougheye/Blackspotted Rockfishes was undertaken in 2013 and indicated the stock complex was above management target levels (Hicks, Wetzel, and Harms (2013)). Recent management performance for Rougheye/Blackspotted Rockfishes is provided in Table 2 for north and south of 40°10'. Rougheye/Blackspotted Rockfishes is managed as a part of the slope rockfish complex.

## 1.8 Fisheries off Canada and Alaska

Rougheye/Blackspotted Rockfishes are distributed throughout Canada and Alaska and are commonly caught in trawl and longline fisheries. Alaska conducts assessments biennially for the Rougheye/Blackspotted complex, and two have been recently done: one for the Bering Sea and Aluetian Islands (Spencer, Ianelli, and Laman 2003) and the other for the Gulf of Alaska (Sullivan et al. 2023). Canada completed an assessment in 2020 (Starr and Haigh 2020). The fisheries and assessments for each country are described below.

Rougheye rockfish have been managed as a bycatch only species in Alaska since 1991 with catches ranging between 130 and 2,418 mt and peaking in the late 1980s and early 1990s (Sullivan et al. 2023). Generally, about 55-75% of the catch are trawl-caught and 30-45% from hook-and-line (mainly, longline) fisheries. Since 2017 the move to pot gear in the sablefish fishery has decreased the longline catches. Discards since 2013 have



ranged from 11.6% (in 2023) and 45% (in 2018). The Rougheye/Blackspotted complex catch levels generally are between 20% and 60% of the Total Allowable Catch since the 2005 when the complex began to be managed separately. The most recent age-structured integrated stock assessments of this complex in the Bering Sea and Aluetian Islands ([Spencer, Ianelli, and Laman 2003](#)) and for the Gulf of Alaska ([Sullivan et al. 2023](#)) do not indicate either overfishing or the stocks being overfished.

Canada identified two species of Rougheye Rockfish (Type I and Type II) in 2007 and designated both species of special concern, which means that they may become threatened or endangered because of a combination of biological characteristics and identified threats ([Report 2007](#)). This designation was given because biomass estimates are uncertain and no strong trends are observed, there is evidence of truncation of the age distribution and overall mortality has doubled, it is a long-lived, low-fecundity *Sebastes* species, which is susceptible to population collapse and slow recovery, and because the difficulty in separating the two species may result in potential impacts on one of the species going unnoticed. Subsequently, the species were identified as rougheye rockfish and blackspotted rockfish and a management plan was created in 2012 with a goal of sustaining the populations of rougheye and blackspotted rockfishes ([Canada 2012](#)). Five high priority and seven low priority actions have been identified to address the threats to the populations and support the management goal.

The first Canadian stock assessment for these species, using a integrated catch-at-age model, was conducted in 2022 to estimated stock status of two Rougheye/Blackspotted (REBS) rockfishes management units (REBS north and REBS south) a the beginning of 2021. The REBS north stock was in the healthy zone in the reference model. The REBS south stock was likely in the healthy zone, but with an elevated possibility of being in the cautious zone.

## 2 Data

Data from a wide range of sources were evaluated within this assessment. Data sources included in the assessment model are summarized in Figure 4. Description of each data source used in the model is provided below.

### 2.1 Fishery-dependent data

Rougheye/Blackspotted Rockfishes are not targeted by a specific fishery, but are desirable and marketable, thus are typically retained when caught. They are often captured in bottom trawl, mid-water trawl, and longline fisheries. They are also commonly bycaught within the at-sea hake fishery. Small numbers have been observed in pot and shrimp trawl. Recreational catch is inconsequential and not accounted for in this assessment.

Rougheye/Blackspotted Rockfishes fishery-dependent data in this assessment are divided among six fleets, which include:

- Fleet 1: Commercial bottom trawl fishery.
- Fleet 2: Dead discard from bottom trawl fishery.
- Fleet 3: Commercial non-trawl (mainly the long-line) fishery.
- Fleet 4: Dead discard from non-trawl fishery.
- Fleet 5: Contemporary mid-water trawl fishery.
- Fleet 6: At-sea hake fishery bycatch.

For details on fleet structure, please refer to Section 3.4.3.

#### 2.1.1 Commercial Fishery Landings

Recent and historical fisheries catches were compiled by state and then combined into the fishing fleets used in the assessment. Time series of catches by fleet are reported in Table 3 and shown in Figure 5. The landings by state are given in Table 4 for bottom trawl, Table 5 for non-trawl, and Table 6 for mid-water trawl fleets.

##### 2.1.1.1 Recent landings

Recent commercial landings of Rougheye/Blackspotted Rockfishes (2000–2024 for Washington, 1987–2024 for Oregon and 1981–2024 for California,) were obtained from Pacific

[Fisheries Information Network \(PacFIN\)](#), a regional fisheries database that manages fishery-dependent information in cooperation with West Coast state agencies and National Marine Fisheries Service (NMFS). Catch data were extracted from PacFIN on April 24, 2025.

#### 2.1.1.2 Historical Landings

Historical landings of Rougheye/Blackspotted Rockfishes were reconstructed by state.

The Washington historical landings (1889–2000) of Rougheye/Blackspotted Rockfishes were provided by Washington Department of Fish and Wildlife (WDFW), who recently conducted historical catch reconstruction for rockfish species, including Rougheye/Blackspotted Rockfishes (pers. comm. T. Tsou, WDFW). The three main sources used in this reconstruction included the US Fish Commission Report (UFSC), Washington Bound Volumes, and Washington Statistical Bulletin. The historical species composition was based on the various historical reports and interviews of fishermen and dockside samplers. The landings between 1981 and 2000 were also provided by WDFW (rather than obtained from PacFIN), since WDFW developed and used a improved method for apportioning unidentified rockfish (URCK) category in fish tickets to the individual species landings. This improved approach relaxed the borrowing rules for missing data used in the WDFW species allocation algorithm that feeds into PacFIN (pers. comm. T. Tsou, WDFW). New Washington historical landings represent improvement to the assessment.

The Oregon historical landings (1896–1986) were obtained from Oregon historical catch reconstruction, conducted by Oregon Department of Fish and Wildlife (ODFW) in collaboration with NWFSC (Karnowski, Gertseva, and Stephens (2014)). The Oregon landings for the period between 1987 and 1999 were also provided by the ODFW. For that period, Oregon PacFIN landings were supplemented with the additional estimates of Rougheye/Blackspotted Rockfishes landings reported within unspecified rockfish market categories (i.e., URCK and POP1; Fish and Wildlife (2017)).

The California historical landings were informed by several sources. Landings from the most recent “historical” period (between 1969 and 1980) were obtained from the California Cooperative Survey (CalCOM) database. Earlier landing records (between 1931 and 1968) were informed by the rockfish historical catch reconstruction conducted by the NOAA’s Southwest Fisheries Science Center (Ralston et al. (2010)).

Comparison of Rougheye/Blackspotted Rockfishes historical landings by state and fleet between this and 2013 assessment is provided in Figure 6. The largest differences in this assessment from 2013 model are in Washington landings (Figure 9), with newly estimated landings being generally lower than those used in previous assessment. The new WDFW catch reconstruction completed by WDFW is considered an improvement.

Historical California and Oregon landings did not change substantially (Figure 7 and Figure 8), with the exception of a few years. Discrepancies in California and Oregon non-trawl landings between the 2013 and 2025 assessments are caused by the fact that non-trawl fleet in 2013 assessment was limited to only fixed gear, when in 2025 assessment non-trawl fleet includes all non-trawl gear groups. Slight discrepancies in Oregon trawl landings between 1987 and 1999, are from adding previously non-reported landings of Rougheye/Blackspotted Rockfishes in the unspecified rockfish market categories (see details above).

The update in historical changes shows only minor differences in model outputs (Figure 37; Figure 38).

#### 2.1.1.3 Bycatch in the foreign POP fishery

Between mid-1960s and mid-1970s, foreign trawl fleets from the former Soviet Union, Japan, Poland, Bulgaria and East Germany targeted aggregations of Pacific ocean perch in the Northeast Pacific Ocean, in the waters off the U.S. West Coast (Love, Yoklavich, and Thorsteinson 2002). J. B. Rogers (2003) estimated removals of rockfish species caught within this foreign POP fishery, including removals of Rougheye/Blackspotted Rockfishes. In the assessment, Rougheye/Blackspotted Rockfishes bycatch in the foreign POP fishery between 1966 and 1976 as estimated by J. B. Rogers (2003) were added to commercial bottom trawl fleet (Table 4).

#### 2.1.1.4 At-Sea Hake Catches

Rougheye/Blackspotted Rockfishes has long been bycaught in the fishery for the coastal population of Pacific hake, which is almost exclusively conducted with mid-water trawls.

Large-scale harvesting of Pacific hake in the United States began in late-1960s, when factory trawlers from the Soviet Union and other countries began targeting this stock. After the 200-mile U.S. Exclusive Economic Zone was declared in 1977, a Joint-Venture fishery was initiated between United States trawlers and Soviet factory trawlers acting as mother-ships (larger, slower ships for fish processing and storage while at sea). By 1989 the U.S. fleet capacity had grown to a level sufficient to harvest the entire quota, and no further foreign fishing was allowed. The Pacific hake fishery is currently 100% observed by the at-sea hake observer program (A-SHOP) and data on bycatch species, including Rougheye/Blackspotted Rockfishes, is being routinely collected.

Annual amounts of Rougheye/Blackspotted Rockfishes bycatch (retained and discarded) in the Pacific hake fishery were obtained from the North Pacific Database Program (NORPAC). That time series covers the period between 1977 and 2024 and include

catches by foreign and domestic fisheries as well as removals during the time of Joint Ventures. Rougheye/Blackspotted Rockfishes catches within the at-sea hake fishery were treated in the model as a separate fleet (Table 4).

### 2.1.2 Discards

#### 2.1.2.1 Historical discard

Historically, little to no discarding was observed for Rougheye/Blackspotted Rockfishes.

The historical discard information comes from Pikitch, Erickson, and Wallace (1988), often referred to as the Pikitch study. The Pikitch study was conducted between 1985 and 1987 between 48°42' and 42°60' N. latitude (Pikitch, Erickson, and Wallace 1988). Participation in the study was voluntary and included vessels using bottom, mid-water and shrimp trawl gears. Observers of normal fishing operations on commercial vessels collected the data, estimated the total weight of the catch by tow, and recorded the weight of species retained and discarded in the sample.

There are no mid-water trawl records of Rougheye/Blackspotted Rockfishes in the Pikitch, Erickson, and Wallace (1988), and only few fish records of bottom trawl catches, based on which discard rate (discard weight over total weight) for bottom trawl was just 0.09%. Therefore, no historical discard was assumed in the model.

#### 2.1.2.2 Recent Discard

With the introduction of trip limits for rockfish in early 2000, limited discard has been observed for Rougheye/Blackspotted Rockfishes in bottom trawl and non-trawl fisheries.

In 2002, the West Coast Groundfish Observer Program (WCGOP) was implemented on the West Coast of the United States, which began with gathering bycatch and discard information for the limited entry trawl and fixed gear fleets. Observer coverage has expanded to include the California halibut trawl, the nearshore fixed gear and pink shrimp trawl fisheries. Since 2011, trawl fisheries have been managed with catch shares under a system of annual individual fishing quotas (IFQs) for the shoreside sector (i.e., vessels delivering to shoreside processors) and harvest cooperatives for the at-sea hake sectors (catcher-processors who catch and process hake at sea; and Motherships, factory processors that take delivery of hake from catcher vessels at sea). Constant monitoring of catch using observers or electronic monitoring (EM) is required to participate in the trawl catch share fishery.

The discard amounts of Rougheye/Blackspotted Rockfishes for the period between 2002 and 2023 were obtained from WCGOP by year and fleet (bottom trawl, mid-water trawl and non-trawl), for both the catch share and the non-catch share sector. The discarding amounts of Rougheye/Blackspotted Rockfishes within bottom trawl and non-trawl fleets were included in the model as separate fleets.

Mid-water trawl discard was not present in non-catch share sector and was extremely minimal (virtually non-existing) in catch-share sector, with discard amounts averaging to 10 kg per year. Therefore, in the model, no discard was assumed for mid-water trawl fleet.

#### 2.1.2.2.1 Bottom Trawl Discard

Bottom trawl discard amounts by year are provided in Table 3 and shown in Figure 5. Prior to 2011, before the start of the IFQ program, the discard of Rougheye/Blackspotted Rockfishes ranged between 1 metric ton and 60 metric tons, averaging at 23 metric tons a year. After 2011, the discard has been very low, not exceeding 0.5 metric ton a year. No discard data were available for 2024, and we used the average discard amount for 2019 - 2023 period to approximate 2024 discards for bottom trawl discard fleet.

#### 2.1.2.2.2 Non-Trawl Discard

Non-trawl discard amounts by year are provided in are provided in Table 3 and shown in Figure 5. Non-trawl discard of Rougheye/Blackspotted Rockfishes were made in both catch share and non-catch share sectors. Discard amounts in these sectors were combined by year to represent total discard within the fleet. The discards within this fleet ranged between 0.5 metric ton and 35 metric tons, with 10 metric tons as average per year. No discard data were available for 2024, and the 2023 discard amount was assumed for 2024 for non-trawl discard fleet.

### 2.1.3 Fishery Length and Age Data

Length bins from 10 to 80 cm in 2 cm increments were used to summarize the length frequency of the catches in each year. The first length bin includes all observations less than 10 cm and the last bin includes all fish 80 cm and longer. Age distributions included bins from age 1 to age 100, with the first bin including all fish ages 0 and 1 and the last bin including all fish age 100 and above.

### 2.1.3.1 Commercial Landings Length and Ages

The fishery length and age data for bottom trawl, non-trawl and mid-water trawl fleets, based on samples collected by port samplers, were obtained from the PacFIN Biological Data System (BDS) database and extracted on April 24, 2025. The number of trips and fish sampled for lengths and ages by state and year are summarized in Table 7 through Table 10, by fishing fleet.

Commercial length-frequency distributions were developed for each fleet and year, for which observations were available

Females and males distributions were treated separately, to track sex-specific differences. For each fleet, the raw observations were expanded to the trip level, to account for differences in samples sizes relative catch weights among trips (first stage expansion). The expanded length observations were then further expanded to state level, to account for differences in sampling intensity of Rougheye/Blackspotted Rockfishes landings among states combined into a single fleet (second stage expansion). The expansion algorithm can be illustrated with the following equation:

$$N_{b,j,y} = \sum_{s=1}^{s=k} \sum_{t=1}^{t=n} L_{b,j,t} \cdot \left( \frac{LC_t}{SC_t} \right) \cdot \left( \frac{LC_{s,y}}{SC_{s,y}} \right)$$

Where  $N_{b,j,y}$  is the number of lengths in each length bin ( $b$ ) by sex ( $j$ ) and year ( $y$ ) within each fleet.  $L_{b,j,t}$  represents an individual length sample by bin ( $b$ ) and sex ( $j$ ) within an individual fishing trip ( $t$ ). In the first stage expansion,  $L_{b,j,t}$  was multiplied by the ratio of landed catch ( $LC_t$ ) within that trip ( $t$ ) to a portion of catch sampled for lengths ( $SC_t$ ) within the same trip ( $t$ ). In the second stage expansion, the individual length sample ( $L_{b,j,t}$ ) was multiplied by the ratio of landed catch ( $LC_{s,y}$ ) within individual state ( $s$ ) and year ( $y$ ) to catch weights sampled for lengths ( $SC_{s,y}$ ) within the same state ( $s$ ) and year ( $y$ ). As the final step, the expanded length samples from the same size bin and sex were summed across all trips and states (combined into a single fleet) within a single year, to obtain the total number of lengths in each length bin by sex, year and fleet ( $N_{b,j,y}$ ). The same calculations were repeated for each length bin, to develop sex specific length frequencies for each fishing fleet by year.

Age distributions were included in the model as conditional-ages-at-length (CAAL) observations. The marginal age compositions were also included, but only for evaluating the implied fits, while the CAAL data were used in the likelihood. The CAAL data were not expanded and were binned according to length, age, sex, and year.

The filtering and processing of the PacFIN length and age composition data were conducted using the Pacific Fisheries Information Network tools (PacFINtools) package in R (Wetzel, Johnson, and Hicks 2025). The filtering steps included removing samples with missing vital information.

Figure 10 through Figure 16 show length frequencies for bottom trawl, non-trawl and mid-water fleets by year, and Figure 21 through Figure 25 show the commercial CAAL distributions by year for the same fleets.

The initial input values for length compositions in this assessment were calculated as a function of the number of trips and number of fish via the Stewart Method (pers.comm. I. Stewart, International Pacific Halibut Commission (IPHC)). The method is based on analysis of the input and model derived effective sample sizes from West Coast groundfish stock assessments. A piece-wise linear regression was used to estimate the increase in effective sample size per sample based on fish-per-sample and the maximum effective sample size for large numbers of individual fish. The resulting equations are:

$$\text{Input effN} = N_{\text{trips}} + 0.138 * N_{\text{fish}} \text{ if } N_{\text{fish}}/N_{\text{trips}} \text{ is } < 44$$

$$\text{Input effN} = 7.06 * N_{\text{trips}} \text{ if } N_{\text{fish}}/N_{\text{trips}} \text{ is } \geq 44$$

The input sample size of CAAL data was set at the number of fish at each length by sex and by year.

#### 2.1.3.1.1 Commercial Discard Lengths

Discard length composition data for both bottom trawl and non-trawl discard fleets were available from WCGOP. The number of trips, hauls and fish sampled for lengths by year are summarized in Table 11, by discard fleet.

Discard length composition data were not sex-specific. Discard raw length observations were expanded to the haul level, to account for differences in catch among hauls (Figure 12 and Figure 15).

The initial input values for length compositions were calculated via the Stewart Method (see above).

No age data were available for discarded fish.

#### 2.1.3.1.2 At-sea hake Fishery Length and Age Compositions

The sex-specific length and age data for at-sea hake fleet were collected by the at-sea hake observer program (a-shop) and available through NORPAC database (Figure 12 and Figure 15). Age distributions were included in the model as CAAL observations, binned according to length, age, sex, and year (Figure 26 through Figure 27).



The number of hauls and fish sampled by year and used to create length frequency and CAAL distributions are summarized in Table 12.

Input sample sizes for length compositions were based on the number of hauls sampled by year. The input sample size of CAAL data was set at the number of fish at each length by sex and by year.

The marginal age compositions were constructed, but only used in the model for evaluating the implied fits, while the CAAL data were used in the likelihood.

## 2.2 Fishery-independent data

Data from four fishery-independent surveys were used in this assessment:

- Survey 1: West Coast Groundfish Bottom Trawl Survey (WCGBTS; 2003-2024)
- Survey 2: Triennial (every three years) Survey (1980-2004)
- Survey 3: Alaska Fishery Science Center (AFSC) Slope Survey (1997-2001)
- Survey 4: Northwest Fisheries Science Center (NWFSC) Slope Survey (1999-2001)

The surveys temporal and spatial coverage is summarized in Table 14.

Information produced by these surveys included indices of relative abundance (all four surveys), length-frequency distributions (WCGBTS and Triennial survey), and age frequency distributions (WCGBTS).

Only the WCGBTS has new data for this assessment, but new methods were applied to all surveys to develop indices of abundance.

In this assessment, geostatistical models of biomass density were fit to survey data using the R package [Species Distribution Models with Template Model Builder \(sdmTMB\)](#) (Anderson et al. 2022). These models can account for latent spatial factors with a constant spatial Gaussian random field and spatiotemporal deviations to evolve as a random walk Gaussian random field (J. T. Thorson et al. 2015). Tweedie, delta-binomial, delta-gamma, and mixture distributions, which allow for extreme catch events, were investigated. Results are only shown for the distribution that led to the best model diagnostics, e.g., similar distributions of theoretical normal quantiles and model quantiles, high precision, lack of extreme predictions that are incompatible with the life history, and low Akaike information criterion (AIC). Estimates of biomass from this best model were predicted using a grid based on available survey locations. The {indexwc} R package (Johnson et al. 2025) was used to conduct the analysis and code to reproduce the analysis is available at <https://pfmc-assessments.github.io/indexwc/>.

Standardized indices for all four surveys overlaid are shown in Figure 31, where each index is rescaled to have mean observation = 1.0.

Description of each survey is provided below; details on methods used to process the data are also discussed.

## 2.2.1 NWFSC West Coast Groundfish Bottom Trawl Survey

### 2.2.1.1 Survey Description

The Northwest Fisheries Science Center West Coast Groundfish Bottom Trawl Survey (WCGBTS) is conducted annually since 2003 Table 14. The survey's design and sampling methods are most recently described in detail in Keller, Wallace, and Methot (2017). The survey is based on a random-grid design, covering the coastal waters from a depth of 100 to 700 fm (183-1280 m). This design generally uses four industry-chartered vessels per year assigned to a roughly equal number of randomly selected grid cells and divided into two 'passes' of the coast. Two vessels fish from north to south during each pass between late May to early October. This design therefore incorporates both vessel-to-vessel differences in catchability, as well as variance associated with selecting a relatively small number (approximately 700) of possible cells from a very large set of possible cells spread from the Mexican to the Canadian borders.

### 2.2.1.2 Abundance Index

The data were truncated to depths less than 875 m prior to modeling given that there were zero positive encounters in depths deeper than 875 m. The prediction grid was also truncated to only include available survey locations in depths between 55–875 m to limit extrapolating beyond the data and edge effects.

The chosen model used a delta model with a lognormal distribution for the catch-rate component. A logit-link was used for encounter probability and a log-link for positive catch rates. The response variable was catch (mt) with an offset of area (km<sup>2</sup>) to account for differences in effort. Fixed effects were estimated for each year. The following additional covariates were included: pass. Vessel-year effects, which were historically used for index standardization of this survey, were not included because the estimated variance for the random effect was close to zero. Vessel-year effects were more prominent when models did not include spatial effects and instead vessel-year terms accounted for the random selection of commercial vessels used during sampling (Helser, Punt, and Methot 2004; J. T. Thorson and Ward 2014).

Spatial and spatiotemporal variation were not included in either the encounter probability nor the positive catch rate model. Spatial variation was approximated using 200 knots, where more knots led to non-estimable standard errors because the positive encounters are too sparse to support the dense spatiotemporal structure.

The biomass estimates produced for this assessment using `sdmTMB` are comparable to the biomass estimates produced in the previous benchmark assessment (Figure 32). The index is relatively flat with high variation (Figure 33).

### 2.2.1.3 Length and Age compositions

Table 15 shows the number of lengths and Table 16 shows the number of ages taken by the survey.

Length compositions were separated into males and females. These length compositions were expanded to account for difference in catch among tows, with further expansion based upon the stratification by depth and latitude using the `{nwfsSurvey}` package in R (Wetzel, Johnson, and Hicks 2025). The stratification for length data expansions are provided in Table 13.

Age distributions were included in the model as CAAL observations. The marginal age compositions were only used for comparing the implied fits, while the CAAL data were used in the likelihood. The CAAL data were not expanded and were binned according to length, age, sex, and year.

Figure 18 shows WCG BTS length frequencies by year, and Figure 28 through Figure 30 show the WCG BTS length and CAAL distributions by year.

The input sample sizes for length composition data were calculated based on Stewart and Hamel (2014) as Input  $N_y = 2.43 * N_{tow}$  where the 2.43 value was estimated for a group of shelf and slope rockfish species.

The input sample size of CAAL data was set at the number of fish at each length by sex and by year.

## 2.2.2 AFSC/NWFSC West Coast Triennial Shelf Survey

### 2.2.2.1 Survey Description

The Triennial Survey was first conducted by the AFSC in 1977 and continued until 2004. The survey's design and sampling methods are most recently described in Weinberg et

**al. (2002).** Its basic design was a series of equally-spaced transects from which searches for tows in a specific depth range were initiated.

The survey spatial coverage and timing has changed over the period of survey duration Table 14.

Haul depths ranged from 91–457 m during the 1977 survey with no hauls shallower than 91 m. The surveys in 1980, 1983, and 1986 covered the West Coast south to 36.8°N latitude and a depth range of 55–366 meters. The surveys in 1989 and 1992 covered the same depth range but extended the southern range to 34.5°N (near Point Conception). From 1995 through 2004, the surveys covered the consistent depth range 55–500 meters and surveyed south to 34.5°N. In the final year of the triennial series (2004), the NWFSC conducted the survey and followed very similar protocols as the AFSC, which conducted surveys in all previous years.

All of the surveys were conducted in the mid-summer through early fall: the 1977 survey was conducted from early July through late September; the surveys from 1980 through 1989 ran from mid-July to late September; the 1992 survey spanned from mid-July through early October; the 1995 survey was conducted from early June to late August; the 1998 survey ran from early June through early August; and the 2001 and 2004 surveys were conducted in May-July.

Water hauls (Zimmermann et al. 2001) and tows located in Canadian waters were also excluded from the analysis of this survey. Given the different depths surveyed during 1977, the data from that year were not included in this assessment.

#### 2.2.2.2 Abundance Index

The Triennial Survey was analyzed as an early series (1980–1992) and a late series (1995–2004) to account for change in spatial coverage and survey timing, as Rough-eye/Blackspotted Rockfishes exhibit ontogenetic movements when individuals gradually shift their distribution toward deeper waters as they grow and mature. Separate catchability parameters were estimated for pre-1995 period and from 1995 forward. Separate selectivity curves were estimated for early and late survey periods as well.

The data for the early series were truncated to depths less than 350 m, and for late series to depths less than 500 m prior to modeling given that there were zero positive encounters in depths deeper than 500 m. The prediction grid was also truncated to only include available survey locations in depths between 55–500 m to limit extrapolating beyond the data and edge effects.

The chosen model used a delta model with a lognormal distribution for the catch-rate component. A logit-link was used for encounter probability and a log-link for positive

catch rates. The response variable was catch (mt) with an offset of area ( $\text{km}^2$ ) to account for differences in effort. Fixed effects were estimated for each year. No other covariates were modeled. Vessel-year effects, which were historically used for index standardization of this survey, were not included because the estimated variance for the random effect was close to zero. Vessel-year effects were more prominent when models did not include spatial effects and instead vessel-year terms accounted for the random selection of commercial vessels used during sampling (Helsler, Punt, and Methot 2004; J. T. Thorson and Ward 2014).

Spatial and spatiotemporal variation were included in the encounter probability but not the positive catch rate model. Spatial variation was approximated using 100 knots, where more knots led to non-estimable standard errors because the positive encounters are too sparse to support the dense spatiotemporal structure.

The estimated index is shown in Figure 34. The index exhibits an increase in biomass from 1995 forward, that corresponds to a change in Triennial Survey depth coverage, when the survey extended to the deeper area Table 14.

### 2.2.2.3 Length Compositions

Table 15 shows the number of lengths taken by the survey.

Length compositions were separated into males and females. These length compositions were expanded to account for difference in catch among tows, with further expansion based upon the stratification by depth and latitude using the `{nwfsSurvey}` package in R (Wetzel, Johnson, and Hicks 2025). The stratification for length data expansions are provided in Table 13. Figure 19 shows Triennial Survey length frequencies by year,

The input sample sizes for length composition data for all fishery-independent surveys were calculated based on Stewart and Hamel (2014) as Input  $N_y = 2.43 * N_{tow}$  where the 2.43 value was estimated for a group of shelf and slope rockfish species.

There are no Rougheye/Blackspotted Rockfishes age data from the Triennial Survey.

## 2.2.3 AFSC Slope Survey

### 2.2.3.1 Survey Description

The AFSC slope survey was initiated in 1984. The survey methods are described in Lauth (2000). Prior to 1997, the survey was conducted in different latitudinal ranges each

year. In this assessment, only data from 1997, 1999, 2000 and 2001 were used – these years were consistent in latitudinal range (from 34°30' N. latitude to the U.S.-Canada border) and depth coverage (183-1280 m; 100-700 fm).

### 2.2.3.2 Abundance Index

The prediction grid was truncated to only include available survey locations in depths deeper than 183 to limit extrapolating beyond the data and edge effects.

The chosen model used a delta model with a lognormal distribution for the catch-rate component. A logit-link was used for encounter probability and a log-link for positive catch rates. The response variable was catch (mt) with an offset of area (km<sup>2</sup>) to account for differences in effort. Fixed effects were estimated for each year. No other covariates were modeled. Vessel-year effects, which were historically used for index standardization of this survey, were not included because the estimated variance for the random effect was close to zero. Vessel-year effects were more prominent when models did not include spatial effects and instead vessel-year terms accounted for the random selection of commercial vessels used during sampling (Helser, Punt, and Methot 2004; J. T. Thorson and Ward 2014).

Spatial and spatiotemporal variation were not included in either the encounter probability nor the positive catch rate model. Spatial variation was approximated using 100 knots, where more knots led to non-estimable standard errors because the positive encounters are too sparse to support the dense spatiotemporal structure.

The AFSC Slope Survey index is shown in Figure 35. The index is short, and does not exhibit significant change over the four year period.

### 2.2.3.3 Length Compositions

Table 15 shows the number of lengths taken by the survey.

Length compositions were separated into males and females. These length compositions were expanded to account for difference in catch among tows, with further expansion based upon the stratification by depth and latitude using the {nwfsurvey} package in R (Wetzel, Johnson, and Hicks 2025). The stratification for length data expansions are provided in Table 13. Figure 20 shows AFSC Slope Survey length frequencies by year.

The input sample sizes for length composition data for all fishery-independent surveys were calculated based on Stewart and Hamel (2014) as  $\text{Input } N_y = 2.43 * N_{\text{tow}}$  where the 2.43 value was estimated for a group of shelf and slope rockfish species.

There are no Rougheye/Blackspotted Rockfishes age data from the AFSC Slope Survey.

## 2.2.4 NWFSC Slope Survey

### 2.2.4.1 Survey Description

The NWFSC slope survey was conducted annually from 1999 to 2002. The survey's design and sampling methods are described in Keller et al.(2007). The surveyed area ranged between 34°50' and 48°07' N. latitude, encompassing the U.S. Vancouver, Columbia, Eureka, Monterey INPFC areas, and a portion of the Conception area, and consistently covered depths from 100 to 700 fm (183-1280 m) (Table 13).

### 2.2.4.2 Abundance Index

The prediction grid was truncated to only include available survey locations in depths deeper than 183 to limit extrapolating beyond the data and edge effects.

The chosen model used a delta model with a lognormal distribution for the catch-rate component. A logit-link was used for encounter probability and a log-link for positive catch rates. The response variable was catch (mt) with an offset of area (km<sup>2</sup>) to account for differences in effort. Fixed effects were estimated for each year. No other covariates were modeled. Vessel-year effects, which were historically used for index standardization of this survey, were not included because the estimated variance for the random effect was close to zero. Vessel-year effects were more prominent when models did not include spatial effects and instead vessel-year terms accounted for the random selection of commercial vessels used during sampling (Helser, Punt, and Methot 2004; J. T. Thorson and Ward 2014).

Spatial and spatiotemporal variation were not included in either the encounter probability nor the positive catch rate model. Spatial variation was approximated using 100 knots, where more knots led to non-estimable standard errors because the positive encounters are too sparse to support the dense spatiotemporal structure.

The NWFSC Slope Survey index is shown in Figure 36. The index is short, and, as in case of AFSC Slope Survey, does not exhibit significant change over the four year period.

There are no Rougheye/Blackspotted Rockfishes length and age data from the NWFSC Slope Survey. Given that spatial coverage of NWFSC Slope Survey is the same of AFSC Slope Survey, selectivity of the NWFSC Slope Survey was assumed the same as selectivity of AFSC Slope Survey (mirrored in the model).

## 2.3 Biological Parameters

The major biological inputs to the models are natural mortality, age and growth parameters, weight-length, maturity and stock-recruitment parameters. The following sections outline the treatment of each section. One change from the previous assessment is moving to a two sex from the one-sex specification from 2013. The 2013 stock assessment one-sex specification was based on the observation that the biology of females and males was very similar, thus justifying the simplifying assumption of one sex. The following sections below demonstrates that females and males do generally have similar growth, though there are differences, but may have different natural mortality values. The current assessment will use a two sex configuration that allows for flexibility to set female and male parameters either equal (i.e., functionally equivalent to a one sex model) and or sex-specific. Figure 39 and Figure 40 show that using a two sex configuration with the same life history parameters for females and males is equivalent to the one sex model. Note that the one sex model sums up both female and male biomass, thus why it is twice the size as the two sex female-only spawning output (Figure 40).

### 2.3.1 Natural Mortality

Natural mortality is a highly influential parameter in age-structured stock assessments. It defines the rate of natural death by age, and thus establishes a stable age-structure and expectation of longevity, and interacts with growth and reproduction to determine stock productivity. It is a very difficult parameter to directly measure, thus empirical relationships based on life history parameters are often used to indirectly determine its value or build prior distributions in belief of what it is in the event we do attempt to estimate it in the model (Cope and Hamel (2022); Hamel and Cope (2022); Maunder et al. (2023)). If length and age data are available, it may be possible to estimate it in the model.

An estimate of maximum age tends to be the most reliable life history parameter related to natural mortality to inform its estimation. Cope and Hamel (2022) ([The Natural Mortality Tool](#)) provide the most up-to-date examination of the relationship between maximum age and natural mortality

$$M = \frac{5.4}{A_{\max}}$$

where  $M$  is natural mortality and  $A_{\max}$  is the assumed maximum age. The prior is defined as a lognormal distribution with mean  $\ln(5.4/A_{\max})$  and standard error = 0.31. This is the equation typically used to estimate a natural mortality point estimate, but is underpinned by the choice of the value of  $A_{\max}$ . This equation assumes that the



proportion of the stable population at this maximum age is 0.4517%. If we take humans as an example, the longest lived human is 122 years. This is not the maximum age, but the oldest ever recorded age. The maximum age that corresponds to 0.4517% of the population is around 100 years. For Rougheye/Blackspotted, the oldest ever aged individual is 205 years with unknown ageing error. We did not consider this as a realistic maximum age.

The 2013 U.S. west coast stock assessment used a prior built around a mean of 0.034 (corresponding to a maximum age of 163), but estimated natural mortality at 0.042 (maximum age between 128-129 years; **Figure M**). The 2023 Gulf of Alaska assessment built a prior conditional on a estimate of natural mortality from their 5 oldest aged individuals that ranged from 126-135 years. This resulted in a mean value of 0.042, similar to the 2013 U.S. west coast stock assessment. The 2023 Bering Sea/Aleutian Islands assessment used  $M = 0.05$  (assumed longevity of 108), and the recent Canadian assessments considered a range of  $M$  values from 0.03 to 0.055 (assumed maximum ages of 180 to 98 years; Figure 41).

We attempt to estimate natural mortality, as was done in the 2013 U.S. West coast assessment. Examining the available age data, the oldest 10 individuals range from 139 to 165 and were all males. For females, the 10 oldest individuals range from 130 to 121 years. If those oldest ages were used in the Hamel and Cope (2022) longevity estimator, these ages would correspond to a range of natural mortality values of 0.033 to 0.039 for males, which include the mean of the prior used in the 2013 assessment. For females, it corresponds to natural mortality values of 0.039 to 0.045. All these assume that the sampled population has enough of an age structure still available for sampling, as opposed to having some level of age truncation from the theoretical unfished stable age distribution.

Related to this issue of possible age truncation, applying a catch curve analysis (taking the log of the abundance of numbers of samples in available age classes) on the aggregated ages across all age sources by sex, the total mortality (Natural + Fishing mortality = Total mortality) is 0.046 for females and 0.035 for males, which may indicate the natural mortality could be lower than that used in the 2013 assessment, but within the range of values considered in other areas (Figure 42). This also indicates the possibility of estimating sex-specific natural mortality, as natural mortality may differ by sex. The two sex model allows for this type of model specification exploration. Further exploration was done by truncating the upper ages considered, with the assumption that the older ages may also not be sampled fully (i.e., dome-shaped selectivity). We considered both 100 (Figure 43) and 80 (Figure 44) as upper age cut-offs. The less older individuals included, the higher the estimate of total mortality, and this a higher natural mortality. But we can see a general overestimate of how many older individuals are expected using these higher  $Z$  values, thus dome-shapeness does not seem to explain the sampling of these older individuals.

One challenge to estimating natural mortality within the model is the interaction of estimating dome-shaped selectivity with estimating natural mortality. If all fleets assume some level of dome-shaped selectivity, it is difficult to determine if the unseen larger, older individuals are due to natural death or fishing mortality. Typically, at least one major fleet needs to achieve full selectivity for the larger, older individuals. The 2013 assessment suggested some dome-shaped selectivity in the two major fleets, thus any natural mortality estimates are evaluated depending on the forms of fleet selectivity.

### 2.3.2 Growth (Length-at-Age)

Age and length data are used to estimate important growth parameters. Figure 45 has the currently available age and length data. Female and male sample sizes are very similar. Estimated growth curves are also presented in Figure 45 and the parameters are provided in Table AL\_1. The West Coast Groundfish Bottom Trawl Survey clearly and importantly samples the smallest, youngest individuals compared to the other two data sources. This allows for a better estimate of the age at size 0 ( $t_0$ ) and growth coefficient ( $k$ ). The female asymptotic size ( $L_\infty$ ) is estimated notably higher from the PacFIN data, though male estimates of  $L_\infty$  are similar across the data sets. The overall externally derived estimates of female and male Rougheye/Blackspotted Rockfishes are

$$\text{Females } L_\infty = 59.03 \text{ cm; } k = 0.07; t_0 = -2.45$$

$$\text{Males } L_\infty = 56.69 \text{ cm; } k = 0.08; t_0 = -2.03$$

The coefficient of variation (CV) of length by age and sex are shown in Figure 46. This is a measure of the variation in length for a given age class. Sample sizes are highest from the youngest ages up to around 70 (females) to 80 (males) years. The smoothed line shows the average response, and indicates similar CVs values for females and males, with the highest at the youngest ages, but generally 0.1. The amount and range of age samples, along with repeated length samples within an age class, allows growth parameters ( $L_\infty$ ,  $k$ ,  $t_0$ , and CVs at age) to be estimated in the model. Ages are conditioned on lengths in the model in order to estimate growth within the model. We also explore sensitivity in growth values by pre-specifying growth to different values.

We note that the growth values being estimated in our data are notably different than those used in Alaska. For instance, the growth parameters for the BSAI stock is  $L_\infty = 51.43$ ,  $k = 0.06$  and  $t_0 = -3.30$  and  $L_\infty = 54.2$  cm,  $k = 0.07$ ,  $t_0 = -1.5$  for the GOA population (both sexes combined). These growth parameters shows a larger size and faster growth of the West Coast stock complex versus those in Alaska, though the West Coast stock complex is more similar to the GOA complex.

### 2.3.3 Ageing Bias and Precision

Counting ages from ageing structures in long-lived, temperate fishes is challenging. Ages derived from these structures can be hard to reproduce within and between readers (i.e., imprecision), and may not contain the true age (i.e., bias). Stock assessment outputs can be affected by bias and imprecision in ageing, thus it is important to quantify and integrate this source of variability when fitting age data in assessments. In Stock Synthesis 3, this is done by including ageing error matrices that include the mean age (row 1) and standard deviation in age (row 2). Ageing bias is implemented when the inputted mean age deviates from the expected middle age for any given age bin (e.g., 1.75 inputted versus 1.5 being the true age for the age 1 bin); ageing imprecision is given as the standard deviation for each age bin.

There are eight primary readers that provided the available ages, two of which often split the ageing duties. Figure 47 shows which reader assignments are given to each year of ages by data source. Reader 7 is the mix of two readers that shared reading duties within years.

Estimation of ageing error matrices used the approach of (2008) in two different forms: one developed in AD Model Builder ([nwfsAgeingError](#) (J. T. Thorson, Stewart, and Punt 2012)) and one adapted to Template Model Builder framework (TMB). The ageing error matrix offers a way to calculate both bias and imprecision in age reads. Reader 1 is always considered unbiased, but may be imprecise. Bias relative to the primary reader is given for the second reader. There were three age readers that were assumed to be unbiased. In those cases, 12 model configurations based on different assumptions of imprecision (constant CV, curvilinear standard deviation, or curvilinear CV, along with an option to either share or independently estimate imprecision between readers) were considered. For the other four age readers that could be biased and/or imprecise, thirty-six total model configurations were explored that included the above imprecision models as well as an exploration of the functional form of bias (e.g., no bias, constant coefficient of variation, or non-linear bias) in the second reader.

Model selection criteria included AIC corrected for small sample size (AICc), which converges to AIC when sample sizes are large, and Bayesian Information Criterion (BIC). Both ADMB and TMB were run using an ([ageing error shiny app](#)). Model selection was then compared between ADMB and TMB, which did not always agree, so model selection criteria was added across the two modeling approaches to get an overall model selection criteria. Ageing error matrices were also inspected for behavior in the best supported models to make sure outrageously large precision or bias was not chosen (effectively rendering the ages worthless, which is not an assumption of the quality of the ages). Figure 48 and Figure 49 show the bias and imprecision assumptions applied for each ageing error (AE) matrix.

### 2.3.4 Length-Weight Relationship

Female and male length-weight relationships were determined using data from the PacFIN database, West Coast Groundfish Bottom Trawl Survey, and ASHOP samples. Samples size by sex were: female (N=13839), males (13625), and unknown sex (53). Each of the data sources estimated very similar length-weight relationships (Figure 50).

The resultant sex-specific length-weight relationships are given in Figure 51, with the following individual values:

- Females:  $W = 0.000008L^{3.15}$
- Males:  $W = 0.000012L^{3.07}$

These values are very similar to the previous assessment that used a combine sex value of  $a=0.0000096$  and  $b=3.12000$  (Figure 51).

### 2.3.5 Maturity

Maturity for the Rougheye/Blackspotted Rockfish complex was estimated using 473 maturity samples collected from 2015 to 2024 on Oregon Department of Fish and Wildlife (ODFW) and Washington Department of Fish and Wildlife (WDFW) surveys and the gls{indexwc} in California, Oregon, and Washington waters (M. Head, pers. comm.). The samples included 194 samples genetically assigned as Rougheye Rockfish, 71 samples genetically assigned as Blackspotted Rockfish, and 208 samples with no genetic assignment. The maturity schedule was assumed to be length-based, as in the 2013 benchmark assessment. This assessment used the functional classification of maturity to describe the maturity schedule, which not only identifies the individuals that are physiologically capable of producing yolk (those that are biologically mature), but also accounts for the occurrence of abortive maturation and skipped spawning, so the functional maturity classification is a more accurate representation of the individuals that may actually spawn in a given year. This is a difference from the 2013 benchmark assessment, which did not explicitly estimate functional maturity, and instead assumed the biological classification of maturity.

Biological maturity and functional maturity observations were fitted in separate models. Biological maturity and functional maturity status observations (0 = immature and 1 = mature) were fitted in a logistic regression model (glm R function, family = binomial, link = "logit"). The estimated model parameters were used to calculate length at 50% maturity (L50%; Table 17) and maturity ogives (Figure 52). The delta method was used to calculate 95% confidence intervals of L50% estimates. The estimated L50% (functional maturity; L50%fxn) was 46.53 cm and the estimated slope of the maturity oogive was

0.25. Sensitivities were run using the estimate of biological maturity and the maturity estimate used in the 2013 benchmark assessment. There was little evidence of skipped spawning, so we did not explore fitting the data with a spline model.

Because there are known life history differences between Rougheye Rockfish and Blackspotted Rockfish, maturity was also estimated for each species, using the samples that were genetically assigned to each species, respectively, using the same methods as above (Table 17 and Figure 53). Two sensitivities were run using the functional maturity L50% (and slope) estimated for 1) Rougheye Rockfish and 2) Blackspotted Rockfish (which mature at larger sizes on average than Rougheye Rockfish).

Sensitivities were run using functional age at 50% maturity estimate for the species complex ( $n = 372$ ) and for each species separately. Age at 50% maturity was estimated using the same methods as for length at 50% maturity (Table 17 and Figure 54).

### 2.3.6 Fecundity

The 2013 U.S. west coast stock assessment assumed that fecundity was proportional to weight. Dick et al. (2017) provided a study on rockfishes showing that rockfishes routinely have a non-proportional relationship of fecundity to weight, with larger individuals producing more eggs than expected only by weight. Neither Rougheye or Blackspotted rockfishes have a species- or subfamily-specific estimate for this relationship, so this stock assessment uses the unobserved Genus *Sebastes* values of  $a = 6.538e-06$  and  $b = 4.043$  using the  $F=aL^b$  relationship. In order to adapt the  $a$  parameter for SS3, the equation  $(a*10^b)/1000$  was used to scale the  $a$  parameter to millions of eggs. This results in  $a = 7.218466e-05$ .

### 2.3.7 Stock-Recruitment Function and Compensation

The Beverton-Holt stock recruit relationship is assumed, as it was in the 2013 assessment, to describe the relationship between spawning biomass and recruitment. The steepness parameter may be considered for estimation, but it is notoriously difficult to estimate in assessment models. The 2013 stock assessment used the previous rockfish steepness mean value of 0.77, but this has subsequently been updated to 0.72, to a value that represents a stock with somewhat lower recruitment compensation. Natural variation in recruitment (i.e., not deterministically taken from the stock-recruit curve) is apparent in the length and age data (as notable length or age classes growing/ageing over time), so deviations in recruitment are estimated.

### 2.3.8 Sex Ratio

No information on the sex ratio at birth was available so it was assumed to be 50:50.

## 2.4 Environmental and ecosystem data

This stock assessment does not explicitly incorporate trophic interactions, habitat factors or environmental factors into the assessment model. More predation, diet and habitat work, and mechanistic linkages to environmental conditions would be needed to incorporate these elements into the stock assessment and should remain a priority. McClure et al. (2023) report the climate vulnerability for several west coast groundfishes, including Rougheye/Blackspotted Rockfishes. Rougheye/Blackspotted Rockfishes demonstrated both high biological sensitivity and high climate exposure risk, to give it an overall high vulnerability score to climate change. This result should also be considered with the fact that, like many rockfishes, periods of low productivity is not unusual to Rougheye/Blackspotted Rockfishes and their extended longevity (though admittedly this seems shorter than previously believed and should be reconsidered) has historically allowed them to wait for advantageous productivity periods. Stressors such as habitat degradation and climate change could bring significant challenges to population sustainability. Regardless, no environmental or ecosystem data are directly incorporated into the stock assessment model.

### 3 Assessment Model

#### 3.1 History of Modeling Approaches

Rougheye Rockfish (not including Blackspotted) on the U.S. Pacific Coast was first evaluated in 2010 by Dick and MacCall (2010) using depletion-based stock reduction analysis (DB-SRA), as Category 3 stock. That model estimated the population had greater than a 50% probability of exceeding the estimated proxy overfishing level in 2010 if the harvest remained at the observed levels. DB-SRA estimated a proxy OFL for Rougheye Rockfish of 78.7 mt with a 95% confidence interval between 4.7-587 metric tons.

The first benchmark assessment for Rougheye/Blackspotted Rockfishes was conducted in 2013 (Hicks, Wetzel, and Harms 2013). A 2013 benchmark stock assessment used Stock Synthesis (version 3.24O) integrated statistical catch-at-age model, which is different from the delay-difference model with an assumed stock status prior DB-SRA analysis used in 2010. The stock assessment has been used for management as a Category 2 stock assessment. The 2013 assessment used a substantially updated catch history, indices of abundance, and biological compositions (lengths and ages). The natural mortality value was also updated to be higher value than the one used in the DB-SRA model. The 2013 assessment also assumed logistic selectivity for all fleets and surveys, except for Triennial Shelf Survey, which was allowed to be dome-shaped. With higher natural mortality and asymptotic selectivity assumptions, the 2013 assessment estimated 2013 spawning biomass to be at 47% relative to unfished equilibrium spawning biomass, with a 95% confidence interval between 30.5% - 64.2%. The 2013 spawning biomass was estimated to be 2,552 metric tons, with a 95% confidence interval between 1,024 - 4,081 metric tons.

With this new benchmark assessment, we re-evaluate all the data sources available for Rougheye/Blackspotted Rockfishes, analyse new and re-analyse previously used data with current statistical methods and best practices, and re-evaluate modelling assumptions. Detailed description of changes made since 2013 benchmark assessment is provided in Section 3.3.

#### 3.2 Response to Most Recent STAR Panel Recommendations

There were several recommendations from the 2013 STAR panel, broken into two categories

### 3.2.1 General recommendations

1. *Investigate data-weighting options.* This has been an ongoing research topic in stock assessments over the last several decades (Francis 2011; James T. Thorson et al. 2017; André E. Punt 2017). In this assessment, we use Francis (2011) method, and explore other methods of data weighting within sensitivity analysis described in Section 3.9.1.
2. *A workshop for constructing abundance indices from survey GLMMs.* This is another topic that has developed greatly since this time (J. T. Thorson et al. 2015). Spatio-temporal models used in this assessment are described in Section 2.2.
3. *Continue collection of ages.* This had been done, and this assessment benefits from several more years of age data.
4. *Exploring historical catches.* This again has been an ongoing topic and addressed for many of our groundfishes. We use the latest estimates in this assessment.
5. *SSC guidance on decision tables.* Decision table discussion evolve after every stock assessment cycle, and we are using the latest approaches to decision tables in this assessment.
6. *Investigate fishery-independent slope surveys, such as submersibles.* These surveys are still not available for slope species.

### 3.2.2 Stock-specific recommendations

1. *Collecting additional age data.* This has been done and included in this stock assessment.
2. *Collecting genetic material to explore distinguishing Rougheye and Blackspotted Rockfishes.* This work has been done as was presented earlier in the document when discussing stock structure decisions.
3. *The cause of the re-occurring decrease in sizes around 40cm.* The data continue to exhibit a lack of 35-40cm fish in fleets that catch smaller fish (WCGBTS and Triennial Survey, and bottom trawl discard fleet), creating bimodal distributions of length data (Figure 66). We found that this pattern is not caused by the relative depth distribution of two stocks within the complex, with Rougheye Rockfish being shallower and Blackspotted Rockfish deeper. Figure 55 (pers. comm. P. Frey, NWFSC) shows the length distribution of Rougheye Rockfish and Blackspotted Rockfish identified to species using genetics, and the Rougheye Rockfish still shows a bimodal length distribution. Further analysis reveals that this bimodal length distribution is not limited to Rougheye/Blackspotted Rockfishes, but is also evident in multiple other rockfish species. We continue to explore data to understand mechanisms behind this pattern, which are potentially related to accessibility of trawl gear to habitats specific to that size group. In this assessment we were able to fit the bimodal length distribution by allowing surveys' selectivity curves more flexibility, and not fixing it asymptotic as in the previous assessment.



4. *Additional maturity and fecundity studies.* While no fecundity studies are available, updated maturity is presented in the maturity section of the document.
5. *Age validation.* While no age validation study has been completed, the age readers are confident what annuli represent a year's worth of growth. Multiple ages are available and ageing error is characterized in this stock assessment.
6. *Understanding stock structure.* Discussed in the Section 1.1 of this document.
7. *Connectivity of stocks across the species ranges.* This is also discussed in the Section 1.1 of the document.

### 3.3 Model Changes from the Last Assessment and Bridging Analysis

The last full assessment of Rougheye/Blackspotted Rockfishes was conducted in 2013. The 2013 assessment model was the starting point for this assessment. We included a number of improvements related to use of data, model structure and modeling techniques. Below, we describe the most important changes made since the last assessment:

- Upgraded the model to Stock Synthesis 3.30.22.1 version. This is standard practice to capitalize on newly developed features and corrections to older versions as well as improvements in computational efficiency. The list of changes made to Stock Synthesis since 2013 can be found in the model [change log](#). No discernible differences were produced by this change. The status (Figure 56) and scale (Figure 57) of both models are exactly the same, as are the estimates of within model uncertainty.
- Specified a two-sex model, instead of one-sex model, to allow sex-specific estimation of natural mortality and growth. No discernible differences were produced by this change either (Figure 39 and Figure 40).
- Included bottom trawl and non-trawl discards as separate fleets (see Section 3.4.3 for details), instead of treating them as part of the same fleets as landings. Results did not impact the model output (Figure 58 and Figure 59).
- Split mid-water trawl catches from bottom trawl landings and treat them as a separate fleet (see Section 3.4.3 for details), to account for gradually increasing contribution of mid-water trawl catches. Results did not impact the model output.
- Updated historical and current fishery removals, to include most up to date information. Since 2013 assessment, WDFW completed historical catch reconstruction of rockfish, and newly estimated landings represent improvement. For the period between 1987 and 1999, Oregon PacFIN landings were supplemented with the additional estimates of Rougheye/Blackspotted Rockfishes landings reported within unspecified rockfish market categories. Results did not impact the model output.
- Recalculated survey abundance indices using sdmTMB geostatistical model. Results did not impact the model output.
- Added more biological compositions, mainly in years since 2013, but also some historical ages. Adding more composition data resulted in slight increase in stock scale (Figure 60).

- Updated input sample sizes associated with fisheries and survey length composition data to using a function of number of trips and number and fish (rather than number of trips and number of hauls, as in previous assessment), to follow current best practices and ensure a consistent treatment of fishery and survey input data.
- Updated ageing error matrices.
- Updated weight-length, maturity and fecundity parameters, to include most up to date and improved information.
- Updated spawn-recruit parameters with Beverton-Holt steepness fixed at 0.72, and recruitment variability at 0.5 for consistency with the calculated recruitment variability in the model.
- Assumed natural mortality for males consistent with maximum ages of observed for Rougheye/Blackspotted Rockfishes, while estimating female natural mortality using the Hamel and Cope (2022) prior. Previously, natural mortality for both sexes was estimated within the model, but estimated values were higher than expected for maximum ages observed for this stock. This change reduced the natural mortality values for both sexes, which resulted in decreased scale of the stock.
- Provided flexibility for the bottom trawl fleet and bottom trawl surveys to estimate dome-shaped selectivity (Figure 132 and Figure 133), previously assumed asymptotic. This change was prompted by the lack of fit to length compositions data. Also, the examination of mean fish lengths by fleet indicated that the bottom trawl fleet capture smaller fish than mid-water trawl and non-trawl gear. This change resulted in a substantial increase of stock scale, and is considered an improvement to the model structure. We also allowed estimated selectivity to vary with time in both the bottom trawl and non-trawl fleets, to account for management changes that can impact selectivity of these fleets.

Bridging analysis was conducted to illustrate the impacts of incremental changes. With the new fecundity parameters, the 2025 model produces spawning output (in millions of eggs) rather than spawning biomass (in mt as in 2013 model); so with different metrics, these outputs were no longer comparable between the two models. However, we ran the 2013 model with new fecundity parameters to allow for direct comparison of the results from the bridging runs. The bridging analysis, with the most influential steps done sequentially, is shown in Figure 60 and Figure 61.

This assessment (compared to 2013 assessment) estimates much higher stock scale, primarily, as discussed above, due to changes in treatment of selectivity parameters and relaxing asymptotic selectivity assumptions for the bottom trawl fleet and bottom trawl surveys. Changes in selectivity assumptions allow to substantially improve fits to length and age composition data in fisheries and surveys. The stock scale was also affected by the changes in treatment of natural mortality, bringing the scale down to a lower level compared to the previously used natural mortality assumption.

### 3.4 General Model Specifications

#### 3.4.1 Modelling Platform

Stock Synthesis statistical catch-at-age modelling framework (Methot and Wetzel 2013), version 3.30.23.1, is used for this assessment. This framework allows the integration of a variety of data types and model specifications. The Stock Assessment Continuum tool (<https://github.com/shcaba/SS-DL-tool>) was also used to explore model efficiency, likelihood profiling, retrospective analyses, and plotting sensitivities. The companion R package `r4ss` (version 1.51.0) along with R version 4.4.3 were used to investigate and plot model fits.

#### 3.4.2 Model Structure

This stock assessment is for the Rougheye/Blackspotted Rockfishes, two species that form one management complex. Assessment area is from the U.S.-Mexican border on the south to the U.S. Canadian border on the north (Figure 1). The assessment excludes consideration of the Puget Sound and Salish Sea. Within this area, the assessment treats the U.S. Rougheye/Blackspotted Rockfishes resource as a single coastwide stock.

This is a sex-specific model. The sex-ratio at birth is assumed to be 1:1. Females and males have separate growth curves (fully estimated within the model) and sex-specific weight-at-length parameters. The model assumes a constant natural mortality of 0.036 yr<sup>-1</sup> for males, while natural mortality for females is estimated based on Hamel and Cope (2022). The length frequency distributions are represented as thirty six 2-cm bins ranging between 10 and 80 cm. Population length bins are defined at a finer 2-cm scale, ranging between 4 and 84 cm. Age data is included as conditional age-at length compositions with bins ranging between 1 and 100 years.

The modeling period begins in 1892, and the stock prior to that is assumed to be in an unfished equilibrium condition.

#### 3.4.3 Fleet Definitions

The model is structured to track six fleets and include data from four surveys.

Defining fleets is largely based on differing fleet selectivity (i.e., how the fishery captures fish by length and/or age). In the stock assessment model, selectivity translates into how the removals are taken via length and/or age out of the population. In this assessment, the following fleet structure is being used to model commercial fishery removals:

- Fleet 1: Commercial bottom trawl fishery.
- Fleet 2: Dead discard from bottom trawl fishery.
- Fleet 3: Commercial non-trawl (mainly the long-line) fishery.
- Fleet 4: Dead discard from non-trawl fishery.
- Fleet 5: Contemporary mid-water trawl fishery.
- Fleet 6: At-sea hake fishery bycatch.

In 2013 assessment ([Hicks, Wetzel, and Harms 2013](#)), fisheries removals were split among three fleets - trawl, hook-and-line and at-sea hake fishery bycatch. For the first two fleets (trawl and hook-and-line), removals were divided between landings and discards, with selectivity and retention curves estimated within the model.

In this assessment, we treat discards in trawl and non-trawl fisheries as separate fleets from landings fleets.

Treating discards as separate fleets from landings provides several advantages, including:

- With separate discard fleets, we can easily track relative amounts of landings and discards within a fishery (they are not being combined into the total catch).
- This approach provides more flexibility to explore different selectivity assumptions for both landed and discarded fish dome-shaped vs asymptotic, mirroring one to the other, etc.
- This approach allows to avoid hard to diagnose issues that come from estimating retention curves (especially with limited amount of data).
- The biological data for landings and discards are collected independently (port sampling vs on-board observers), and using different sampling approaches. Treating landings and discards as separate fleets in the model allows us to weight those data sources separately as well, to balance the representation of samples.

The change in treating discards as separate fleets does not impact model results (Figure 58 and Figure 59), regardless of the selectivity form being assumed for the discard fleets. Historically, no discarding was observed for Rougheye/Blackspotted Rockfishes ([Pikitch, Erickson, and Wallace 1988](#)), see Section 2.1.2.1 for details. The 2013 assessment estimated zero historical discard for both trawl and fixed gear fleets, based on the available data. In this assessment, therefore, we assume no discard until early 2000, when the first Rougheye/Blackspotted Rockfishes was observed after the introduction of trip limits for rockfish.

We also split trawl fishery data into bottom trawl and mid-water trawl fleets. Catch data indicates that contribution of mid-water trawl catches gradually grew over the past 20 years, and now they represent majority of the trawl removals (Figure 5). Historical information on mid-water catches of Rougheye/Blackspotted Rockfishes comes from

Pikitch, Erickson, and Wallace (1988), which has no records of Rougheye/Blackspotted Rockfishes mid-water trawl catches, neither retained nor discarded. Also, Oregon historical catch reconstruction (Karnowski, Gertseva, and Stephens 2014) has only one record of 0.0002 metric tons of Rougheye/Blackspotted Rockfishes taken in 1985, even though the mid-water trawl catches had their own market category in Oregon since the early-1980s (Karnowski, Gertseva, and Stephens 2014), and multiple rockfish species are reported as caught by this gear. This information suggest that historically Rougheye/Blackspotted Rockfishes mid-water catches were negligible.

As reported in Section 2.2, the following surveys are included in the model:

- Survey 1: West Coast Groundfish Bottom Trawl Survey (WCGBTS; 2003-2024)
- Survey 2: Triennial (every three years) Survey (1980-2004)
- Survey 3: Alaska Fishery Science Center (AFSC) Slope Survey (1997-2001)
- Survey 4: Northwest Fisheries Science Center (NWFSC) Slope Survey (1999-2001)

We use length-based selectivity curves for all fleets for this stock assessment model (as was done in the 2013 assessment), as there is no evidence that significant age-based selectivity is occurring. We considered logistic and dome-shaped selectivity options for various combinations of fleets and time periods during model development.

#### 3.4.4 Model Likelihood Components

There are five primary likelihood components for each assessment model:

1. Fit to length composition samples.
2. Fit to age composition samples (all fit as conditional age-at-length).
3. Fit to survey indices of abundance.
4. Penalties on recruitment deviations (specified differently for each model).
5. Prior distribution penalties

In addition, there is a catch component to the likelihood, but catches are essentially fit without error. Additionally, there is a crash penalty that is invoked if true catches would cause the stock to go extinct. The penalty would alter catches to avoid extinction, but any presence of a crash penalty is used as an indication that the model has been misspecified, so this likelihood contribution should always be 0.

### 3.5 Model Parameters

#### 3.5.1 Estimated and Fixed Parameters

The full list of estimated and fixed parameters are found in Table 20.

All growth parameters were estimable and did not change across the large majority of explored model scenarios, so they were estimated in the reference model. Natural mortality ( $M$ ) was not estimable for both sexes. When attempted, both values were estimated at values that caused the scale to approach the higher end of reasonable values, and thus not a risk neutral option. In order to balance model fit and reality, a likelihood profile was conducted on natural mortality for males (females  $M$  being estimated) in order to find the lowest supported (i.e., within 2 negative log likelihood units) by the data male  $M$  value. The profile shows conflicting information in the data, where lengths support higher natural mortality values and ages support lower natural mortality (Figure 62). It is expected that ages would be more informative to natural mortality, which encourages considering just the age component likelihood. Most of the age components are not well informed for natural mortality, though the at-sea-hake fishery sampled age data does seem to be informative. This fishery has a logisitic selectivity, thus obtaining large and old individuals. Using this component likelihood, the value of 0.036 for male  $M$  is the lowest value supported. The reference model this fixes male  $M$  to this value and estimates female  $M$ . Length-at-maturity, fecundity-weight, and length-weight relationship were all fixed, as is the only treatment option in SS3.

For recruitment, steepness ( $h$ ) was not estimable and was fixed to the rockfish prior of 0.72. Recruitment variability was set at 0.5 and checked for consistency with the calculated recruitment variability in the model. Recruitment deviations were estimated as a deviation not constrained to sum to 0, and initially estimated for the full time series. If the full time series was not estimated, the scale and status of the stock increased to unrealistically high levels, so the full time series estimation was retained. Given the longevity of these species, information on recruitment going deeper into the time series is not unreasonable.

#### 3.5.2 Selectivity Assumptions

The selectivity of all fisheries and surveys were estimated either as logistic or dome-shaped selectivity. Blocks were also added as described in the data section. In the attempt to fit the biological data, it was found that bottom trawl fisheries, just as the trawl surveys were treated, only fit the data if the selectivity was domed. All fisheries that had final dome-shaped selectivity were given the flexibility to be logistic if it led to a better fit. The midwater, at-sea-hake and the final block of the non-trawl fisheries all required

logistic selectivity to fit the data. The use of dome-shaped selectivity for the bottom trawl was a major difference from the previous stock assessment. The choice of selectivity for the bottom trawl survey changed the scale and status of the stock and therefore a major source of sensitivity.

### 3.5.3 Data Weighting

Initial sample sizes for the length and conditional age-at-length compositions were also considered for additional data-weighting. The method of Francis (2011), specifically equation TA1.8, was used to re-weight the length and conditional age-at-length composition data against other inputs and likelihood components. The Francis method treats mean length and age as indices, with effective sample size defining the variance around the mean. If the variability around the mean does not encompass model predictions, the data should be down-weighted until predictions fit within the intervals. This method accounts for correlation in the data (i.e., the multinomial distribution), but can be sensitive to years that are outliers, as the amount of down-weighting is applied to all years within a data source, and are not year-specific. Sensitivities were performed examining different data-weighting treatments: 1) the Dirichlet-Multinomial approach (James T. Thorson et al. 2017), 2) the McAllister-Ianelli Harmonic Mean approach (McAllister and Ianelli 1997), or 3) no additional data-weighting.

The ability to estimate additional variance for indices allows the model to balance model fit to that data while acknowledging that variances may be underestimated in the index standardization. Given the large inputted variances and the limited contrast in the index trends did not require the consideration of further variance estimation. Removal of the index data was explored to demonstrate the limited influence of this data in the model.

## 3.6 Model Selection and Key Assumptions

The reference model for Rougheye/Blackspotted Rockfishes was developed to balance parsimony and realism, and the goal was to estimate a risk neutral spawning output trajectory and relative stock status for the stocks of Rougheye/Blackspotted Rockfishes in state and federal waters off the U.S. West Coast. To achieve the above goals, the model uses different data types and sources to estimate reality, but relies on simplifying assumptions when the data are not informative to parameters. A series of investigative model runs were done to achieve the final reference model. Constructing integrated models (i.e., those fitting many data types) takes considerable model exploration using different configurations of the following treatments:

- Data types and weighting

- Parameter treatments: which parameter can, cannot and do not need to be estimated
- Phasing of parameter estimation
- Exploration of local minima vs global minimum (see Model Convergence and Acceptability section below)

Regarding data types, different biological data (i.e., length and/or age composition) with and without the catch time series (and no additional data weighting) were first included to obtain an understanding of the signal of stock status coming from the data (Figure XXXXXX). The length and age only models assume fixed life history values (growth fixed to external estimates, natural mortality assume the reference model values) and constant catch over the entire time series, while estimating the selectivity of each fleet. Under this constraint, the lengths suggest a stock status lower than the reference model, while the ages consider the stock is less depleted than supported by the ages (with no ageing error), and more similar to the reference model. Adding ageing error, Putting the two data sources together produce an intermediate stock status in the lower precautionary zone. Adding the catch time series substantially changes the stock status trajectory, with length or age only model above the reference stocks status. Combining the two came out just under the reference model. Only one model includes recruitment deviations, and demonstrates more dynamics behavior similar to that seen when biological compositions are unweighted (see Model Specification Sensitivities section ).

Stock scale was comparable once removal history was included, and demonstrates a large sensitivity to the scale of the stock given the data with no additional weighting included (Figure ).

Numerous exploratory models that included all data types and a variety of model specifications were subsequently explored and too numerous to fully report. In summary, the estimation of which life history parameters to estimate and fix was liberally explored.

The following is a list of things that were explored, typically in combination with one another

- Estimate or fix  $M$
- Estimate or fix any of the three growth parameter for each sex
- Estimate or fix the stock-recruit relationship
- Estimate or assume constant recruitment. If estimating recruitment, for what years?
- Estimate additional survey variance, and for which survey?
- Logistic or dome-shaped selectivity?
- Estimate or fix selectivity parameters

The biggest uncertainty was in the treatment of sex-specific  $M$  and the selectivity of the bottom trawl fishery. The combination of these two sources covered the extent of all



other sources of uncertainty observed and presented in the “Characterizing uncertainty” section of the document. The parameters uncertainty is different than the uncertainty derived from data treatment, such as ageing error and data-weighting. While these issues cause large uncertainty in stock scale and status estimates, the choice of treatments are based on the common challenge of balancing information content (i.e., what should the data be informing in the stock assessment) in the data within an integrated statistical frameworks. Those explorations are also provided in the “Characterizing uncertainty” section.

General attributes of the reference model are that indices of abundance are assumed to have lognormal measurement errors. Length compositions and conditional age at length samples are all assumed to follow a multinomial sampling distribution, where the sample size is fixed at the input sample size calculated during compositional example, and where this input sample size is subsequently reweighted to account for additional sources of overdispersion (see below). Recruitment deviations were also estimated are assumed to follow a lognormal distribution, where the standard deviation of this distribution is tuned as noted above.

Sensitivity scenarios and likelihood profiles (on  $\ln R_0$ , steepness, and natural mortality) were used to explore uncertainty in the above model specifications and are reported in the “Characterizing uncertainty” section.

### 3.7 Model Diagnostics

#### 3.7.1 Model Convergence and Acceptability

While there is no definitive measure of model convergence, several measures are routinely applied. These criteria include a low maximum gradient (0), inversion of the Hessian (passed), acceptable fits to data (passed), and reasonable parameter values (passed).

Model efficiency was explored by doing a short run Bayesian analysis using the Random Walk Metropolis with 2,000 draws, keeping all the draws and examining the fast mixing parameters. Those estimated parameters that do not move much from the initial values slow the model down and are recommended to be fixed at the starting value ([Monnahan et al. 2019](#)). No additional parameters were fixed based on this analysis (Figure 63).

An extra effort was given to ensure the model did not rest on a local likelihood minimum. This was done by starting the minimization process from dispersed parameter values away from the maximum likelihood estimates to determine if the approach found a better model fit (i.e., minimum negative log-likelihood value). Starting parameters used a jitter shift value of 0.01 and 0.05. Both jitter scenarios were repeated 100 times with 78 out of 100 (jitter 0.01) and 49 out of 100 (jitter 0.05) runs returned to the reference model likelihood (Figure 64 and Figure 65). Out of the combined 200 jitter runs, a better fit, lower negative log-likelihood model was not found in any of the remaining runs. The reference model did not experience convergence issues when provided reasonable starting values. Through the jittering and likelihood profiles, the present reference model represents the best fit to the data given the assumptions.

#### 3.7.2 Retrospective Analysis

A five-year retrospective analysis was conducted by running the model and sequentially removing one year of data up through minus 5 years. Retrospective spawning output (Figure 142) and relative stock status (Figure 143) show a distinct sensitivity to the removal of the latest data. Both the scale and status drop when missing the latest model. Once two years are removed, the model returns to something more similar to the reference model. After several additional model runs to explore why this sensitivity is occurring, it was clear that removal of the most recent age data in all fleets causes this pattern. Looking at each of the fisheries mean age data time series, all 4 sources of age data, which have different selectivities, but all of which to some degree include bigger and older individuals, show the consistent pattern of decreasing mean age from 2022 to 2023, and an increase in 2024. These changes, except for the large increase in scale for the -4 and -5, are within the uncertainty of the reference model. The Mohn's rho evaluation of the degree of retrospective pattern is given in Table and shown in Figure . The relative error in the data peels are below significant levels.

### 3.7.3 Fits to the Data

#### 3.7.3.1 Indices of Abundance

The fits to the 4 available indices of abundance demonstrate little information content in the survey indices (Figure 126 to Figure 129). They are all mostly flat with large uncertainty. Such lack of contrast and high uncertainty in the abundance measure indicate the indices contribute little influence to the model (see data sensitivities section below for more details).

#### 3.7.3.2 Lengths

Fits to the length data are examined based on the Pearson residuals-at-length, the annual mean lengths, and aggregated length composition data for the commercial and recreational fleets. The aggregate fit to each length composition demonstrates acceptable fits to each fleet and survey source (Figure 66). One noticeable behavior is the trade-off in fit between the fit in the bottom trawl fishery versus the fit in the at-sea-hake (ASHOP) fishery. This current model specification was the best trade off of fits between the two.

Fits to the annual length composition are provided for the following fisheries and surveys:

##### Fishery

- Bottom trawl (Figure 67 and Figure 68)
- Bottom trawl discard (Figure 69)
- Non-trawl (Figure 70 and Figure 71)
- Non-trawl discard (Figure 72)
- Midwater trawl (Figure 73)
- At-sea-hake (Figure 74)

##### Surveys

- Triennial (Figure 75)
- Alaska slope (Figure 76)
- West Coast Groundfish Bottom Trawl (Figure 77)

Pearson residuals of fits to the fishery (Figure 78) and survey (Figure 79) length data are reasonably small with no distinct patterns.

Model fits to the mean lengths, assuming Francis data-weighting of 1 and blocking patterns, demonstrate fits within the error bars of most years and no strong residual

patterns (Figure 80 to Figure 88). A notable observation in the means lengths within blocks generally show a lack of trend, not surprising given how many years an individual may be at its maximum size. This is in contrast with the age data that do show more nuances in age structure trend (see next section on age fits for more detail). This demonstrates the general lack of contrast in the length data.

### 3.7.3.3 Ages

#### 3.7.3.3.1 Conditional Age at Length

Fits to the sex-specific conditional age at length data are examined based on the age-at-length Pearson residuals, the annual mean ages, and mean age at length by year for the four fishery and one survey source. Pearson residuals were of reasonable size with no distinct patterns (Figure 89 to Figure 98), as most of the residuals were small and not noteworthy and demonstrate the expected shape of the growth curve. There is more contrast in the age data compared to the length data (Figure 99 to Figure 103). While the mean age for fisheries varied by gear selectivity (25-35 years for bottom trawl; 30-40 years for non-trawl; ~40 for midwater trawl; 40-50 years for the at-sea-hake fishery), one commonality was the increase in mean age in the last year of the model. This consistent increase in mean age across fisheries with different selectivities is important to remember when interpreting the retrospective pattern of the model. Mean age for the West Coast Groundfish Bottom Trawl survey, which catch much smaller and less larger individuals, fluctuated around 20 and did not show an mean age increase in the final year. Fits to the mean ages by length bins show acceptable fits consistent with model expectations (Figure 104 to Figure 120).

#### 3.7.3.3.2 Marginal Age

Marginal age compositions are not fit in the model, but they are included in order to see how well they fit the reference model without influencing the likelihood (Figure 121 to Figure 125). Marginal length and age composition cannot be used in the same model because of the overlap of fish in both samples. This is why ages conditioned on lengths are often used with the length compositions. But it still stands that age compositions, instead of lengths, could be used. So adding the marginal age compositions passively (i.e., not contributing to the overall likelihood of the model) can offer insight into how consistent they are with the current model fit. Overall the realized fits are good.

### 3.8 Model Results

As a supplement to the model results figures included in this report and described below, a full set of diagnostic plots created by the `{r4ss}` package is available at <https://github.com/shcaba/REBS-2025> along with the Stock Synthesis input files.

#### 3.8.1 Parameter Estimates

Estimated parameters by category are given in Table 19. The reference model parameter estimates along with asymptotic standard errors are shown in Table 20 and the likelihood components are shown in Table 21. Estimates of derived outputs and reference points and approximate 95 percent asymptotic confidence intervals are provided in Table 22.

The estimate of female natural mortality is higher than the assumed value for males, which fits the expectation given the oldest individuals in the population are all males, and within reason (0.039) given the oldest individual aged female sampled.

Estimated growth parameters values are similar to the externally estimated values (Table 20 and Figure 130), though with some important difference. The estimated  $L_\infty$  and  $k$  for both sexes were slightly greater and lower than the values estimated externally, respectively. This is not surprising, given external fits assume all variability is in the length at age, while the model incorporates ageing error. Both females and males reach their maximum size at relatively young ages ( $<$  half their presumed longevity), thus possibly limiting the information content of lengths on the underlying age structure.

Estimated ending selectivity curves for each fleet and survey (Figure 131) are a mix of dome-shaped (for bottom trawl gears) and logistic (for midwater gears) and look plausible given the biology (i.e., as a model convergence check for realism, the selectivity curves must look plausible). The surveys show the greatest degree of dome-shapeness, while the fisheries selectivities included sampling of at least some of the larger individuals. Time-varying selectivity showed mostly the same functional form for each fleet, despite changes in the selectivity, except for the non-trawl fishery, which changed from dome-shaped in the earlier blocks to logistic in the most recent time period (Figure 132 and Figure 133). The realized age selectivity based on the length-based selectivity show even more truncated sampling of older individuals in some of the dome-shaped fleets (Figure 134). Values for the estimated selectivity parameters are in Table 20.

The estimate of initial recruitment ( $\ln R_0$ ) is much higher than the previous assessment (7 vs. 6.20). While this is a large increase in the scale of the stock, a value of 7 is not unusual for shelf and/or slope rockfish species. The estimate of ( $\ln R_0$ ) for Rougheye/Blackspotted Rockfishes is well within the range of other groundfishes in similar habitat (Figure 135).

And given this assessment is for two species, the estimate for  $(\ln R_0)$  is reasonable. There is also a very large variability estimated for this parameter (Coefficient of variation = 0.62), thus scale is generally very poorly informed in this model.

### 3.8.2 Population Trajectory

The predicted spawning output (in millions of eggs) is provided in Table 23 and plotted in Figure 136. Estimated spawning output shows a decline in the early part of the time series due to poor recruitment, but rise before 1980 from offsetting recruitments (all not well informed), followed by a slight decline during the heaviest period of fishing, though moderated by recruitments in several years over the past 4 decades. The uncertainty around the estimate of spawning output is enormous, highlighting a major feature of the model output.

Relative spawning output never declined below the management target ( $SO_{40\%}$ ) and currently is estimated well above the target (Figure 137; 0.87 in 2025). The uncertainty in stocks status also does not support the stock being below the management target of  $SO_{40\%}$ . This uncertainty is based only on the asymptotic estimation of variance from one risk neutral model. Further uncertainty exploration is needed to capture a fuller range of uncertainty (see Sensitivity section).

The time series of estimated annual recruitment deviations are shown in Figure 138. The bias adjustment plot (Figure 139) indicates that the most informed recruitment deviations occur after 1980. While post-1980 is the most informed, the recruitment deviations before that time period, while less assure, are important for the model to prepare the population structure for the upcoming increase in fishing while reconciling weak, but still present, signals in recruitment from the biological compositions that can contain information on recruitment for decades given the long-liveness of these species. Sensitivities to when recruitment estimation begins in the model is shown in the Sensitivity section. Numbers of age-0 individuals indicate those years of particularly strong recruitment (Figure 140). Noticeable recruitment years are seen in 1988, 1993, 1999, 2008, 2010, 2012, and 2017 (Figure 141). This amounts to roughly two notable recruitments per decade. Given this assessment is tracking two species, it is hard to tell whether the species are synchronized or showing their own recruitment pulse in this signal.

### 3.9 Characterizing uncertainty

#### 3.9.1 Sensitivity Analyses

Sensitivity analyses were conducted to evaluate model sensitivity to alternative data treatment and model specifications.

##### 3.9.1.1 Data treatment sensitivities

Data treatments explored the removal of data types (indices and biological compositions), data weighting options and ageing error treatment.

###### 3.9.1.1.1 Removal of data sources

The following data source sensitivities were explored:

- Remove a single index of abundance (Triennial, Alaska slope, NW slope, WCGBTS, all indices)
- For each fleet and index, including discard fleets, removal of length composition data
- For each fleet and the WCGBTS, removal of age composition data

Likelihood values and estimates of key parameters and derived quantities from each data source removal sensitivity are available in Table 24 to Table 27. Additionally, time series of spawning output and relative spawning output are shown in Figure 144 to Figure 151. Derived quantities relative of all data removal sensitivities to the reference model are summarized in Figure 152. Generally, the removal of indices of abundance decreased both the spawning output and the relative spawning output as compared to the base model (Figure 144 and Figure 145). Of these, the WCGBTS appears to have the most significant impact. Removal of the bottom trawl (bottom trawl + discards) and non-trawl (non-trawl + discards) fishery length compositions decreased the spawning output and the relative spawning output, whereas removal of the mid-water trawl and the at-sea hake length compositions increased the spawning output and relative spawning output (Figure 146 and Figure 147). Of the length composition removals, the most influential data sources appear to be the fishery fleets and removal of the survey length compositions had a limited impact on both spawning output and relative spawning output compared to the base model (Figure 148 and Figure 149). Finally, the removal of any age composition data source decreased the spawning output, with the exception of the mid-water trawl ages, and decreased the relative spawning output (Figure 150 and Figure 151). Again, the WCGBTS appears to be the most influential of the data sources with age composition information.

#### 3.9.1.1.2 Data weighting

The following data weighting sensitivities were explored:

- No data-weighting
- Dirichlet data-weighting
- McAllister-Ianelli data weighting

Likelihood values and estimates of key parameters and derived quantities from each data weighting sensitivity are available in Table 28. Comparison plots for the results relative to the reference model are in Figure 153 and Figure 154. Each data weighting scenario up-weighted the influence of the length data relative to the age data. This resulted in a large drop in the stock scale, but little change in the high stock status. These changes were within the uncertainty of the base model, though at the low end for the stock scale.

#### 3.9.1.1.3 Ageing error

The following ageing error sensitivities were explored:

- No Ageing error for all sources (unbiased and precise ages)
- No ageing error for each source at a time

Likelihood values and estimates of key parameters and derived quantities from each ageing error sensitivity are available in Table 29. Comparison plots for the results relative to the reference model are in Figure 155 and Figure 156. A drop in both stock scale and stock status is generally observed when the ageing error is removed. One notable observation is that the recruitment events become less resolved and variance increases in lengths at age when ages are assumed biased and precise. When the ages are taken as true, all variation between lengths and ages are interpreted in the lengths, while the recruitment must interpret the age composition precisely, thus assigning positive recruitment deviations to years adjacent to strong recruitment years seen in the reference model. Having more positive recruitment can compensate for a lower initial stock size, causing the initial stock size to decrease. All in stock scale and status changes were within the uncertainty of the base model.

#### 3.9.1.2 Sensitivities to Model Specification

Model specifications looked at the treatment of life history parameters selectivity and the estimation of recruitment. All scenarios match the reference model specifications in all other aspects unless otherwise stated.



- Life history parameters
  - Natural mortality ( $M$ )
    1. Estimate male  $M$
    2. Fix  $M$  to 2013 value
    3. Lorenzen age varying  $M$
  - Growth parameters
    4. Fix all growth parameters to external values
  - Reproductive Biology
    5. Proportional fecundity to weight
    6. Functional maturity at age
    7. Functional maturity at length for Blackspotted Rockfish
    8. Functional maturity at length for Rougheye Rockfish
    9. Biological maturity at length

Summaries of the model results for these sensitivities are presented in Table 30, Figure 157 and Figure 158. Additionally, the scale comparisons minus the estimation of  $M$  scenario is provided in Figure 159 as that model scenario showed such higher estimates of scale to obscure all other models. Note also that the proportional fecundity scale measure is in metric tons, not millions of eggs, and thus are not comparable in scale to the other models.

None of the sensitivities were found to be outside of the reference model uncertainty for scale or stock status (Figure 158). In particular, the stock status was highly conserved over these sensitivities, while both the initial and final stock scales were more sensitive (but still within the uncertainty of the reference model). In general, higher natural mortality assumptions reduced the stock scale when the bottom trawl and non-trawl fisheries were dome-shaped, as did the fixing of growth to the external model estimates. For reproductive biology, scale decreased if either age at maturity of functional length at maturity for Blackspotted Rockfish only was used. But scale increased if either functional maturity for Rougheye Rockfish only or biological maturity was used. The reference model, a mix of both species maturity at length, settled the scale inbetween each individual species-specific functional maturity model. For this long-lived species and future consideration, age may be a more precise determination of maturity (because there is a wide range of lengths per age) than length; however, age data is more limited and further information is encouraged to build this relationship.

- Recruitment estimation

13. No recruitment estimation
  14. Estimate from 1940 onward
  15. Estimate from 1980 onward
- Selectivity
    16. Logistic selectivity for all fisheries and time blocks
    17. Logistic selectivity for bottom trawl fishery
    18. Logistic selectivity for non-trawl fishery

Likelihood values and estimates of key parameters and derived quantities from each sensitivity are available in Table 31. Comparison plots for the results relative to the reference model are in Figure 160 and Figure 161. Few (starting in 1980) or no recruitment deviation estimation leads to larger stock scale required to make up for the catch history and lack of recruitment compensation. There is a drop in stocks status when starting recruitment deviations in 1940, but more similar to the reference model. Stock status is steady in these scenarios and all uncertainty is included in the reference model uncertainty (Figure 161).

The model is much more sensitive to the assumption that fishery selectivity for the bottom trawl and/or non-trawl are logistic. These produce the most sensitive of all the explored alternative models scenarios in both scale (lower) and stock status. In all of these instances, the uncertainty is outside that of the reference model Figure 161. The reason these are less likely scenarios can be seen in the notably degraded fits to the bottom trawl and/or non-trawl length compositions (Figure 162). This supports the choice for the change in the reference model from the previous stock assessment to consider dome-shaped selectivity for those two fisheries. Given the only scenarios to fall below the low uncertainty treat the bottom trawl as logistic selectivity, it is a good candidate for a lower state of nature (and not a risk neutral model).

### 3.9.2 Likelihood profiles

Likelihood profiles were conducted for the the log of initial recruitment ( $\ln(R_0)$ ), steepness ( $h$ ), and female male natural mortality ( $M$ ) with female natural mortality estimated. Likelihood profiles were conducted by fixing the featured parameter(s) at specific values across a range of values and estimating all remaining parameters. A likelihood profile offers insight into model sensitivity to changing model parameter values, while providing an additional way to describe uncertainty in the parameter by identifying the range of parameters within 1.96 likelihood units of the reference model.

### 3.9.2.1 Initial unfished recruitment ( $\ln R_0$ )

The profile on the assumption of  $\ln R_0$  (that sets the initial scale of the population) demonstrates the data and current model specification support value of  $\ln R_0$  greater than 6.25, with a steep increase in likelihood difference as  $\ln R_0$  goes lower (Figure 163). There is the expectant increase in scale across increasing  $\ln R_0$  values for both initial and ending spawning output. The final spawning output goes down quicker than the initial spawning output, leading to a lower stock status. Even at very low  $\ln R_0$  values, the stock does not drop below the target reference stock status.

The component likelihoods show a clear conflict in the length and age data, with lengths pulling for lower  $\ln R_0$  values and ages for higher  $\ln R_0$  values (Figure 164). While this is a consistent signal in the ages, the individual length sources also show conflict, with the bottom trawl length data pulling for higher  $\ln R_0$  values.

### 3.9.2.2 Steepness ( $h$ )

The profile on the assumption of  $h$  (defining productivity of the stock) demonstrates the data and current model specification do not demonstrate support for a particular steepness value (Figure 165). The change in scale and stock status is small and mostly flat through a large range of  $h$  values. The lack of information on  $h$  is seen across the data likelihood components (Figure 166). Given the strong signal of a health stock with little contrast in historical population trajectory, these are expected results. This strongly supports the pre-specifying of this value, but also demonstrates the low sensitivity to pre-specifying this value.

### 3.9.2.3 Male natural mortality ( $M$ )

The profile on the assumption of male  $M$  (while still estimating female  $M$ ) demonstrates the data and current model specification support a higher value of male  $M$  than what is being used in the reference model (Figure 167). This uncertainty has a much larger expression in the scale than in the stock status, and one major reason the model estimated uncertainty in scale is so large. It is also a reason why only female  $M$  was estimated. Comparing the sex-specific values of  $M$ , females are always higher than males (Figure 168). Regarding stock status, there were no explored  $M$  values that caused the stock to go below the target reference point.

The component likelihoods show a some conflict in the length and age data, with lengths (mostly the non-trawl lengths; most of the other length contain no information) pulling for higher  $M$  values and ages for lower  $M$  values (Figure 169). As noted when building the reference model, ages are believed to contain more information on natural mortality

than lengths. Consider that component likelihood profile, the reference model value for male  $M$  is the lowest support by the age data, and was chosen to avoid the fast increasing scale estimates. The integrated assessment approach does not allow for ages only to define the estimation of  $M$ , and thus the choice to pre-specify one  $M$  value and estimate the other in order to anchor to compensate for the incursion of the length data signal in the estimation of  $M$ .

### 3.9.3 Summary of Uncertainty source and states of nature

Over the many different sources of potential uncertainty, patterns in how these sources affect scale and status emerge (Table 32). These sources highlight conflicts in data and selectivity assumptions, while also emphasizing key uncertainty in natural mortality. Specifically, there are many ways to make the reference model at a higher or lower scale and/or stock status (Table 32). In building states of nature and around this risk neutral reference model, the two key axes of uncertainty are based around the treatment of natural mortality and selectivity. The high state takes the estimation of selectivity (allowing dome-shaped if chosen) and adding the estimation of  $M$  for both sexes (Model specification sensitivity scenario #1). The low state of nature assume the reference value of male  $M$  with an estimated female  $M$  and constraining the selectivity of bottom trawl and non-trawl fisheries to logistic (Model specification sensitivity scenario #16). These produce a very wide set of uncertainties around both scale and stock status (Figure 170). If these are treated as an ensemble, with high and low states of nature assuming a weight of 12.5% and the reference model the remaining 75%, the uncertainty range can be expressed as one model output (Figure 171).

### 3.9.4 Unresolved Problems and Major Uncertainties

Rougheye/Blackspotted Rockfishes are one of the longest lived species of rockfish on the West Coast, with maximum ages of 169 years in this assessment, Individual as old as 205 years are reported in literature. It is, therefore, likely that natural mortality for Rougheye/Blackspotted Rockfishes is lower than for other rockfish species. With length and age data available only for years after 1994, there is limited ability to monitor the long-term changes of aging cohorts. Therefore, estimates of natural mortality are uncertain. This assessment attempts to capture uncertainty in this parameters and integrate it into the derived biomass estimates by estimating natural mortality for females, while fixing the male natural mortality at the value from meta-analytical study. Model sensitivities and profiles over  $M$  showed that current stock status was sensitive to the assumption about natural mortality.

Model is also sensitive to assumptions about shape of the selectivity curves, especially for the bottom trawl and non-trawl fleet. In the previous assessment all fleets (except for

Triennial Survey) were assumed asymptotic. In this assessment, we provided flexibility for the bottom trawl fleet and bottom trawl surveys to estimate dome-shaped selectivity curves. Changes in selectivity assumptions allow to substantially improve fits to length and age composition data in fisheries and surveys, but also resulted in a substantial increase of stock scale.

## 4 Management

### 4.1 Reference Points

Reference points were calculated using the estimated fishery selectivity and removals in the most recent year of the model (2024, Table 22). Reference points were based on the rockfish  $F_{MSY\%}$  proxy ( $SPR_{50\%}$ ), target relative biomass (40%), and estimated selectivity and catch for each fleet (Table 22). The proxy MSY values of management quantities are by definition more conservative compared to the estimated MSY and MSY relative to 40% of unfished spawning output because of the assumed steepness value. Sustainable total yield, removals, using the proxy  $SPR_{50\%}$  is 526.17 mt. The spawning output equivalent to 40% of the unfished spawning output ( $SO_{40\%}$ ) calculated using the SPR target ( $SPR_{50\%}$ ) was  $2.51973 \times 10^6$  millions of eggs.

The 2025 spawning biomass relative to unfished equilibrium spawning biomass, based on the 2024 fishing year, is 87%, above the management target of 40% of unfished spawning output (Figure 172). The fishing intensity,  $1 - SPR$ , was below or just slightly above harvest rate limit ( $SPR_{50\%}$ ) from the mid-1980s through the 1990s. Removals since 2000 have been below the point estimate of potential long-term yields calculated using an  $SPR_{50\%}$  reference point (Table 23 and Figure 173). The equilibrium estimates of yield relative to biomass based on a steepness value fixed at 0.72 are provided in Figure 174, where vertical dashed lines indicate the estimate of fraction unfished at the start of 2025 (current) and the estimated management targets calculated based on the relative target biomass (B target), the SPR target, and the maximum sustainable yield (MSY).

The relative biomass and the ratio of the estimated SPR to the management target ( $SPR_{50\%}$ ) across all model years are shown in Figure 175 where cooler colors (purple) represent early years and warmer colors (yellow) represent recent years. The stock status has not decreased below the target relative biomass, and fishing intensity (except for in 1995) has been below the target fishing intensity based on  $SPR_{50\%}$ .

### 4.2 Management performance

In the last ten years total dead catches of Rougheye/Blackspotted Rockfishes have been below the annual catch limit. The last ten years total dead catches for Rougheye/Blackspotted Rockfishes against the overfishing limits (OFLs), the acceptable biological catches (ABCs), the annual catch limits (ACLs) are shown in Table 33.

### 4.3 Harvest Projections and Decision Tables

The Rougheye/Blackspotted Rockfishes assessment is being considered as a category 1 assessment with a  $P^* = 0.45$ ,  $\sigma = 0.68$  (estimated within the model, see Section 4.4) and a time-varying buffer applied to set the ABC below the OFL. These multipliers are also combined with the rockfish MSY proxy of  $SPR_{50}$  and the 40-10 harvest control rule to calculate OFLs and ACLs. Projections of the overfishing limit, acceptable biological catch, and annual catch limit, all based on a  $P^*$  of 0.45 and a log-space standard deviation of the overfishing limit ( $\sigma$ ) of 0.50 are included in Table 34. Assumed catches for 2025 and 2026 for this projection were provided by the PFMC Groundfish Management Team, and catches from 2027 onward assume full attainment of the acceptable biological catch.

No decision table needed in draft assessments undergoing review.

### 4.4 Evaluation of Scientific Uncertainty

The model estimated uncertainty around the 2025 spawning output for is  $\sigma = 0.6836465$ . The uncertainty around the OFL is  $\sigma = 0.6713586$ . These  $\sigma$  values underestimate the overall uncertainty as they do not incorporate the model structural uncertainty and do not account for any time-varying dynamics other than recruitment. The estimated uncertainty values are higher than the Category 1 default  $\sigma = 0.5$ , so all projections will use the estimated  $\sigma$ . The alternative states of nature used to bracket uncertainty in the decision table assist with encapsulating model structure uncertainty.

### 4.5 Research and Data Needs

There are many areas of research that could be improved to benefit the understanding and assessment of Rougheye/Blackspotted Rockfishes. Below, we identify several of them that we consider particularly important.

- Understanding the stock structure and biology of Rougheye and Blackspotted Rockfishes: This assessment reports the status of Rougheye/Blackspotted Rockfishes as a pooled complex because it is extremely difficult to separate the catches of each species even in recent data, and attempting to do so would greatly increase the uncertainty in the predictions. Because little is known about the respective biology and catch histories of the two species, it is unclear whether managing them as a complex may place one species at disproportionate risk of overfishing relative to the other. Additional research that will provide insight into the distribution, life history, biological characteristics, and catch and discard profiles of the two species

is recommended. Such an endeavor would like require the efforts of at sea observers in all fleets, biologists aboard fishery-independent surveys, and port samplers along the entire West Coast requiring broad, inter-agency collaboration.

- Understanding of coastwide stock structure, connectivity, and distribution: This is a stock assessment for Rougheye/Blackspotted Rockfishes off of the west coast of the U.S. and does not consider data from British Columbia or Alaska. Further investigating and comparing the data and predictions from British Columbia and Alaska to determine if there are similarities with the U.S. West Coast observations would help to define the connectivity between rougheye rockfish north of the U.S.-Canada border.
- Natural mortality: Uncertainty in natural mortality translates into uncertain estimates of status and sustainable fishing levels for Rougheye/Blackspotted Rockfishes. Rougheye/Blackspotted Rockfishes are one of the longest lived species of rockfish (with maximum age as old as 205 years reported in the literature). Assessment model was able to estimate female natural mortality, consistent with maximum age based meta-analytical prior, but with male natural mortality fixed at the median of the prior. The collection of additional age data and further research of the life-history of Rougheye/Blackspotted Rockfishes may improve our understanding of Rougheye/Blackspotted Rockfishes natural mortality.
- Historical landings and discards, including investigation of fisheries selectivity: The substantial progress has been made in reconstructing historical landings of rockfishes on the U.S. West Coast, including those for Rougheye/Blackspotted Rockfishes. This assessment highlighted the importance of understanding of fishery selectivity assumptions associated with removals and how fishery selectivity changes throughout the years. Further understanding of this area would help reduce uncertainty in estimated scale of the stock.



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## 7 Tables

Table 1: Major management actions since 2000 that have impacted the management of the Rougheye/Blackspotted complex.

| Year | Management Action                                                                                                                                                                                                                                     |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2000 | Minor slope rockfish complex formed north and south of 40° 10' N and is subject to bi-monthly vessel limits. New limited entry trawl gear restrictions implemented for large footrope trawl gear, small footrope trawl gear, and midwater trawl gear. |
| 2002 | Rockfish Conservation Areas (RCA) established. Large footrope gear prohibited inside 275 m. Open access trip limits revised for the minor slope rockfish complex.                                                                                     |
| 2005 | Selective flatfish trawl required shoreward of the RCA north of 40° 10' N                                                                                                                                                                             |
| 2006 | Amendment 19 established essential fish habitat (EFH) boundaries and conservation areas.                                                                                                                                                              |
| 2007 | Seasonal changes of trawl RCA boundaries and periodic closures within certain latitude boundaries (e.g., north of Cape Alava at 48°10' N. latitude to the U.S.- Canada border) started in 2007.                                                       |
| 2011 | Trawl rationalization began, establishing the IFQ fishery.                                                                                                                                                                                            |
| 2015 | Canary rockfish rebuilt; increased attainment of mid-water rockfishes when implemented in 2017.                                                                                                                                                       |
| 2020 | Trawl RCA off OR and CA removed (WA RCA remained in place)                                                                                                                                                                                            |
| 2023 | Non-bottom contact hook and line gear allowed in non-trawl RCA south of WA-OR border.                                                                                                                                                                 |
| 2024 | Multiple non-trawl RCA south of WA-OR border rule updates, ultimately pushing fixed gear effort onto the shelf/slope.                                                                                                                                 |

Table 2: Recent trend in the overfishing limits (OFL), the acceptable biological catches (ABCs), the annual catch limits (ACLs), and the total dead catch (landings + discards) all in metric tons (mt) for the Rougheye/Blackspotted complex north and south of 40°10'.

|      | North of 40°10' |          |          | South of 40°10' |          |          | Combined |          |          | Catch (mt) |
|------|-----------------|----------|----------|-----------------|----------|----------|----------|----------|----------|------------|
|      | OFL (mt)        | ABC (mt) | ACL (mt) | OFL (mt)        | ABC (mt) | ACL (mt) | OFL (mt) | ABC (mt) | ACL (mt) |            |
| 2015 | 201.9           | 184.3    | 184.3    | 4.1             | 3.8      | 3.8      | 206      | 188.1    | 188.1    | 132.1      |
| 2016 | 206.8           | 188.8    | 188.8    | 4.2             | 3.9      | 3.9      | 211      | 192.7    | 192.7    | 149.0      |
| 2017 | 210.7           | 192.4    | 192.4    | 4.3             | 3.9      | 3.9      | 215      | 196.3    | 196.3    | 156.0      |
| 2018 | 214.6           | 195.9    | 195.9    | 4.4             | 4.0      | 4.0      | 219      | 199.9    | 199.9    | 241.5      |
| 2019 | 217.6           | 198.7    | 198.7    | 4.4             | 4.0      | 4.0      | 222      | 202.7    | 202.7    | 226.6      |
| 2020 | 219.5           | 200.4    | 200.4    | 4.5             | 4.1      | 4.1      | 224      | 204.5    | 204.5    | 106.1      |
| 2021 | 232.3           | 191.8    | 191.8    | 4.7             | 3.9      | 3.9      | 237      | 195.8    | 195.8    | 92.8       |
| 2022 | 233.2           | 190.8    | 190.8    | 4.8             | 3.9      | 3.9      | 238      | 194.7    | 194.7    | 117.4      |
| 2023 | 233.2           | 188.9    | 188.9    | 4.8             | 3.9      | 3.9      | 238      | 192.8    | 192.8    | 97.6       |
| 2024 | 233.2           | 190.8    | 190.8    | 4.8             | 3.9      | 3.9      | 238      | 194.7    | 194.7    | 118.9      |



Table 3: Landings in metric tons (mt) by year for each fleet.

| Year | Total (mt) | Trawl (mt) | Trawl Discard (mt) | Non-trawl (mt) | Non-trawl discard (mt) | Midwater trawl (mt) | At-sea-Hake (mt) |
|------|------------|------------|--------------------|----------------|------------------------|---------------------|------------------|
| 1891 | 0          | 0          | 0                  | 0              | 0                      | 0                   | 0                |
| 1892 | 19         | 0          | 0                  | 19             | 0                      | 0                   | 0                |
| 1893 | 19         | 0          | 0                  | 19             | 0                      | 0                   | 0                |
| 1894 | 19         | 0          | 0                  | 19             | 0                      | 0                   | 0                |
| 1895 | 5          | 0          | 0                  | 5              | 0                      | 0                   | 0                |
| 1896 | 1          | 0          | 0                  | 1              | 0                      | 0                   | 0                |
| 1897 | 1          | 0          | 0                  | 1              | 0                      | 0                   | 0                |
| 1898 | 1          | 0          | 0                  | 1              | 0                      | 0                   | 0                |
| 1899 | 1          | 0          | 0                  | 1              | 0                      | 0                   | 0                |
| 1900 | 2          | 0          | 0                  | 2              | 0                      | 0                   | 0                |
| 1901 | 2          | 0          | 0                  | 2              | 0                      | 0                   | 0                |
| 1902 | 3          | 0          | 0                  | 3              | 0                      | 0                   | 0                |
| 1903 | 3          | 0          | 0                  | 3              | 0                      | 0                   | 0                |
| 1904 | 4          | 0          | 0                  | 4              | 0                      | 0                   | 0                |
| 1905 | 4          | 0          | 0                  | 4              | 0                      | 0                   | 0                |
| 1906 | 4          | 0          | 0                  | 4              | 0                      | 0                   | 0                |
| 1907 | 5          | 0          | 0                  | 5              | 0                      | 0                   | 0                |
| 1908 | 8          | 0          | 0                  | 8              | 0                      | 0                   | 0                |
| 1909 | 6          | 0          | 0                  | 6              | 0                      | 0                   | 0                |
| 1910 | 6          | 0          | 0                  | 6              | 0                      | 0                   | 0                |
| 1911 | 7          | 0          | 0                  | 7              | 0                      | 0                   | 0                |
| 1912 | 7          | 0          | 0                  | 7              | 0                      | 0                   | 0                |
| 1913 | 8          | 0          | 0                  | 8              | 0                      | 0                   | 0                |
| 1914 | 8          | 0          | 0                  | 8              | 0                      | 0                   | 0                |
| 1915 | 10         | 0          | 0                  | 10             | 0                      | 0                   | 0                |
| 1916 | 9          | 0          | 0                  | 9              | 0                      | 0                   | 0                |
| 1917 | 10         | 0          | 0                  | 10             | 0                      | 0                   | 0                |
| 1918 | 55         | 0          | 0                  | 55             | 0                      | 0                   | 0                |
| 1919 | 26         | 0          | 0                  | 26             | 0                      | 0                   | 0                |
| 1920 | 23         | 0          | 0                  | 23             | 0                      | 0                   | 0                |
| 1921 | 23         | 0          | 0                  | 23             | 0                      | 0                   | 0                |
| 1922 | 18         | 0          | 0                  | 18             | 0                      | 0                   | 0                |
| 1923 | 20         | 0          | 0                  | 20             | 0                      | 0                   | 0                |
| 1924 | 32         | 0          | 0                  | 32             | 0                      | 0                   | 0                |
| 1925 | 38         | 0          | 0                  | 38             | 0                      | 0                   | 0                |
| 1926 | 54         | 0          | 0                  | 54             | 0                      | 0                   | 0                |
| 1927 | 69         | 0          | 0                  | 69             | 0                      | 0                   | 0                |
| 1928 | 72         | 0          | 0                  | 72             | 0                      | 0                   | 0                |
| 1929 | 66         | 0          | 0                  | 66             | 0                      | 0                   | 0                |
| 1930 | 67         | 0          | 0                  | 67             | 0                      | 0                   | 0                |
| 1931 | 44         | 0          | 0                  | 44             | 0                      | 0                   | 0                |
| 1932 | 25         | 0          | 0                  | 25             | 0                      | 0                   | 0                |
| 1933 | 35         | 0          | 0                  | 35             | 0                      | 0                   | 0                |
| 1934 | 42         | 0          | 0                  | 42             | 0                      | 0                   | 0                |
| 1935 | 34         | 0          | 0                  | 34             | 0                      | 0                   | 0                |
| 1936 | 61         | 0          | 0                  | 61             | 0                      | 0                   | 0                |
| 1937 | 53         | 0          | 0                  | 53             | 0                      | 0                   | 0                |
| 1938 | 55         | 0          | 0                  | 55             | 0                      | 0                   | 0                |

|      |     |     |   |     |   |   |    |
|------|-----|-----|---|-----|---|---|----|
| 1939 | 28  | 0   | 0 | 28  | 0 | 0 | 0  |
| 1940 | 60  | 1   | 0 | 59  | 0 | 0 | 0  |
| 1941 | 102 | 1   | 0 | 101 | 0 | 0 | 0  |
| 1942 | 126 | 2   | 0 | 124 | 0 | 0 | 0  |
| 1943 | 258 | 7   | 0 | 251 | 0 | 0 | 0  |
| 1944 | 85  | 11  | 0 | 74  | 0 | 0 | 0  |
| 1945 | 50  | 20  | 0 | 30  | 0 | 0 | 0  |
| 1946 | 69  | 11  | 0 | 58  | 0 | 0 | 0  |
| 1947 | 42  | 7   | 0 | 35  | 0 | 0 | 0  |
| 1948 | 44  | 5   | 0 | 39  | 0 | 0 | 0  |
| 1949 | 31  | 5   | 0 | 26  | 0 | 0 | 0  |
| 1950 | 52  | 6   | 0 | 46  | 0 | 0 | 0  |
| 1951 | 59  | 6   | 0 | 53  | 0 | 0 | 0  |
| 1952 | 38  | 6   | 0 | 32  | 0 | 0 | 0  |
| 1953 | 21  | 5   | 0 | 16  | 0 | 0 | 0  |
| 1954 | 36  | 6   | 0 | 30  | 0 | 0 | 0  |
| 1955 | 32  | 6   | 0 | 26  | 0 | 0 | 0  |
| 1956 | 21  | 8   | 0 | 13  | 0 | 0 | 0  |
| 1957 | 35  | 9   | 0 | 26  | 0 | 0 | 0  |
| 1958 | 15  | 7   | 0 | 8   | 0 | 0 | 0  |
| 1959 | 23  | 7   | 0 | 16  | 0 | 0 | 0  |
| 1960 | 23  | 10  | 0 | 13  | 0 | 0 | 0  |
| 1961 | 26  | 11  | 0 | 15  | 0 | 0 | 0  |
| 1962 | 32  | 14  | 0 | 18  | 0 | 0 | 0  |
| 1963 | 24  | 13  | 0 | 11  | 0 | 0 | 0  |
| 1964 | 31  | 11  | 0 | 20  | 0 | 0 | 0  |
| 1965 | 31  | 23  | 0 | 8   | 0 | 0 | 0  |
| 1966 | 117 | 111 | 0 | 6   | 0 | 0 | 0  |
| 1967 | 108 | 98  | 0 | 10  | 0 | 0 | 0  |
| 1968 | 172 | 165 | 0 | 7   | 0 | 0 | 0  |
| 1969 | 50  | 25  | 0 | 25  | 0 | 0 | 0  |
| 1970 | 23  | 19  | 0 | 4   | 0 | 0 | 0  |
| 1971 | 68  | 67  | 0 | 1   | 0 | 0 | 0  |
| 1972 | 76  | 75  | 0 | 1   | 0 | 0 | 0  |
| 1973 | 75  | 69  | 0 | 6   | 0 | 0 | 0  |
| 1974 | 76  | 58  | 0 | 18  | 0 | 0 | 0  |
| 1975 | 43  | 35  | 0 | 5   | 0 | 0 | 3  |
| 1976 | 19  | 16  | 0 | 2   | 0 | 0 | 1  |
| 1977 | 166 | 1   | 0 | 164 | 0 | 0 | 1  |
| 1978 | 69  | 33  | 0 | 36  | 0 | 0 | 0  |
| 1979 | 185 | 63  | 0 | 121 | 0 | 0 | 1  |
| 1980 | 99  | 56  | 0 | 43  | 0 | 0 | 0  |
| 1981 | 131 | 61  | 0 | 68  | 0 | 0 | 2  |
| 1982 | 167 | 99  | 0 | 68  | 0 | 0 | 0  |
| 1983 | 126 | 55  | 0 | 70  | 0 | 0 | 1  |
| 1984 | 144 | 75  | 0 | 67  | 0 | 0 | 2  |
| 1985 | 298 | 139 | 0 | 158 | 0 | 0 | 1  |
| 1986 | 428 | 154 | 0 | 273 | 0 | 0 | 1  |
| 1987 | 570 | 198 | 0 | 368 | 0 | 0 | 4  |
| 1988 | 351 | 173 | 0 | 162 | 0 | 0 | 16 |
| 1989 | 418 | 287 | 0 | 131 | 0 | 0 | 0  |
| 1990 | 244 | 167 | 0 | 76  | 0 | 0 | 1  |
| 1991 | 299 | 235 | 0 | 59  | 0 | 0 | 5  |
| 1992 | 306 | 186 | 0 | 110 | 0 | 0 | 10 |

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|      |     |     |    |     |    |    |     |
|------|-----|-----|----|-----|----|----|-----|
| 1993 | 327 | 166 | 0  | 159 | 0  | 0  | 2   |
| 1994 | 306 | 127 | 0  | 173 | 0  | 0  | 6   |
| 1995 | 744 | 165 | 0  | 576 | 0  | 0  | 3   |
| 1996 | 339 | 127 | 0  | 204 | 0  | 0  | 8   |
| 1997 | 303 | 107 | 0  | 186 | 0  | 0  | 10  |
| 1998 | 441 | 110 | 0  | 313 | 0  | 0  | 18  |
| 1999 | 256 | 81  | 0  | 166 | 0  | 0  | 9   |
| 2000 | 183 | 79  | 0  | 29  | 0  | 4  | 71  |
| 2001 | 114 | 74  | 0  | 18  | 0  | 1  | 21  |
| 2002 | 74  | 31  | 14 | 27  | 1  | 0  | 1   |
| 2003 | 100 | 58  | 15 | 23  | 2  | 0  | 2   |
| 2004 | 115 | 58  | 3  | 34  | 5  | 1  | 14  |
| 2005 | 137 | 45  | 1  | 50  | 5  | 0  | 36  |
| 2006 | 127 | 48  | 12 | 59  | 1  | 0  | 7   |
| 2007 | 187 | 60  | 27 | 59  | 10 | 2  | 29  |
| 2008 | 219 | 54  | 29 | 56  | 3  | 1  | 76  |
| 2009 | 228 | 67  | 45 | 104 | 1  | 2  | 9   |
| 2010 | 263 | 79  | 60 | 71  | 25 | 6  | 22  |
| 2011 | 210 | 53  | 0  | 63  | 9  | 4  | 81  |
| 2012 | 244 | 47  | 0  | 74  | 20 | 49 | 54  |
| 2013 | 156 | 64  | 0  | 59  | 12 | 3  | 18  |
| 2014 | 91  | 34  | 0  | 37  | 10 | 4  | 6   |
| 2015 | 133 | 31  | 0  | 47  | 14 | 19 | 22  |
| 2016 | 150 | 31  | 0  | 60  | 13 | 16 | 30  |
| 2017 | 155 | 22  | 0  | 59  | 34 | 2  | 38  |
| 2018 | 242 | 16  | 0  | 47  | 15 | 3  | 161 |
| 2019 | 226 | 22  | 0  | 39  | 31 | 9  | 125 |
| 2020 | 106 | 10  | 0  | 24  | 1  | 29 | 42  |
| 2021 | 92  | 10  | 0  | 21  | 2  | 21 | 38  |
| 2022 | 118 | 12  | 0  | 19  | 3  | 19 | 65  |
| 2023 | 96  | 13  | 0  | 19  | 0  | 26 | 38  |
| 2024 | 118 | 10  | 0  | 10  | 0  | 69 | 29  |

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Table 4: Landings (metric tons) for botton trawl fleet by state and bycatch within POP historcial fishery.

| Year | California | Oregon | Washington | POP_Fishery_Bycatch | Total |
|------|------------|--------|------------|---------------------|-------|
| 1932 | 0          | 0      | 0          | 0                   | 0     |
| 1933 | 0          | 0      | 0          | 0                   | 0     |
| 1934 | 0          | 0      | 0          | 0                   | 0     |
| 1935 | 0          | 0      | 0          | 0                   | 0     |
| 1936 | 0          | 0      | 0          | 0                   | 0     |
| 1937 | 0          | 0      | 0          | 0                   | 0     |
| 1938 | 0          | 0      | 0          | 0                   | 0     |
| 1939 | 0          | 0      | 0          | 0                   | 0     |
| 1940 | 0          | 0      | 0          | 0                   | 1     |
| 1941 | 0          | 1      | 0          | 0                   | 1     |
| 1942 | 0          | 1      | 0          | 0                   | 2     |
| 1943 | 0          | 5      | 2          | 0                   | 7     |
| 1944 | 0          | 9      | 3          | 0                   | 11    |
| 1945 | 0          | 14     | 6          | 0                   | 20    |
| 1946 | 0          | 8      | 3          | 0                   | 11    |
| 1947 | 0          | 5      | 1          | 0                   | 7     |
| 1948 | 0          | 3      | 2          | 0                   | 5     |
| 1949 | 0          | 3      | 1          | 0                   | 5     |
| 1950 | 0          | 4      | 2          | 0                   | 6     |
| 1951 | 0          | 4      | 2          | 0                   | 6     |
| 1952 | 0          | 5      | 1          | 0                   | 6     |
| 1953 | 0          | 3      | 1          | 0                   | 5     |
| 1954 | 0          | 4      | 2          | 0                   | 6     |
| 1955 | 0          | 4      | 2          | 0                   | 6     |
| 1956 | 0          | 6      | 2          | 0                   | 8     |
| 1957 | 0          | 7      | 2          | 0                   | 9     |
| 1958 | 0          | 6      | 2          | 0                   | 7     |
| 1959 | 0          | 5      | 1          | 0                   | 7     |
| 1960 | 0          | 7      | 3          | 0                   | 10    |
| 1961 | 0          | 7      | 4          | 0                   | 11    |
| 1962 | 0          | 8      | 6          | 0                   | 14    |
| 1963 | 0          | 6      | 7          | 0                   | 13    |
| 1964 | 0          | 6      | 6          | 0                   | 11    |
| 1965 | 0          | 17     | 7          | 0                   | 23    |
| 1966 | 0          | 7      | 6          | 98                  | 111   |
| 1967 | 0          | 7      | 6          | 85                  | 98    |
| 1968 | 0          | 4      | 114        | 47                  | 165   |
| 1969 | 0          | 9      | 2          | 15                  | 25    |
| 1970 | 0          | 0      | 1          | 17                  | 19    |
| 1971 | 0          | 11     | 7          | 49                  | 67    |
| 1972 | 0          | 6      | 1          | 68                  | 75    |
| 1973 | 0          | 3      | 4          | 63                  | 69    |
| 1974 | 0          | 2      | 11         | 45                  | 58    |
| 1975 | 0          | 4      | 4          | 27                  | 35    |
| 1976 | 0          | 0      | 3          | 12                  | 16    |
| 1977 | 0          | 0      | 0          | 0                   | 1     |
| 1978 | 0          | 28     | 6          | 0                   | 33    |
| 1979 | 0          | 16     | 47         | 0                   | 63    |
| 1980 | 0          | 30     | 26         | 0                   | 56    |

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|      |   |     |    |   |     |
|------|---|-----|----|---|-----|
| 1981 | 0 | 45  | 16 | 0 | 61  |
| 1982 | 0 | 52  | 47 | 0 | 99  |
| 1983 | 0 | 44  | 11 | 0 | 55  |
| 1984 | 0 | 45  | 30 | 0 | 75  |
| 1985 | 0 | 96  | 44 | 0 | 139 |
| 1986 | 0 | 137 | 17 | 0 | 154 |
| 1987 | 1 | 132 | 66 | 0 | 198 |
| 1988 | 0 | 144 | 29 | 0 | 173 |
| 1989 | 0 | 240 | 46 | 0 | 287 |
| 1990 | 2 | 154 | 11 | 0 | 167 |
| 1991 | 1 | 213 | 21 | 0 | 235 |
| 1992 | 2 | 155 | 29 | 0 | 186 |
| 1993 | 0 | 164 | 3  | 0 | 166 |
| 1994 | 7 | 115 | 5  | 0 | 127 |
| 1995 | 5 | 113 | 48 | 0 | 165 |
| 1996 | 2 | 97  | 29 | 0 | 127 |
| 1997 | 0 | 71  | 36 | 0 | 107 |
| 1998 | 1 | 89  | 21 | 0 | 110 |
| 1999 | 0 | 60  | 21 | 0 | 81  |
| 2000 | 2 | 56  | 21 | 0 | 79  |
| 2001 | 0 | 64  | 10 | 0 | 74  |
| 2002 | 0 | 22  | 8  | 0 | 31  |
| 2003 | 1 | 47  | 10 | 0 | 58  |
| 2004 | 0 | 51  | 8  | 0 | 58  |
| 2005 | 0 | 38  | 7  | 0 | 45  |
| 2006 | 0 | 35  | 13 | 0 | 48  |
| 2007 | 1 | 50  | 10 | 0 | 60  |
| 2008 | 0 | 45  | 8  | 0 | 54  |
| 2009 | 0 | 51  | 15 | 0 | 67  |
| 2010 | 0 | 63  | 16 | 0 | 79  |
| 2011 | 0 | 43  | 10 | 0 | 53  |
| 2012 | 0 | 32  | 15 | 0 | 47  |
| 2013 | 1 | 54  | 8  | 0 | 64  |
| 2014 | 0 | 27  | 7  | 0 | 34  |
| 2015 | 0 | 30  | 1  | 0 | 31  |
| 2016 | 0 | 28  | 3  | 0 | 31  |
| 2017 | 0 | 20  | 2  | 0 | 22  |
| 2018 | 1 | 15  | 1  | 0 | 16  |
| 2019 | 1 | 21  | 1  | 0 | 22  |
| 2020 | 0 | 10  | 0  | 0 | 10  |
| 2021 | 0 | 9   | 1  | 0 | 10  |
| 2022 | 0 | 10  | 1  | 0 | 12  |
| 2023 | 0 | 13  | 0  | 0 | 13  |
| 2024 | 0 | 10  | 0  | 0 | 10  |

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Table 5: Landings (metric tons) for non-trawl fleet by state.

| Year | California | Oregon | California.1 | Total |
|------|------------|--------|--------------|-------|
| 1892 | 0          | 19     | 0            | 19    |
| 1893 | 0          | 19     | 0            | 19    |
| 1894 | 0          | 19     | 0            | 19    |
| 1895 | 0          | 5      | 0            | 5     |
| 1896 | 0          | 1      | 0            | 1     |
| 1897 | 0          | 1      | 0            | 1     |
| 1898 | 0          | 1      | 0            | 1     |
| 1899 | 0          | 1      | 0            | 1     |
| 1900 | 0          | 2      | 0            | 2     |
| 1901 | 0          | 2      | 0            | 2     |
| 1902 | 0          | 3      | 0            | 3     |
| 1903 | 0          | 3      | 0            | 3     |
| 1904 | 0          | 4      | 0            | 4     |
| 1905 | 0          | 4      | 0            | 4     |
| 1906 | 0          | 4      | 0            | 4     |
| 1907 | 0          | 5      | 0            | 5     |
| 1908 | 0          | 5      | 3            | 8     |
| 1909 | 0          | 6      | 0            | 6     |
| 1910 | 0          | 6      | 0            | 6     |
| 1911 | 0          | 7      | 0            | 7     |
| 1912 | 0          | 7      | 0            | 7     |
| 1913 | 0          | 8      | 0            | 8     |
| 1914 | 0          | 8      | 0            | 8     |
| 1915 | 0          | 9      | 2            | 10    |
| 1916 | 0          | 9      | 0            | 9     |
| 1917 | 0          | 10     | 0            | 10    |
| 1918 | 0          | 10     | 45           | 55    |
| 1919 | 0          | 11     | 15           | 26    |
| 1920 | 0          | 11     | 12           | 23    |
| 1921 | 0          | 12     | 11           | 23    |
| 1922 | 0          | 12     | 6            | 18    |
| 1923 | 0          | 13     | 8            | 20    |
| 1924 | 0          | 13     | 19           | 32    |
| 1925 | 0          | 14     | 24           | 38    |
| 1926 | 0          | 14     | 40           | 54    |
| 1927 | 0          | 15     | 54           | 69    |
| 1928 | 0          | 24     | 48           | 72    |
| 1929 | 0          | 38     | 28           | 66    |
| 1930 | 0          | 32     | 35           | 67    |
| 1931 | 0          | 26     | 17           | 44    |
| 1932 | 0          | 10     | 15           | 25    |
| 1933 | 0          | 14     | 20           | 35    |
| 1934 | 0          | 16     | 26           | 42    |
| 1935 | 0          | 15     | 18           | 34    |
| 1936 | 0          | 34     | 27           | 61    |
| 1937 | 0          | 33     | 20           | 53    |
| 1938 | 0          | 31     | 24           | 55    |
| 1939 | 0          | 10     | 19           | 28    |
| 1940 | 0          | 37     | 22           | 59    |
| 1941 | 0          | 61     | 39           | 101   |

|      |   |     |     |     |
|------|---|-----|-----|-----|
| 1942 | 0 | 87  | 36  | 124 |
| 1943 | 0 | 235 | 15  | 251 |
| 1944 | 0 | 42  | 32  | 74  |
| 1945 | 0 | 16  | 14  | 30  |
| 1946 | 0 | 23  | 35  | 58  |
| 1947 | 0 | 17  | 18  | 35  |
| 1948 | 0 | 24  | 15  | 39  |
| 1949 | 0 | 8   | 18  | 26  |
| 1950 | 0 | 19  | 27  | 46  |
| 1951 | 0 | 14  | 39  | 53  |
| 1952 | 0 | 6   | 26  | 32  |
| 1953 | 0 | 6   | 10  | 16  |
| 1954 | 0 | 9   | 21  | 30  |
| 1955 | 0 | 6   | 19  | 26  |
| 1956 | 0 | 5   | 7   | 13  |
| 1957 | 0 | 11  | 14  | 26  |
| 1958 | 0 | 3   | 4   | 8   |
| 1959 | 0 | 6   | 10  | 16  |
| 1960 | 0 | 2   | 11  | 13  |
| 1961 | 0 | 9   | 7   | 15  |
| 1962 | 0 | 10  | 7   | 18  |
| 1963 | 0 | 6   | 5   | 11  |
| 1964 | 0 | 14  | 6   | 20  |
| 1965 | 0 | 3   | 5   | 8   |
| 1966 | 0 | 3   | 3   | 6   |
| 1967 | 0 | 8   | 2   | 10  |
| 1968 | 0 | 6   | 1   | 7   |
| 1969 | 0 | 21  | 4   | 25  |
| 1970 | 0 | 4   | 0   | 4   |
| 1971 | 0 | 1   | 0   | 1   |
| 1972 | 0 | 1   | 0   | 1   |
| 1973 | 0 | 6   | 0   | 6   |
| 1974 | 0 | 18  | 0   | 18  |
| 1975 | 0 | 5   | 0   | 5   |
| 1976 | 0 | 2   | 0   | 2   |
| 1977 | 0 | 152 | 12  | 164 |
| 1978 | 0 | 20  | 16  | 36  |
| 1979 | 0 | 66  | 55  | 121 |
| 1980 | 0 | 19  | 23  | 43  |
| 1981 | 2 | 51  | 15  | 68  |
| 1982 | 0 | 51  | 17  | 68  |
| 1983 | 0 | 53  | 17  | 70  |
| 1984 | 0 | 39  | 28  | 67  |
| 1985 | 0 | 71  | 87  | 158 |
| 1986 | 0 | 185 | 88  | 273 |
| 1987 | 9 | 245 | 114 | 368 |
| 1988 | 0 | 103 | 60  | 162 |
| 1989 | 0 | 98  | 33  | 131 |
| 1990 | 0 | 51  | 24  | 76  |
| 1991 | 0 | 34  | 25  | 59  |
| 1992 | 1 | 63  | 46  | 110 |
| 1993 | 0 | 34  | 125 | 159 |
| 1994 | 7 | 39  | 127 | 173 |
| 1995 | 1 | 131 | 444 | 576 |

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|      |   |    |     |     |
|------|---|----|-----|-----|
| 1996 | 1 | 85 | 119 | 204 |
| 1997 | 0 | 47 | 139 | 186 |
| 1998 | 1 | 86 | 226 | 313 |
| 1999 | 0 | 41 | 125 | 166 |
| 2000 | 1 | 2  | 26  | 29  |
| 2001 | 2 | 2  | 13  | 18  |
| 2002 | 1 | 4  | 23  | 27  |
| 2003 | 1 | 3  | 18  | 23  |
| 2004 | 0 | 2  | 31  | 34  |
| 2005 | 3 | 6  | 40  | 50  |
| 2006 | 2 | 5  | 52  | 59  |
| 2007 | 3 | 6  | 49  | 59  |
| 2008 | 1 | 11 | 44  | 56  |
| 2009 | 5 | 21 | 77  | 104 |
| 2010 | 2 | 22 | 46  | 71  |
| 2011 | 2 | 20 | 40  | 63  |
| 2012 | 5 | 23 | 45  | 74  |
| 2013 | 3 | 19 | 37  | 59  |
| 2014 | 3 | 6  | 29  | 37  |
| 2015 | 1 | 14 | 32  | 47  |
| 2016 | 3 | 16 | 41  | 60  |
| 2017 | 0 | 13 | 46  | 59  |
| 2018 | 0 | 7  | 39  | 47  |
| 2019 | 1 | 8  | 30  | 39  |
| 2020 | 1 | 11 | 13  | 24  |
| 2021 | 1 | 9  | 12  | 21  |
| 2022 | 1 | 4  | 14  | 19  |
| 2023 | 5 | 5  | 9   | 19  |
| 2024 | 2 | 2  | 5   | 10  |

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Table 6: Landings (metric tons) for mid-water trawl fleet by state.

| Year | California | Oregon | Washington | Total |
|------|------------|--------|------------|-------|
| 1985 | 0          | 0      | 0          | 0     |
| 1986 | 0          | 0      | 0          | 0     |
| 1987 | 0          | 0      | 0          | 0     |
| 1988 | 0          | 0      | 0          | 0     |
| 1989 | 0          | 0      | 0          | 0     |
| 1990 | 0          | 0      | 0          | 0     |
| 1991 | 0          | 0      | 0          | 0     |
| 1992 | 0          | 0      | 0          | 0     |
| 1993 | 0          | 0      | 0          | 0     |
| 1994 | 0          | 0      | 0          | 0     |
| 1995 | 0          | 0      | 0          | 0     |
| 1996 | 0          | 0      | 0          | 0     |
| 1997 | 0          | 0      | 0          | 0     |
| 1998 | 0          | 0      | 0          | 0     |
| 1999 | 0          | 0      | 0          | 0     |
| 2000 | 0          | 2      | 3          | 4     |
| 2001 | 0          | 0      | 1          | 1     |
| 2002 | 0          | 0      | 0          | 0     |
| 2003 | 0          | 0      | 0          | 0     |
| 2004 | 0          | 0      | 1          | 1     |
| 2005 | 0          | 0      | 0          | 0     |
| 2006 | 0          | 0      | 0          | 0     |
| 2007 | 0          | 0      | 2          | 2     |
| 2008 | 0          | 0      | 1          | 1     |
| 2009 | 0          | 0      | 2          | 2     |
| 2010 | 0          | 3      | 3          | 6     |
| 2011 | 0          | 3      | 1          | 4     |
| 2012 | 0          | 42     | 7          | 49    |
| 2013 | 0          | 3      | 0          | 3     |
| 2014 | 0          | 2      | 2          | 4     |
| 2015 | 0          | 12     | 7          | 19    |
| 2016 | 0          | 11     | 4          | 16    |
| 2017 | 0          | 1      | 1          | 2     |
| 2018 | 0          | 1      | 1          | 3     |
| 2019 | 0          | 4      | 5          | 9     |
| 2020 | 0          | 13     | 16         | 29    |
| 2021 | 0          | 11     | 10         | 21    |
| 2022 | 0          | 7      | 11         | 19    |
| 2023 | 0          | 13     | 13         | 26    |
| 2024 | 0          | 43     | 27         | 69    |

Table 7: Summary of fishery sampling effort (number of trips and fish sampled) by state used to create length frequency distributions of the bottom trawl fleet.

| Year | N_Trips_CA | N_Fish_CA | N_Trips_OR | N_Fish_OR | N_Trips_WA | N_Fish_WA | Input_N |
|------|------------|-----------|------------|-----------|------------|-----------|---------|
| 1995 | 3          | 4         | 1          | 22        | 0          | 0         | 8       |
| 1996 | 6          | 15        | 2          | 44        | 0          | 0         | 16      |
| 1997 | 1          | 1         | 1          | 24        | 0          | 0         | 5       |
| 1998 | 0          | 0         | 0          | 0         | 24         | 509       | 94      |
| 1999 | 2          | 3         | 1          | 33        | 19         | 404       | 83      |
| 2000 | 4          | 11        | 1          | 12        | 28         | 573       | 115     |
| 2001 | 1          | 1         | 5          | 111       | 18         | 334       | 86      |
| 2002 | 3          | 3         | 1          | 5         | 26         | 378       | 83      |
| 2003 | 4          | 6         | 8          | 46        | 31         | 794       | 160     |
| 2004 | 1          | 1         | 20         | 318       | 13         | 300       | 119     |
| 2005 | 1          | 1         | 20         | 258       | 9          | 184       | 91      |
| 2006 | 5          | 5         | 29         | 254       | 9          | 297       | 120     |
| 2007 | 13         | 17        | 53         | 815       | 21         | 722       | 301     |
| 2008 | 15         | 32        | 42         | 659       | 14         | 673       | 259     |
| 2009 | 11         | 16        | 65         | 891       | 14         | 448       | 277     |
| 2010 | 8          | 12        | 60         | 711       | 8          | 354       | 225     |
| 2011 | 9          | 12        | 37         | 386       | 16         | 263       | 153     |
| 2012 | 15         | 25        | 46         | 409       | 18         | 612       | 223     |
| 2013 | 11         | 15        | 45         | 515       | 7          | 195       | 163     |
| 2014 | 11         | 15        | 61         | 507       | 7          | 113       | 167     |
| 2015 | 12         | 20        | 63         | 778       | 15         | 382       | 253     |
| 2016 | 19         | 28        | 42         | 631       | 8          | 144       | 180     |
| 2017 | 19         | 38        | 76         | 728       | 8          | 87        | 221     |
| 2018 | 14         | 19        | 61         | 650       | 12         | 169       | 203     |
| 2019 | 21         | 59        | 81         | 730       | 7          | 92        | 231     |
| 2020 | 3          | 3         | 51         | 381       | 9          | 101       | 130     |
| 2021 | 15         | 33        | 53         | 353       | 11         | 135       | 151     |
| 2022 | 6          | 19        | 33         | 373       | 11         | 230       | 136     |
| 2023 | 6          | 6         | 20         | 254       | 8          | 296       | 111     |
| 2024 | 21         | 38        | 23         | 357       | 22         | 569       | 199     |

Table 8: Summary of fishery sampling effort (number of trips and fish sampled) by state used to create length frequency distributions of the non-trawl fleet.

| Year | N_Trips_CA | N_Fish_CA | N_Trips_OR | N_Fish_OR | N_Trips_WA | N_Fish_WA | Input_N |
|------|------------|-----------|------------|-----------|------------|-----------|---------|
| 1996 | 0          | 0         | 5          | 123       | 0          | 0         | 22      |
| 1998 | 0          | 0         | 2          | 44        | 11         | 678       | 92      |
| 1999 | 0          | 0         | 3          | 69        | 10         | 392       | 77      |
| 2000 | 1          | 9         | 1          | 23        | 0          | 0         | 6       |
| 2001 | 0          | 0         | 1          | 24        | 5          | 62        | 18      |
| 2002 | 2          | 13        | 0          | 0         | 17         | 268       | 58      |
| 2003 | 2          | 7         | 1          | 2         | 50         | 967       | 188     |
| 2004 | 0          | 0         | 0          | 0         | 33         | 741       | 135     |
| 2005 | 13         | 57        | 5          | 22        | 42         | 1167      | 232     |
| 2006 | 9          | 113       | 10         | 177       | 49         | 1323      | 291     |
| 2007 | 4          | 9         | 5          | 88        | 27         | 747       | 152     |
| 2008 | 8          | 47        | 7          | 121       | 34         | 953       | 204     |
| 2009 | 11         | 92        | 17         | 257       | 27         | 1115      | 257     |
| 2010 | 12         | 93        | 38         | 499       | 21         | 742       | 255     |
| 2011 | 15         | 60        | 44         | 544       | 32         | 1034      | 317     |
| 2012 | 4          | 17        | 34         | 318       | 27         | 847       | 228     |
| 2013 | 4          | 26        | 27         | 488       | 29         | 919       | 258     |
| 2014 | 1          | 2         | 24         | 176       | 31         | 882       | 202     |
| 2015 | 0          | 0         | 47         | 342       | 33         | 927       | 255     |
| 2016 | 3          | 15        | 49         | 459       | 53         | 1236      | 341     |
| 2017 | 0          | 0         | 61         | 441       | 73         | 1061      | 341     |
| 2018 | 0          | 0         | 88         | 477       | 93         | 1293      | 425     |
| 2019 | 0          | 0         | 68         | 379       | 117        | 1456      | 438     |
| 2020 | 1          | 1         | 40         | 242       | 33         | 489       | 175     |
| 2021 | 1          | 10        | 34         | 232       | 83         | 931       | 280     |
| 2022 | 0          | 0         | 53         | 344       | 90         | 881       | 312     |
| 2023 | 3          | 10        | 25         | 190       | 73         | 871       | 249     |
| 2024 | 27         | 102       | 29         | 162       | 65         | 731       | 258     |

Table 9: Summary of fishery sampling effort (number of trips and fish sampled) by state used to create length frequency distributions of the mid-water trawl fleet.

| Year | N_Trips_OR | N_Fish_OR | Input_N |
|------|------------|-----------|---------|
| 2000 | 1          | 28        | 5       |
| 2008 | 2          | 13        | 4       |
| 2010 | 3          | 48        | 10      |
| 2011 | 4          | 8         | 5       |
| 2012 | 19         | 460       | 82      |
| 2013 | 4          | 29        | 8       |
| 2014 | 9          | 19        | 12      |
| 2015 | 32         | 324       | 77      |
| 2016 | 16         | 147       | 36      |
| 2017 | 9          | 63        | 18      |
| 2018 | 6          | 44        | 12      |
| 2019 | 8          | 24        | 11      |
| 2020 | 15         | 124       | 32      |
| 2021 | 32         | 178       | 57      |
| 2022 | 17         | 154       | 38      |
| 2023 | 20         | 237       | 53      |
| 2024 | 26         | 327       | 71      |

Table 10: Summary of fishery sampling effort (number of trips and fish sampled) by state used to create conditional ages-at-length compositions.

| Year | Fleet          | N_Trips_OR | N_Fish_OR | N_Trips_WA | N_Fish_WA |
|------|----------------|------------|-----------|------------|-----------|
| 2008 | Bottom Trawl   | 38         | 630       | 0          | 0         |
| 2009 | Bottom Trawl   | 0          | 0         | 3          | 87        |
| 2010 | Bottom Trawl   | 0          | 0         | 7          | 217       |
| 2011 | Bottom Trawl   | 36         | 384       | 2          | 39        |
| 2012 | Bottom Trawl   | 0          | 0         | 9          | 191       |
| 2013 | Bottom Trawl   | 0          | 0         | 2          | 66        |
| 2017 | Bottom Trawl   | 49         | 257       | 0          | 0         |
| 2021 | Bottom Trawl   | 39         | 172       | 0          | 0         |
| 2022 | Bottom Trawl   | 26         | 174       | 0          | 0         |
| 2023 | Bottom Trawl   | 20         | 253       | 0          | 0         |
| 2024 | Bottom Trawl   | 22         | 182       | 6          | 202       |
| 2008 | Non-trawl      | 4          | 51        | 3          | 41        |
| 2009 | Non-trawl      | 0          | 0         | 10         | 406       |
| 2010 | Non-trawl      | 0          | 0         | 17         | 562       |
| 2011 | Non-trawl      | 1          | 3         | 9          | 234       |
| 2012 | Non-trawl      | 0          | 0         | 22         | 710       |
| 2013 | Non-trawl      | 0          | 0         | 6          | 179       |
| 2017 | Non-trawl      | 39         | 118       | 0          | 0         |
| 2021 | Non-trawl      | 31         | 132       | 0          | 0         |
| 2022 | Non-trawl      | 37         | 160       | 0          | 0         |
| 2023 | Non-trawl      | 25         | 190       | 0          | 0         |
| 2024 | Non-trawl      | 25         | 77        | 21         | 278       |
| 2011 | Midwater Trawl | 4          | 8         | 0          | 0         |
| 2017 | Midwater Trawl | 8          | 25        | 0          | 0         |
| 2021 | Midwater Trawl | 25         | 95        | 0          | 0         |
| 2022 | Midwater Trawl | 15         | 66        | 0          | 0         |
| 2023 | Midwater Trawl | 20         | 235       | 0          | 0         |
| 2024 | Midwater Trawl | 25         | 141       | 0          | 0         |

Table 11: Summary of fishery sampling effort (number of trips, number of hauls and fish sampled) used to create length frequency distributions of discard fleets.

| year | Fleet                | N_Fish | N_Hauls | N_Trips | Input_N |
|------|----------------------|--------|---------|---------|---------|
| 2004 | Bottom Trawl Discard | 51     | 10      | 7       | 14      |
| 2005 | Bottom Trawl Discard | 36     | 20      | 16      | 21      |
| 2006 | Bottom Trawl Discard | 88     | 26      | 18      | 30      |
| 2007 | Bottom Trawl Discard | 456    | 123     | 40      | 103     |
| 2008 | Bottom Trawl Discard | 1104   | 324     | 95      | 247     |
| 2009 | Bottom Trawl Discard | 1026   | 295     | 101     | 243     |
| 2010 | Bottom Trawl Discard | 655    | 166     | 49      | 139     |
| 2011 | Bottom Trawl Discard | 27     | 14      | 14      | 18      |
| 2012 | Bottom Trawl Discard | 92     | 39      | 24      | 37      |
| 2013 | Bottom Trawl Discard | 115    | 46      | 27      | 43      |
| 2014 | Bottom Trawl Discard | 60     | 30      | 23      | 31      |
| 2015 | Bottom Trawl Discard | 5      | 4       | 4       | 5       |
| 2016 | Bottom Trawl Discard | 18     | 8       | 7       | 9       |
| 2019 | Bottom Trawl Discard | 14     | 4       | 3       | 5       |
| 2020 | Bottom Trawl Discard | 10     | 4       | 4       | 5       |
| 2022 | Bottom Trawl Discard | 13     | 6       | 6       | 8       |
| 2023 | Bottom Trawl Discard | 24     | 5       | 5       | 8       |
| 2004 | Non-trawl Discard    | 22     | 5       | 3       | 6       |
| 2005 | Non-trawl Discard    | 202    | 21      | 7       | 35      |
| 2006 | Non-trawl Discard    | 34     | 10      | 4       | 9       |
| 2007 | Non-trawl Discard    | 92     | 26      | 9       | 22      |
| 2008 | Non-trawl Discard    | 33     | 18      | 9       | 14      |
| 2009 | Non-trawl Discard    | 19     | 11      | 9       | 12      |
| 2010 | Non-trawl Discard    | 180    | 49      | 16      | 41      |
| 2011 | Non-trawl Discard    | 400    | 100     | 12      | 67      |
| 2012 | Non-trawl Discard    | 395    | 114     | 18      | 73      |
| 2013 | Non-trawl Discard    | 112    | 26      | 5       | 20      |
| 2014 | Non-trawl Discard    | 108    | 26      | 8       | 23      |
| 2015 | Non-trawl Discard    | 254    | 52      | 15      | 50      |
| 2016 | Non-trawl Discard    | 154    | 39      | 16      | 37      |
| 2017 | Non-trawl Discard    | 340    | 55      | 17      | 64      |
| 2018 | Non-trawl Discard    | 320    | 60      | 19      | 63      |
| 2019 | Non-trawl Discard    | 753    | 123     | 21      | 125     |
| 2020 | Non-trawl Discard    | 21     | 6       | 5       | 8       |
| 2021 | Non-trawl Discard    | 49     | 11      | 9       | 16      |
| 2022 | Non-trawl Discard    | 76     | 22      | 8       | 18      |
| 2023 | Non-trawl Discard    | 39     | 11      | 8       | 13      |

Table 12: Summary of fishery sampling effort (number of hauls and fish sampled) used to create length frequency distributions and CAAL of At-sea Hake fleet.

| Year | N_Hauls | N_Lengths | N_Ages |
|------|---------|-----------|--------|
| 2003 | 66      | 338       | 0      |
| 2004 | 425     | 2132      | 0      |
| 2005 | 461     | 3102      | 0      |
| 2006 | 305     | 1029      | 0      |
| 2007 | 572     | 5135      | 0      |
| 2008 | 893     | 7547      | 555    |
| 2009 | 284     | 1093      | 0      |
| 2010 | 380     | 1956      | 0      |
| 2011 | 1092    | 8672      | 508    |
| 2012 | 591     | 5176      | 0      |
| 2013 | 446     | 2233      | 0      |
| 2014 | 278     | 744       | 283    |
| 2015 | 424     | 2044      | 0      |
| 2016 | 794     | 3092      | 507    |
| 2017 | 507     | 2746      | 0      |
| 2018 | 524     | 2900      | 0      |
| 2019 | 317     | 2488      | 311    |
| 2020 | 189     | 1213      | 185    |
| 2021 | 282     | 1356      | 262    |
| 2022 | 519     | 2363      | 450    |
| 2023 | 260     | 776       | 373    |
| 2024 | 306     | 898       | 221    |

Table 13: Stratifications used for generate survey length compositions.

| Survey          | Strata     | Depth1 | Depth2 | Latitude1 | Latitude2 |
|-----------------|------------|--------|--------|-----------|-----------|
| WCGBTS          | Shallow OR | 55     | 183    | 42        | 46        |
|                 | Middle OR  | 183    | 350    | 42        | 46        |
|                 | Deep OR    | 350    | 549    | 42        | 46        |
|                 | Shallow WA | 55     | 183    | 46        | 49        |
|                 | Middle WA  | 183    | 350    | 46        | 49        |
|                 | Deep WA    | 350    | 549    | 46        | 49        |
| Triennial early | Shallow    | 55     | 183    | 42        | 49        |
|                 | Deep       | 183    | 350    | 42        | 49        |
| Triennial late  | Shallow    | 55     | 183    | 42        | 49        |
|                 | Deep       | 183    | 500    | 42        | 49        |
| AFSC Slope      | Shallow    | 183    | 549    | 42        | 46        |
|                 | Deep       | 183    | 549    | 46        | 49        |



Table 14: Stratifications used for generate survey length compositions.

| Survey               | Year        | Latitudes                | Depths..fm. | Depths..m. |
|----------------------|-------------|--------------------------|-------------|------------|
| WCGBTS               | 2003 - 2019 | 32o 34'- 48o 27'         | 30-700      | 55-1,280   |
|                      | 2020 - 2024 | 32o 34'- 48o 27'         | 30-700      | 55-1,280   |
| AFSC/NWFSC Triennial | 1977        | 34o 00'- Canadian border | 50-250      | 91-457     |
|                      | 1980        | 36o 48'- 49o 15'         | 30-200      | 55-366     |
|                      | 1983        | 36o 48'- 49o 15'         | 30-200      | 55-366     |
|                      | 1986        | 36o 48'- Border          | 30-200      | 55-366     |
|                      | 1989        | 34o 30'- 49o 40'         | 30-200      | 55-366     |
|                      | 1992        | 34o 30'- 49o 40'         | 30-200      | 55-366     |
|                      | 1995        | 34o 30'- 49o 40'         | 30-275      | 55-500     |
|                      | 1998        | 34o 30'- 49o 40'         | 30-275      | 55-500     |
|                      | 2001        | 34o 30'- 49o 40'         | 30-275      | 55-500     |
|                      | 2004        | 34o 30'- Canadian border | 30-275      | 55-500     |
| AFSC Slope           | 1988        | 44o 05'- 45o 30'         | 100-700     | 182-1,280  |
|                      | 1990        | 44o 30'- 40o 30'         | 100-700     | 182-1,280  |
|                      | 1991        | 38o 20'- 40o 30'         | 100-700     | 182-1,280  |
|                      | 1992        | 45o 30'- Border          | 100-700     | 182-1,280  |
|                      | 1993        | 43o 00'- 45o 30'         | 100-700     | 182-1,280  |
|                      | 1995        | 40o 30'- 43o 00'         | 100-700     | 182-1,280  |
|                      | 1996        | 43o 00'- Canadian border | 100-700     | 182-1,280  |
|                      | 1997        | 34o 00'- Canadian border | 100-700     | 182-1,280  |
|                      | 1999        | 34o 00'- Canadian border | 100-700     | 182-1,280  |
|                      | 2000        | 34o 00'- Canadian border | 100-700     | 182-1,280  |
|                      | 2001        | 34o 00'- Canadian border | 100-700     | 182-1,280  |
| NWFSC Slope          | 1999        | 34o 50'- 48o 10'         | 100-700     | 182-1,280  |
|                      | 2000        | 34o 50'- 48o 10'         | 100-700     | 182-1,280  |
|                      | 2001        | 34o 50'- 48o 10'         | 100-700     | 182-1,280  |
|                      | 2002        | 34o 50'- 48o 10'         | 100-700     | 182-1,280  |

Table 15: Summary of surveys' sampling effort (number of hauls and fish sampled) used to create length frequency distributions.

| Year | Survey     | N_Hauls | N_Fish | Input_N |
|------|------------|---------|--------|---------|
| 2003 | WCGBTS     | 33      | 111    | 80      |
| 2004 | WCGBTS     | 24      | 111    | 58      |
| 2005 | WCGBTS     | 27      | 259    | 65      |
| 2006 | WCGBTS     | 34      | 99     | 82      |
| 2007 | WCGBTS     | 35      | 106    | 85      |
| 2008 | WCGBTS     | 35      | 120    | 85      |
| 2009 | WCGBTS     | 26      | 125    | 63      |
| 2010 | WCGBTS     | 29      | 89     | 70      |
| 2011 | WCGBTS     | 29      | 113    | 70      |
| 2012 | WCGBTS     | 21      | 82     | 51      |
| 2013 | WCGBTS     | 21      | 67     | 51      |
| 2014 | WCGBTS     | 20      | 39     | 39      |
| 2015 | WCGBTS     | 35      | 181    | 85      |
| 2016 | WCGBTS     | 31      | 103    | 75      |
| 2017 | WCGBTS     | 26      | 173    | 63      |
| 2018 | WCGBTS     | 25      | 97     | 60      |
| 2019 | WCGBTS     | 10      | 57     | 24      |
| 2021 | WCGBTS     | 29      | 139    | 70      |
| 2022 | WCGBTS     | 19      | 41     | 41      |
| 2023 | WCGBTS     | 28      | 78     | 68      |
| 2024 | WCGBTS     | 28      | 59     | 59      |
| 1980 | Triennial  | 2       | 70     | 4       |
| 1986 | Triennial  | 10      | 94     | 24      |
| 1989 | Triennial  | 24      | 276    | 58      |
| 1992 | Triennial  | 17      | 290    | 41      |
| 1995 | Triennial  | 54      | 432    | 131     |
| 1998 | Triennial  | 49      | 240    | 119     |
| 2001 | Triennial  | 50      | 287    | 121     |
| 2004 | Triennial  | 46      | 315    | 111     |
| 1997 | AFSC Slope | 10      | 32     | 24      |
| 1999 | AFSC Slope | 11      | 25     | 25      |
| 2000 | AFSC Slope | 13      | 128    | 31      |
| 2001 | AFSC Slope | 10      | 99     | 24      |

Table 16: Summary of WCGBTS sampling effort (number of hauls and fish sampled) used to create conditional ages-at-length distributions.

| Year | Survey | N_Hauls | N_Fish | Input_N |
|------|--------|---------|--------|---------|
| 2003 | WCGBTS | 17      | 56     | 56      |
| 2004 | WCGBTS | 24      | 74     | 74      |
| 2005 | WCGBTS | 27      | 139    | 139     |
| 2006 | WCGBTS | 31      | 94     | 94      |
| 2007 | WCGBTS | 35      | 105    | 105     |
| 2008 | WCGBTS | 35      | 119    | 119     |
| 2009 | WCGBTS | 24      | 93     | 93      |
| 2010 | WCGBTS | 29      | 89     | 89      |
| 2011 | WCGBTS | 27      | 110    | 110     |
| 2012 | WCGBTS | 19      | 78     | 78      |
| 2013 | WCGBTS | 21      | 67     | 67      |
| 2014 | WCGBTS | 11      | 22     | 22      |
| 2015 | WCGBTS | 34      | 169    | 169     |
| 2016 | WCGBTS | 31      | 103    | 103     |
| 2017 | WCGBTS | 25      | 128    | 128     |
| 2018 | WCGBTS | 25      | 97     | 97      |
| 2019 | WCGBTS | 10      | 57     | 57      |
| 2021 | WCGBTS | 28      | 138    | 138     |
| 2022 | WCGBTS | 19      | 41     | 41      |
| 2023 | WCGBTS | 27      | 76     | 76      |
| 2024 | WCGBTS | 28      | 59     | 59      |

Table 17: Length and age at biological (L50 bio, A50 bio) and functional maturity (L50 fxn, A50 fxn) for the Rougheye and Blackspotted rockfish complex (RE/BS), and genetically confirmed Rougheye and Blackspotted separately. Only certain maturity determinations were used in this analysis.

| Data used                   | n(length) | L50 bio      | L50 fxn      | n(age) | A50 bio      | A50 fxn      |
|-----------------------------|-----------|--------------|--------------|--------|--------------|--------------|
| RE/BS all data              | 473       | 42.92 (1.28) | 46.53 (1.11) | 372    | 21.23 (2.05) | 26.02 (2.12) |
| RE/BS genetically confirmed | 265       | 43.98 (1.86) | 46.83 (1.64) | 258    | 19.45 (2.16) | 21.85 (2.14) |
| Rougheye Rockfish           | 194       | 39.86 (2.18) | 43.57 (2.02) | 188    | 13.89 (1.77) | 15.80 (1.84) |
| Blackspotted Rockfish       | 71        | 49.46 (2.09) | 51.45 (2.35) | 70     | 26.18 (4.40) | 29.27 (3.75) |

Table 18: Specifications and structure of the model.

| Section                   | Configuration         |
|---------------------------|-----------------------|
| Maximum age               | 140                   |
| Sexes                     | Females, males        |
| Population bins           | 4-84 cm by 2 cm bins  |
| Summary biomass (mt) age  | 26+                   |
| Number of areas           | 1                     |
| Number of seasons         | 1                     |
| Number of growth patterns | 1                     |
| Start year                | 1892                  |
| End year                  | 2024                  |
| Data length bins          | 10-80 cm by 2 cm bins |
| Data age bins             | 1-100 by 1 year       |

Table 19: Estimated parameters in the model.

| Type                            | Count |
|---------------------------------|-------|
| Natural Mortality (M)           | 1     |
| M time-variation                | 0     |
| Growth mean                     | 6     |
| Growth variability              | 4     |
| Growth time-variation           | 0     |
| Stock-recruit                   | 1     |
| Stock-recruit variation         | 0     |
| Rec. dev. time series           | 133   |
| Rec. dev. initial age           | 0     |
| Rec. dev. forecast              | 12    |
| Index                           | 1     |
| Index time-variation            | 1     |
| Size selectivity                | 30    |
| Size selectivity time-variation | 24    |
| Retention                       | 0     |
| Retention time-variation        | 0     |
| Age selectivity                 | 0     |
| Age selectivity time-variation  | 0     |

\begin{landscape}

\end{landscape}

Table 20: Parameter estimates, estimation phase, parameter bounds, estimation status, estimated standard deviation (SD), prior information [distribution(mean, SD)] used in the base model.

| Label                                | Value    | Phase | Bounds       | Status | SD       | Prior                   |
|--------------------------------------|----------|-------|--------------|--------|----------|-------------------------|
| NatM_-<br>uniform_-<br>Fem_-<br>GP_1 | 0.0391   | 1     | (0.001, 0.2) | ok     | 0.000835 | lognormal(0.034, 0.310) |
| L_at_-<br>Amin_-<br>Fem_-<br>GP_1    | -3.1     | 2     | (-100, 25)   | ok     | 0.621    | none                    |
| L_at_-<br>Amax_-<br>Fem_-<br>GP_1    | 60.1     | 2     | (40, 90)     | ok     | 0.351    | none                    |
| VonBert_-<br>K_Fem_-<br>GP_1         | 0.0786   | 2     | (0.01, 0.15) | ok     | 0.00179  | none                    |
| CV_-<br>young_-<br>Fem_-<br>GP_1     | 0.0513   | 2     | (1e-06, 1)   | ok     | 0.0143   | none                    |
| CV_old_-<br>Fem_-<br>GP_1            | 0.0936   | 2     | (1e-06, 1)   | ok     | 0.00305  | none                    |
| Wtlen_-<br>1_Fem_-<br>GP_1           | 8.78e-06 | -3    | (-3, 3)      | fixed  | 0        | none                    |
| Wtlen_-<br>2_Fem_-<br>GP_1           | 3.15     | -3    | (-3, 4)      | fixed  | 0        | none                    |
| Mat50%_-<br>Fem_-<br>GP_1            | 46.5     | -3    | (1, 60)      | fixed  | 0        | none                    |
| Mat_-<br>slope_-<br>Fem_-<br>GP_1    | -0.254   | -3    | (-30, 3)     | fixed  | 0        | none                    |
| Eggs_-<br>scalar_-<br>Fem_-<br>GP_1  | 7.22e-05 | -3    | (-3, 3)      | fixed  | 0        | none                    |

|                                      |          |     |              |       |         |                         |
|--------------------------------------|----------|-----|--------------|-------|---------|-------------------------|
| Eggs_-<br>exp_len_-<br>Fem_-<br>GP_1 | 4.04     | -3  | (-3, 5)      | fixed | 0       | none                    |
| NatM_-<br>uniform_-<br>Mal_-<br>GP_1 | 0.036    | -2  | (0.001, 0.2) | fixed | 0       | lognormal(0.034, 0.310) |
| L_at_-<br>Amin_-<br>Mal_-<br>GP_1    | -2.68    | 2   | (-100, 25)   | ok    | 1.05    | none                    |
| L_at_-<br>Amax_-<br>Mal_-<br>GP_1    | 57.8     | 2   | (40, 90)     | ok    | 0.315   | none                    |
| VonBert_-<br>K_Mal_-<br>GP_1         | 0.0837   | 2   | (0.01, 0.15) | ok    | 0.00253 | none                    |
| CV_-<br>young_-<br>Mal_-<br>GP_1     | 0.0911   | 2   | (1e-06, 1)   | ok    | 0.0197  | none                    |
| CV_old_-<br>Mal_-<br>GP_1            | 0.085    | 2   | (1e-06, 1)   | ok    | 0.00296 | none                    |
| Wtlen_-<br>1_Mal_-<br>GP_1           | 1.18e-05 | -3  | (-3, 3)      | fixed | 0       | none                    |
| Wtlen_-<br>2_Mal_-<br>GP_1           | 3.07     | -3  | (-3, 4)      | fixed | 0       | none                    |
| Cohort-<br>GrowDev                   | 1        | -4  | (0, 1)       | fixed | 0       | none                    |
| FracFe-<br>male_-<br>GP_1            | 0.5      | -5  | (1e-06, 1)   | fixed | 0       | none                    |
| SR_-<br>LN(R0)                       | 7        | 1   | (1, 15)      | ok    | 0.621   | none                    |
| SR_BH_-<br>steep                     | 0.72     | -3  | (0.25, 0.99) | fixed | 0       | beta(0.718, 0.152)      |
| SR_-<br>sigmaR                       | 0.5      | -4  | (0, 2)       | fixed | 0       | none                    |
| SR_-<br>regime                       | 0        | -4  | (-5, 5)      | fixed | 0       | none                    |
| SR_-<br>autocorr                     | 0        | -99 | (0, 0)       | fixed | 0       | none                    |
| Main_-<br>RecrDev_-<br>1892          | -0.0795  | 1   | (-5, 5)      | dev   | 0.481   | normal(0.00, 0.50)      |
| Main_-<br>RecrDev_-<br>1893          | -0.0812  | 1   | (-5, 5)      | dev   | 0.48    | normal(0.00, 0.50)      |



|                             |         |   |         |     |       |                            |
|-----------------------------|---------|---|---------|-----|-------|----------------------------|
| Main_-<br>RecrDev_-<br>1894 | -0.0829 | 1 | (-5, 5) | dev | 0.48  | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1895 | -0.0846 | 1 | (-5, 5) | dev | 0.48  | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1896 | -0.0862 | 1 | (-5, 5) | dev | 0.479 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1897 | -0.0878 | 1 | (-5, 5) | dev | 0.479 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1898 | -0.0893 | 1 | (-5, 5) | dev | 0.479 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1899 | -0.0908 | 1 | (-5, 5) | dev | 0.478 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1900 | -0.0922 | 1 | (-5, 5) | dev | 0.478 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1901 | -0.0934 | 1 | (-5, 5) | dev | 0.478 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1902 | -0.0946 | 1 | (-5, 5) | dev | 0.477 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1903 | -0.0957 | 1 | (-5, 5) | dev | 0.477 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1904 | -0.0967 | 1 | (-5, 5) | dev | 0.477 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1905 | -0.0975 | 1 | (-5, 5) | dev | 0.477 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1906 | -0.0982 | 1 | (-5, 5) | dev | 0.476 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1907 | -0.0988 | 1 | (-5, 5) | dev | 0.476 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1908 | -0.0991 | 1 | (-5, 5) | dev | 0.476 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1909 | -0.0993 | 1 | (-5, 5) | dev | 0.476 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1910 | -0.0993 | 1 | (-5, 5) | dev | 0.476 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1911 | -0.0991 | 1 | (-5, 5) | dev | 0.476 | nor-<br>mal(0.00,<br>0.50) |

|                             |         |   |         |     |       |                            |
|-----------------------------|---------|---|---------|-----|-------|----------------------------|
| Main_-<br>RecrDev_-<br>1912 | -0.0986 | 1 | (-5, 5) | dev | 0.476 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1913 | -0.098  | 1 | (-5, 5) | dev | 0.476 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1914 | -0.0971 | 1 | (-5, 5) | dev | 0.476 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1915 | -0.0959 | 1 | (-5, 5) | dev | 0.477 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1916 | -0.0945 | 1 | (-5, 5) | dev | 0.477 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1917 | -0.0927 | 1 | (-5, 5) | dev | 0.477 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1918 | -0.0907 | 1 | (-5, 5) | dev | 0.478 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1919 | -0.0884 | 1 | (-5, 5) | dev | 0.478 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1920 | -0.0857 | 1 | (-5, 5) | dev | 0.479 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1921 | -0.0828 | 1 | (-5, 5) | dev | 0.479 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1922 | -0.0796 | 1 | (-5, 5) | dev | 0.48  | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1923 | -0.0761 | 1 | (-5, 5) | dev | 0.48  | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1924 | -0.0725 | 1 | (-5, 5) | dev | 0.481 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1925 | -0.0687 | 1 | (-5, 5) | dev | 0.482 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1926 | -0.0649 | 1 | (-5, 5) | dev | 0.482 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1927 | -0.0611 | 1 | (-5, 5) | dev | 0.483 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1928 | -0.0574 | 1 | (-5, 5) | dev | 0.484 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1929 | -0.0539 | 1 | (-5, 5) | dev | 0.484 | nor-<br>mal(0.00,<br>0.50) |

|                             |          |   |         |     |       |                            |
|-----------------------------|----------|---|---------|-----|-------|----------------------------|
| Main_-<br>RecrDev_-<br>1930 | -0.0506  | 1 | (-5, 5) | dev | 0.485 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1931 | -0.0477  | 1 | (-5, 5) | dev | 0.485 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1932 | -0.0452  | 1 | (-5, 5) | dev | 0.486 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1933 | -0.0431  | 1 | (-5, 5) | dev | 0.486 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1934 | -0.0412  | 1 | (-5, 5) | dev | 0.486 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1935 | -0.0395  | 1 | (-5, 5) | dev | 0.487 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1936 | -0.0378  | 1 | (-5, 5) | dev | 0.487 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1937 | -0.036   | 1 | (-5, 5) | dev | 0.487 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1938 | -0.0338  | 1 | (-5, 5) | dev | 0.487 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1939 | -0.031   | 1 | (-5, 5) | dev | 0.488 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1940 | -0.0273  | 1 | (-5, 5) | dev | 0.489 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1941 | -0.0225  | 1 | (-5, 5) | dev | 0.49  | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1942 | -0.0164  | 1 | (-5, 5) | dev | 0.491 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1943 | -0.00858 | 1 | (-5, 5) | dev | 0.493 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1944 | 0.00111  | 1 | (-5, 5) | dev | 0.495 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1945 | 0.0129   | 1 | (-5, 5) | dev | 0.497 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1946 | 0.0271   | 1 | (-5, 5) | dev | 0.501 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1947 | 0.0436   | 1 | (-5, 5) | dev | 0.504 | nor-<br>mal(0.00,<br>0.50) |

|                             |        |   |         |     |       |                            |
|-----------------------------|--------|---|---------|-----|-------|----------------------------|
| Main_-<br>RecrDev_-<br>1948 | 0.0627 | 1 | (-5, 5) | dev | 0.509 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1949 | 0.0843 | 1 | (-5, 5) | dev | 0.514 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1950 | 0.108  | 1 | (-5, 5) | dev | 0.521 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1951 | 0.134  | 1 | (-5, 5) | dev | 0.527 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1952 | 0.16   | 1 | (-5, 5) | dev | 0.535 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1953 | 0.186  | 1 | (-5, 5) | dev | 0.542 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1954 | 0.211  | 1 | (-5, 5) | dev | 0.549 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1955 | 0.232  | 1 | (-5, 5) | dev | 0.556 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1956 | 0.249  | 1 | (-5, 5) | dev | 0.561 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1957 | 0.259  | 1 | (-5, 5) | dev | 0.564 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1958 | 0.263  | 1 | (-5, 5) | dev | 0.566 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1959 | 0.262  | 1 | (-5, 5) | dev | 0.565 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1960 | 0.257  | 1 | (-5, 5) | dev | 0.563 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1961 | 0.247  | 1 | (-5, 5) | dev | 0.559 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1962 | 0.232  | 1 | (-5, 5) | dev | 0.554 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1963 | 0.212  | 1 | (-5, 5) | dev | 0.547 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1964 | 0.186  | 1 | (-5, 5) | dev | 0.539 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1965 | 0.156  | 1 | (-5, 5) | dev | 0.53  | nor-<br>mal(0.00,<br>0.50) |

|                             |        |   |         |     |       |                            |
|-----------------------------|--------|---|---------|-----|-------|----------------------------|
| Main_-<br>RecrDev_-<br>1966 | 0.125  | 1 | (-5, 5) | dev | 0.521 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1967 | 0.0968 | 1 | (-5, 5) | dev | 0.514 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1968 | 0.0744 | 1 | (-5, 5) | dev | 0.507 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1969 | 0.0591 | 1 | (-5, 5) | dev | 0.503 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1970 | 0.0494 | 1 | (-5, 5) | dev | 0.5   | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1971 | 0.042  | 1 | (-5, 5) | dev | 0.497 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1972 | 0.0361 | 1 | (-5, 5) | dev | 0.495 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1973 | 0.0352 | 1 | (-5, 5) | dev | 0.493 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1974 | 0.0403 | 1 | (-5, 5) | dev | 0.492 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1975 | 0.0422 | 1 | (-5, 5) | dev | 0.492 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1976 | 0.0554 | 1 | (-5, 5) | dev | 0.492 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1977 | 0.0757 | 1 | (-5, 5) | dev | 0.493 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1978 | 0.107  | 1 | (-5, 5) | dev | 0.495 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1979 | 0.108  | 1 | (-5, 5) | dev | 0.492 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1980 | 0.098  | 1 | (-5, 5) | dev | 0.484 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1981 | 0.0848 | 1 | (-5, 5) | dev | 0.476 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1982 | 0.0217 | 1 | (-5, 5) | dev | 0.466 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1983 | 0.0285 | 1 | (-5, 5) | dev | 0.46  | nor-<br>mal(0.00,<br>0.50) |

|                             |         |   |         |     |       |                            |
|-----------------------------|---------|---|---------|-----|-------|----------------------------|
| Main_-<br>RecrDev_-<br>1984 | 0.181   | 1 | (-5, 5) | dev | 0.457 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1985 | 0.169   | 1 | (-5, 5) | dev | 0.436 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1986 | -0.0205 | 1 | (-5, 5) | dev | 0.438 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1987 | 0.125   | 1 | (-5, 5) | dev | 0.462 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1988 | 0.657   | 1 | (-5, 5) | dev | 0.388 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1989 | 0.223   | 1 | (-5, 5) | dev | 0.439 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1990 | -0.113  | 1 | (-5, 5) | dev | 0.416 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1991 | 0.035   | 1 | (-5, 5) | dev | 0.394 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1992 | 0.154   | 1 | (-5, 5) | dev | 0.394 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1993 | 0.305   | 1 | (-5, 5) | dev | 0.365 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1994 | 0.0239  | 1 | (-5, 5) | dev | 0.353 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1995 | -0.489  | 1 | (-5, 5) | dev | 0.351 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1996 | -0.617  | 1 | (-5, 5) | dev | 0.343 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1997 | -0.604  | 1 | (-5, 5) | dev | 0.345 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1998 | -0.27   | 1 | (-5, 5) | dev | 0.35  | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>1999 | 0.548   | 1 | (-5, 5) | dev | 0.268 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>2000 | 0.187   | 1 | (-5, 5) | dev | 0.34  | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>2001 | 0.11    | 1 | (-5, 5) | dev | 0.316 | nor-<br>mal(0.00,<br>0.50) |

|                             |         |   |         |     |       |                            |
|-----------------------------|---------|---|---------|-----|-------|----------------------------|
| Main_-<br>RecrDev_-<br>2002 | -0.218  | 1 | (-5, 5) | dev | 0.338 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>2003 | -0.525  | 1 | (-5, 5) | dev | 0.373 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>2004 | -0.0497 | 1 | (-5, 5) | dev | 0.362 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>2005 | 0.0375  | 1 | (-5, 5) | dev | 0.37  | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>2006 | -0.213  | 1 | (-5, 5) | dev | 0.411 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>2007 | 0.223   | 1 | (-5, 5) | dev | 0.391 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>2008 | 0.577   | 1 | (-5, 5) | dev | 0.356 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>2009 | 0.0826  | 1 | (-5, 5) | dev | 0.427 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>2010 | 0.647   | 1 | (-5, 5) | dev | 0.332 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>2011 | 0.0457  | 1 | (-5, 5) | dev | 0.383 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>2012 | 1.04    | 1 | (-5, 5) | dev | 0.273 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>2013 | -0.0412 | 1 | (-5, 5) | dev | 0.371 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>2014 | -0.364  | 1 | (-5, 5) | dev | 0.397 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>2015 | -0.445  | 1 | (-5, 5) | dev | 0.422 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>2016 | -0.172  | 1 | (-5, 5) | dev | 0.429 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>2017 | 0.738   | 1 | (-5, 5) | dev | 0.381 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>2018 | 0.251   | 1 | (-5, 5) | dev | 0.416 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>2019 | 0.132   | 1 | (-5, 5) | dev | 0.413 | nor-<br>mal(0.00,<br>0.50) |

|                             |          |   |         |     |       |                            |
|-----------------------------|----------|---|---------|-----|-------|----------------------------|
| Main_-<br>RecrDev_-<br>2020 | -0.22    | 1 | (-5, 5) | dev | 0.44  | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>2021 | -0.138   | 1 | (-5, 5) | dev | 0.465 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>2022 | -0.0461  | 1 | (-5, 5) | dev | 0.491 | nor-<br>mal(0.00,<br>0.50) |
| Main_-<br>RecrDev_-<br>2023 | -0.00314 | 1 | (-5, 5) | dev | 0.499 | nor-<br>mal(0.00,<br>0.50) |
| Late_-<br>RecrDev_-<br>2024 | 0        | 5 | (-5, 5) | dev | 0.5   | nor-<br>mal(0.00,<br>0.50) |
| For-<br>eRecr_-<br>2025     | 0        | 5 | (-5, 5) | dev | 0.5   | nor-<br>mal(0.00,<br>0.50) |
| For-<br>eRecr_-<br>2026     | 0        | 5 | (-5, 5) | dev | 0.5   | nor-<br>mal(0.00,<br>0.50) |
| For-<br>eRecr_-<br>2027     | 0        | 5 | (-5, 5) | dev | 0.5   | nor-<br>mal(0.00,<br>0.50) |
| For-<br>eRecr_-<br>2028     | 0        | 5 | (-5, 5) | dev | 0.5   | nor-<br>mal(0.00,<br>0.50) |
| For-<br>eRecr_-<br>2029     | 0        | 5 | (-5, 5) | dev | 0.5   | nor-<br>mal(0.00,<br>0.50) |
| For-<br>eRecr_-<br>2030     | 0        | 5 | (-5, 5) | dev | 0.5   | nor-<br>mal(0.00,<br>0.50) |
| For-<br>eRecr_-<br>2031     | 0        | 5 | (-5, 5) | dev | 0.5   | nor-<br>mal(0.00,<br>0.50) |
| For-<br>eRecr_-<br>2032     | 0        | 5 | (-5, 5) | dev | 0.5   | nor-<br>mal(0.00,<br>0.50) |
| For-<br>eRecr_-<br>2033     | 0        | 5 | (-5, 5) | dev | 0.5   | nor-<br>mal(0.00,<br>0.50) |
| For-<br>eRecr_-<br>2034     | 0        | 5 | (-5, 5) | dev | 0.5   | nor-<br>mal(0.00,<br>0.50) |
| For-<br>eRecr_-<br>2035     | 0        | 5 | (-5, 5) | dev | 0.5   | nor-<br>mal(0.00,<br>0.50) |
| For-<br>eRecr_-<br>2036     | 0        | 5 | (-5, 5) | dev | 0.5   | nor-<br>mal(0.00,<br>0.50) |



|                                                                     |       |    |             |       |       |      |
|---------------------------------------------------------------------|-------|----|-------------|-------|-------|------|
| LnQ_-<br>base_TRI-<br>EN-<br>NIAL(7)                                | -1.49 | 1  | (-10, 2)    | ok    | 0.743 | none |
| LnQ_-<br>base_-<br>AK_-<br>SLOPE(8)                                 | -3.11 | -1 | (-15, 15)   | fixed | 0     | none |
| LnQ_-<br>base_-<br>NW_-<br>SLOPE(9)                                 | -2.04 | -1 | (-15, 15)   | fixed | 0     | none |
| LnQ_-<br>base_-<br>WCG-<br>BTS(10)                                  | -2.61 | -1 | (-15, 15)   | fixed | 0     | none |
| LnQ_-<br>base_TRI-<br>EN-<br>NIAL(7)_-<br>BLK3repl_-<br>1892        | -2.27 | 1  | (-10, 2)    | ok    | 0.77  | none |
| Size_-<br>DblN_-<br>peak_-<br>BOT-<br>TOM_-<br>TRAWL(1)             | 49.3  | 3  | (15, 79)    | ok    | 1.02  | none |
| Size_-<br>DblN_-<br>top_-<br>logit_-<br>BOT-<br>TOM_-<br>TRAWL(1)   | -11.5 | -4 | (-15, 20)   | fixed | 0     | none |
| Size_-<br>DblN_as-<br>cend_se_-<br>BOT-<br>TOM_-<br>TRAWL(1)        | 4.57  | 3  | (-15, 12)   | ok    | 0.207 | none |
| Size_-<br>DblN_de-<br>scend_-<br>se_BOT-<br>TOM_-<br>TRAWL(1)       | 2.95  | 4  | (-15, 20)   | ok    | 0.694 | none |
| Size_-<br>DblN_-<br>start_-<br>logit_-<br>BOT-<br>TOM_-<br>TRAWL(1) | -999  | -3 | (-1000, 20) | fixed | 0     | none |

|                                                                                       |        |    |           |       |       |      |
|---------------------------------------------------------------------------------------|--------|----|-----------|-------|-------|------|
| Size_-<br>DbLN_-<br>end_-<br>logit_-<br>BOT-<br>TOM_-<br>TRAWL(1)                     | -0.639 | 4  | (-15, 20) | ok    | 0.344 | none |
| Size_-<br>DbLN_-<br>peak_-<br>BOT-<br>TOM_-<br>TRAWL_-<br>DIS-<br>CARD(2)             | 25.3   | 3  | (15, 79)  | ok    | 5.37  | none |
| Size_-<br>DbLN_-<br>top_-<br>logit_-<br>BOT-<br>TOM_-<br>TRAWL_-<br>DIS-<br>CARD(2)   | -15    | -4 | (-15, 20) | fixed | 0     | none |
| Size_-<br>DbLN_as-<br>cend_se_-<br>BOT-<br>TOM_-<br>TRAWL_-<br>DIS-<br>CARD(2)        | 4.89   | 3  | (-15, 12) | ok    | 2.42  | none |
| Size_-<br>DbLN_de-<br>scend_-<br>se_BOT-<br>TOM_-<br>TRAWL_-<br>DIS-<br>CARD(2)       | 3.84   | 4  | (-15, 20) | ok    | 1.49  | none |
| Size_-<br>DbLN_-<br>start_-<br>logit_-<br>BOT-<br>TOM_-<br>TRAWL_-<br>DIS-<br>CARD(2) | -15    | -3 | (-15, 20) | fixed | 0     | none |

|                                                                                     |       |    |             |       |       |      |
|-------------------------------------------------------------------------------------|-------|----|-------------|-------|-------|------|
| Size_-<br>DbLN_-<br>end_-<br>logit_-<br>BOT-<br>TOM_-<br>TRAWL_-<br>DIS-<br>CARD(2) | -3.82 | 4  | (-15, 20)   | ok    | 1.31  | none |
| Size_-<br>DbLN_-<br>peak_-<br>NON_-<br>TRAWL(3)                                     | 50.3  | 3  | (15, 70)    | ok    | 1.25  | none |
| Size_-<br>DbLN_-<br>top_-<br>logit_-<br>NON_-<br>TRAWL(3)                           | -15   | -4 | (-15, 20)   | fixed | 0     | none |
| Size_-<br>DbLN_as-<br>cend_se_-<br>NON_-<br>TRAWL(3)                                | 3.86  | 3  | (-15, 12)   | ok    | 0.296 | none |
| Size_-<br>DbLN_de-<br>scend_-<br>se_NON_-<br>TRAWL(3)                               | 20    | -4 | (-15, 20)   | fixed | 0     | none |
| Size_-<br>DbLN_-<br>start_-<br>logit_-<br>NON_-<br>TRAWL(3)                         | -999  | -3 | (-1000, 20) | fixed | 0     | none |
| Size_-<br>DbLN_-<br>end_-<br>logit_-<br>NON_-<br>TRAWL(3)                           | 4.6   | -4 | (-15, 20)   | fixed | 0     | none |
| Size_-<br>DbLN_-<br>peak_-<br>NON_-<br>TRAWL_-<br>DIS-<br>CARD(4)                   | 49.9  | 3  | (15, 70)    | ok    | 1.5   | none |

|                                                                               |        |    |             |       |       |      |
|-------------------------------------------------------------------------------|--------|----|-------------|-------|-------|------|
| Size_-<br>DbLN_-<br>top_-<br>logit_-<br>NON_-<br>TRAWL_-<br>DIS-<br>CARD(4)   | -15    | -4 | (-15, 20)   | fixed | 0     | none |
| Size_-<br>DbLN_as-<br>cend_se_-<br>NON_-<br>TRAWL_-<br>DIS-<br>CARD(4)        | 3.75   | 3  | (-15, 12)   | ok    | 0.439 | none |
| Size_-<br>DbLN_de-<br>scend_-<br>se_NON_-<br>TRAWL_-<br>DIS-<br>CARD(4)       | 2.66   | 4  | (-15, 20)   | ok    | 0.95  | none |
| Size_-<br>DbLN_-<br>start_-<br>logit_-<br>NON_-<br>TRAWL_-<br>DIS-<br>CARD(4) | -999   | -3 | (-1000, 20) | fixed | 0     | none |
| Size_-<br>DbLN_-<br>end_-<br>logit_-<br>NON_-<br>TRAWL_-<br>DIS-<br>CARD(4)   | -0.192 | 4  | (-15, 20)   | ok    | 0.419 | none |
| Size_-<br>DbLN_-<br>peak_-<br>MIDWA-<br>TER_-<br>TRAWL(5)                     | 52.2   | 3  | (15, 79)    | ok    | 2.31  | none |
| Size_-<br>DbLN_-<br>top_-<br>logit_-<br>MIDWA-<br>TER_-<br>TRAWL(5)           | -15    | -4 | (-15, 20)   | fixed | 0     | none |

|                                                              |      |    |             |       |       |      |
|--------------------------------------------------------------|------|----|-------------|-------|-------|------|
| Size_<br>DbLN_ascend_<br>se_<br>MIDWA-<br>TER_<br>TRAWL(5)   | 4.57 | 3  | (-15, 12)   | ok    | 0.384 | none |
| Size_<br>DbLN_descend_<br>se_<br>MID-<br>WATER_<br>TRAWL(5)  | 20   | -4 | (-15, 20)   | fixed | 0     | none |
| Size_<br>DbLN_start_<br>logit_<br>MIDWA-<br>TER_<br>TRAWL(5) | -999 | -3 | (-1000, 20) | fixed | 0     | none |
| Size_<br>DbLN_end_<br>logit_<br>MIDWA-<br>TER_<br>TRAWL(5)   | -999 | -4 | (-1000, 20) | fixed | 0     | none |
| Size_<br>DbLN_peak_<br>AT_<br>SEA_<br>HAKE(6)                | 49.7 | 3  | (15, 70)    | ok    | 1.41  | none |
| Size_<br>DbLN_top_<br>logit_<br>AT_<br>SEA_<br>HAKE(6)       | -15  | -4 | (-15, 20)   | fixed | 0     | none |
| Size_<br>DbLN_ascend_<br>se_<br>AT_<br>SEA_<br>HAKE(6)       | 3.59 | 3  | (-15, 12)   | ok    | 0.419 | none |
| Size_<br>DbLN_descend_<br>se_<br>AT_<br>SEA_<br>HAKE(6)      | 20   | -4 | (-15, 20)   | fixed | 0     | none |

|                                                                    |       |    |             |       |       |      |
|--------------------------------------------------------------------|-------|----|-------------|-------|-------|------|
| Size_-<br>DbLN_-<br>start_-<br>logit_-<br>AT_-<br>SEA_-<br>HAKE(6) | -999  | -2 | (-1000, 20) | fixed | 0     | none |
| Size_-<br>DbLN_-<br>end_-<br>logit_-<br>AT_-<br>SEA_-<br>HAKE(6)   | -999  | -4 | (-1000, 20) | fixed | 0     | none |
| Size_-<br>DbLN_-<br>peak_TRI-<br>EN-<br>NIAL(7)                    | 21.8  | 3  | (13, 50)    | ok    | 1.94  | none |
| Size_-<br>DbLN_-<br>top_-<br>logit_TRI-<br>EN-<br>NIAL(7)          | -8.6  | -4 | (-15, 20)   | fixed | 0     | none |
| Size_-<br>DbLN_as-<br>cend_se_-<br>TRIEN-<br>NIAL(7)               | 3.52  | 3  | (-15, 12)   | ok    | 0.587 | none |
| Size_-<br>DbLN_de-<br>scend_-<br>se_TRI-<br>EN-<br>NIAL(7)         | 3.95  | 4  | (-15, 20)   | ok    | 0.553 | none |
| Size_-<br>DbLN_-<br>start_-<br>logit_TRI-<br>EN-<br>NIAL(7)        | -999  | -2 | (-1000, 20) | fixed | 0     | none |
| Size_-<br>DbLN_-<br>end_-<br>logit_TRI-<br>EN-<br>NIAL(7)          | -2.58 | 4  | (-15, 20)   | ok    | 0.321 | none |
| Size_-<br>DbLN_-<br>peak_-<br>AK_-<br>SLOPE(8)                     | 37.3  | 3  | (13, 50)    | ok    | 2.38  | none |

|                                                            |       |    |             |       |       |      |
|------------------------------------------------------------|-------|----|-------------|-------|-------|------|
| Size_-<br>DbLN_-<br>top_-<br>logit_-<br>AK_-<br>SLOPE(8)   | -15   | -4 | (-15, 20)   | fixed | 0     | none |
| Size_-<br>DbLN_-<br>ascend_-<br>se_AK_-<br>SLOPE(8)        | 4.94  | 3  | (-15, 12)   | ok    | 0.425 | none |
| Size_-<br>DbLN_-<br>descend_-<br>se_AK_-<br>SLOPE(8)       | 4.63  | 4  | (-15, 20)   | ok    | 0.381 | none |
| Size_-<br>DbLN_-<br>start_-<br>logit_-<br>AK_-<br>SLOPE(8) | -999  | -2 | (-1000, 20) | fixed | 0     | none |
| Size_-<br>DbLN_-<br>end_-<br>logit_-<br>AK_-<br>SLOPE(8)   | -10.5 | 4  | (-15, 20)   | ok    | 64.5  | none |
| Size_-<br>DbLN_-<br>peak_-<br>WCG-<br>BTS(10)              | 21.1  | 3  | (13, 50)    | ok    | 3.76  | none |
| Size_-<br>DbLN_-<br>top_-<br>logit_-<br>WCG-<br>BTS(10)    | -15   | -4 | (-15, 20)   | fixed | 0     | none |
| Size_-<br>DbLN_as-<br>cend_se_-<br>WCG-<br>BTS(10)         | 3.39  | 3  | (-15, 12)   | ok    | 1.08  | none |
| Size_-<br>DbLN_de-<br>scend_-<br>se_WCG-<br>BTS(10)        | 4.65  | 4  | (-15, 20)   | ok    | 1.05  | none |

|            |        |    |             |       |       |      |
|------------|--------|----|-------------|-------|-------|------|
| Size_-     | -999   | -2 | (-1000, 20) | fixed | 0     | none |
| DblN_-     |        |    |             |       |       |      |
| start_-    |        |    |             |       |       |      |
| logit_-    |        |    |             |       |       |      |
| WCG-       |        |    |             |       |       |      |
| BTS(10)    |        |    |             |       |       |      |
| Size_-     | -0.834 | 4  | (-15, 20)   | ok    | 0.31  | none |
| DblN_-     |        |    |             |       |       |      |
| end_-      |        |    |             |       |       |      |
| logit_-    |        |    |             |       |       |      |
| WCG-       |        |    |             |       |       |      |
| BTS(10)    |        |    |             |       |       |      |
| Size_-     | 44.7   | 3  | (15, 70)    | ok    | 2.95  | none |
| DblN_-     |        |    |             |       |       |      |
| peak_-     |        |    |             |       |       |      |
| BOT-       |        |    |             |       |       |      |
| TOM_-      |        |    |             |       |       |      |
| TRAWL(1)_- |        |    |             |       |       |      |
| BLK4repl_- |        |    |             |       |       |      |
| 1892       |        |    |             |       |       |      |
| Size_-     | 47.6   | 3  | (15, 70)    | ok    | 1.07  | none |
| DblN_-     |        |    |             |       |       |      |
| peak_-     |        |    |             |       |       |      |
| BOT-       |        |    |             |       |       |      |
| TOM_-      |        |    |             |       |       |      |
| TRAWL(1)_- |        |    |             |       |       |      |
| BLK4repl_- |        |    |             |       |       |      |
| 2002       |        |    |             |       |       |      |
| Size_-     | 5.03   | 3  | (-15, 12)   | ok    | 0.51  | none |
| DblN_as-   |        |    |             |       |       |      |
| cend_se_-  |        |    |             |       |       |      |
| BOT-       |        |    |             |       |       |      |
| TOM_-      |        |    |             |       |       |      |
| TRAWL(1)_- |        |    |             |       |       |      |
| BLK4repl_- |        |    |             |       |       |      |
| 1892       |        |    |             |       |       |      |
| Size_-     | 4.1    | 3  | (-15, 12)   | ok    | 0.306 | none |
| DblN_as-   |        |    |             |       |       |      |
| cend_se_-  |        |    |             |       |       |      |
| BOT-       |        |    |             |       |       |      |
| TOM_-      |        |    |             |       |       |      |
| TRAWL(1)_- |        |    |             |       |       |      |
| BLK4repl_- |        |    |             |       |       |      |
| 2002       |        |    |             |       |       |      |
| Size_-     | 2.65   | 4  | (-15, 20)   | ok    | 1.7   | none |
| DblN_de-   |        |    |             |       |       |      |
| scend_-    |        |    |             |       |       |      |
| se_BOT-    |        |    |             |       |       |      |
| TOM_-      |        |    |             |       |       |      |
| TRAWL(1)_- |        |    |             |       |       |      |
| BLK4repl_- |        |    |             |       |       |      |
| 1892       |        |    |             |       |       |      |



|                                                                                                       |        |   |           |    |       |      |
|-------------------------------------------------------------------------------------------------------|--------|---|-----------|----|-------|------|
| Size_-<br>DbLN_de-<br>scend_-<br>se_BOT-<br>TOM_-<br>TRAWL(1)-<br>BLK4repl_-<br>2002                  | 2.75   | 4 | (-15, 20) | ok | 0.751 | none |
| Size_-<br>DbLN_-<br>end_-<br>logit_-<br>BOT-<br>TOM_-<br>TRAWL(1)-<br>BLK4repl_-<br>1892              | -1.59  | 4 | (-15, 20) | ok | 0.585 | none |
| Size_-<br>DbLN_-<br>end_-<br>logit_-<br>BOT-<br>TOM_-<br>TRAWL(1)-<br>BLK4repl_-<br>2002              | -0.934 | 4 | (-15, 20) | ok | 0.324 | none |
| Size_-<br>DbLN_-<br>peak_-<br>BOT-<br>TOM_-<br>TRAWL_-<br>DIS-<br>CARD(2)-<br>BLK1repl_-<br>1892      | 47.7   | 3 | (15, 79)  | ok | 2.49  | none |
| Size_-<br>DbLN_as-<br>cend_se_-<br>BOT-<br>TOM_-<br>TRAWL_-<br>DIS-<br>CARD(2)-<br>BLK1repl_-<br>1892 | 6.32   | 3 | (-15, 12) | ok | 0.627 | none |

|                                                                                                             |      |   |           |    |       |      |
|-------------------------------------------------------------------------------------------------------------|------|---|-----------|----|-------|------|
| Size_-<br>DblN_de-<br>scend_-<br>se_BOT-<br>TOM_-<br>TRAWL_-<br>DIS-<br>CARD(2)_-<br>BLK1repl_-<br>1892     | 3.01 | 4 | (-15, 20) | ok | 1.19  | none |
| Size_-<br>DblN_-<br>end_-<br>logit_-<br>BOT-<br>TOM_-<br>TRAWL_-<br>DIS-<br>CARD(2)_-<br>BLK1repl_-<br>1892 | -1.7 | 4 | (-15, 20) | ok | 0.77  | none |
| Size_-<br>DblN_-<br>peak_-<br>NON_-<br>TRAWL(3)_-<br>BLK2repl_-<br>1892                                     | 46.8 | 3 | (15, 70)  | ok | 0.484 | none |
| Size_-<br>DblN_-<br>peak_-<br>NON_-<br>TRAWL(3)_-<br>BLK2repl_-<br>2011                                     | 49.5 | 3 | (15, 70)  | ok | 0.556 | none |
| Size_-<br>DblN_as-<br>cend_se_-<br>NON_-<br>TRAWL(3)_-<br>BLK2repl_-<br>1892                                | 3.05 | 3 | (-15, 12) | ok | 0.207 | none |
| Size_-<br>DblN_as-<br>cend_se_-<br>NON_-<br>TRAWL(3)_-<br>BLK2repl_-<br>2011                                | 3.81 | 3 | (-15, 12) | ok | 0.154 | none |

|                                                                                    |        |   |           |    |       |      |
|------------------------------------------------------------------------------------|--------|---|-----------|----|-------|------|
| Size_-<br>DbLN_de-<br>scend_-<br>se_NON_-<br>TRAWL(3)_-<br>BLK2repl_-<br>1892      | 3.17   | 4 | (-15, 20) | ok | 0.242 | none |
| Size_-<br>DbLN_de-<br>scend_-<br>se_NON_-<br>TRAWL(3)_-<br>BLK2repl_-<br>2011      | 2.32   | 4 | (-15, 20) | ok | 0.52  | none |
| Size_-<br>DbLN_-<br>end_-<br>logit_-<br>NON_-<br>TRAWL(3)_-<br>BLK2repl_-<br>1892  | -2.29  | 4 | (-15, 20) | ok | 0.258 | none |
| Size_-<br>DbLN_-<br>end_-<br>logit_-<br>NON_-<br>TRAWL(3)_-<br>BLK2repl_-<br>2011  | -0.626 | 4 | (-15, 20) | ok | 0.193 | none |
| Size_-<br>DbLN_-<br>peak_TRI-<br>EN-<br>NIAL(7)_-<br>BLK3repl_-<br>1892            | 17.4   | 3 | (13, 50)  | ok | 2.64  | none |
| Size_-<br>DbLN_as-<br>cend_se_-<br>TRIEN-<br>NIAL(7)_-<br>BLK3repl_-<br>1892       | 2.09   | 3 | (-15, 12) | ok | 1.39  | none |
| Size_-<br>DbLN_de-<br>scend_-<br>se_TRI-<br>EN-<br>NIAL(7)_-<br>BLK3repl_-<br>1892 | 5.11   | 4 | (-15, 20) | ok | 0.636 | none |

|            |       |   |           |    |      |      |
|------------|-------|---|-----------|----|------|------|
| Size_-     | -4.29 | 4 | (-15, 20) | ok | 1.69 | none |
| DblN_-     |       |   |           |    |      |      |
| end_-      |       |   |           |    |      |      |
| logit_TRI- |       |   |           |    |      |      |
| EN-        |       |   |           |    |      |      |
| NIAL(7)_-  |       |   |           |    |      |      |
| BLK3repl_- |       |   |           |    |      |      |
| 1892       |       |   |           |    |      |      |

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Table 21: Likelihood components by source.

| Label                | Total   |
|----------------------|---------|
| TOTAL                | 7,334.0 |
| Catch                | 0.0     |
| Equil catch          | 0.0     |
| Survey               | -26.8   |
| Length comp          | 574.8   |
| Age comp             | 6,783.5 |
| Recruitment          | 2.3     |
| InitEQ Regime        | 0.0     |
| Forecast Recruitment | 0.0     |
| Parm priors          | 0.1     |
| Parm softbounds      | 0.0     |
| Parm devs            | 0.0     |
| Crash Pen            | 0.0     |

Table 22: Summary of reference points and management quantities, including estimates of the 95 percent confidence intervals. SO is spawning output, SPR is the spawning potential ratio, and MSY is maximum sustainable yield.

| Reference Point                                | Estimate  | Lower Interval | Upper Interval |
|------------------------------------------------|-----------|----------------|----------------|
| Unfished Spawning output                       | 5,647,680 | -1,285,908     | 12,581,268     |
| Unfished Age 26+ Biomass (mt)                  | 33,631    | -7,664         | 74,925         |
| Unfished Recruitment (R0)                      | 1,092     | -237           | 2,420          |
| 2025 Spawning output                           | 4,929,140 | -2,527,955     | 12,386,235     |
| 2025 Fraction Unfished                         | 0.873     | 0.615          | 1.130          |
| Reference Points Based SO40%                   | —         | —              | —              |
| Proxy Spawning output SO40%                    | 2,259,070 | -514,357       | 5,032,497      |
| SPR Resulting in SO40%                         | 0.458     | 0.458          | 0.458          |
| Exploitation Rate Resulting in SO40%           | 0.048     | 0.045          | 0.051          |
| Yield with SPR Based On SO40% (mt)             | 553       | -119           | 1,224          |
| Reference Points Based on SPR Proxy for MSY    | —         | —              | —              |
| Proxy Spawning output (SPR50)                  | 2,519,730 | -573,720       | 5,613,180      |
| SPR50                                          | 0.500     | —              | —              |
| Exploitation Rate Corresponding to SPR50       | 0.040     | 0.038          | 0.042          |
| Yield with SPR50 at SO SPR (mt)                | 526       | -113           | 1,165          |
| Reference Points Based on Estimated MSY Values | —         | —              | —              |
| Spawning output at MSY (SO MSY)                | 1,497,170 | -343,318       | 3,337,658      |
| SPR MSY                                        | 0.337     | 0.334          | 0.339          |
| Exploitation Rate Corresponding to SPR MSY     | 0.085     | 0.079          | 0.090          |
| MSY (mt)                                       | 592       | -127           | 1,311          |

Table 23: Time series of population estimates from the base model.

| Year | Total<br>Biomass<br>(mt) | Spawning<br>output | Total<br>Biomass<br>26+ (mt) | Fraction<br>Unfished | Age-0<br>Recruits<br>(1,000s) | Total<br>Mortality<br>(mt) | 1-SPR | Exploita-<br>tion Rate |
|------|--------------------------|--------------------|------------------------------|----------------------|-------------------------------|----------------------------|-------|------------------------|
| 1892 | 47710.4                  | 5647680            | 33630.9                      | 1.000                | 1008                          | 19                         | 0.018 | 0.001                  |
| 1893 | 47688.5                  | 5645340            | 33617.5                      | 1.000                | 1006                          | 19                         | 0.018 | 0.001                  |
| 1894 | 47665.6                  | 5642950            | 33603.7                      | 0.999                | 1005                          | 19                         | 0.018 | 0.001                  |
| 1895 | 47640.9                  | 5640540            | 33589.5                      | 0.999                | 1003                          | 5                          | 0.005 | 0.000                  |
| 1896 | 47629.5                  | 5639850            | 33584.8                      | 0.999                | 1001                          | 1                          | 0.001 | 0.000                  |
| 1897 | 47618.6                  | 5639620            | 33582.6                      | 0.999                | 1000                          | 1                          | 0.001 | 0.000                  |
| 1898 | 47603.1                  | 5639420            | 33580.2                      | 0.999                | 998                           | 1                          | 0.001 | 0.000                  |
| 1899 | 47583.3                  | 5639290            | 33578.1                      | 0.999                | 997                           | 1                          | 0.001 | 0.000                  |
| 1900 | 47557.6                  | 5639110            | 33575.8                      | 0.998                | 995                           | 2                          | 0.002 | 0.000                  |
| 1901 | 47525.7                  | 5638850            | 33573.1                      | 0.998                | 994                           | 2                          | 0.002 | 0.000                  |
| 1902 | 47487.6                  | 5638450            | 33570.3                      | 0.998                | 993                           | 3                          | 0.002 | 0.000                  |
| 1903 | 47443.3                  | 5637830            | 33567.3                      | 0.998                | 992                           | 3                          | 0.003 | 0.000                  |
| 1904 | 47393.0                  | 5636880            | 33564.4                      | 0.998                | 991                           | 4                          | 0.003 | 0.000                  |
| 1905 | 47336.9                  | 5635450            | 33561.5                      | 0.998                | 990                           | 4                          | 0.004 | 0.000                  |
| 1906 | 47275.4                  | 5633410            | 33558.7                      | 0.997                | 989                           | 4                          | 0.004 | 0.000                  |
| 1907 | 47208.7                  | 5630640            | 33555.9                      | 0.997                | 989                           | 5                          | 0.005 | 0.000                  |
| 1908 | 47137.2                  | 5627050            | 33553.0                      | 0.996                | 988                           | 8                          | 0.007 | 0.000                  |
| 1909 | 47058.6                  | 5622270            | 33548.1                      | 0.996                | 988                           | 6                          | 0.006 | 0.000                  |
| 1910 | 46979.0                  | 5616900            | 33544.6                      | 0.995                | 988                           | 6                          | 0.006 | 0.000                  |
| 1911 | 46895.9                  | 5610650            | 33540.7                      | 0.993                | 988                           | 7                          | 0.006 | 0.000                  |
| 1912 | 46809.8                  | 5603570            | 33536.4                      | 0.992                | 988                           | 7                          | 0.007 | 0.000                  |
| 1913 | 46721.1                  | 5595700            | 33531.6                      | 0.991                | 989                           | 8                          | 0.007 | 0.000                  |
| 1914 | 46630.3                  | 5587100            | 33526.4                      | 0.989                | 990                           | 8                          | 0.008 | 0.000                  |
| 1915 | 46537.7                  | 5577850            | 33520.8                      | 0.988                | 991                           | 10                         | 0.010 | 0.000                  |
| 1916 | 46441.9                  | 5567820            | 33513.5                      | 0.986                | 992                           | 9                          | 0.009 | 0.000                  |
| 1917 | 46346.9                  | 5557450            | 33506.9                      | 0.984                | 993                           | 10                         | 0.010 | 0.000                  |
| 1918 | 46251.3                  | 5546640            | 33427.7                      | 0.982                | 995                           | 55                         | 0.052 | 0.002                  |
| 1919 | 46105.1                  | 5529890            | 33314.7                      | 0.979                | 997                           | 26                         | 0.025 | 0.001                  |
| 1920 | 45992.9                  | 5516430            | 33221.1                      | 0.977                | 1000                          | 23                         | 0.023 | 0.001                  |
| 1921 | 45884.8                  | 5503070            | 33128.4                      | 0.974                | 1002                          | 23                         | 0.022 | 0.001                  |
| 1922 | 45778.9                  | 5489620            | 33035.2                      | 0.972                | 1005                          | 18                         | 0.018 | 0.001                  |
| 1923 | 45680.5                  | 5476710            | 32945.2                      | 0.970                | 1008                          | 20                         | 0.020 | 0.001                  |

|      |         |         |         |       |      |     |       |       |
|------|---------|---------|---------|-------|------|-----|-------|-------|
| 1924 | 45581.9 | 5463510 | 32853.3 | 0.967 | 1012 | 32  | 0.031 | 0.001 |
| 1925 | 45472.9 | 5448910 | 32753.2 | 0.965 | 1015 | 38  | 0.037 | 0.001 |
| 1926 | 45359.9 | 5433620 | 32648.6 | 0.962 | 1019 | 54  | 0.052 | 0.002 |
| 1927 | 45231.9 | 5416390 | 32532.5 | 0.959 | 1023 | 69  | 0.066 | 0.002 |
| 1928 | 45091.3 | 5397490 | 32406.2 | 0.956 | 1026 | 72  | 0.069 | 0.002 |
| 1929 | 44951.8 | 5378370 | 32278.0 | 0.952 | 1029 | 66  | 0.064 | 0.002 |
| 1930 | 44823.4 | 5360170 | 32154.2 | 0.949 | 1032 | 67  | 0.065 | 0.002 |
| 1931 | 44699.0 | 5342140 | 32030.5 | 0.946 | 1035 | 44  | 0.043 | 0.001 |
| 1932 | 44605.8 | 5327290 | 31924.1 | 0.943 | 1037 | 25  | 0.025 | 0.001 |
| 1933 | 44538.8 | 5315180 | 31832.5 | 0.941 | 1039 | 35  | 0.035 | 0.001 |
| 1934 | 44465.8 | 5302350 | 31735.8 | 0.939 | 1041 | 42  | 0.042 | 0.001 |
| 1935 | 44389.7 | 5289120 | 31636.0 | 0.937 | 1042 | 34  | 0.034 | 0.001 |
| 1936 | 44328.1 | 5277430 | 31544.2 | 0.934 | 1044 | 61  | 0.060 | 0.002 |
| 1937 | 44241.6 | 5263020 | 31436.5 | 0.932 | 1046 | 53  | 0.052 | 0.002 |
| 1938 | 44169.1 | 5250110 | 31337.1 | 0.930 | 1048 | 56  | 0.055 | 0.002 |
| 1939 | 44099.3 | 5237520 | 31239.6 | 0.927 | 1050 | 28  | 0.029 | 0.001 |
| 1940 | 44064.9 | 5228850 | 31164.7 | 0.926 | 1054 | 59  | 0.058 | 0.002 |
| 1941 | 44001.0 | 5217130 | 31072.7 | 0.924 | 1059 | 102 | 0.097 | 0.003 |
| 1942 | 43895.0 | 5200900 | 30955.4 | 0.921 | 1065 | 125 | 0.119 | 0.004 |
| 1943 | 43767.9 | 5182370 | 30824.9 | 0.918 | 1073 | 257 | 0.227 | 0.008 |
| 1944 | 43499.1 | 5148380 | 30606.2 | 0.912 | 1083 | 85  | 0.085 | 0.003 |
| 1945 | 43429.3 | 5135720 | 30507.5 | 0.909 | 1095 | 50  | 0.053 | 0.002 |
| 1946 | 43404.6 | 5128160 | 30437.5 | 0.908 | 1111 | 69  | 0.070 | 0.002 |
| 1947 | 43364.6 | 5118970 | 30357.1 | 0.906 | 1129 | 41  | 0.042 | 0.001 |
| 1948 | 43361.7 | 5113790 | 30299.1 | 0.905 | 1151 | 44  | 0.045 | 0.001 |
| 1949 | 43362.2 | 5109020 | 30243.7 | 0.905 | 1176 | 31  | 0.031 | 0.001 |
| 1950 | 43385.0 | 5106620 | 30202.6 | 0.904 | 1204 | 53  | 0.053 | 0.002 |
| 1951 | 43391.4 | 5102330 | 30152.7 | 0.903 | 1235 | 59  | 0.059 | 0.002 |
| 1952 | 43400.1 | 5098000 | 30104.8 | 0.903 | 1268 | 38  | 0.039 | 0.001 |
| 1953 | 43442.0 | 5096810 | 30077.1 | 0.902 | 1302 | 20  | 0.021 | 0.001 |
| 1954 | 43514.8 | 5098420 | 30068.2 | 0.903 | 1334 | 36  | 0.037 | 0.001 |
| 1955 | 43583.2 | 5098940 | 30056.4 | 0.903 | 1363 | 32  | 0.032 | 0.001 |
| 1956 | 43670.3 | 5100690 | 30054.1 | 0.903 | 1385 | 20  | 0.022 | 0.001 |
| 1957 | 43785.8 | 5104620 | 30065.7 | 0.904 | 1400 | 35  | 0.035 | 0.001 |
| 1958 | 43902.3 | 5107740 | 30073.0 | 0.904 | 1406 | 15  | 0.016 | 0.000 |
| 1959 | 44058.6 | 5114140 | 30097.3 | 0.906 | 1405 | 22  | 0.023 | 0.001 |
| 1960 | 44225.0 | 5120720 | 30119.7 | 0.907 | 1397 | 23  | 0.024 | 0.001 |
| 1961 | 44409.3 | 5128490 | 30144.9 | 0.908 | 1383 | 26  | 0.027 | 0.001 |



|      |         |         |         |       |      |     |       |       |
|------|---------|---------|---------|-------|------|-----|-------|-------|
| 1962 | 44608.4 | 5137260 | 30170.8 | 0.910 | 1363 | 32  | 0.033 | 0.001 |
| 1963 | 44818.7 | 5146960 | 30195.8 | 0.911 | 1337 | 23  | 0.024 | 0.001 |
| 1964 | 45054.4 | 5159360 | 30228.8 | 0.914 | 1303 | 32  | 0.032 | 0.001 |
| 1965 | 45294.8 | 5172690 | 30259.1 | 0.916 | 1265 | 31  | 0.032 | 0.001 |
| 1966 | 45547.6 | 5188400 | 30295.0 | 0.919 | 1227 | 117 | 0.116 | 0.004 |
| 1967 | 45714.8 | 5197660 | 30292.8 | 0.920 | 1192 | 108 | 0.107 | 0.004 |
| 1968 | 45895.3 | 5209770 | 30298.4 | 0.922 | 1166 | 172 | 0.164 | 0.006 |
| 1969 | 46006.3 | 5217540 | 30278.3 | 0.924 | 1149 | 51  | 0.049 | 0.002 |
| 1970 | 46246.7 | 5239140 | 30319.8 | 0.928 | 1138 | 22  | 0.023 | 0.001 |
| 1971 | 46513.7 | 5266190 | 30386.7 | 0.932 | 1130 | 68  | 0.067 | 0.002 |
| 1972 | 46724.1 | 5290950 | 30443.7 | 0.937 | 1124 | 76  | 0.074 | 0.003 |
| 1973 | 46914.5 | 5316590 | 30509.1 | 0.941 | 1124 | 75  | 0.072 | 0.002 |
| 1974 | 47093.4 | 5343710 | 30589.5 | 0.946 | 1130 | 77  | 0.072 | 0.003 |
| 1975 | 47256.4 | 5371480 | 30684.5 | 0.951 | 1133 | 43  | 0.041 | 0.001 |
| 1976 | 47442.1 | 5403430 | 30816.1 | 0.957 | 1149 | 19  | 0.018 | 0.001 |
| 1977 | 47638.9 | 5438220 | 30982.3 | 0.963 | 1173 | 166 | 0.133 | 0.005 |
| 1978 | 47658.6 | 5455410 | 31079.3 | 0.966 | 1211 | 69  | 0.062 | 0.002 |
| 1979 | 47771.9 | 5483420 | 31261.9 | 0.971 | 1212 | 185 | 0.154 | 0.006 |
| 1980 | 47744.8 | 5497010 | 31398.1 | 0.973 | 1198 | 99  | 0.089 | 0.003 |
| 1981 | 47801.4 | 5518740 | 31603.5 | 0.977 | 1178 | 131 | 0.115 | 0.004 |
| 1982 | 47811.8 | 5534670 | 31801.4 | 0.980 | 1101 | 167 | 0.147 | 0.005 |
| 1983 | 47773.5 | 5544590 | 31987.0 | 0.982 | 1104 | 126 | 0.111 | 0.004 |
| 1984 | 47771.4 | 5556240 | 32190.6 | 0.984 | 1280 | 144 | 0.128 | 0.004 |
| 1985 | 47742.1 | 5563660 | 32377.5 | 0.985 | 1260 | 299 | 0.245 | 0.009 |
| 1986 | 47534.9 | 5550970 | 32456.6 | 0.983 | 1038 | 428 | 0.326 | 0.013 |
| 1987 | 47180.0 | 5520180 | 32427.4 | 0.977 | 1194 | 571 | 0.409 | 0.018 |
| 1988 | 46663.6 | 5469770 | 32273.7 | 0.968 | 2024 | 351 | 0.289 | 0.011 |
| 1989 | 46393.3 | 5442980 | 32226.9 | 0.964 | 1304 | 418 | 0.345 | 0.013 |
| 1990 | 46055.4 | 5408210 | 32114.3 | 0.958 | 928  | 244 | 0.223 | 0.008 |
| 1991 | 45918.2 | 5391250 | 32063.6 | 0.955 | 1071 | 299 | 0.269 | 0.009 |
| 1992 | 45729.4 | 5367770 | 31946.6 | 0.950 | 1199 | 306 | 0.269 | 0.010 |
| 1993 | 45538.9 | 5341880 | 31782.8 | 0.946 | 1389 | 327 | 0.283 | 0.010 |
| 1994 | 45330.0 | 5312720 | 31575.6 | 0.941 | 1044 | 306 | 0.265 | 0.010 |
| 1995 | 45151.3 | 5285150 | 31357.5 | 0.936 | 625  | 745 | 0.507 | 0.024 |
| 1996 | 44488.4 | 5204820 | 30828.9 | 0.922 | 549  | 339 | 0.292 | 0.011 |
| 1997 | 44277.1 | 5172580 | 30560.4 | 0.916 | 555  | 303 | 0.266 | 0.010 |
| 1998 | 44097.4 | 5145200 | 30307.9 | 0.911 | 775  | 441 | 0.355 | 0.015 |
| 1999 | 43752.0 | 5102490 | 29962.1 | 0.903 | 1756 | 256 | 0.231 | 0.009 |

|      |         |         |         |       |      |     |       |       |
|------|---------|---------|---------|-------|------|-----|-------|-------|
| 2000 | 43593.6 | 5083220 | 29743.6 | 0.900 | 1223 | 183 | 0.164 | 0.006 |
| 2001 | 43514.7 | 5074280 | 29572.1 | 0.898 | 1132 | 114 | 0.112 | 0.004 |
| 2002 | 43495.7 | 5076390 | 29469.9 | 0.899 | 816  | 74  | 0.072 | 0.003 |
| 2003 | 43511.0 | 5084450 | 29410.8 | 0.900 | 600  | 99  | 0.093 | 0.003 |
| 2004 | 43492.7 | 5091030 | 29371.2 | 0.901 | 965  | 115 | 0.103 | 0.004 |
| 2005 | 43449.2 | 5096310 | 29332.3 | 0.902 | 1054 | 138 | 0.119 | 0.005 |
| 2006 | 43373.2 | 5098700 | 29280.8 | 0.903 | 820  | 127 | 0.115 | 0.004 |
| 2007 | 43295.9 | 5101840 | 29242.9 | 0.903 | 1268 | 187 | 0.162 | 0.006 |
| 2008 | 43148.7 | 5095850 | 29123.3 | 0.902 | 1807 | 218 | 0.183 | 0.007 |
| 2009 | 42967.2 | 5083210 | 28996.5 | 0.900 | 1102 | 228 | 0.202 | 0.008 |
| 2010 | 42771.2 | 5067430 | 29021.7 | 0.897 | 1938 | 261 | 0.226 | 0.009 |
| 2011 | 42552.0 | 5045000 | 29015.7 | 0.893 | 1061 | 210 | 0.168 | 0.007 |
| 2012 | 42420.9 | 5025520 | 28861.5 | 0.890 | 2857 | 244 | 0.192 | 0.008 |
| 2013 | 42280.5 | 5000830 | 28812.2 | 0.885 | 977  | 156 | 0.134 | 0.005 |
| 2014 | 42264.1 | 4987020 | 29489.1 | 0.883 | 711  | 91  | 0.082 | 0.003 |
| 2015 | 42350.0 | 4980900 | 29647.2 | 0.882 | 659  | 132 | 0.114 | 0.004 |
| 2016 | 42423.7 | 4969060 | 29475.7 | 0.880 | 870  | 149 | 0.127 | 0.005 |
| 2017 | 42502.0 | 4954890 | 29408.6 | 0.877 | 2172 | 156 | 0.133 | 0.005 |
| 2018 | 42587.5 | 4940000 | 29441.8 | 0.875 | 1341 | 242 | 0.188 | 0.008 |
| 2019 | 42610.9 | 4914900 | 29559.4 | 0.870 | 1197 | 227 | 0.180 | 0.008 |
| 2020 | 42665.6 | 4893960 | 29413.4 | 0.867 | 846  | 106 | 0.090 | 0.004 |
| 2021 | 42856.9 | 4891660 | 29028.1 | 0.866 | 923  | 93  | 0.079 | 0.003 |
| 2022 | 43067.6 | 4895290 | 28596.8 | 0.867 | 1017 | 117 | 0.098 | 0.004 |
| 2023 | 43255.1 | 4900660 | 28158.0 | 0.868 | 1067 | 98  | 0.083 | 0.003 |
| 2024 | 43457.9 | 4913840 | 27923.1 | 0.870 | 1076 | 119 | 0.098 | 0.004 |
| 2025 | 43632.6 | 4929140 | 28487.5 | 0.873 | 1076 | 141 | 0.114 | 0.005 |
| 2026 | 43775.5 | 4945630 | 28597.4 | 0.876 | 1077 | 174 | 0.136 | 0.006 |
| 2027 | 43875.4 | 4960900 | 28610.8 | 0.878 | 1077 | 894 | 0.474 | 0.031 |
| 2028 | 43263.3 | 4888090 | 27828.0 | 0.866 | 1075 | 878 | 0.473 | 0.032 |
| 2029 | 42664.7 | 4818790 | 26900.8 | 0.853 | 1074 | 862 | 0.471 | 0.032 |
| 2030 | 42080.7 | 4752560 | 26297.5 | 0.842 | 1072 | 846 | 0.469 | 0.032 |
| 2031 | 41511.4 | 4688990 | 25785.3 | 0.830 | 1070 | 830 | 0.467 | 0.032 |
| 2032 | 40957.1 | 4627620 | 25113.8 | 0.819 | 1069 | 815 | 0.466 | 0.032 |
| 2033 | 40417.5 | 4567870 | 24804.2 | 0.809 | 1067 | 799 | 0.464 | 0.032 |
| 2034 | 39894.2 | 4509380 | 24904.8 | 0.798 | 1065 | 784 | 0.462 | 0.031 |
| 2035 | 39387.0 | 4451710 | 24483.9 | 0.788 | 1064 | 769 | 0.460 | 0.031 |
| 2036 | 38897.6 | 4394830 | 24681.2 | 0.778 | 1062 | 755 | 0.459 | 0.031 |

Table 24: Base model sensitivity to the removal of data sources (indices).

| Label                                      | Base    | - Triennial | - AK Slope | -NW slope | - WCGBTS | No indices |
|--------------------------------------------|---------|-------------|------------|-----------|----------|------------|
| <b>Diff. in likelihood from base model</b> |         |             |            |           |          |            |
| Total                                      | 0       | -15.45      | -52.23     | 1.03      | -2058.14 | -2129.36   |
| Index                                      | 0       | 7.784       | 0.176      | 1.032     | 17.58    | NA         |
| Length comp                                | 0       | -21.986     | -52.07     | 0         | -92.246  | -173.251   |
| Age comp                                   | 0       | -0.12       | -0.22      | 0         | -1976.5  | -1975.13   |
| Recruitment                                | 0       | -1.132      | -0.128     | -0.001    | -6.988   | -7.802     |
| Parm priors                                | 0       | 0.001       | 0.005      | 0         | 0.002    | 0.006      |
| <b>Estimates of key parameters</b>         |         |             |            |           |          |            |
| Recruitment unfished millions              | 1.092   | 0.93        | 1.075      | 1.088     | 0.674    | 0.631      |
| log(R0)                                    | 6.995   | 6.836       | 6.98       | 6.992     | 6.513    | 6.447      |
| NatM uniform Female                        | 0.039   | 0.039       | 0.039      | 0.039     | 0.039    | 0.039      |
| NatM uniform Male                          | 0.036   | 0.036       | 0.036      | 0.036     | 0.036    | 0.036      |
| L at Amax Female                           | 60.1    | 60.1        | 60.1       | 60.1      | 61.9     | 62.4       |
| L at Amax Male                             | 57.8    | 57.9        | 57.8       | 57.8      | 58.7     | 58.7       |
| <b>Estimates of derived quantities</b>     |         |             |            |           |          |            |
| Unfished age 8+ bio 1000 mt                | 33.631  | 28.678      | 33.004     | 33.509    | 20.708   | 19.333     |
| B0 millions of eggs                        | 5647660 | 4816400     | 5526870    | 5627100   | 3465540  | 3241330    |
| B2025 millions of eggs                     | 4929120 | 4011930     | 4796650    | 4907010   | 2573150  | 2313870    |
| Fraction unfished 2025                     | 0.873   | 0.833       | 0.868      | 0.872     | 0.742    | 0.714      |
| Fishing intensity 2024                     | 0.098   | 0.119       | 0.1        | 0.099     | 0.182    | 0.2        |
| Catchability for WCGBTS                    | 0.074   | 0.089       | 0.075      | 0.074     | NA       | NA         |

Table 25: Base model sensitivity to the removal of data sources (length compositions by fleet).

| Label                                      | Base    | - bottom<br>trawl | - non-<br>trawl | - mid-<br>water<br>trawl | -<br>ASHOP | -<br>Triennial | - AK<br>slope | - NW<br>slope | - WCG-<br>BTS |
|--------------------------------------------|---------|-------------------|-----------------|--------------------------|------------|----------------|---------------|---------------|---------------|
| <b>Diff. in likelihood from base model</b> |         |                   |                 |                          |            |                |               |               |               |
| Total                                      | 0       | -128.29           | -251.07         | -59.31                   | -26.57     | -25.13         | -52.49        | 0             | -83.81        |
| Index                                      | 0       | 1.361             | 0.211           | -0.117                   | 0.006      | -2.543         | -0.048        | 0             | 0.019         |
| Length comp                                | 0       | -115.595          | -222.555        | -58.149                  | -25.251    | -22.257        | -51.94        | 0             | -82.399       |
| Age comp                                   | 0       | -11.77            | -26.82          | -0.63                    | -1.39      | 0.37           | -0.29         | 0             | -1.49         |
| Recruitment                                | 0       | -2.186            | -1.854          | -0.423                   | 0.073      | -0.716         | -0.227        | 0             | 0.065         |
| Parm priors                                | 0       | -0.108            | -0.062          | 0.005                    | -0.004     | 0              | 0.005         | 0             | -0.006        |
| <b>Estimates of key parameters</b>         |         |                   |                 |                          |            |                |               |               |               |
| Recruitment unfished millions              | 1.092   | 0.277             | 0.401           | 1.536                    | 1.628      | 0.972          | 1.076         | 1.092         | 1.166         |
| log(R0)                                    | 6.995   | 5.623             | 5.995           | 7.337                    | 7.395      | 6.88           | 6.981         | 6.995         | 7.061         |
| NatM uniform Female                        | 0.039   | 0.035             | 0.037           | 0.039                    | 0.039      | 0.039          | 0.039         | 0.039         | 0.039         |
| NatM uniform Male                          | 0.036   | 0.036             | 0.036           | 0.036                    | 0.036      | 0.036          | 0.036         | 0.036         | 0.036         |
| L at Amax Female                           | 60.1    | 60.2              | 59.1            | 60.2                     | 60.5       | 60.1           | 60.1          | 60.1          | 60.1          |
| L at Amax Male                             | 57.8    | 57.5              | 57.4            | 58                       | 58.2       | 57.8           | 57.8          | 57.8          | 57.9          |
| <b>Estimates of derived quantities</b>     |         |                   |                 |                          |            |                |               |               |               |
| Unfished age 8+ bio 1000 mt                | 33.631  | 9.566             | 12.602          | 47.494                   | 51.097     | 29.983         | 33.04         | 33.631        | 36.135        |
| B0 millions of eggs                        | 5647660 | 1774240           | 2140660         | 7962190                  | 8658500    | 5038190        | 5532480       | 5647660       | 6097240       |
| B2025 millions of eggs                     | 4929120 | 616218            | 1203260         | 7357030                  | 7991000    | 4255270        | 4801130       | 4929120       | 5416980       |
| Fraction unfished 2025                     | 0.873   | 0.347             | 0.562           | 0.924                    | 0.923      | 0.845          | 0.868         | 0.873         | 0.888         |
| Fishing intensity 2024                     | 0.098   | 0.536             | 0.313           | 0.072                    | 0.063      | 0.113          | 0.1           | 0.098         | 0.091         |
| Catchability for WCG BTS                   | 0.074   | 0.349             | 0.236           | 0.051                    | 0.047      | 0.085          | 0.075         | 0.074         | 0.068         |

Table 26: Base model sensitivity to the removal of data sources (length compositions by source).

| Label                                      | Base    | - fishery | - survey | no lengths |
|--------------------------------------------|---------|-----------|----------|------------|
| <b>Diff. in likelihood from base model</b> |         |           |          |            |
| Total                                      | 0       | -454.05   | -162.74  | -617.35    |
| Index                                      | 0       | 1.049     | -3.924   | -2.848     |
| Length comp                                | 0       | -415.616  | -157.698 | -574.836   |
| Age comp                                   | 0       | -37.58    | -0.45    | -38.19     |
| Recruitment                                | 0       | -1.929    | -0.678   | -2.429     |
| Parm priors                                | 0       | 0.011     | -0.001   | 0.938      |
| <b>Estimates of key parameters</b>         |         |           |          |            |
| Recruitment unfished thousands             | 1091.58 | 359.453   | 1042.2   | 769.18     |
| log(R0)                                    | 6.995   | 5.885     | 6.949    | 6.645      |
| NatM uniform Female                        | 0.039   | 0.039     | 0.039    | 0.053      |
| NatM uniform Male                          | 0.036   | 0.036     | 0.036    | 0.036      |
| L at Amax Female                           | 60.1    | 57.4      | 60.2     | 56         |
| L at Amax Male                             | 57.8    | 56.5      | 57.9     | 56.7       |
| <b>Estimates of derived quantities</b>     |         |           |          |            |
| Unfished age 8+ bio 1000 mt                | 33.631  | 10.074    | 32.222   | 16.412     |
| B0 millions of eggs                        | 5647660 | 1559520   | 5425890  | 1688620    |
| B2025 millions of eggs                     | 4929120 | 745994    | 4674580  | 1214190    |
| Fraction unfished 2025                     | 0.873   | 0.478     | 0.862    | 0.719      |
| Fishing intensity 2024                     | 0.098   | 0.403     | 0.104    | 0.175      |
| Catchability for WCGBTS                    | 0.074   | 0.286     | 6.426    | 7.108      |

Table 27: Base model sensitivity to the removal of data sources (age compositions by fleet).

| Label                                      | Base    | - bottom<br>trawl | - non-<br>trawl | - mid-<br>water<br>trawl | -<br>ASHOP | -<br>WGCBTS | - fishery | no ages       |
|--------------------------------------------|---------|-------------------|-----------------|--------------------------|------------|-------------|-----------|---------------|
| <b>Diff. in likelihood from base model</b> |         |                   |                 |                          |            |             |           |               |
| Total                                      | 0       | -467.49           | -332.87         | -182.78                  | -3886.07   | -1988.26    | -4868.13  | -<br>6840.569 |
| Index                                      | 0       | 0.219             | 0.016           | 0.09                     | -0.455     | 0.006       | -0.204    | -1.149        |
| Length comp                                | 0       | -0.58             | -2.269          | 0.085                    | -30.757    | -5.665      | -37.129   | -49.98        |
| Age comp                                   | 0       | -467.56           | -330.26         | -183.28                  | -3854.02   | -1975.69    | -4829.43  | -6783.49      |
| Recruitment                                | 0       | 0.428             | -0.36           | 0.329                    | -0.817     | -6.922      | -1.329    | -5.862        |
| Parm priors                                | 0       | -0.001            | -0.003          | -0.003                   | -0.03      | 0.008       | -0.045    | -0.092        |
| <b>Estimates of key parameters</b>         |         |                   |                 |                          |            |             |           |               |
| Recruitment unfished thousands             | 1091.58 | 925.774           | 1152.03         | 1062.2                   | 486.651    | 669.58      | 504.669   | 647.515       |
| log(R0)                                    | 6.995   | 6.831             | 7.049           | 6.968                    | 6.188      | 6.507       | 6.224     | 6.473         |
| NatM uniform Female                        | 0.039   | 0.039             | 0.039           | 0.039                    | 0.038      | 0.039       | 0.038     | 0.036         |
| NatM uniform Male                          | 0.036   | 0.036             | 0.036           | 0.036                    | 0.036      | 0.036       | 0.036     | 0.036         |
| L at Amax Female                           | 60.1    | 60                | 60              | 60.1                     | 59.6       | 61.7        | 58.4      | 55.3          |
| L at Amax Male                             | 57.8    | 57.7              | 57.7            | 57.8                     | 57.4       | 58.7        | 56.9      | 54.2          |
| <b>Estimates of derived quantities</b>     |         |                   |                 |                          |            |             |           |               |
| Unfished age 8+ bio 1000 mt                | 33.631  | 28.572            | 35.428          | 32.801                   | 15.219     | 20.495      | 15.741    | 19.027        |
| B0 millions of eggs                        | 5647660 | 4819970           | 5957400         | 5527370                  | 2628770    | 3412570     | 3933420   | 3265070       |
| B2025 millions of eggs                     | 4929120 | 3968100           | 5184400         | 4772870                  | 1844350    | 2507090     | 1714540   | 2402180       |
| Fraction unfished 2025                     | 0.873   | 0.823             | 0.87            | 0.863                    | 0.702      | 0.735       | 0.436     | 0.736         |
| Fishing intensity 2024                     | 0.098   | 0.119             | 0.094           | 0.101                    | 0.232      | 0.184       | 0.169     | 0.165         |
| Catchability for WGCBTS                    | 0.074   | 0.089             | 0.07            | 0.076                    | 0.205      | 0.094       | 0.214     | 0.18          |

Table 28: Likelihood values and estimates of key parameters and derived quantities from each data weighting sensitivity.

| Label                | Type                 | Reference | No data<br>weighting | McAllister<br>& Ianelli | Dirichlet |
|----------------------|----------------------|-----------|----------------------|-------------------------|-----------|
| Survey<br>likelihood | ALL                  | -26.81    | -27.30               | -27.44                  | -27.28    |
|                      | TRIENNIAL            | -7.76     | -7.97                | -8.15                   | -7.95     |
|                      | AK_SLOPE             | -0.53     | -0.56                | -0.55                   | -0.56     |
|                      | NW_SLOPE             | -1.03     | -1.03                | -1.04                   | -1.03     |
|                      | WCGBTS               | -17.49    | -17.73               | -17.70                  | -17.74    |
| Lt<br>likelihood     | ALL                  | 574.84    | 3,390.72             | 1,911.36                | 9,924.11  |
|                      | BOTTOM_TRAWL         | 79.10     | 603.67               | 316.54                  | 2,083.61  |
|                      | BOTTOM_TRAWL_DISCARD | 27.15     | 310.71               | 48.27                   | 578.49    |
|                      | NON_TRAWL            | 179.67    | 580.15               | 301.85                  | 1,993.14  |
|                      | NON_TRAWL_DISCARD    | 46.62     | 188.12               | 116.18                  | 463.26    |
|                      | MIDWATER_TRAWL       | 58.95     | 147.40               | 118.80                  | 492.88    |
|                      | AT_SEA_HAKE          | 25.42     | 670.55               | 323.73                  | 2,157.88  |
|                      | TRIENNIAL            | 22.42     | 135.04               | 85.53                   | 451.19    |
|                      | AK_SLOPE             | 52.30     | 52.49                | 52.48                   | 138.45    |
|                      | NW_SLOPE             | 0.00      | 0.00                 | 0.00                    | 0.00      |
| Age<br>likelihood    | WCGBTS               | 83.19     | 702.60               | 548.00                  | 1,565.23  |
|                      | ALL                  | 6,783.49  | 18,052.20            | 6,726.98                | 25,656.00 |
|                      | BOTTOM_TRAWL         | 464.25    | 4,116.95             | 1,274.70                | 5,974.62  |
|                      | BOTTOM_TRAWL_DISCARD | 0.00      | 0.00                 | 0.00                    | 0.00      |
|                      | NON_TRAWL            | 331.38    | 4,013.75             | 1,089.14                | 5,922.32  |
|                      | NON_TRAWL_DISCARD    | 0.00      | 0.00                 | 0.00                    | 0.00      |
|                      | MIDWATER_TRAWL       | 182.47    | 1,186.38             | 559.08                  | 1,538.06  |
|                      | AT_SEA_HAKE          | 3,847.50  | 5,317.71             | 1,597.41                | 8,046.53  |
|                      | TRIENNIAL            | 0.00      | 0.00                 | 0.00                    | 0.00      |
|                      | AK_SLOPE             | 0.00      | 0.00                 | 0.00                    | 0.00      |
|                      | NW_SLOPE             | 0.00      | 0.00                 | 0.00                    | 0.00      |
|                      | WCGBTS               | 1,957.88  | 3,417.40             | 2,206.66                | 4,174.46  |

## Parameters

|                         |         |         |         |         |
|-------------------------|---------|---------|---------|---------|
| NatM_uniform_Fem_GP_1   | 0.04    | 0.04    | 0.04    | 0.04    |
| L_at_Amin_Fem_GP_1      | -3.10   | -1.88   | -3.27   | -1.77   |
| L_at_Amax_Fem_GP_1      | 60.08   | 59.82   | 59.23   | 59.76   |
| VonBert_K_Fem_GP_1      | 0.08    | 0.08    | 0.08    | 0.08    |
| CV_young_Fem_GP_1       | 0.05    | 0.08    | 0.04    | 0.08    |
| CV_old_Fem_GP_1         | 0.09    | 0.10    | 0.10    | 0.10    |
| NatM_uniform_Mal_GP_1   | 0.04    | 0.04    | 0.04    | 0.04    |
| L_at_Amin_Mal_GP_1      | -2.68   | -0.98   | -2.80   | -0.86   |
| L_at_Amax_Mal_GP_1      | 57.83   | 57.81   | 57.35   | 57.76   |
| VonBert_K_Mal_GP_1      | 0.08    | 0.08    | 0.09    | 0.08    |
| CV_young_Mal_GP_1       | 0.09    | 0.11    | 0.07    | 0.11    |
| CV_old_Mal_GP_1         | 0.08    | 0.09    | 0.09    | 0.09    |
| SR_LN(R0)               | 7.00    | 6.30    | 6.29    | 6.28    |
| LnQ_base_TRIENNIAL(7)   | -1.49   | -1.09   | -0.90   | -1.08   |
| LnQ_base_AK_SLOPE(8)    | -3.11   | -2.80   | -2.74   | -2.79   |
| LnQ_base_NW_SLOPE(9)    | -2.04   | -1.74   | -1.68   | -1.73   |
| LnQ_base_WCGBTS(10)     | -2.61   | -1.85   | -1.73   | -1.83   |
| LnQ_base_TRIENNIAL(7)_- | -2.27   | -2.06   | -1.94   | -2.06   |
| BLK3repl_1892           |         |         |         |         |
| Size_DblN_peak_BOTTOM_- | 49.34   | 49.49   | 49.62   | 49.62   |
| TRAWL(1)                |         |         |         |         |
| Size_DblN_top_logit_-   | -11.52  | -11.52  | -11.52  | -11.52  |
| BOTTOM_TRAWL(1)         |         |         |         |         |
| Size_DblN_ascend_se_-   | 4.57    | 4.75    | 4.76    | 4.76    |
| BOTTOM_TRAWL(1)         |         |         |         |         |
| Size_DblN_descend_se_-  | 2.95    | 2.87    | 2.93    | 2.84    |
| BOTTOM_TRAWL(1)         |         |         |         |         |
| Size_DblN_start_logit_- | -999.00 | -999.00 | -999.00 | -999.00 |
| BOTTOM_TRAWL(1)         |         |         |         |         |
| Size_DblN_end_logit_-   | -0.64   | -0.22   | -0.16   | -0.17   |
| BOTTOM_TRAWL(1)         |         |         |         |         |
| Size_DblN_peak_BOTTOM_- | 25.26   | 24.65   | 24.82   | 24.71   |
| TRAWL_DISCARD(2)        |         |         |         |         |
| Size_DblN_top_logit_-   | -15.00  | -15.00  | -15.00  | -15.00  |
| BOTTOM_TRAWL_-          |         |         |         |         |
| DISCARD(2)              |         |         |         |         |



|                                                         |         |         |         |         |
|---------------------------------------------------------|---------|---------|---------|---------|
| Size_DblN_ascend_se_-<br>BOTTOM_TRAWL_-<br>DISCARD(2)   | 4.89    | 4.75    | 4.78    | 4.66    |
| Size_DblN_descend_se_-<br>BOTTOM_TRAWL_-<br>DISCARD(2)  | 3.84    | 3.80    | 3.80    | 3.75    |
| Size_DblN_start_logit_-<br>BOTTOM_TRAWL_-<br>DISCARD(2) | -15.00  | -15.00  | -15.00  | -15.00  |
| Size_DblN_end_logit_-<br>BOTTOM_TRAWL_-<br>DISCARD(2)   | -3.82   | -4.13   | -4.10   | -3.73   |
| Size_DblN_peak_NON_-<br>TRAWL(3)                        | 50.27   | 50.26   | 50.73   | 50.46   |
| Size_DblN_top_logit_NON_-<br>TRAWL(3)                   | -15.00  | -15.00  | -15.00  | -15.00  |
| Size_DblN_ascend_se_NON_-<br>TRAWL(3)                   | 3.86    | 4.01    | 4.04    | 4.04    |
| Size_DblN_descend_se_NON_-<br>TRAWL(3)                  | 20.00   | 20.00   | 20.00   | 20.00   |
| Size_DblN_start_logit_NON_-<br>TRAWL(3)                 | -999.00 | -999.00 | -999.00 | -999.00 |
| Size_DblN_end_logit_NON_-<br>TRAWL(3)                   | 4.60    | 4.60    | 4.60    | 4.60    |
| Size_DblN_peak_NON_-<br>TRAWL_DISCARD(4)                | 49.94   | 51.33   | 51.43   | 51.62   |
| Size_DblN_top_logit_NON_-<br>TRAWL_DISCARD(4)           | -15.00  | -15.00  | -15.00  | -15.00  |
| Size_DblN_ascend_se_NON_-<br>TRAWL_DISCARD(4)           | 3.75    | 4.08    | 4.10    | 4.15    |
| Size_DblN_descend_se_NON_-<br>TRAWL_DISCARD(4)          | 2.66    | 1.88    | 1.86    | 1.62    |
| Size_DblN_start_logit_NON_-<br>TRAWL_DISCARD(4)         | -999.00 | -999.00 | -999.00 | -999.00 |
| Size_DblN_end_logit_NON_-<br>TRAWL_DISCARD(4)           | -0.19   | 0.33    | 0.45    | 0.39    |
| Size_DblN_peak_MIDWATER_-<br>TRAWL(5)                   | 52.17   | 54.04   | 54.71   | 54.54   |

|                                              |         |         |         |         |
|----------------------------------------------|---------|---------|---------|---------|
| Size_DblN_top_logit_-<br>MIDWATER_TRAWL(5)   | -15.00  | -15.00  | -15.00  | -15.00  |
| Size_DblN_ascend_se_-<br>MIDWATER_TRAWL(5)   | 4.57    | 4.87    | 4.92    | 4.92    |
| Size_DblN_descend_se_-<br>MIDWATER_TRAWL(5)  | 20.00   | 20.00   | 20.00   | 20.00   |
| Size_DblN_start_logit_-<br>MIDWATER_TRAWL(5) | -999.00 | -999.00 | -999.00 | -999.00 |
| Size_DblN_end_logit_-<br>MIDWATER_TRAWL(5)   | -999.00 | -999.00 | -999.00 | -999.00 |
| Size_DblN_peak_AT_SEA_-<br>HAKE(6)           | 49.71   | 51.29   | 51.71   | 51.45   |
| Size_DblN_top_logit_AT_-<br>SEA_HAKE(6)      | -15.00  | -15.00  | -15.00  | -15.00  |
| Size_DblN_ascend_se_AT_-<br>SEA_HAKE(6)      | 3.59    | 3.85    | 3.93    | 3.87    |
| Size_DblN_descend_se_AT_-<br>SEA_HAKE(6)     | 20.00   | 20.00   | 20.00   | 20.00   |
| Size_DblN_start_logit_AT_-<br>SEA_HAKE(6)    | -999.00 | -999.00 | -999.00 | -999.00 |
| Size_DblN_end_logit_AT_-<br>SEA_HAKE(6)      | -999.00 | -999.00 | -999.00 | -999.00 |
| Size_DblN_peak_-<br>TRIENNIAL(7)             | 21.78   | 21.79   | 21.46   | 21.85   |
| Size_DblN_top_logit_-<br>TRIENNIAL(7)        | -8.60   | -8.60   | -8.60   | -8.60   |
| Size_DblN_ascend_se_-<br>TRIENNIAL(7)        | 3.52    | 3.56    | 3.46    | 3.57    |
| Size_DblN_descend_se_-<br>TRIENNIAL(7)       | 3.95    | 3.79    | 3.82    | 3.77    |
| Size_DblN_start_logit_-<br>TRIENNIAL(7)      | -999.00 | -999.00 | -999.00 | -999.00 |
| Size_DblN_end_logit_-<br>TRIENNIAL(7)        | -2.58   | -2.27   | -2.38   | -2.24   |
| Size_DblN_peak_AK_-<br>SLOPE(8)              | 37.29   | 38.04   | 37.40   | 38.25   |
| Size_DblN_top_logit_AK_-<br>SLOPE(8)         | -15.00  | -15.00  | -15.00  | -15.00  |

|                                                              |         |         |         |         |
|--------------------------------------------------------------|---------|---------|---------|---------|
| Size_DblN_ascend_se_AK_-<br>SLOPE(8)                         | 4.94    | 5.02    | 5.04    | 5.05    |
| Size_DblN_descend_se_AK_-<br>SLOPE(8)                        | 4.63    | 4.75    | 4.87    | 4.70    |
| Size_DblN_start_logit_AK_-<br>SLOPE(8)                       | -999.00 | -999.00 | -999.00 | -999.00 |
| Size_DblN_end_logit_AK_-<br>SLOPE(8)                         | -10.49  | -10.15  | -10.05  | -4.10   |
| Size_DblN_peak_WCGBTS(10)                                    | 21.08   | 21.45   | 21.56   | 21.50   |
| Size_DblN_top_logit_-<br>WCGBTS(10)                          | -15.00  | -15.00  | -15.00  | -15.00  |
| Size_DblN_ascend_se_-<br>WCGBTS(10)                          | 3.39    | 3.56    | 3.55    | 3.57    |
| Size_DblN_descend_se_-<br>WCGBTS(10)                         | 4.65    | 4.42    | 4.41    | 4.38    |
| Size_DblN_start_logit_-<br>WCGBTS(10)                        | -999.00 | -999.00 | -999.00 | -999.00 |
| Size_DblN_end_logit_-<br>WCGBTS(10)                          | -0.83   | -1.01   | -1.01   | -0.98   |
| Size_DblN_peak_BOTTOM_-<br>TRAWL(1)_BLK4repl_1892            | 44.70   | 45.85   | 46.25   | 45.90   |
| Size_DblN_peak_BOTTOM_-<br>TRAWL(1)_BLK4repl_2002            | 47.64   | 48.70   | 50.06   | 50.14   |
| Size_DblN_ascend_se_-<br>BOTTOM_TRAWL(1)_-<br>BLK4repl_1892  | 5.03    | 5.10    | 5.19    | 5.10    |
| Size_DblN_ascend_se_-<br>BOTTOM_TRAWL(1)_-<br>BLK4repl_2002  | 4.10    | 4.22    | 4.49    | 4.44    |
| Size_DblN_descend_se_-<br>BOTTOM_TRAWL(1)_-<br>BLK4repl_1892 | 2.65    | 2.15    | 1.84    | 2.11    |
| Size_DblN_descend_se_-<br>BOTTOM_TRAWL(1)_-<br>BLK4repl_2002 | 2.75    | 2.30    | -0.18   | -0.35   |
| Size_DblN_end_logit_-<br>BOTTOM_TRAWL(1)_-<br>BLK4repl_1892  | -1.59   | -1.07   | -0.89   | -1.04   |

|                                                                      |       |       |       |       |
|----------------------------------------------------------------------|-------|-------|-------|-------|
| Size_DblN_end_logit_-<br>BOTTOM_TRAWL(1)_-<br>BLK4repl_2002          | -0.93 | -0.34 | -0.15 | -0.26 |
| Size_DblN_peak_BOTTOM_-<br>TRAWL_DISCARD(2)_-<br>BLK1repl_1892       | 47.67 | 48.66 | 48.47 | 52.98 |
| Size_DblN_ascend_se_-<br>BOTTOM_TRAWL_-<br>DISCARD(2)_BLK1repl_1892  | 6.32  | 6.76  | 6.93  | 6.93  |
| Size_DblN_descend_se_-<br>BOTTOM_TRAWL_-<br>DISCARD(2)_BLK1repl_1892 | 3.01  | 2.86  | 2.93  | -9.39 |
| Size_DblN_end_logit_-<br>BOTTOM_TRAWL_-<br>DISCARD(2)_BLK1repl_1892  | -1.70 | -1.16 | -1.08 | -0.95 |
| Size_DblN_peak_NON_-<br>TRAWL(3)_BLK2repl_1892                       | 46.77 | 47.32 | 47.35 | 47.37 |
| Size_DblN_peak_NON_-<br>TRAWL(3)_BLK2repl_2011                       | 49.49 | 50.04 | 50.10 | 51.03 |
| Size_DblN_ascend_se_NON_-<br>TRAWL(3)_BLK2repl_1892                  | 3.05  | 3.13  | 3.16  | 3.14  |
| Size_DblN_ascend_se_NON_-<br>TRAWL(3)_BLK2repl_2011                  | 3.81  | 3.97  | 3.99  | 4.12  |
| Size_DblN_descend_se_NON_-<br>TRAWL(3)_BLK2repl_1892                 | 3.17  | 3.15  | 3.19  | 3.13  |
| Size_DblN_descend_se_NON_-<br>TRAWL(3)_BLK2repl_2011                 | 2.32  | 1.88  | 1.86  | -9.27 |
| Size_DblN_end_logit_NON_-<br>TRAWL(3)_BLK2repl_1892                  | -2.29 | -1.84 | -1.75 | -1.81 |
| Size_DblN_end_logit_NON_-<br>TRAWL(3)_BLK2repl_2011                  | -0.63 | -0.16 | -0.08 | -0.08 |
| Size_DblN_peak_-<br>TRIENNIAL(7)_BLK3repl_1892                       | 17.36 | 17.26 | 17.03 | 17.31 |
| Size_DblN_ascend_se_-<br>TRIENNIAL(7)_BLK3repl_1892                  | 2.09  | 2.13  | 2.01  | 2.16  |
| Size_DblN_descend_se_-<br>TRIENNIAL(7)_BLK3repl_1892                 | 5.11  | 5.13  | 5.17  | 5.12  |

|                    |                                                     |              |              |              |              |
|--------------------|-----------------------------------------------------|--------------|--------------|--------------|--------------|
| Derived quantities | Size_DblN_end_logit_-<br>TRIENNIAL(7)_BLK3repl_1892 | -4.29        | -3.50        | -3.59        | -3.43        |
|                    | SB0                                                 | 5,647,660.00 | 2,746,610.00 | 2,714,590.00 | 2,662,210.00 |
|                    | SSB_2025                                            | 4,929,120.00 | 2,536,090.00 | 2,236,850.00 | 2,434,820.00 |
|                    | Bratio_2025                                         | 0.87         | 0.92         | 0.82         | 0.91         |
|                    | MSY_SPR                                             | 526.16       | 260.02       | 261.95       | 254.01       |
|                    | F_SPR                                               | 0.04         | 0.04         | 0.04         | 0.04         |

Table 29: Likelihood values and estimates of key parameters and derived quantities from each ageing error sensitivity.

| Label                | Type                 | Reference | No ageing<br>error | No trawl<br>AE | No<br>non-trawl<br>AE | No<br>midwater<br>AE | No<br>ASHOP<br>AE | No<br>WCG BTS<br>AE |
|----------------------|----------------------|-----------|--------------------|----------------|-----------------------|----------------------|-------------------|---------------------|
| Survey<br>likelihood |                      |           |                    |                |                       |                      |                   |                     |
|                      | ALL                  | -26.81    | -25.50             | -26.61         | -26.75                | -26.76               | -26.06            | -26.01              |
|                      | TRIENNIAL            | -7.76     | -6.92              | -7.60          | -7.78                 | -7.73                | -7.23             | -7.23               |
|                      | AK_SLOPE             | -0.53     | -0.47              | -0.52          | -0.51                 | -0.53                | -0.50             | -0.50               |
|                      | NW_SLOPE             | -1.03     | -1.01              | -1.03          | -1.02                 | -1.03                | -1.02             | -1.02               |
|                      | WCG BTS              | -17.49    | -17.10             | -17.46         | -17.44                | -17.48               | -17.31            | -17.26              |
| Lt<br>likelihood     |                      |           |                    |                |                       |                      |                   |                     |
|                      | ALL                  | 574.84    | 581.11             | 574.57         | 573.27                | 575.18               | 583.00            | 585.41              |
|                      | BOTTOM_TRAWL         | 79.10     | 80.00              | 78.97          | 78.76                 | 79.14                | 79.57             | 80.81               |
|                      | BOTTOM_TRAWL_DISCARD | 27.15     | 27.19              | 27.11          | 27.08                 | 27.14                | 27.08             | 27.14               |
|                      | NON_TRAWL            | 179.67    | 177.09             | 177.88         | 178.32                | 179.87               | 180.01            | 181.55              |
|                      | NON_TRAWL_DISCARD    | 46.62     | 46.60              | 46.44          | 46.49                 | 46.64                | 46.91             | 46.90               |
|                      | MIDWATER_TRAWL       | 58.95     | 58.49              | 58.74          | 58.49                 | 58.92                | 58.83             | 59.16               |
|                      | AT_SEA_HAKE          | 25.42     | 26.72              | 25.16          | 24.85                 | 25.50                | 26.86             | 26.56               |
|                      | TRIENNIAL            | 22.42     | 25.79              | 24.01          | 23.16                 | 22.42                | 25.73             | 25.14               |
|                      | AK_SLOPE             | 52.30     | 53.38              | 53.01          | 53.06                 | 52.29                | 53.04             | 53.29               |
|                      | NW_SLOPE             | 0.00      | 0.00               | 0.00           | 0.00                  | 0.00                 | 0.00              | 0.00                |
|                      | WCG BTS              | 83.19     | 85.85              | 83.25          | 83.05                 | 83.26                | 84.98             | 84.87               |
| Age<br>likelihood    |                      |           |                    |                |                       |                      |                   |                     |
|                      | ALL                  | 6,783.49  | 6,641.85           | 6,773.06       | 6,781.03              | 6,785.55             | 6,764.29          | 6,696.85            |
|                      | BOTTOM_TRAWL         | 464.25    | 450.06             | 447.40         | 463.74                | 464.91               | 469.79            | 460.45              |
|                      | BOTTOM_TRAWL_DISCARD | 0.00      | 0.00               | 0.00           | 0.00                  | 0.00                 | 0.00              | 0.00                |
|                      | NON_TRAWL            | 331.38    | 324.78             | 332.28         | 323.66                | 331.55               | 332.18            | 328.35              |
|                      | NON_TRAWL_DISCARD    | 0.00      | 0.00               | 0.00           | 0.00                  | 0.00                 | 0.00              | 0.00                |
|                      | MIDWATER_TRAWL       | 182.47    | 186.06             | 181.89         | 181.81                | 185.22               | 186.59            | 183.16              |
|                      | AT_SEA_HAKE          | 3,847.50  | 3,794.09           | 3,848.04       | 3,847.44              | 3,846.76             | 3,796.20          | 3,852.57            |
|                      | TRIENNIAL            | 0.00      | 0.00               | 0.00           | 0.00                  | 0.00                 | 0.00              | 0.00                |
|                      | AK_SLOPE             | 0.00      | 0.00               | 0.00           | 0.00                  | 0.00                 | 0.00              | 0.00                |
|                      | NW_SLOPE             | 0.00      | 0.00               | 0.00           | 0.00                  | 0.00                 | 0.00              | 0.00                |

| Parameters              | WCGBTS  | 1,957.88 | 1,886.85 | 1,963.45 | 1,964.38 | 1,957.11 | 1,979.54 | 1,872.32 |
|-------------------------|---------|----------|----------|----------|----------|----------|----------|----------|
| NatM_uniform_Fem_GP_1   | 0.04    | 0.04     | 0.04     | 0.04     | 0.04     | 0.04     | 0.04     | 0.04     |
| L_at_Amin_Fem_GP_1      | -3.10   | -2.30    | -2.66    | -2.52    | -3.14    | -3.29    | -1.98    |          |
| L_at_Amax_Fem_GP_1      | 60.08   | 59.94    | 60.10    | 60.07    | 60.08    | 59.77    | 60.27    |          |
| VonBert_K_Fem_GP_1      | 0.08    | 0.08     | 0.08     | 0.08     | 0.08     | 0.08     | 0.08     |          |
| CV_young_Fem_GP_1       | 0.05    | 0.16     | 0.05     | 0.05     | 0.05     | 0.12     | 0.14     |          |
| CV_old_Fem_GP_1         | 0.09    | 0.10     | 0.09     | 0.10     | 0.09     | 0.09     | 0.09     |          |
| NatM_uniform_Mal_GP_1   | 0.04    | 0.04     | 0.04     | 0.04     | 0.04     | 0.04     | 0.04     |          |
| L_at_Amin_Mal_GP_1      | -2.68   | -2.04    | -1.84    | -1.71    | -2.78    | -2.82    | -1.67    |          |
| L_at_Amax_Mal_GP_1      | 57.83   | 57.78    | 57.87    | 57.84    | 57.82    | 57.72    | 58.02    |          |
| VonBert_K_Mal_GP_1      | 0.08    | 0.08     | 0.08     | 0.08     | 0.08     | 0.09     | 0.08     |          |
| CV_young_Mal_GP_1       | 0.09    | 0.19     | 0.10     | 0.10     | 0.09     | 0.16     | 0.17     |          |
| CV_old_Mal_GP_1         | 0.08    | 0.09     | 0.09     | 0.09     | 0.08     | 0.09     | 0.08     |          |
| SR_LN(R0)               | 7.00    | 6.42     | 6.79     | 6.73     | 6.98     | 6.92     | 6.56     |          |
| LnQ_base_TRIENNIAL(7)   | -1.49   | -0.77    | -1.27    | -1.15    | -1.46    | -1.36    | -1.03    |          |
| LnQ_base_AK_SLOPE(8)    | -3.11   | -2.35    | -2.90    | -2.84    | -3.08    | -2.93    | -2.58    |          |
| LnQ_base_NW_SLOPE(9)    | -2.04   | -1.27    | -1.83    | -1.77    | -2.02    | -1.85    | -1.50    |          |
| LnQ_base_WCGBTS(10)     | -2.61   | -2.03    | -2.41    | -2.37    | -2.59    | -2.54    | -2.19    |          |
| LnQ_base_TRIENNIAL(7)_- | -2.27   | -1.75    | -2.10    | -2.06    | -2.25    | -2.20    | -1.88    |          |
| BLK3repl_1892           |         |          |          |          |          |          |          |          |
| Size_DblN_peak_BOTTOM_- | 49.34   | 49.64    | 49.49    | 49.54    | 49.33    | 49.32    | 49.46    |          |
| TRAWL(1)                |         |          |          |          |          |          |          |          |
| Size_DblN_top_logit_-   | -11.52  | -11.52   | -11.52   | -11.52   | -11.52   | -11.52   | -11.52   |          |
| BOTTOM_TRAWL(1)         |         |          |          |          |          |          |          |          |
| Size_DblN_ascend_se_-   | 4.57    | 4.49     | 4.55     | 4.54     | 4.56     | 4.53     | 4.51     |          |
| BOTTOM_TRAWL(1)         |         |          |          |          |          |          |          |          |
| Size_DblN_descend_se_-  | 2.95    | 2.97     | 2.95     | 2.92     | 2.95     | 2.97     | 2.95     |          |
| BOTTOM_TRAWL(1)         |         |          |          |          |          |          |          |          |
| Size_DblN_start_logit_- | -999.00 | -999.00  | -999.00  | -999.00  | -999.00  | -999.00  | -999.00  |          |
| BOTTOM_TRAWL(1)         |         |          |          |          |          |          |          |          |
| Size_DblN_end_logit_-   | -0.64   | -0.68    | -0.61    | -0.59    | -0.64    | -0.71    | -0.69    |          |
| BOTTOM_TRAWL(1)         |         |          |          |          |          |          |          |          |
| Size_DblN_peak_BOTTOM_- | 25.26   | 25.81    | 25.31    | 25.36    | 25.33    | 25.60    | 25.77    |          |
| TRAWL_DISCARD(2)        |         |          |          |          |          |          |          |          |

|                                                         |         |         |         |         |         |         |         |
|---------------------------------------------------------|---------|---------|---------|---------|---------|---------|---------|
| Size_DblN_top_logit_-<br>BOTTOM_TRAWL_-<br>DISCARD(2)   | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  |
| Size_DblN_ascend_se_-<br>BOTTOM_TRAWL_-<br>DISCARD(2)   | 4.89    | 4.57    | 4.84    | 4.86    | 4.89    | 4.70    | 4.58    |
| Size_DblN_descend_se_-<br>BOTTOM_TRAWL_-<br>DISCARD(2)  | 3.84    | 3.76    | 3.83    | 3.82    | 3.83    | 3.80    | 3.72    |
| Size_DblN_start_logit_-<br>BOTTOM_TRAWL_-<br>DISCARD(2) | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  |
| Size_DblN_end_logit_-<br>BOTTOM_TRAWL_-<br>DISCARD(2)   | -3.82   | -3.63   | -3.73   | -3.68   | -3.81   | -3.78   | -3.76   |
| Size_DblN_peak_NON_-<br>TRAWL(3)                        | 50.27   | 50.71   | 50.49   | 50.62   | 50.27   | 50.35   | 50.17   |
| Size_DblN_top_logit_NON_-<br>TRAWL(3)                   | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  |
| Size_DblN_ascend_se_NON_-<br>TRAWL(3)                   | 3.86    | 3.80    | 3.86    | 3.86    | 3.85    | 3.82    | 3.76    |
| Size_DblN_descend_se_NON_-<br>TRAWL(3)                  | 20.00   | 20.00   | 20.00   | 20.00   | 20.00   | 20.00   | 20.00   |
| Size_DblN_start_logit_NON_-<br>TRAWL(3)                 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 |
| Size_DblN_end_logit_NON_-<br>TRAWL(3)                   | 4.60    | 4.60    | 4.60    | 4.60    | 4.60    | 4.60    | 4.60    |
| Size_DblN_peak_NON_-<br>TRAWL_DISCARD(4)                | 49.94   | 50.03   | 50.07   | 50.06   | 49.92   | 49.78   | 50.01   |
| Size_DblN_top_logit_NON_-<br>TRAWL_DISCARD(4)           | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  |
| Size_DblN_ascend_se_NON_-<br>TRAWL_DISCARD(4)           | 3.75    | 3.72    | 3.76    | 3.75    | 3.74    | 3.70    | 3.74    |
| Size_DblN_descend_se_NON_-<br>TRAWL_DISCARD(4)          | 2.66    | 2.69    | 2.64    | 2.64    | 2.67    | 2.72    | 2.66    |
| Size_DblN_start_logit_NON_-<br>TRAWL_DISCARD(4)         | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 |



|                                               |         |         |         |         |         |         |         |
|-----------------------------------------------|---------|---------|---------|---------|---------|---------|---------|
| Size_DblN_end_logit_NON_-<br>TRAWL_DISCARD(4) | -0.19   | -0.25   | -0.16   | -0.14   | -0.20   | -0.29   | -0.26   |
| Size_DblN_peak_MIDWATER_-<br>TRAWL(5)         | 52.17   | 52.46   | 52.45   | 52.52   | 52.15   | 52.01   | 52.06   |
| Size_DblN_top_logit_-<br>MIDWATER_TRAWL(5)    | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  |
| Size_DblN_ascend_se_-<br>MIDWATER_TRAWL(5)    | 4.57    | 4.49    | 4.57    | 4.56    | 4.56    | 4.52    | 4.48    |
| Size_DblN_descend_se_-<br>MIDWATER_TRAWL(5)   | 20.00   | 20.00   | 20.00   | 20.00   | 20.00   | 20.00   | 20.00   |
| Size_DblN_start_logit_-<br>MIDWATER_TRAWL(5)  | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 |
| Size_DblN_end_logit_-<br>MIDWATER_TRAWL(5)    | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 |
| Size_DblN_peak_AT_SEA_-<br>HAKE(6)            | 49.71   | 49.74   | 49.87   | 49.92   | 49.69   | 49.46   | 49.65   |
| Size_DblN_top_logit_AT_-<br>SEA_HAKE(6)       | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  |
| Size_DblN_ascend_se_AT_-<br>SEA_HAKE(6)       | 3.59    | 3.55    | 3.61    | 3.61    | 3.59    | 3.55    | 3.55    |
| Size_DblN_descend_se_AT_-<br>SEA_HAKE(6)      | 20.00   | 20.00   | 20.00   | 20.00   | 20.00   | 20.00   | 20.00   |
| Size_DblN_start_logit_AT_-<br>SEA_HAKE(6)     | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 |
| Size_DblN_end_logit_AT_-<br>SEA_HAKE(6)       | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 |
| Size_DblN_peak_-<br>TRIENNIAL(7)              | 21.78   | 21.95   | 21.96   | 21.91   | 21.78   | 21.97   | 22.01   |
| Size_DblN_top_logit_-<br>TRIENNIAL(7)         | -8.60   | -8.60   | -8.60   | -8.60   | -8.60   | -8.60   | -8.60   |
| Size_DblN_ascend_se_-<br>TRIENNIAL(7)         | 3.52    | 3.61    | 3.60    | 3.58    | 3.52    | 3.61    | 3.61    |
| Size_DblN_descend_se_-<br>TRIENNIAL(7)        | 3.95    | 3.94    | 3.90    | 3.89    | 3.95    | 3.97    | 3.93    |
| Size_DblN_start_logit_-<br>TRIENNIAL(7)       | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 |
| Size_DblN_end_logit_-<br>TRIENNIAL(7)         | -2.58   | -2.56   | -2.55   | -2.60   | -2.59   | -2.62   | -2.49   |

|                            |         |         |         |         |         |         |         |
|----------------------------|---------|---------|---------|---------|---------|---------|---------|
| Size_DblN_peak_AK_-        | 37.29   | 36.69   | 37.31   | 36.84   | 37.26   | 36.73   | 37.29   |
| SLOPE(8)                   |         |         |         |         |         |         |         |
| Size_DblN_top_logit_AK_-   | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  |
| SLOPE(8)                   |         |         |         |         |         |         |         |
| Size_DblN_ascend_se_AK_-   | 4.94    | 4.85    | 4.97    | 4.97    | 4.94    | 4.84    | 4.90    |
| SLOPE(8)                   |         |         |         |         |         |         |         |
| Size_DblN_descend_se_AK_-  | 4.63    | 4.70    | 4.65    | 4.73    | 4.63    | 4.65    | 4.64    |
| SLOPE(8)                   |         |         |         |         |         |         |         |
| Size_DblN_start_logit_AK_- | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 |
| SLOPE(8)                   |         |         |         |         |         |         |         |
| Size_DblN_end_logit_AK_-   | -10.49  | -10.28  | -10.39  | -10.54  | -10.49  | -10.35  | -10.47  |
| SLOPE(8)                   |         |         |         |         |         |         |         |
| Size_DblN_peak_WCGBTS(10)  | 21.08   | 21.57   | 21.33   | 21.37   | 21.07   | 20.79   | 21.39   |
| Size_DblN_top_logit_-      | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  |
| WCGBTS(10)                 |         |         |         |         |         |         |         |
| Size_DblN_ascend_se_-      | 3.39    | 3.48    | 3.46    | 3.47    | 3.38    | 3.30    | 3.46    |
| WCGBTS(10)                 |         |         |         |         |         |         |         |
| Size_DblN_descend_se_-     | 4.65    | 4.44    | 4.55    | 4.55    | 4.65    | 4.74    | 4.49    |
| WCGBTS(10)                 |         |         |         |         |         |         |         |
| Size_DblN_start_logit_-    | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 |
| WCGBTS(10)                 |         |         |         |         |         |         |         |
| Size_DblN_end_logit_-      | -0.83   | -0.49   | -0.73   | -0.68   | -0.83   | -0.78   | -0.59   |
| WCGBTS(10)                 |         |         |         |         |         |         |         |
| Size_DblN_peak_BOTTOM_-    | 44.70   | 44.60   | 44.88   | 45.18   | 44.69   | 44.10   | 44.76   |
| TRAWL(1)_BLK4repl_1892     |         |         |         |         |         |         |         |
| Size_DblN_peak_BOTTOM_-    | 47.64   | 47.61   | 47.72   | 47.65   | 47.62   | 47.45   | 47.70   |
| TRAWL(1)_BLK4repl_2002     |         |         |         |         |         |         |         |
| Size_DblN_ascend_se_-      | 5.03    | 5.06    | 5.05    | 5.13    | 5.03    | 4.98    | 5.03    |
| BOTTOM_TRAWL(1)_-          |         |         |         |         |         |         |         |
| BLK4repl_1892              |         |         |         |         |         |         |         |
| Size_DblN_ascend_se_-      | 4.10    | 4.06    | 4.11    | 4.13    | 4.10    | 4.06    | 4.05    |
| BOTTOM_TRAWL(1)_-          |         |         |         |         |         |         |         |
| BLK4repl_2002              |         |         |         |         |         |         |         |
| Size_DblN_descend_se_-     | 2.65    | 2.64    | 2.56    | 2.41    | 2.66    | 2.88    | 2.59    |
| BOTTOM_TRAWL(1)_-          |         |         |         |         |         |         |         |
| BLK4repl_1892              |         |         |         |         |         |         |         |

|                                                                      |       |       |       |       |       |       |       |
|----------------------------------------------------------------------|-------|-------|-------|-------|-------|-------|-------|
| Size_DblN_descend_se_-<br>BOTTOM_TRAWL(1)_-<br>BLK4repl_2002         | 2.75  | 2.76  | 2.74  | 2.76  | 2.75  | 2.76  | 2.75  |
| Size_DblN_end_logit_-<br>BOTTOM_TRAWL(1)_-<br>BLK4repl_1892          | -1.59 | -1.36 | -1.46 | -1.38 | -1.59 | -1.68 | -1.45 |
| Size_DblN_end_logit_-<br>BOTTOM_TRAWL(1)_-<br>BLK4repl_2002          | -0.93 | -0.97 | -0.89 | -0.86 | -0.94 | -1.03 | -0.98 |
| Size_DblN_peak_BOTTOM_-<br>TRAWL_DISCARD(2)_-<br>BLK1repl_1892       | 47.67 | 47.78 | 47.75 | 47.61 | 47.64 | 47.52 | 48.03 |
| Size_DblN_ascend_se_-<br>BOTTOM_TRAWL_-<br>DISCARD(2)_BLK1repl_1892  | 6.32  | 6.33  | 6.29  | 6.23  | 6.32  | 6.35  | 6.39  |
| Size_DblN_descend_se_-<br>BOTTOM_TRAWL_-<br>DISCARD(2)_BLK1repl_1892 | 3.01  | 2.99  | 3.01  | 3.03  | 3.02  | 3.03  | 2.92  |
| Size_DblN_end_logit_-<br>BOTTOM_TRAWL_-<br>DISCARD(2)_BLK1repl_1892  | -1.70 | -1.72 | -1.66 | -1.66 | -1.71 | -1.79 | -1.70 |
| Size_DblN_peak_NON_-<br>TRAWL(3)_BLK2repl_1892                       | 46.77 | 46.74 | 46.83 | 46.81 | 46.76 | 46.61 | 46.81 |
| Size_DblN_peak_NON_-<br>TRAWL(3)_BLK2repl_2011                       | 49.49 | 49.57 | 49.57 | 49.57 | 49.48 | 49.40 | 49.57 |
| Size_DblN_ascend_se_NON_-<br>TRAWL(3)_BLK2repl_1892                  | 3.05  | 3.02  | 3.06  | 3.07  | 3.05  | 3.01  | 3.02  |
| Size_DblN_ascend_se_NON_-<br>TRAWL(3)_BLK2repl_2011                  | 3.81  | 3.79  | 3.80  | 3.79  | 3.81  | 3.79  | 3.81  |
| Size_DblN_descend_se_NON_-<br>TRAWL(3)_BLK2repl_1892                 | 3.17  | 3.19  | 3.18  | 3.19  | 3.17  | 3.18  | 3.16  |
| Size_DblN_descend_se_NON_-<br>TRAWL(3)_BLK2repl_2011                 | 2.32  | 2.39  | 2.31  | 2.29  | 2.33  | 2.40  | 2.35  |
| Size_DblN_end_logit_NON_-<br>TRAWL(3)_BLK2repl_1892                  | -2.29 | -2.29 | -2.25 | -2.22 | -2.29 | -2.35 | -2.31 |
| Size_DblN_end_logit_NON_-<br>TRAWL(3)_BLK2repl_2011                  | -0.63 | -0.66 | -0.60 | -0.59 | -0.63 | -0.71 | -0.67 |

|                       |                                                      |              |              |              |              |              |              |              |
|-----------------------|------------------------------------------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| Derived<br>quantities | Size_DblN_peak_-<br>TRIENNIAL(7)_BLK3repl_1892       | 17.36        | 17.69        | 17.69        | 17.68        | 17.33        | 17.71        | 17.72        |
|                       | Size_DblN_ascend_se_-<br>TRIENNIAL(7)_BLK3repl_1892  | 2.09         | 2.15         | 2.16         | 2.18         | 2.08         | 2.14         | 2.14         |
|                       | Size_DblN_descend_se_-<br>TRIENNIAL(7)_BLK3repl_1892 | 5.11         | 4.86         | 5.01         | 5.04         | 5.10         | 4.92         | 4.94         |
|                       | Size_DblN_end_logit_-<br>TRIENNIAL(7)_BLK3repl_1892  | -4.29        | -3.70        | -4.07        | -4.11        | -4.27        | -3.94        | -3.89        |
|                       | SB0                                                  | 5,647,660.00 | 3,078,080.00 | 4,528,730.00 | 4,242,460.00 | 5,550,590.00 | 5,254,360.00 | 3,596,610.00 |
|                       | SSB_2025                                             | 4,929,120.00 | 2,095,990.00 | 3,728,710.00 | 3,428,260.00 | 4,817,120.00 | 4,392,990.00 | 2,696,790.00 |
|                       | Bratio_2025                                          | 0.87         | 0.68         | 0.82         | 0.81         | 0.87         | 0.84         | 0.75         |
|                       | MSY_SPR                                              | 526.16       | 299.77       | 425.16       | 399.88       | 517.04       | 501.96       | 340.37       |
|                       | F_SPR                                                | 0.04         | 0.04         | 0.04         | 0.04         | 0.04         | 0.04         | 0.04         |
|                       |                                                      |              |              |              |              |              |              |              |
|                       |                                                      |              |              |              |              |              |              |              |
|                       |                                                      |              |              |              |              |              |              |              |

Table 30: Likelihood values and estimates of key parameters and derived quantities from each life history parameter sensitivity.

| Label                | Type                  | Reference | Est female M | Fixed_M_-<br>2013 | Lorenzen M | Fixed growth<br>ext | Prop fecund | Fxn mat age | Fxn mat<br>Bsp | Fxn mat<br>Reye |
|----------------------|-----------------------|-----------|--------------|-------------------|------------|---------------------|-------------|-------------|----------------|-----------------|
| AIC                  | AIC                   | 15,093.92 | 15,086.58    | 15,103.72         | 15,123.46  | 15,580.08           | 15,093.90   | 15,094.00   | 15,093.96      | 15,093.90       |
|                      | deltaAIC              | 0.00      | -7.34        | 9.80              | 29.54      | 486.16              | -0.02       | 0.08        | 0.04           | -0.02           |
| Survey<br>likelihood | ALL                   | -26.81    | -27.07       | -26.66            | -26.75     | -27.02              | -26.81      | -26.82      | -26.81         | -26.81          |
|                      | TRIENNIAL             | -7.76     | -8.05        | -7.64             | -7.66      | -7.98               | -7.76       | -7.76       | -7.76          | -7.76           |
|                      | AK_SLOPE              | -0.53     | -0.54        | -0.52             | -0.52      | -0.53               | -0.53       | -0.53       | -0.53          | -0.53           |
|                      | NW_SLOPE              | -1.03     | -1.03        | -1.03             | -1.03      | -1.03               | -1.03       | -1.03       | -1.03          | -1.03           |
|                      | WCGBTS                | -17.49    | -17.45       | -17.47            | -17.54     | -17.48              | -17.49      | -17.49      | -17.49         | -17.49          |
| Survey<br>lambda     |                       |           |              |                   |            |                     |             |             |                |                 |
| Lt likelihood        | ALL                   |           |              |                   |            |                     |             |             |                |                 |
|                      | ALL                   | 574.84    | 573.20       | 585.04            | 573.83     | 605.80              | 574.83      | 574.86      | 574.83         | 574.83          |
|                      | BOTTOM_TRAWL          | 79.10     | 79.75        | 78.31             | 79.67      | 79.10               | 79.10       | 79.10       | 79.10          | 79.10           |
|                      | BOTTOM_TRAWL_DISCARD  | 27.15     | 27.18        | 27.24             | 27.09      | 28.35               | 27.15       | 27.15       | 27.15          | 27.15           |
|                      | NON_TRAWL             | 179.67    | 178.16       | 188.34            | 179.62     | 194.51              | 179.66      | 179.68      | 179.68         | 179.66          |
|                      | NON_TRAWL_DISCARD     | 46.62     | 46.71        | 47.17             | 46.47      | 48.93               | 46.62       | 46.62       | 46.62          | 46.62           |
|                      | MIDWATER_TRAWL        | 58.95     | 58.49        | 59.98             | 58.82      | 59.11               | 58.95       | 58.96       | 58.96          | 58.95           |
|                      | AT_SEA_HAKE           | 25.42     | 25.03        | 25.98             | 24.58      | 23.50               | 25.42       | 25.43       | 25.43          | 25.42           |
|                      | TRIENNIAL             | 22.42     | 22.40        | 22.40             | 22.38      | 25.37               | 22.42       | 22.42       | 22.42          | 22.42           |
|                      | AK_SLOPE              | 52.30     | 52.19        | 52.92             | 51.86      | 52.86               | 52.30       | 52.30       | 52.30          | 52.30           |
|                      | NW_SLOPE              | 0.00      | 0.00         | 0.00              | 0.00       | 0.00                | 0.00        | 0.00        | 0.00           | 0.00            |
|                      | WCGBTS                | 83.19     | 83.29        | 82.70             | 83.36      | 90.48               | 83.19       | 83.19       | 83.19          | 83.19           |
|                      |                       |           |              |                   |            |                     |             |             |                |                 |
| Age<br>likelihood    | ALL                   | 6,783.49  | 6,781.98     | 6,779.34          | 6,795.59   | 6,981.03            | 6,783.49    | 6,783.49    | 6,783.49       | 6,783.49        |
|                      | BOTTOM_TRAWL          | 464.25    | 463.72       | 464.13            | 466.54     | 462.65              | 464.25      | 464.26      | 464.25         | 464.25          |
|                      | BOTTOM_TRAWL_DISCARD  | 0.00      | 0.00         | 0.00              | 0.00       | 0.00                | 0.00        | 0.00        | 0.00           | 0.00            |
|                      | NON_TRAWL             | 331.38    | 330.63       | 331.05            | 332.27     | 331.61              | 331.38      | 331.38      | 331.38         | 331.38          |
|                      | NON_TRAWL_DISCARD     | 0.00      | 0.00         | 0.00              | 0.00       | 0.00                | 0.00        | 0.00        | 0.00           | 0.00            |
|                      | MIDWATER_TRAWL        | 182.47    | 182.42       | 182.21            | 183.34     | 183.76              | 182.47      | 182.47      | 182.47         | 182.47          |
|                      | AT_SEA_HAKE           | 3,847.50  | 3,848.16     | 3,843.86          | 3,853.80   | 3,862.56            | 3,847.50    | 3,847.49    | 3,847.50       | 3,847.50        |
|                      | TRIENNIAL             | 0.00      | 0.00         | 0.00              | 0.00       | 0.00                | 0.00        | 0.00        | 0.00           | 0.00            |
|                      | AK_SLOPE              | 0.00      | 0.00         | 0.00              | 0.00       | 0.00                | 0.00        | 0.00        | 0.00           | 0.00            |
|                      | NW_SLOPE              | 0.00      | 0.00         | 0.00              | 0.00       | 0.00                | 0.00        | 0.00        | 0.00           | 0.00            |
|                      | WCGBTS                | 1,957.88  | 1,957.05     | 1,958.10          | 1,959.64   | 2,140.44            | 1,957.88    | 1,957.89    | 1,957.88       | 1,957.88        |
|                      |                       |           |              |                   |            |                     |             |             |                |                 |
|                      |                       |           |              |                   |            |                     |             |             |                |                 |
| Age lambda           |                       |           |              |                   |            |                     |             |             |                |                 |
| Parameters           | ALL                   |           |              |                   |            |                     |             |             |                |                 |
|                      | NatM_uniform_Fem_GP_1 | 0.04      | 0.04         | 0.04              |            | 0.04                | 0.04        | 0.04        | 0.04           | 0.04            |
|                      | L_at_Amin_Fem_GP_1    | -3.10     | -3.03        | -3.19             | -2.64      | 0.00                | -3.10       | -3.10       | -3.10          | -3.10           |
|                      | L_at_Amax_Fem_GP_1    | 60.08     | 60.08        | 59.84             | 60.04      | 59.03               | 60.08       | 60.08       | 60.08          | 60.08           |
|                      | VonBert_K_Fem_GP_1    | 0.08      | 0.08         | 0.08              | 0.08       | 0.07                | 0.08        | 0.08        | 0.08           | 0.08            |
|                      | CV_young_Fem_GP_1     | 0.05      | 0.05         | 0.05              | 0.05       | 0.13                | 0.05        | 0.05        | 0.05           | 0.05            |
|                      | CV_old_Fem_GP_1       | 0.09      | 0.09         | 0.09              | 0.09       | 0.09                | 0.09        | 0.09        | 0.09           | 0.09            |
|                      | NatM_uniform_Mal_GP_1 | 0.04      | 0.04         | 0.04              |            | 0.04                | 0.04        | 0.04        | 0.04           | 0.04            |
|                      | L_at_Amin_Mal_GP_1    | -2.68     | -2.67        | -2.38             | -5.11      | 1.71                | -2.68       | -2.68       | -2.68          | -2.68           |
|                      | L_at_Amax_Mal_GP_1    | 57.83     | 57.76        | 57.87             | 57.52      | 56.69               | 57.83       | 57.83       | 57.83          | 57.83           |
|                      | VonBert_K_Mal_GP_1    | 0.08      | 0.08         | 0.08              | 0.09       | 0.08                | 0.08        | 0.08        | 0.08           | 0.08            |
|                      |                       |           |              |                   |            |                     |             |             |                |                 |
|                      |                       |           |              |                   |            |                     |             |             |                |                 |

|                                                |         |         |         |         |         |         |         |         |         |         |
|------------------------------------------------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| CV_young_Mal_GP_1                              | 0.09    | 0.09    | 0.09    | 0.09    | 0.19    | 0.09    | 0.09    | 0.09    | 0.09    | 0.09    |
| CV_old_Mal_GP_1                                | 0.08    | 0.08    | 0.09    | 0.08    | 0.08    | 0.08    | 0.08    | 0.08    | 0.08    | 0.08    |
| SR_LN(R0)                                      | 7.00    | 10.90   | 6.86    | 7.39    | 6.78    | 6.99    | 7.00    | 7.00    | 7.00    | 6.99    |
| SR_BH_steep                                    | 0.72    | 0.72    | 0.72    | 0.72    | 0.72    | 0.72    | 0.72    | 0.72    | 0.72    | 0.72    |
| LnQ_base_TRIENNIAL(7)                          | -1.49   | -5.31   | -1.28   | -1.22   | -1.57   | -1.48   | -1.49   | -1.49   | -1.49   | -1.48   |
| LnQ_base_AK_SLOPE(8)                           | -3.11   | -6.89   | -2.90   | -2.73   | -3.12   | -3.11   | -3.11   | -3.11   | -3.11   | -3.11   |
| LnQ_base_NW_SLOPE(9)                           | -2.04   | -5.83   | -1.83   | -1.66   | -2.06   | -2.06   | -2.05   | -2.05   | -2.05   | -2.04   |
| LnQ_base_WCGBTS(10)                            | -2.61   | -6.46   | -2.41   | -2.28   | -2.57   | -2.61   | -2.61   | -2.61   | -2.61   | -2.61   |
| LnQ_base_TRIENNIAL(7)_BLK3repl_-1892           | -2.27   | -6.00   | -2.09   | -2.11   | -2.27   | -2.27   | -2.27   | -2.27   | -2.27   | -2.27   |
| Size_DblN_peak_BOTTOM_TRAWL(1)                 | 49.34   | 49.42   | 49.49   | 49.30   | 50.17   | 49.34   | 49.34   | 49.34   | 49.34   | 49.34   |
| Size_DblN_top_logit_BOTTOM_-TRAWL(1)           | -11.52  | -11.52  | -11.52  | -11.52  | -11.52  | -11.52  | -11.52  | -11.52  | -11.52  | -11.52  |
| Size_DblN_ascend_se_BOTTOM_-TRAWL(1)           | 4.57    | 4.56    | 4.56    | 4.55    | 4.65    | 4.57    | 4.57    | 4.57    | 4.57    | 4.57    |
| Size_DblN_descend_se_BOTTOM_-TRAWL(1)          | 2.95    | 2.93    | 2.86    | 2.93    | 1.86    | 2.95    | 2.95    | 2.95    | 2.95    | 2.95    |
| Size_DblN_start_logit_BOTTOM_-TRAWL(1)         | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 |
| Size_DblN_end_logit_BOTTOM_-TRAWL(1)           | -0.64   | -0.60   | -0.53   | -0.61   | 0.28    | -0.64   | -0.64   | -0.64   | -0.64   | -0.64   |
| Size_DblN_peak_BOTTOM_TRAWL_-DISCARD(2)        | 25.26   | 25.39   | 25.38   | 25.72   | 26.10   | 25.26   | 25.26   | 25.26   | 25.26   | 25.26   |
| Size_DblN_top_logit_BOTTOM_-TRAWL_DISCARD(2)   | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  |
| Size_DblN_ascend_se_BOTTOM_-TRAWL_DISCARD(2)   | 4.89    | 4.88    | 4.88    | 4.75    | 7.87    | 4.89    | 4.89    | 4.89    | 4.89    | 4.89    |
| Size_DblN_descend_se_BOTTOM_-TRAWL_DISCARD(2)  | 3.84    | 3.82    | 3.82    | 3.79    | 3.70    | 3.84    | 3.84    | 3.84    | 3.84    | 3.84    |
| Size_DblN_start_logit_BOTTOM_-TRAWL_DISCARD(2) | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  |
| Size_DblN_end_logit_BOTTOM_-TRAWL_DISCARD(2)   | -3.82   | -3.74   | -3.74   | -3.69   | -3.47   | -3.82   | -3.82   | -3.82   | -3.82   | -3.82   |
| Size_DblN_peak_NON_TRAWL(3)                    | 50.27   | 50.39   | 50.56   | 50.22   | 51.68   | 50.27   | 50.27   | 50.27   | 50.27   | 50.27   |
| Size_DblN_top_logit_NON_TRAWL(3)               | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  |
| Size_DblN_ascend_se_NON_TRAWL(3)               | 3.86    | 3.86    | 3.87    | 3.86    | 4.02    | 3.86    | 3.86    | 3.86    | 3.86    | 3.86    |
| Size_DblN_descend_se_NON_-TRAWL(3)             | 20.00   | 20.00   | 20.00   | 20.00   | 20.00   | 20.00   | 20.00   | 20.00   | 20.00   | 20.00   |
| Size_DblN_start_logit_NON_TRAWL(3)             | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 |
| Size_DblN_end_logit_NON_TRAWL(3)               | 4.60    | 4.60    | 4.60    | 4.60    | 4.60    | 4.60    | 4.60    | 4.60    | 4.60    | 4.60    |
| Size_DblN_peak_NON_TRAWL_-DISCARD(4)           | 49.94   | 49.99   | 50.05   | 49.94   | 51.10   | 49.95   | 49.94   | 49.94   | 49.94   | 49.95   |
| Size_DblN_top_logit_NON_TRAWL_-DISCARD(4)      | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  |
| Size_DblN_ascend_se_NON_TRAWL_-DISCARD(4)      | 3.75    | 3.75    | 3.75    | 3.74    | 3.95    | 3.75    | 3.75    | 3.75    | 3.75    | 3.75    |
| Size_DblN_descend_se_NON_TRAWL_-DISCARD(4)     | 2.66    | 2.64    | 2.60    | 2.64    | 1.71    | 2.66    | 2.66    | 2.66    | 2.66    | 2.66    |
| Size_DblN_start_logit_NON_TRAWL_-DISCARD(4)    | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 |
| Size_DblN_end_logit_NON_TRAWL_-DISCARD(4)      | -0.19   | -0.15   | -0.10   | -0.14   | 0.69    | -0.19   | -0.19   | -0.19   | -0.19   | -0.19   |
| Size_DblN_peak_MIDWATER_-TRAWL(5)              | 52.17   | 52.35   | 52.63   | 52.16   | 55.82   | 52.17   | 52.17   | 52.17   | 52.17   | 52.17   |
| Size_DblN_top_logit_MIDWATER_-TRAWL(5)         | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  |
| Size_DblN_ascend_se_MIDWATER_-TRAWL(5)         | 4.57    | 4.57    | 4.59    | 4.56    | 4.89    | 4.57    | 4.57    | 4.57    | 4.57    | 4.57    |

|                                                                 |         |         |         |         |         |         |         |         |         |         |
|-----------------------------------------------------------------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| Size_DbIN_descend_se_MIDWATER_-<br>TRAWL(5)                     | 20.00   | 20.00   | 20.00   | 20.00   | 20.00   | 20.00   | 20.00   | 20.00   | 20.00   | 20.00   |
| Size_DbIN_start_logit_MIDWATER_-<br>TRAWL(5)                    | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 |
| Size_DbIN_end_logit_MIDWATER_-<br>TRAWL(5)                      | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 |
| Size_DbIN_peak_AT_SEA_HAKE(6)                                   | 49.71   | 49.78   | 49.92   | 49.80   | 51.50   | 49.71   | 49.71   | 49.71   | 49.71   | 49.71   |
| Size_DbIN_top_logit_AT_SEA_-<br>HAKE(6)                         | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  |
| Size_DbIN_ascend_se_AT_SEA_-<br>HAKE(6)                         | 3.59    | 3.60    | 3.61    | 3.60    | 3.84    | 3.59    | 3.59    | 3.59    | 3.59    | 3.59    |
| Size_DbIN_descend_se_AT_SEA_-<br>HAKE(6)                        | 20.00   | 20.00   | 20.00   | 20.00   | 20.00   | 20.00   | 20.00   | 20.00   | 20.00   | 20.00   |
| Size_DbIN_start_logit_AT_SEA_-<br>HAKE(6)                       | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 |
| Size_DbIN_end_logit_AT_SEA_-<br>HAKE(6)                         | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 |
| Size_DbIN_peak_TRIENNIAL(7)                                     | 21.78   | 21.89   | 21.82   | 22.14   | 22.70   | 21.78   | 21.78   | 21.78   | 21.78   | 21.78   |
| Size_DbIN_top_logit_TRIENNIAL(7)                                | -8.60   | -8.60   | -8.60   | -8.60   | -8.60   | -8.60   | -8.60   | -8.60   | -8.60   | -8.60   |
| Size_DbIN_ascend_se_TRIENNIAL(7)                                | 3.52    | 3.54    | 3.54    | 3.52    | 4.09    | 3.52    | 3.52    | 3.52    | 3.52    | 3.52    |
| Size_DbIN_descend_se_TRIENNIAL(7)                               | 3.95    | 3.95    | 3.93    | 3.90    | 3.72    | 3.95    | 3.95    | 3.95    | 3.95    | 3.95    |
| Size_DbIN_start_logit_TRIENNIAL(7)                              | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 |
| Size_DbIN_end_logit_TRIENNIAL(7)                                | -2.58   | -2.54   | -2.55   | -2.37   | -2.23   | -2.58   | -2.58   | -2.58   | -2.58   | -2.58   |
| Size_DbIN_peak_AK_SLOPE(8)                                      | 37.29   | 37.34   | 37.34   | 37.79   | 37.43   | 37.29   | 37.29   | 37.29   | 37.29   | 37.29   |
| Size_DbIN_top_logit_AK_SLOPE(8)                                 | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  |
| Size_DbIN_ascend_se_AK_SLOPE(8)                                 | 4.94    | 4.92    | 4.95    | 4.92    | 4.97    | 4.94    | 4.94    | 4.94    | 4.94    | 4.94    |
| Size_DbIN_descend_se_AK_SLOPE(8)                                | 4.63    | 4.62    | 4.64    | 4.61    | 4.71    | 4.63    | 4.63    | 4.63    | 4.63    | 4.63    |
| Size_DbIN_start_logit_AK_SLOPE(8)                               | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 |
| Size_DbIN_end_logit_AK_SLOPE(8)                                 | -10.49  | -10.52  | -10.15  | -10.49  | -10.60  | -10.48  | -10.48  | -10.48  | -10.48  | -10.48  |
| Size_DbIN_peak_WCGBTS(10)                                       | 21.08   | 21.29   | 21.34   | 21.91   | 22.02   | 21.09   | 21.08   | 21.08   | 21.08   | 21.09   |
| Size_DbIN_top_logit_WCGBTS(10)                                  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  |
| Size_DbIN_ascend_se_WCGBTS(10)                                  | 3.39    | 3.43    | 3.45    | 3.50    | 4.28    | 3.39    | 3.39    | 3.39    | 3.39    | 3.39    |
| Size_DbIN_descend_se_WCGBTS(10)                                 | 4.65    | 4.60    | 4.54    | 4.55    | 4.10    | 4.65    | 4.65    | 4.65    | 4.65    | 4.65    |
| Size_DbIN_start_logit_WCGBTS(10)                                | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 |
| Size_DbIN_end_logit_WCGBTS(10)                                  | -0.83   | -0.73   | -0.74   | -0.62   | -0.57   | -0.83   | -0.83   | -0.83   | -0.83   | -0.83   |
| Size_DbIN_peak_BOTTOM_-<br>TRAWL(1)_BLK4repl_1892               | 44.70   | 44.39   | 44.73   | 45.23   | 45.42   | 44.70   | 44.70   | 44.70   | 44.70   | 44.70   |
| Size_DbIN_peak_BOTTOM_-<br>TRAWL(1)_BLK4repl_2002               | 47.64   | 47.64   | 47.73   | 47.71   | 48.25   | 47.64   | 47.64   | 47.64   | 47.64   | 47.64   |
| Size_DbIN_ascend_se_BOTTOM_-<br>TRAWL(1)_BLK4repl_1892          | 5.03    | 4.98    | 5.03    | 5.03    | 5.08    | 5.03    | 5.03    | 5.03    | 5.03    | 5.03    |
| Size_DbIN_ascend_se_BOTTOM_-<br>TRAWL(1)_BLK4repl_2002          | 4.10    | 4.09    | 4.11    | 4.10    | 4.11    | 4.10    | 4.10    | 4.10    | 4.10    | 4.10    |
| Size_DbIN_descend_se_BOTTOM_-<br>TRAWL(1)_BLK4repl_1892         | 2.65    | 2.84    | 2.67    | 2.24    | 2.13    | 2.65    | 2.65    | 2.65    | 2.65    | 2.65    |
| Size_DbIN_descend_se_BOTTOM_-<br>TRAWL(1)_BLK4repl_2002         | 2.75    | 2.73    | 2.68    | 2.70    | 1.92    | 2.75    | 2.75    | 2.75    | 2.75    | 2.75    |
| Size_DbIN_end_logit_BOTTOM_-<br>TRAWL(1)_BLK4repl_1892          | -1.59   | -1.79   | -1.59   | -1.14   | -1.11   | -1.59   | -1.59   | -1.59   | -1.59   | -1.59   |
| Size_DbIN_end_logit_BOTTOM_-<br>TRAWL(1)_BLK4repl_2002          | -0.93   | -0.92   | -0.86   | -0.86   | -0.33   | -0.93   | -0.94   | -0.93   | -0.93   | -0.93   |
| Size_DbIN_peak_BOTTOM_TRAWL_-<br>DISCARD(2)_BLK1repl_1892       | 47.67   | 47.74   | 47.79   | 47.77   | 48.58   | 47.67   | 47.67   | 47.67   | 47.67   | 47.67   |
| Size_DbIN_ascend_se_BOTTOM_-<br>TRAWL_DISCARD(2)_BLK1repl_1892  | 6.32    | 6.25    | 6.30    | 6.16    | 6.34    | 6.32    | 6.32    | 6.32    | 6.32    | 6.32    |
| Size_DbIN_descend_se_BOTTOM_-<br>TRAWL_DISCARD(2)_BLK1repl_1892 | 3.01    | 2.98    | 2.98    | 2.97    | 2.63    | 3.01    | 3.01    | 3.01    | 3.01    | 3.01    |

|                    |                                    |              |                |              |              |              |           |              |              |              |
|--------------------|------------------------------------|--------------|----------------|--------------|--------------|--------------|-----------|--------------|--------------|--------------|
| Derived quantities | Size_DblN_end_logit_BOTTOM_-       | -1.70        | -1.68          | -1.65        | -1.63        | -1.08        | -1.70     | -1.70        | -1.70        | -1.70        |
|                    | TRAWL_DISCARD(2)_BLK1repl_1892     |              |                |              |              |              |           |              |              |              |
|                    | Size_DblN_peak_NON_TRAWL(3)_-      | 46.77        | 46.76          | 46.82        | 46.84        | 47.08        | 46.77     | 46.77        | 46.77        | 46.77        |
|                    | BLK2repl_1892                      |              |                |              |              |              |           |              |              |              |
|                    | Size_DblN_peak_NON_TRAWL(3)_-      | 49.49        | 49.54          | 49.60        | 49.49        | 51.00        | 49.49     | 49.49        | 49.49        | 49.49        |
|                    | BLK2repl_2011                      |              |                |              |              |              |           |              |              |              |
|                    | Size_DblN_ascend_se_NON_-          | 3.05         | 3.04           | 3.05         | 3.05         | 3.08         | 3.05      | 3.05         | 3.05         | 3.05         |
|                    | TRAWL(3)_BLK2repl_1892             |              |                |              |              |              |           |              |              |              |
|                    | Size_DblN_ascend_se_NON_-          | 3.81         | 3.81           | 3.81         | 3.80         | 4.02         | 3.81      | 3.81         | 3.81         | 3.81         |
|                    | TRAWL(3)_BLK2repl_2011             |              |                |              |              |              |           |              |              |              |
|                    | Size_DblN_descend_se_NON_-         | 3.17         | 3.18           | 3.16         | 3.16         | 3.01         | 3.17      | 3.17         | 3.17         | 3.17         |
|                    | TRAWL(3)_BLK2repl_1892             |              |                |              |              |              |           |              |              |              |
|                    | Size_DblN_descend_se_NON_-         | 2.32         | 2.29           | 2.21         | 2.29         | -13.02       | 2.32      | 2.32         | 2.32         | 2.32         |
|                    | TRAWL(3)_BLK2repl_2011             |              |                |              |              |              |           |              |              |              |
|                    | Size_DblN_end_logit_NON_-          | -2.29        | -2.29          | -2.23        | -2.19        | -1.75        | -2.28     | -2.29        | -2.29        | -2.28        |
|                    | TRAWL(3)_BLK2repl_1892             |              |                |              |              |              |           |              |              |              |
|                    | Size_DblN_end_logit_NON_-          | -0.63        | -0.59          | -0.54        | -0.59        | 0.03         | -0.63     | -0.63        | -0.63        | -0.63        |
|                    | TRAWL(3)_BLK2repl_2011             |              |                |              |              |              |           |              |              |              |
|                    | Size_DblN_peak_TRIENNIAL(7)_-      | 17.36        | 17.50          | 17.40        | 17.66        | 17.19        | 17.36     | 17.36        | 17.36        | 17.36        |
|                    | BLK3repl_1892                      |              |                |              |              |              |           |              |              |              |
|                    | Size_DblN_ascend_se_TRIENNIAL(7)_- | 2.09         | 2.13           | 2.10         | 2.16         | 2.02         | 2.09      | 2.09         | 2.09         | 2.09         |
|                    | BLK3repl_1892                      |              |                |              |              |              |           |              |              |              |
|                    | Size_DblN_descend_se_-             | 5.11         | 5.16           | 5.08         | 5.19         | 5.12         | 5.11      | 5.11         | 5.11         | 5.11         |
|                    | TRIENNIAL(7)_BLK3repl_1892         |              |                |              |              |              |           |              |              |              |
|                    | Size_DblN_end_logit_TRIENNIAL(7)_- | -4.29        | -4.46          | -4.20        | -4.05        | -4.04        | -4.29     | -4.29        | -4.29        | -4.29        |
|                    | BLK3repl_1892                      |              |                |              |              |              |           |              |              |              |
|                    | NatM_Lorenzen_Fem_GP_1             |              |                |              | 0.04         |              |           |              |              |              |
|                    | NatM_Lorenzen_Mal_GP_1             |              |                |              | 0.04         |              |           |              |              |              |
|                    | Eggs/kg_inter_Fem_GP_1             |              |                |              |              |              | 1.00      |              |              |              |
|                    | Eggs/kg_slope_wt_Fem_GP_1          |              |                |              |              |              | 0.00      |              |              |              |
| Derived quantities | SB0                                | 5,647,660.00 | 228,262,000.00 | 4,203,990.00 | 3,672,360.00 | 4,001,980.00 | 18,434.60 | 4,951,840.00 | 4,889,410.00 | 6,176,980.00 |
|                    | SSB_2025                           | 4,929,120.00 | 220,424,000.00 | 3,416,730.00 | 3,065,150.00 | 3,700,840.00 | 16,181.90 | 4,282,080.00 | 4,215,800.00 | 5,425,420.00 |
|                    | Bratio_2025                        | 0.87         | 0.97           | 0.81         | 0.83         | 0.92         | 0.88      | 0.86         | 0.86         | 0.88         |
|                    | MSY_SPR                            | 526.16       | 23,228.70      | 450.29       | 313.36       | 418.21       | 539.52    | 484.71       | 501.82       | 540.14       |
|                    | F_SPR                              | 0.04         | 0.05           | 0.04         | 0.04         | 0.04         | 0.04      | 0.03         | 0.04         | 0.04         |



Table 31: Likelihood values and estimates of key parameters and derived quantities from each recruitment and selectivity sensitivity.

| Label                | Type                 | Reference | No recdevs | 1940<br>recdevs | 1980<br>recdevs | Logistic<br>fisheries | Logistic<br>btrawl | Logistic<br>nontrawl |
|----------------------|----------------------|-----------|------------|-----------------|-----------------|-----------------------|--------------------|----------------------|
| AIC                  | AIC                  | 15,093.92 | 14,899.82  | 15,038.28       | 14,966.68       | 15,393.56             | 15,185.10          | 15,357.90            |
|                      | deltaAIC             | 0.00      | -194.10    | -55.64          | -127.24         | 299.64                | 91.18              | 263.98               |
| Survey<br>likelihood | ALL                  | -26.81    | -26.63     | -26.73          | -27.06          | -27.18                | -26.73             | -26.74               |
|                      | TRIENNIAL            | -7.76     | -7.45      | -7.67           | -8.00           | -8.22                 | -7.75              | -7.80                |
|                      | AK_SLOPE             | -0.53     | -0.52      | -0.52           | -0.55           | -0.54                 | -0.52              | -0.52                |
|                      | NW_SLOPE             | -1.03     | -1.02      | -1.03           | -1.04           | -1.03                 | -1.03              | -1.02                |
|                      | WCGBTS               | -17.49    | -17.63     | -17.50          | -17.47          | -17.40                | -17.43             | -17.40               |
| Lt<br>likelihood     | ALL                  | 574.84    | 590.58     | 574.64          | 576.89          | 702.16                | 617.53             | 683.72               |
|                      | BOTTOM_TRAWL         | 79.10     | 80.13      | 83.67           | 83.29           | 106.01                | 123.64             | 82.59                |
|                      | BOTTOM_TRAWL_DISCARD | 27.15     | 28.13      | 28.97           | 28.96           | 27.68                 | 27.31              | 27.65                |
|                      | NON_TRAWL            | 179.67    | 179.75     | 172.00          | 173.42          | 276.01                | 181.88             | 282.83               |
|                      | NON_TRAWL_DISCARD    | 46.62     | 47.02      | 46.19           | 46.24           | 49.95                 | 47.56              | 49.48                |
|                      | MIDWATER_TRAWL       | 58.95     | 60.49      | 57.37           | 57.87           | 58.59                 | 57.37              | 58.36                |
|                      | AT_SEA_HAKE          | 25.42     | 27.06      | 30.10           | 30.71           | 24.94                 | 22.45              | 24.37                |
|                      | TRIENNIAL            | 22.42     | 28.29      | 22.09           | 22.09           | 21.73                 | 21.97              | 21.72                |
|                      | AK_SLOPE             | 52.30     | 53.05      | 52.36           | 52.38           | 52.57                 | 52.39              | 52.54                |
|                      | NW_SLOPE             | 0.00      | 0.00       | 0.00            | 0.00            | 0.00                  | 0.00               | 0.00                 |
|                      | WCGBTS               | 83.19     | 86.66      | 81.91           | 81.93           | 84.69                 | 82.95              | 84.17                |
| Age<br>likelihood    | ALL                  | 6,783.49  | 6,817.81   | 6,803.70        | 6,808.00        | 6,818.06              | 6,793.11           | 6,812.24             |
|                      | BOTTOM_TRAWL         | 464.25    | 464.97     | 464.35          | 464.17          | 462.87                | 463.82             | 462.83               |
|                      | BOTTOM_TRAWL_DISCARD | 0.00      | 0.00       | 0.00            | 0.00            | 0.00                  | 0.00               | 0.00                 |
|                      | NON_TRAWL            | 331.38    | 335.01     | 331.04          | 331.30          | 333.46                | 331.62             | 333.10               |
|                      | NON_TRAWL_DISCARD    | 0.00      | 0.00       | 0.00            | 0.00            | 0.00                  | 0.00               | 0.00                 |
|                      | MIDWATER_TRAWL       | 182.47    | 182.05     | 181.83          | 181.59          | 180.88                | 181.84             | 180.98               |
|                      | AT_SEA_HAKE          | 3,847.50  | 3,860.07   | 3,845.56        | 3,849.93        | 3,884.81              | 3,858.47           | 3,878.72             |

|            |                         |          |          |          |          |          |          |          |
|------------|-------------------------|----------|----------|----------|----------|----------|----------|----------|
| Parameters | TRIENNIAL               | 0.00     | 0.00     | 0.00     | 0.00     | 0.00     | 0.00     | 0.00     |
|            | AK_SLOPE                | 0.00     | 0.00     | 0.00     | 0.00     | 0.00     | 0.00     | 0.00     |
|            | NW_SLOPE                | 0.00     | 0.00     | 0.00     | 0.00     | 0.00     | 0.00     | 0.00     |
|            | WCGBTS                  | 1,957.88 | 1,975.71 | 1,980.92 | 1,981.01 | 1,956.05 | 1,957.36 | 1,956.62 |
|            | NatM_uniform_Fem_GP_1   | 0.04     | 0.04     | 0.04     | 0.04     | 0.04     | 0.04     | 0.04     |
|            | L_at_Amin_Fem_GP_1      | -3.10    | -3.34    | -3.24    | -3.30    | -3.43    | -3.13    | -3.35    |
|            | L_at_Amax_Fem_GP_1      | 60.08    | 60.11    | 59.77    | 59.81    | 57.59    | 59.16    | 57.84    |
|            | VonBert_K_Fem_GP_1      | 0.08     | 0.08     | 0.08     | 0.08     | 0.08     | 0.08     | 0.08     |
|            | CV_young_Fem_GP_1       | 0.05     | 0.04     | 0.05     | 0.05     | 0.03     | 0.04     | 0.03     |
|            | CV_old_Fem_GP_1         | 0.09     | 0.09     | 0.09     | 0.09     | 0.10     | 0.10     | 0.10     |
|            | NatM_uniform_Mal_GP_1   | 0.04     | 0.04     | 0.04     | 0.04     | 0.04     | 0.04     | 0.04     |
|            | L_at_Amin_Mal_GP_1      | -2.68    | -3.63    | -2.95    | -3.04    | -4.07    | -2.95    | -3.84    |
|            | L_at_Amax_Mal_GP_1      | 57.83    | 57.82    | 57.56    | 57.64    | 55.23    | 56.91    | 55.50    |
|            | VonBert_K_Mal_GP_1      | 0.08     | 0.09     | 0.08     | 0.09     | 0.09     | 0.09     | 0.09     |
|            | CV_young_Mal_GP_1       | 0.09     | 0.06     | 0.09     | 0.09     | 0.05     | 0.08     | 0.05     |
|            | CV_old_Mal_GP_1         | 0.08     | 0.09     | 0.08     | 0.08     | 0.09     | 0.09     | 0.09     |
|            | SR_LN(R0)               | 7.00     | 8.29     | 6.71     | 7.91     | 6.07     | 6.21     | 6.05     |
|            | LnQ_base_TRIENNIAL(7)   | -1.49    | -2.83    | -1.19    | -2.43    | -0.35    | -0.55    | -0.34    |
|            | LnQ_base_AK_SLOPE(8)    | -3.11    | -4.28    | -2.83    | -4.05    | -2.15    | -2.25    | -2.11    |
|            | LnQ_base_NW_SLOPE(9)    | -2.04    | -3.21    | -1.77    | -2.99    | -1.09    | -1.18    | -1.04    |
|            | LnQ_base_WCGBTS(10)     | -2.61    | -3.81    | -2.30    | -3.56    | -1.46    | -1.67    | -1.44    |
|            | LnQ_base_TRIENNIAL(7)_- | -2.27    | -3.48    | -2.01    | -3.15    | -1.08    | -1.34    | -1.16    |
|            | BLK3repl_1892           |          |          |          |          |          |          |          |
|            | Size_DblN_peak_BOTTOM_- | 49.34    | 49.32    | 49.43    | 49.29    | 50.78    | 49.15    | 52.27    |
|            | TRAWL(1)                |          |          |          |          |          |          |          |
|            | Size_DblN_top_logit_-   | -11.52   | -11.52   | -11.52   | -11.52   | -15.00   | -15.00   | -11.52   |
|            | BOTTOM_TRAWL(1)         |          |          |          |          |          |          |          |
|            | Size_DblN_ascend_se_-   | 4.57     | 4.66     | 4.58     | 4.58     | 4.64     | 4.52     | 4.84     |
|            | BOTTOM_TRAWL(1)         |          |          |          |          |          |          |          |
|            | Size_DblN_descend_se_-  | 2.95     | 3.04     | 2.82     | 2.84     | 20.00    | 20.00    | -1.20    |
|            | BOTTOM_TRAWL(1)         |          |          |          |          |          |          |          |
|            | Size_DblN_start_logit_- | -999.00  | -999.00  | -999.00  | -999.00  | -999.00  | -999.00  | -999.00  |
|            | BOTTOM_TRAWL(1)         |          |          |          |          |          |          |          |
|            | Size_DblN_end_logit_-   | -0.64    | -0.68    | -0.48    | -0.53    | -999.00  | -999.00  | 0.96     |
|            | BOTTOM_TRAWL(1)         |          |          |          |          |          |          |          |

|                                                         |         |         |         |         |         |         |         |
|---------------------------------------------------------|---------|---------|---------|---------|---------|---------|---------|
| Size_DblN_peak_BOTTOM_-<br>TRAWL_DISCARD(2)             | 25.26   | 23.61   | 25.28   | 25.21   | 25.46   | 25.45   | 25.50   |
| Size_DblN_top_logit_-<br>BOTTOM_TRAWL_-<br>DISCARD(2)   | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  |
| Size_DblN_ascend_se_-<br>BOTTOM_TRAWL_-<br>DISCARD(2)   | 4.89    | 4.82    | 4.89    | 4.90    | 5.04    | 4.92    | 5.03    |
| Size_DblN_descend_se_-<br>BOTTOM_TRAWL_-<br>DISCARD(2)  | 3.84    | 4.07    | 3.83    | 3.84    | 3.73    | 3.78    | 3.72    |
| Size_DblN_start_logit_-<br>BOTTOM_TRAWL_-<br>DISCARD(2) | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  |
| Size_DblN_end_logit_-<br>BOTTOM_TRAWL_-<br>DISCARD(2)   | -3.82   | -4.06   | -3.79   | -3.85   | -3.40   | -3.57   | -3.41   |
| Size_DblN_peak_NON_-<br>TRAWL(3)                        | 50.27   | 50.20   | 50.54   | 50.28   | 54.16   | 51.61   | 53.75   |
| Size_DblN_top_logit_NON_-<br>TRAWL(3)                   | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  |
| Size_DblN_ascend_se_NON_-<br>TRAWL(3)                   | 3.86    | 4.00    | 3.89    | 3.88    | 4.25    | 3.96    | 4.19    |
| Size_DblN_descend_se_NON_-<br>TRAWL(3)                  | 20.00   | 20.00   | 20.00   | 20.00   | 20.00   | 20.00   | 20.00   |
| Size_DblN_start_logit_NON_-<br>TRAWL(3)                 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 |
| Size_DblN_end_logit_NON_-<br>TRAWL(3)                   | 4.60    | 4.60    | 4.60    | 4.60    | -999.00 | 4.60    | -999.00 |
| Size_DblN_peak_NON_-<br>TRAWL_DISCARD(4)                | 49.94   | 50.08   | 50.01   | 49.89   | 51.62   | 51.08   | 51.59   |
| Size_DblN_top_logit_NON_-<br>TRAWL_DISCARD(4)           | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  |
| Size_DblN_ascend_se_NON_-<br>TRAWL_DISCARD(4)           | 3.75    | 3.79    | 3.76    | 3.75    | 3.98    | 3.94    | 3.98    |
| Size_DblN_descend_se_NON_-<br>TRAWL_DISCARD(4)          | 2.66    | 2.65    | 2.57    | 2.60    | 1.25    | 1.94    | 1.32    |

|                                                 |         |         |         |         |         |         |         |
|-------------------------------------------------|---------|---------|---------|---------|---------|---------|---------|
| Size_DblN_start_logit_NON_-<br>TRAWL_DISCARD(4) | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 |
| Size_DblN_end_logit_NON_-<br>TRAWL_DISCARD(4)   | -0.19   | -0.24   | -0.03   | -0.09   | 2.23    | 0.42    | 1.74    |
| Size_DblN_peak_MIDWATER_-<br>TRAWL(5)           | 52.17   | 52.17   | 52.73   | 52.35   | 62.39   | 55.01   | 60.49   |
| Size_DblN_top_logit_-<br>MIDWATER_TRAWL(5)      | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  |
| Size_DblN_ascend_se_-<br>MIDWATER_TRAWL(5)      | 4.57    | 4.66    | 4.62    | 4.60    | 5.33    | 4.79    | 5.19    |
| Size_DblN_descend_se_-<br>MIDWATER_TRAWL(5)     | 20.00   | 20.00   | 20.00   | 20.00   | 20.00   | 20.00   | 20.00   |
| Size_DblN_start_logit_-<br>MIDWATER_TRAWL(5)    | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 |
| Size_DblN_end_logit_-<br>MIDWATER_TRAWL(5)      | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 |
| Size_DblN_peak_AT_SEA_-<br>HAKE(6)              | 49.71   | 49.56   | 49.34   | 49.21   | 54.29   | 51.18   | 53.66   |
| Size_DblN_top_logit_AT_-<br>SEA_HAKE(6)         | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  |
| Size_DblN_ascend_se_AT_-<br>SEA_HAKE(6)         | 3.59    | 3.58    | 3.57    | 3.57    | 4.22    | 3.79    | 4.13    |
| Size_DblN_descend_se_AT_-<br>SEA_HAKE(6)        | 20.00   | 20.00   | 20.00   | 20.00   | 20.00   | 20.00   | 20.00   |
| Size_DblN_start_logit_AT_-<br>SEA_HAKE(6)       | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 |
| Size_DblN_end_logit_AT_-<br>SEA_HAKE(6)         | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 |
| Size_DblN_peak_-<br>TRIENNIAL(7)                | 21.78   | 22.07   | 21.78   | 21.80   | 21.82   | 21.82   | 21.82   |
| Size_DblN_top_logit_-<br>TRIENNIAL(7)           | -8.60   | -8.60   | -8.60   | -8.60   | -8.60   | -8.60   | -8.60   |
| Size_DblN_ascend_se_-<br>TRIENNIAL(7)           | 3.52    | 3.58    | 3.52    | 3.52    | 3.53    | 3.54    | 3.54    |
| Size_DblN_descend_se_-<br>TRIENNIAL(7)          | 3.95    | 3.99    | 3.94    | 3.96    | 3.80    | 3.87    | 3.81    |
| Size_DblN_start_logit_-<br>TRIENNIAL(7)         | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 |

|                                                              |         |         |         |         |         |         |         |
|--------------------------------------------------------------|---------|---------|---------|---------|---------|---------|---------|
| Size_DblN_end_logit_-<br>TRIENNIAL(7)                        | -2.58   | -2.56   | -2.57   | -2.57   | -2.34   | -2.47   | -2.34   |
| Size_DblN_peak_AK_-<br>SLOPE(8)                              | 37.29   | 36.46   | 37.22   | 37.20   | 37.03   | 37.30   | 37.20   |
| Size_DblN_top_logit_AK_-<br>SLOPE(8)                         | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  |
| Size_DblN_ascend_se_AK_-<br>SLOPE(8)                         | 4.94    | 4.68    | 4.94    | 4.93    | 4.96    | 4.97    | 4.97    |
| Size_DblN_descend_se_AK_-<br>SLOPE(8)                        | 4.63    | 4.65    | 4.65    | 4.64    | 4.85    | 4.71    | 4.81    |
| Size_DblN_start_logit_AK_-<br>SLOPE(8)                       | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 |
| Size_DblN_end_logit_AK_-<br>SLOPE(8)                         | -10.49  | -10.08  | -10.56  | -10.48  | -10.31  | -10.51  | -10.34  |
| Size_DblN_peak_WCGBTS(10)                                    | 21.08   | 20.65   | 21.20   | 20.99   | 21.49   | 21.60   | 21.51   |
| Size_DblN_top_logit_-<br>WCGBTS(10)                          | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  | -15.00  |
| Size_DblN_ascend_se_-<br>WCGBTS(10)                          | 3.39    | 3.24    | 3.42    | 3.37    | 3.49    | 3.52    | 3.50    |
| Size_DblN_descend_se_-<br>WCGBTS(10)                         | 4.65    | 4.69    | 4.57    | 4.67    | 4.15    | 4.31    | 4.15    |
| Size_DblN_start_logit_-<br>WCGBTS(10)                        | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 | -999.00 |
| Size_DblN_end_logit_-<br>WCGBTS(10)                          | -0.83   | -1.05   | -0.81   | -0.87   | -0.44   | -0.59   | -0.44   |
| Size_DblN_peak_BOTTOM_-<br>TRAWL(1)_BLK4repl_1892            | 44.70   | 42.94   | 44.87   | 44.53   |         |         | 45.70   |
| Size_DblN_peak_BOTTOM_-<br>TRAWL(1)_BLK4repl_2002            | 47.64   | 47.51   | 49.00   | 49.00   |         |         | 49.92   |
| Size_DblN_ascend_se_-<br>BOTTOM_TRAWL(1)_-<br>BLK4repl_1892  | 5.03    | 4.77    | 5.05    | 5.01    |         |         | 5.13    |
| Size_DblN_ascend_se_-<br>BOTTOM_TRAWL(1)_-<br>BLK4repl_2002  | 4.10    | 4.01    | 4.29    | 4.30    |         |         | 4.44    |
| Size_DblN_descend_se_-<br>BOTTOM_TRAWL(1)_-<br>BLK4repl_1892 | 2.65    | 3.36    | 2.52    | 2.73    |         |         | 1.72    |

|                            |       |       |        |        |       |       |       |
|----------------------------|-------|-------|--------|--------|-------|-------|-------|
| Size_DblN_descend_se_-     | 2.75  | 2.75  | -12.39 | -12.33 |       |       | -0.20 |
| BOTTOM_TRAWL(1)_-          |       |       |        |        |       |       |       |
| BLK4repl_2002              |       |       |        |        |       |       |       |
| Size_DblN_end_logit_-      | -1.59 | -1.86 | -1.42  | -1.63  |       |       | -0.75 |
| BOTTOM_TRAWL(1)_-          |       |       |        |        |       |       |       |
| BLK4repl_1892              |       |       |        |        |       |       |       |
| Size_DblN_end_logit_-      | -0.93 | -1.05 | -0.62  | -0.67  |       |       | 0.13  |
| BOTTOM_TRAWL(1)_-          |       |       |        |        |       |       |       |
| BLK4repl_2002              |       |       |        |        |       |       |       |
| Size_DblN_peak_BOTTOM_-    | 47.67 | 47.88 | 50.96  | 50.96  | 48.70 | 48.12 | 48.69 |
| TRAWL_DISCARD(2)_-         |       |       |        |        |       |       |       |
| BLK1repl_1892              |       |       |        |        |       |       |       |
| Size_DblN_ascend_se_-      | 6.32  | 6.24  | 6.59   | 6.58   | 6.37  | 6.34  | 6.36  |
| BOTTOM_TRAWL_-             |       |       |        |        |       |       |       |
| DISCARD(2)_BLK1repl_1892   |       |       |        |        |       |       |       |
| Size_DblN_descend_se_-     | 3.01  | 2.93  | -8.19  | -8.12  | 2.60  | 2.87  | 2.63  |
| BOTTOM_TRAWL_-             |       |       |        |        |       |       |       |
| DISCARD(2)_BLK1repl_1892   |       |       |        |        |       |       |       |
| Size_DblN_end_logit_-      | -1.70 | -1.76 | -1.17  | -1.21  | -0.65 | -1.30 | -0.75 |
| BOTTOM_TRAWL_-             |       |       |        |        |       |       |       |
| DISCARD(2)_BLK1repl_1892   |       |       |        |        |       |       |       |
| Size_DblN_peak_NON_-       | 46.77 | 46.61 | 46.82  | 46.75  | 45.83 | 47.05 | 45.75 |
| TRAWL(3)_BLK2repl_1892     |       |       |        |        |       |       |       |
| Size_DblN_peak_NON_-       | 49.49 | 49.62 | 49.59  | 49.51  | 50.02 | 50.02 | 49.84 |
| TRAWL(3)_BLK2repl_2011     |       |       |        |        |       |       |       |
| Size_DblN_ascend_se_NON_-  | 3.05  | 2.97  | 3.06   | 3.05   | 2.98  | 3.11  | 2.96  |
| TRAWL(3)_BLK2repl_1892     |       |       |        |        |       |       |       |
| Size_DblN_ascend_se_NON_-  | 3.81  | 3.83  | 3.83   | 3.82   | 3.91  | 3.87  | 3.88  |
| TRAWL(3)_BLK2repl_2011     |       |       |        |        |       |       |       |
| Size_DblN_descend_se_NON_- | 3.17  | 3.18  | 3.14   | 3.16   |       | 3.10  |       |
| TRAWL(3)_BLK2repl_1892     |       |       |        |        |       |       |       |
| Size_DblN_descend_se_NON_- | 2.32  | 2.35  | 2.15   | 2.20   |       | 1.73  |       |
| TRAWL(3)_BLK2repl_2011     |       |       |        |        |       |       |       |
| Size_DblN_end_logit_NON_-  | -2.29 | -2.39 | -2.17  | -2.23  |       | -1.88 |       |
| TRAWL(3)_BLK2repl_1892     |       |       |        |        |       |       |       |
| Size_DblN_end_logit_NON_-  | -0.63 | -0.64 | -0.49  | -0.54  |       | -0.10 |       |
| TRAWL(3)_BLK2repl_2011     |       |       |        |        |       |       |       |

|                    |                                                             |              |               |              |               |              |              |              |
|--------------------|-------------------------------------------------------------|--------------|---------------|--------------|---------------|--------------|--------------|--------------|
| Derived quantities | Size_DblN_peak_-<br>TRIENNIAL(7)_BLK3repl_1892              | 17.36        | 17.15         | 17.31        | 17.37         | 16.94        | 17.31        | 16.95        |
|                    | Size_DblN_ascend_se_-<br>TRIENNIAL(7)_BLK3repl_1892         | 2.09         | 1.94          | 2.07         | 2.09          | 1.88         | 2.06         | 1.90         |
|                    | Size_DblN_descend_se_-<br>TRIENNIAL(7)_BLK3repl_1892        | 5.11         | 5.07          | 5.10         | 5.14          | 5.17         | 5.09         | 5.13         |
|                    | Size_DblN_end_logit_-<br>TRIENNIAL(7)_BLK3repl_1892         | -4.29        | -4.20         | -4.24        | -4.40         | -4.74        | -4.43        | -4.42        |
|                    | Size_DblN_peak_BOTTOM_-<br>TRAWL(1)_BLK2repl_1892           |              |               |              |               | 47.94        | 46.60        |              |
|                    | Size_DblN_peak_BOTTOM_-<br>TRAWL(1)_BLK2repl_2011           |              |               |              |               | 51.07        | 49.15        |              |
|                    | Size_DblN_ascend_se_-<br>BOTTOM_TRAWL(1)_-<br>BLK2repl_1892 |              |               |              |               | 4.89         | 4.84         |              |
|                    | Size_DblN_ascend_se_-<br>BOTTOM_TRAWL(1)_-<br>BLK2repl_2011 |              |               |              |               | 4.79         | 4.68         |              |
|                    | SB0                                                         | 5,647,660.00 | 20,342,100.00 | 4,072,700.00 | 13,532,200.00 | 1,817,430.00 | 2,365,820.00 | 1,804,090.00 |
|                    | SSB_2025                                                    | 4,929,120.00 | 19,323,100.00 | 3,426,350.00 | 13,002,700.00 | 1,028,680.00 | 1,501,040.00 | 1,003,810.00 |
|                    | Bratio_2025                                                 | 0.87         | 0.95          | 0.84         | 0.96          | 0.57         | 0.63         | 0.56         |
|                    | MSY_SPR                                                     | 526.16       | 1,906.71      | 394.31       | 1,309.06      | 206.63       | 239.03       | 203.05       |
|                    | F_SPR                                                       | 0.04         | 0.04          | 0.02         | 0.02          | 0.05         | 0.04         | 0.05         |
|                    |                                                             |              |               |              |               |              |              |              |
|                    |                                                             |              |               |              |               |              |              |              |

Table 32: PLACEHOLDER.

| Source                   | Change                    | Scale | Status |
|--------------------------|---------------------------|-------|--------|
| Natural mortality        | Increase                  | Up    | Up     |
| Bottom Trawl Selectivity |                           |       |        |
| Logistic                 |                           | Down  | Down   |
| Dome-shaped              |                           | Up    | Up     |
| Ageing Error             | Less bias and imprecision | Down  | Same   |
| Data weighting           |                           |       |        |
| Lengths                  | Higher weight             | Down  | Down   |
| Ages                     | Higher weight             | Up    | Up     |
| Recruitment estimation   | estimate fewer years      | Up    | Up     |



Table 33: Recent trend in the overfishing limits (OFL), the acceptable biological catches (ABCs), the annual catch limits (ACLs), and the total dead catch (landings + discards) all in metric tons (mt). Specifications are combined across north and south of 40°10.

| Year | OFL (mt) | ABC (mt) | ACL (mt) | Catch (mt) |
|------|----------|----------|----------|------------|
| 2015 | 206      | 188.1    | 188.1    | 132.1      |
| 2016 | 211      | 192.7    | 192.7    | 149.0      |
| 2017 | 215      | 196.3    | 196.3    | 156.0      |
| 2018 | 219      | 199.9    | 199.9    | 241.5      |
| 2019 | 222      | 202.7    | 202.7    | 226.6      |
| 2020 | 224      | 204.5    | 204.5    | 106.1      |
| 2021 | 237      | 195.8    | 195.8    | 92.8       |
| 2022 | 238      | 194.7    | 194.7    | 117.4      |
| 2023 | 238      | 192.8    | 192.8    | 97.6       |
| 2024 | 238      | 194.7    | 194.7    | 118.9      |

Table 34: Potential OFLs (mt), ABCs (mt), ACLs (mt), the buffer between the OFL and ABC, estimated spawning output, and fraction unfished with adopted OFLs and ACLs and assumed catch for the first two years of the projection period.

| Year | Adopted<br>OFL (mt) | Adopted<br>ACL (mt) | Assumed<br>Catch<br>(mt) | OFL (mt) | Buffer | ABC (mt) | ACL (mt) | Spawning<br>output | Fraction<br>Unfished |
|------|---------------------|---------------------|--------------------------|----------|--------|----------|----------|--------------------|----------------------|
| 2025 | —                   | —                   | 141                      | —        | —      | —        | —        | 4,929,140.000      | 0.873                |
| 2026 | —                   | —                   | 174                      | —        | —      | —        | —        | 4,945,630.000      | 0.876                |
| 2027 | —                   | —                   | —                        | 980      | 0.913  | 894      | 894      | 4,960,900.000      | 0.878                |
| 2028 | —                   | —                   | —                        | 968      | 0.908  | 878      | 878      | 4,888,090.000      | 0.866                |
| 2029 | —                   | —                   | —                        | 955      | 0.902  | 862      | 862      | 4,818,790.000      | 0.853                |
| 2030 | —                   | —                   | —                        | 944      | 0.896  | 846      | 846      | 4,752,560.000      | 0.842                |
| 2031 | —                   | —                   | —                        | 932      | 0.890  | 830      | 830      | 4,688,990.000      | 0.830                |
| 2032 | —                   | —                   | —                        | 921      | 0.885  | 815      | 815      | 4,627,620.000      | 0.819                |
| 2033 | —                   | —                   | —                        | 909      | 0.879  | 799      | 799      | 4,567,870.000      | 0.809                |
| 2034 | —                   | —                   | —                        | 898      | 0.874  | 784      | 784      | 4,509,380.000      | 0.798                |
| 2035 | —                   | —                   | —                        | 886      | 0.868  | 769      | 769      | 4,451,710.000      | 0.788                |
| 2036 | —                   | —                   | —                        | 874      | 0.863  | 755      | 755      | 4,394,830.000      | 0.778                |

## 8 Figures

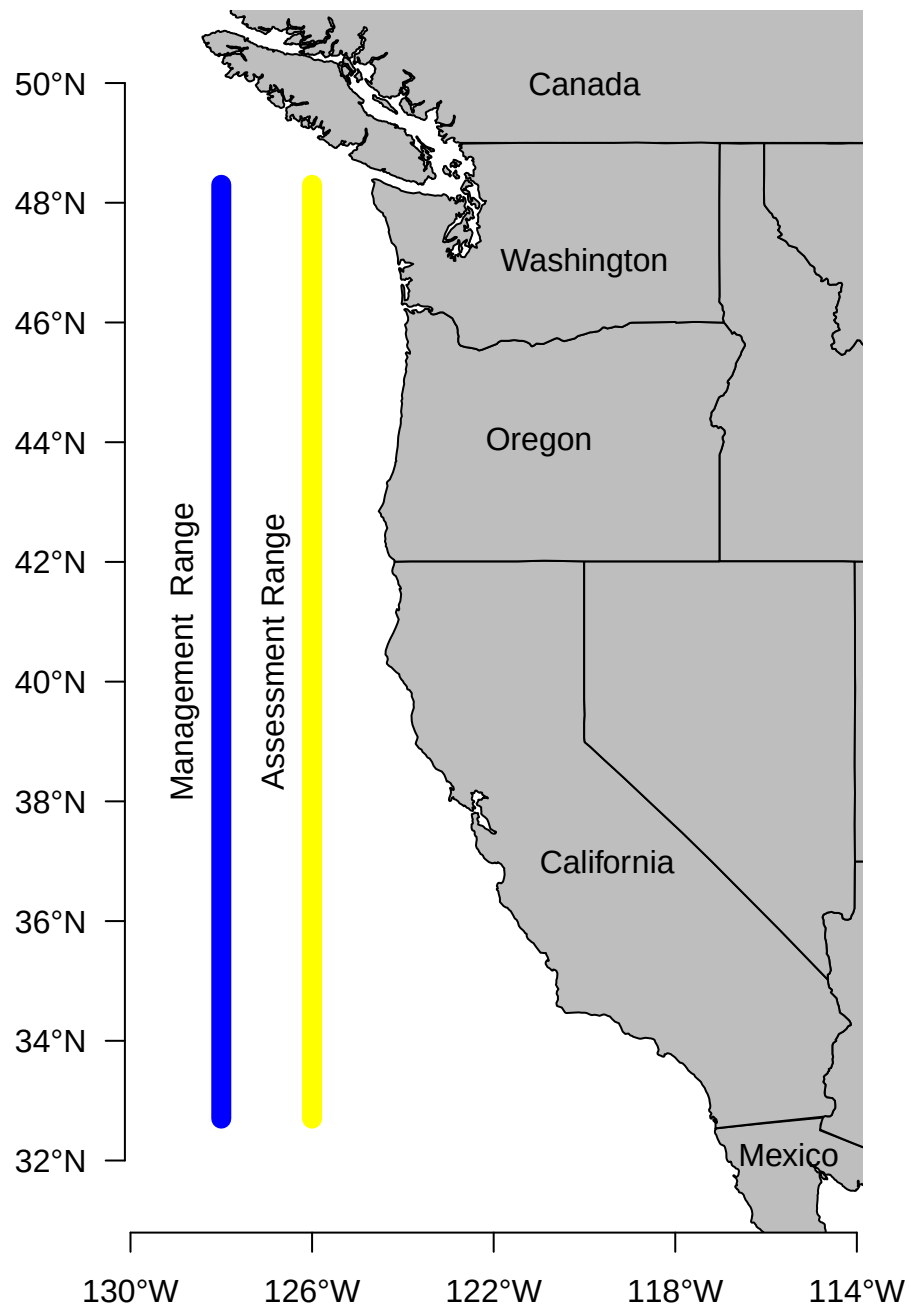


Figure 1: Map of the assessment area.

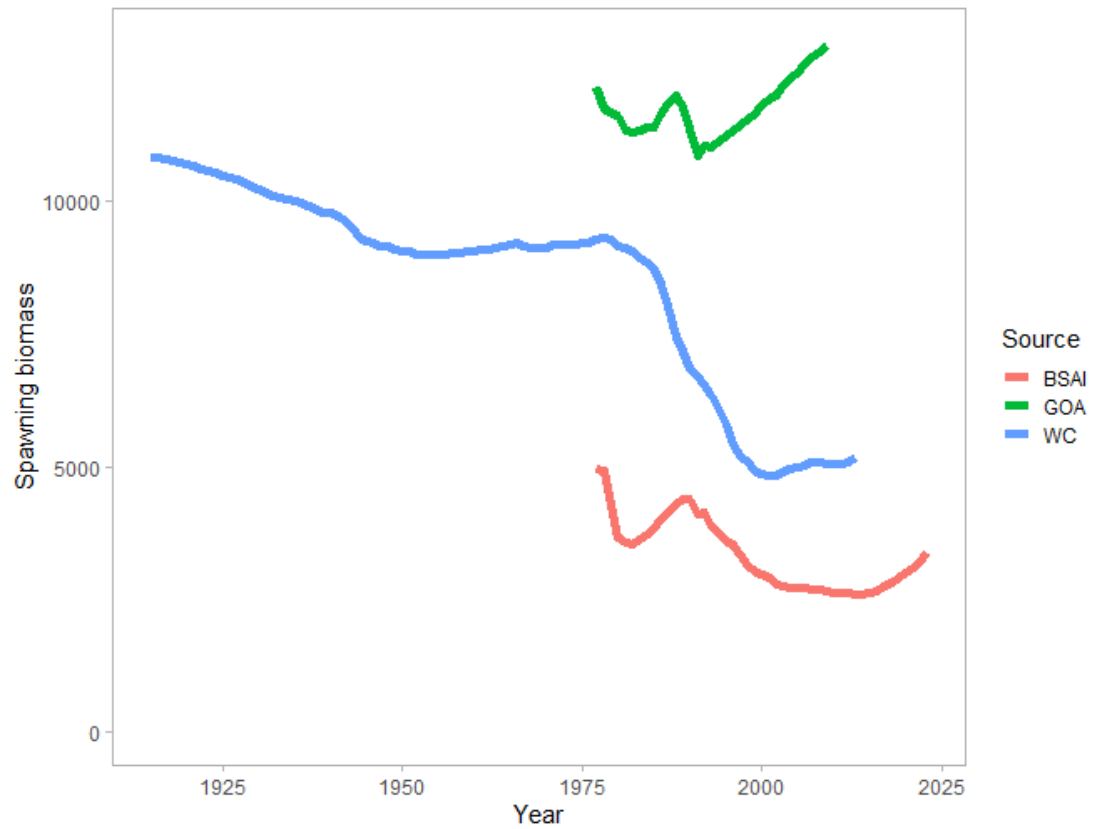


Figure 2: Estimates of spawning biomass (current spawning output/unfished spawning output) for the Rougheye/Blackspotted rockfish complex from the two most recent Alaska (Bering Sea/Aleutian Islands (BSAI) and Gulf of Alaska (GOA)) and the 2013 U.S. west coast stock assessment.

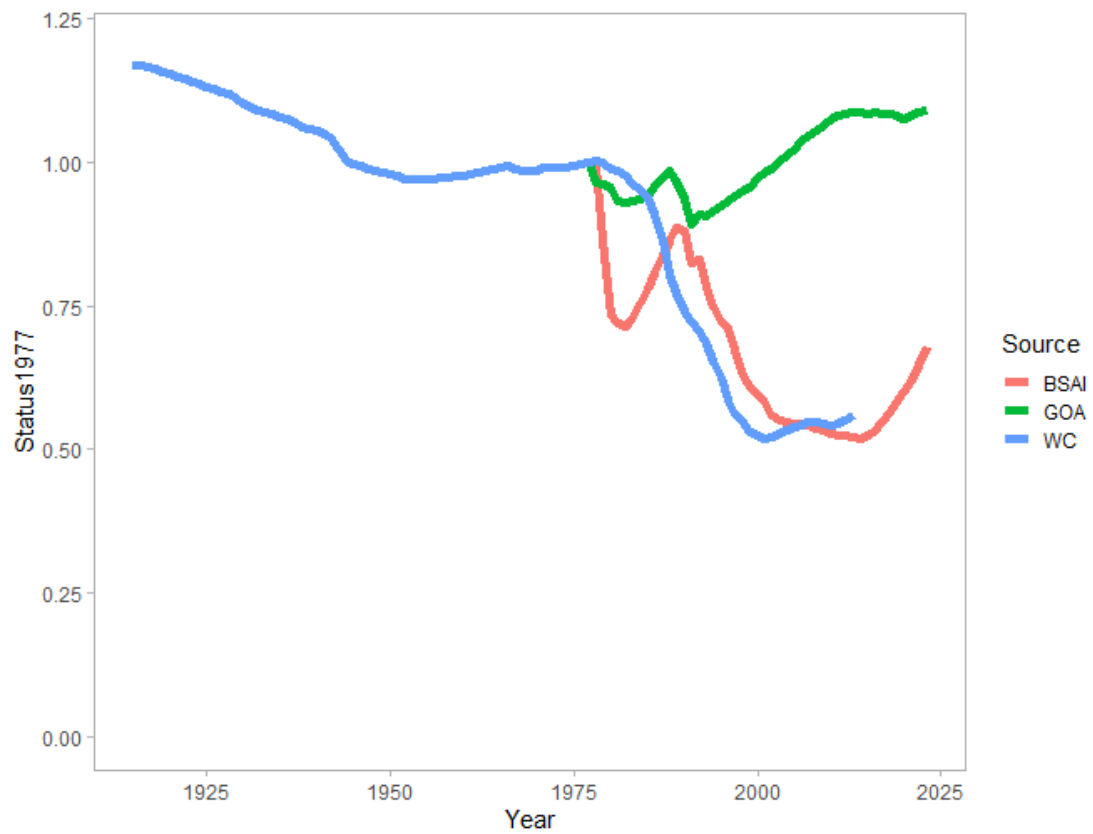


Figure 3: Estimates of relative stock size (current spawning output/unfished spawning output) relative to 1977 (the common year in all stock assessments compared) for the Rougheye/Blackspotted rockfish complex from the two most recent Alaska (Bering Sea/Aleutian Islands (BSAI) and Gulf of Alaska (GOA)) and the 2013 U.S. west coast stock assessment.

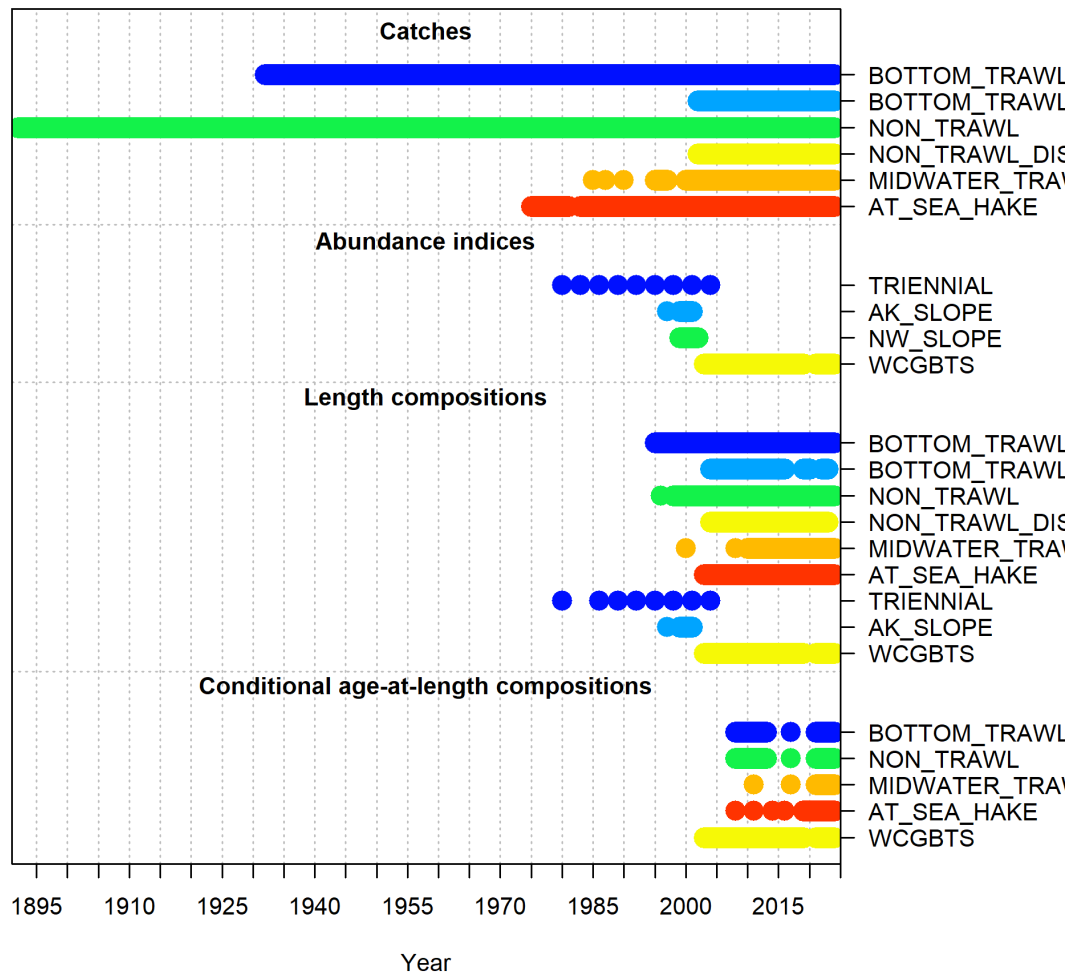


Figure 4: Data used in the base model.

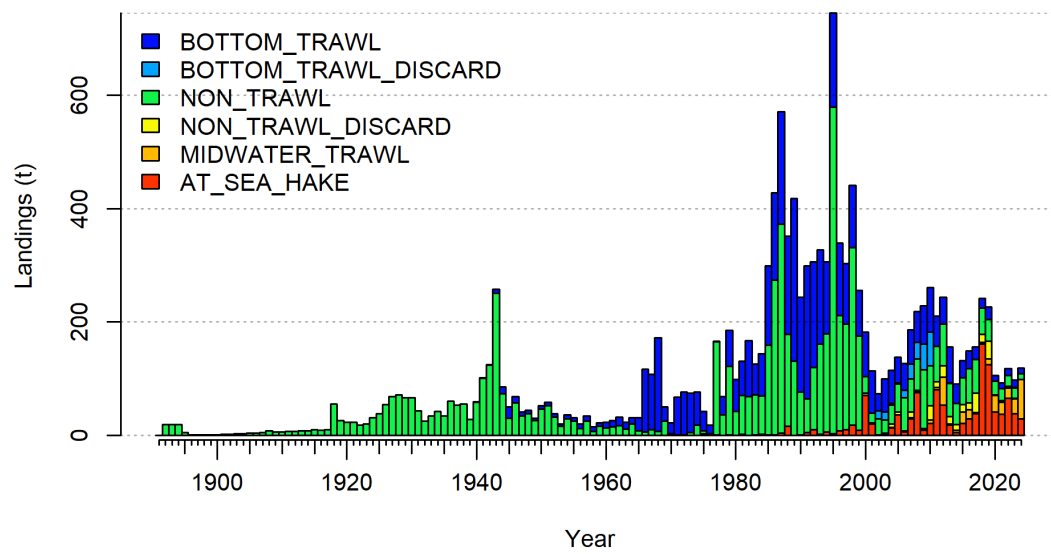


Figure 5: Landings by fleet.

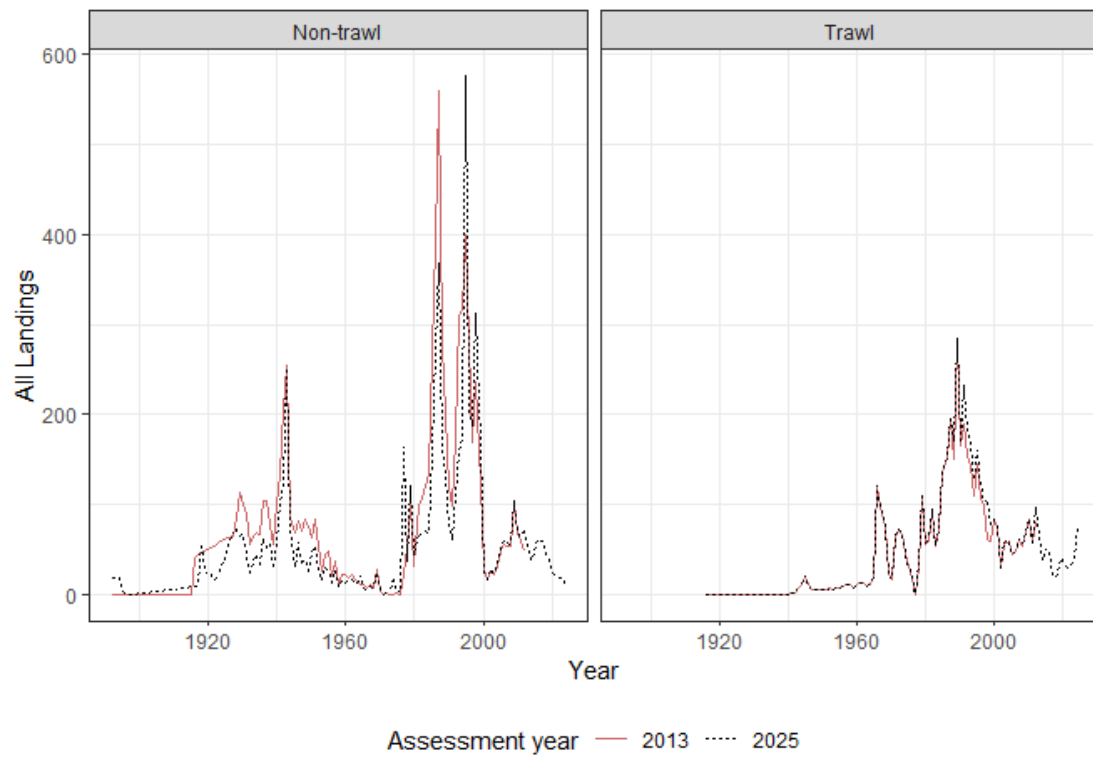


Figure 6: Landings across all states for non-trawl and trawl fisheries compared between the 2013 assessment and updated landings for the 2025 stock assessment model.



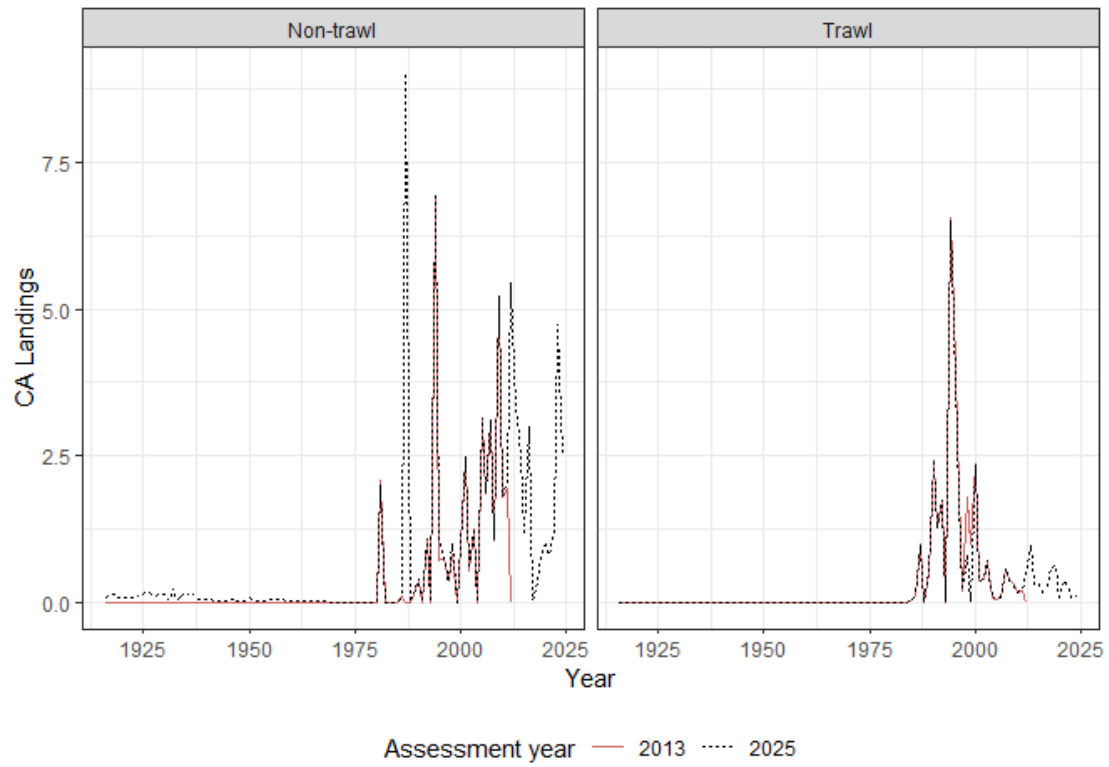


Figure 7: California state landings for non-trawl and trawl fisheries compared between the 2013 assessment and updated landings for the 2025 stock assessment model.

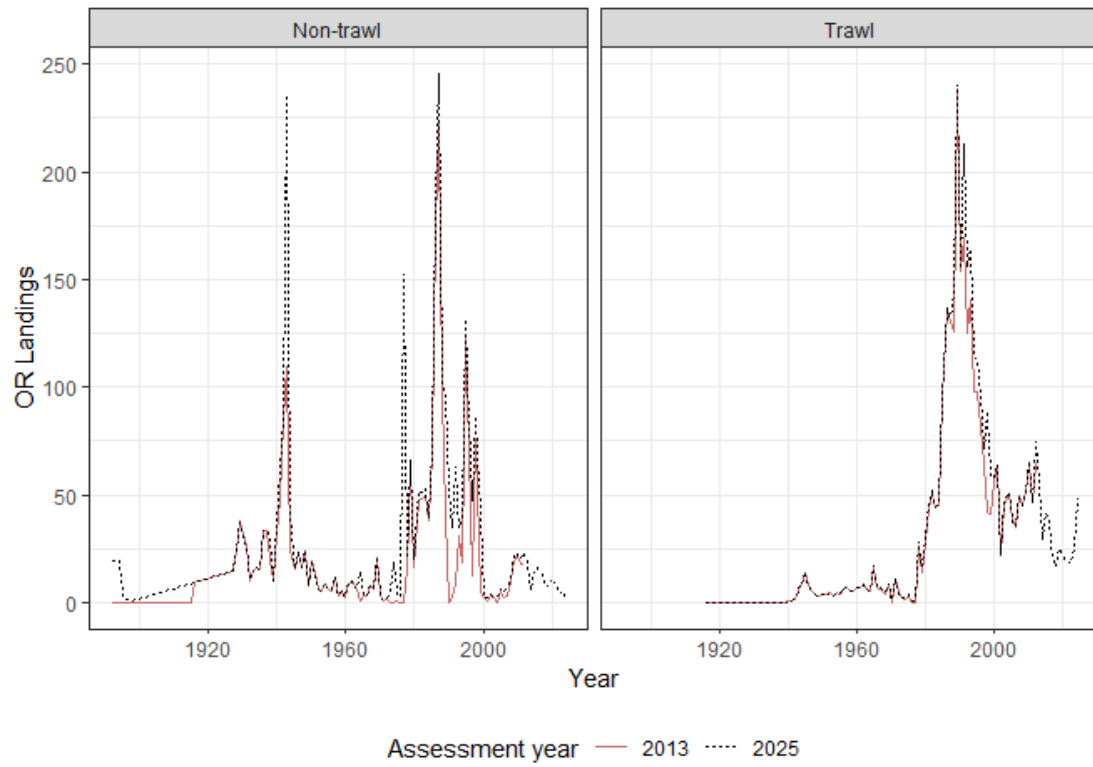


Figure 8: Oregon state landings for non-trawl and trawl fisheries compared between the 2013 assessment and updated landings for the 2025 stock assessment model.

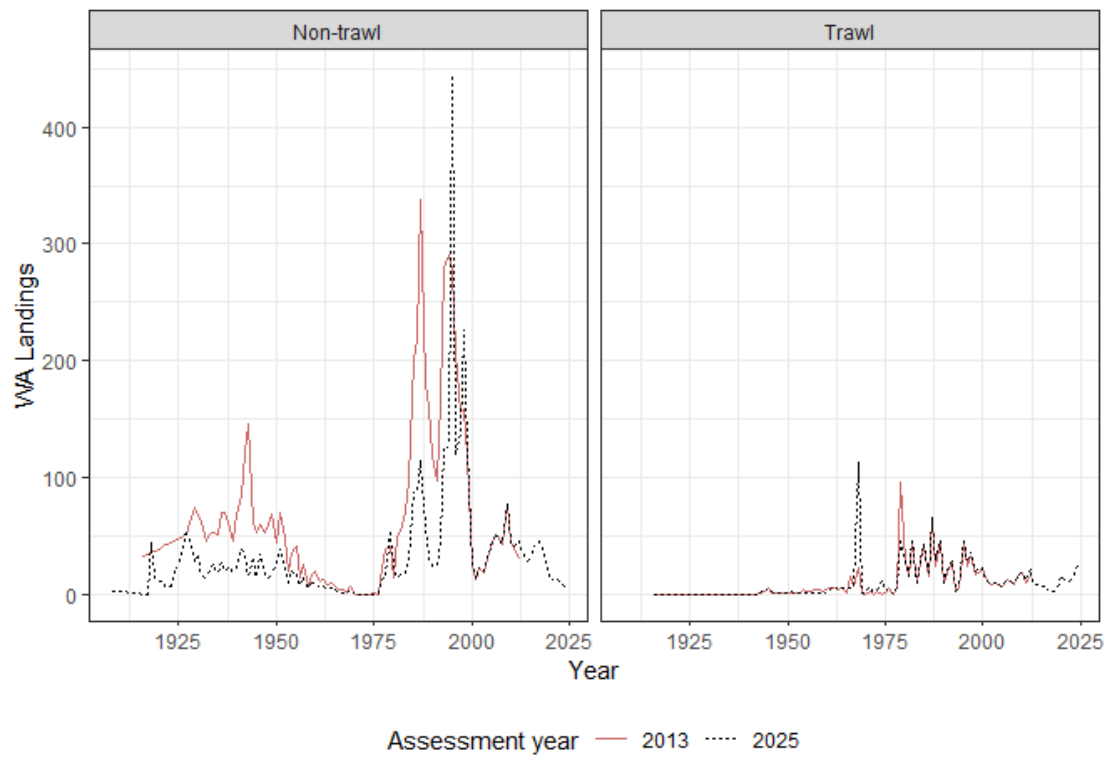


Figure 9: WA. Washington state landings for non-trawl and trawl fisheries compared between the 2013 assessment and updated landings for the 2025 stock assessment model.

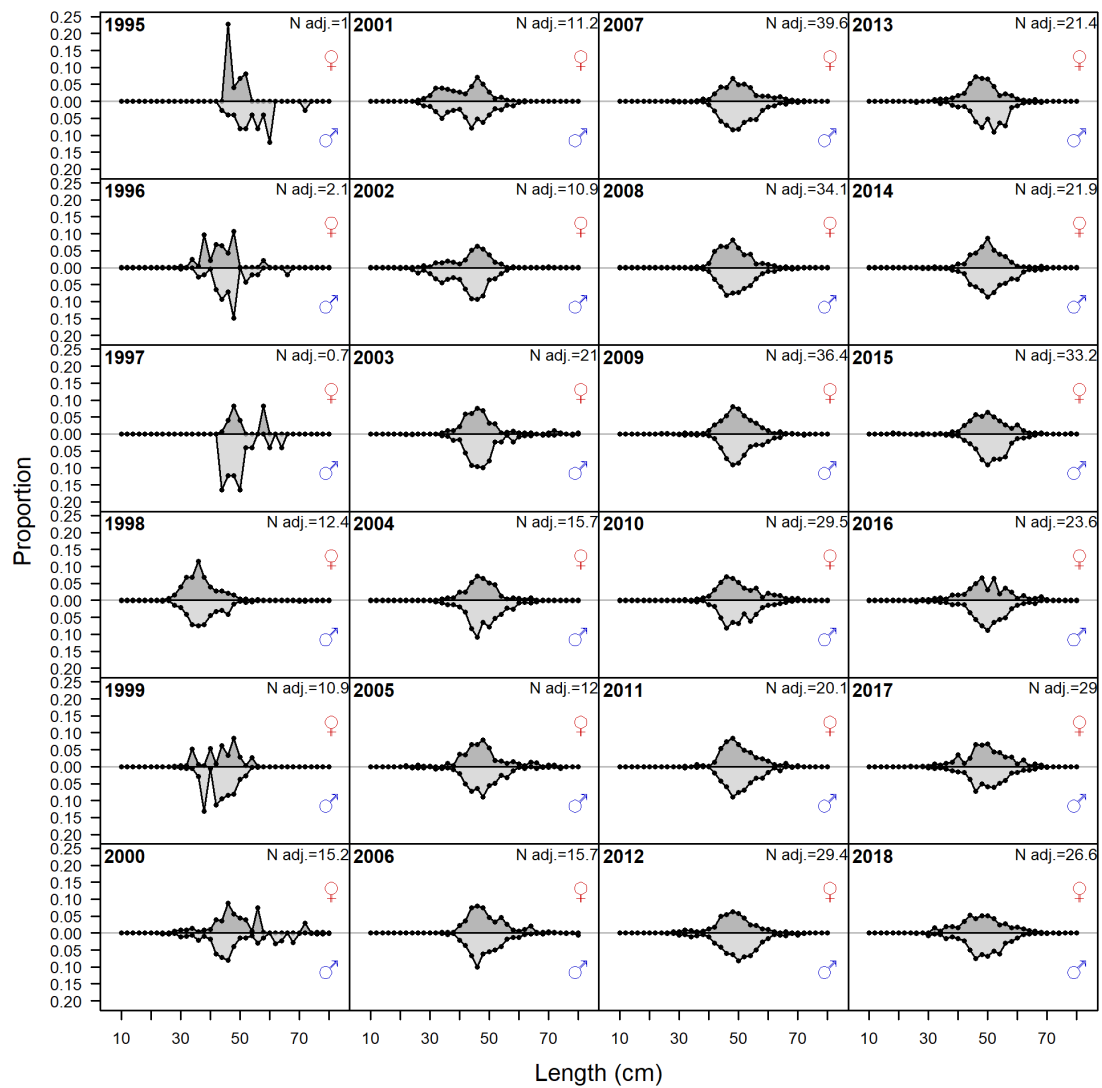


Figure 10: Length composition data for bottom trawl fleet.

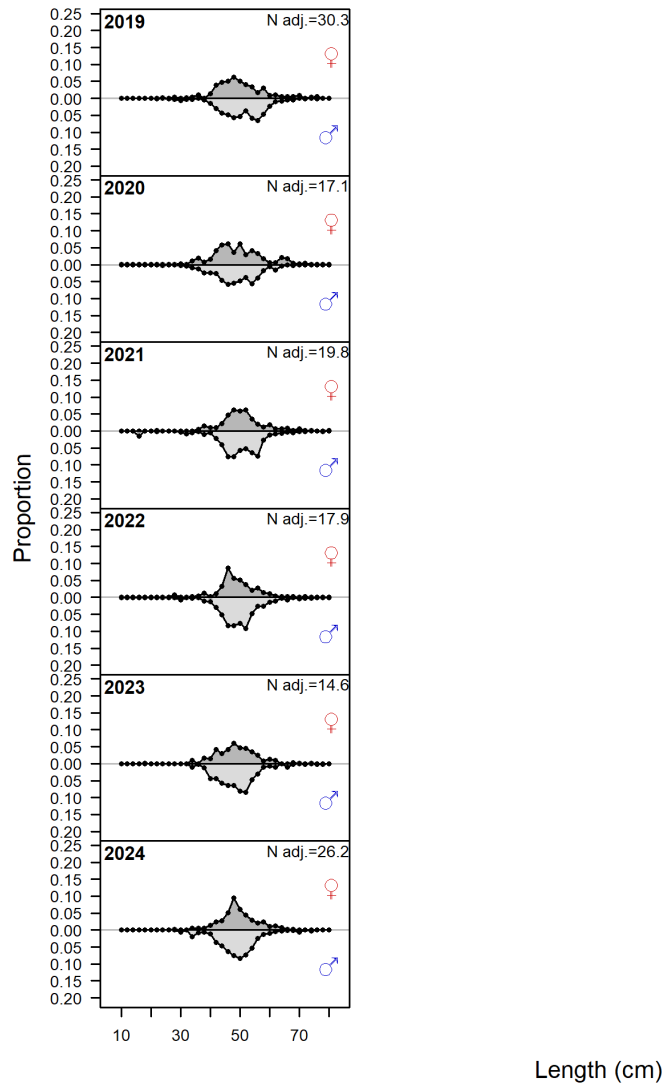


Figure 11: Length composition data for bottom trawl fleet, continued.

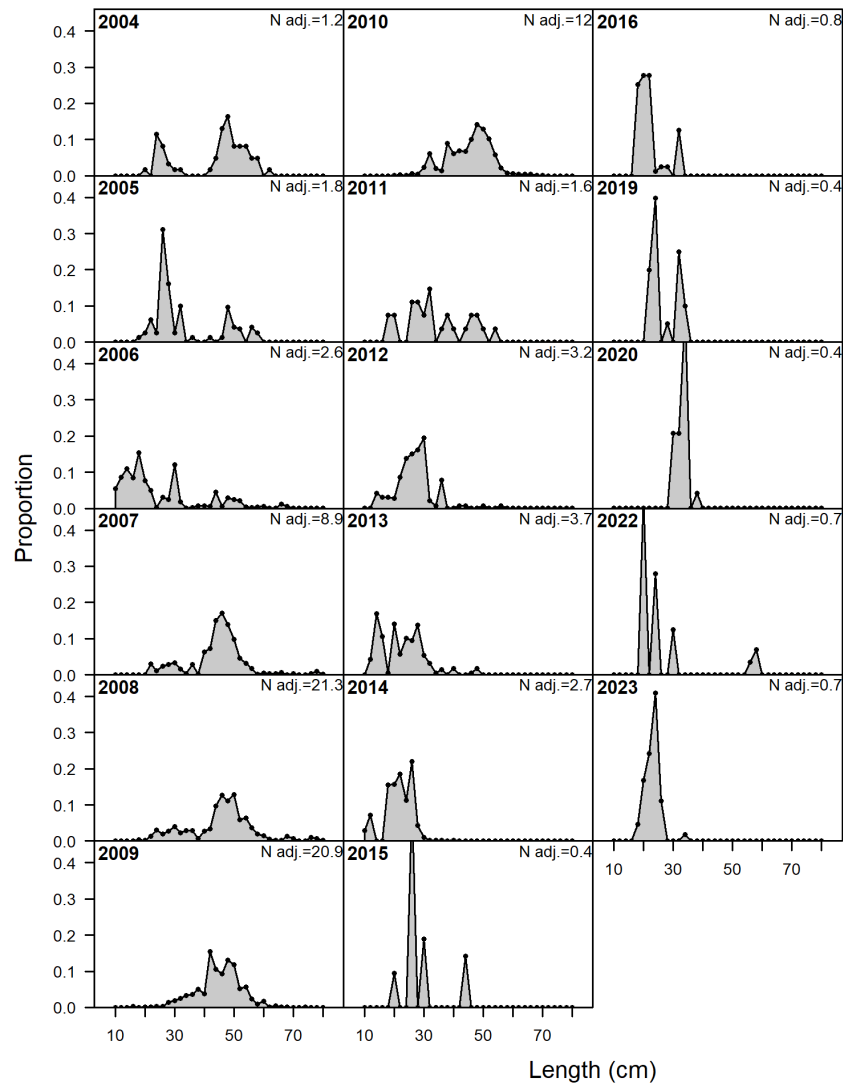


Figure 12: Length composition data for bottom trawl discard fleet.

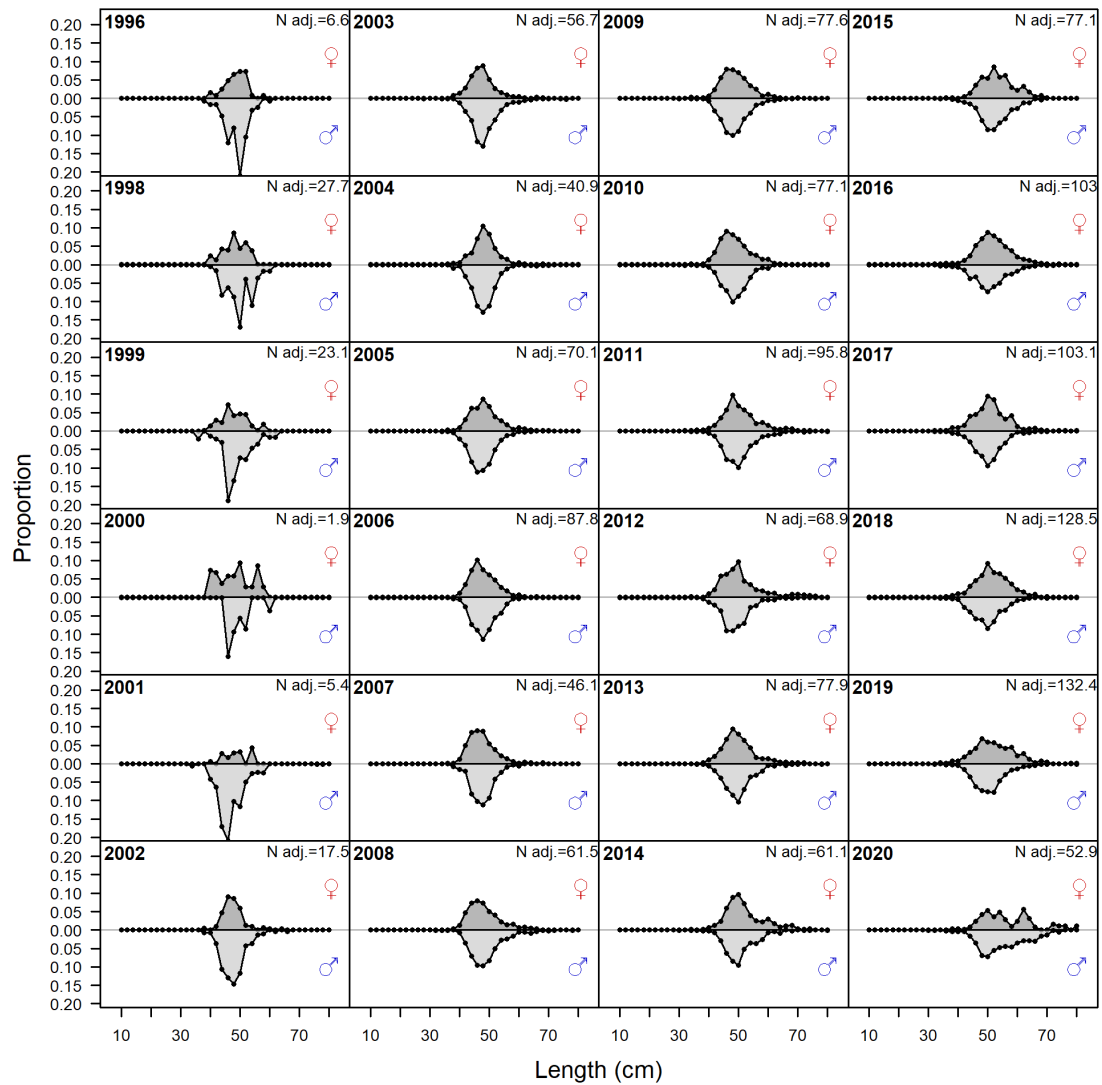


Figure 13: Length composition data for non-trawl fleet.

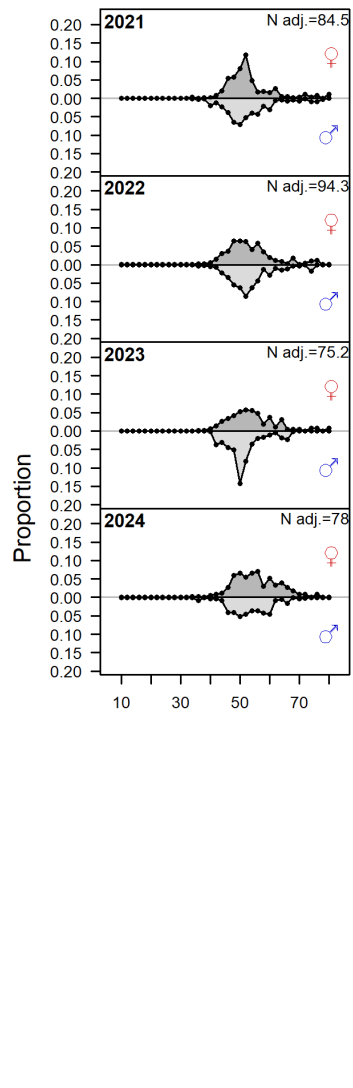


Figure 14: Length composition data for non-trawl fleet, continued.



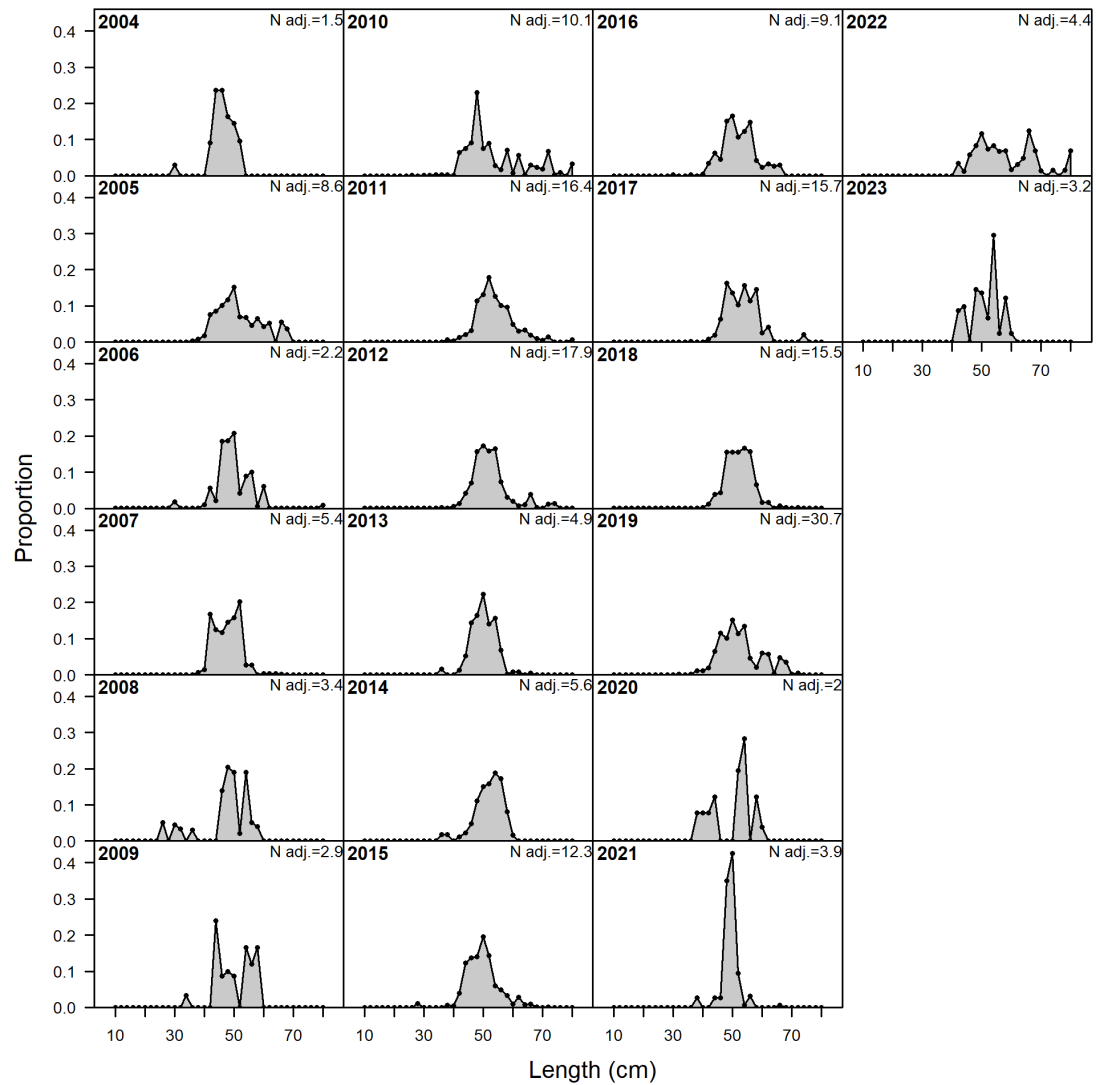


Figure 15: Length composition data for non-trawl discard fleet.

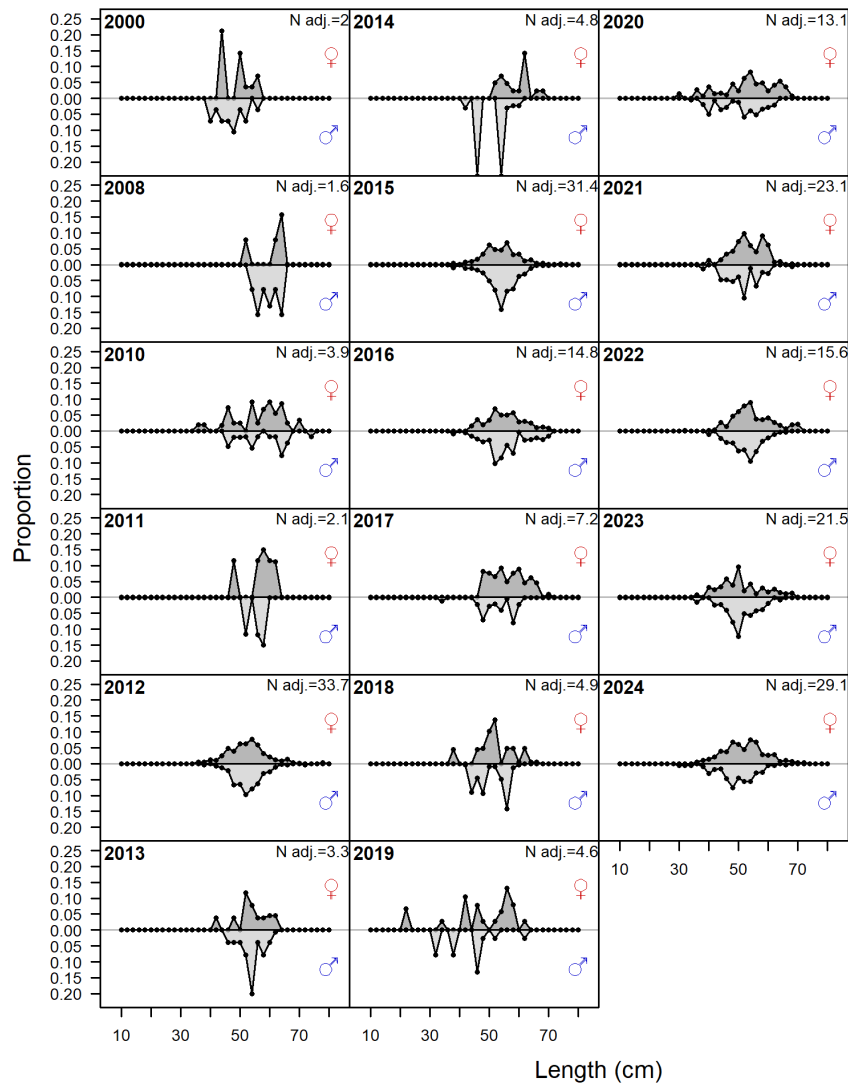


Figure 16: Length composition data for mid-water trawl fleet.

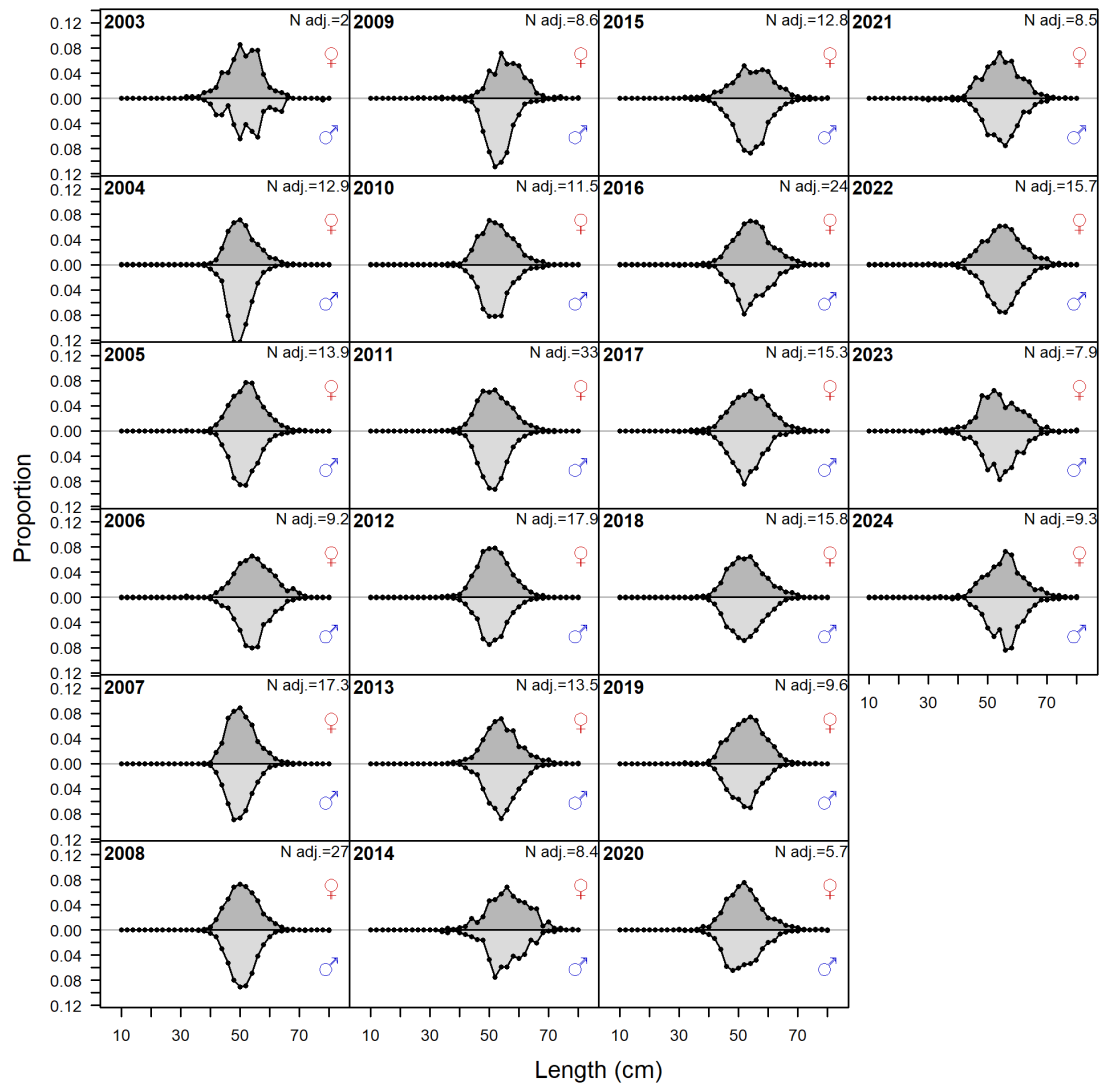


Figure 17: Length composition data for At-sea Hake fleet.

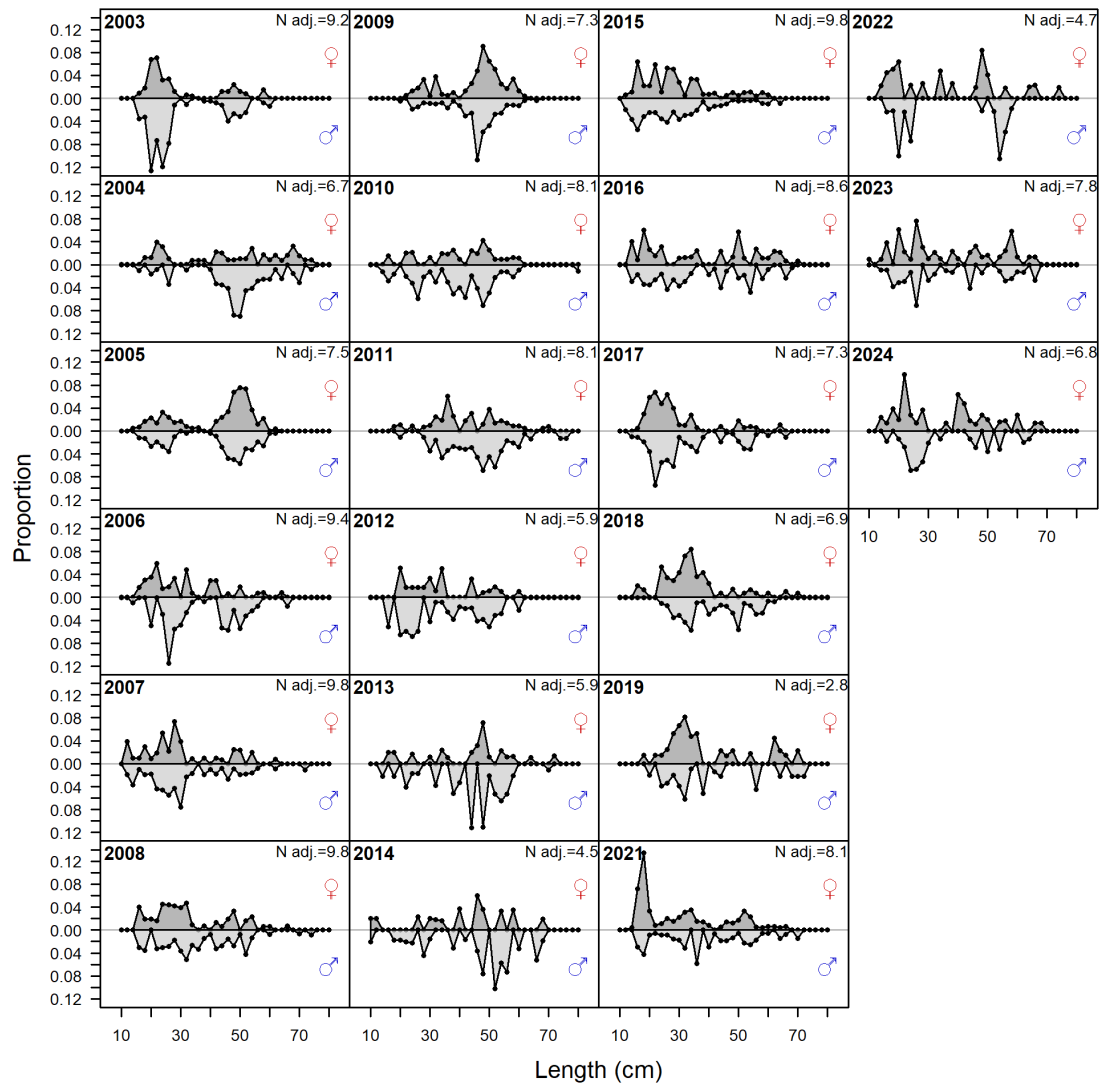


Figure 18: Length composition data for WCGBTs.

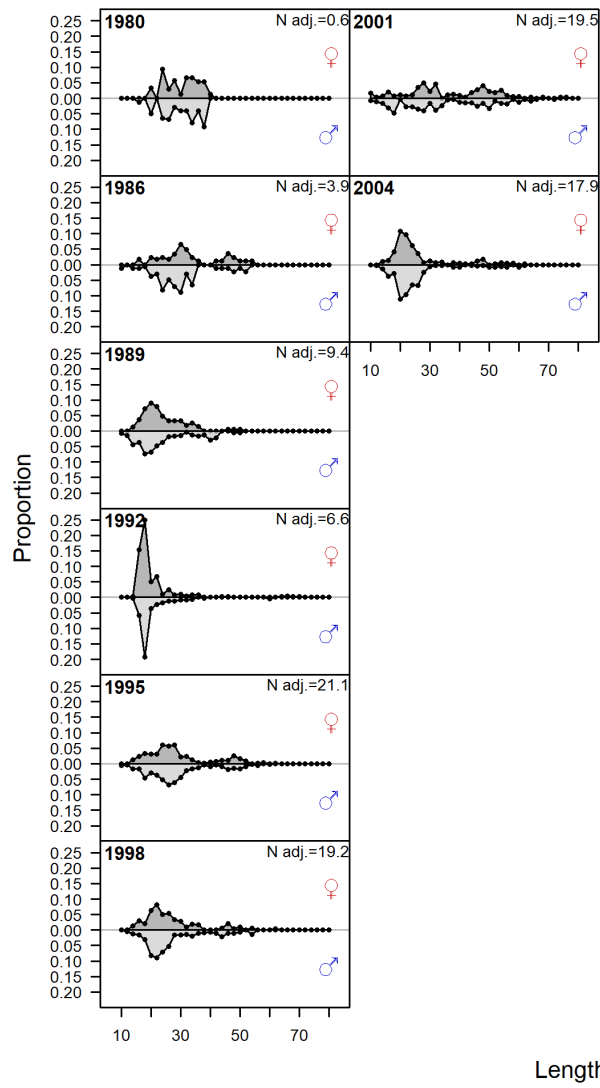


Figure 19: Length composition data for Triennial Survey.

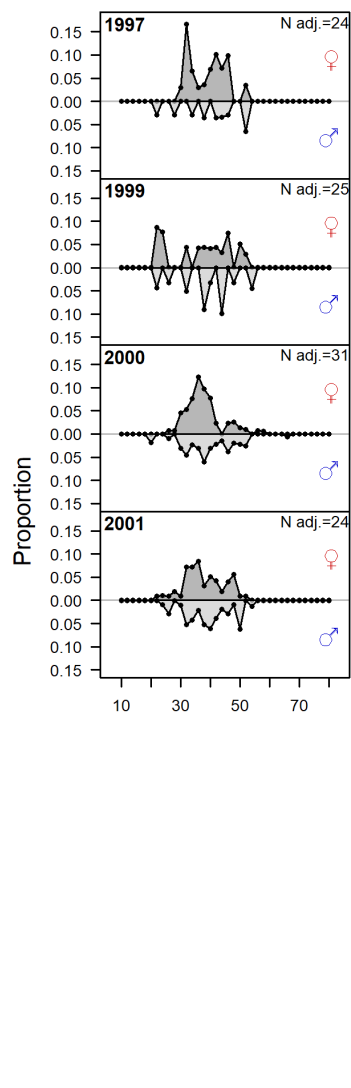


Figure 20: Length composition data for AFSC Slope Survey.

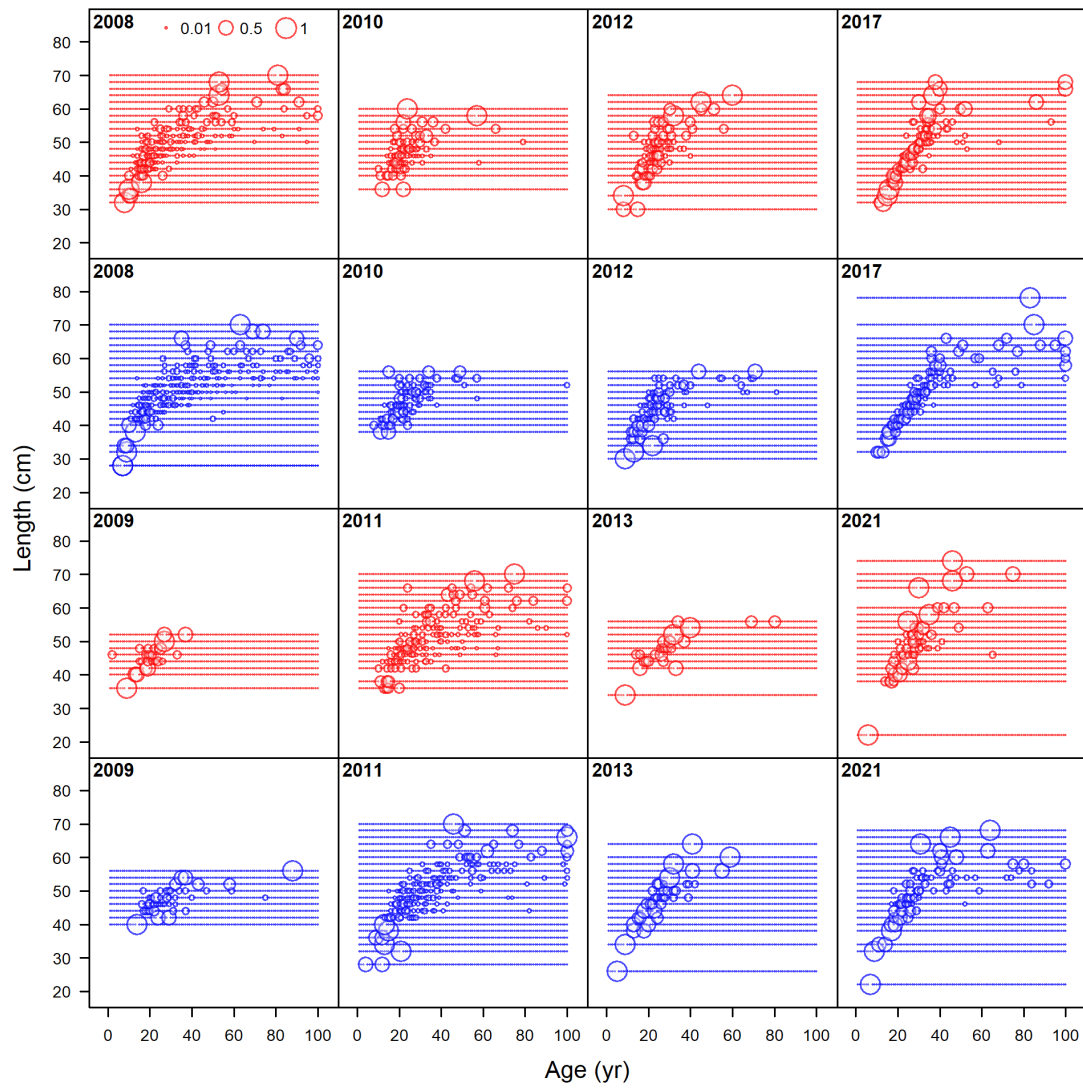


Figure 21: Length composition data for bottom trawl fleet.

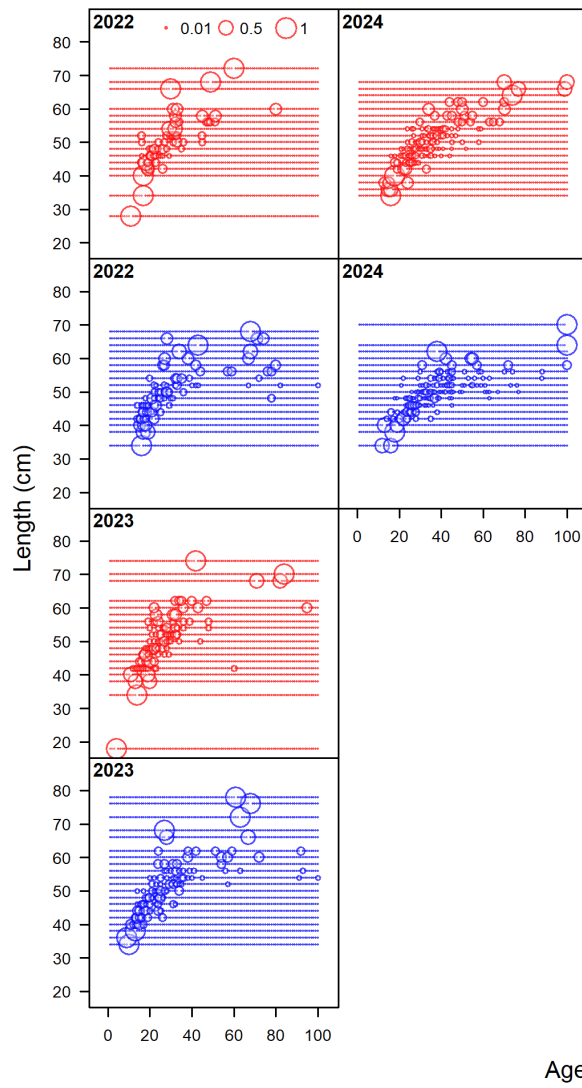


Figure 22: Length composition data for bottom trawl fleet, continued.



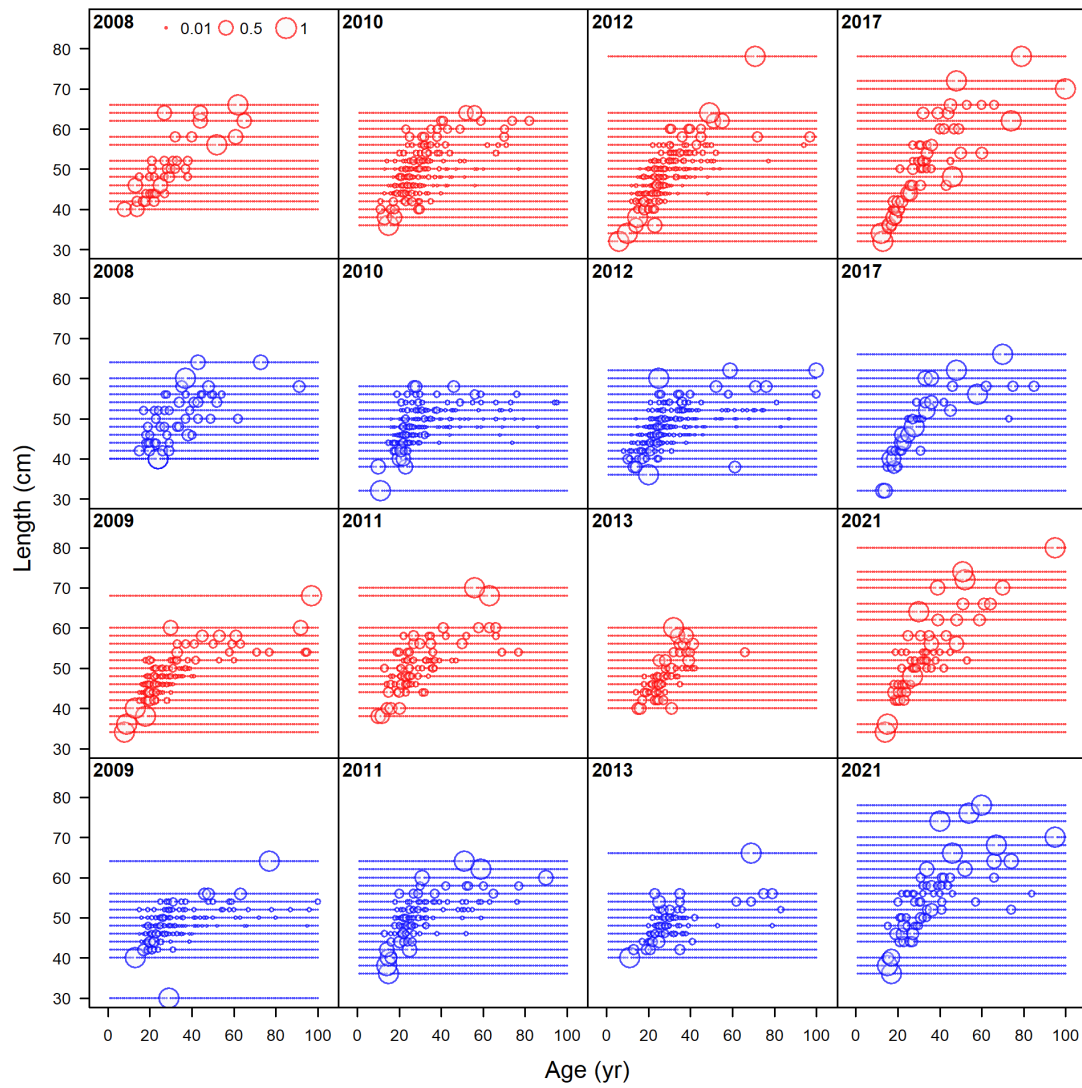


Figure 23: Conditional ages-at-length composition data for non-trawl fleet.

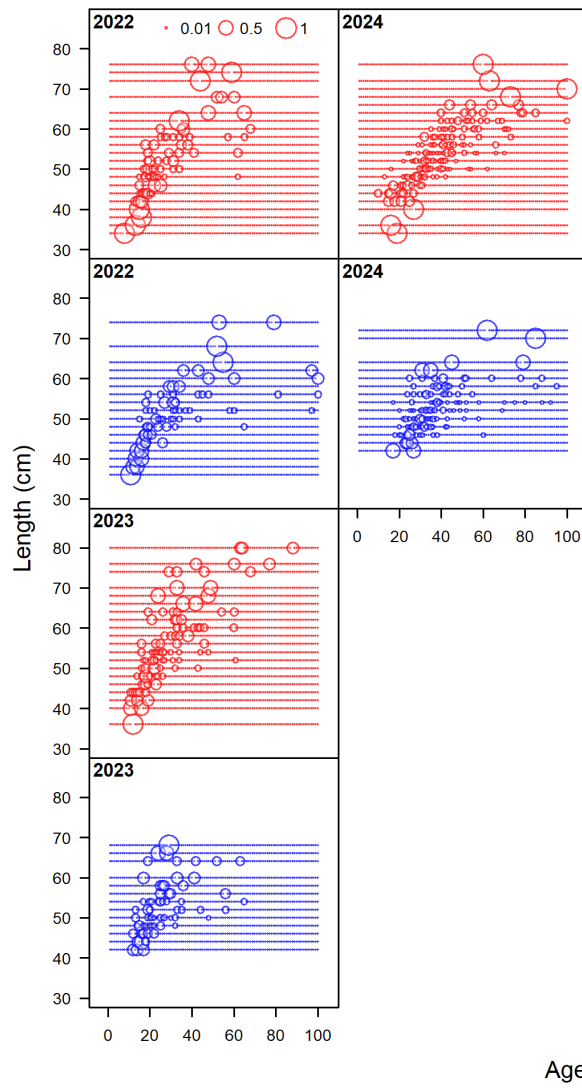


Figure 24: Conditional ages-at-length composition data for non-trawl fleet, continued.

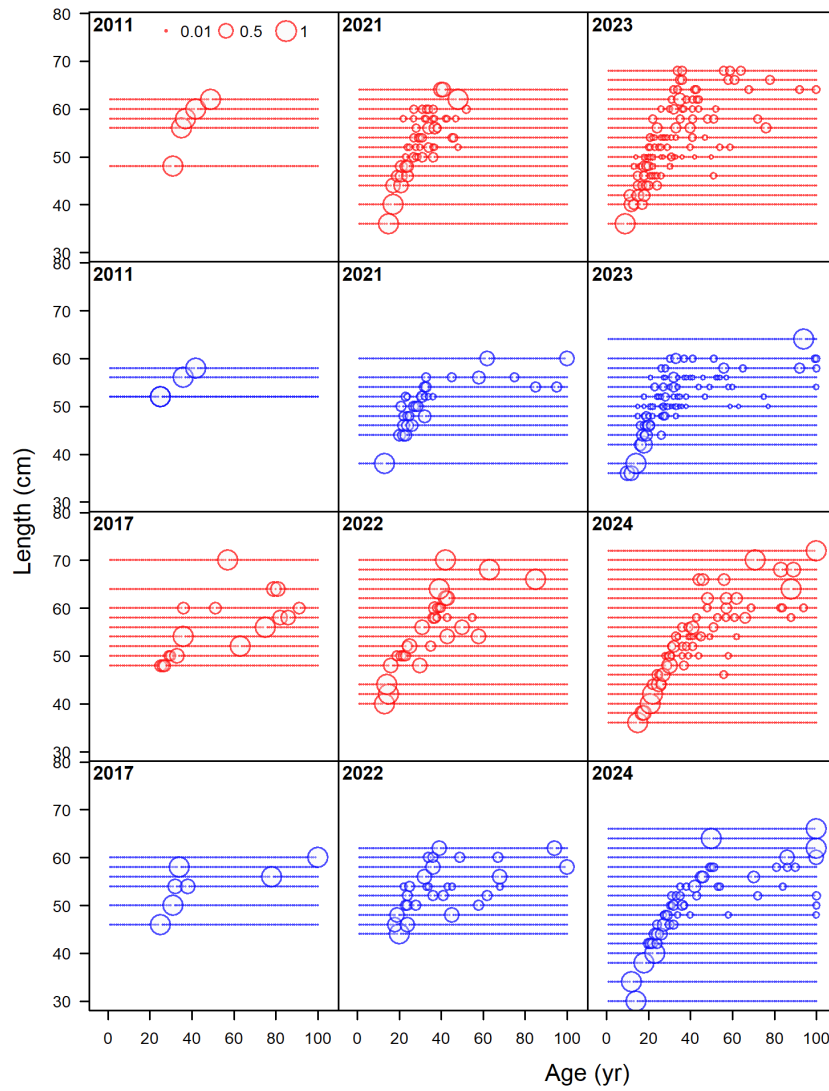


Figure 25: Conditional ages-at-length composition data for Midwater trawl fleet.

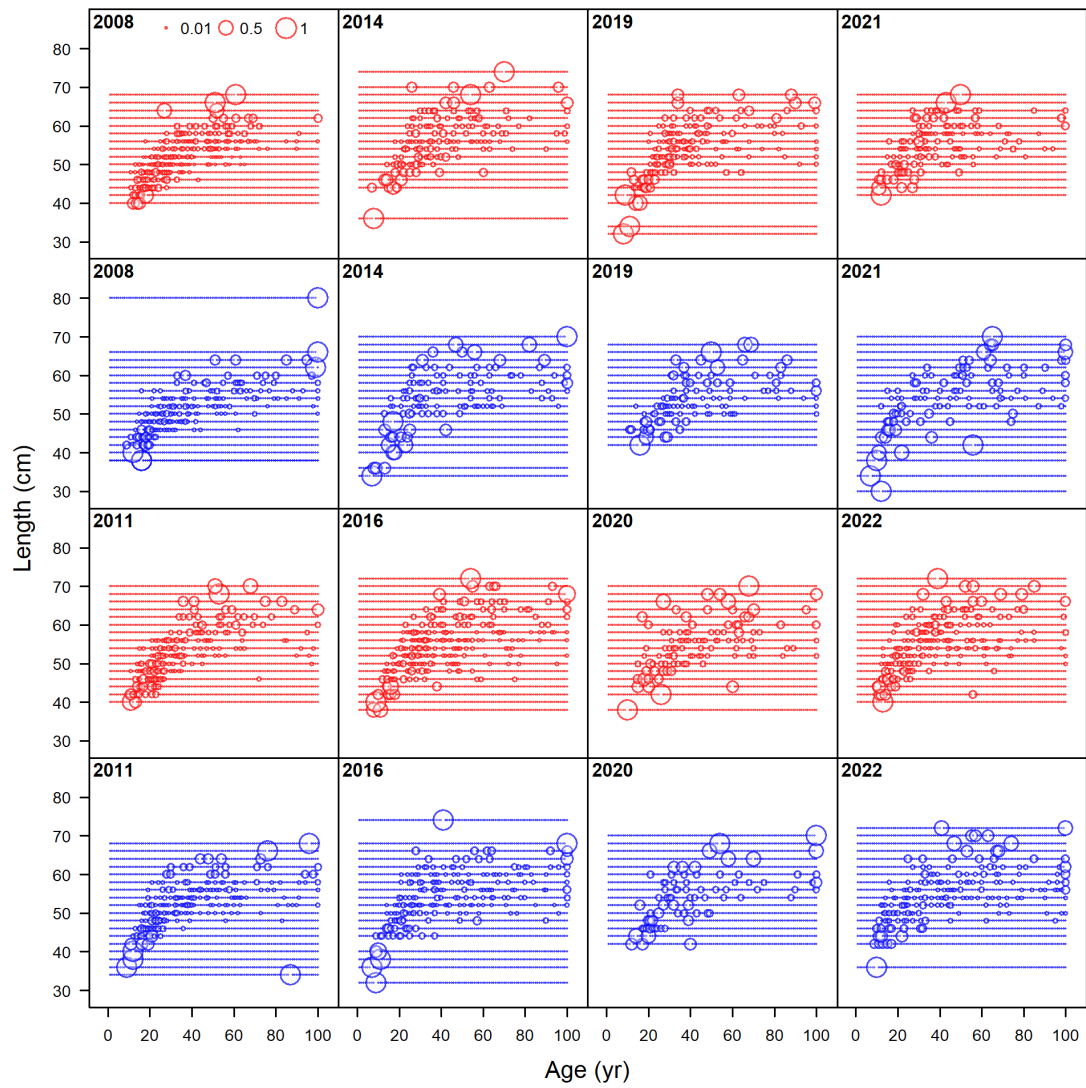


Figure 26: Conditional ages-at-length composition data for At-Sea Hake fleet.

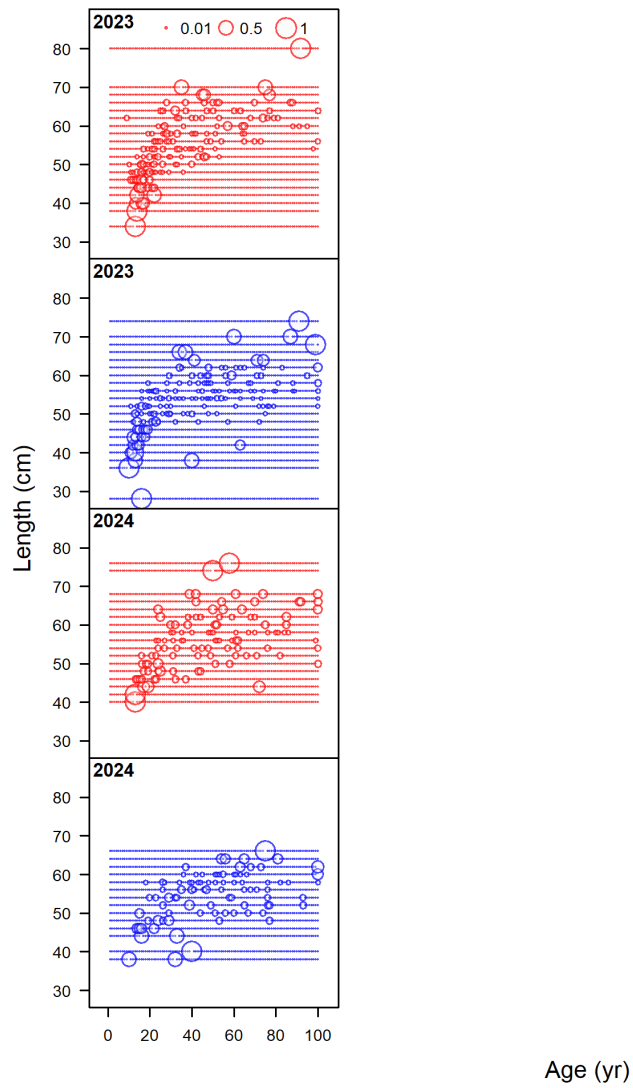


Figure 27: Conditional ages-at-length composition data for At-Sea Hake fleet, continued.

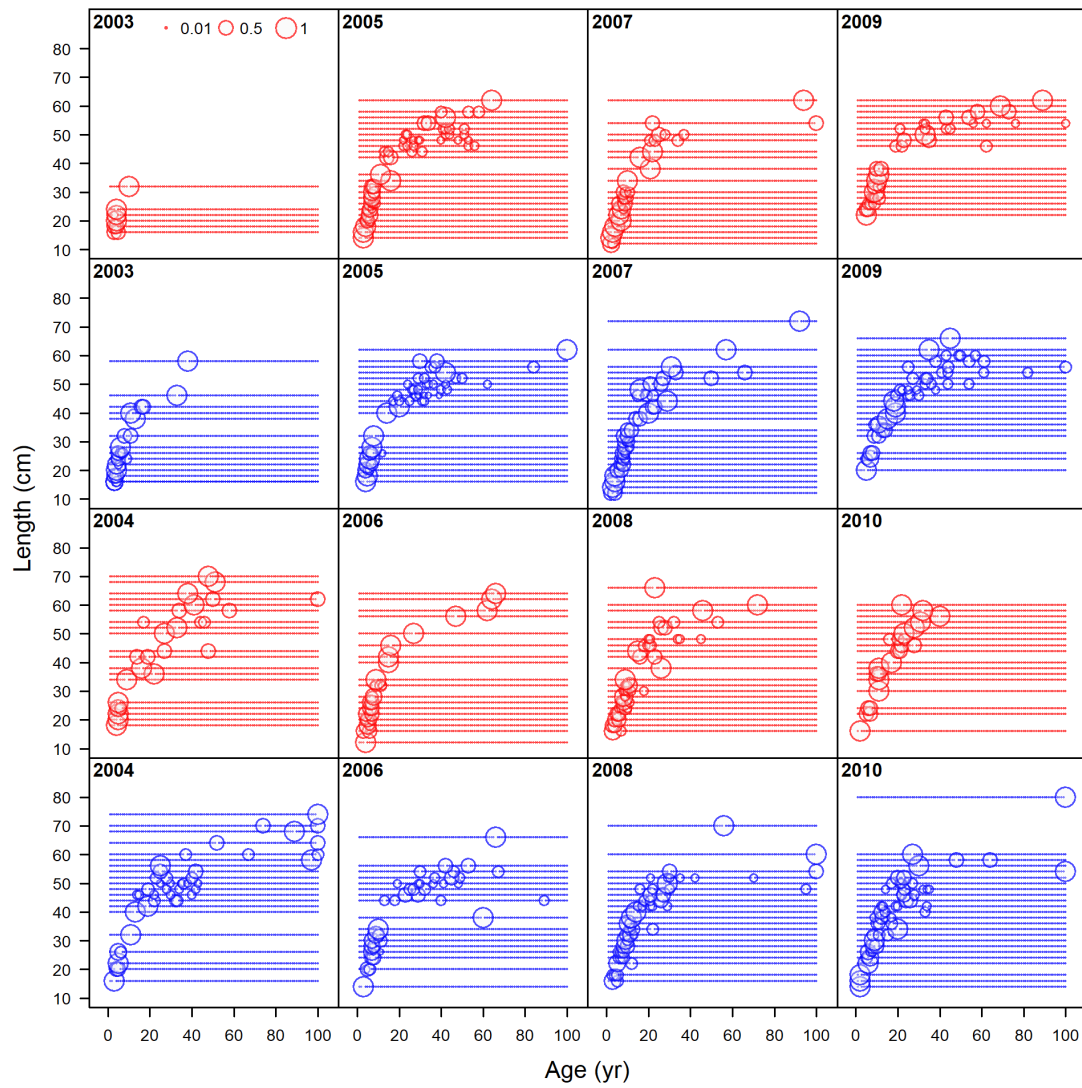


Figure 28: Conditional ages-at-length composition data for WCGBTS.

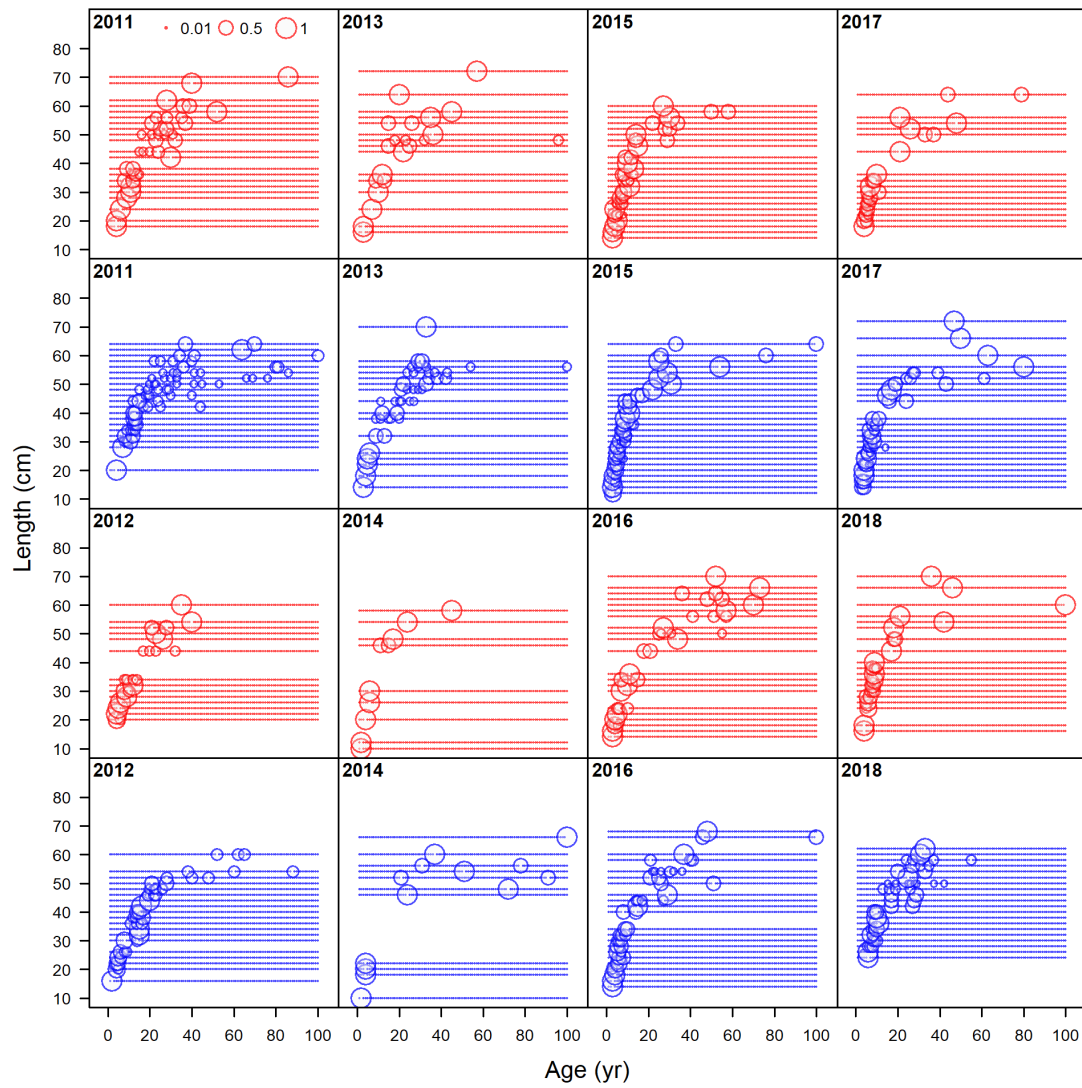


Figure 29: Conditional ages-at-length composition data for WCGBTS, continued.

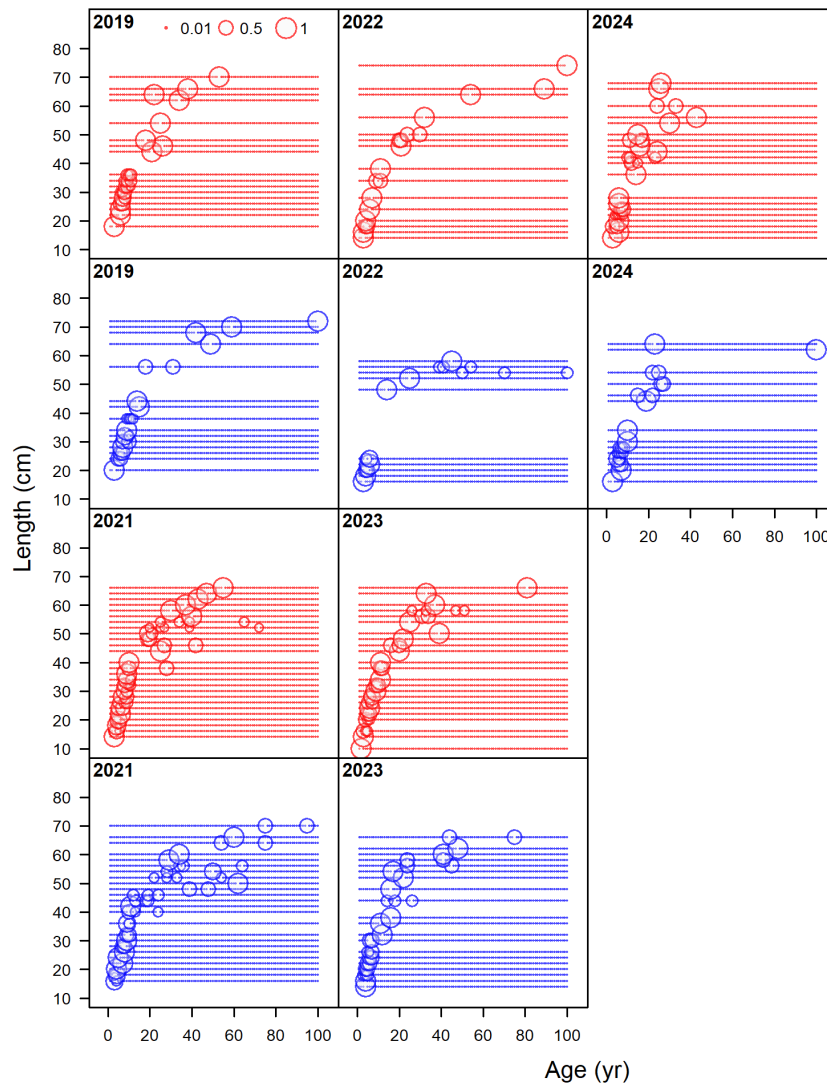


Figure 30: Conditional ages-at-length composition data for WCGBTS, continued.



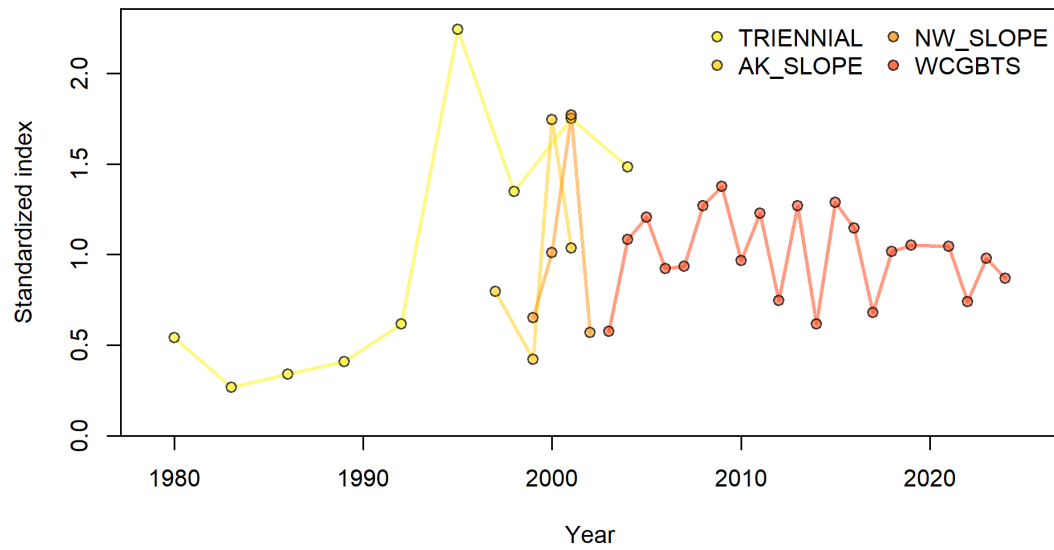


Figure 31: Standardized indices.

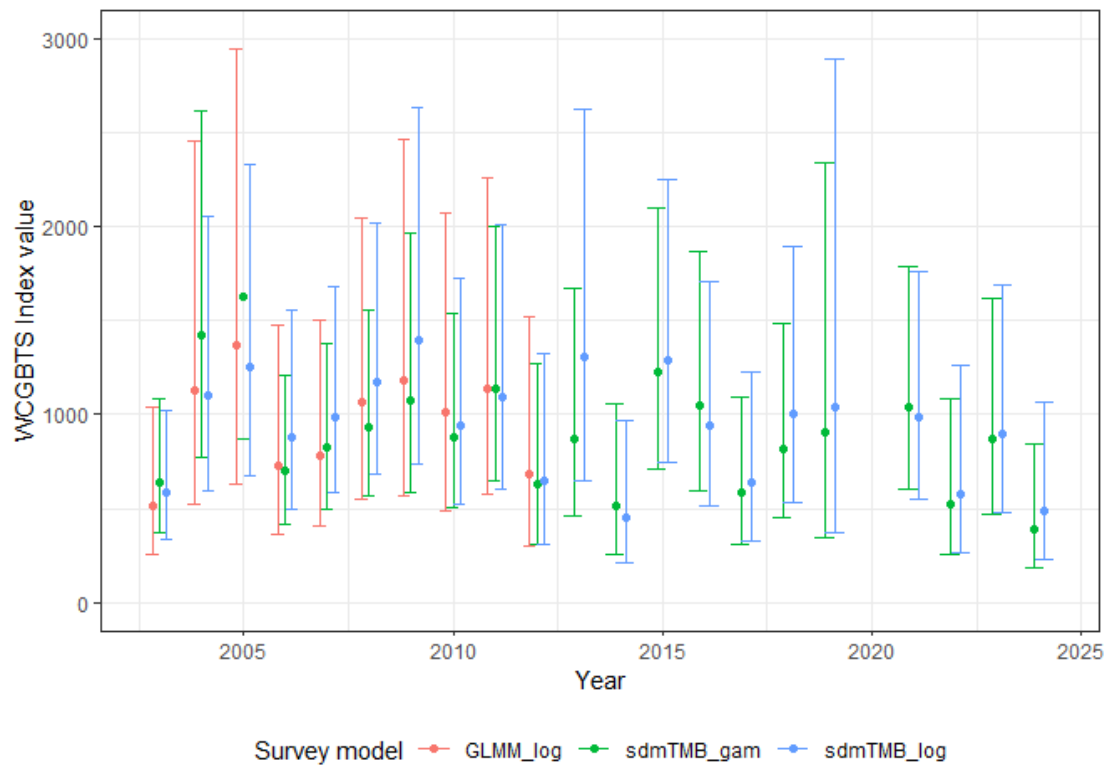


Figure 32: Comparison of the West Coast Groundfish Bottom Trawl Survey (WCGBTS) index from the previous assessment (GLMM) and the WCGBTS index of abundance used in this assessment (sdmTMB, lognormal distribution).

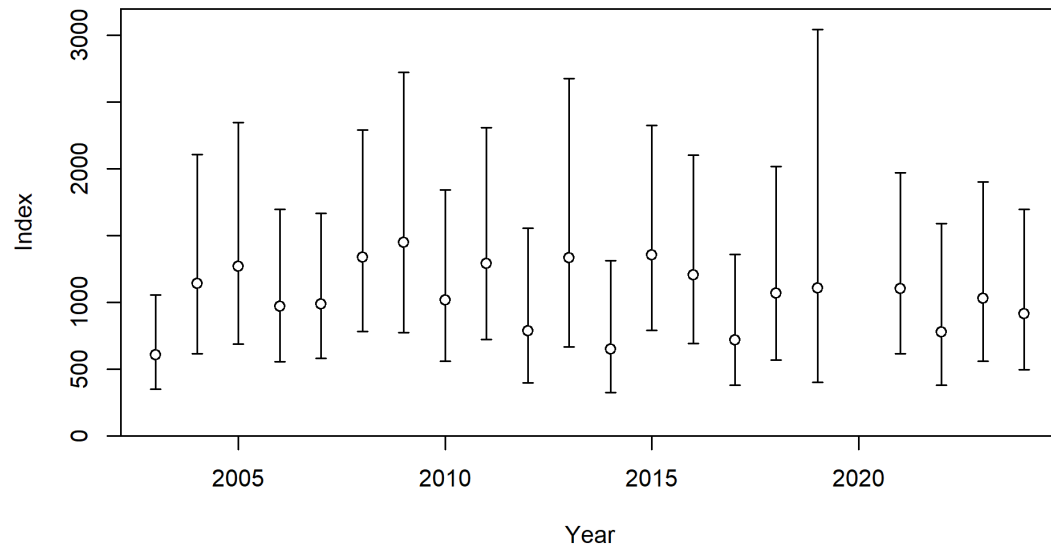


Figure 33: WCGBTS index.

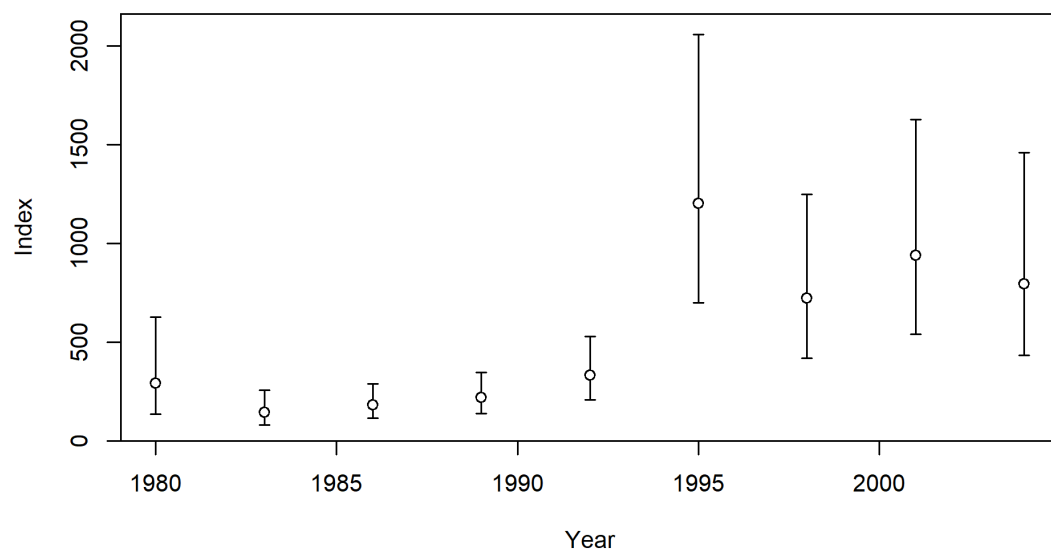


Figure 34: Triennial Survey index.

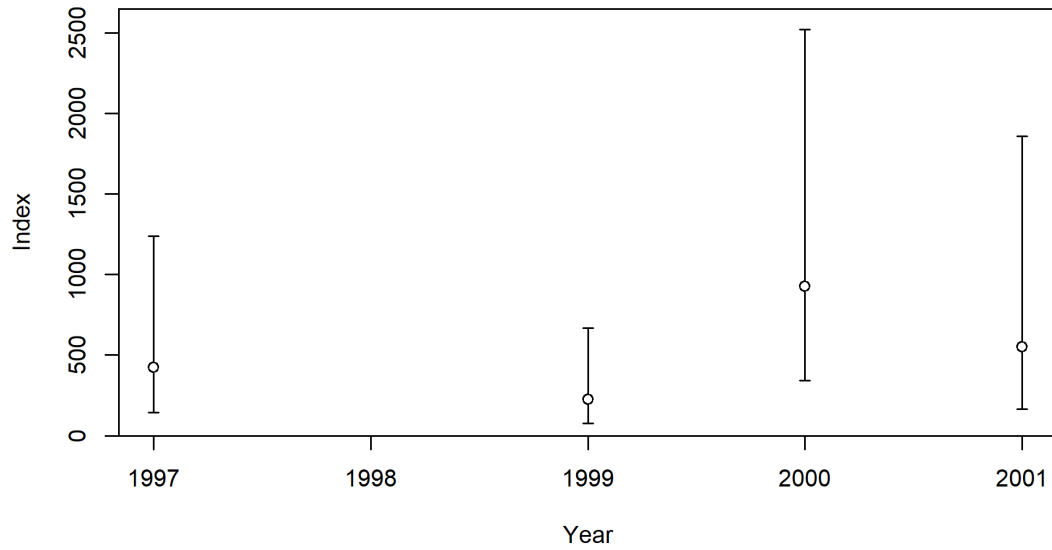


Figure 35: AFSC Slope Survey index.

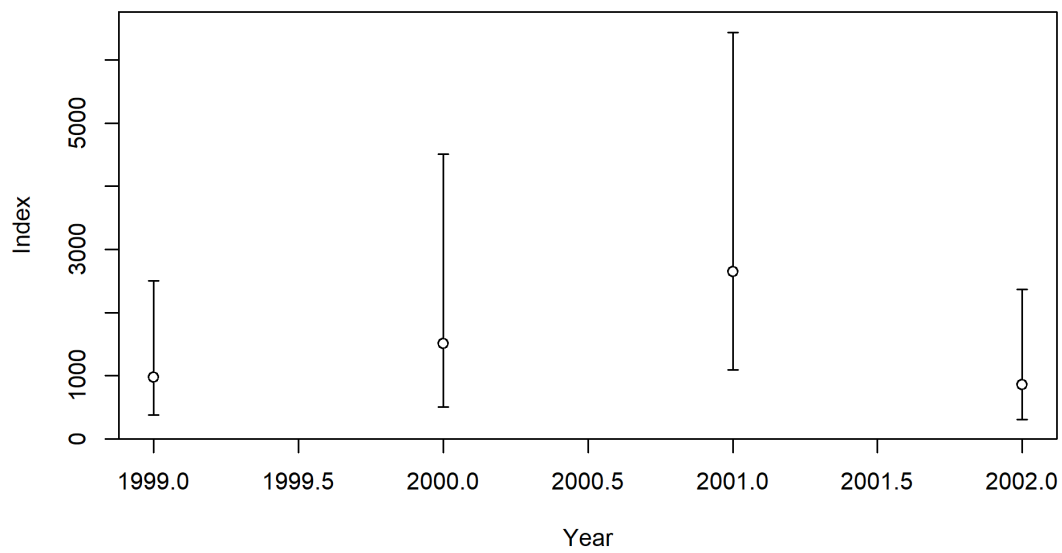


Figure 36: NWFSC Slope Survey index.

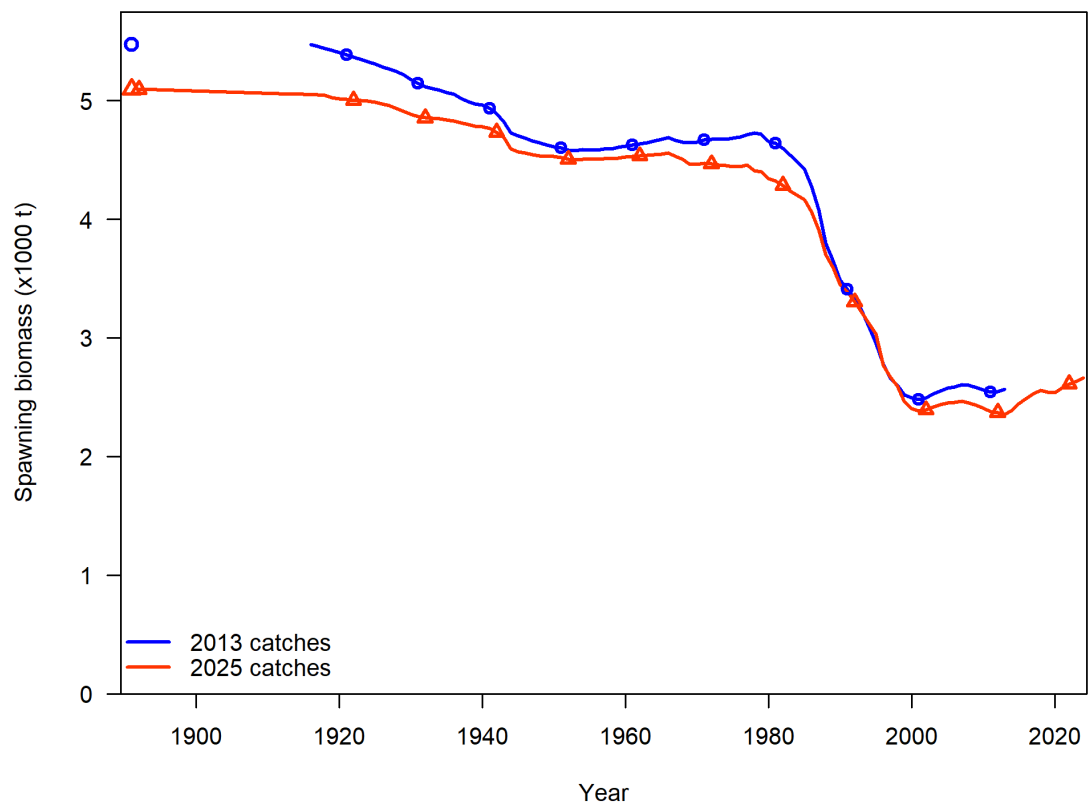


Figure 37: Comparison of spawning output using updated catches vs using catches from the 2013 Rougheye/Blackspotted Rockfishes assessment.

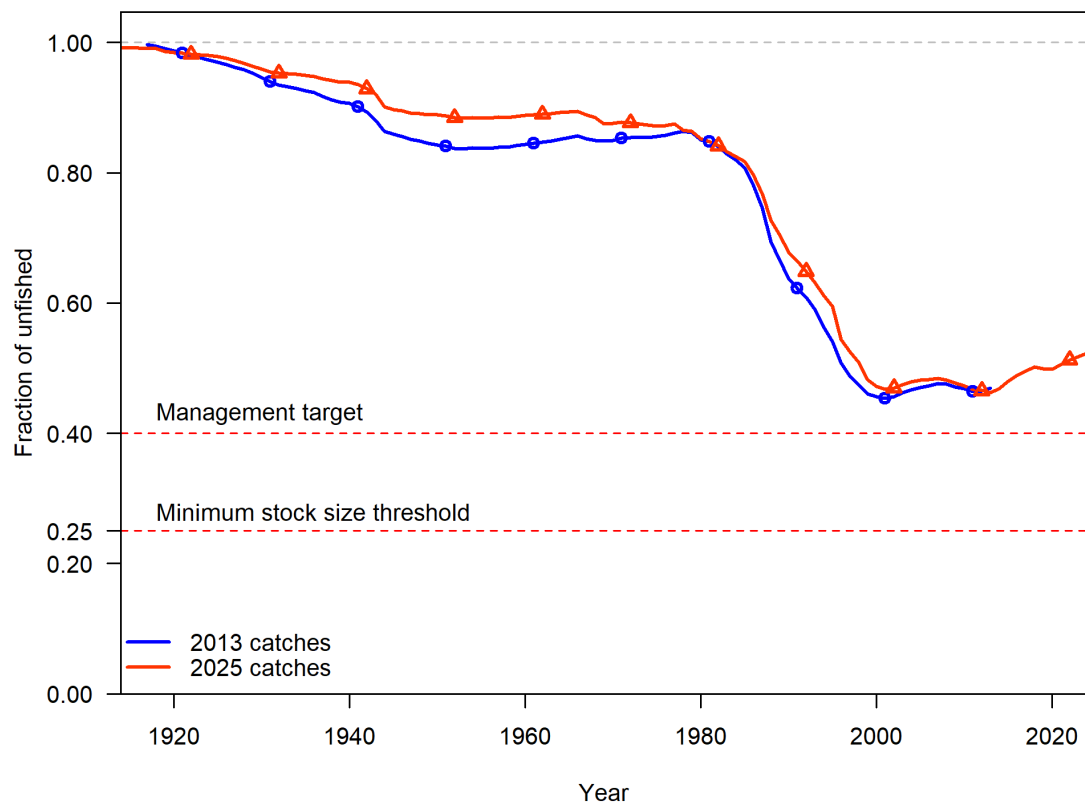


Figure 38: Comparison of relative spawning output using updated catches vs using catches from the 2013 Rougheye/Blackspotted Rockfishes assessment.

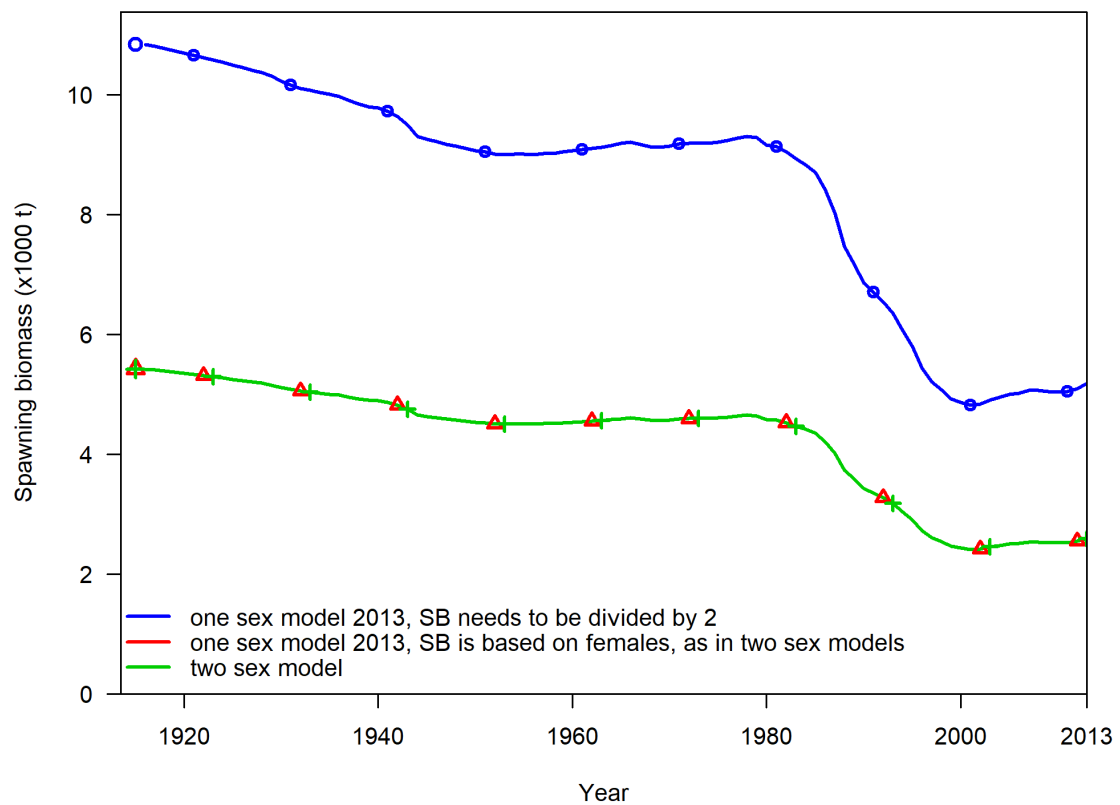


Figure 39: Comparison of spawning output using the 1 sex and 2 sexes set to equal values models based on the 2013 Rougheye/Blackspotted Rockfishes assessment data. The 1 sex model has double the biomass because it includes both females and males.

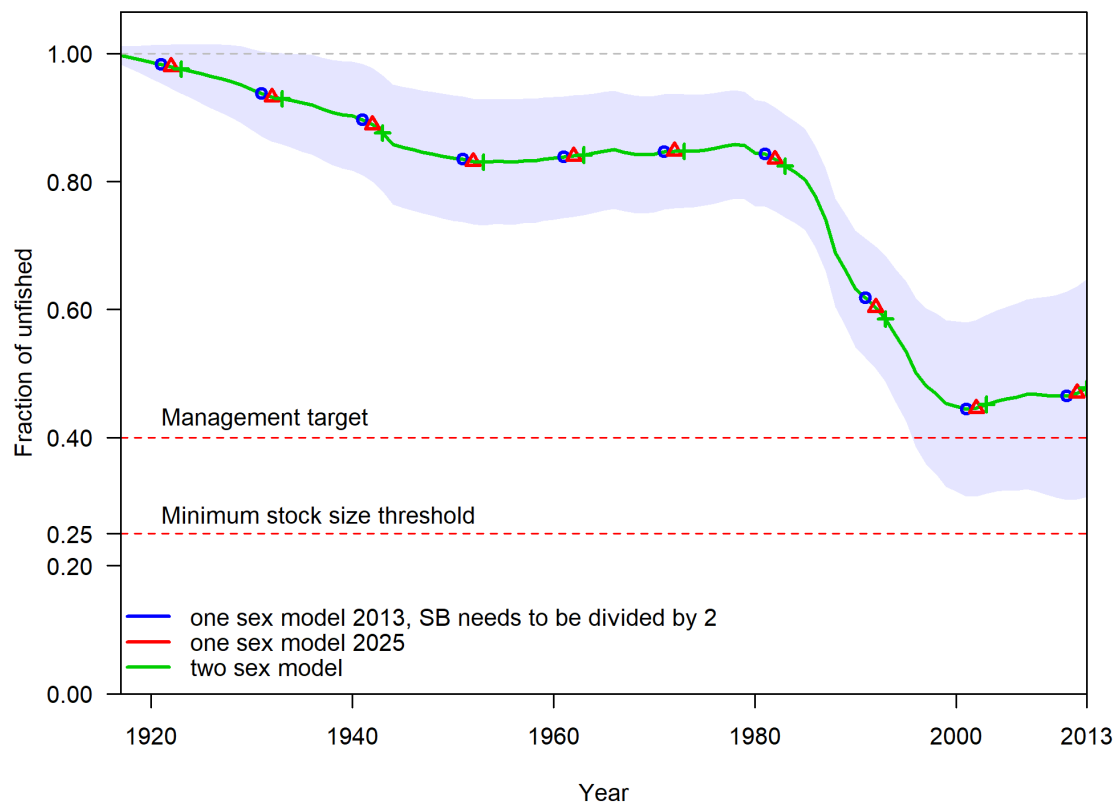


Figure 40: Comparison of spawning output using the 1 sex and 2 sexes set to equal values models based on the 2013 Rougheye/Blackspotted Rockfishes assessment data.



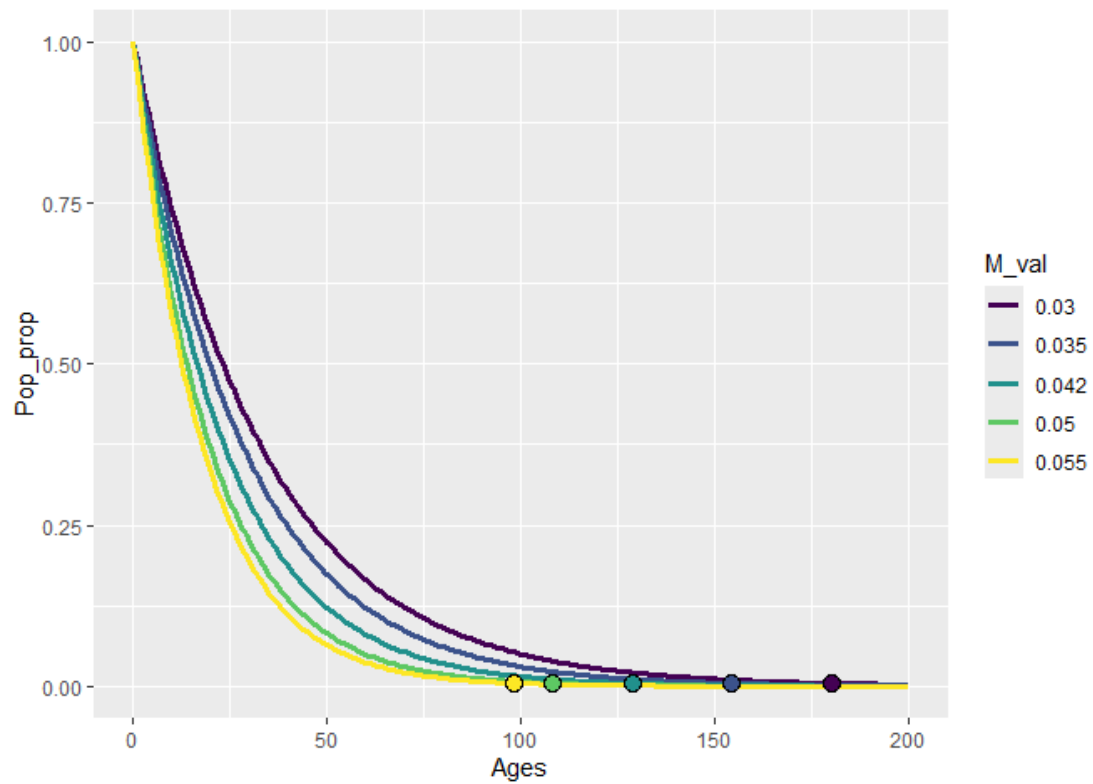


Figure 41: Natural mortality curves by age in years for values of natural mortality used in various Rougheye/Blackspotted Rockfish stock assessments. Dots indicate the range of assumed maximum ages using the equation from Hamel and Cope 2022.

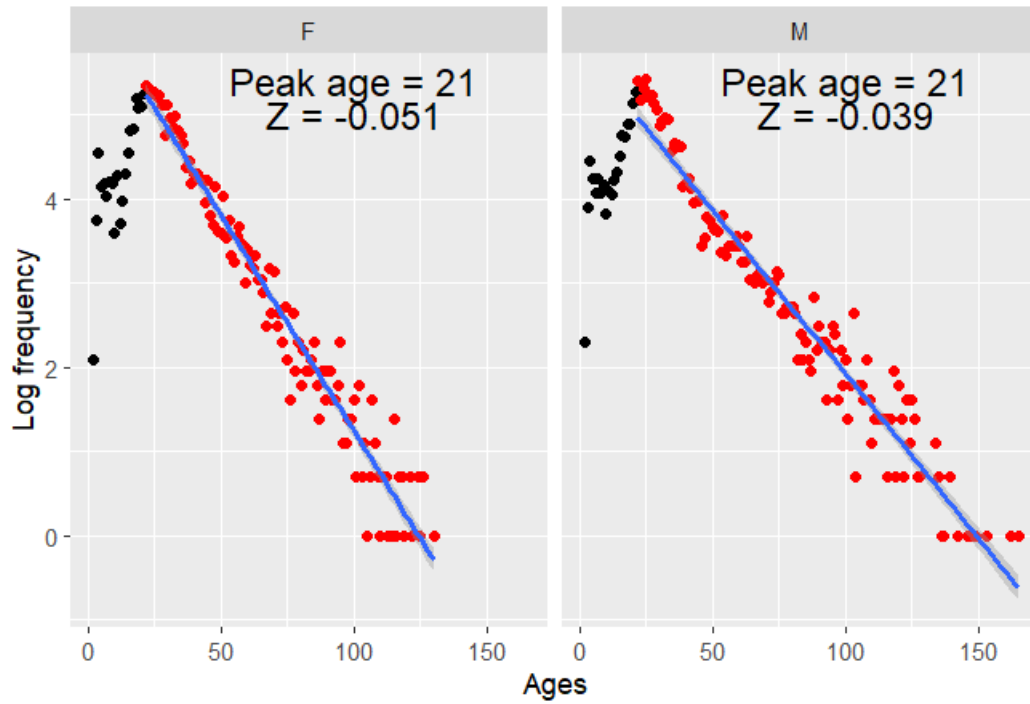


Figure 42: Catch curve (log abundance by age) analysis on aggregated ages over all age sources by sex (black points). The peak selected age was 21 for both sexes, so the linear model was run from age 21 until the oldest age (red points). The slope of the linear model is equal to the estimate of an aggregate total mortality ( $Z$ ).

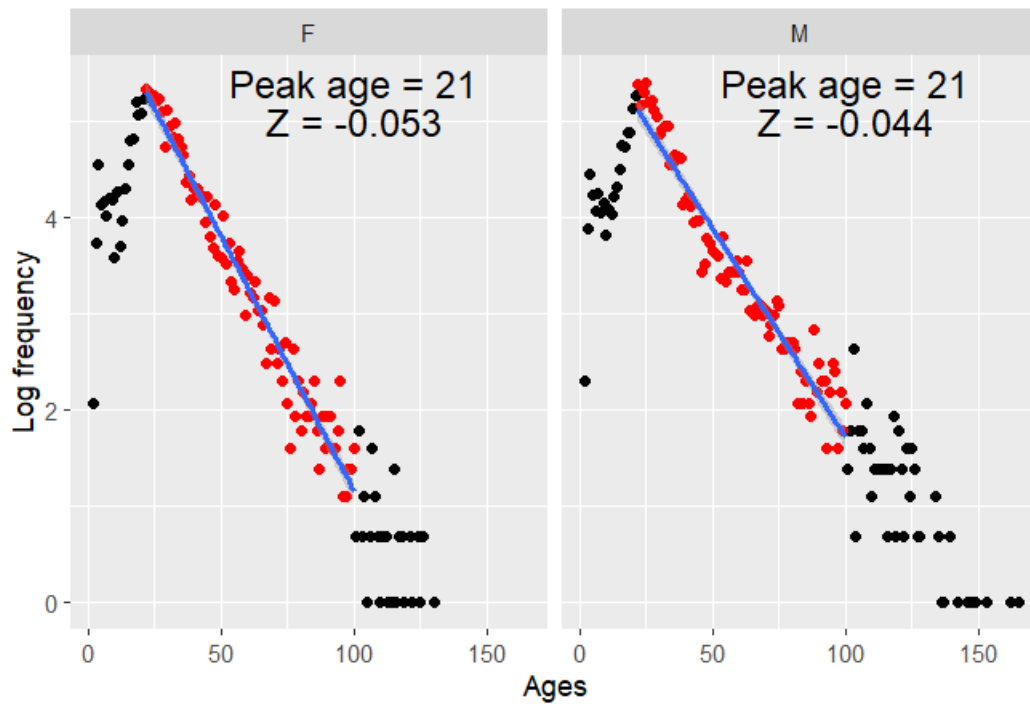


Figure 43: Catch curve (log abundance by age) analysis on aggregated ages over all age sources by sex (black points). The peak selected age was 21 for both sexes with a max age of 100, so the linear model was run from age 21 until age 100 (red points). The slope of the linear model is equal to the estimate of an aggregate total mortality ( $Z$ ).

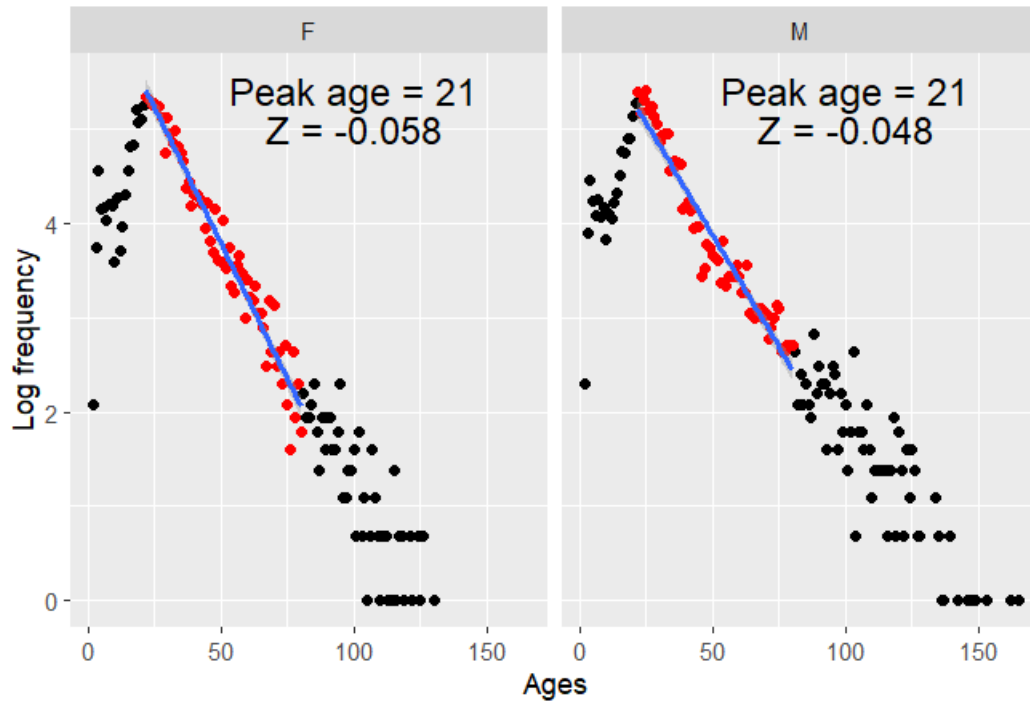


Figure 44: Catch curve (log abundance by age) analysis on aggregated ages over all age sources by sex (black points). The peak selected age was 21 for both sexes with a max age of 80, so the linear model was run from age 21 until age 80 (red points). The slope of the linear model is equal to the estimate of an aggregate total mortality ( $Z$ ).

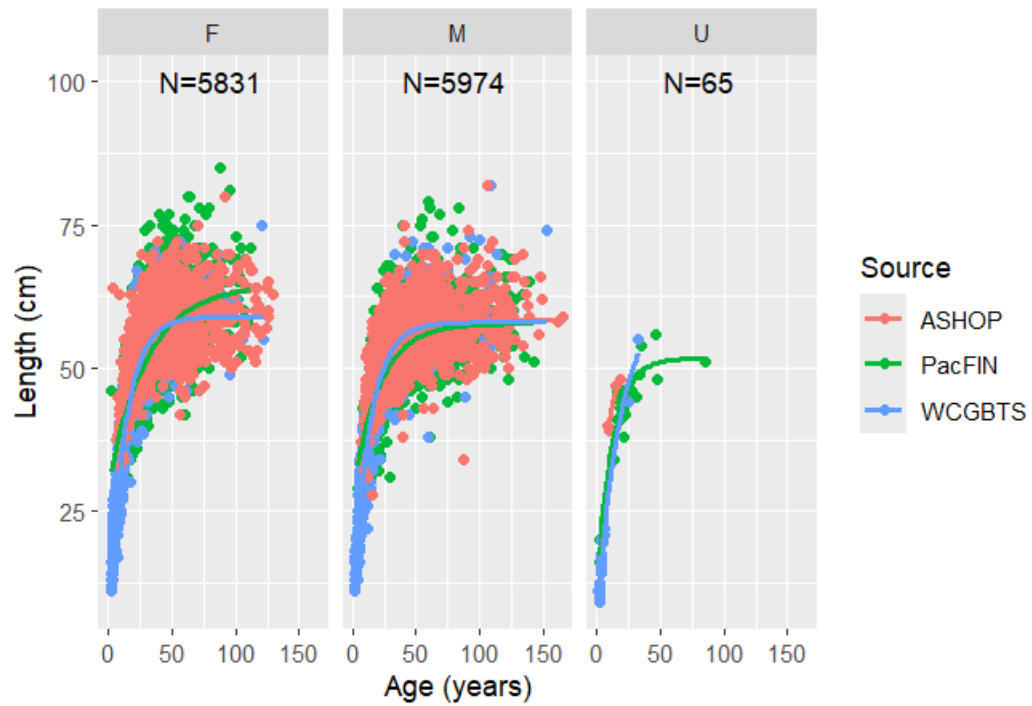


Figure 45: Age and length data, with fitted von Bertalanffy growth curves, by sex and data source for the Rougheye/Blackspotted rockfish complex. Sample sizes (N) are also provided.

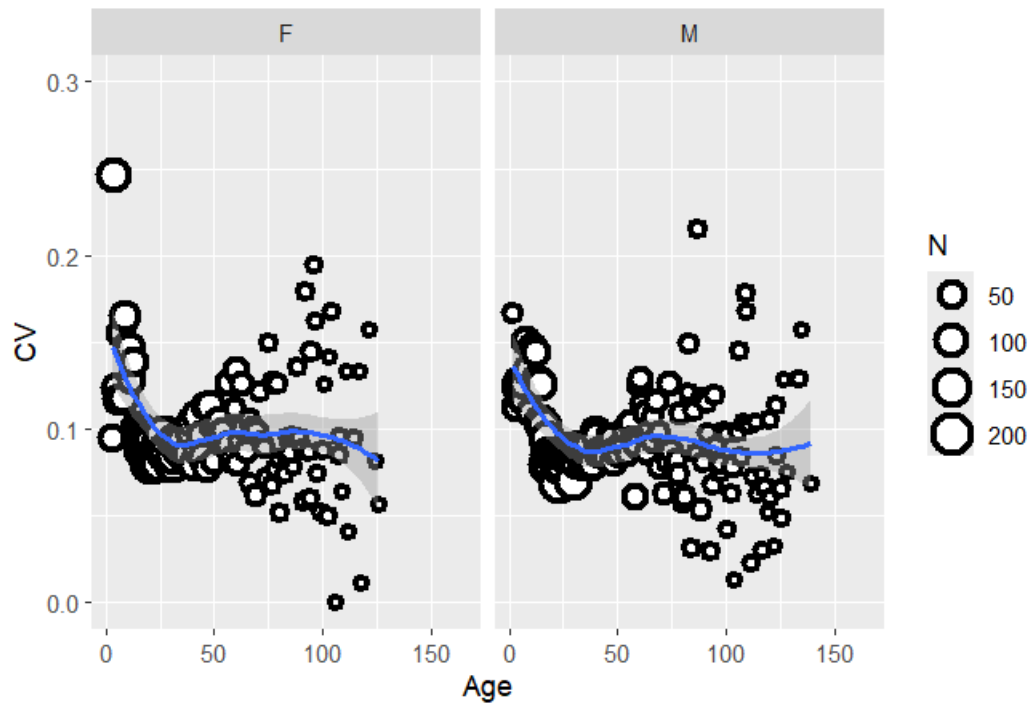


Figure 46: Coefficient of variation by age and sex for all sources of Rougheye/Blackspotted rockfishes ages. Sample sizes (N) are also indicated by size of the point. The line is a smoothed loess (polynomial) line that gives a moving average of CV by age and sex.

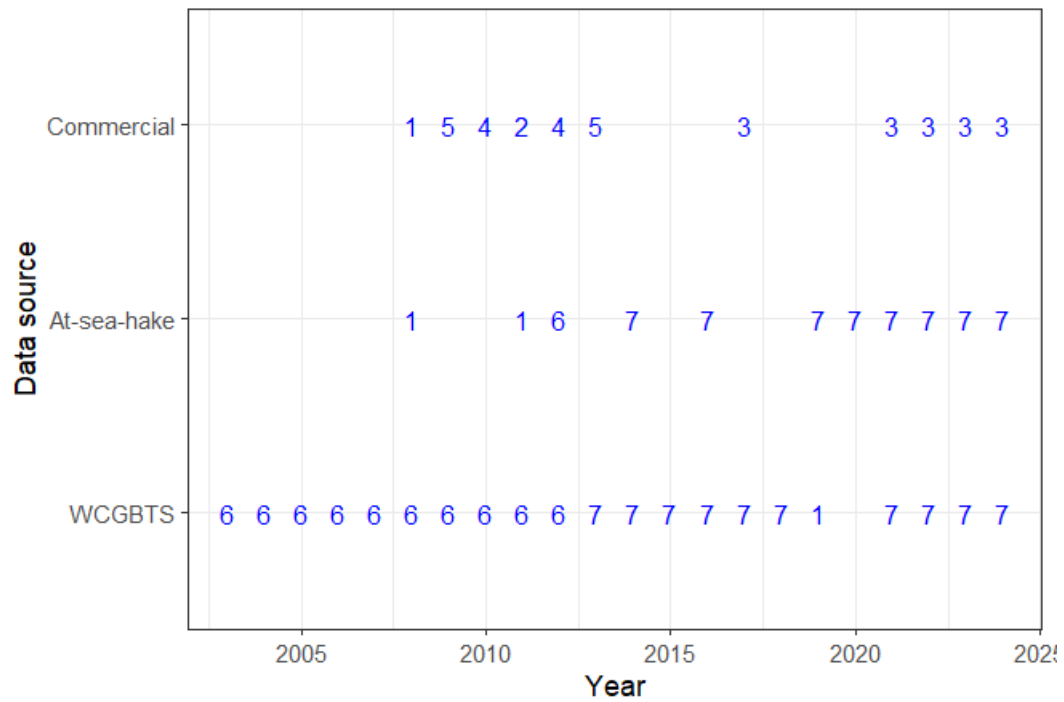


Figure 47: Ageing error matrix assignments by year and data source. The number indicates which ageing error matrix was used for conditional ages within those years and data sources. 'Commercial' is a combination of all commercial fleets.

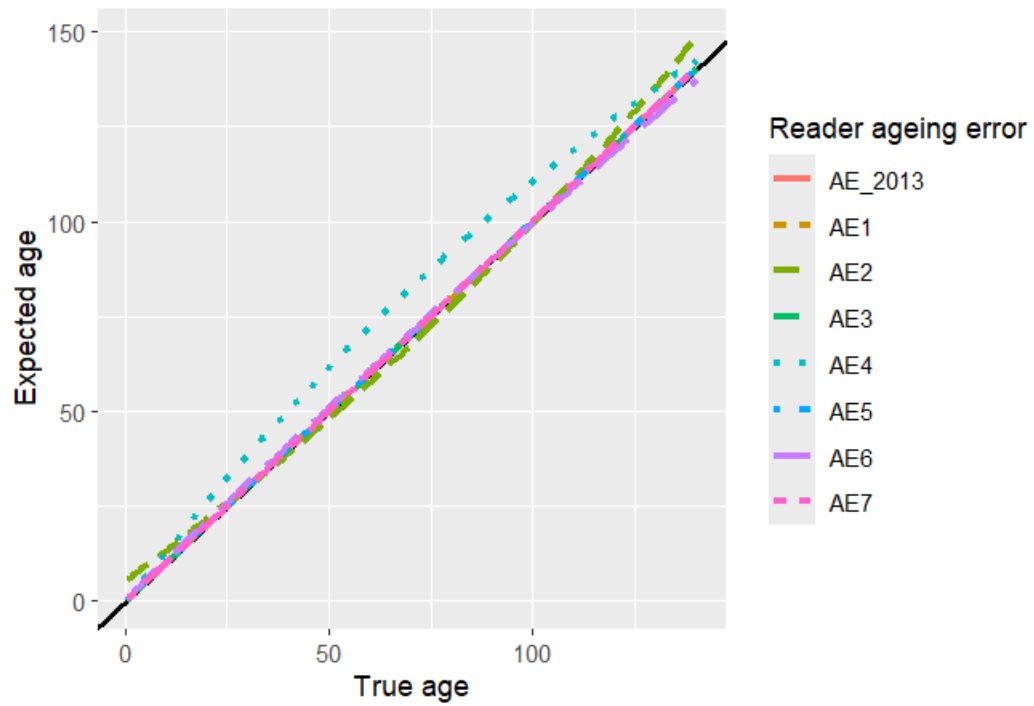


Figure 48: Estimated bias used for each of the seven ageing error matrices.



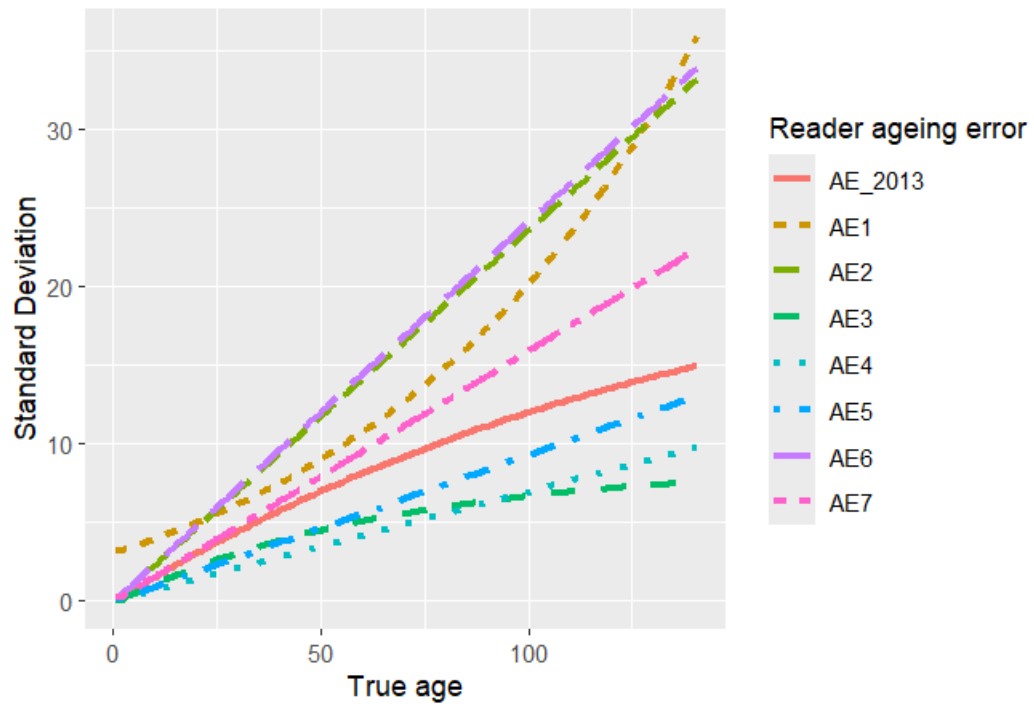


Figure 49: Estimated imprecision (as a standard deviation) used for each of the seven ageing error matrices.

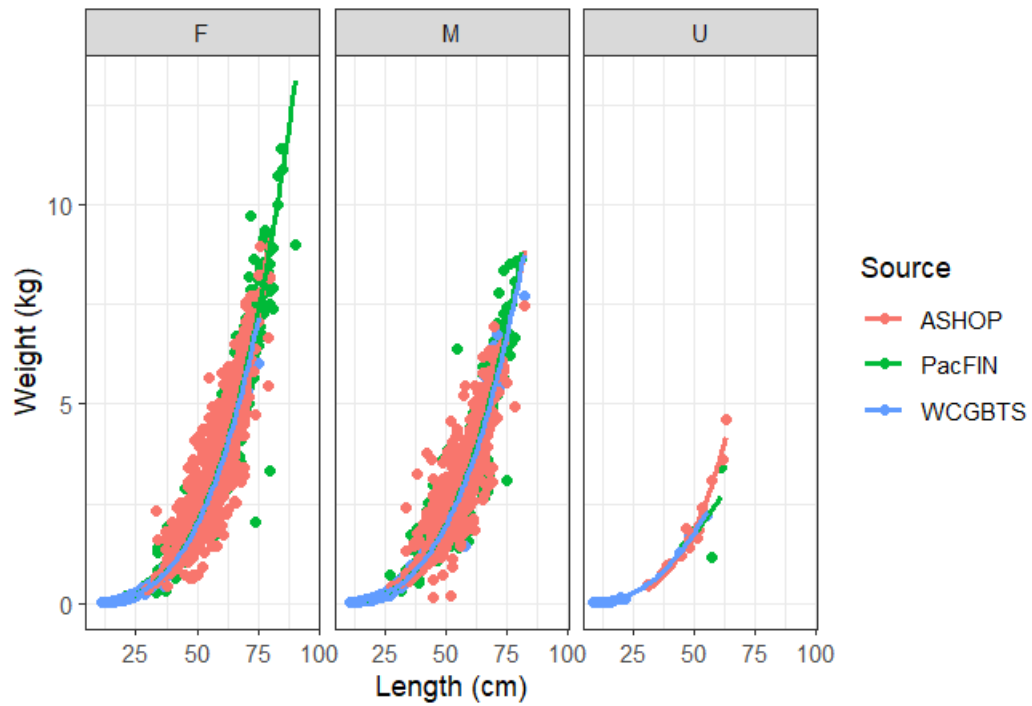


Figure 50: Length and weight samples by sex and data source. Lines are the power function fits by data source.

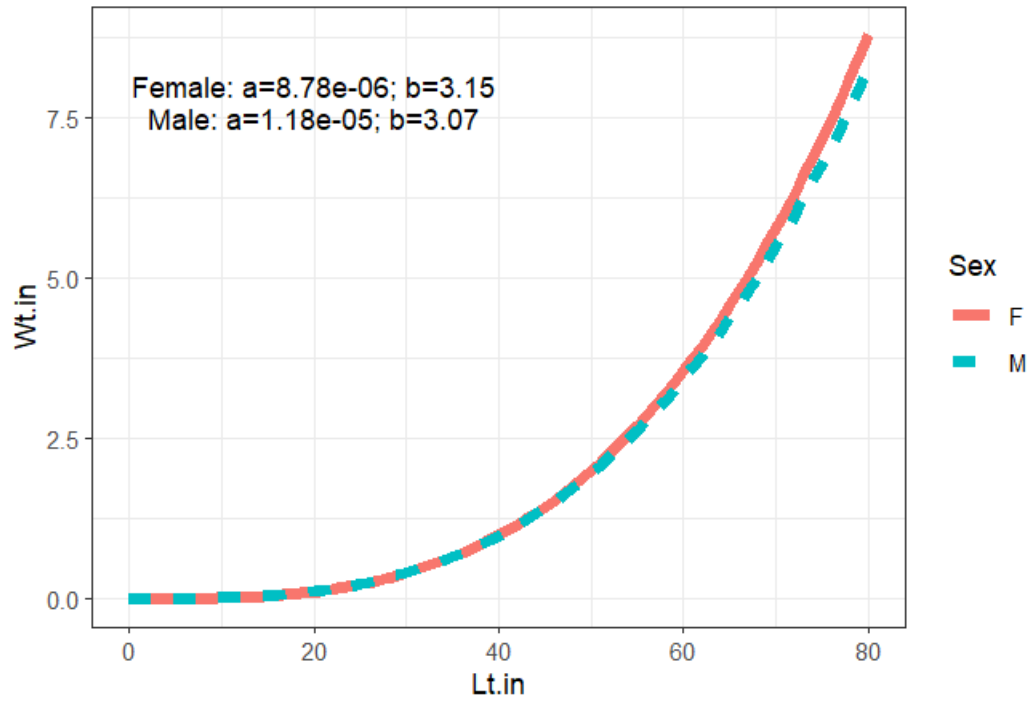


Figure 51: Realized length and weight relationships for female and male Rough-eye/Blackspotted Rockfishes.

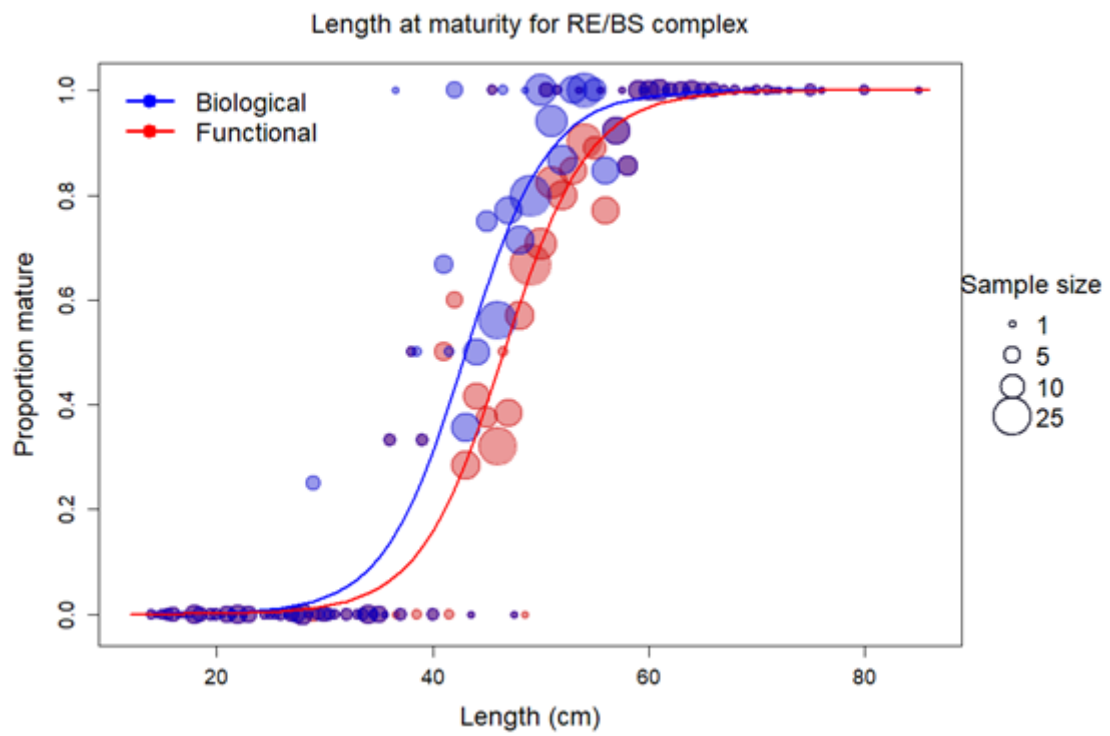


Figure 52: Placeholder.

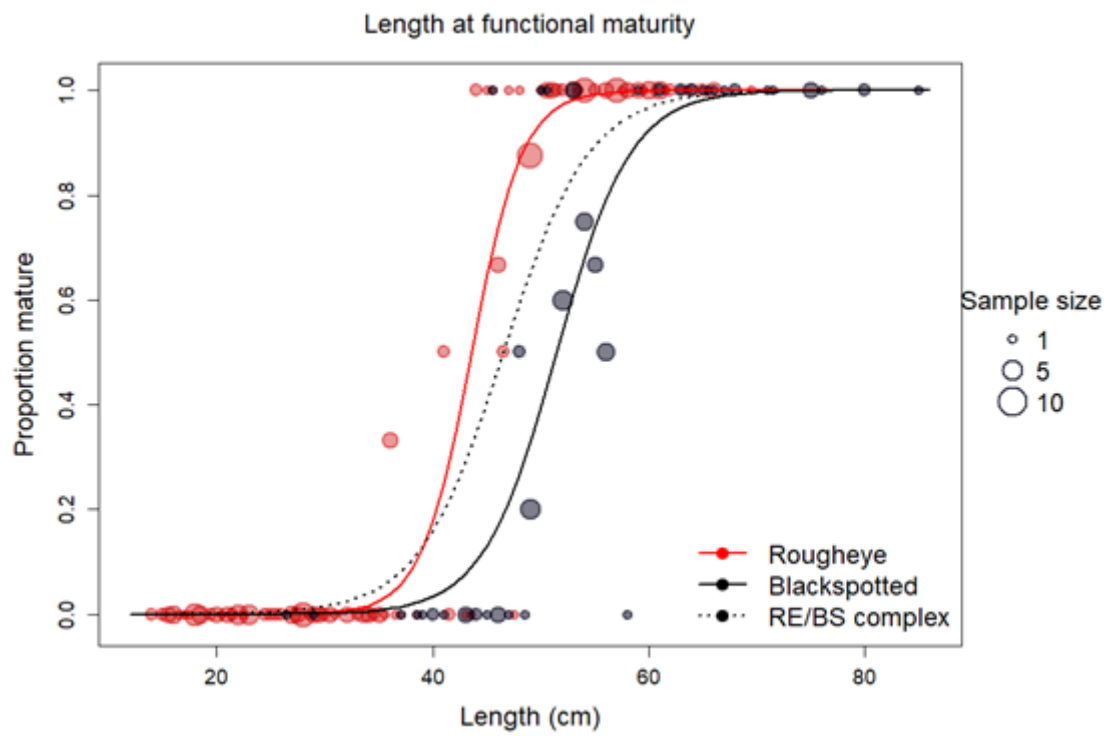


Figure 53: Placeholder.

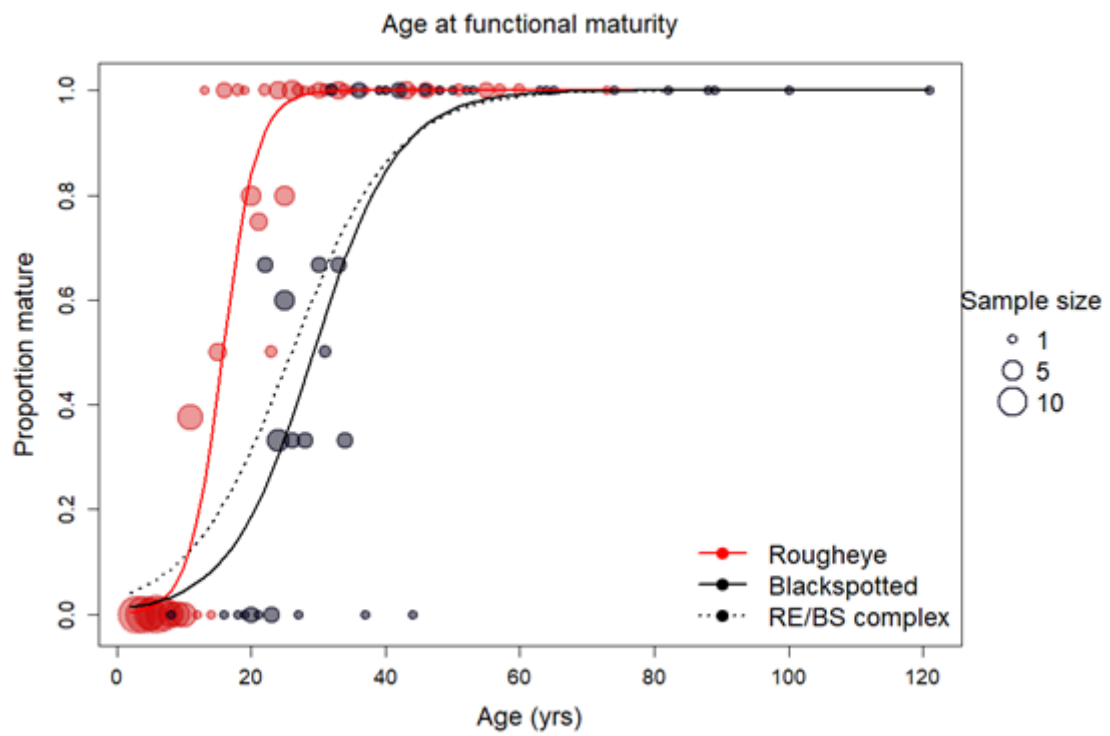


Figure 54: Placeholder.

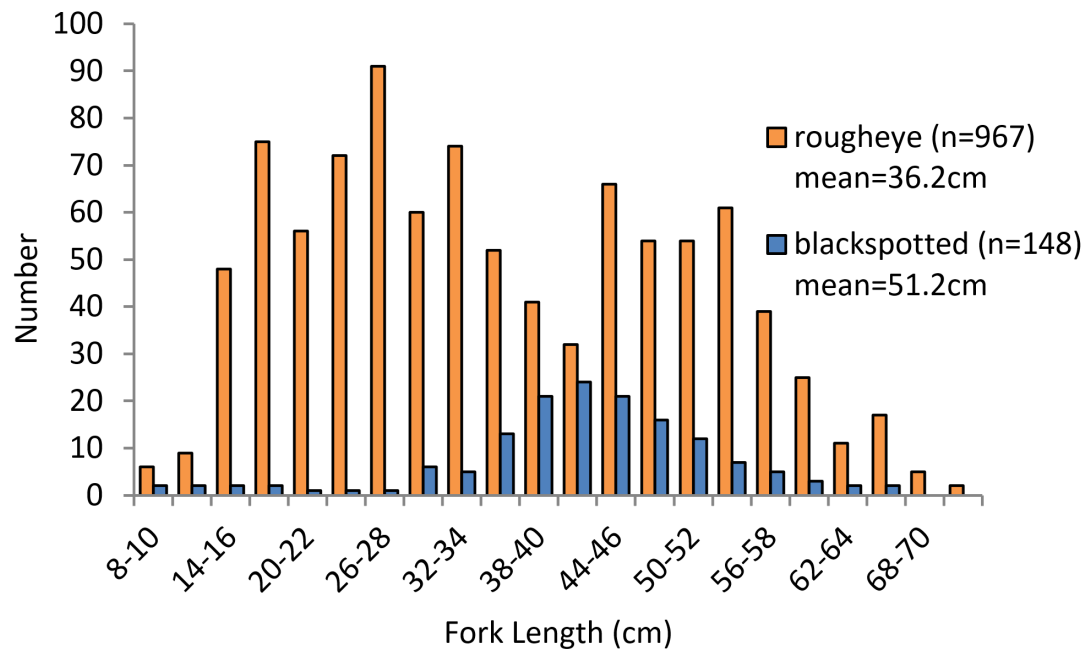


Figure 55: Length distribution of Rougheye Rockfish and Blackspotted Rockfish identified to species using genetics (pers. comm. P. Prey, NWFSC).

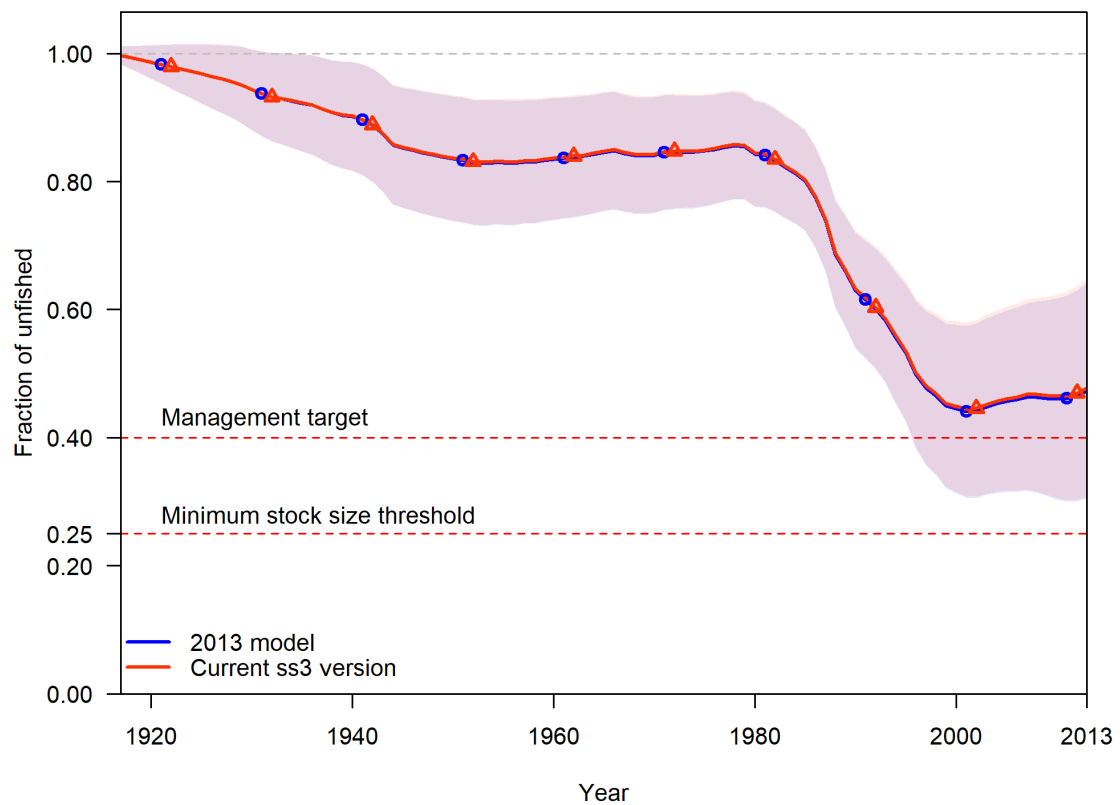


Figure 56: Estimates of relative stock size (current spawning output/unfished spawning output) for the Rougheye/Blackspotted rockfish complex in U.S. west coast waters from the 2013 assessment, and compared to the using the same data in the newest version of SS3 (3.30.22.1).



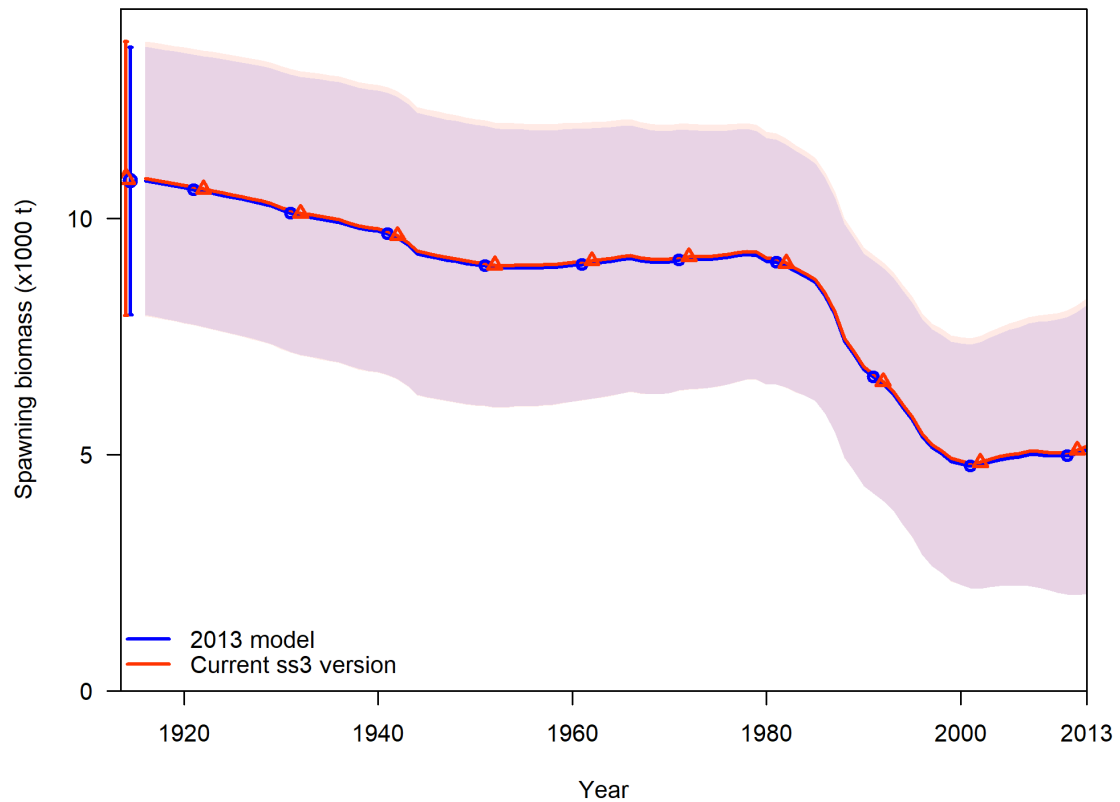


Figure 57: Estimates of spawning output for the Rougheye/Blackspotted rockfish complex in U.S. west coast waters from the 2013 assessment, and compared to the same data in the newest version of SS3 (3.30.22.1). Shading denotes 95% confidence intervals. Shading denotes 95% confidence intervals.

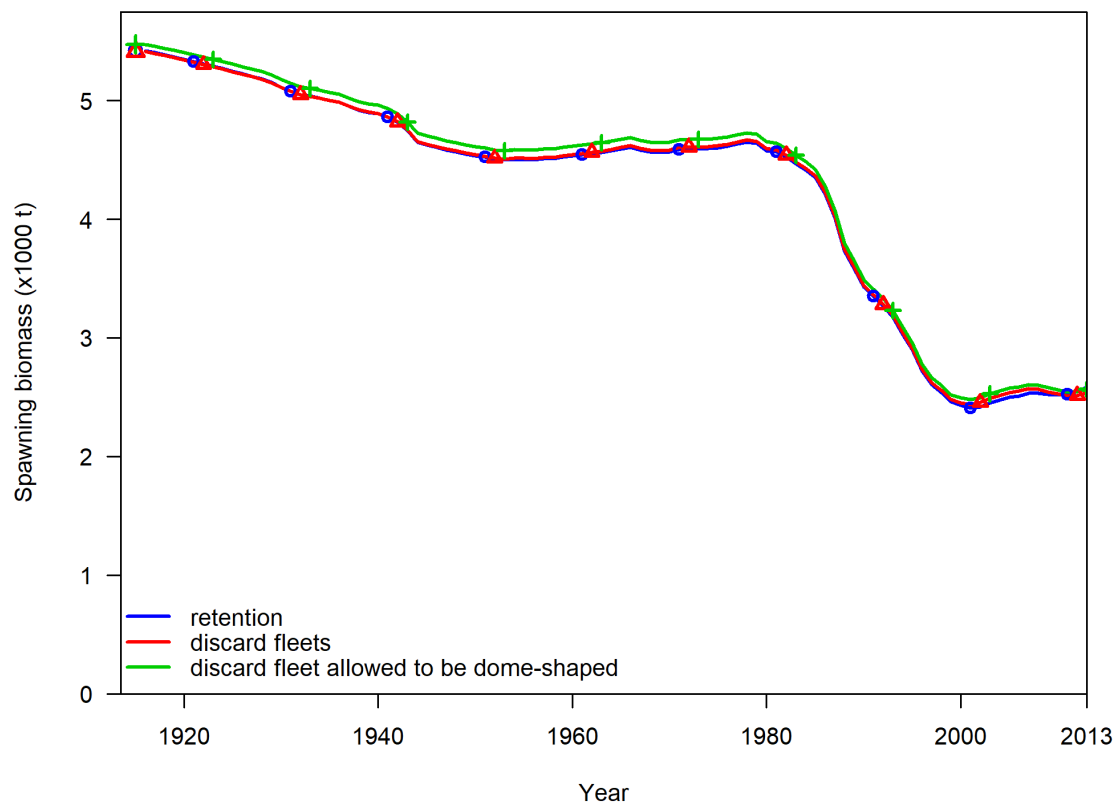


Figure 58: Comparison of spawning output using retention curves or discard fleets using the 2013 Rougheye/Blackspotted Rockfishes assessment.

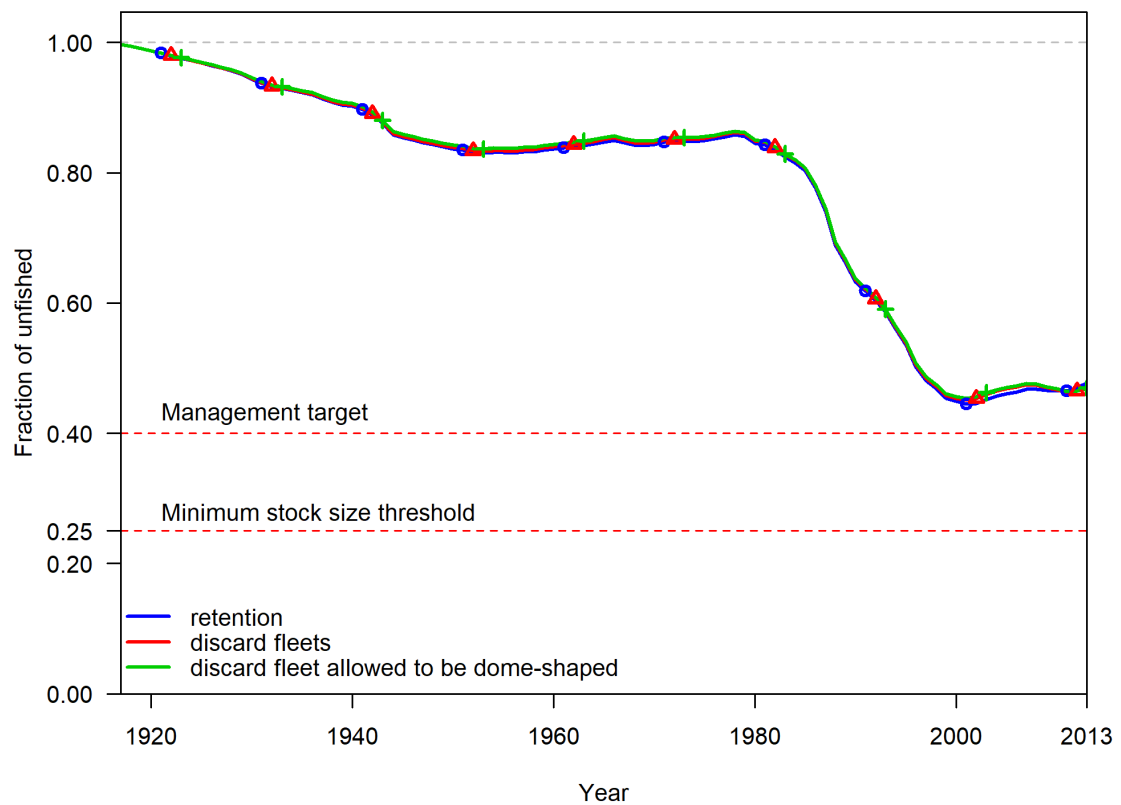


Figure 59: Comparison of relative spawning output using retention curves or discard fleets using the 2013 Rougheye/Blackspotted Rockfishes assessment.

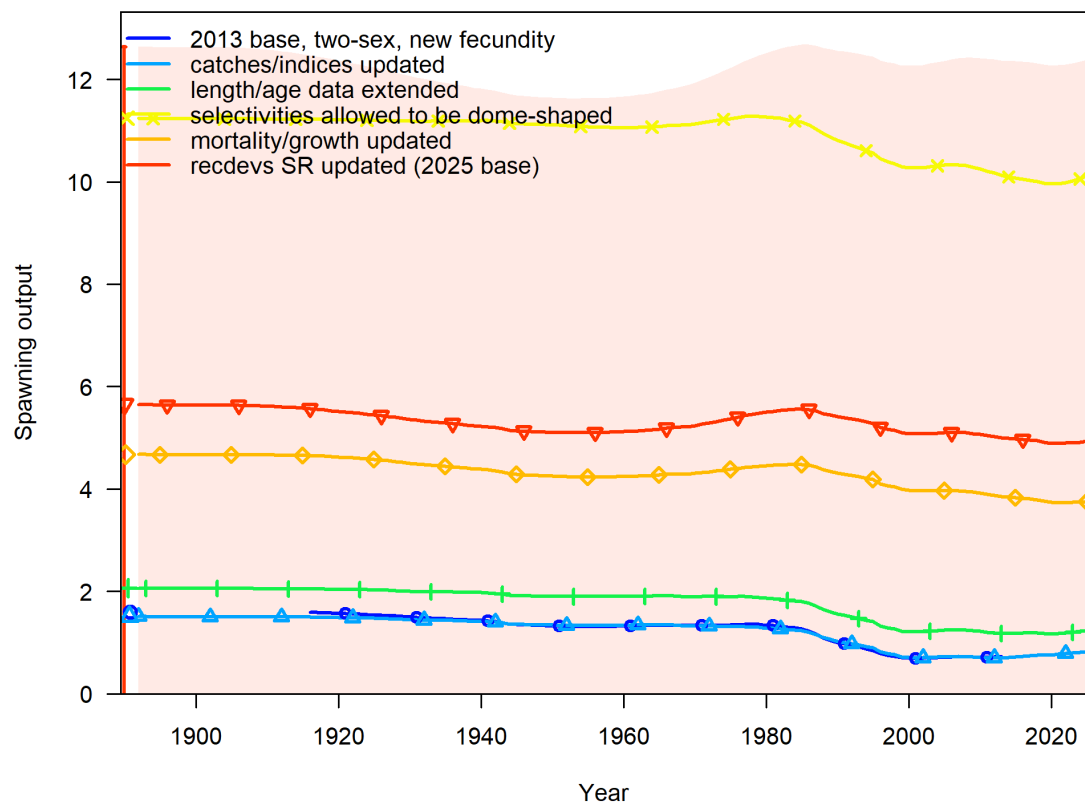


Figure 60: Time series of estimated spawning output for bridging analysis.

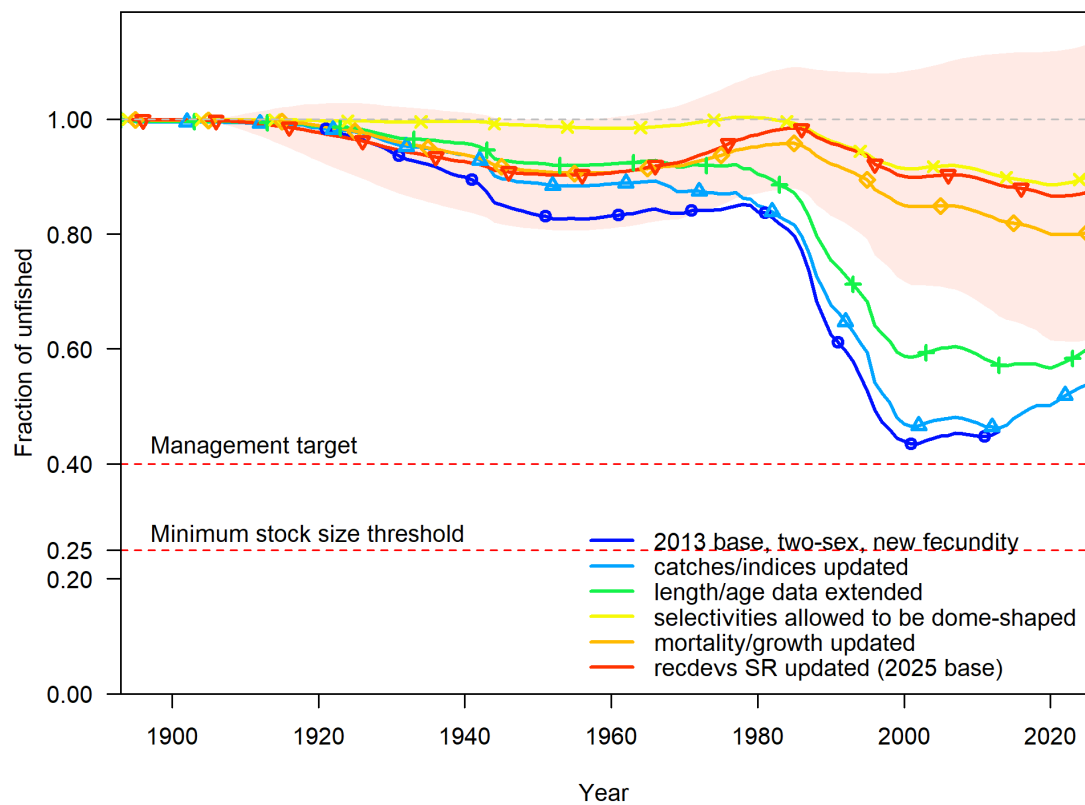


Figure 61: Time series of fraction of unfished spawning output for bridging analysis.

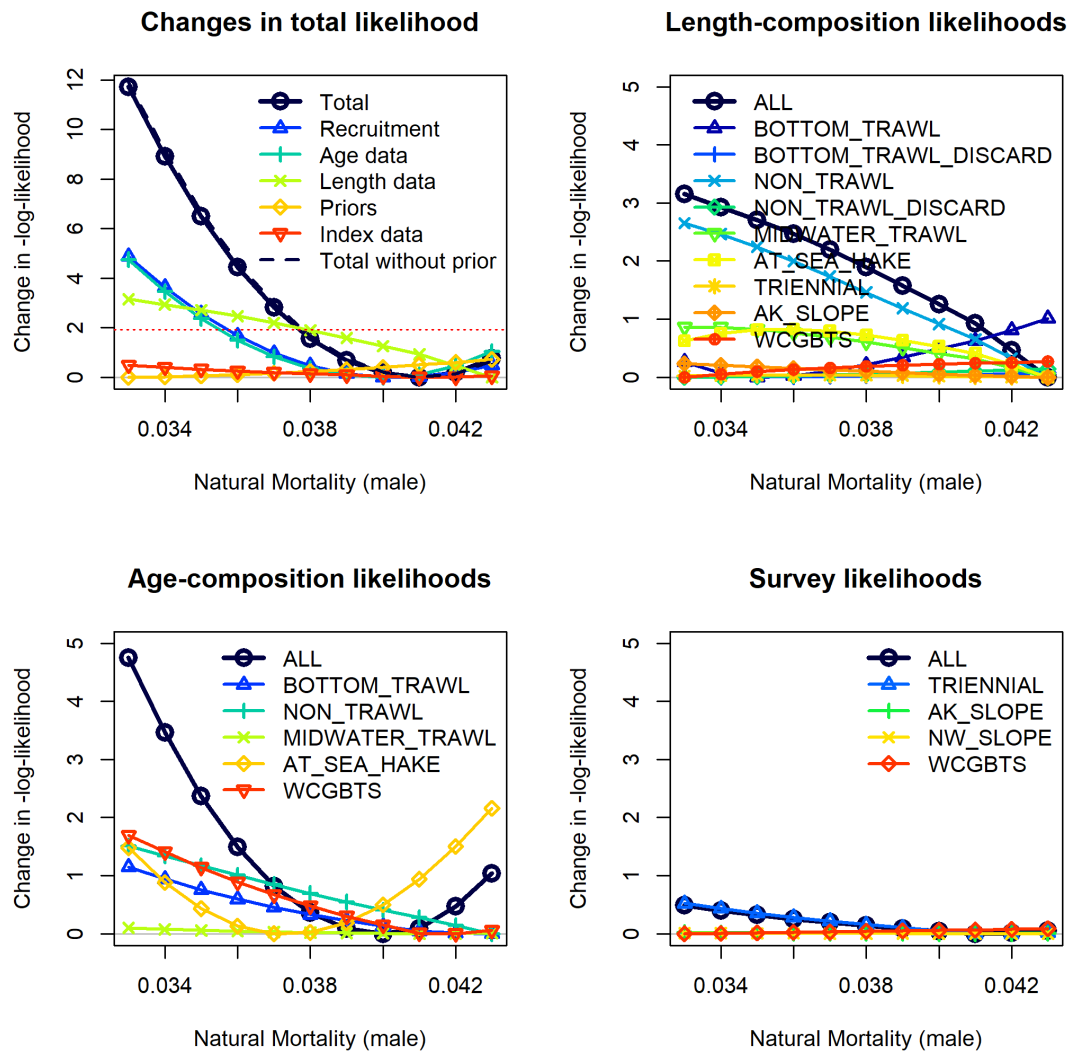


Figure 62: Likelihood profile and component likelihoods used to establish a fixed value for male natural mortality.

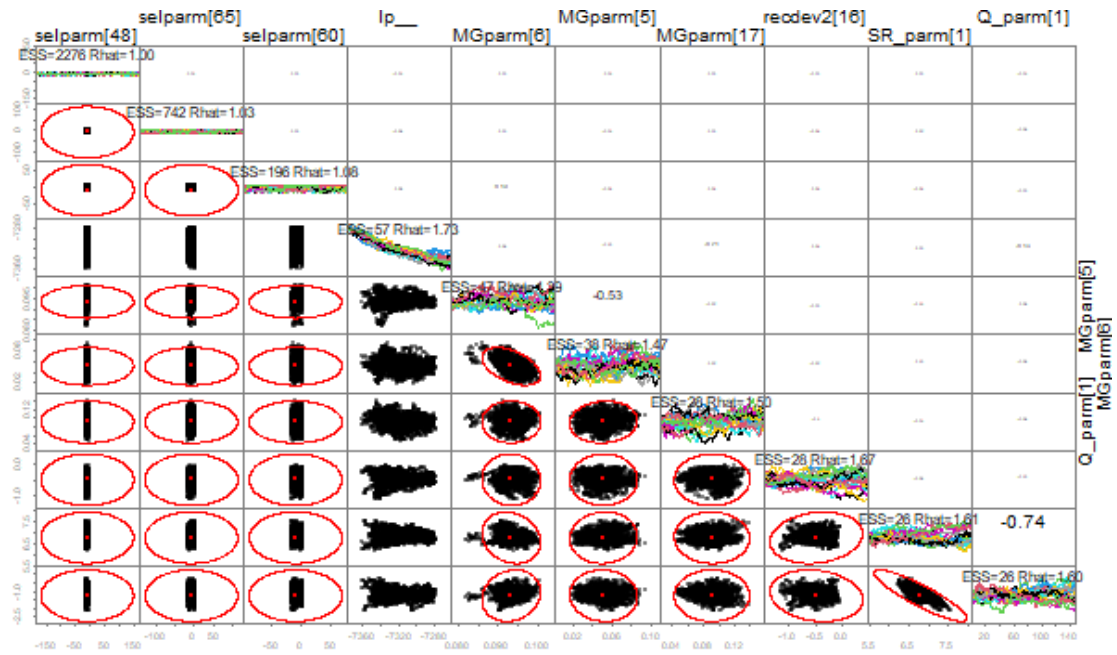


Figure 63: Pairs plots of the fastest mixing parameters from running 2000 posterior draws (keeping every draw) using the random walk Metropolis algorithm. Parameters that show little to no movement are recommended to be fixed to improve model speed and efficiency.

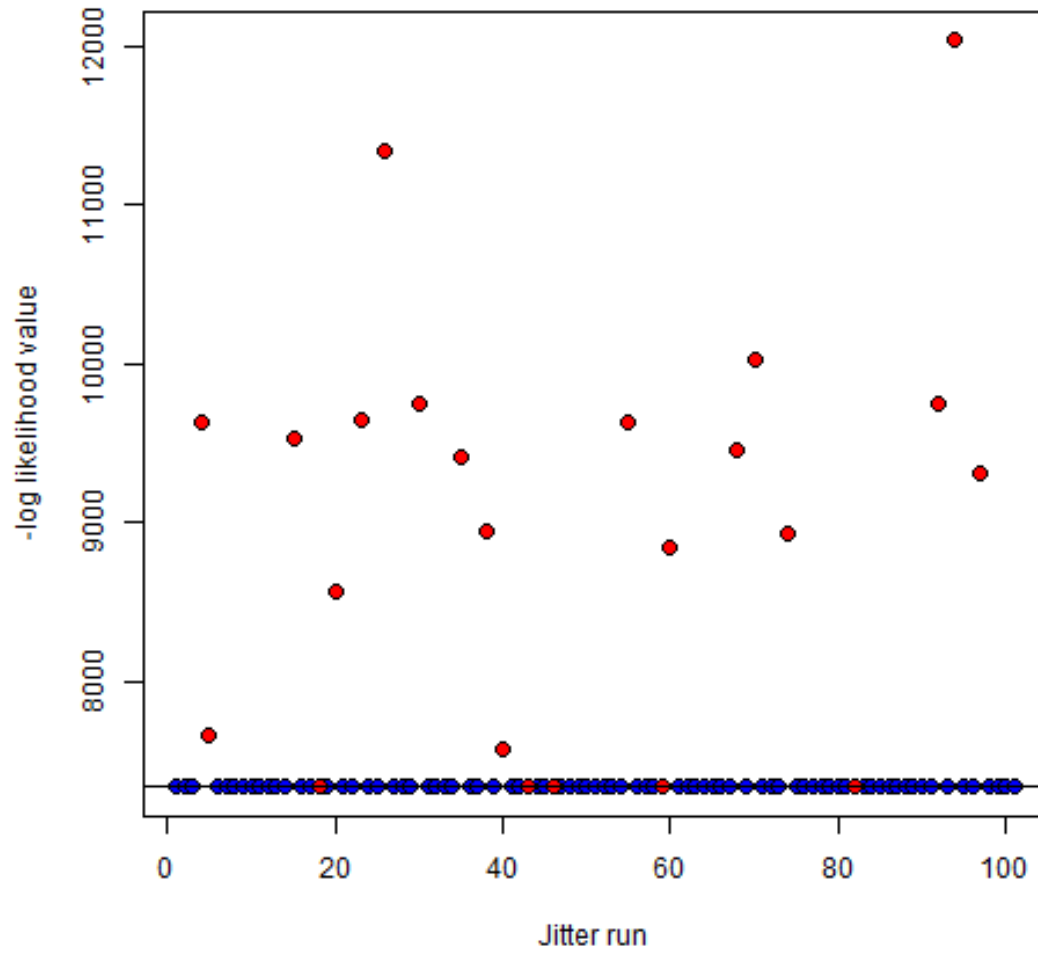


Figure 64: Jitter runs (using a value of 0.01) for the reference model, with jitter run number on the x-axis and -log likelihood value on the y-axis. Blue dot are models that match the likelihood value of the reference model, while red dots deviate from the reference model. All red dots are above the blue dots, indicating no better fit to the reference model was found.



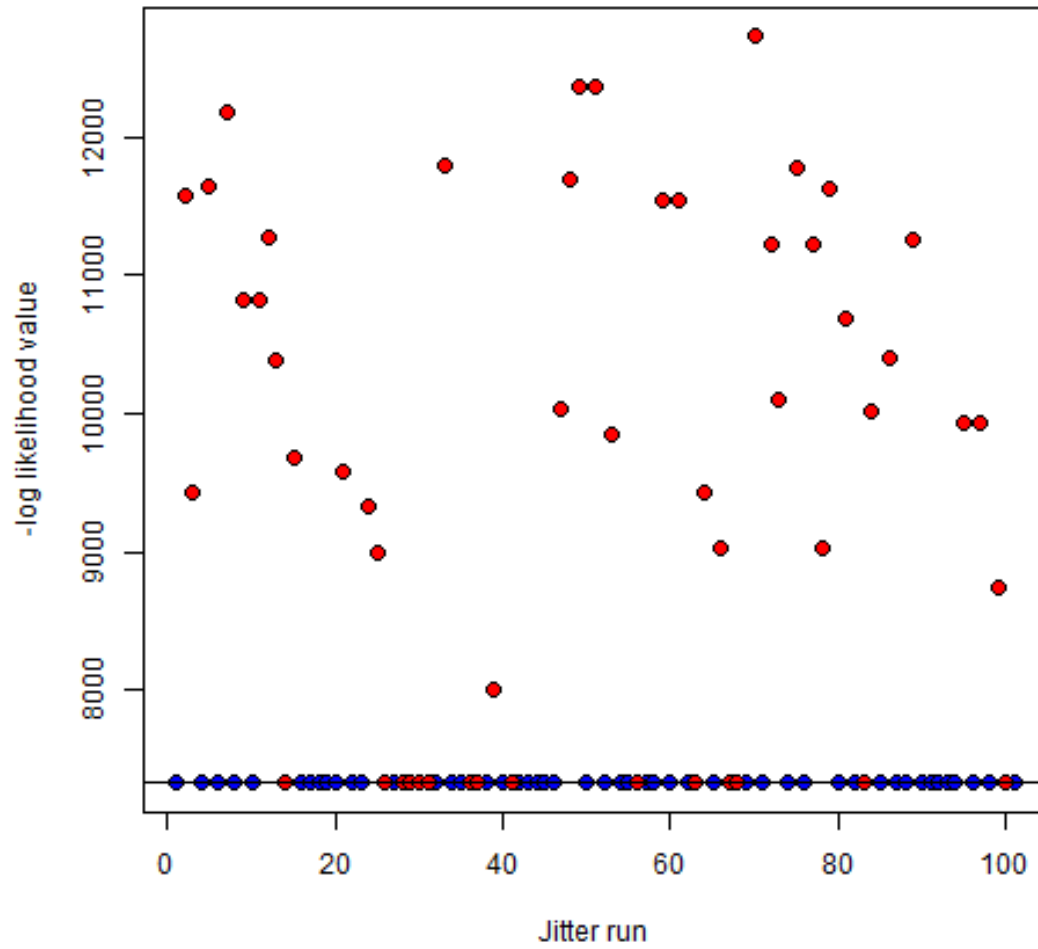


Figure 65: Jitter runs (using a value of 0.05) for the reference model, with jitter run number on the x-axis and -log likelihood value on the y-axis. Blue dot are models that match the likelihood value of the reference model, while red dots deviate from the reference model. All red dots are above the blue dots, indicating no better fit to the reference model was found.

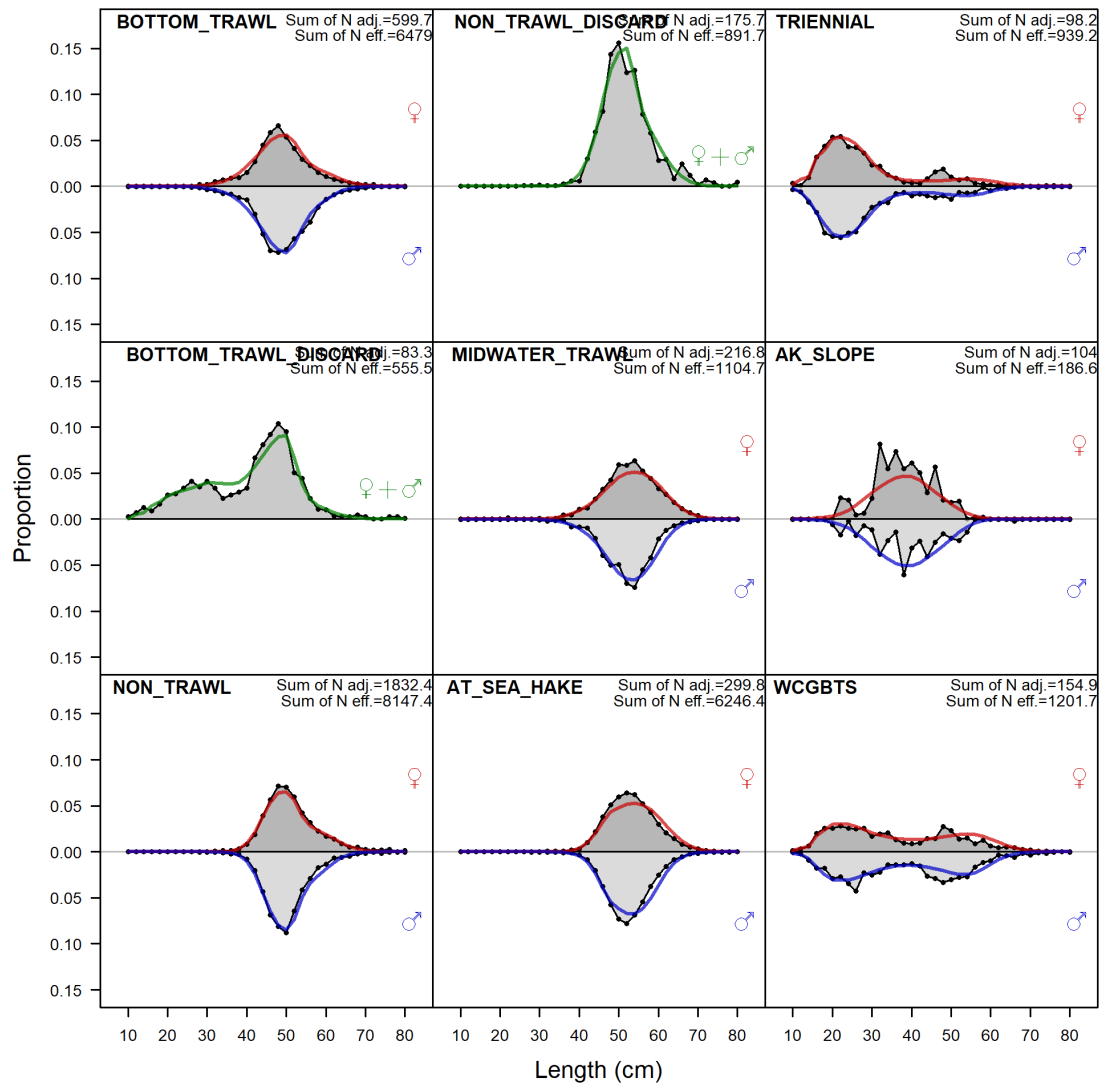


Figure 66: Aggregated length (cm) compositions over all years.

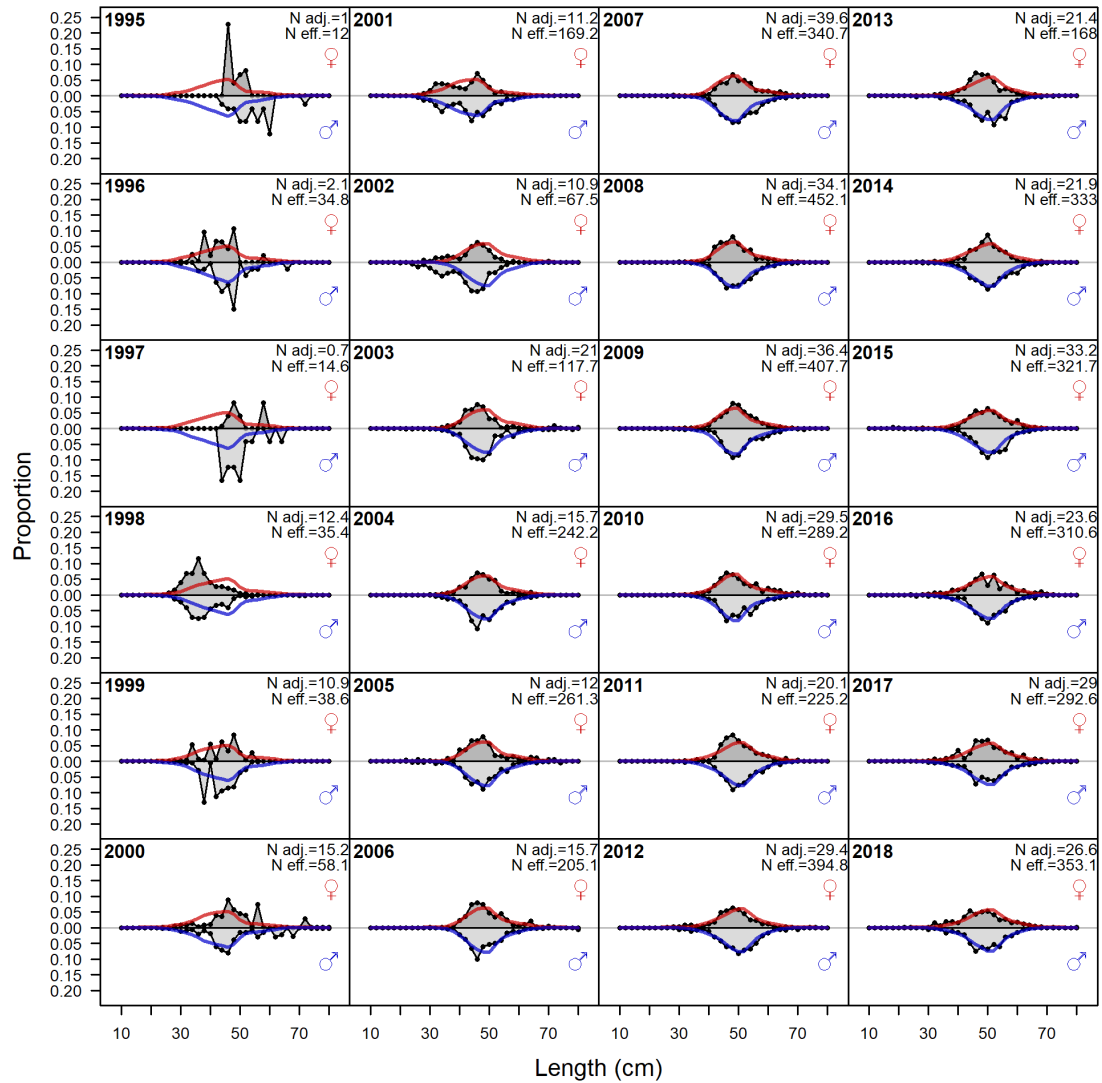


Figure 67: Observed (gray density plot) and expected (density lines by sex) length compositions by year for the bottom trawl fishery in years available between 1995-2018. 'N adj.' is the input sample size after data-weighting adjustment. N eff. is the calculated effective sample size used in the McAllister-Ianelli tuning method.

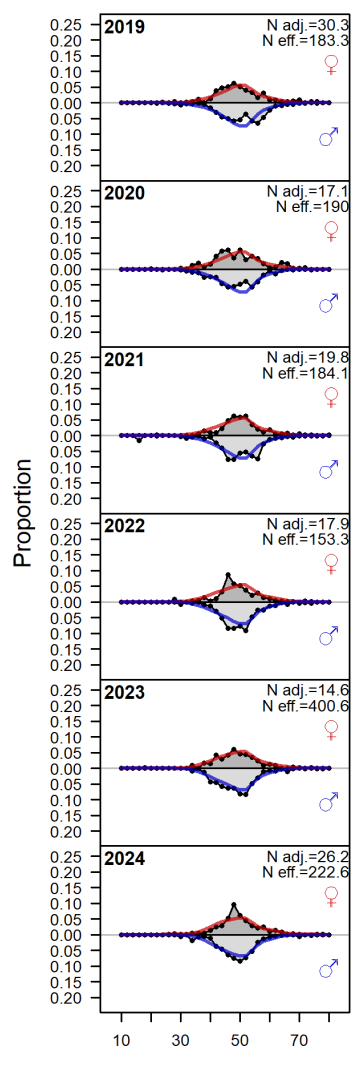


Figure 68: Observed (gray density plot) and expected (density lines by sex) length compositions by year for the bottom trawl fishery in years available between 2019-2024. 'N adj.' is the input sample size after data-weighting adjustment. N eff. is the calculated effective sample size used in the McAllister-Ianelli tuning method.

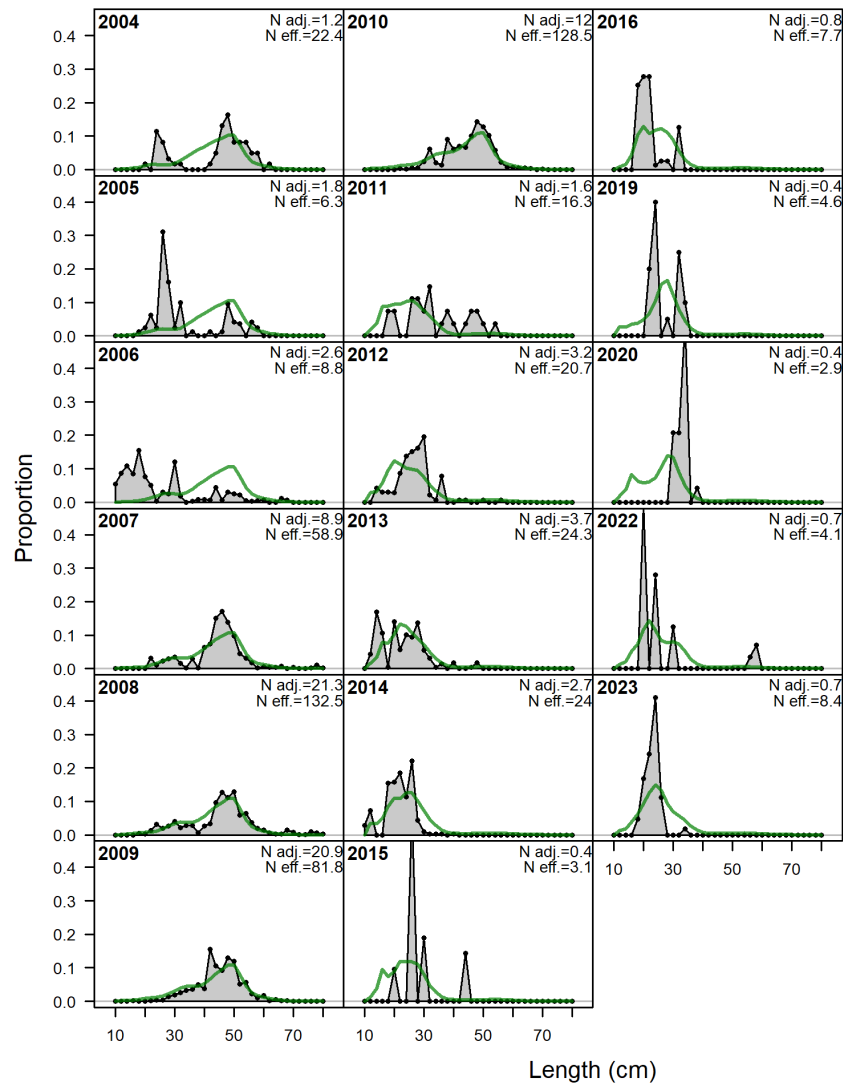


Figure 69: Observed (gray density plot) and expected (density lines) length compositions by year for the bottom trawl discard fishery in years available between 2004-2023. 'N adj.' is the input sample size after data-weighting adjustment. N eff. is the calculated effective sample size used in the McAllister-Ianelli tuning method.

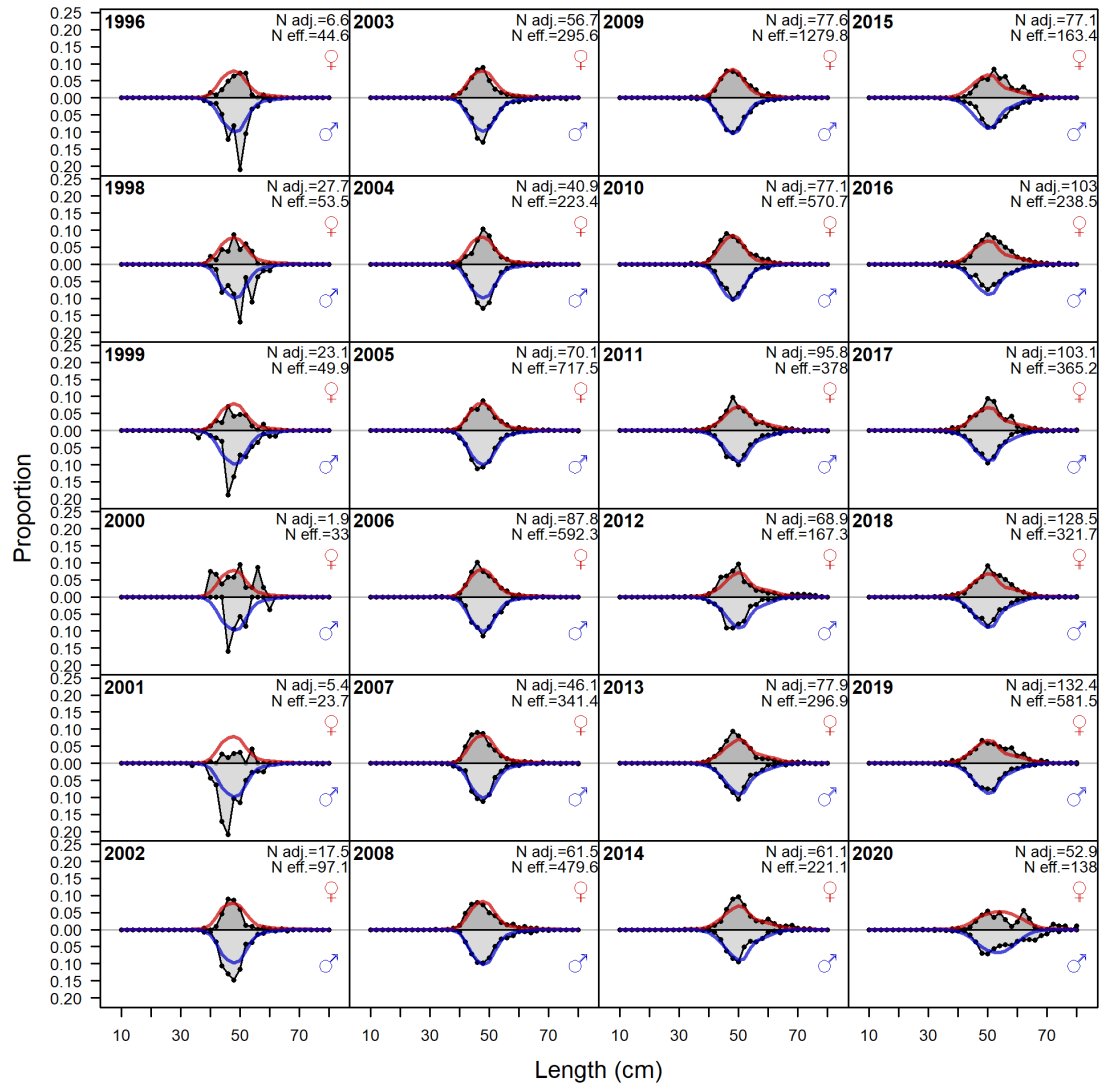


Figure 70: Observed (gray density plot) and expected (density lines by sex) length compositions by year for the non-trawl fishery in years available between 1996-2020. 'N adj.' is the input sample size after data-weighting adjustment. N eff. is the calculated effective sample size used in the McAllister-Ianelli tuning method.

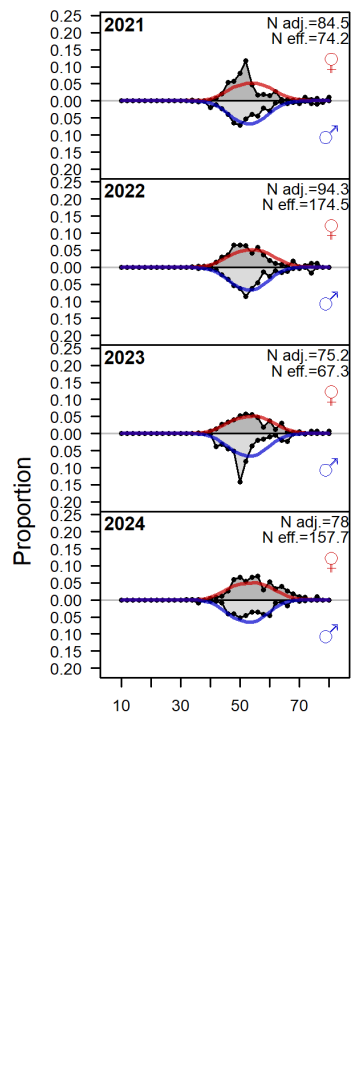


Figure 71: Observed (gray density plot) and expected (density lines by sex) length compositions by year for the non-trawl fishery in years available between 2021-2024. 'N adj.' is the input sample size after data-weighting adjustment. N eff. is the calculated effective sample size used in the McAllister-Ianelli tuning method.

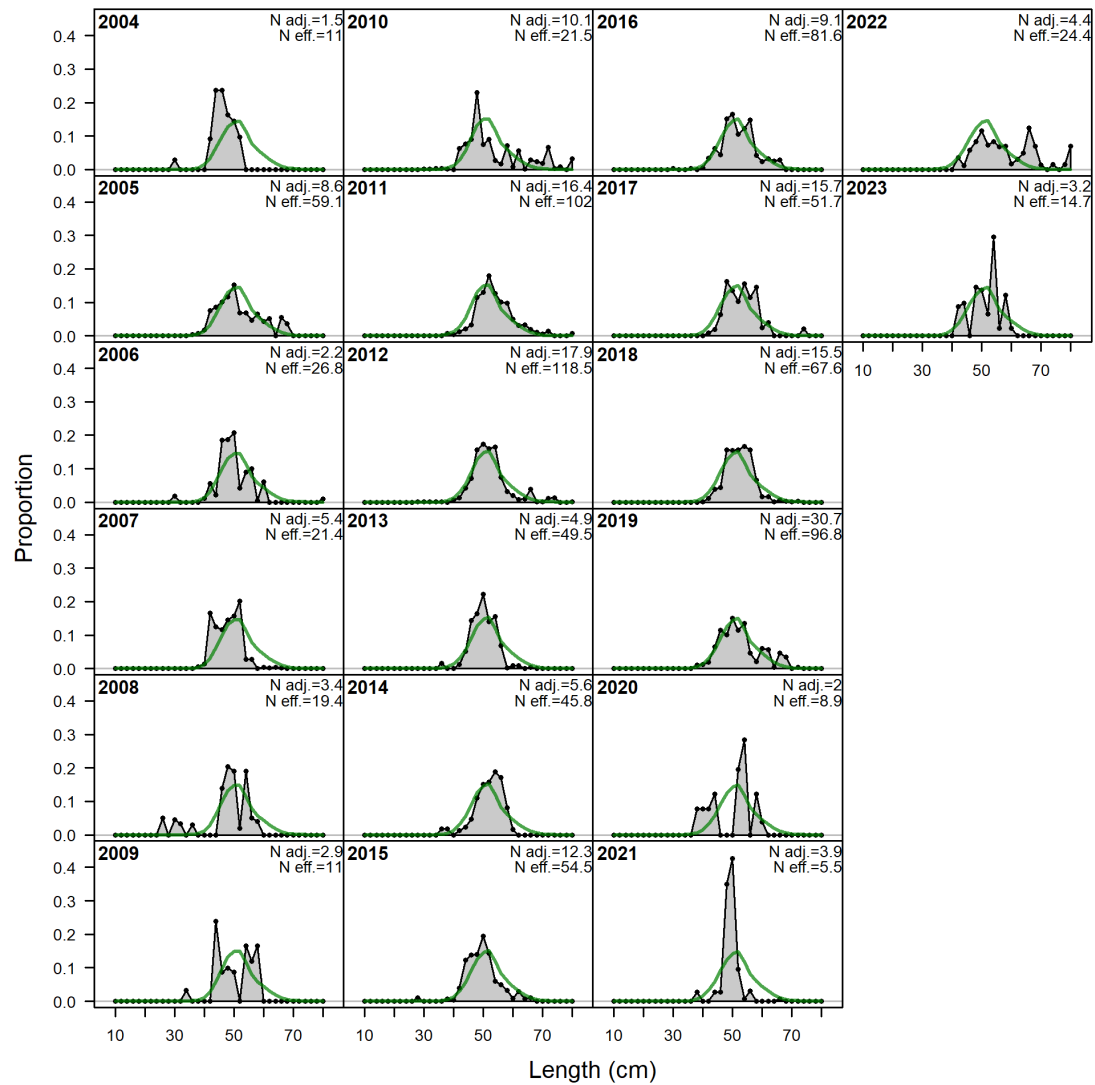


Figure 72: Observed (gray density plot) and expected (density lines) length compositions by year for the midwater trawl fishery in years available between 2004-2024. 'N adj.' is the input sample size after data-weighting adjustment. N eff. is the calculated effective sample size used in the McAllister-Ianelli tuning method.



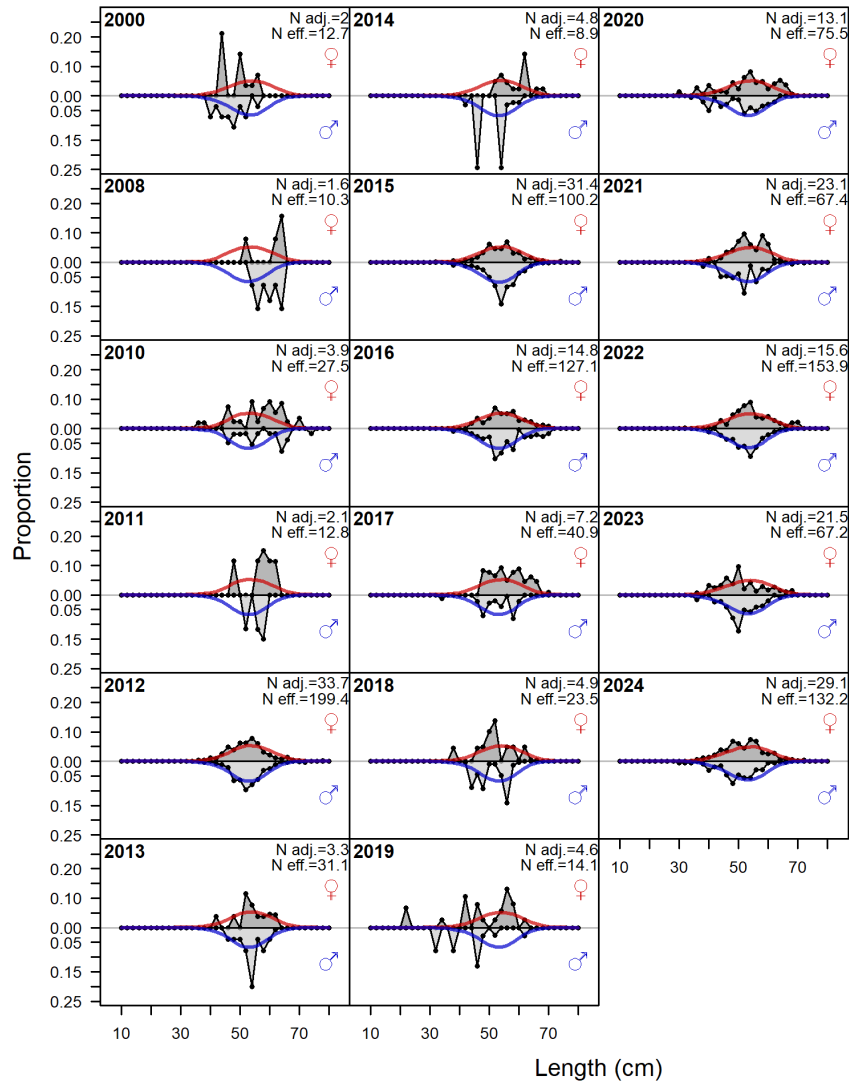


Figure 73: Observed (gray density plot) and expected (density lines by sex) length compositions by year for the non-trawl discard fishery in years available between 2000-2023. 'N adj.' is the input sample size after data-weighting adjustment. N eff. is the calculated effective sample size used in the McAllister-Ianelli tuning method.

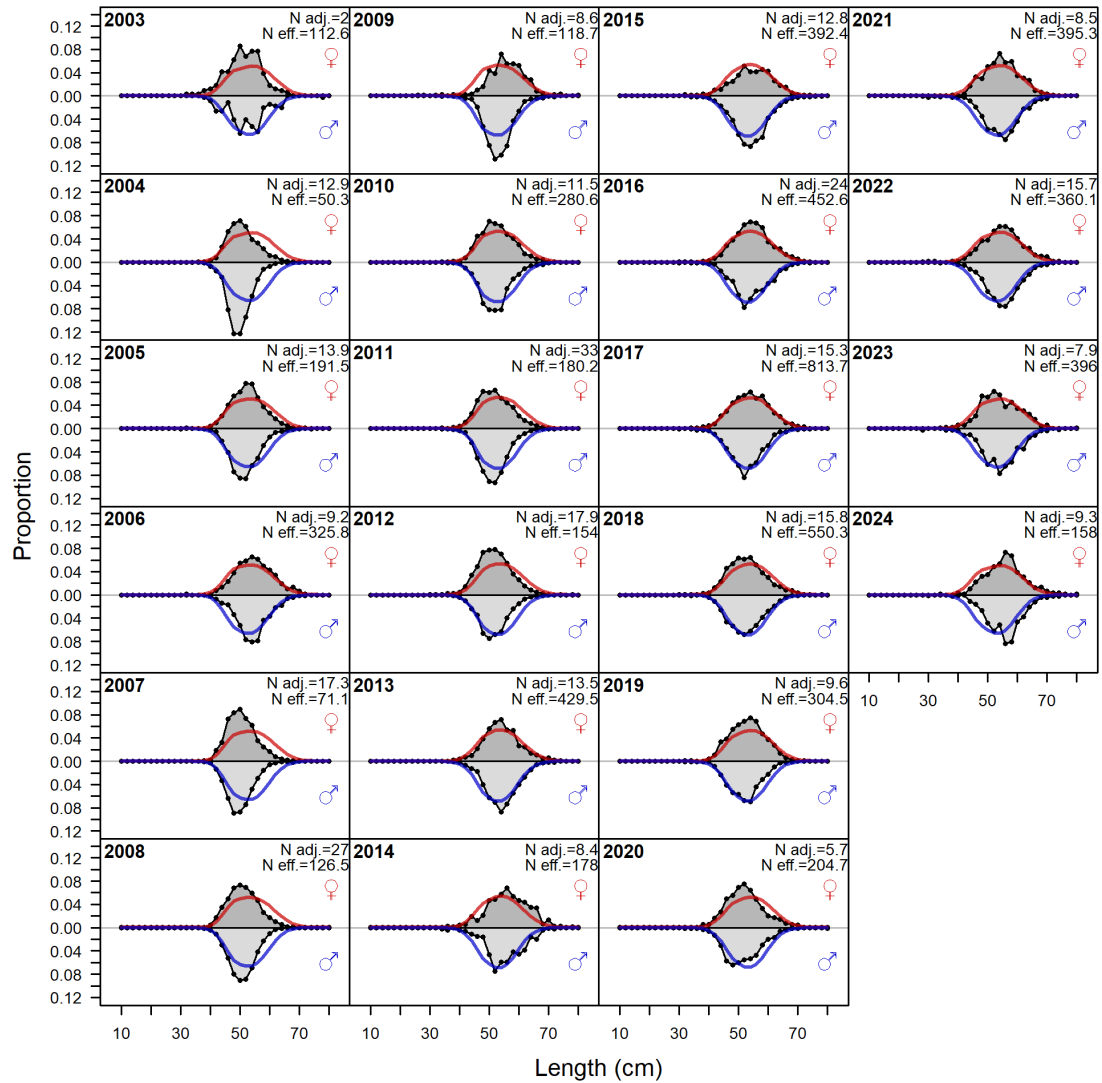


Figure 74: Observed (gray density plot) and expected (density lines by sex) length compositions by year for the at-sea-hake fishery in years available between 2003-2024. 'N adj.' is the input sample size after data-weighting adjustment. N eff. is the calculated effective sample size used in the McAllister-Ianelli tuning method.

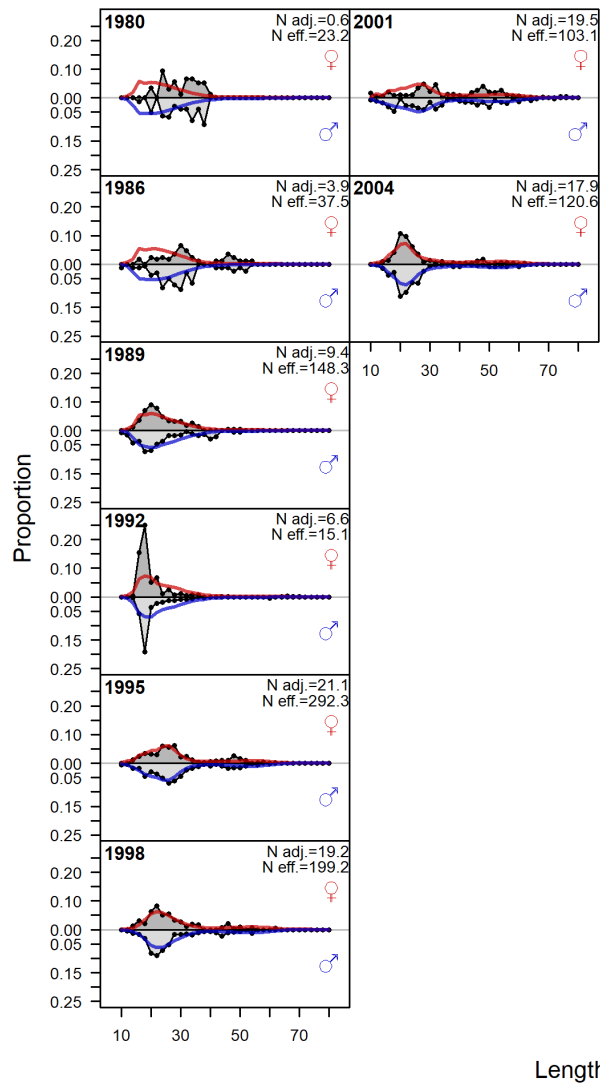


Figure 75: Observed (gray density plot) and expected (density lines by sex) length compositions by year for the Triennial Bottom Trawl Survey in years available between 1980-2004. 'N adj.' is the input sample size after data-weighting adjustment. N eff. is the calculated effective sample size used in the McAllister-Ianelli tuning method.

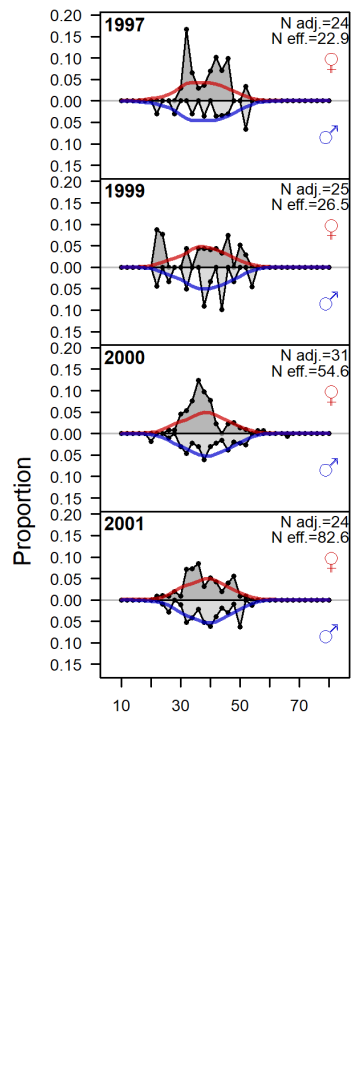


Figure 76: Observed (gray density plot) and expected (density lines by sex) length compositions by year for the Alaskan Slope Bottom Trawl Survey in years available between 1997-2001. 'N adj.' is the input sample size after data-weighting adjustment. N eff. is the calculated effective sample size used in the McAllister-Ianelli tuning method.

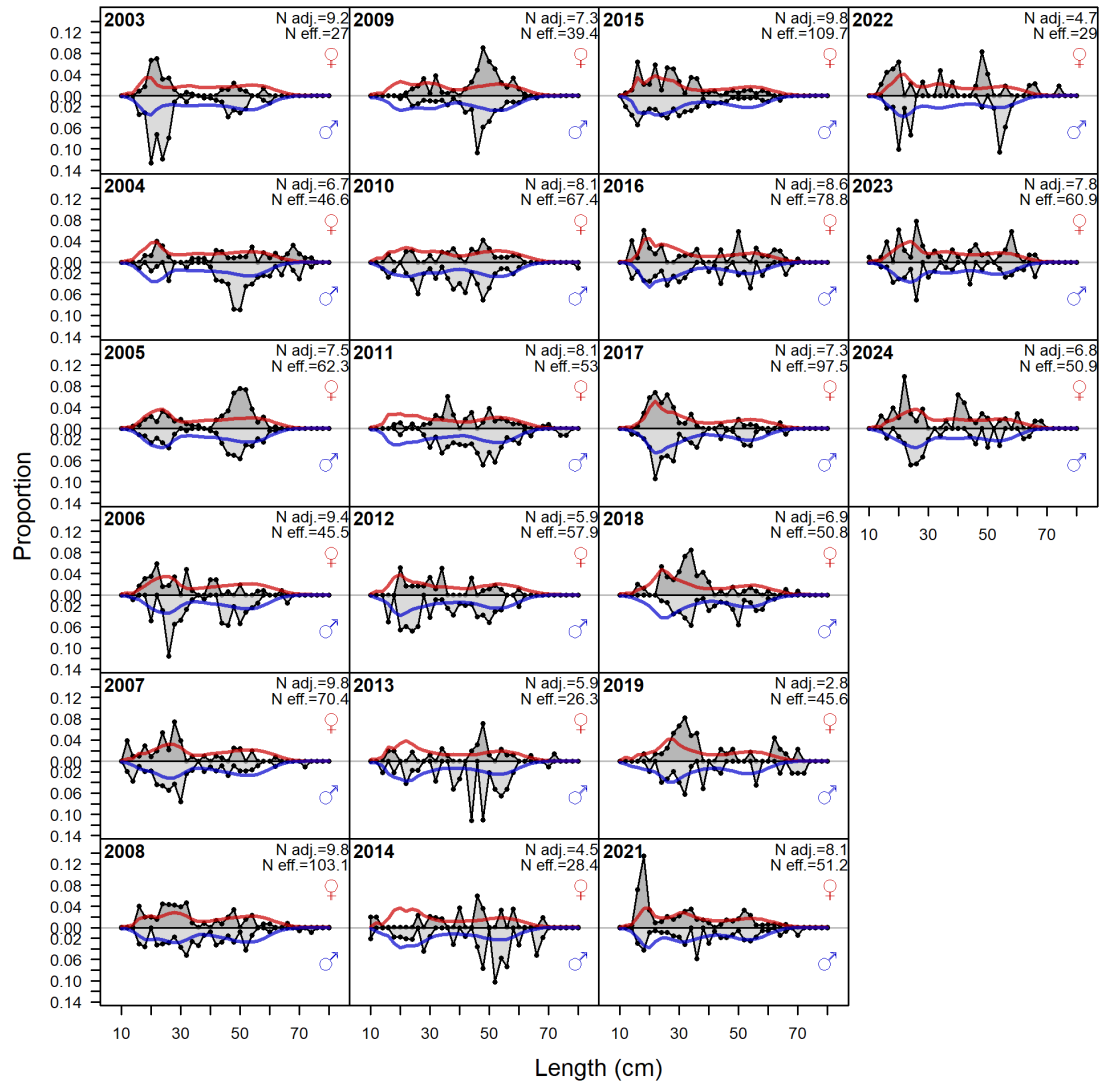


Figure 77: Observed (gray density plot) and expected (density lines by sex) length compositions by year for the West Coast Groundfish Bottom Trawl Survey in years available between 1980-2004. 'N adj.' is the input sample size after data-weighting adjustment. N eff. is the calculated effective sample size used in the McAllister-Ianelli tuning method.

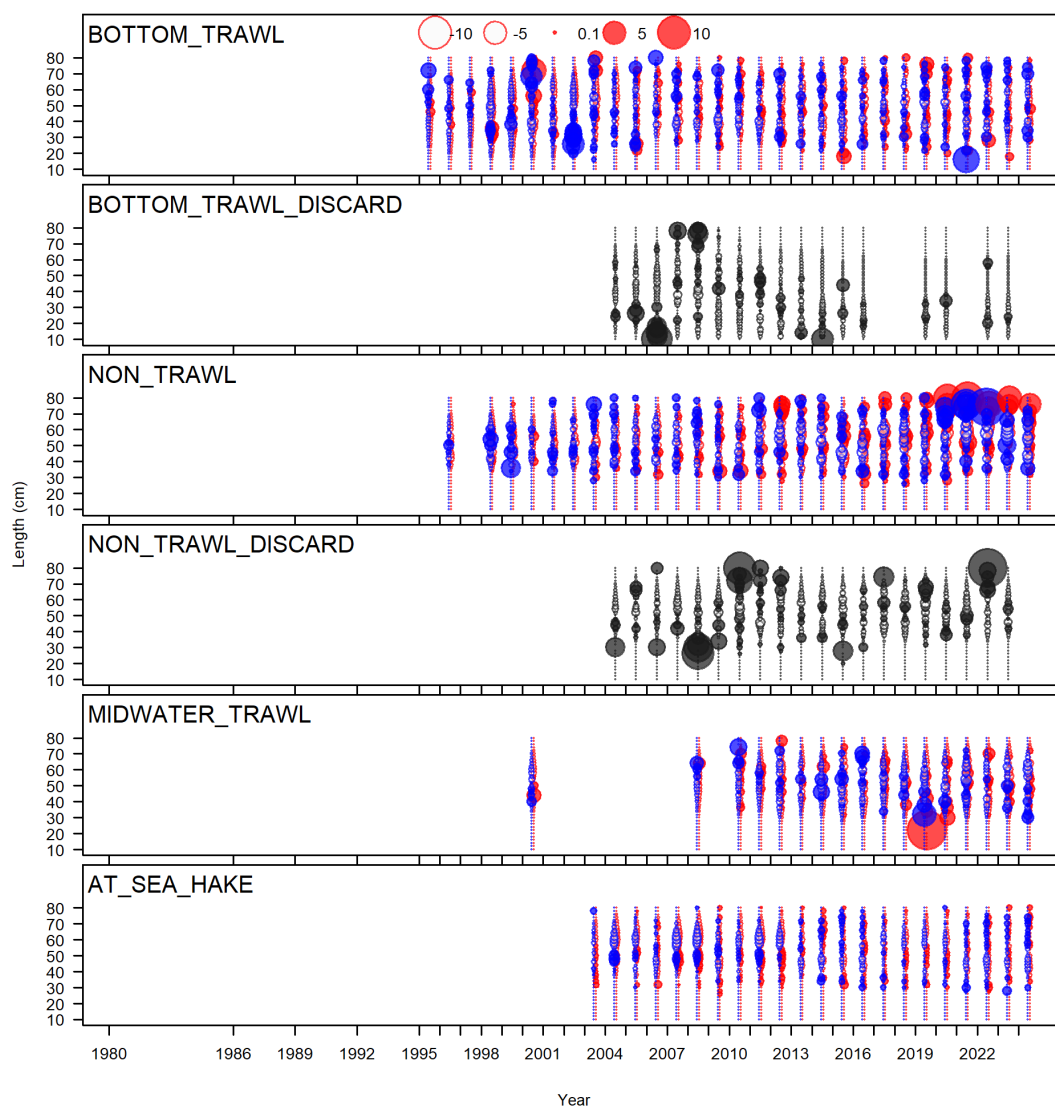


Figure 78: Pearson residuals of length fits for each fishing fleet. Closed bubble are positive residuals (observed > expected) and open bubbles are negative residuals (observed < expected).

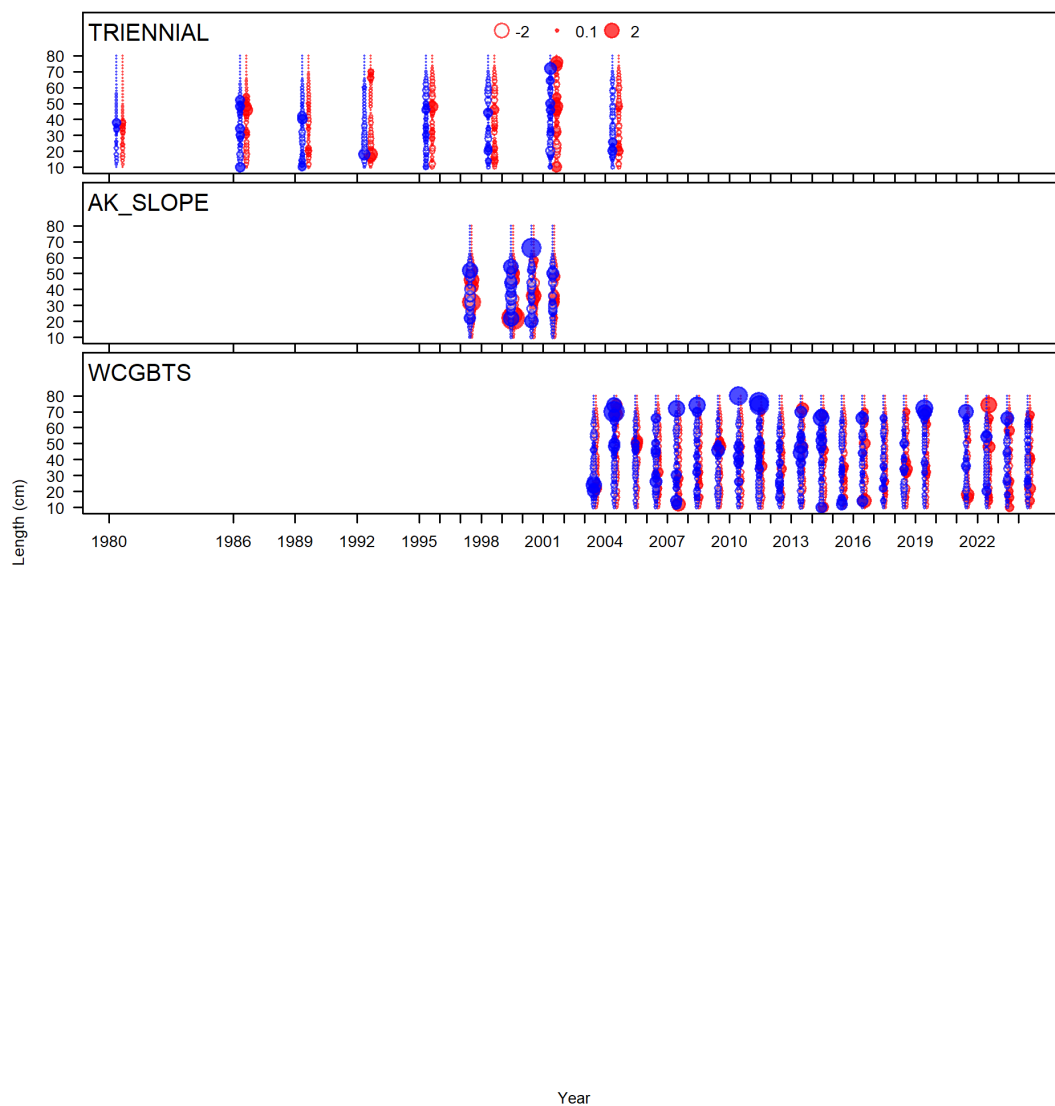


Figure 79: Pearson residuals of length fits for each survey. Closed bubble are positive residuals (observed > expected) and open bubbles are negative residuals (observed < expected).

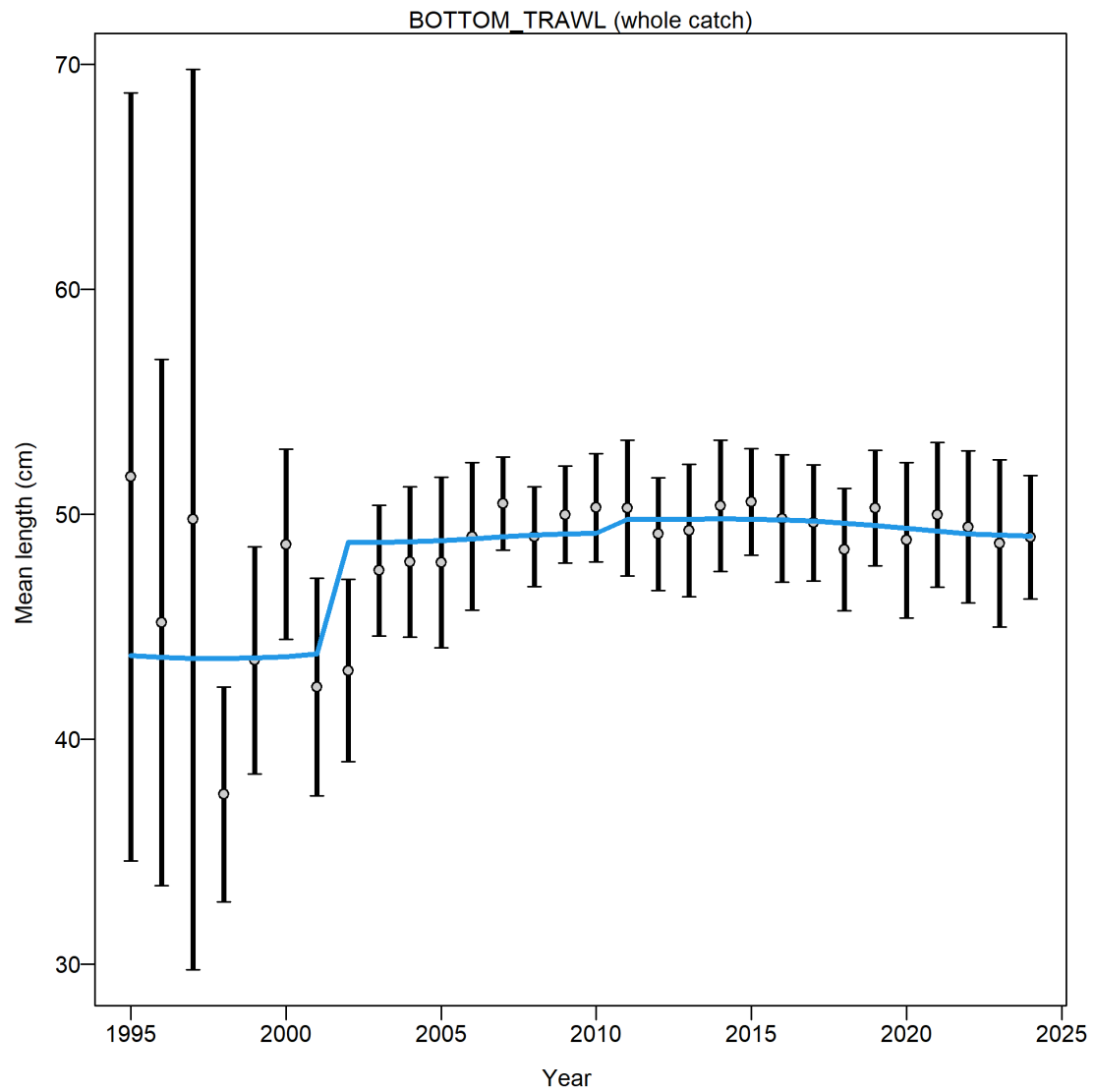


Figure 80: Mean length (cm) index from the bottom trawl fishery with 95 percent confidence intervals based on sample sizes and data weighting.



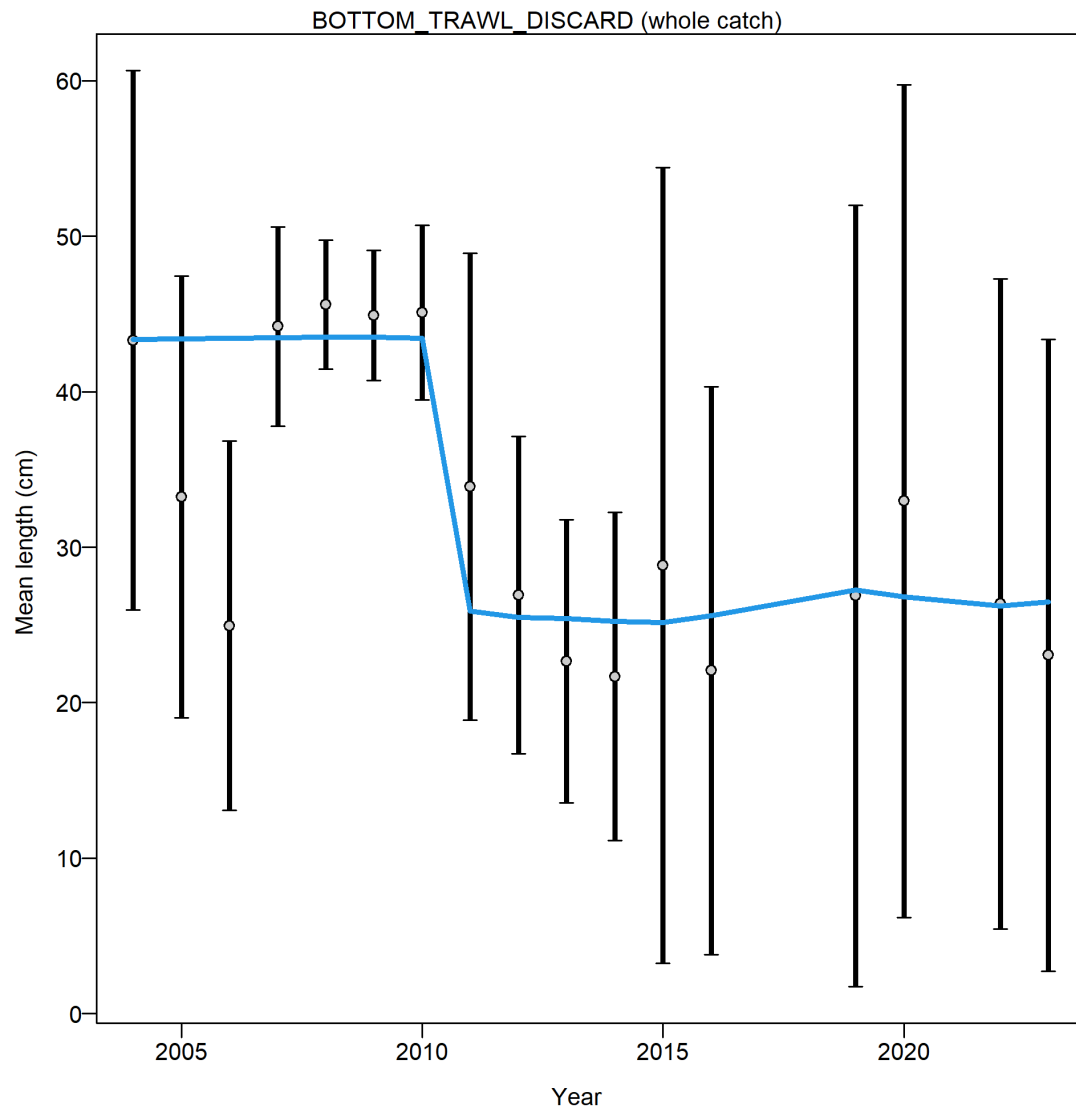


Figure 81: Mean length (cm) index from the bottom trawl discard fishery with 95 percent confidence intervals based on sample sizes and data weighting.

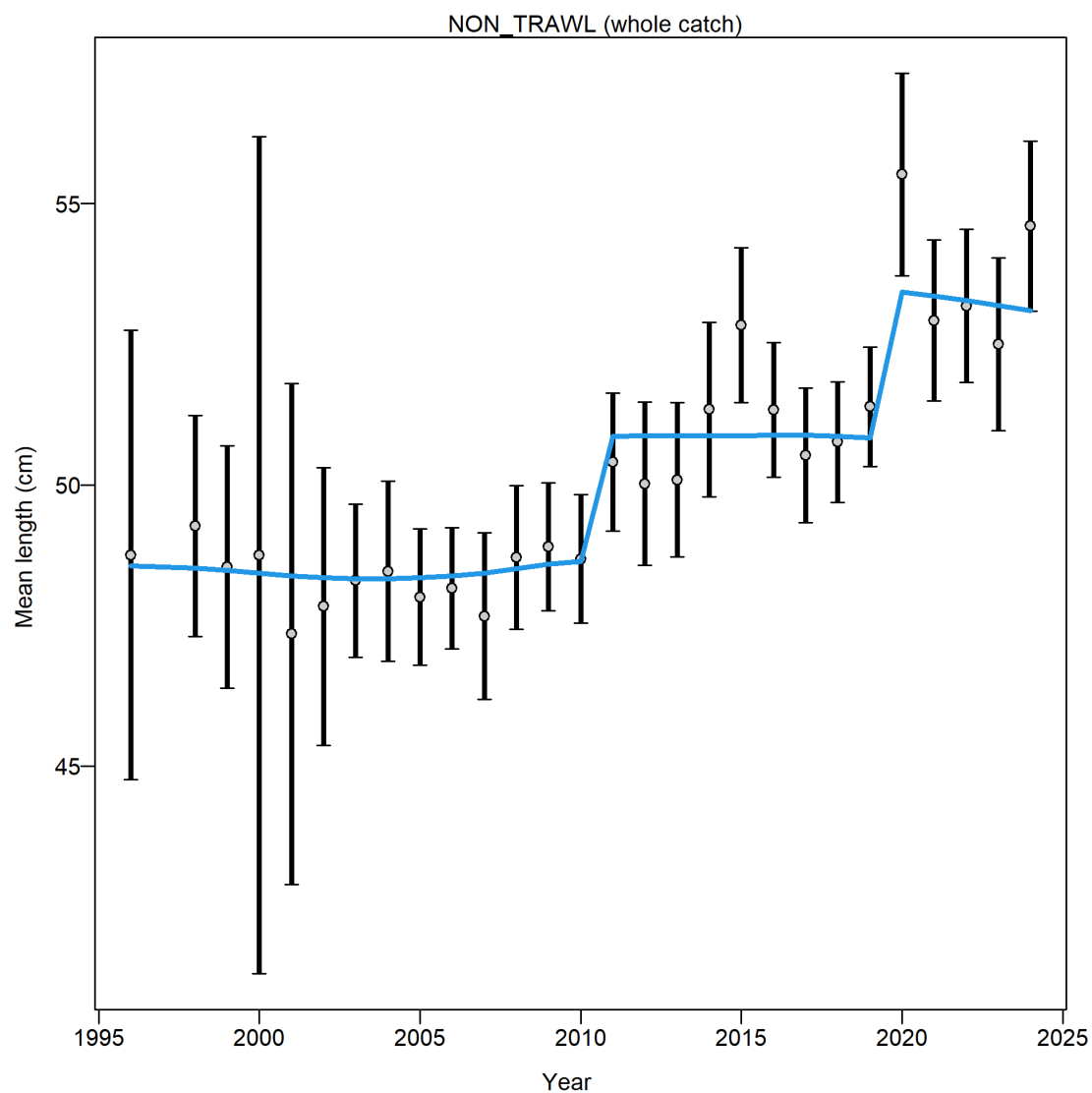


Figure 82: Mean length (cm) index from the non-trawl fishery with 95 percent confidence intervals based on sample sizes and data weighting.

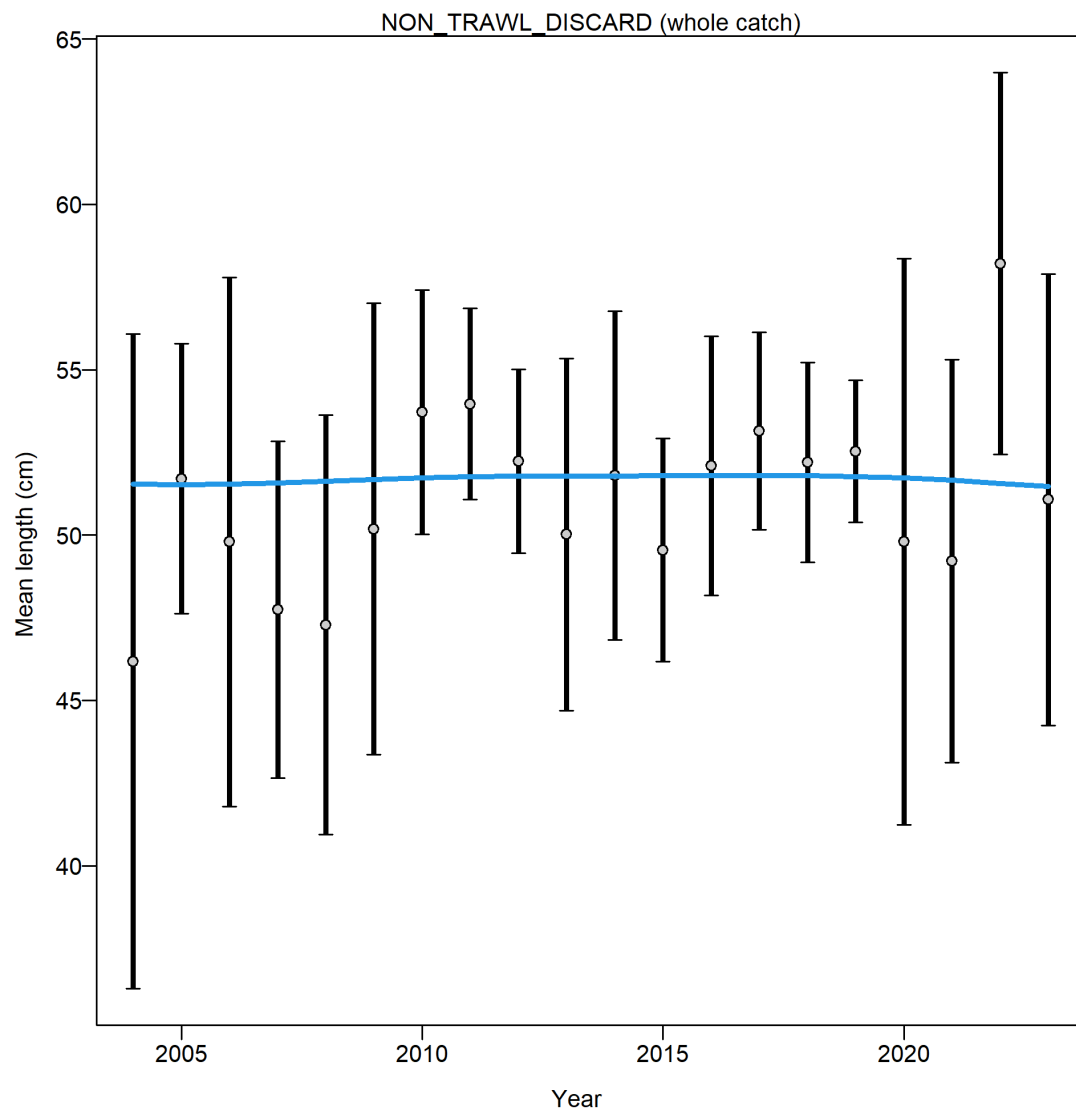


Figure 83: Mean length (cm) index from the non-trawl discard fishery with 95 percent confidence intervals based on sample sizes and data weighting.

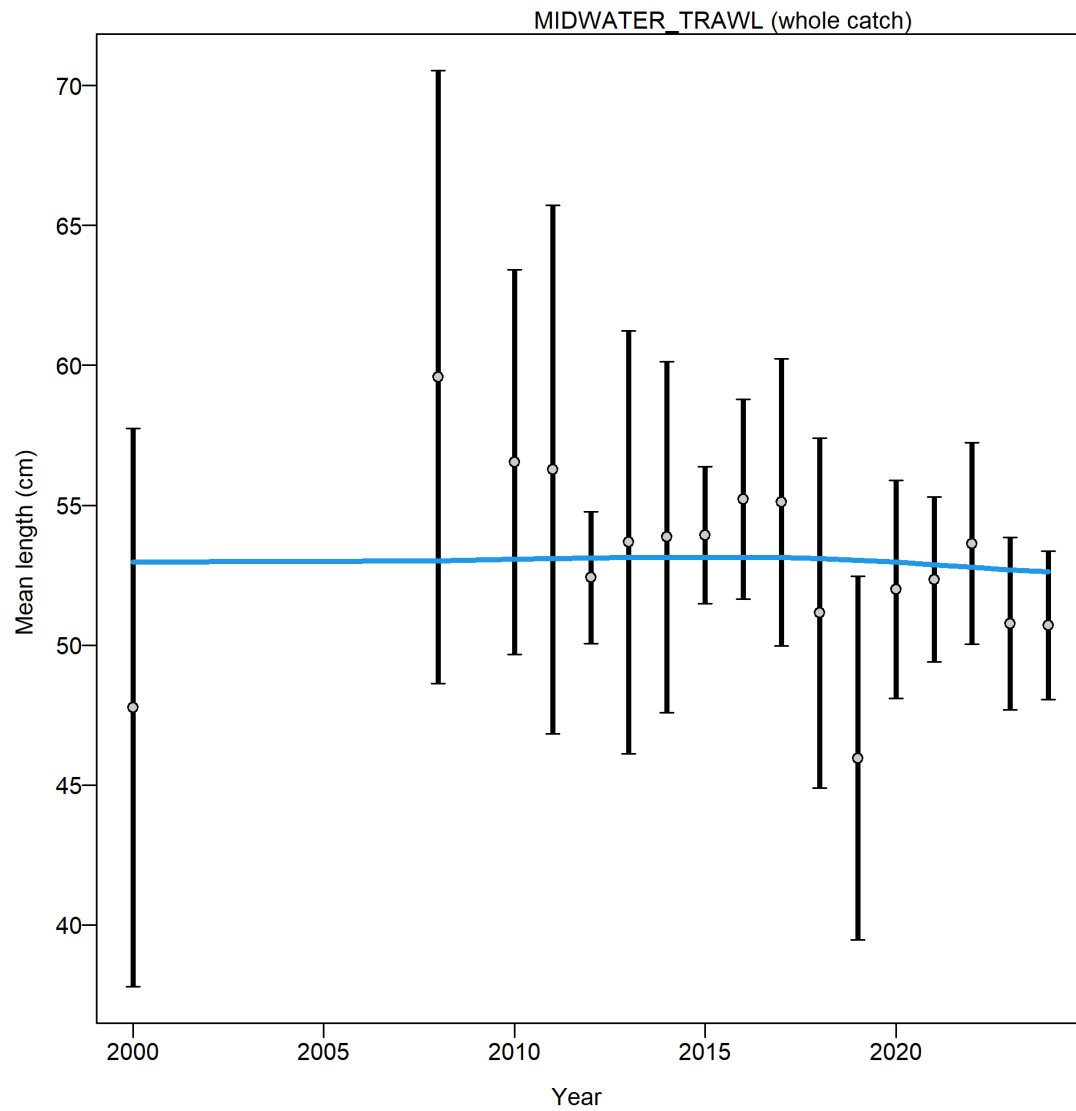


Figure 84: Mean length (cm) index from the midwater trawl fishery with 95 percent confidence intervals based on sample sizes and data weighting.

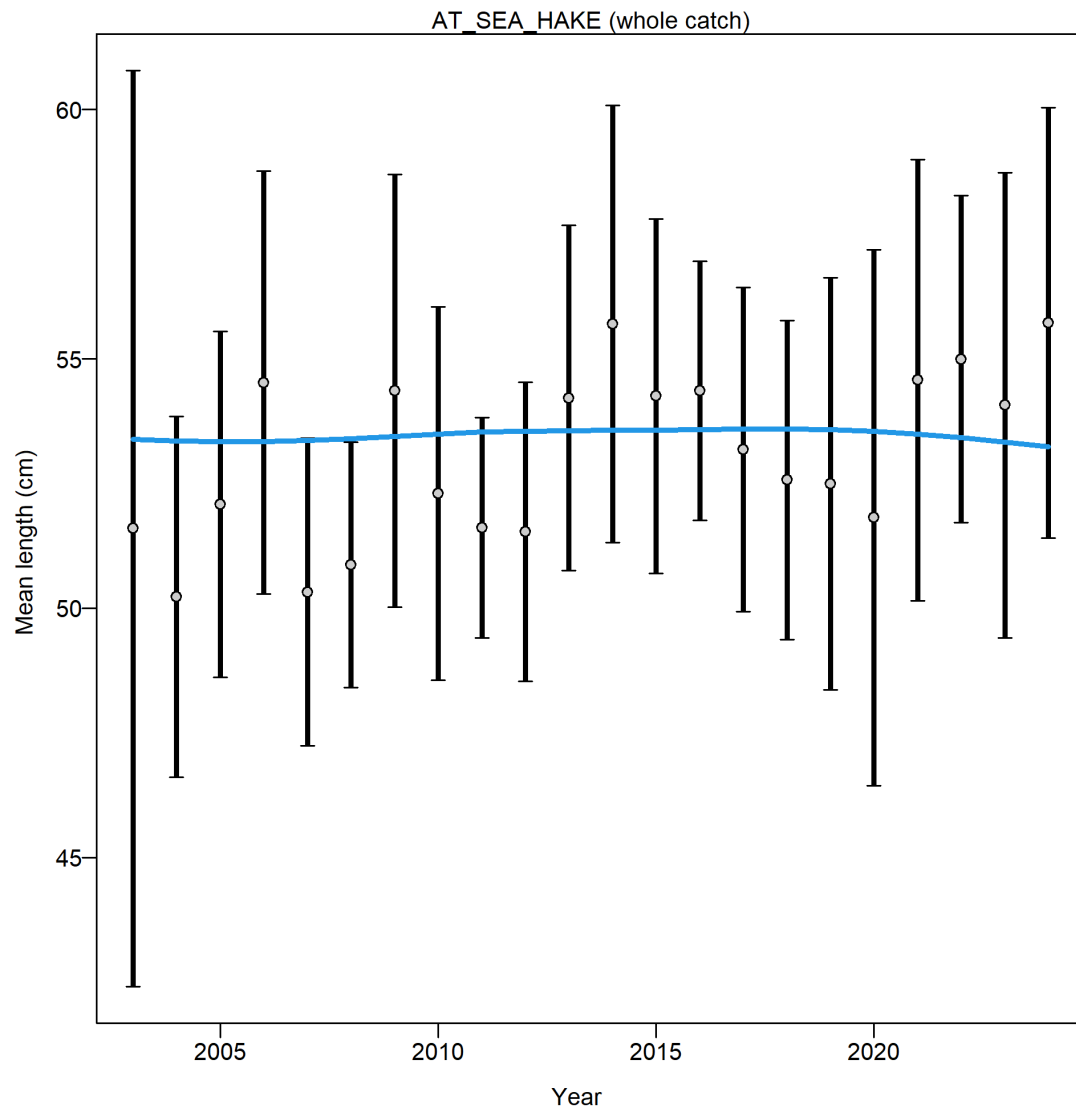


Figure 85: Mean length (cm) index from the at-sea-hake fishery with 95 percent confidence intervals based on sample sizes and data weighting.

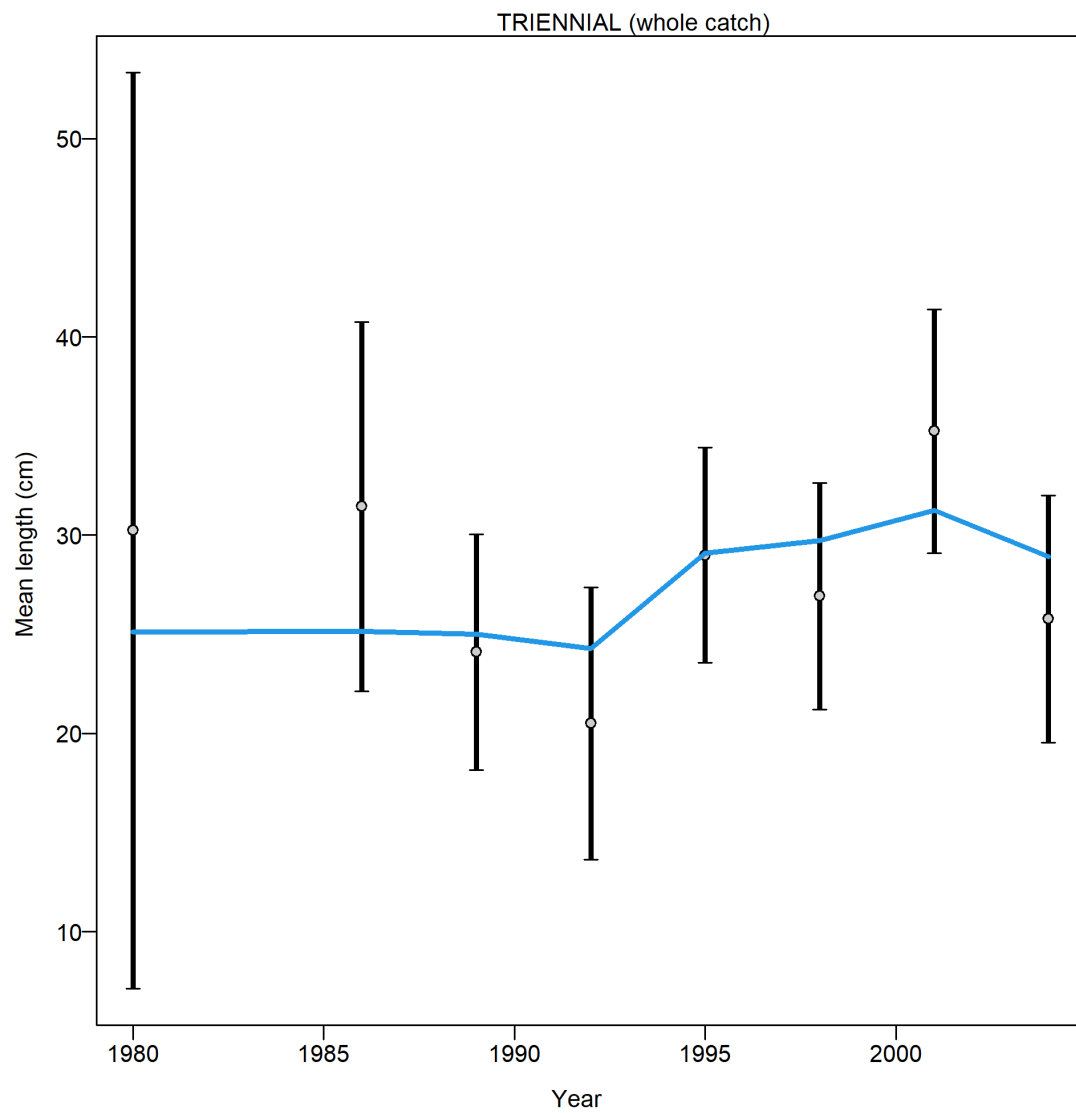


Figure 86: Mean length (cm) index from the Triennial survey with 95 percent confidence intervals based on sample sizes and data weighting.

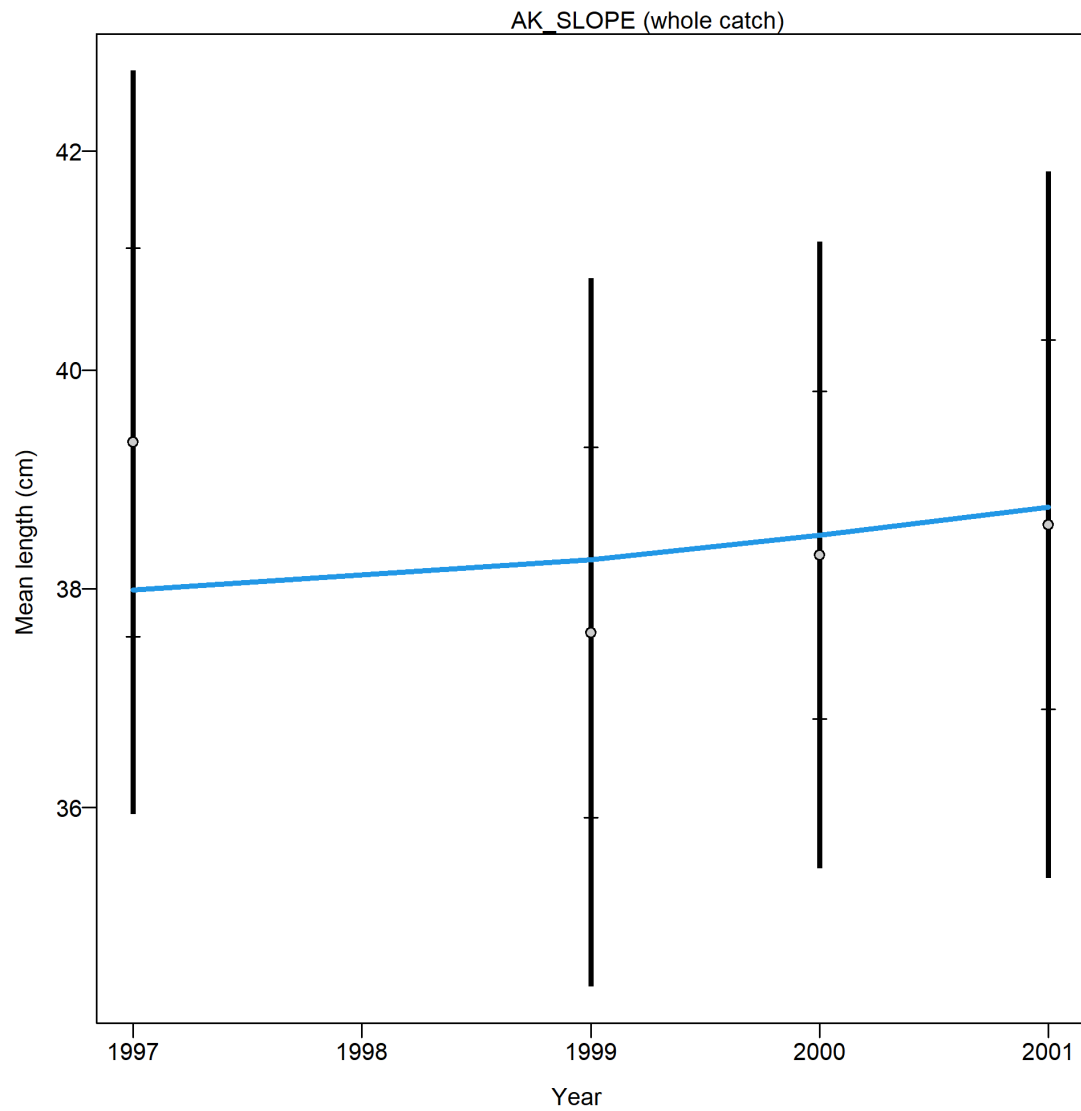


Figure 87: Mean length (cm) index from the Alaskan slope survey with 95 percent confidence intervals based on sample sizes and data weighting.

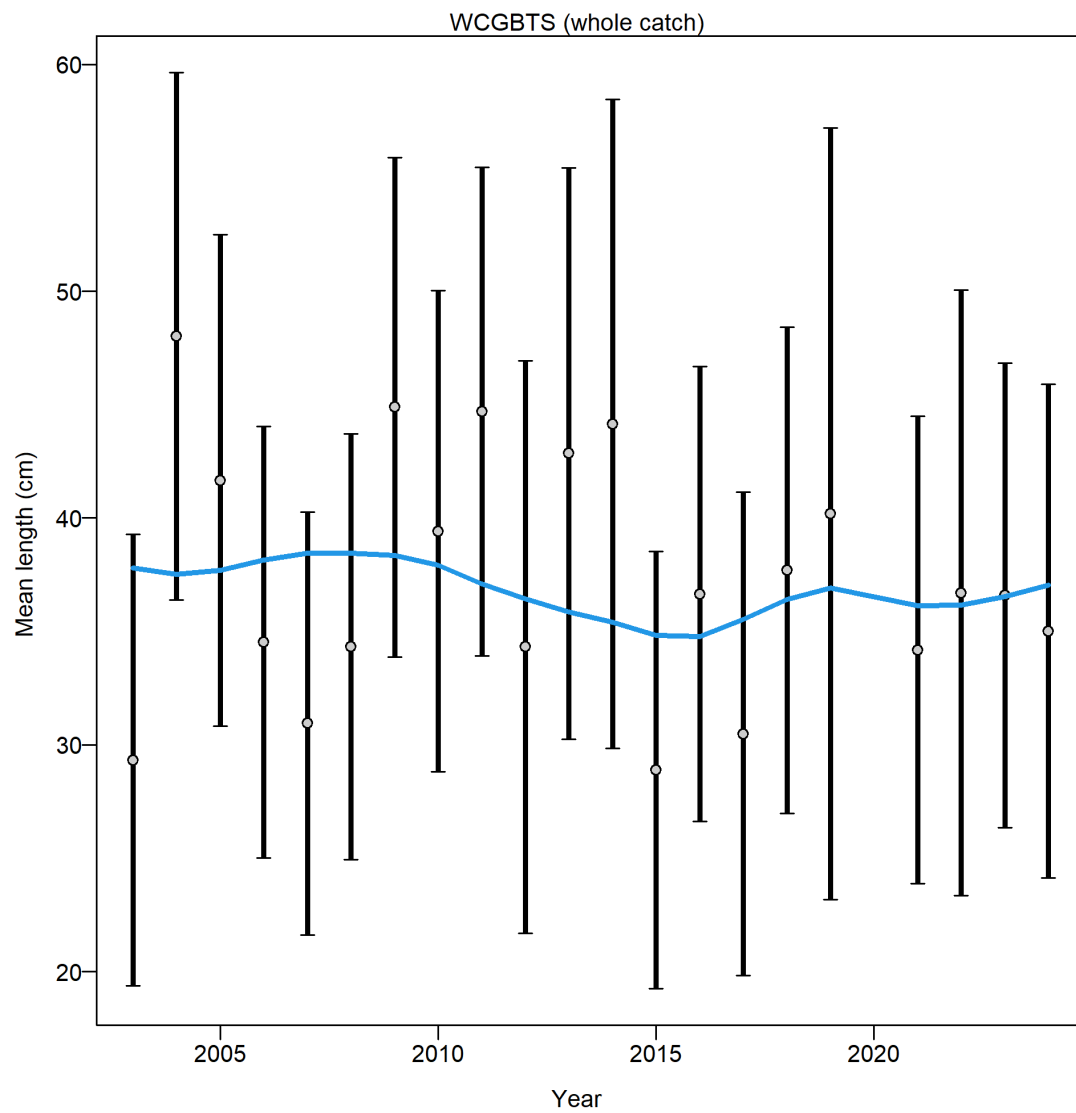


Figure 88: Mean length (cm) index from the West Coast Groundfish Bottom Trawl survey with 95 percent confidence intervals based on sample sizes and data weighting.



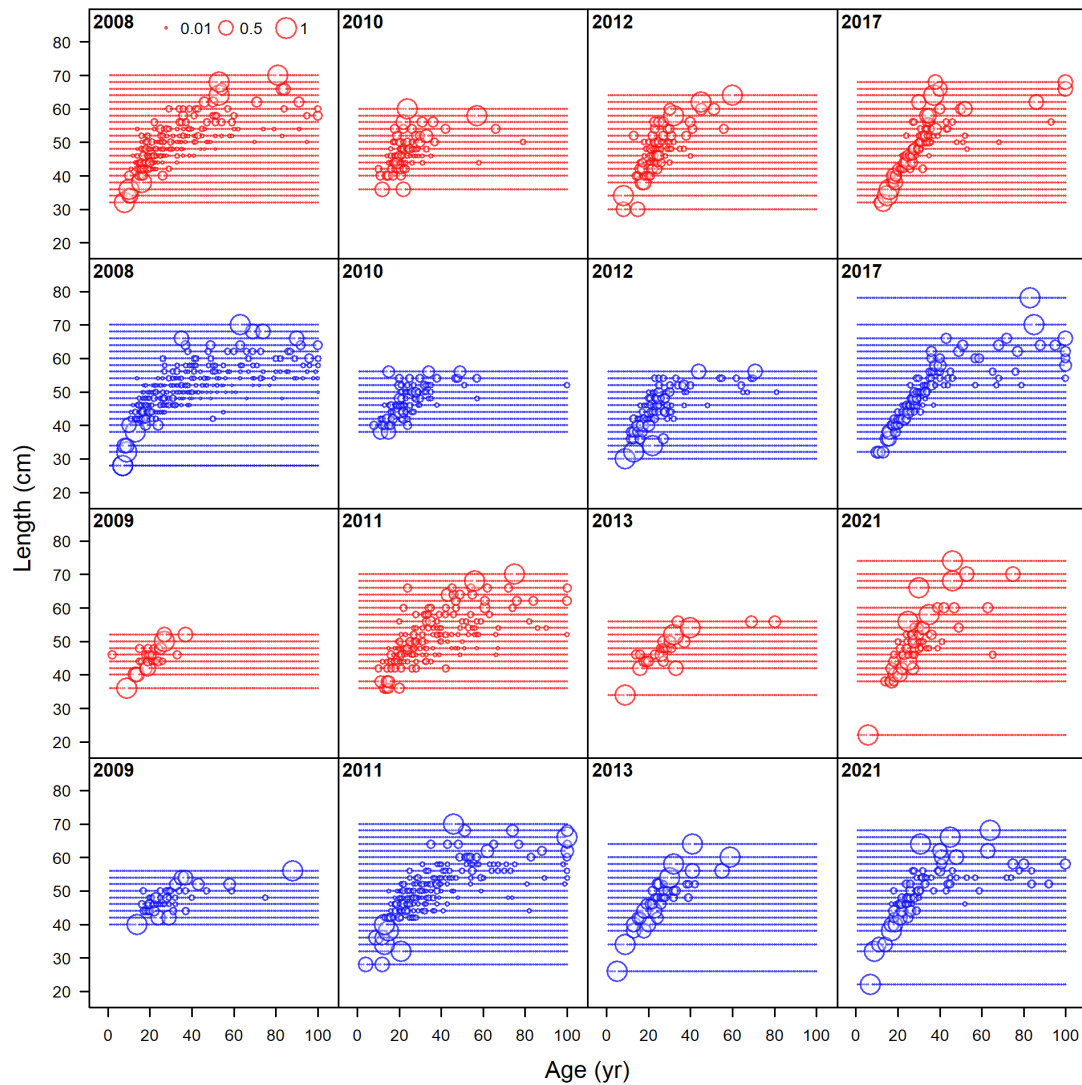


Figure 89: Pearson residuals of conditional age at length fits for the bottom trawl fishery in the years 2008-2021. Closed bubble are positive residuals (observed > expected) and open bubbles are negative residuals (observed < expected).

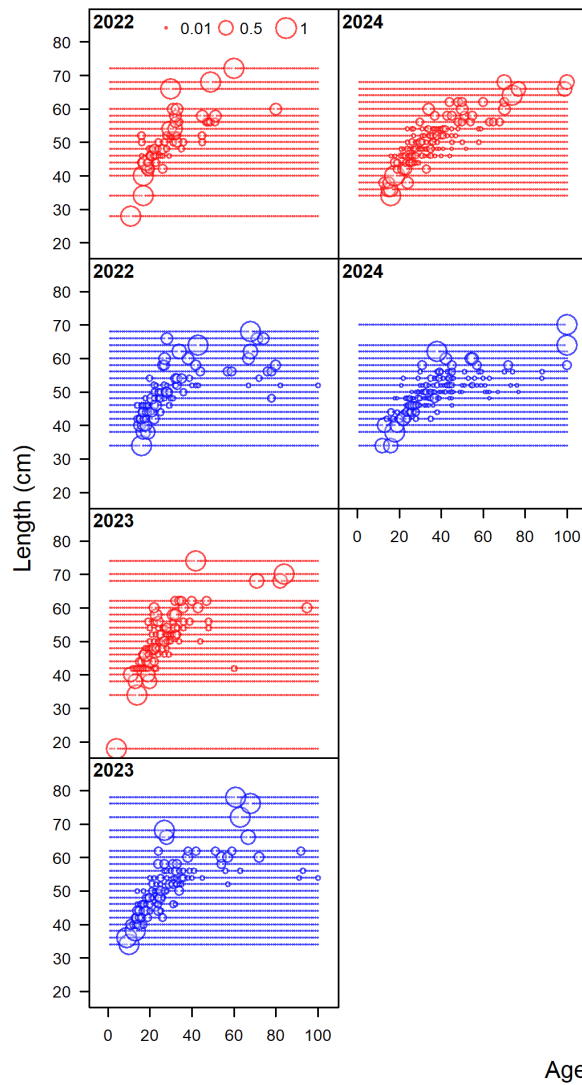


Figure 90: Pearson residuals of conditional age at length fits for the bottom trawl fishery in the years 2022-2024. Closed bubble are positive residuals (observed > expected) and open bubbles are negative residuals (observed < expected).

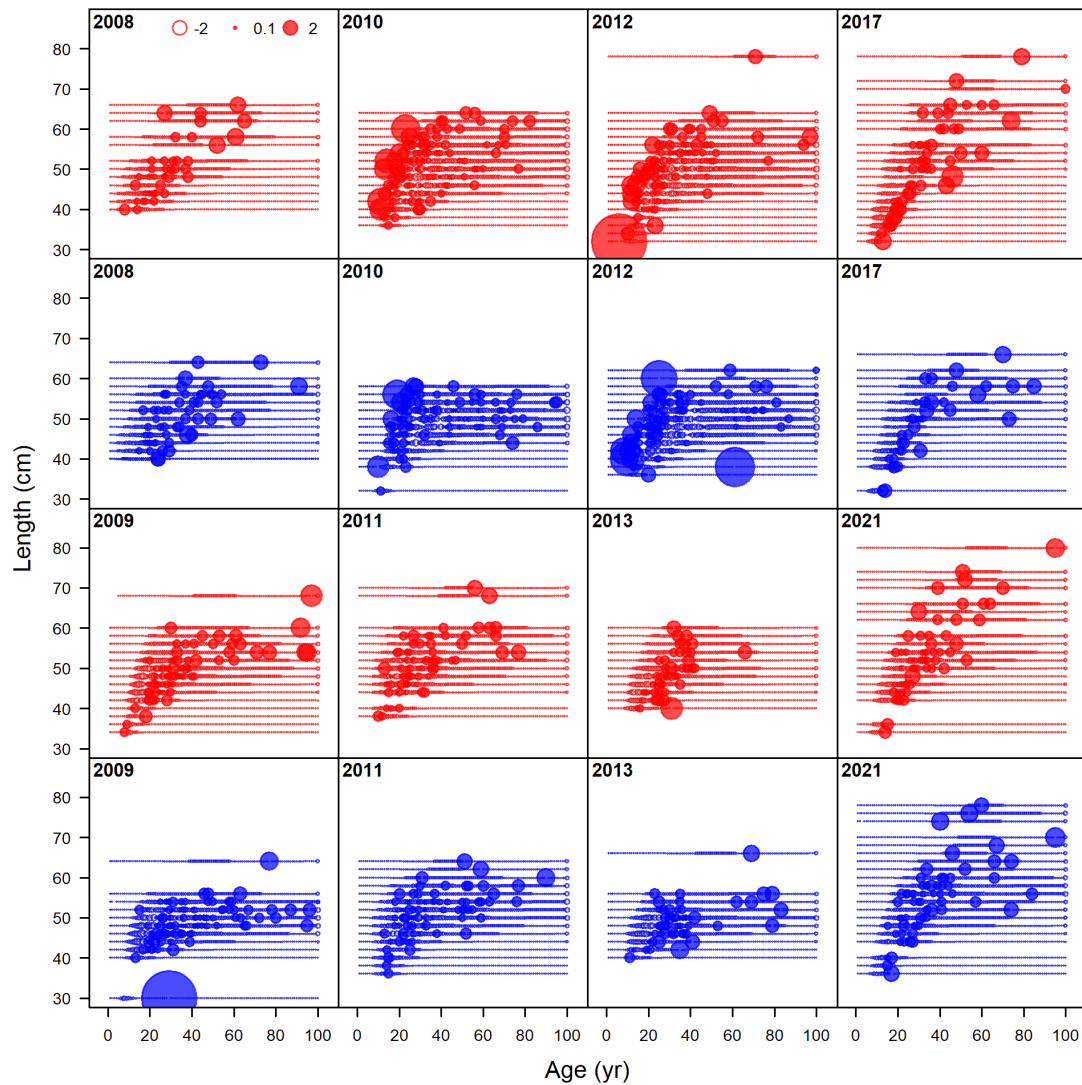


Figure 91: Pearson residuals of conditional age at length fits for the non-trawl fishery in the years 2008-2021. Closed bubble are positive residuals (observed > expected) and open bubbles are negative residuals (observed < expected).

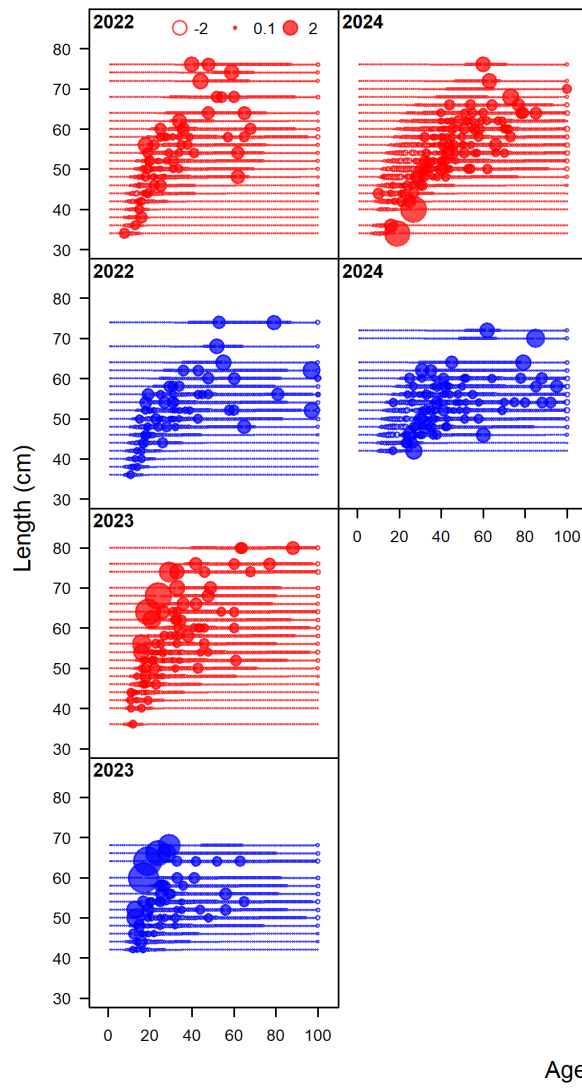


Figure 92: Pearson residuals of conditional age at length fits for the non-trawl fishery in the years 2022-2024. Closed bubble are positive residuals (observed > expected) and open bubbles are negative residuals (observed < expected).

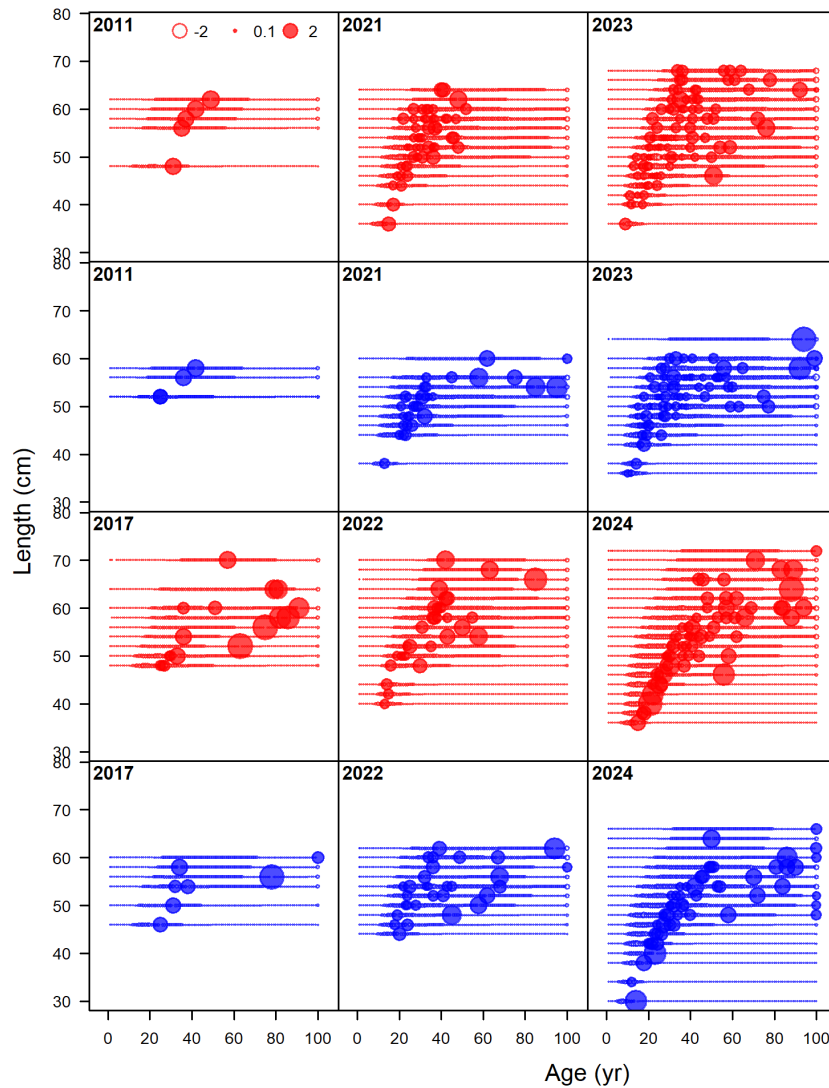


Figure 93: Pearson residuals of conditional age at length fits for the midwater trawl fishery. Closed bubble are positive residuals (observed > expected) and open bubbles are negative residuals (observed < expected).

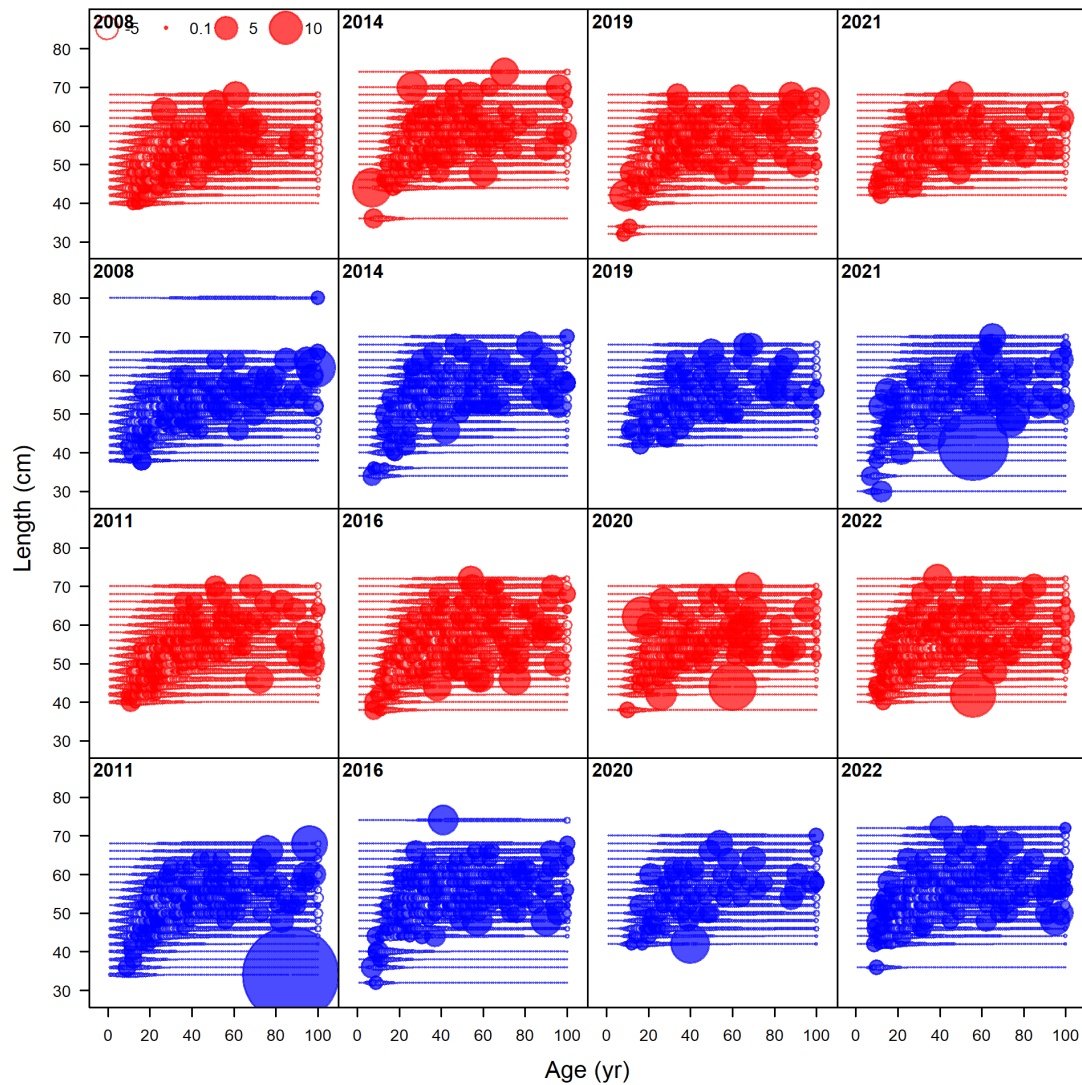


Figure 94: Pearson residuals of conditional age at length fits for the at-sea-hake fishery in the years 2008-2022. Closed bubble are positive residuals (observed > expected) and open bubbles are negative residuals (observed < expected).

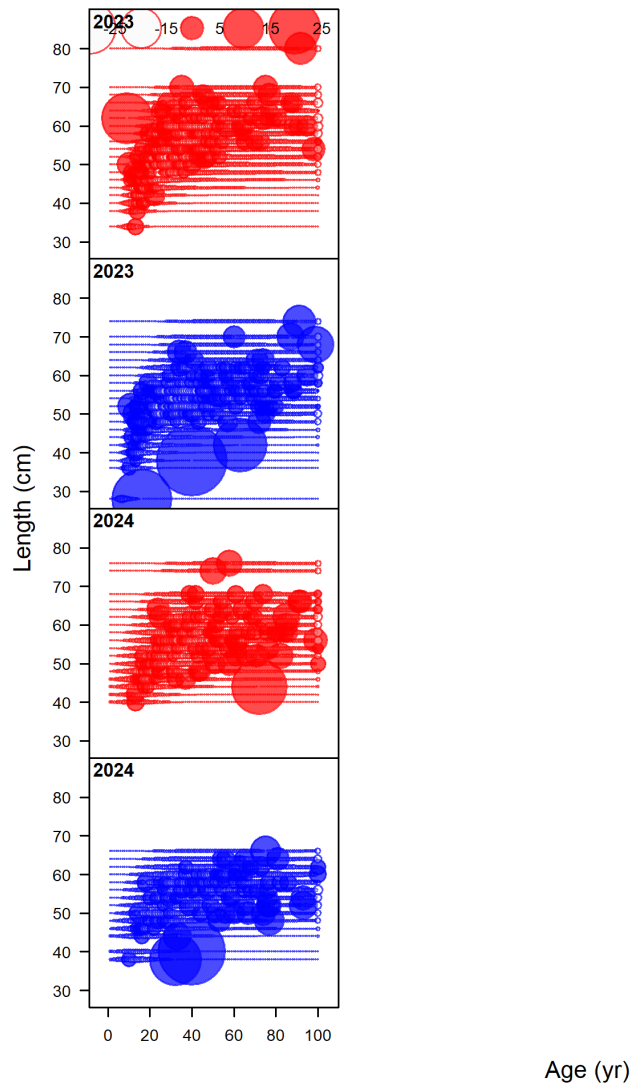


Figure 95: Pearson residuals of conditional age at length fits for the at-sea-hake fishery in the years 2023-2024. Closed bubble are positive residuals (observed > expected) and open bubbles are negative residuals (observed < expected).

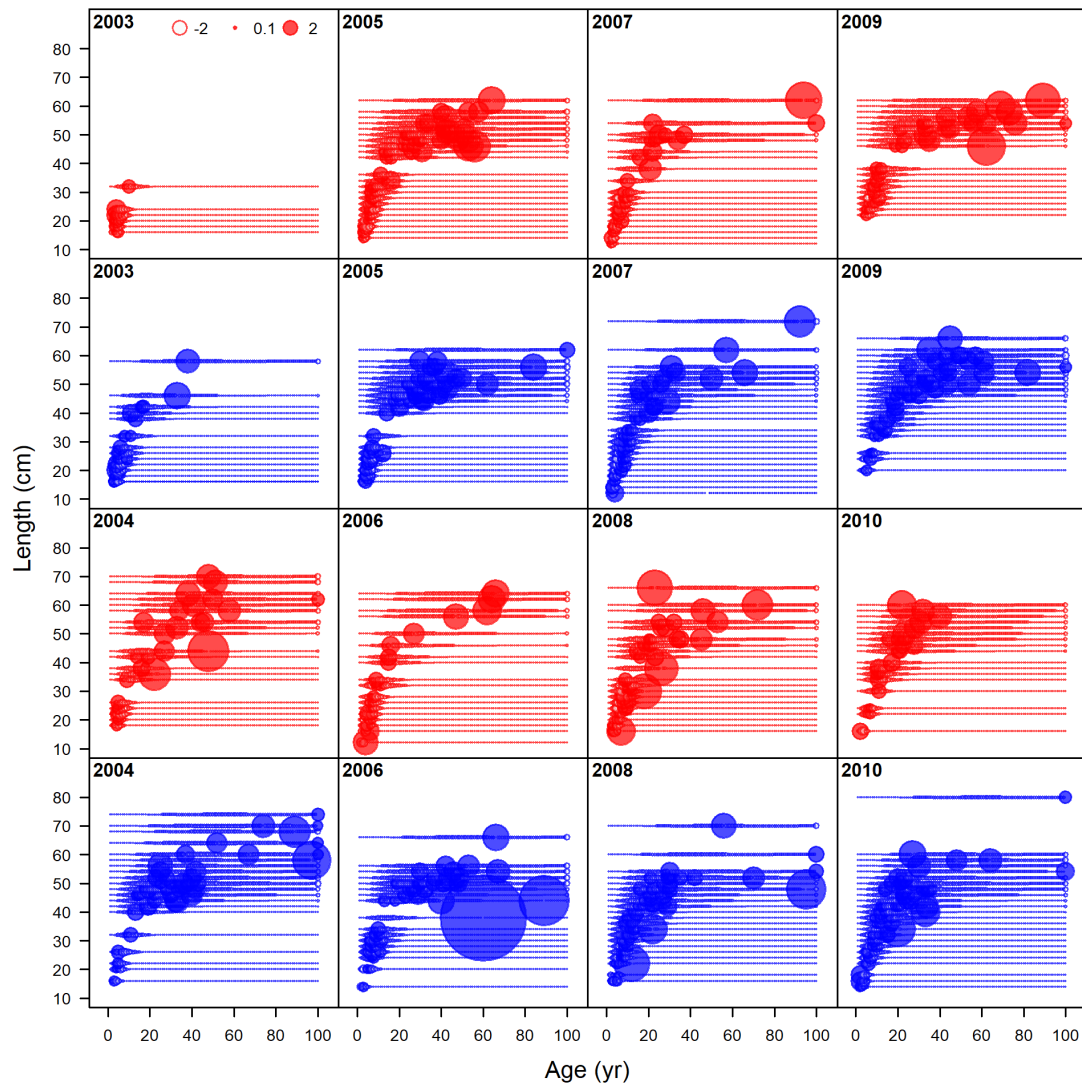


Figure 96: Pearson residuals of conditional age at length fits for the West Coast Groundfish Bottom Trawl survey in the years 2003-2010. Closed bubble are positive residuals (observed > expected) and open bubbles are negative residuals (observed < expected).



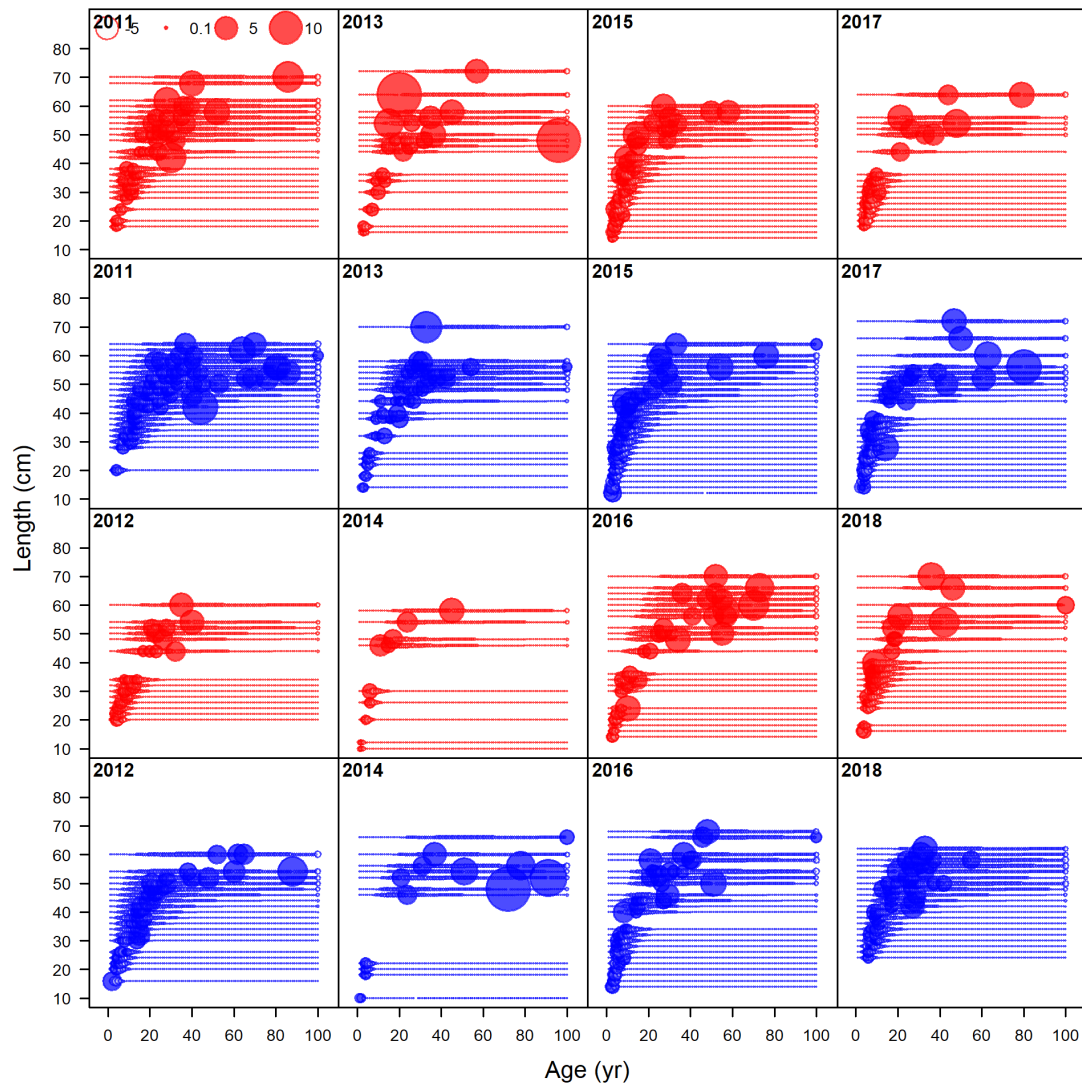


Figure 97: Pearson residuals of conditional age at length fits for the West Coast Groundfish Bottom Trawl survey in the years 2011-2018. Closed bubble are positive residuals (observed > expected) and open bubbles are negative residuals (observed < expected).

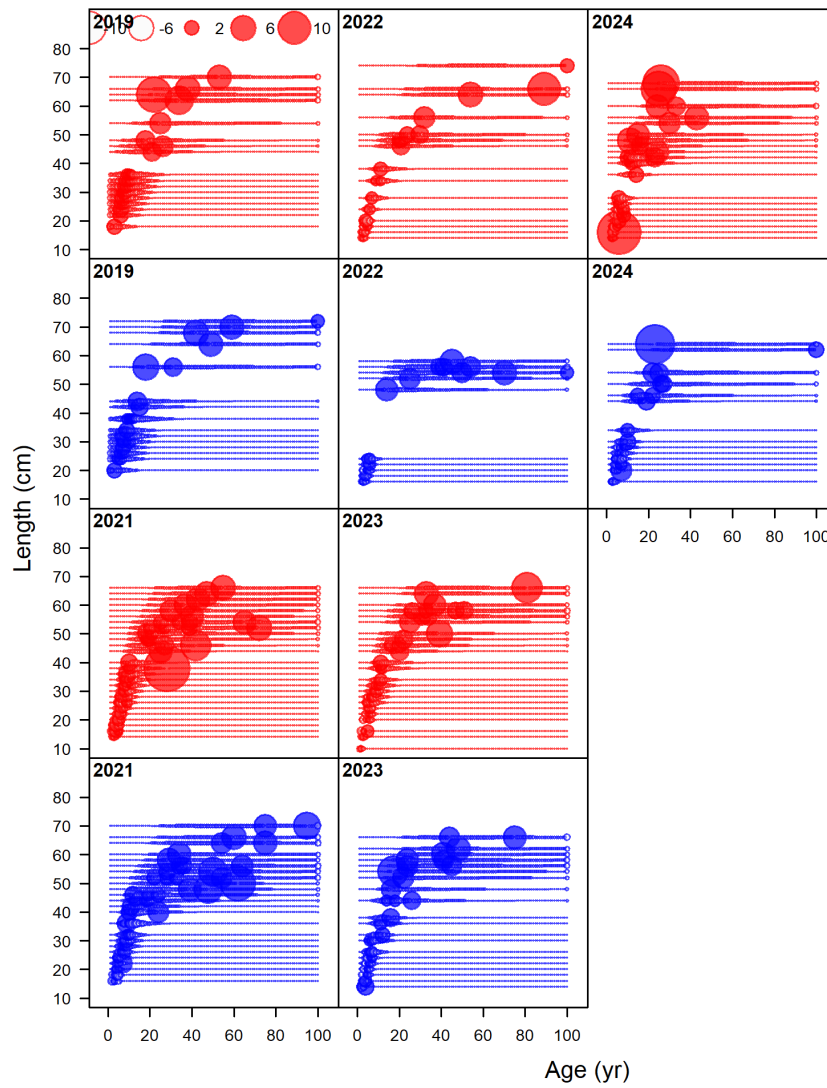


Figure 98: Pearson residuals of conditional age at length fits for the West Coast Groundfish Bottom Trawl survey in the years 2019-2024. Closed bubble are positive residuals (observed > expected) and open bubbles are negative residuals (observed < expected).

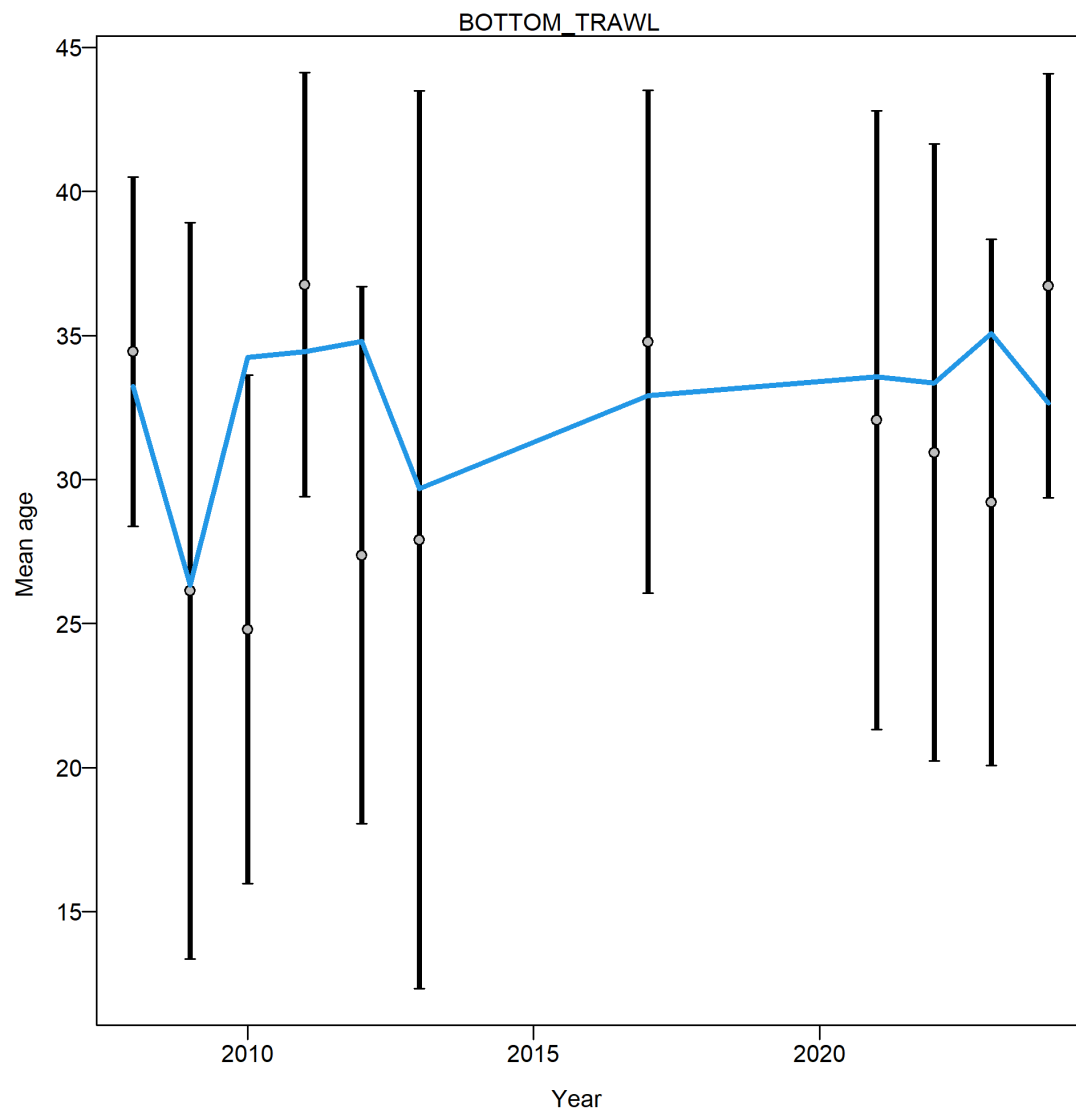


Figure 99: Mean age from conditional age-at-length data for the trawl fishery with 95% confidence intervals based on current samples sizes and data weighting.

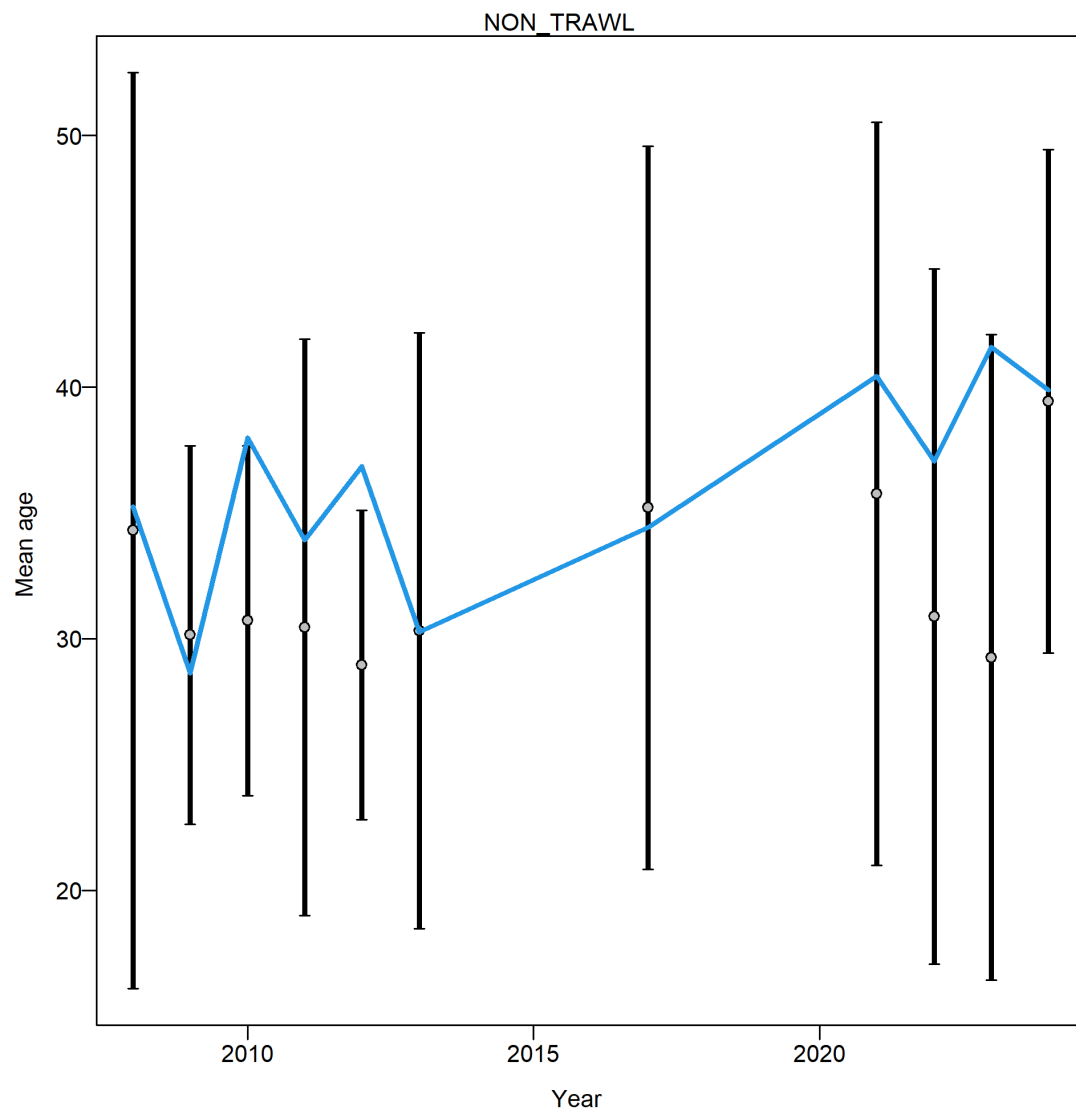


Figure 100: Mean age from conditional age-at-length data for the non-trawl fishery with 95% confidence intervals based on current samples sizes and data weighting.

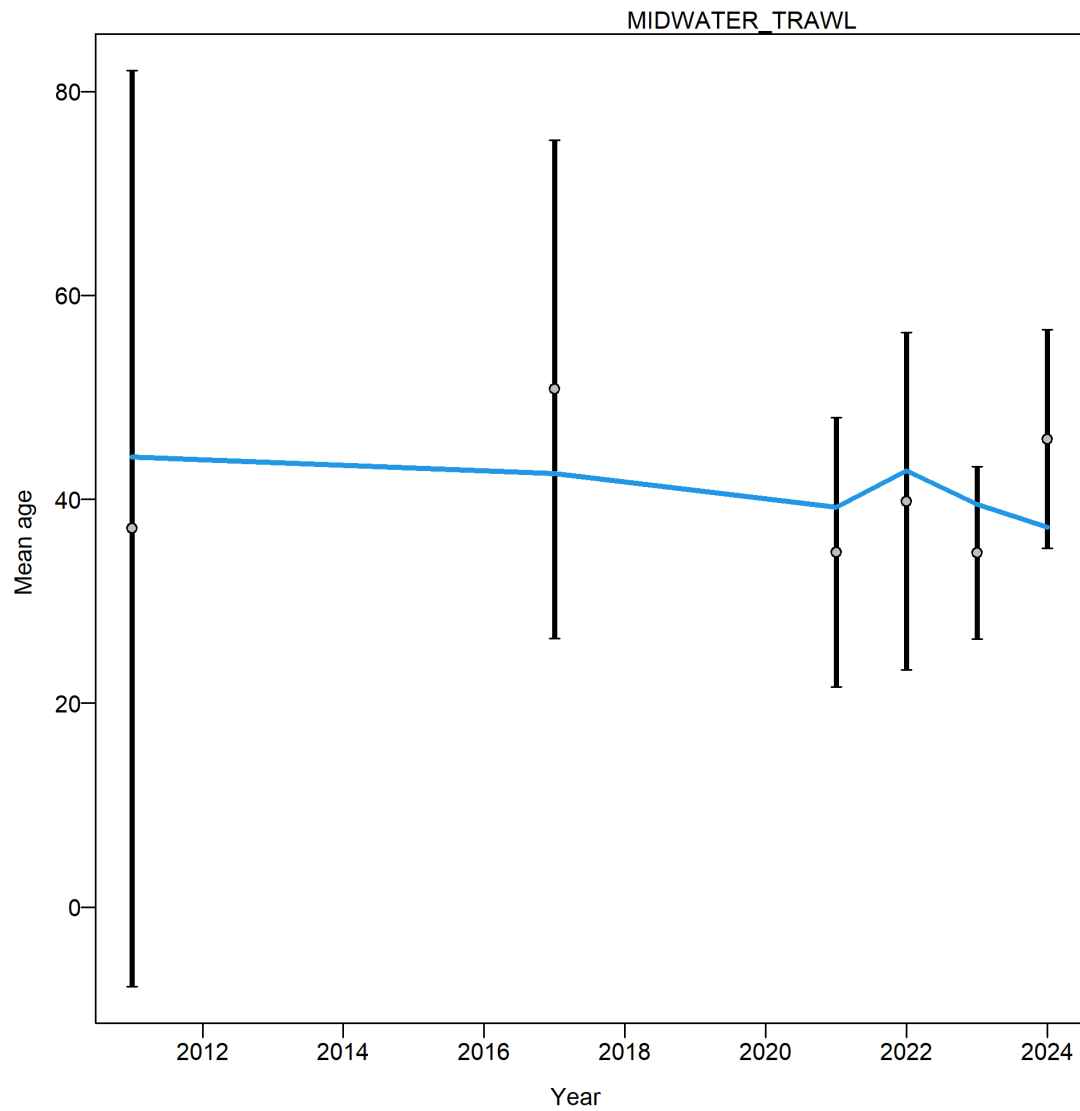


Figure 101: Mean age from conditional age-at-length data for the midwater trawl fishery with 95% confidence intervals based on current samples sizes and data weighting.

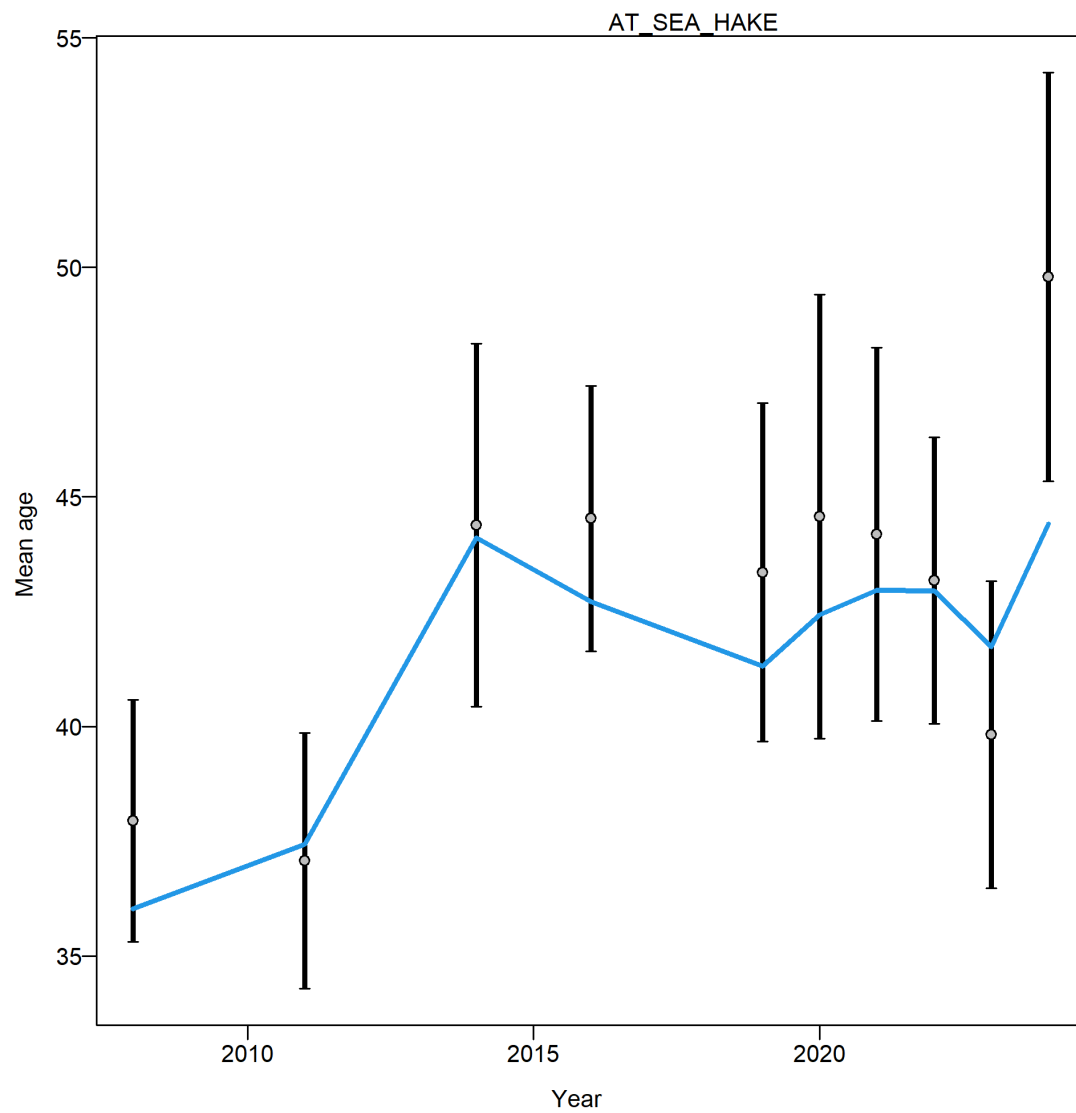


Figure 102: Mean age from conditional age-at-length data for the at-sea-hake fishery with 95% confidence intervals based on current samples sizes and data weighting.

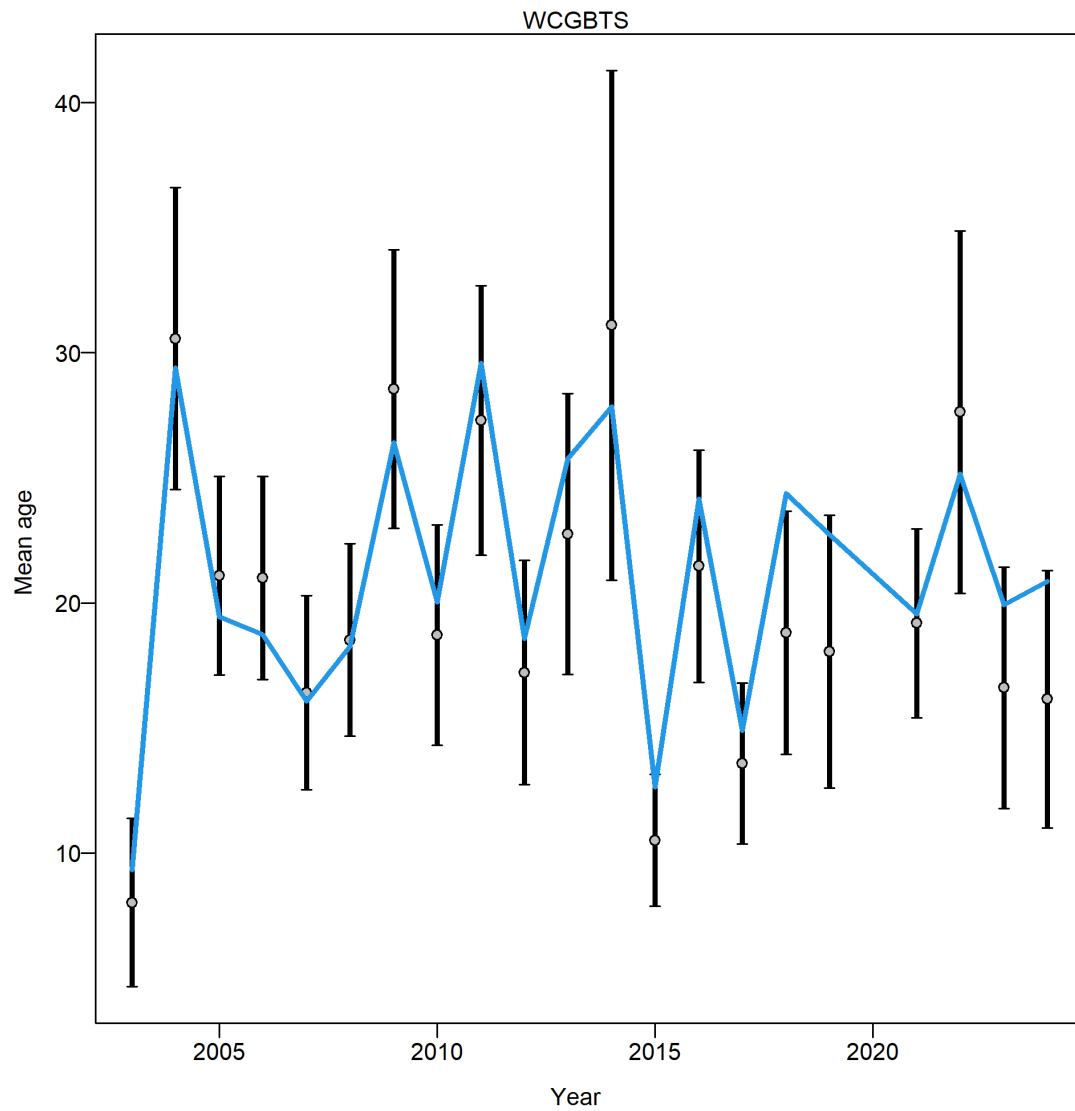


Figure 103: Mean age from conditional age-at-length data for the West Coast Groundfish Bottom Trawl survey with 95% confidence intervals based on current samples sizes and data weighting.

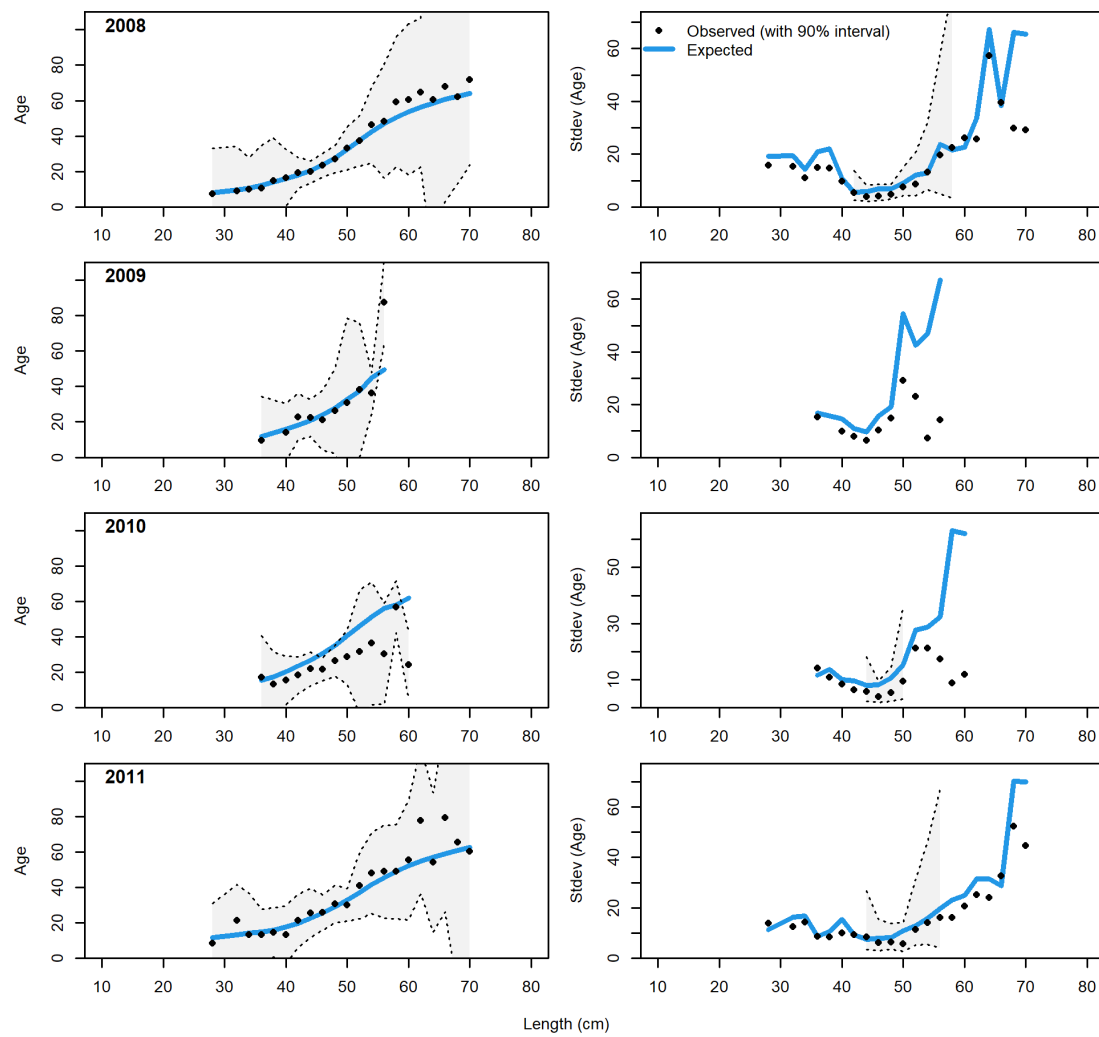


Figure 104: Conditional age at length plot showing mean age (left panels) and standard deviation (right panels) for the bottom trawl fishery in years 2008-2011. The CIs are 90% based on adding 1.64 SE of mean to the data for the left panels and the standard error of mean age at length (obs. and exp.) with 90% CIs based on the chi-square distribution.



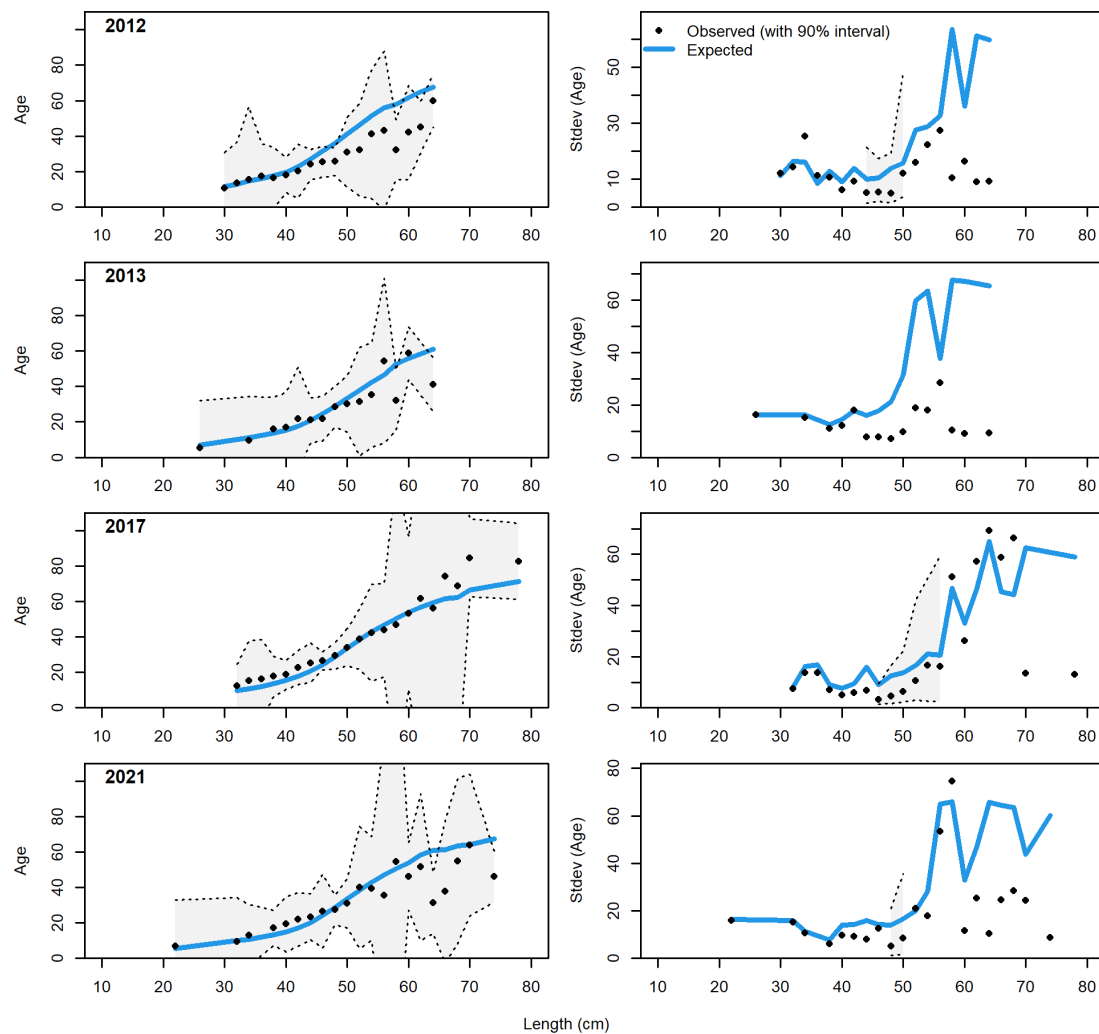


Figure 105: Conditional age at length plot showing mean age (left panels) and standard deviation (right panels) for the bottom trawl fishery in years 2012-2021. The CIs are 90% based on adding 1.64 SE of mean to the data for the left panels and the standard error of mean age at length (obs. and exp.) with 90% CIs based on the chi-square distribution.

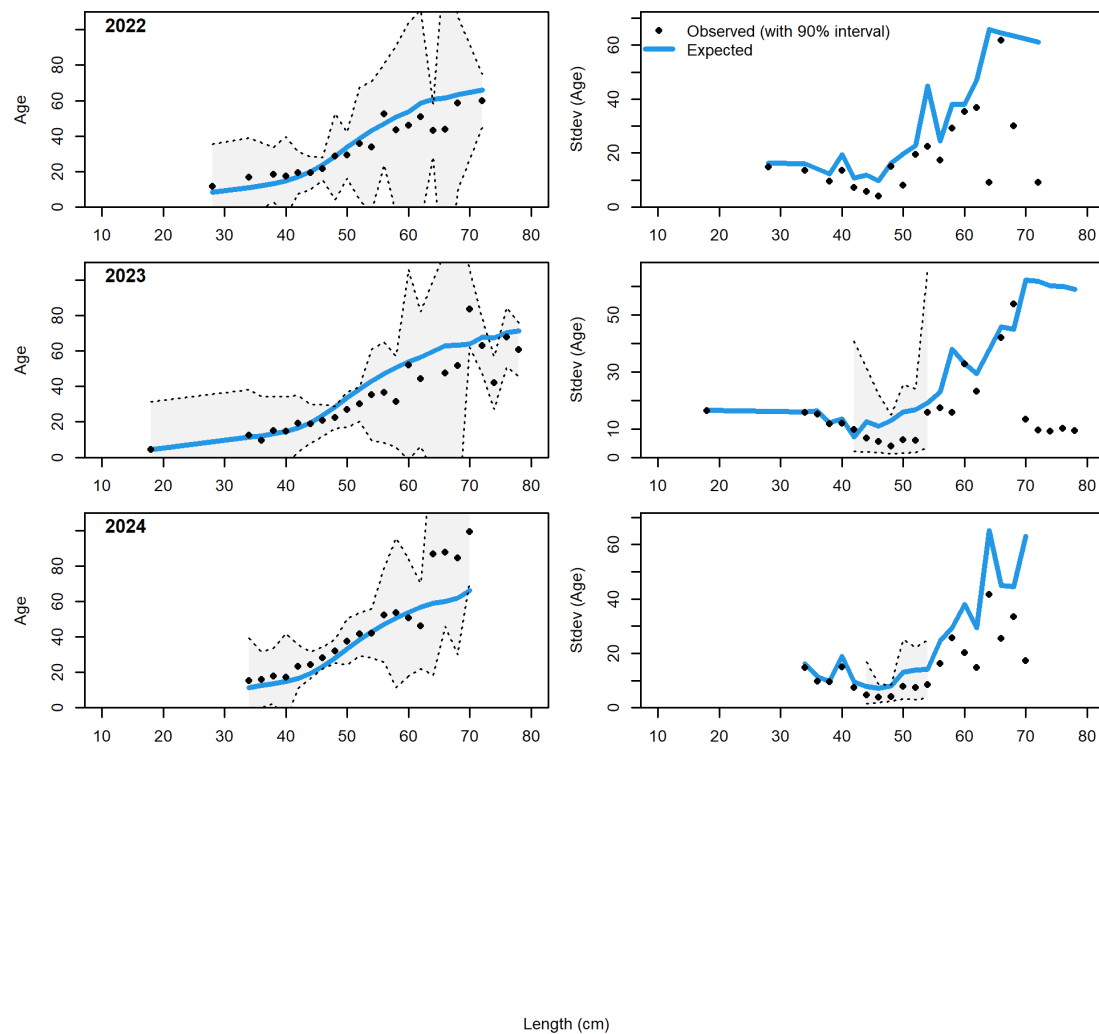


Figure 106: Conditional age at length plot showing mean age (left panels) and standard deviation (right panels) for the bottom trawl fishery in years 2022-2024. The CIs are 90% based on adding 1.64 SE of mean to the data for the left panels and the standard error of mean age at length (obs. and exp.) with 90% CIs based on the chi-square distribution.

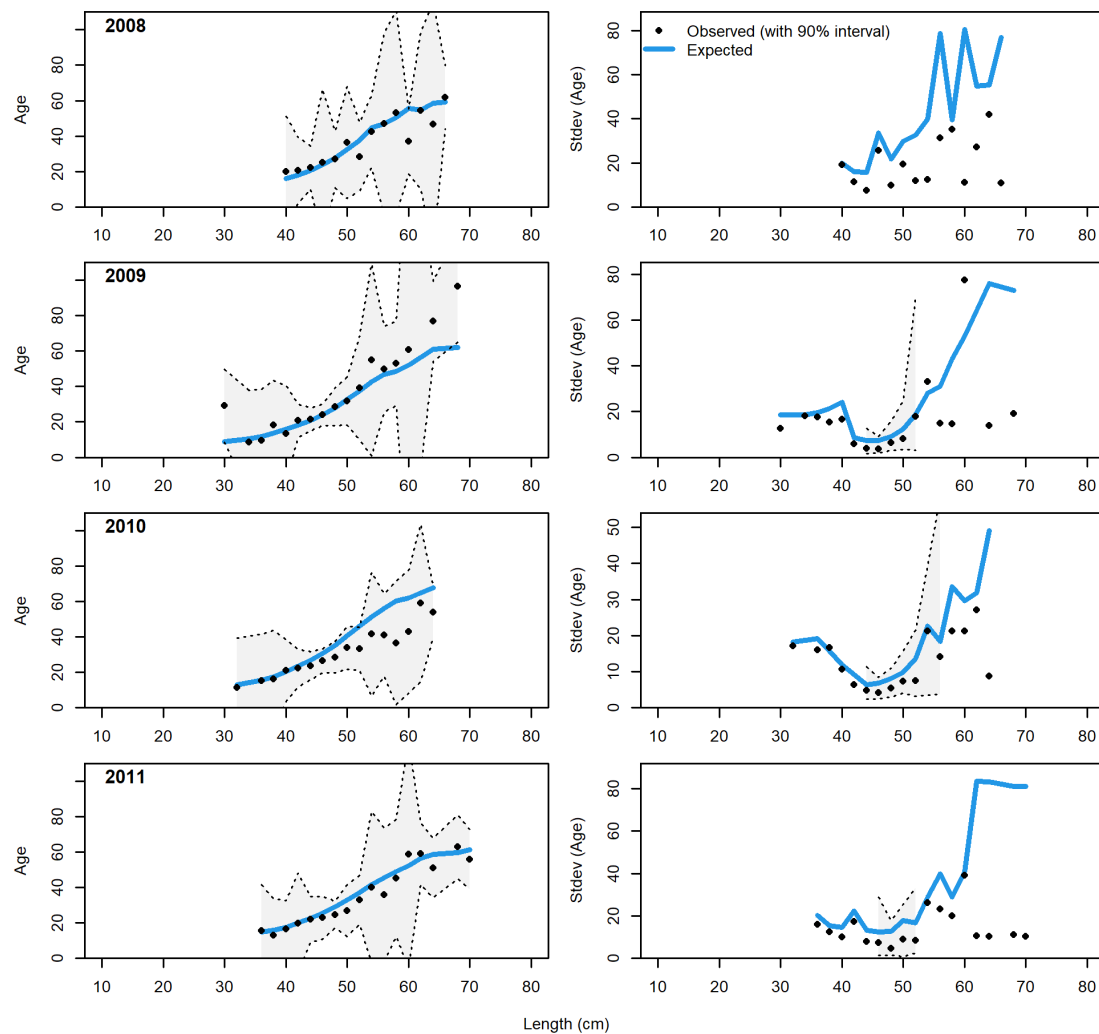


Figure 107: Conditional age at length plot showing mean age (left panels) and standard deviation (right panels) for the non-trawl fishery in years 2008-2011. The CIs are 90% based on adding 1.64 SE of mean to the data for the left panels and the standard error of mean age at length (obs. and exp.) with 90% CIs based on the chi-square distribution.

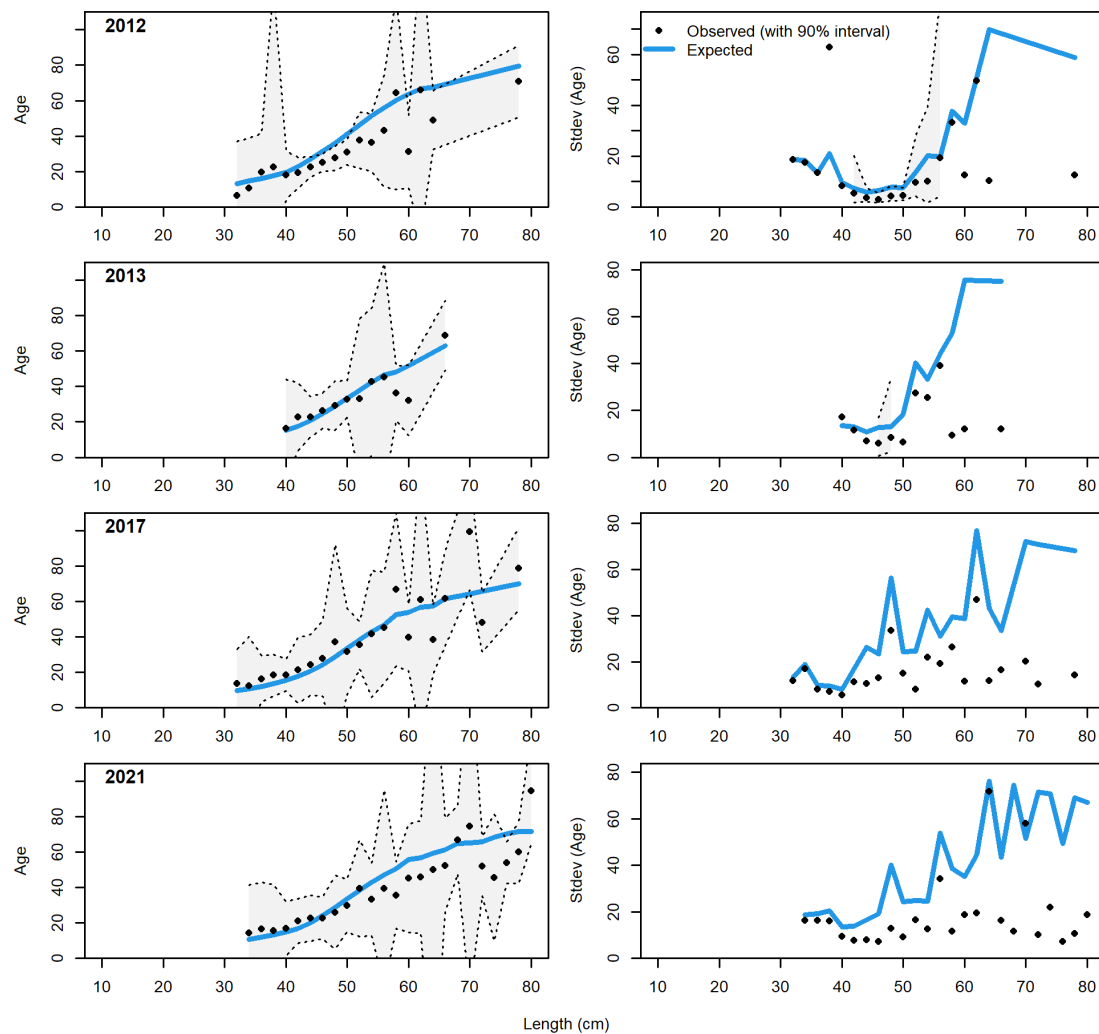


Figure 108: Conditional age at length plot showing mean age (left panels) and standard deviation (right panels) for the non-trawl fishery in years 2012-2021. The CIs are 90% based on adding 1.64 SE of mean to the data for the left panels and the standard error of mean age at length (obs. and exp.) with 90% CIs based on the chi-square distribution.

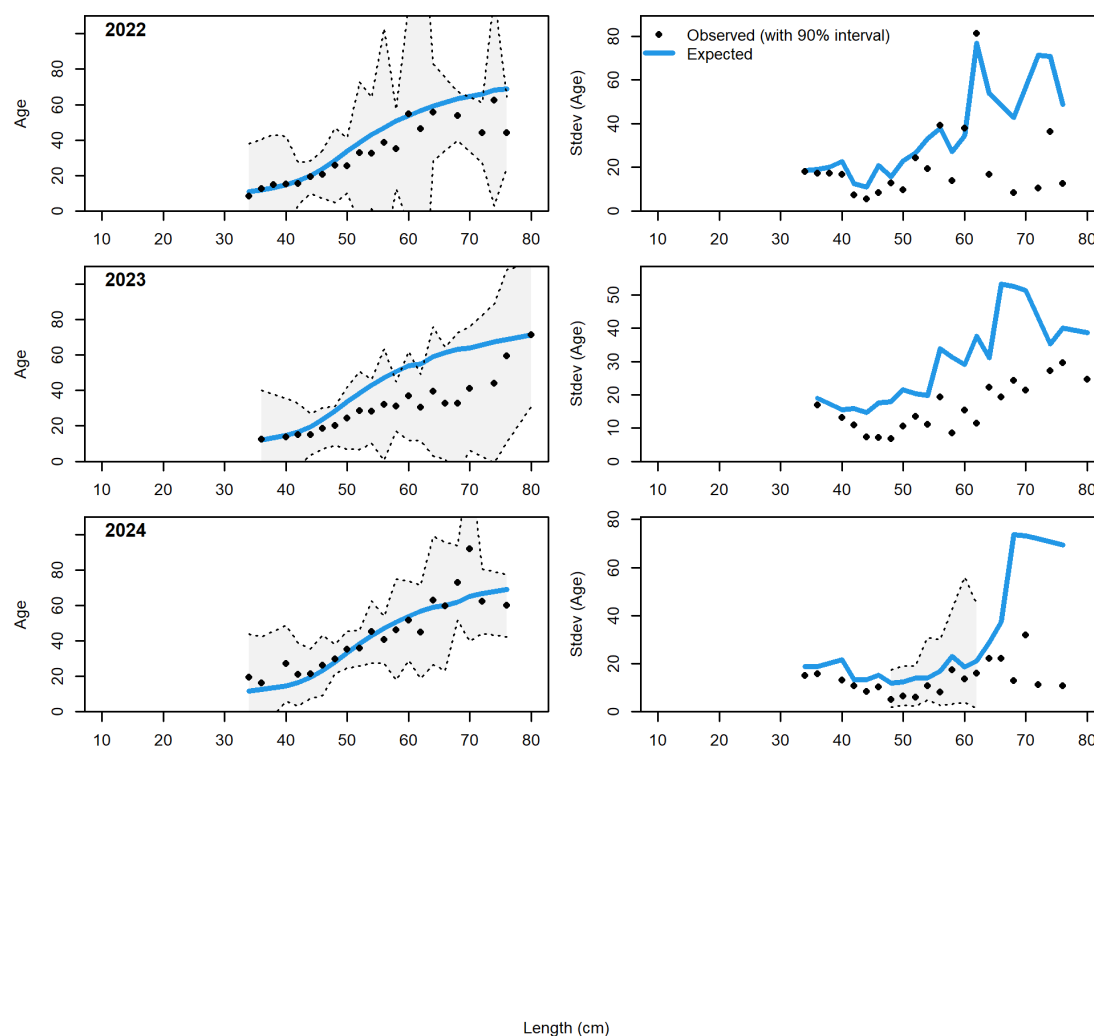


Figure 109: Conditional age at length plot showing mean age (left panels) and standard deviation (right panels) for the non-trawl fishery in years 2022-2024. The CIs are 90% based on adding 1.64 SE of mean to the data for the left panels and the standard error of mean age at length (obs. and exp.) with 90% CIs based on the chi-square distribution.

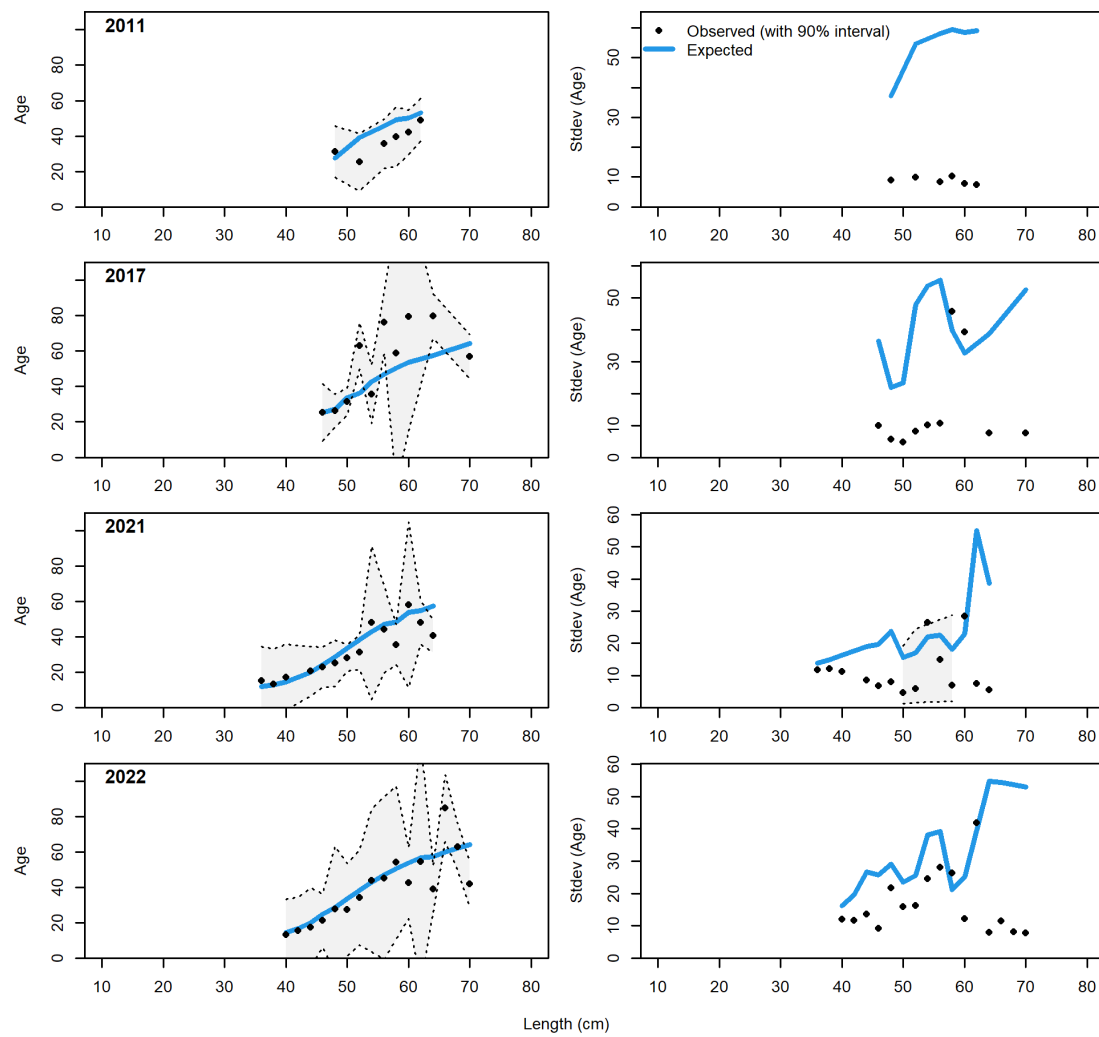


Figure 110: Conditional age at length plot showing mean age (left panels) and standard deviation (right panels) for the midwater trawl fishery in years 2011-2022. The CIs are 90% based on adding 1.64 SE of mean to the data for the left panels and the standard error of mean age at length (obs. and exp.) with 90% CIs based on the chi-square distribution.

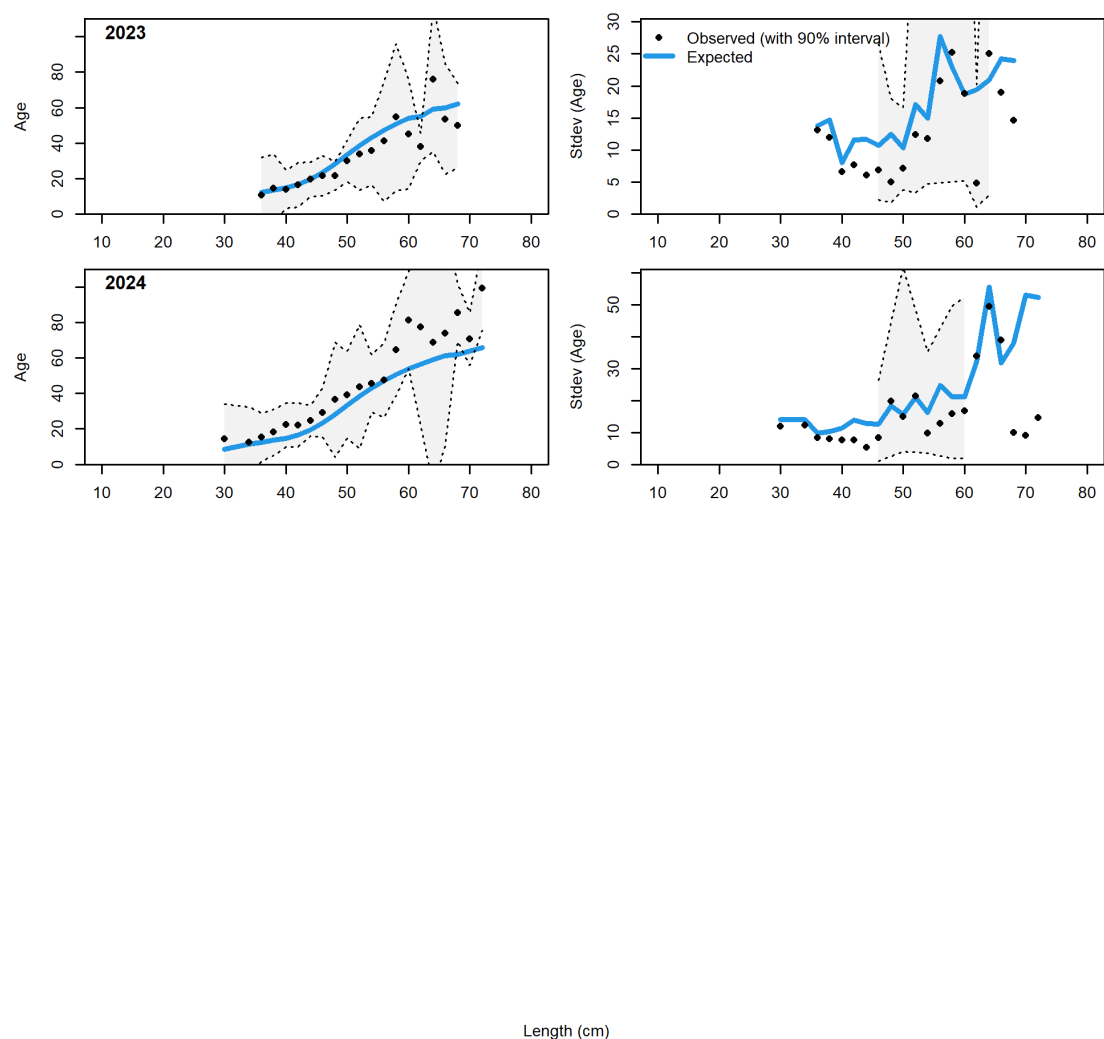


Figure 111: Conditional age at length plot showing mean age (left panels) and standard deviation (right panels) for the non-trawl fishery in years 2023-2024. The CIs are 90% based on adding 1.64 SE of mean to the data for the left panels and the standard error of mean age at length (obs. and exp.) with 90% CIs based on the chi-square distribution.

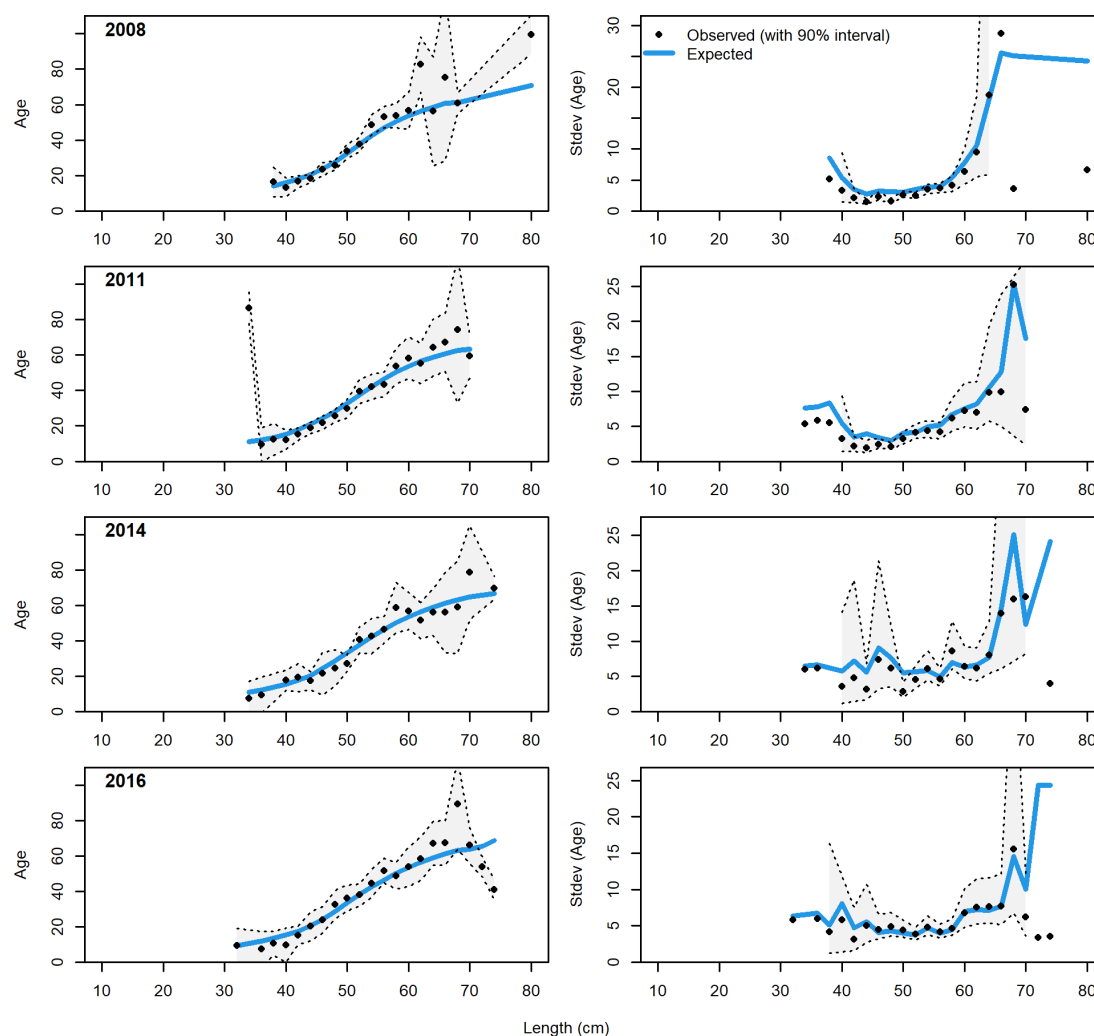


Figure 112: Conditional age at length plot showing mean age (left panels) and standard deviation (right panels) for the at-sea-hake fishery in years 2008-2016. The CIs are 90% based on adding 1.64 SE of mean to the data for the left panels and the standard error of mean age at length (obs. and exp.) with 90% CIs based on the chi-square distribution.



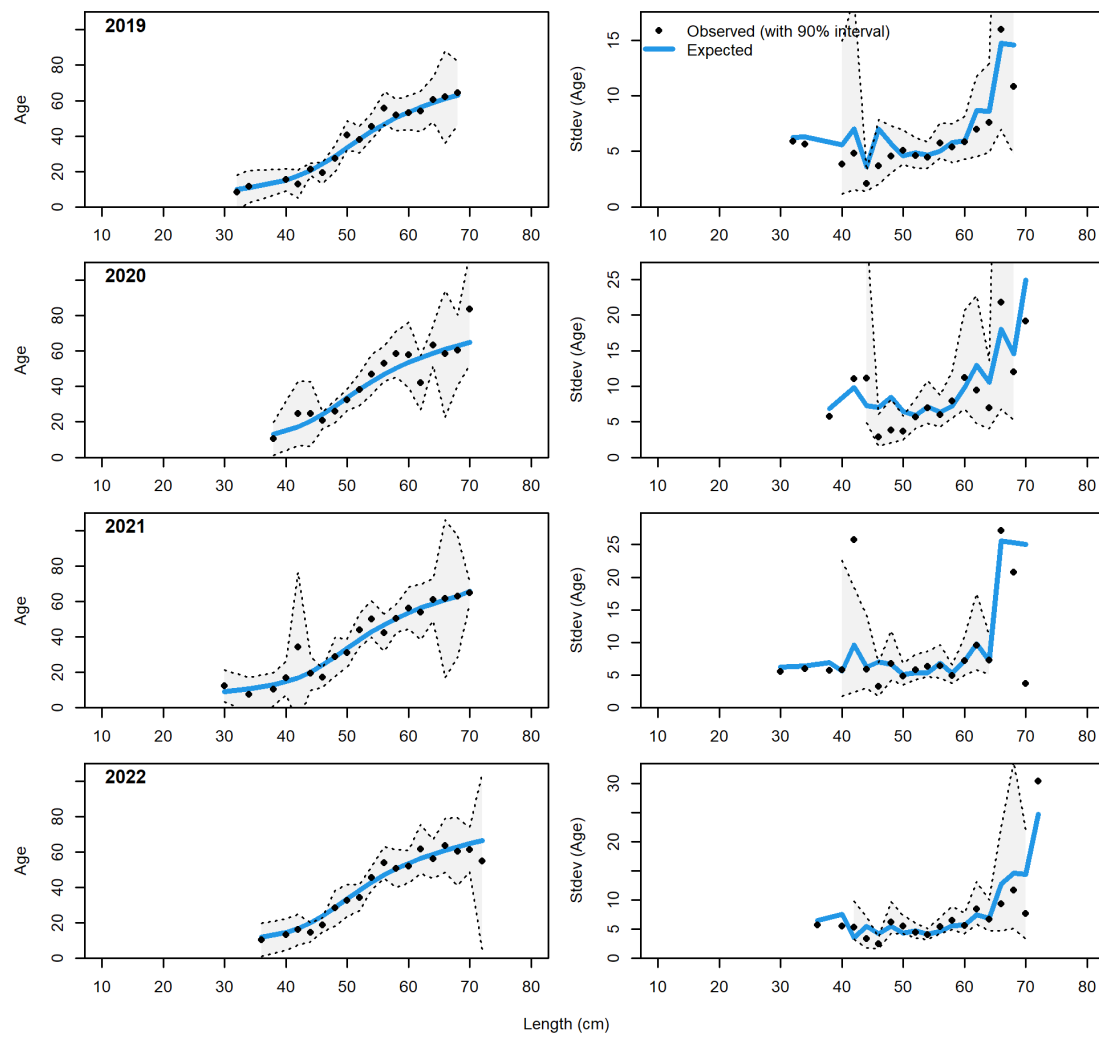


Figure 113: Conditional age at length plot showing mean age (left panels) and standard deviation (right panels) for the at-sea-hake fishery in years 2019-2022. The CIs are 90% based on adding 1.64 SE of mean to the data for the left panels and the standard error of mean age at length (obs. and exp.) with 90% CIs based on the chi-square distribution.

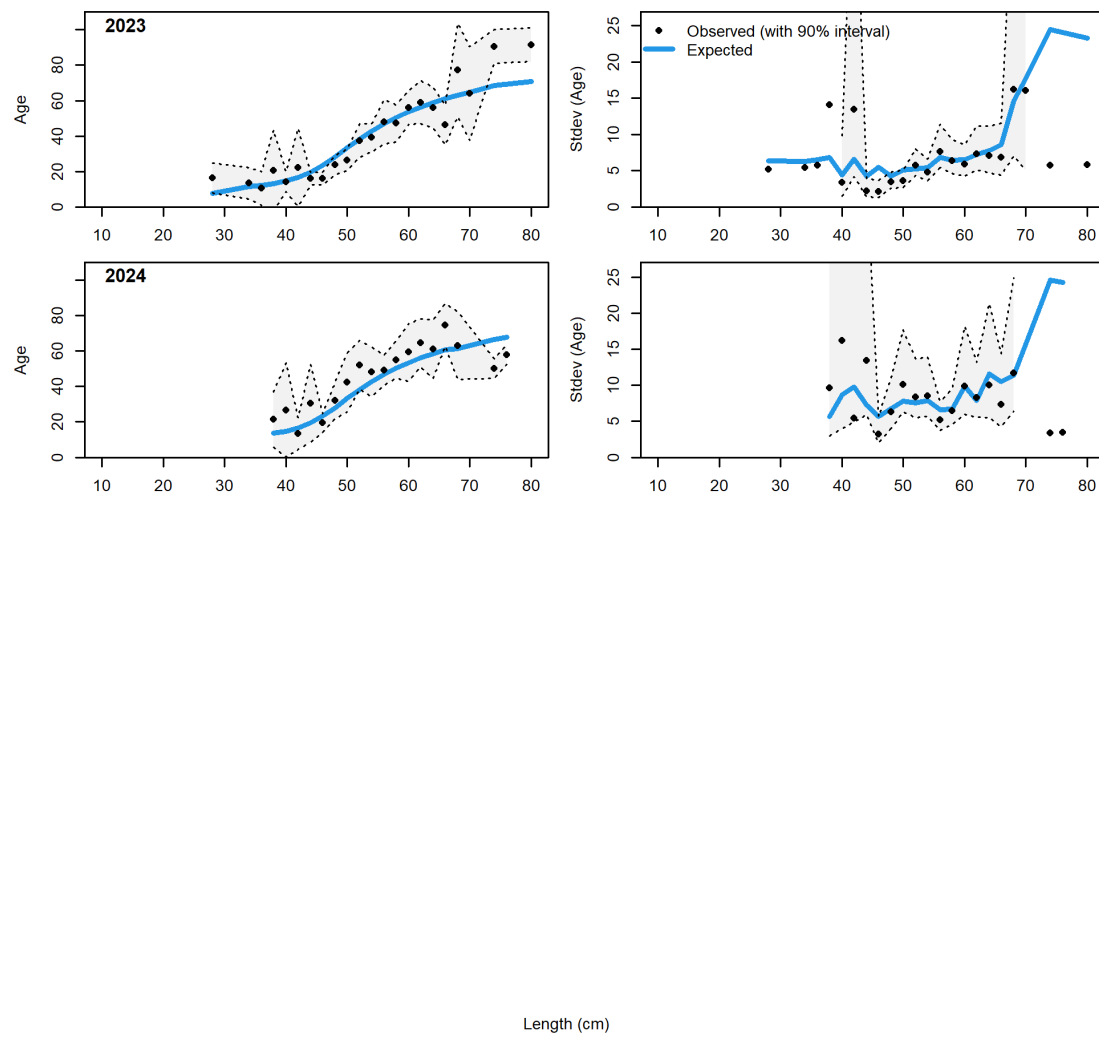


Figure 114: Conditional age at length plot showing mean age (left panels) and standard deviation (right panels) for the at-sea-hake fishery in years 2023-2024. The CIs are 90% based on adding 1.64 SE of mean to the data for the left panels and the standard error of mean age at length (obs. and exp.) with 90% CIs based on the chi-square distribution.

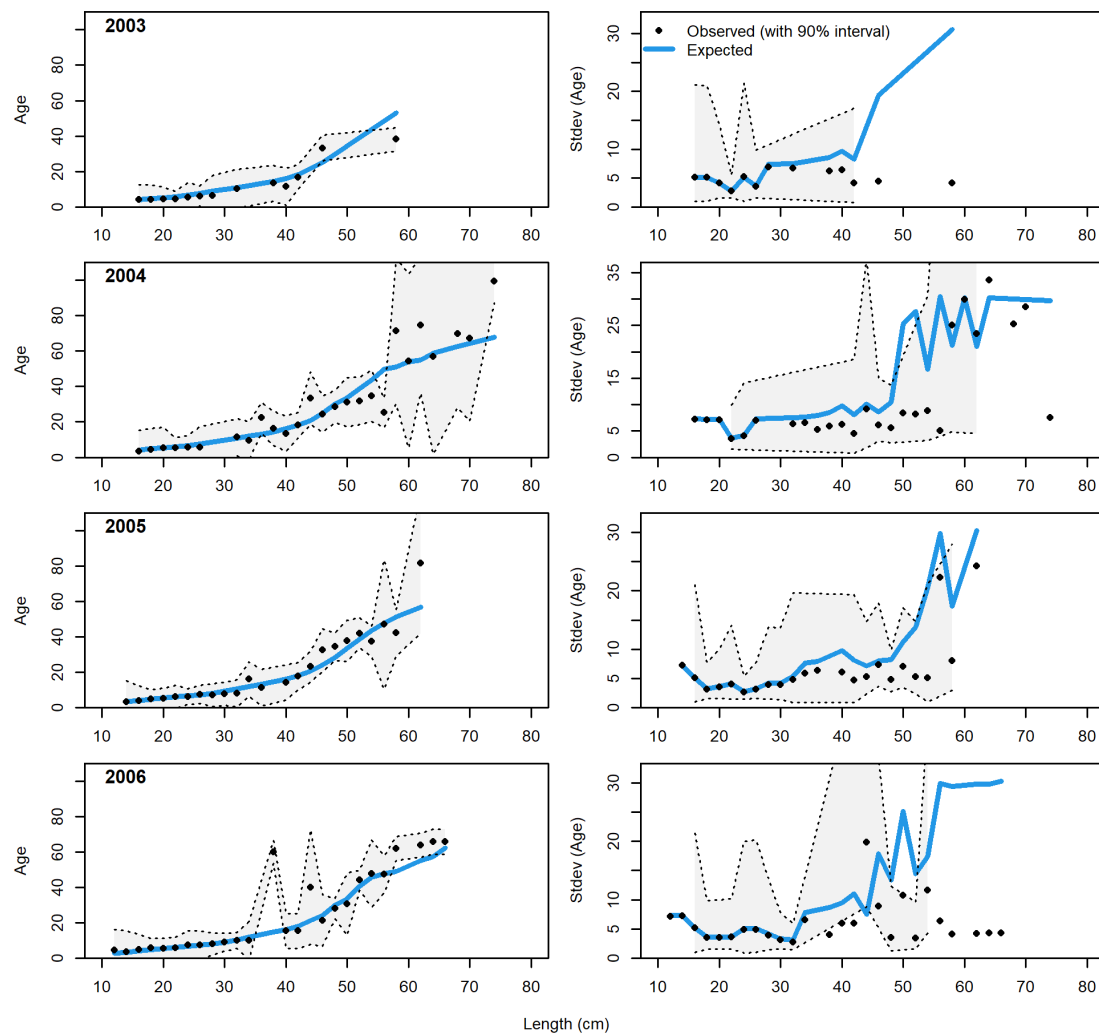


Figure 115: Conditional age at length plot showing mean age (left panels) and standard deviation (right panels) for the West Coast Groundfish Bottom Trawl survey in years 2003-2006. The CIs are 90% based on adding 1.64 SE of mean to the data for the left panels and the standard error of mean age at length (obs. and exp.) with 90% CIs based on the chi-square distribution.

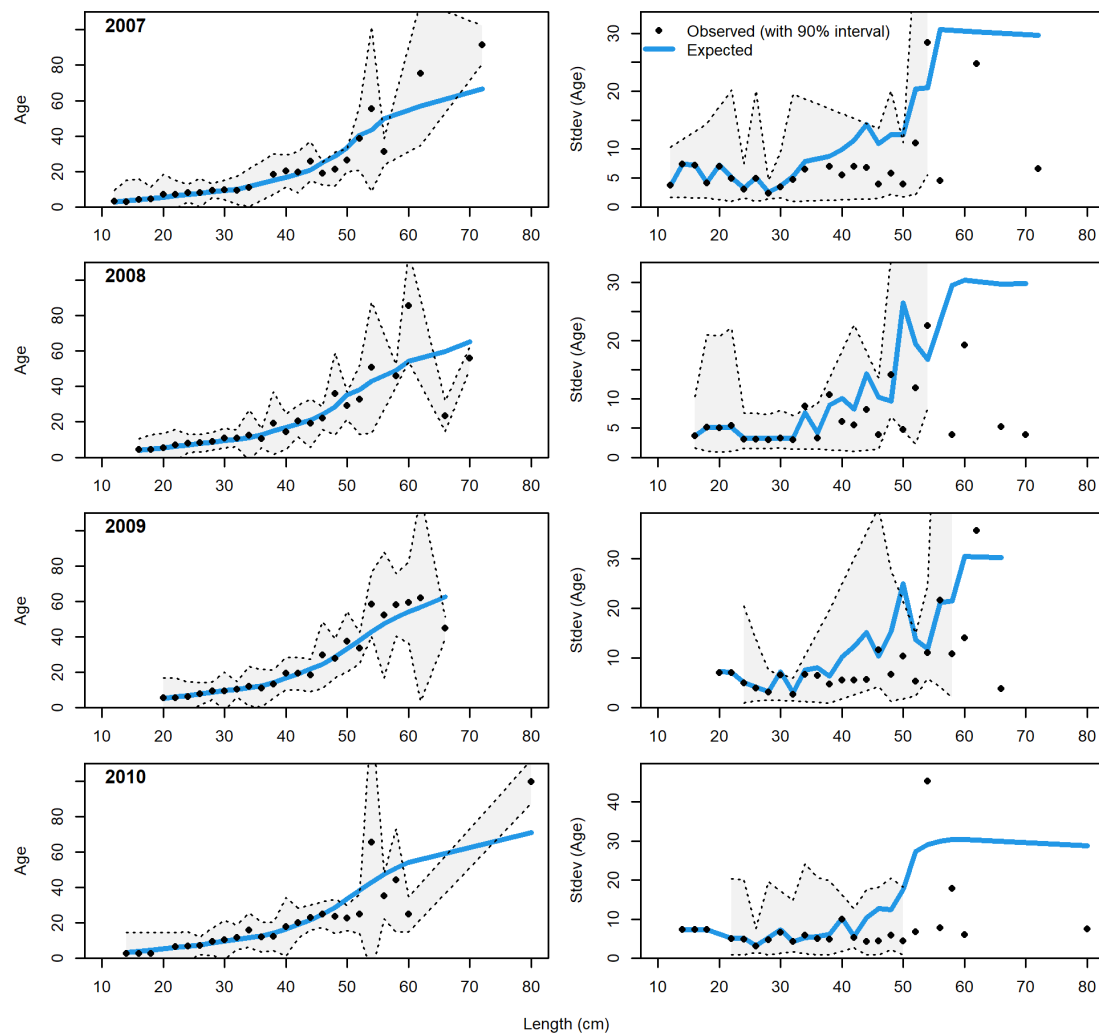


Figure 116: Conditional age at length plot showing mean age (left panels) and standard deviation (right panels) for the West Coast Groundfish Bottom Trawl survey in years 2007-2010. The CIs are 90% based on adding 1.64 SE of mean to the data for the left panels and the standard error of mean age at length (obs. and exp.) with 90% CIs based on the chi-square distribution.

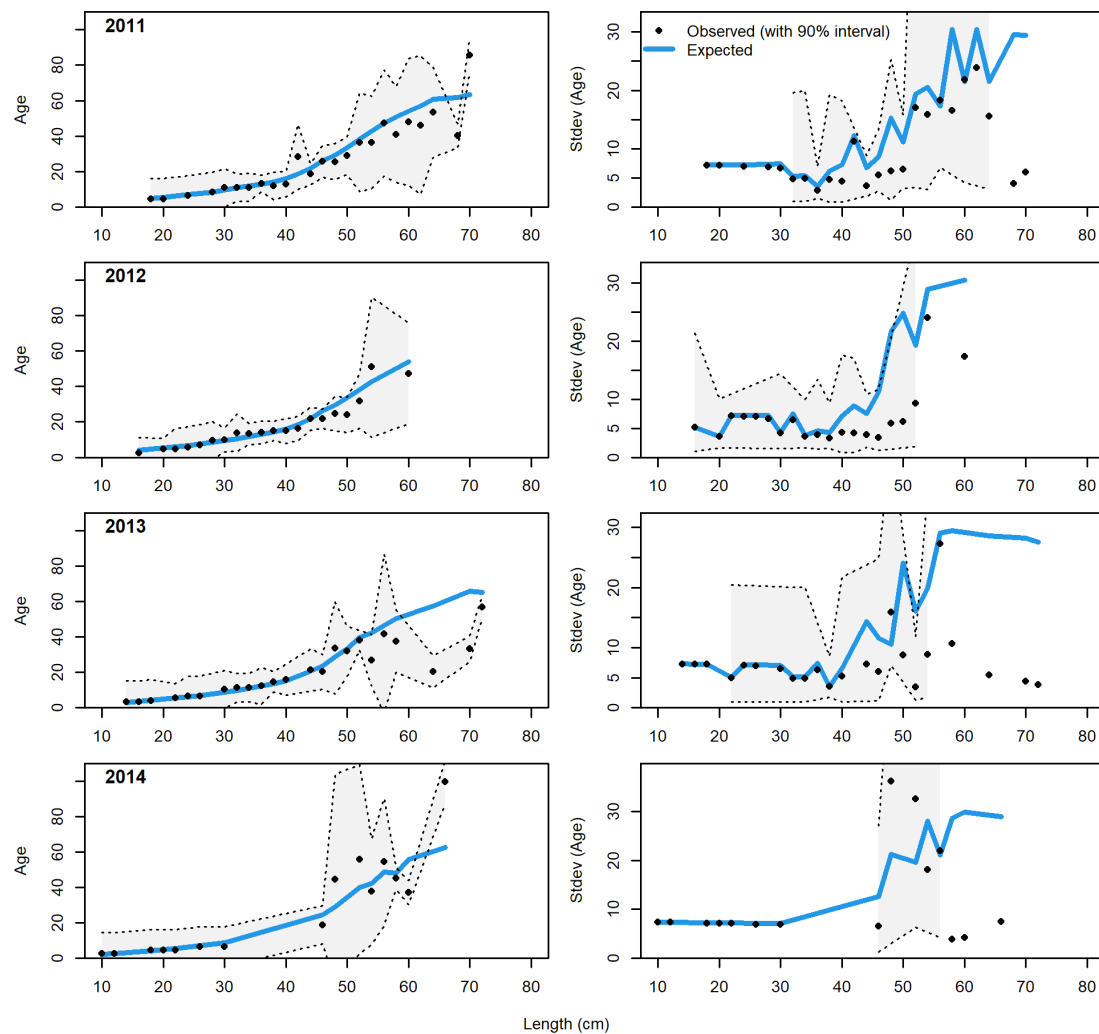


Figure 117: Conditional age at length plot showing mean age (left panels) and standard deviation (right panels) for the West Coast Groundfish Bottom Trawl survey in years 2011-2014. The CIs are 90% based on adding 1.64 SE of mean to the data for the left panels and the standard error of mean age at length (obs. and exp.) with 90% CIs based on the chi-square distribution.

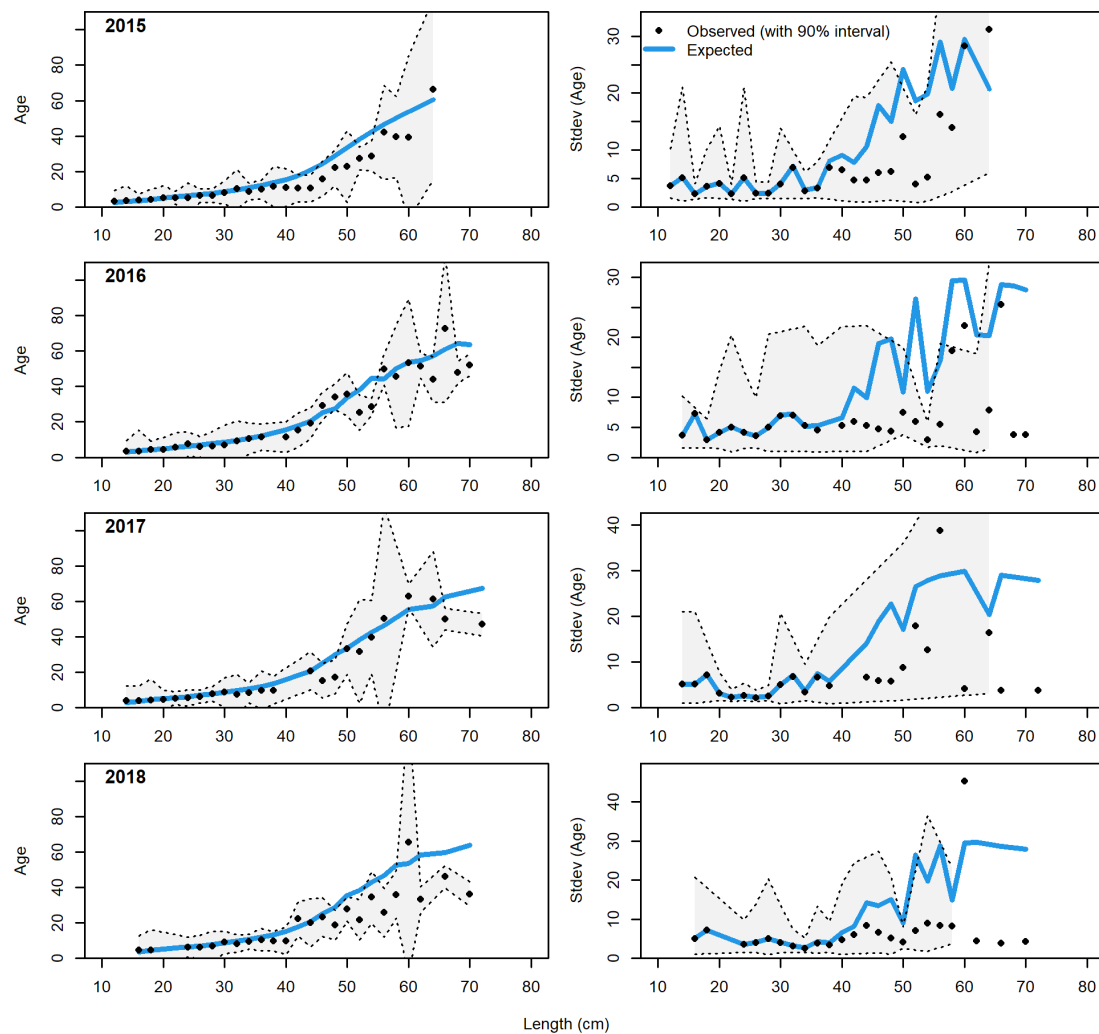


Figure 118: Conditional age at length plot showing mean age (left panels) and standard deviation (right panels) for the West Coast Groundfish Bottom Trawl survey in years 2015-2018. The CIs are 90% based on adding 1.64 SE of mean to the data for the left panels and the standard error of mean age at length (obs. and exp.) with 90% CIs based on the chi-square distribution.

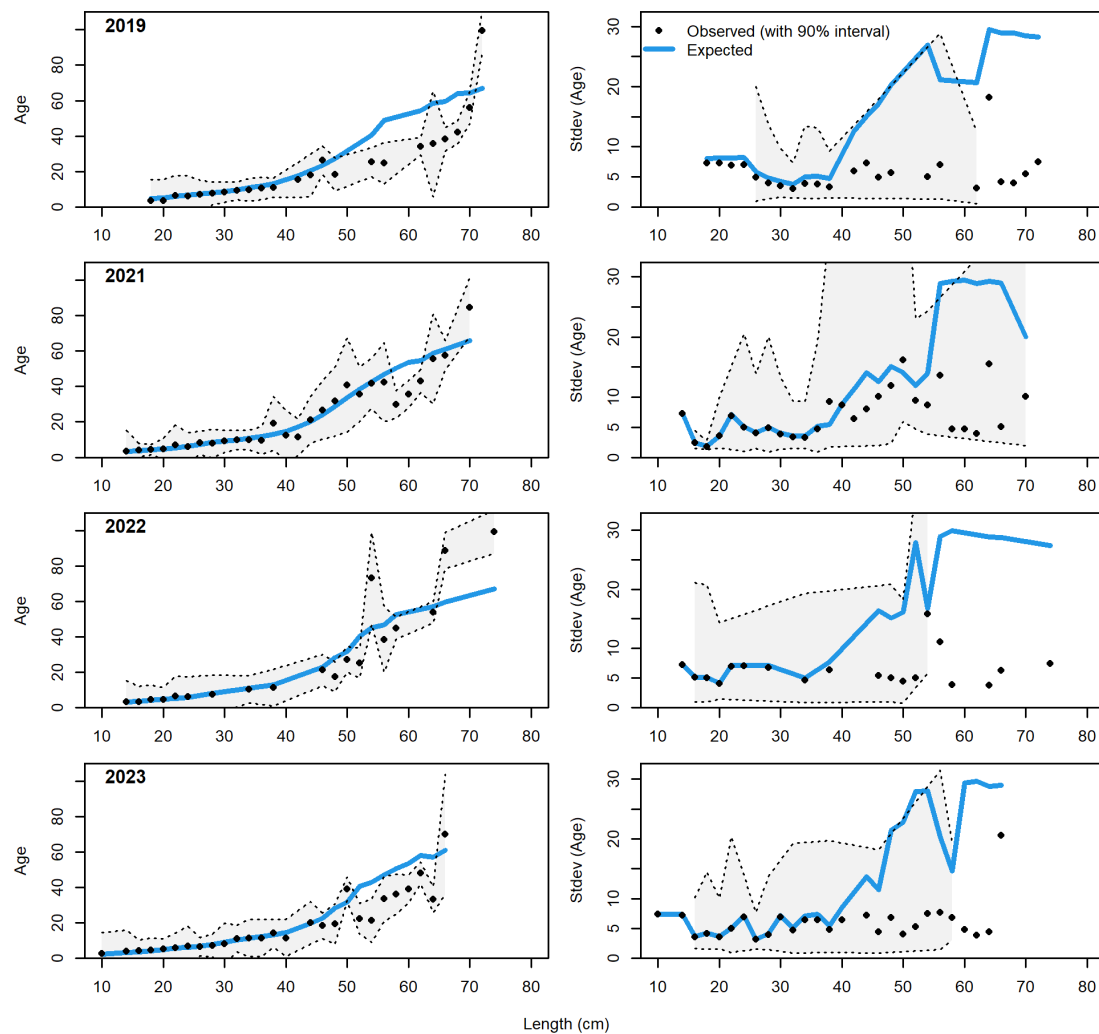


Figure 119: Conditional age at length plot showing mean age (left panels) and standard deviation (right panels) for the West Coast Groundfish Bottom Trawl survey in years 2019-2023. The CIs are 90% based on adding 1.64 SE of mean to the data for the left panels and the standard error of mean age at length (obs. and exp.) with 90% CIs based on the chi-square distribution.

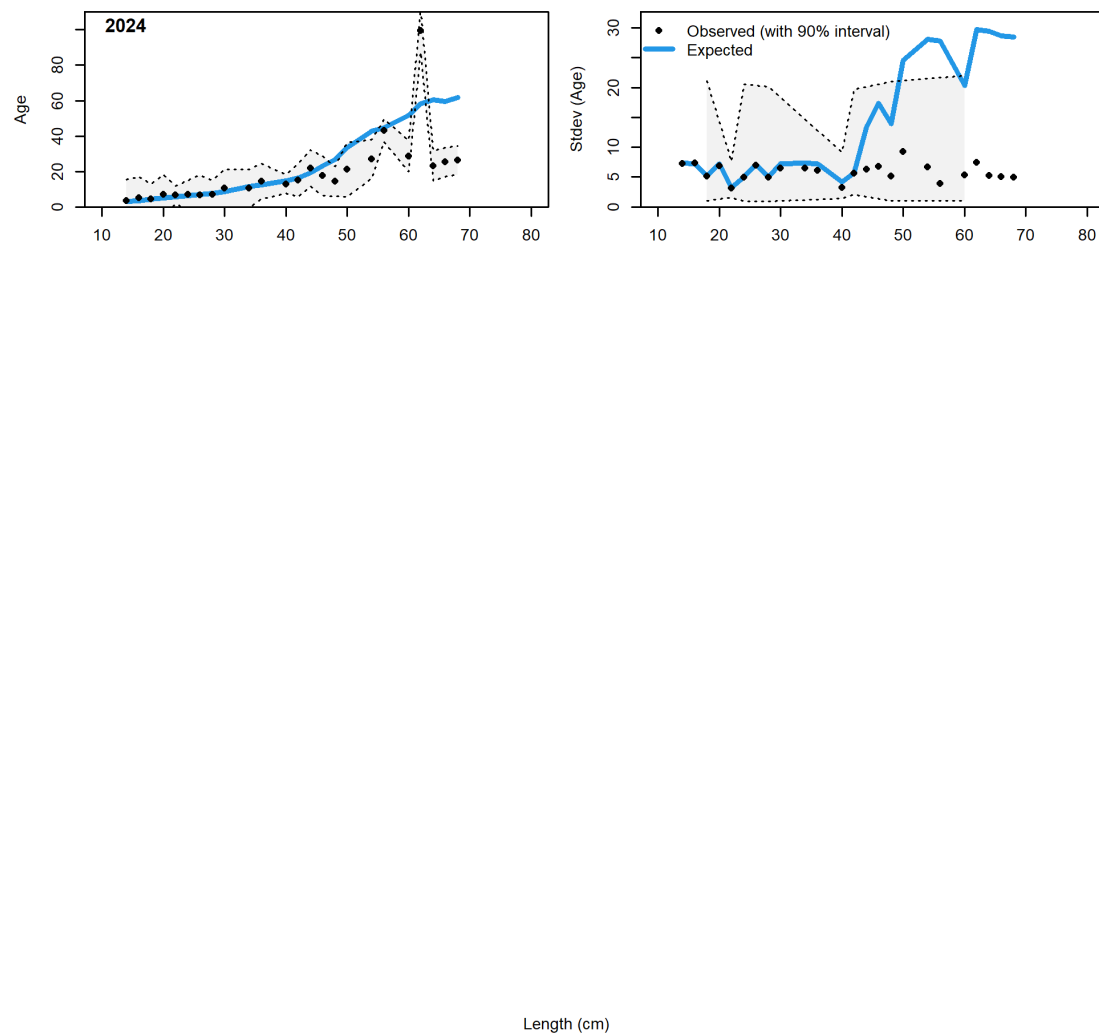


Figure 120: Conditional age at length plot showing mean age (left panels) and standard deviation (right panels) for the West Coast Groundfish Bottom Trawl survey in 2024. The CIs are 90% based on adding 1.64 SE of mean to the data for the left panels and the standard error of mean age at length (obs. and exp.) with 90% CIs based on the chi-square distribution.



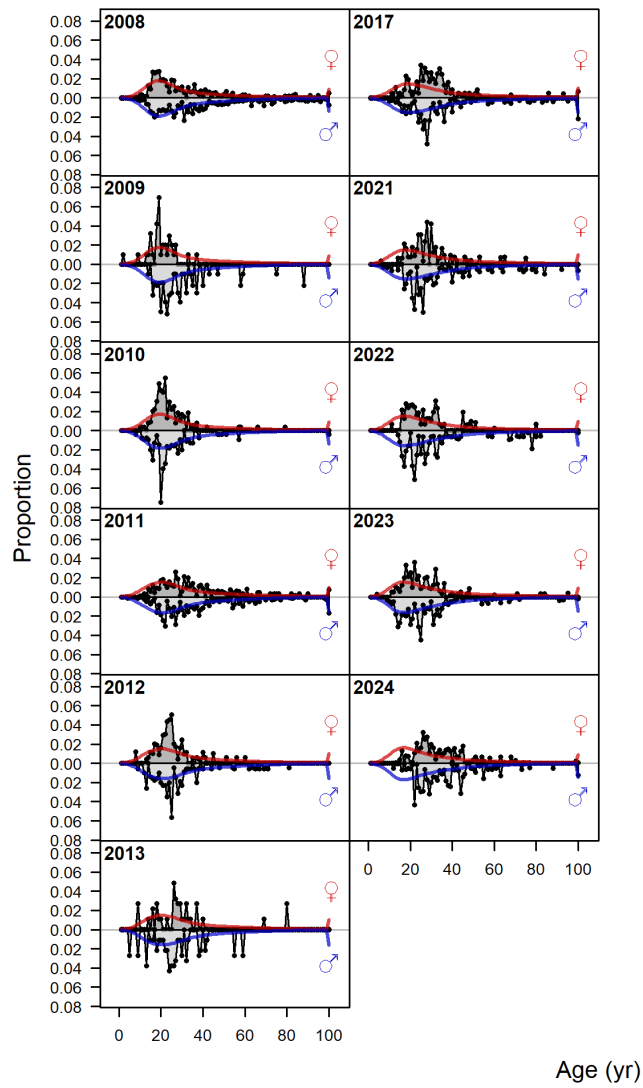


Figure 121: Realized fits (lines) to the marginal age composition data (density) for the bottom trawl fishery.

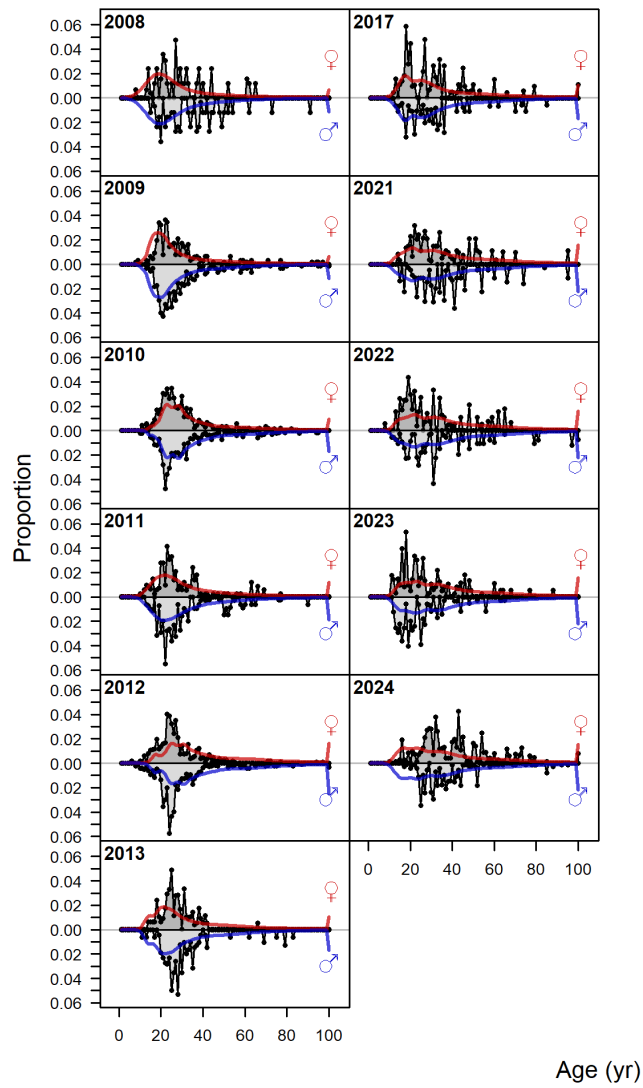


Figure 122: Realized fits (lines) to the marginal age composition data (density) for the non-trawl fishery.

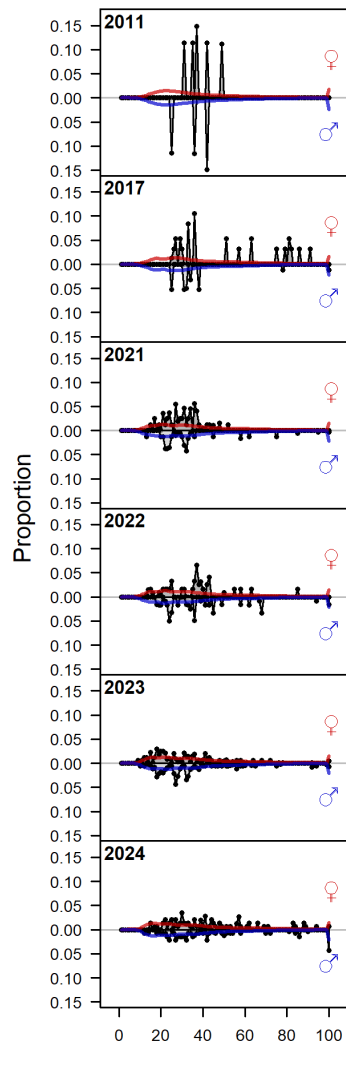


Figure 123: Realized fits (lines) to the marginal age composition data (density) for the midwater trawl fishery.

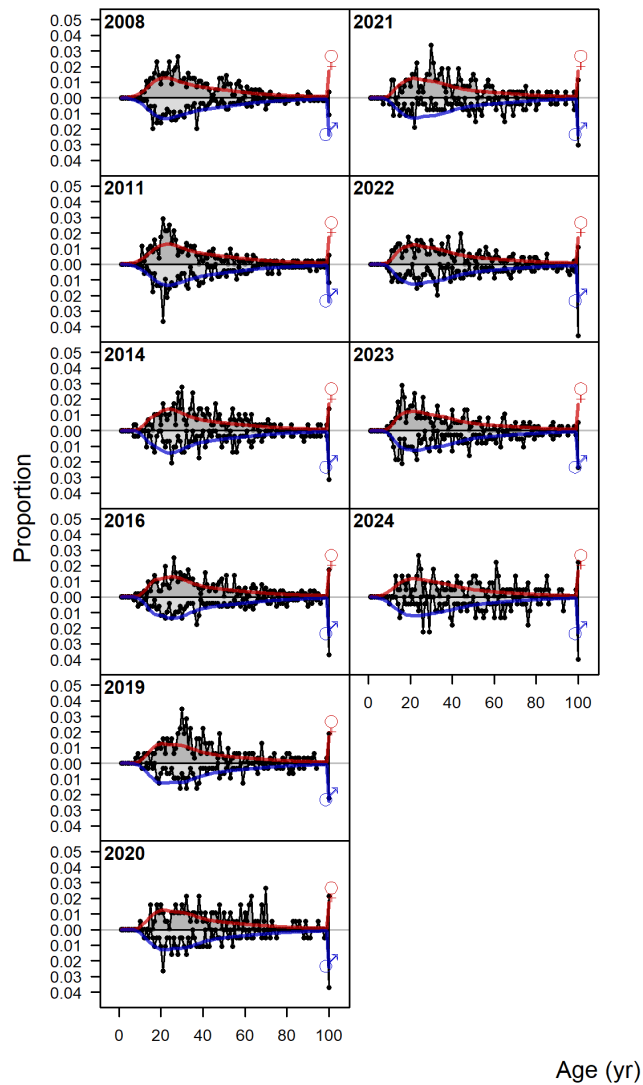


Figure 124: Realized fits (lines) to the marginal age composition data (density) for the at-sea-hake fishery.

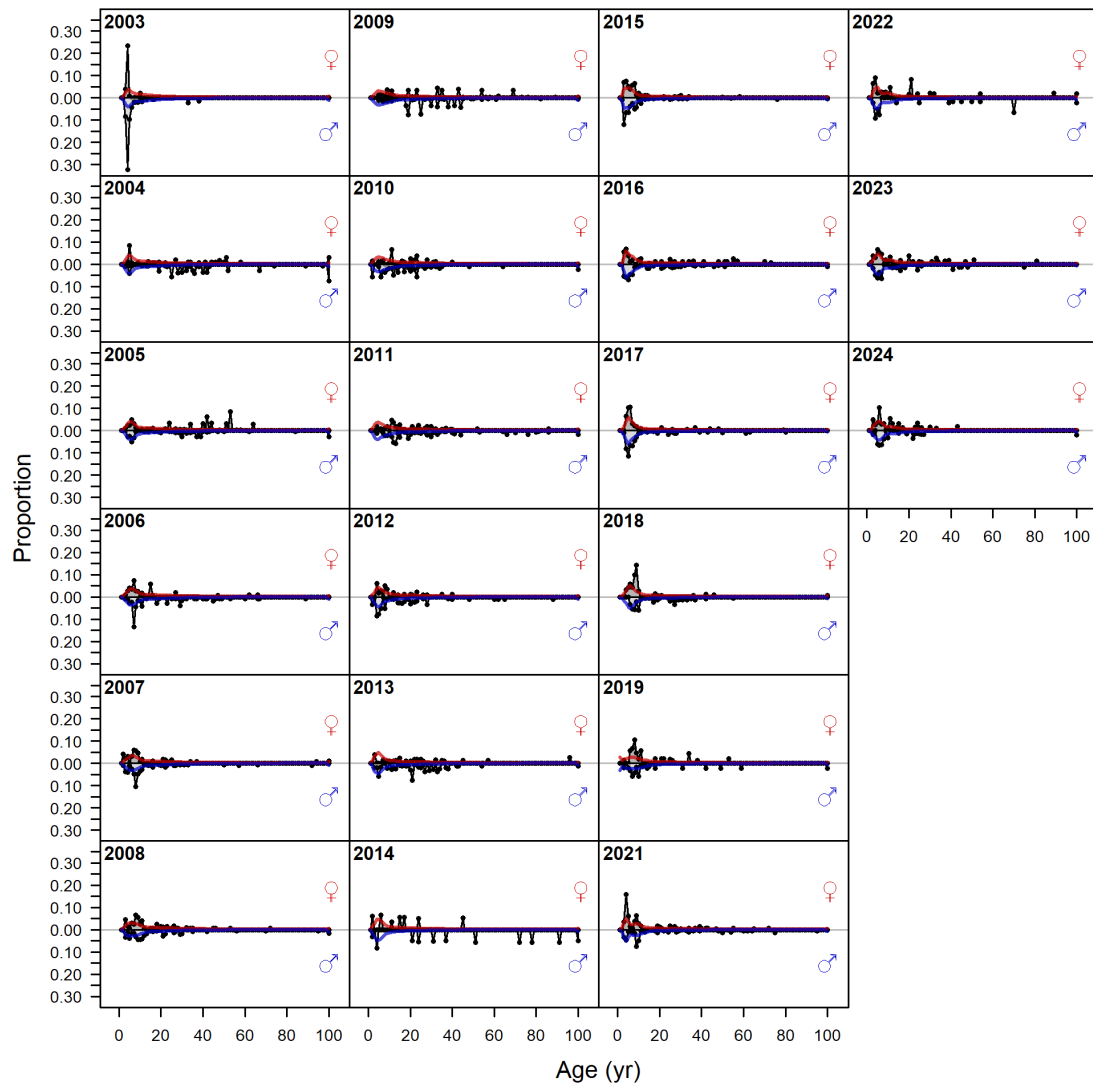


Figure 125: Realized fits (lines) to the marginal age composition data (density) for the West Coast Groundfish Bottom Trawl survey.

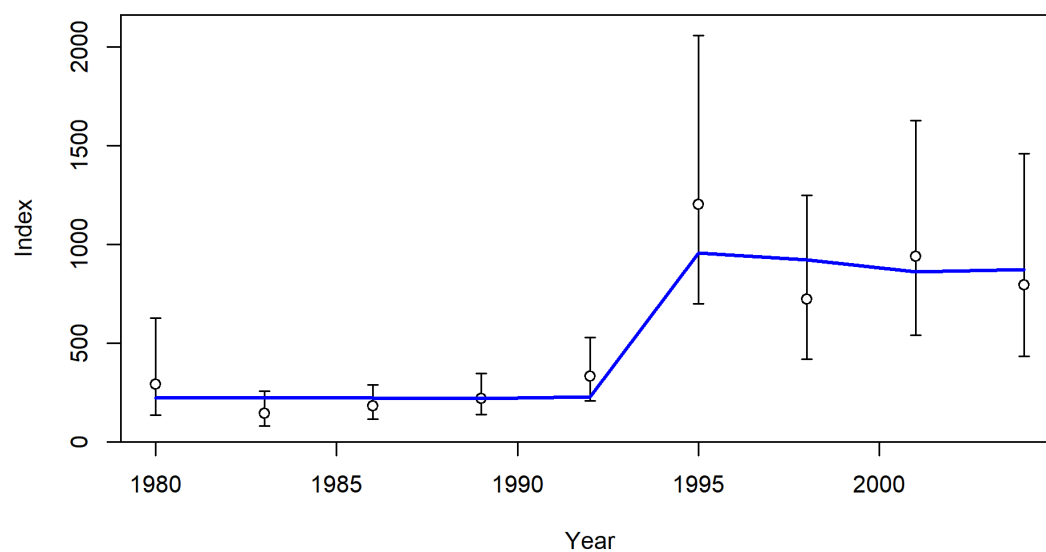


Figure 126: Fit to index data for Triennial survey. Lines indicate 95% uncertainty interval around index values based on the assumption of lognormal error.

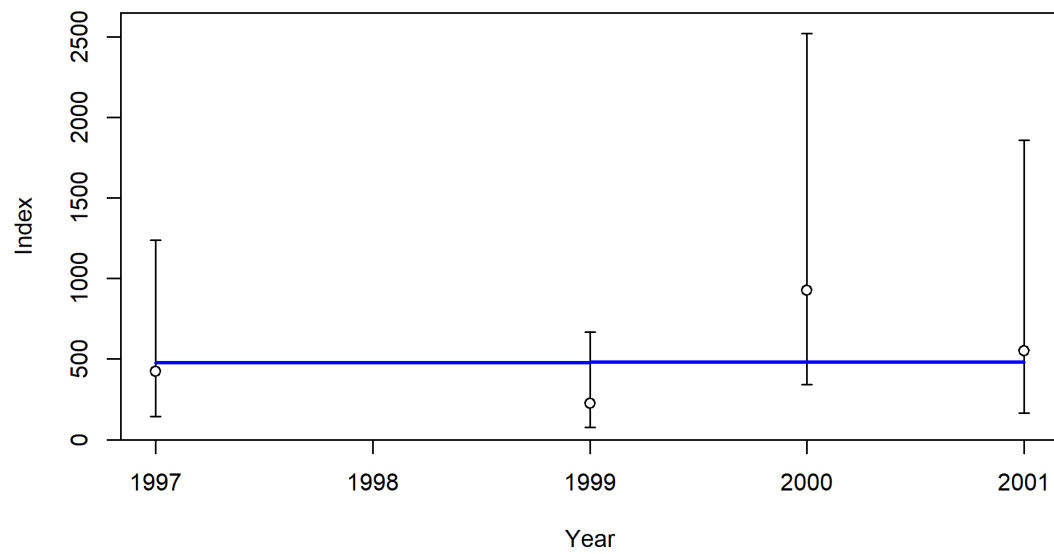


Figure 127: Fit to index data for the Alaska Slope survey. Lines indicate 95% uncertainty interval around index values based on the assumption of lognormal error.

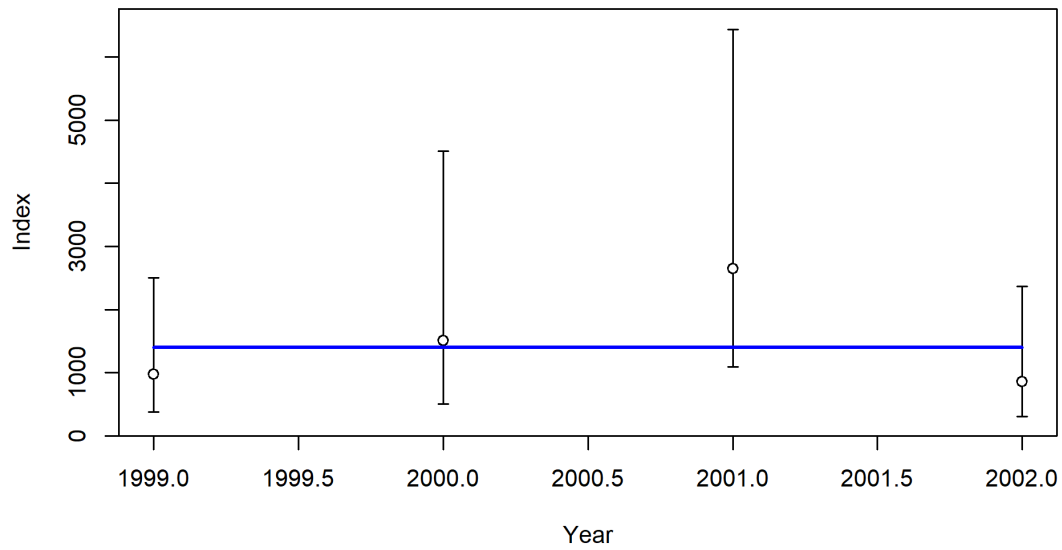


Figure 128: Fit to index data for Northwest Fisheries Science Center Slope survey. Lines indicate 95% uncertainty interval around index values based on the assumption of lognormal error.



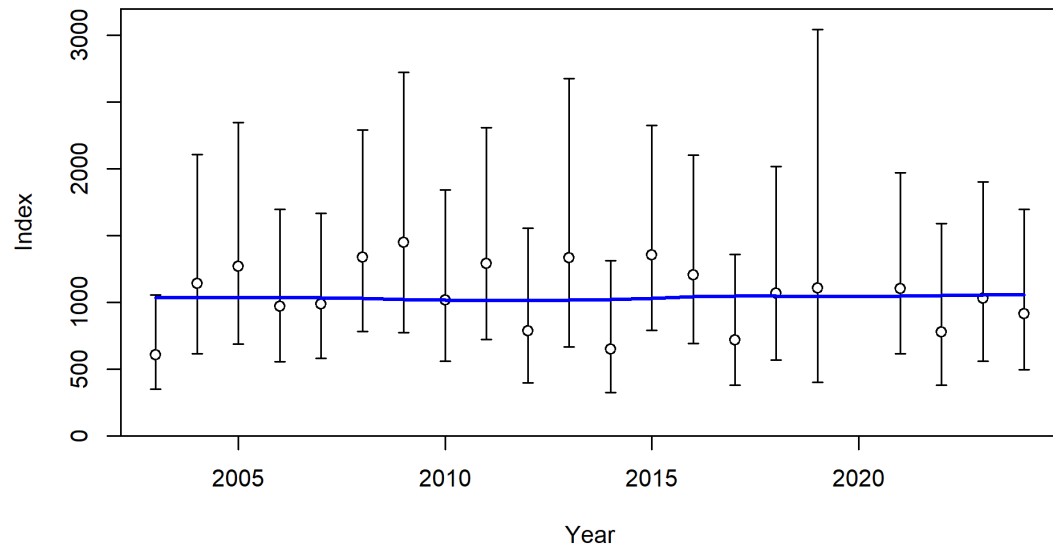


Figure 129: Fit to index data for West Coast Groundfish Bottom Trawl survey. Lines indicate 95% uncertainty interval around index values based on the assumption of lognormal error.

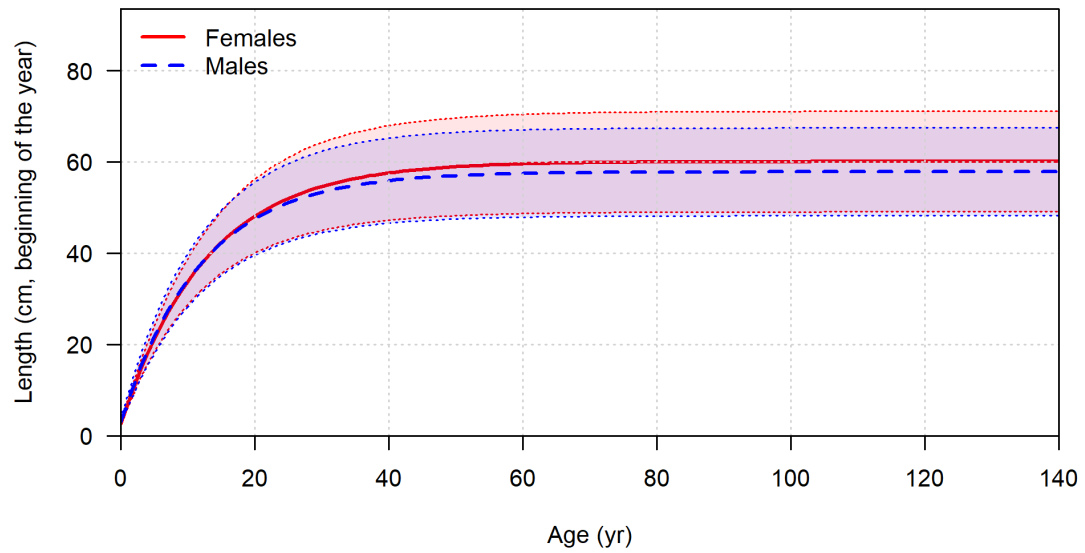


Figure 130: Estimated age and growth relationship in the reference model .Shaded area indicates 95% distribution of length at age around estimated growth curve.

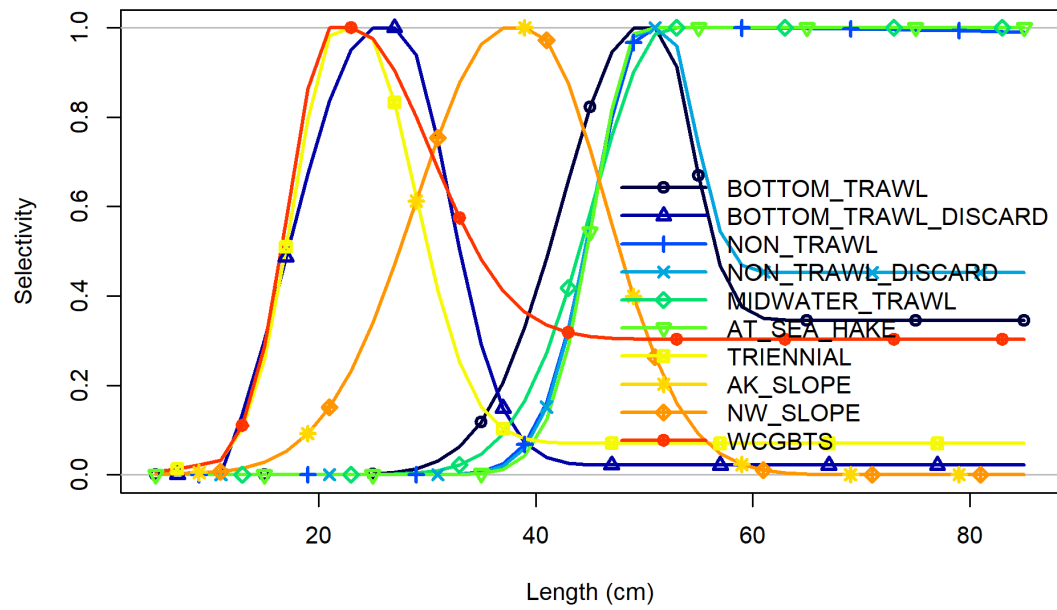


Figure 131: Ending selectivity at length for each fishery and survey.

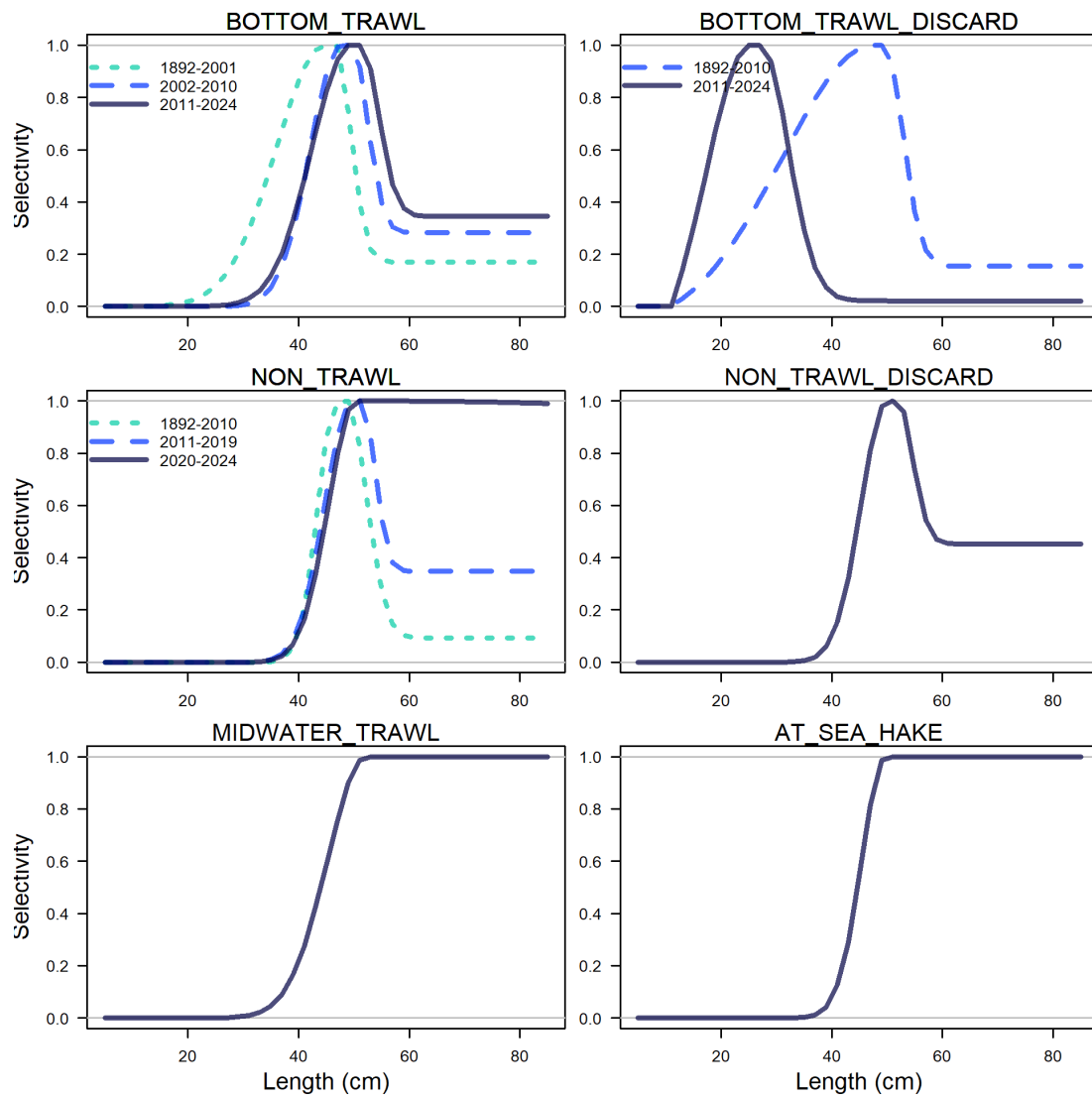


Figure 132: Time-varying selectivity for fishery fleets with time blocks.

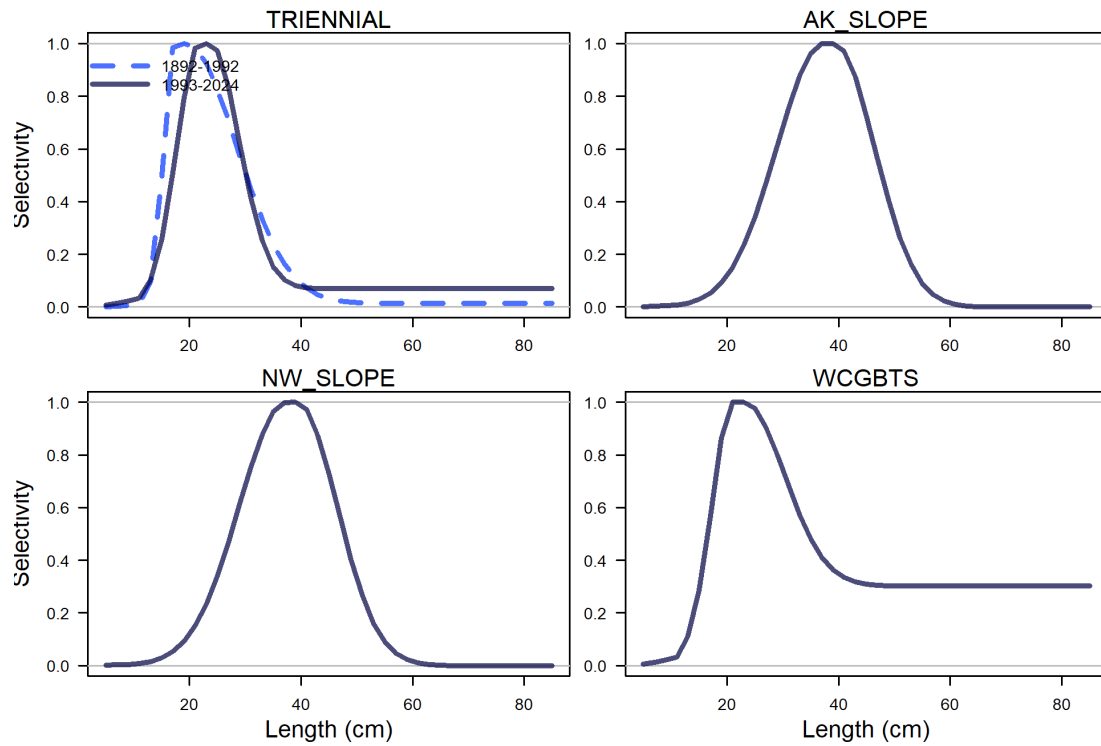


Figure 133: Time-varying selectivity for surveys with time blocks.

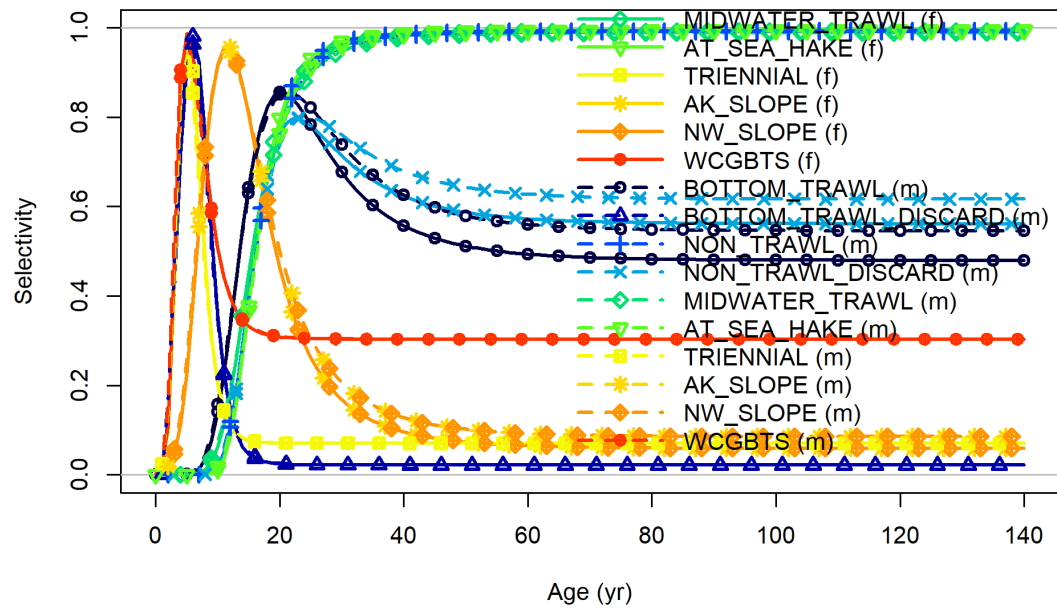


Figure 134: Ending selectivity at age derived from lengths for each fishery and survey.

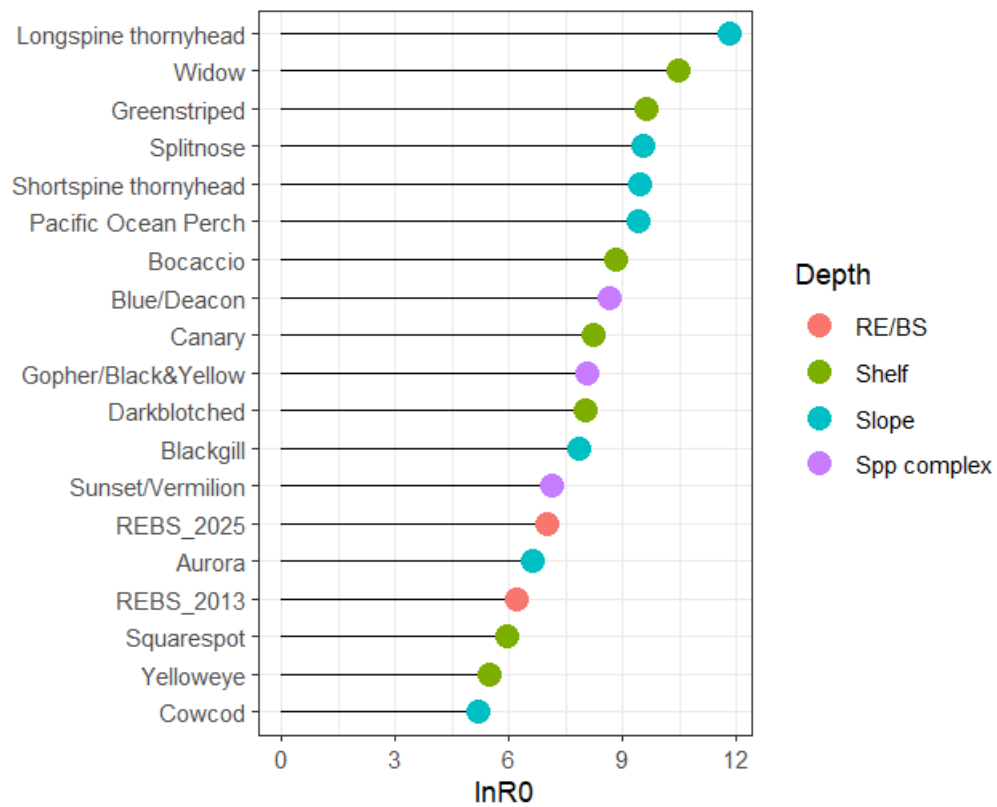


Figure 135: Estimated unfished recruitment ( $\ln R_0$ ) for several assessed shelf and slope rockfishes.

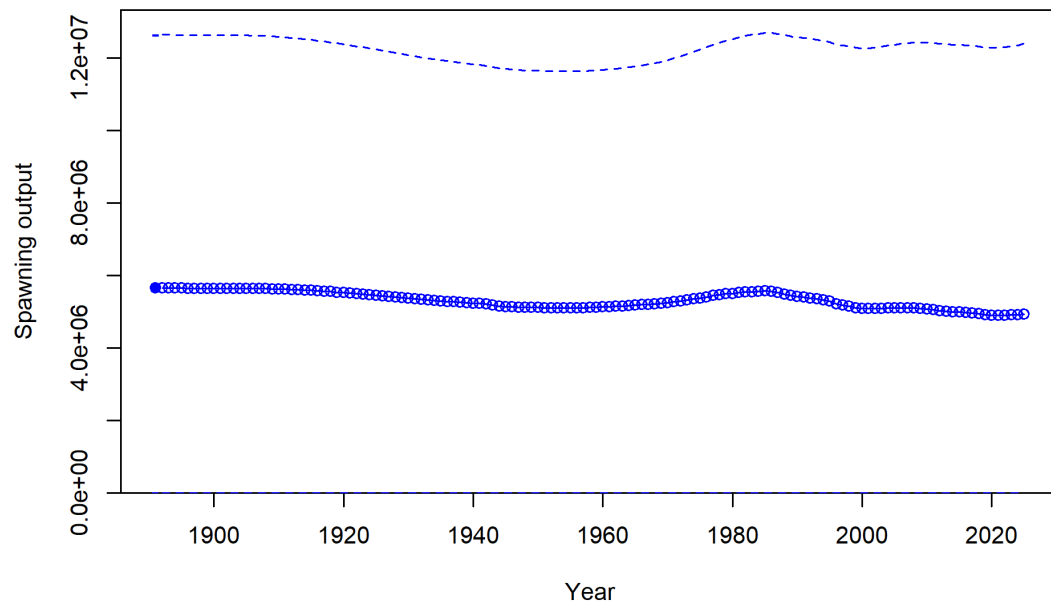


Figure 136: Estimated time series of spawning output (in millions of eggs) for the reference model.



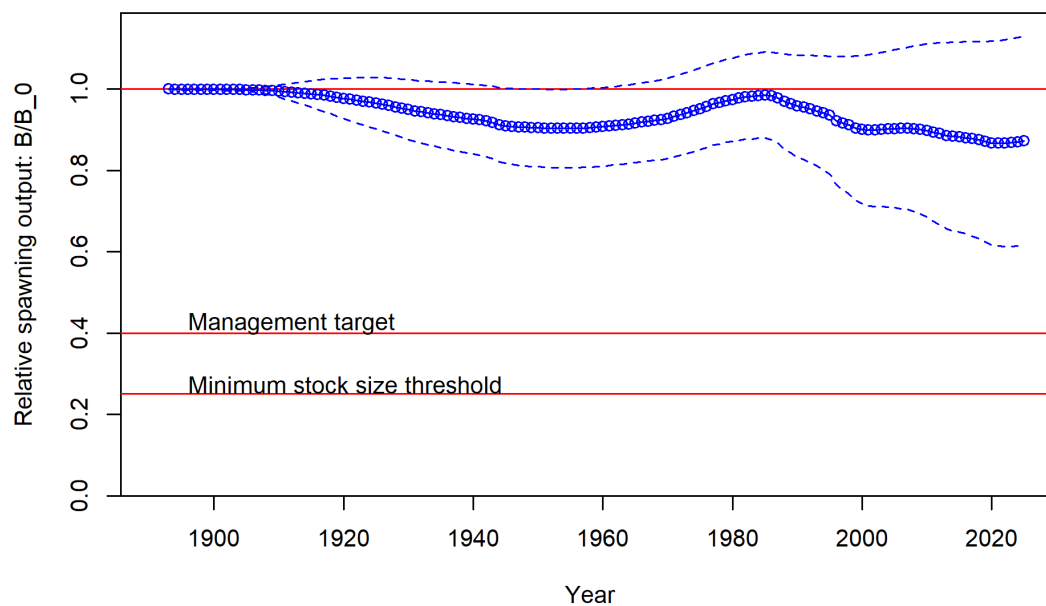


Figure 137: Estimated time series of fraction of unfished spawning output for the reference model.

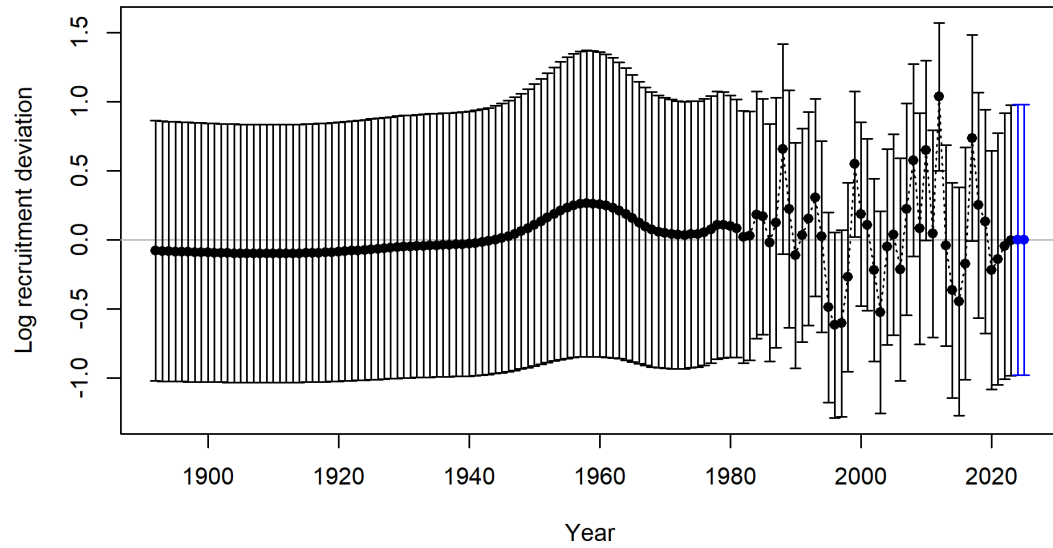


Figure 138: Estimated time series of recruitment deviations for the base model.

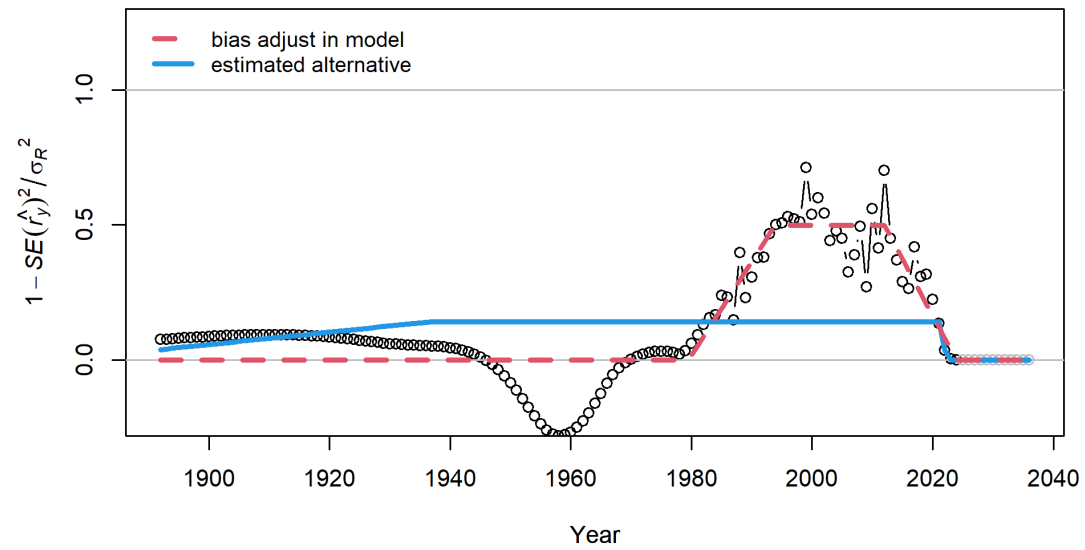


Figure 139: Bias adjustment applied to the recruitment deviations (red line). Points are transformed variances relative to the assumed variance of recruitment.

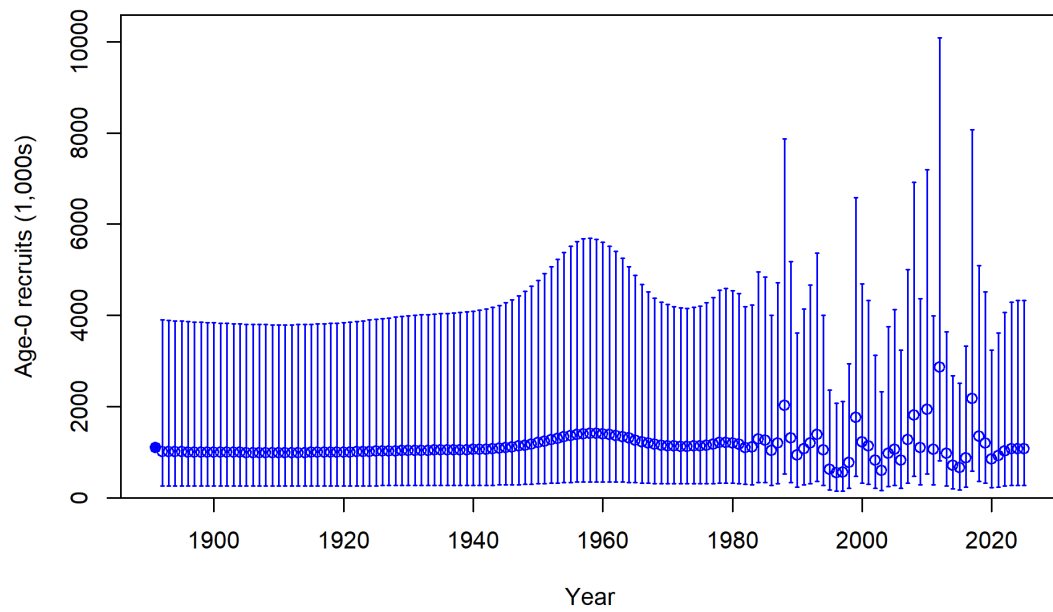


Figure 140: Estimated time series of age-0 recruits for the base model.

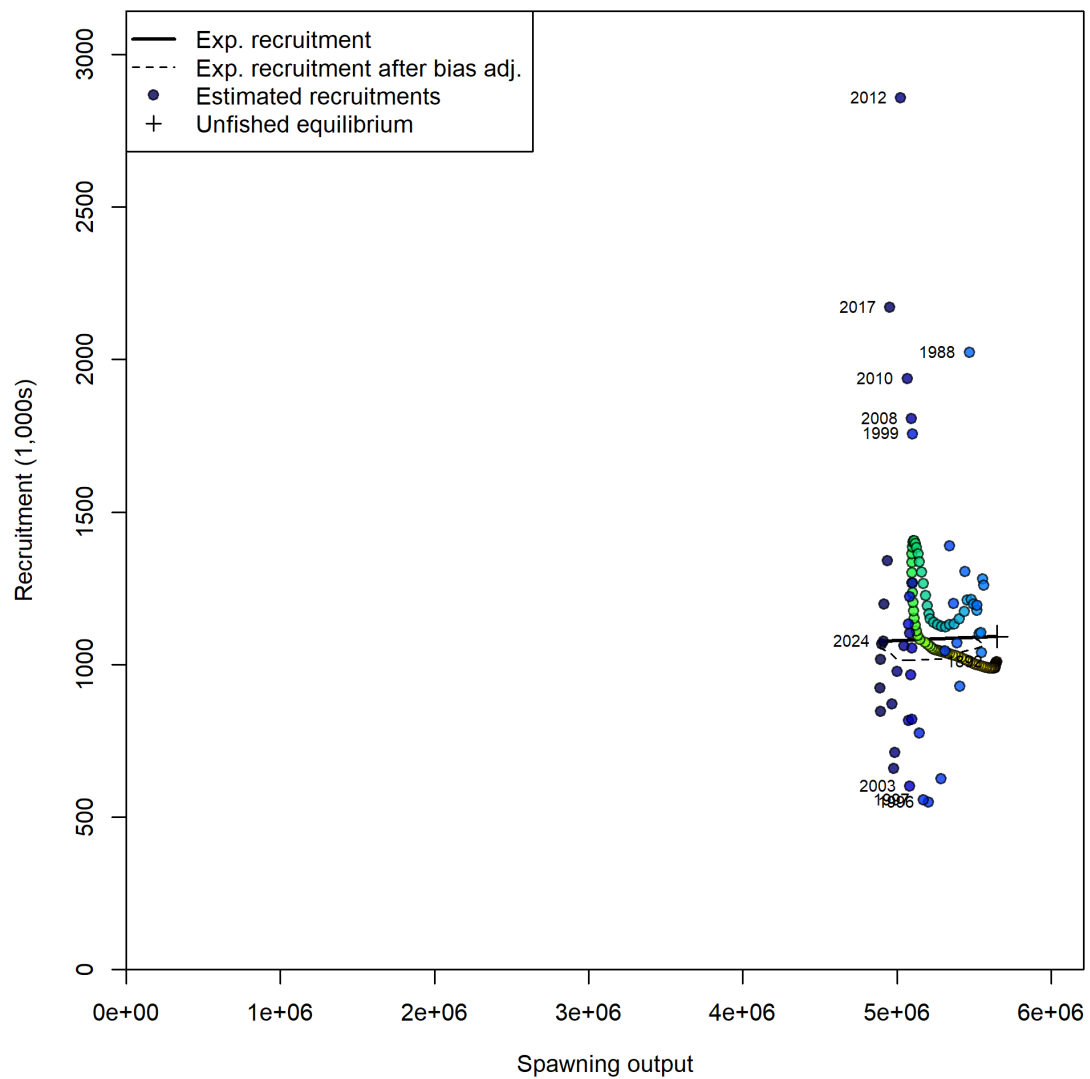


Figure 141: Stock-recruit curve with labels on first, last, and years with (log) deviations  $> 0.5$ . Point colors indicate year, with warmer colors indicating earlier years and cooler colors in showing later years.

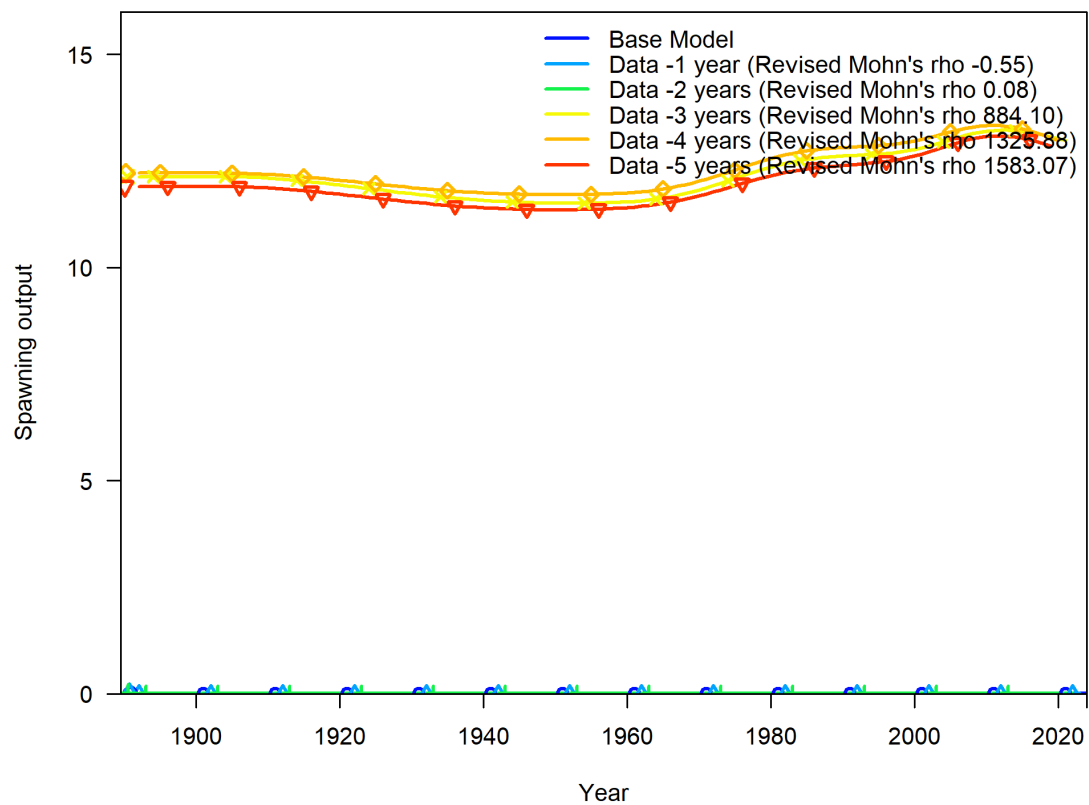


Figure 142: Change in the estimate of spawning output (millions of eggs) when the five most recent years of data area removed sequentially.

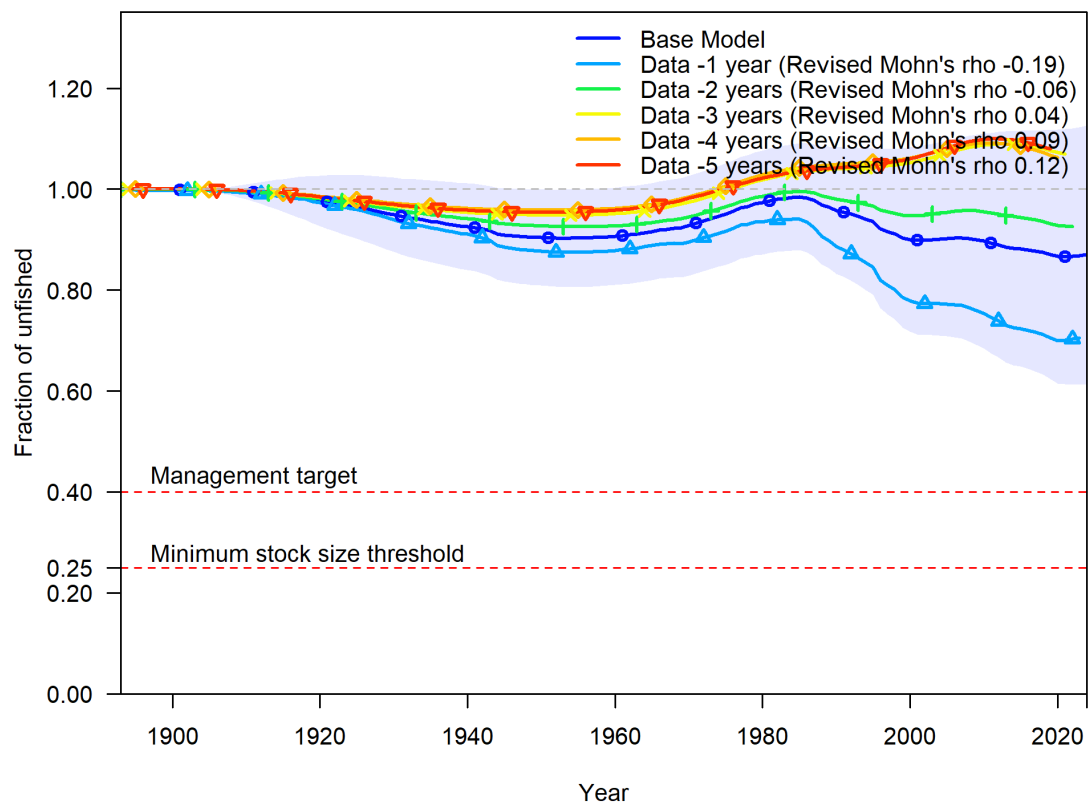


Figure 143: Change in the estimate of relative stock status when the five most recent years of data area removed sequentially.

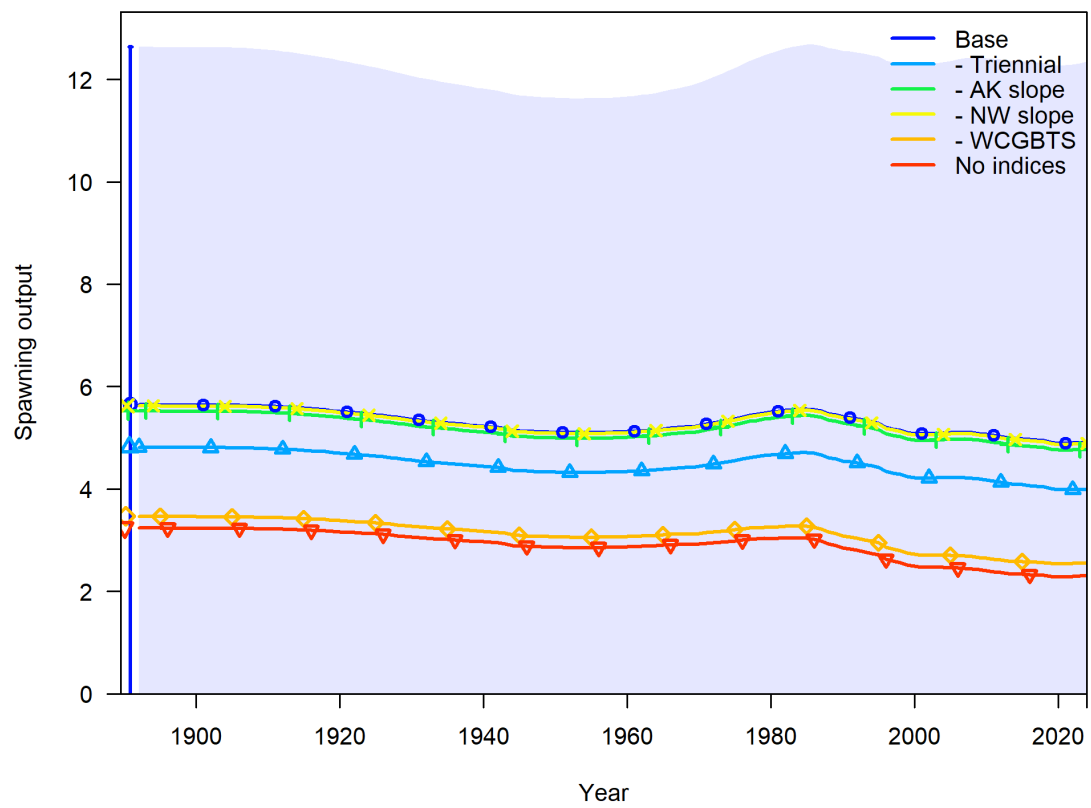


Figure 144: Spawning output (millions of eggs) across data removal sensitivities (indices).



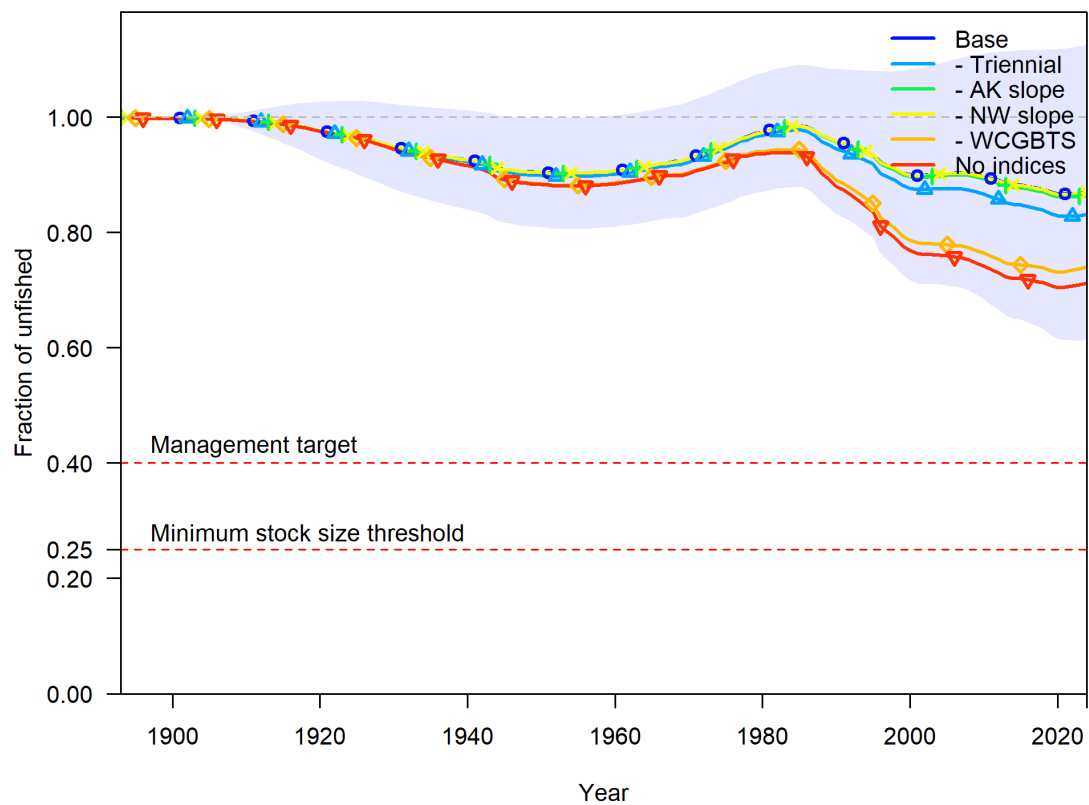


Figure 145: Relative spawning output (fraction unfished) across data removal sensitivities (indices).

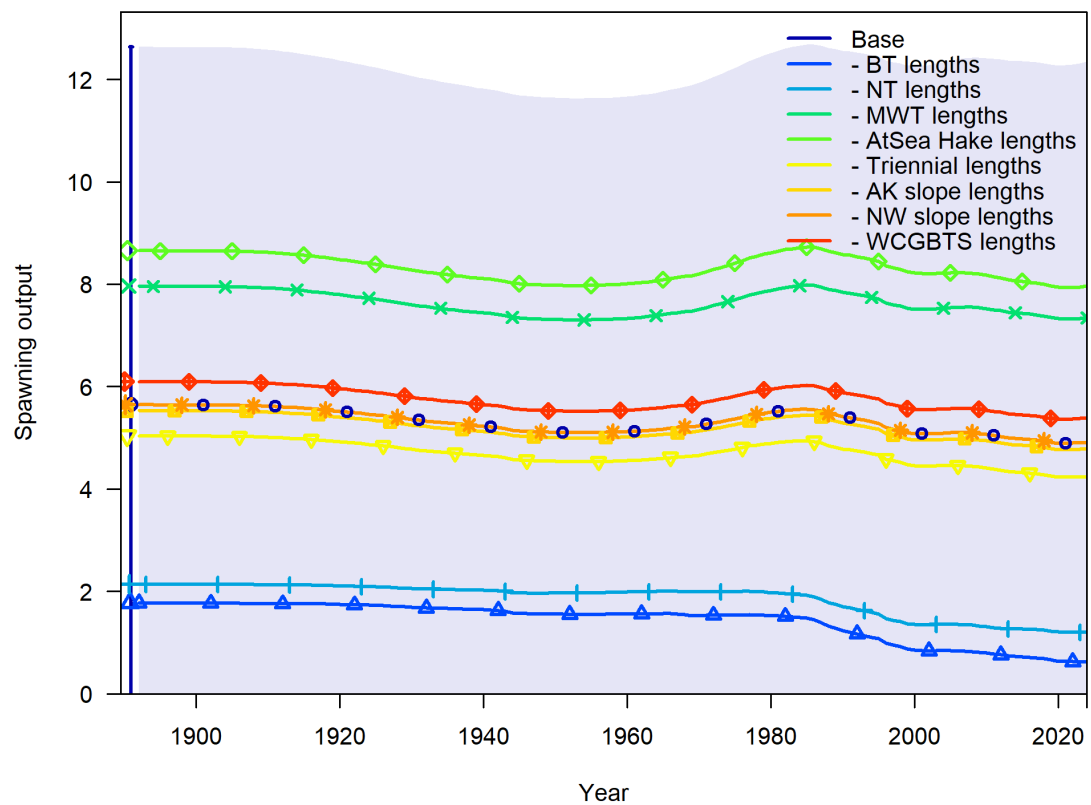


Figure 146: Spawning output (millions of eggs) across data removal sensitivities (length compositions by fleet).

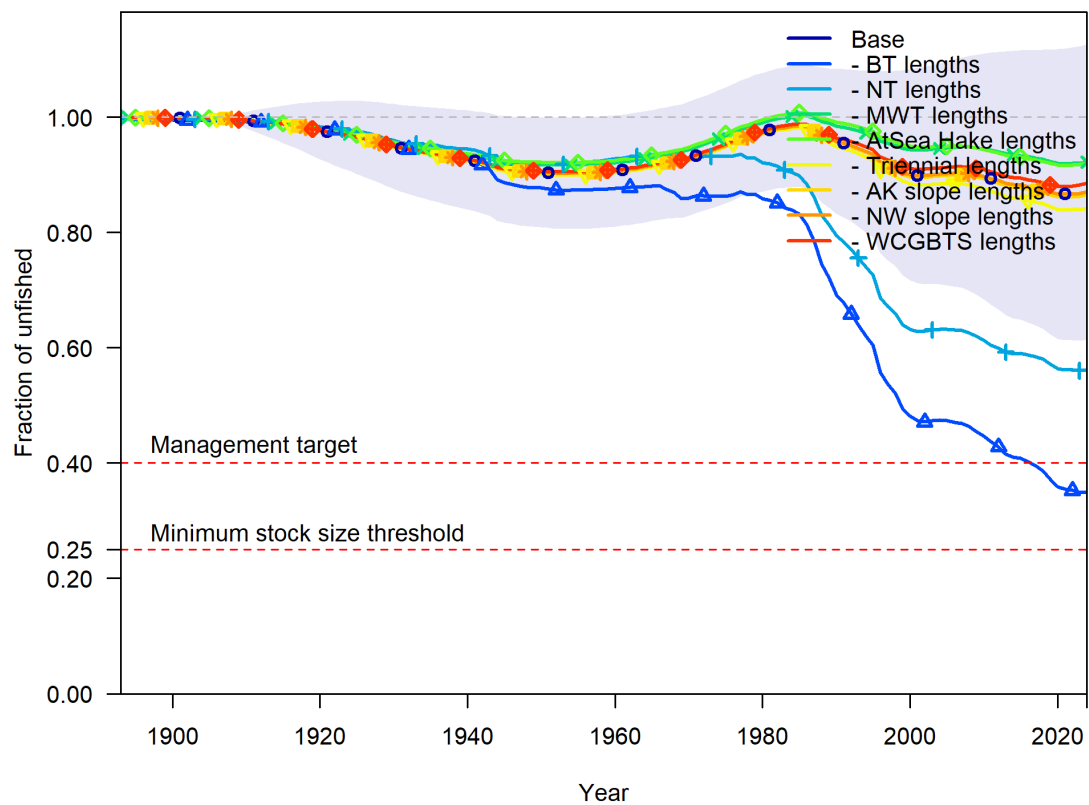


Figure 147: Relative spawning output (fraction unfished) across data removal sensitivities (length compositions by fleet).

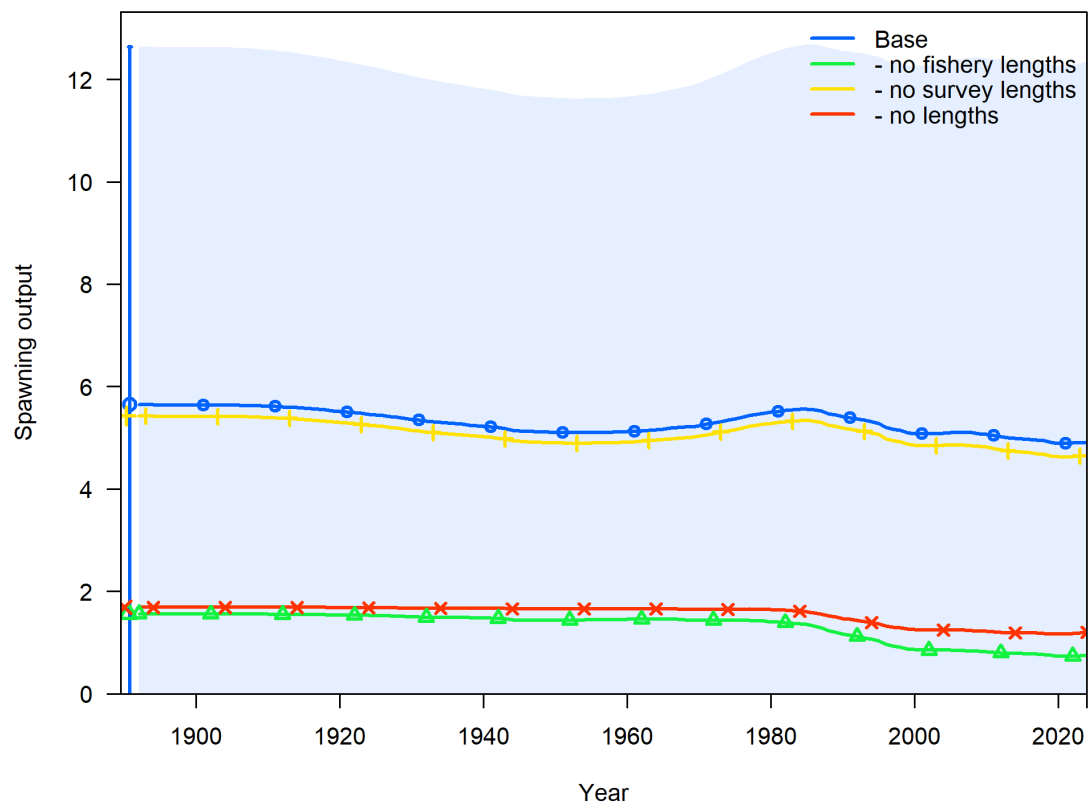


Figure 148: Spawning output (millions of eggs) across data removal sensitivities (length compositions by source).

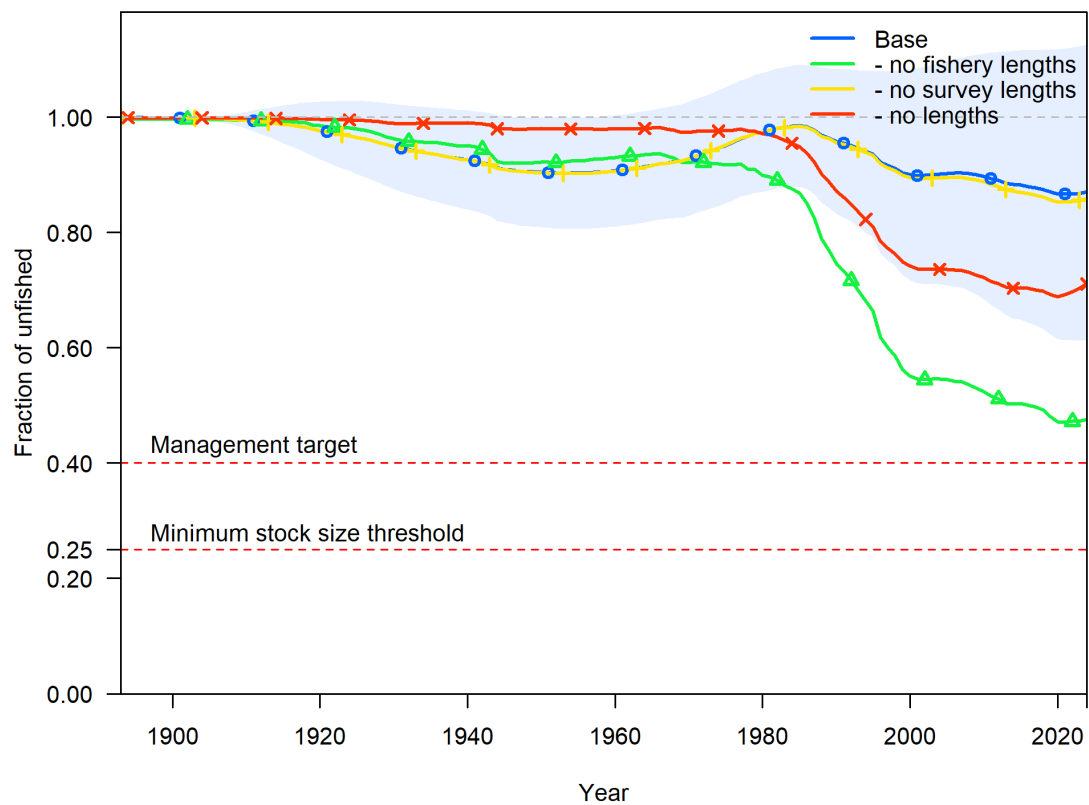


Figure 149: Relative spawning output (fraction unfished) across data removal sensitivities (length compositions by source).

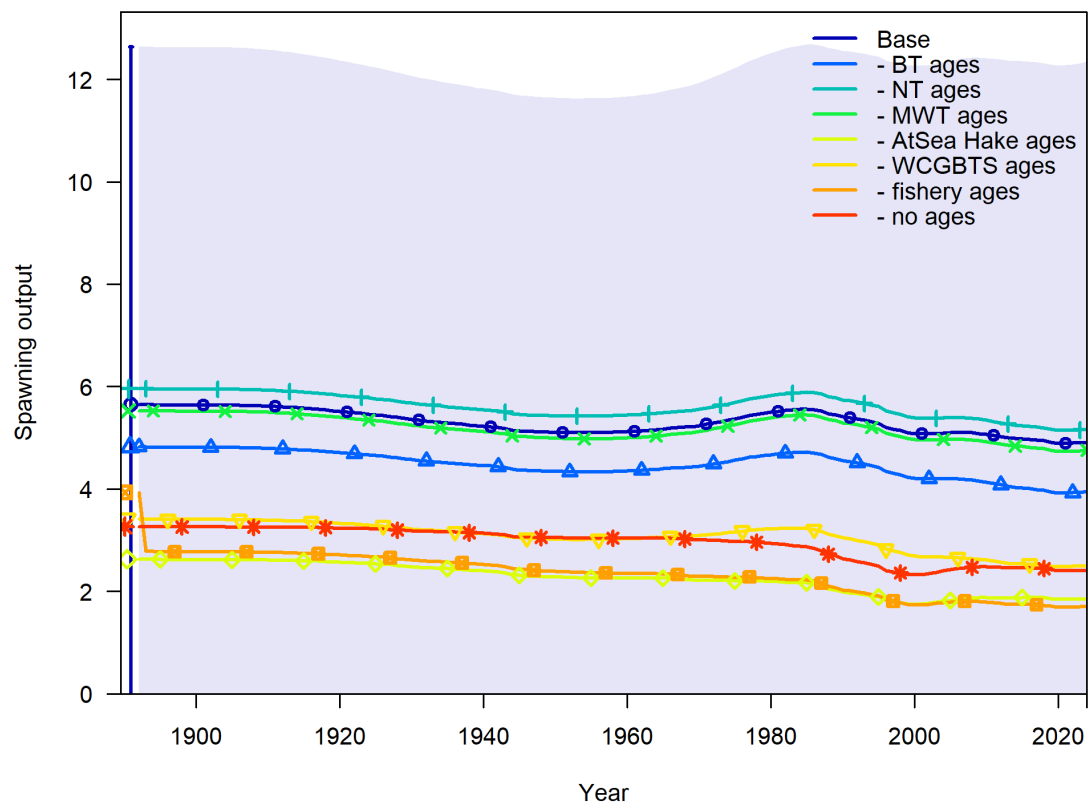


Figure 150: Spawning output (millions of eggs) across data removal sensitivities (age compositions by fleet).

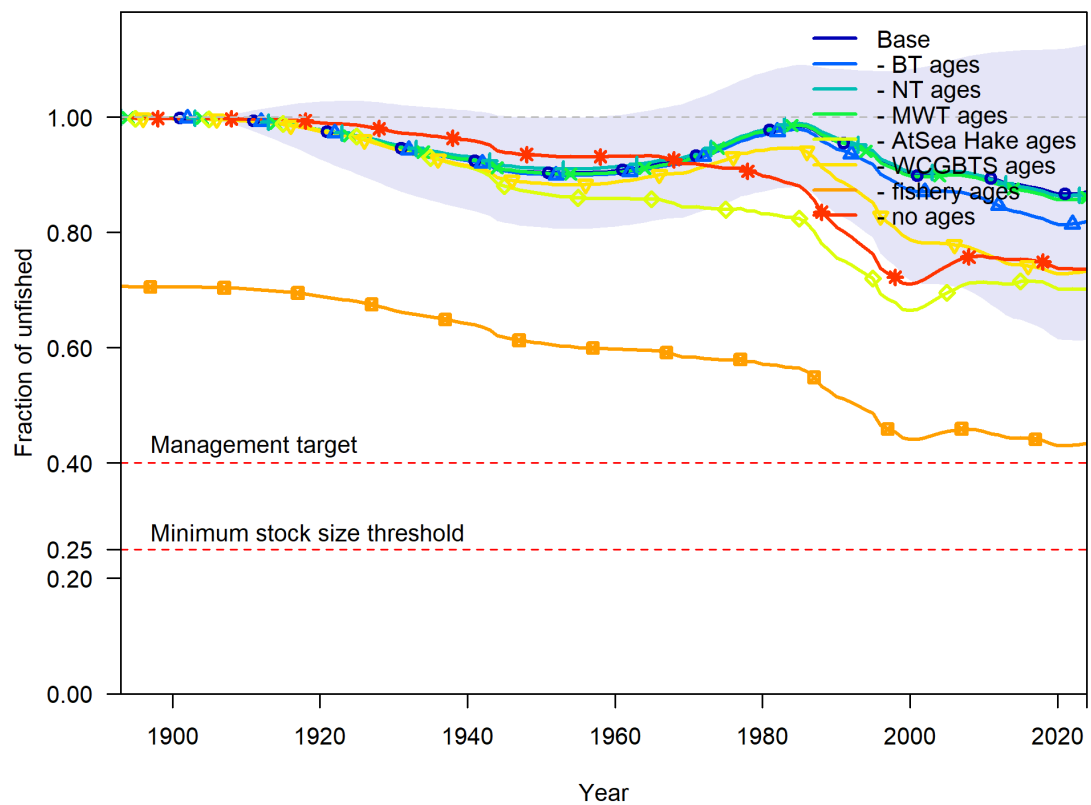


Figure 151: Relative spawning output (fraction unfished) across data removal sensitivities (age compositions by fleet).

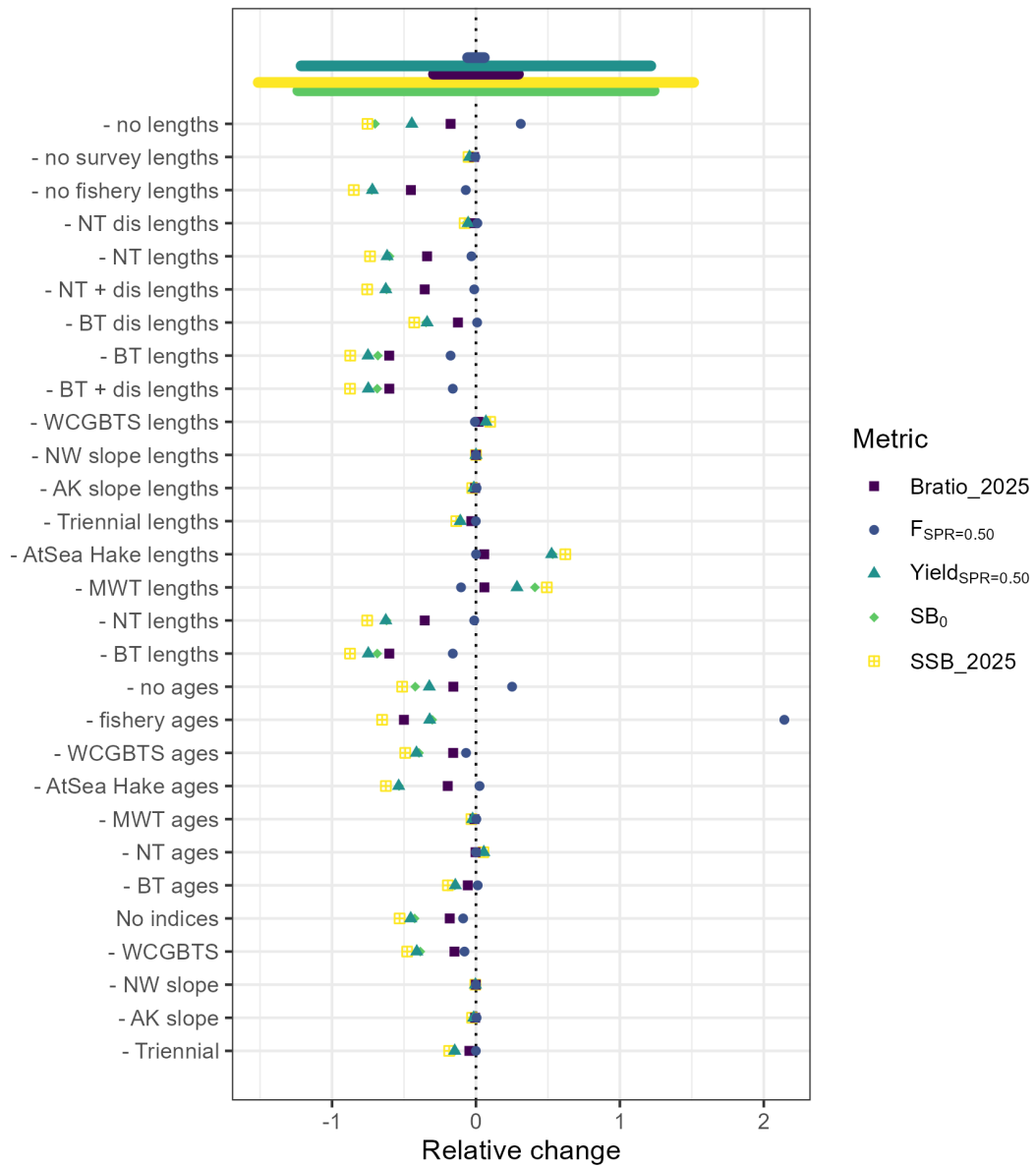


Figure 152: Relative change in management quantities across models conducted as sensitivities (removal of data sources).



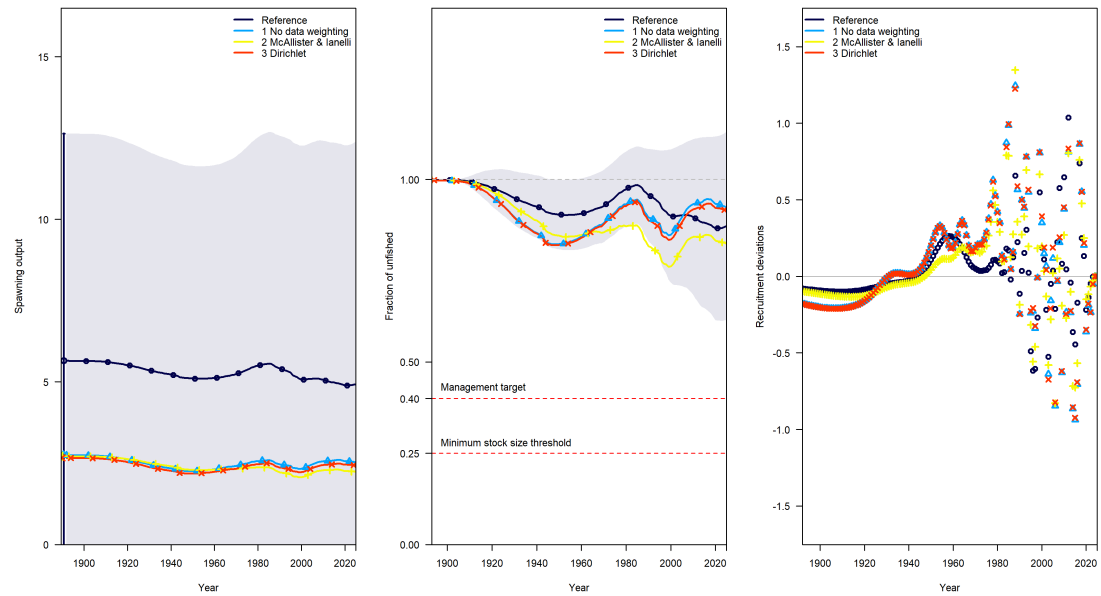


Figure 153: Comparison of spawning output (left panel), relative stock status (middle panel) and recruitment estimation (right panel) for models under different data weighting scenarios. The reference model uses the Francis method.

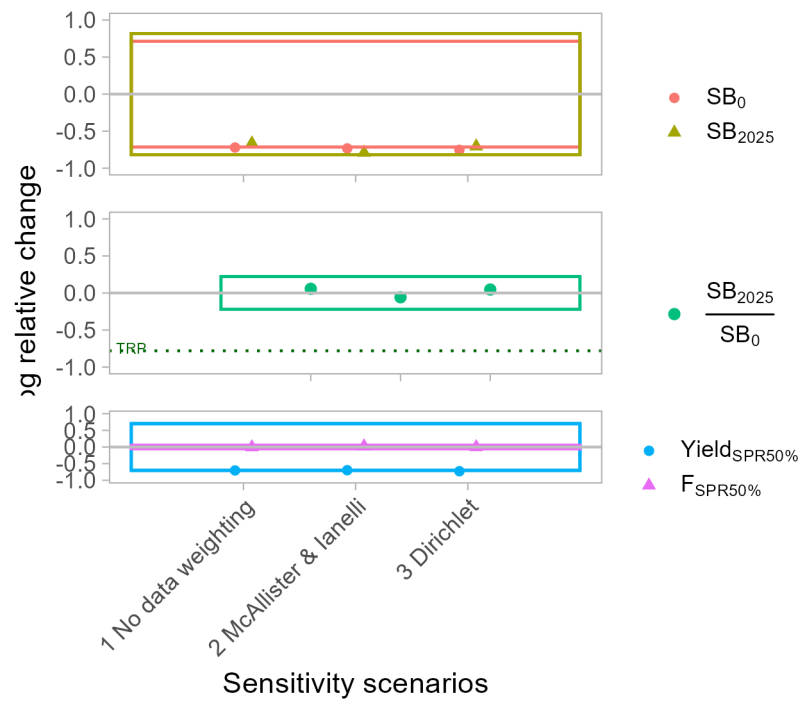


Figure 154: Log relative change ( $\log((\text{Model\_sensi}-\text{Model\_ref})/\text{Model\_ref}))$  in data weighting treatment for 5 derived quantities. Colored boxes indicate 95 percent confidence interval of the reference model. See ‘Sensitivity Analysis’ section for more details on each scenario.

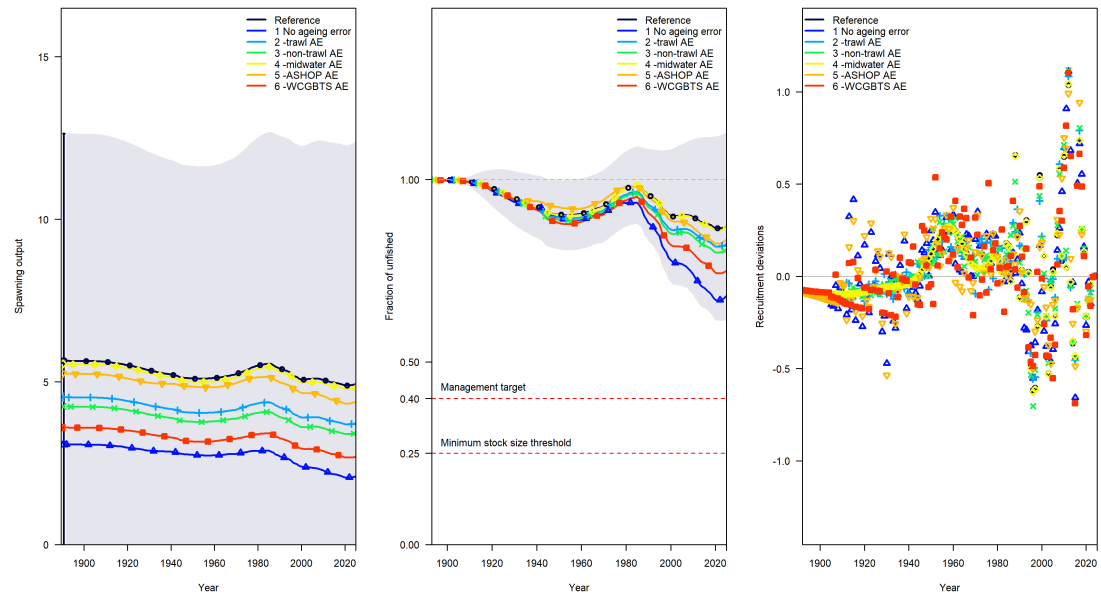


Figure 155: Comparison of spawning output (left panel), relative stock status (middle panel) and recruitment estimation (right panel) for models under different ageing error scenarios. The reference model uses ageing error matrices for all age sources.

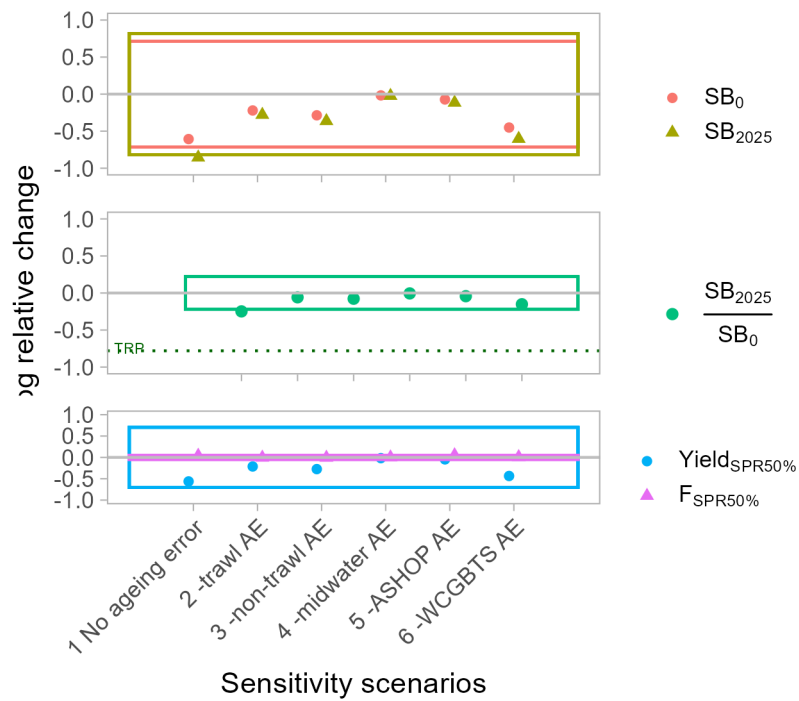


Figure 156: Log relative change ( $\log((\text{Model\_sensi}-\text{Model\_ref})/\text{Model\_ref})$ ) in ageing error scenarios for 5 derived quantities. Colored boxes indicate 95 percent confidence interval of the reference model. See ‘Sensitivity Analysis’ section for more details on each scenario.

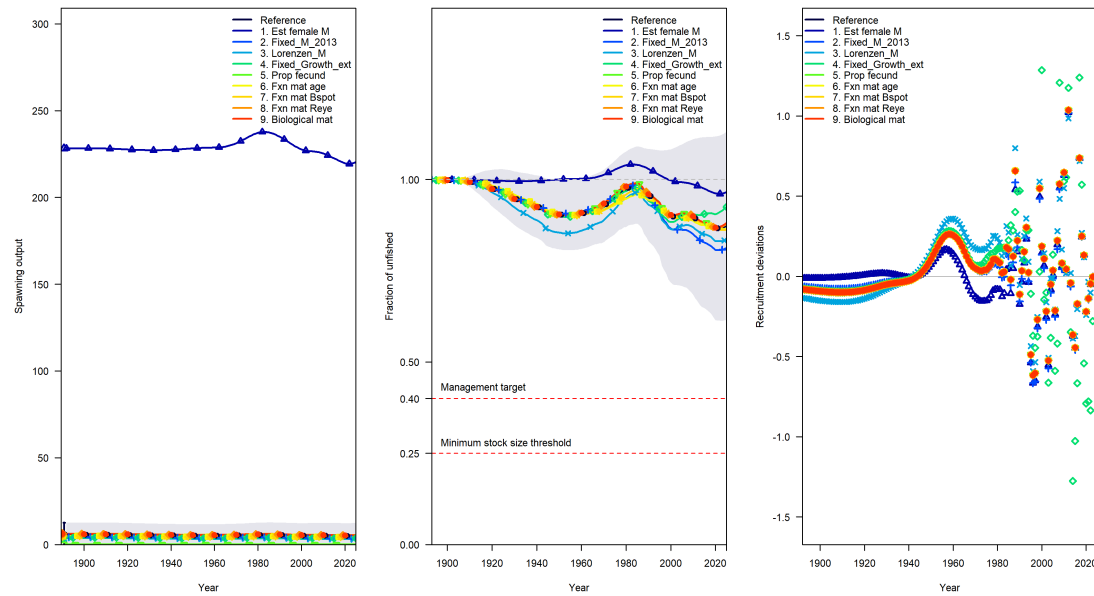


Figure 157: Comparison of spawning output (left panel), relative stock status (middle panel) and recruitment estimation (right panel) for models under different natural mortality, growth and reproductive biology scenarios.

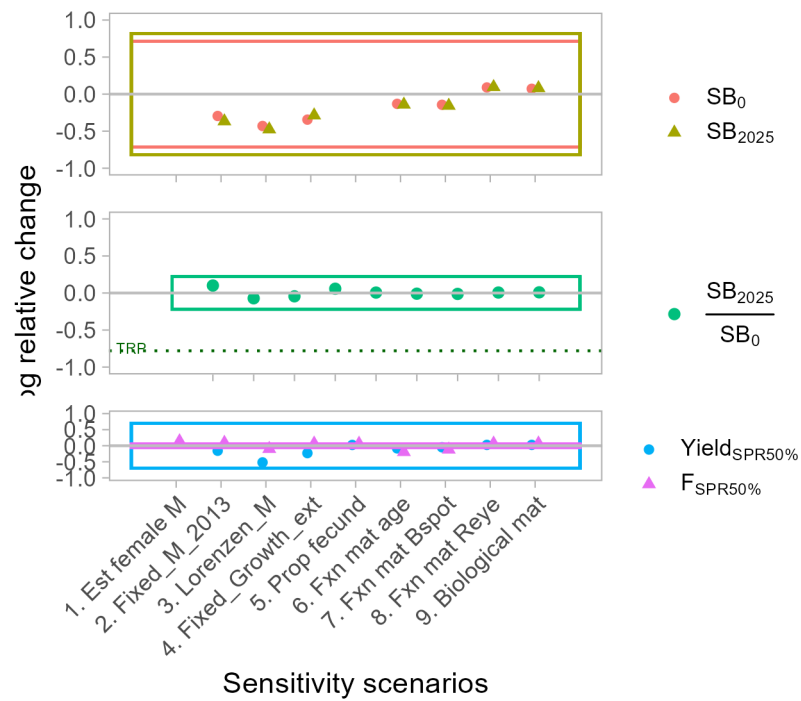


Figure 158: Log relative change ( $\log((\text{Model\_sensi}-\text{Model\_ref})/\text{Model\_ref}))$ ) in natural mortality, growth and reproductive biology scenarios for 5 derived quantities. Colored boxes indicate 95 percent confidence interval of the reference model. See ‘Sensitivity Analysis’ section for more details on each scenario.

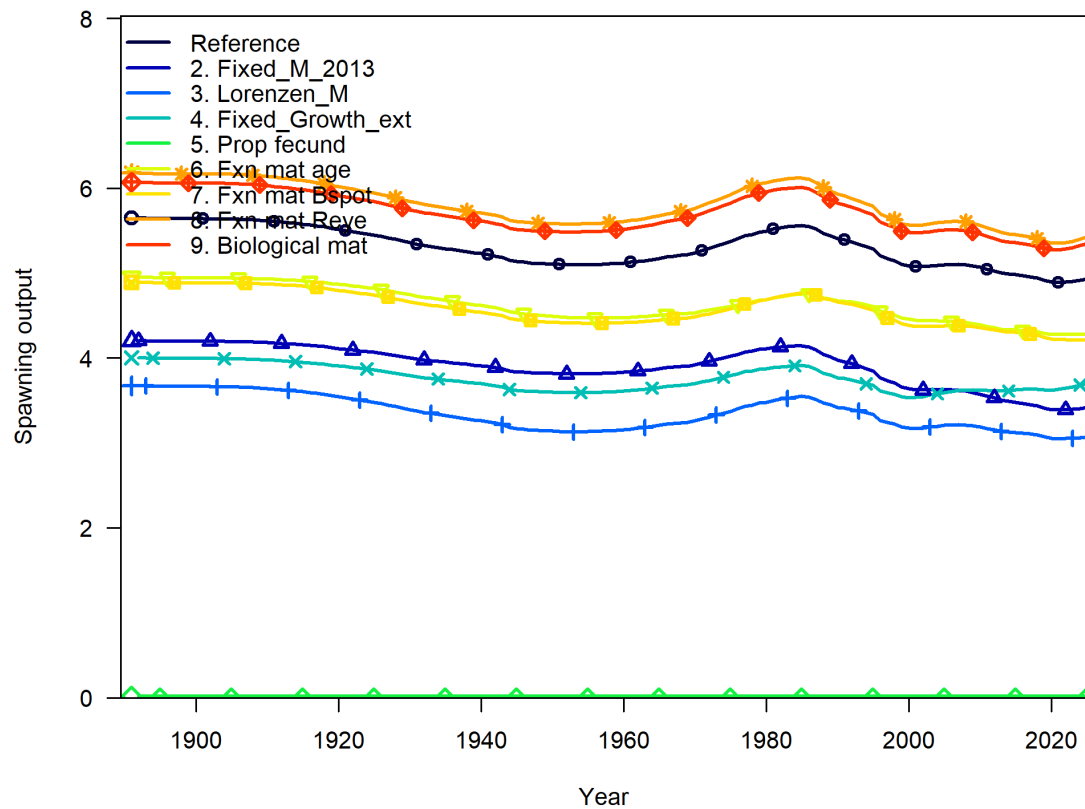


Figure 159: Comparison of spawning output for models under different natural mortality, growth and reproductive biology scenarios. This comparison excludes the model estimating natural mortality, as the scale goes extremely high.

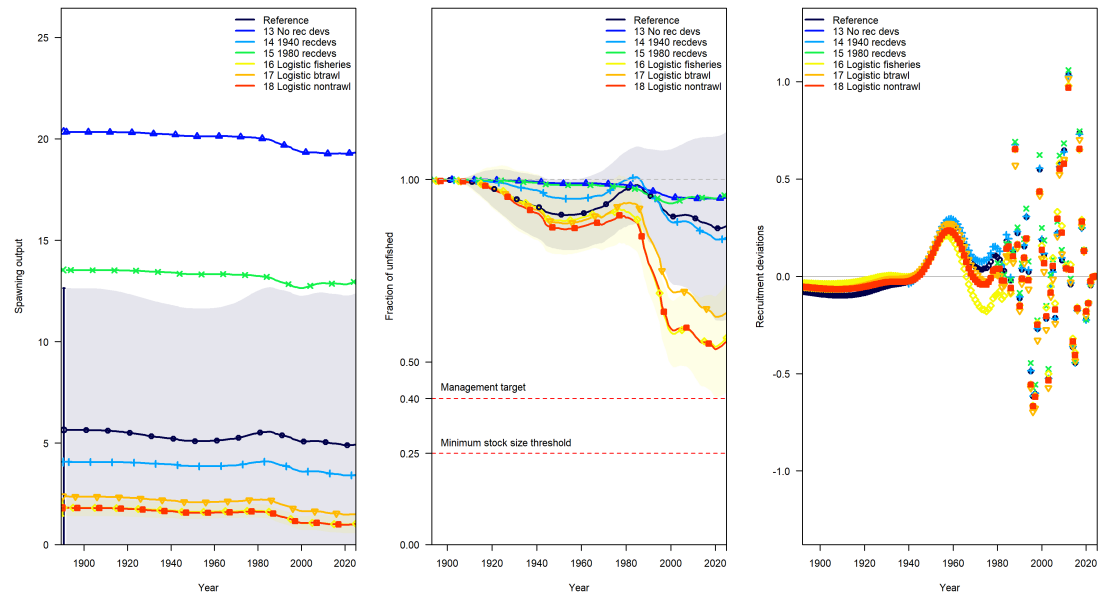


Figure 160: Comparison of spawning output (left panel), relative stock status (middle panel) and recruitment estimation (right panel) for models under different recruitment and selectivity scenarios.



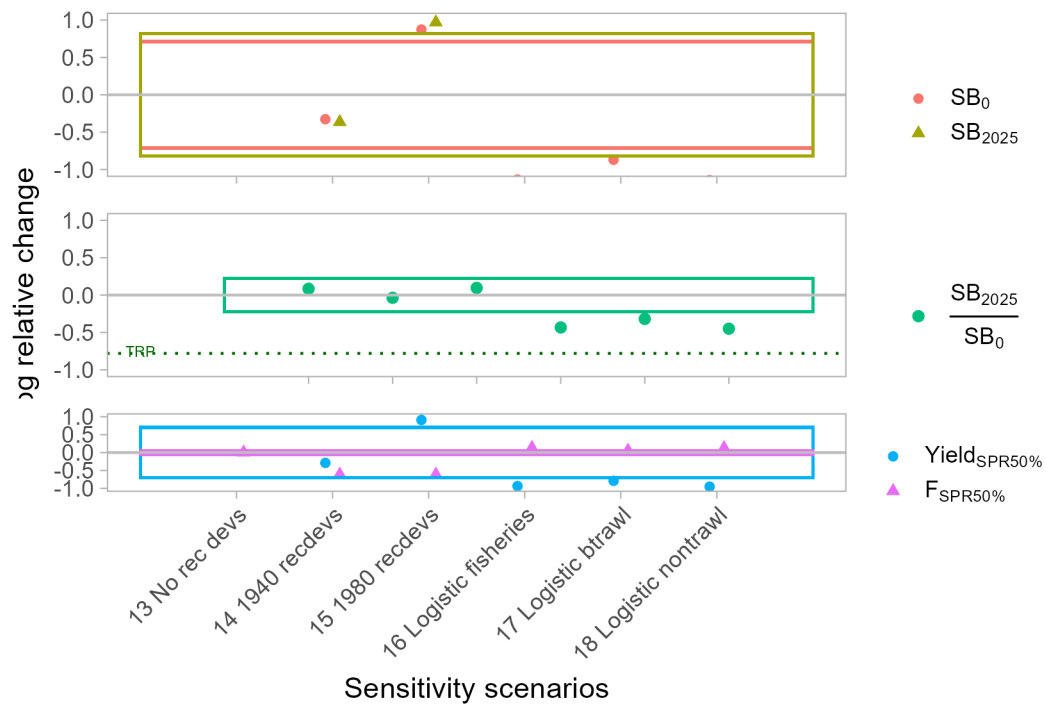


Figure 161: Log relative change ( $\log((\text{Model\_sensi}-\text{Model\_ref})/\text{Model\_ref}))$  in recruitment and selectivity scenarios for 5 derived quantities. Colored boxes indicate 95 percent confidence interval of the reference model. See ‘Sensitivity Analysis’ section for more details on each scenario.

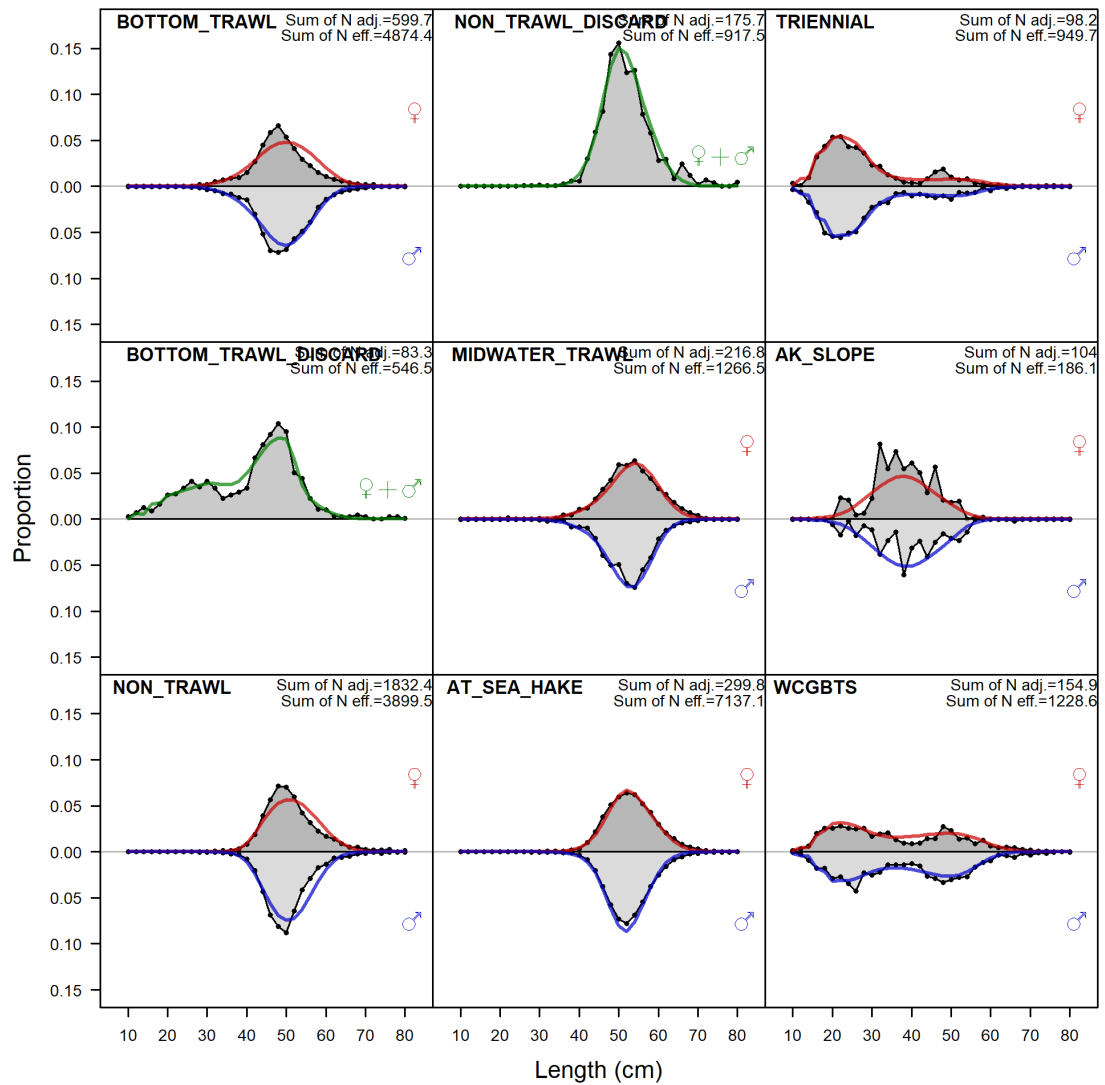


Figure 162: Aggregate fit to length compositions for the scenario that assume logistic selectivity for all fisheries and time blocks.

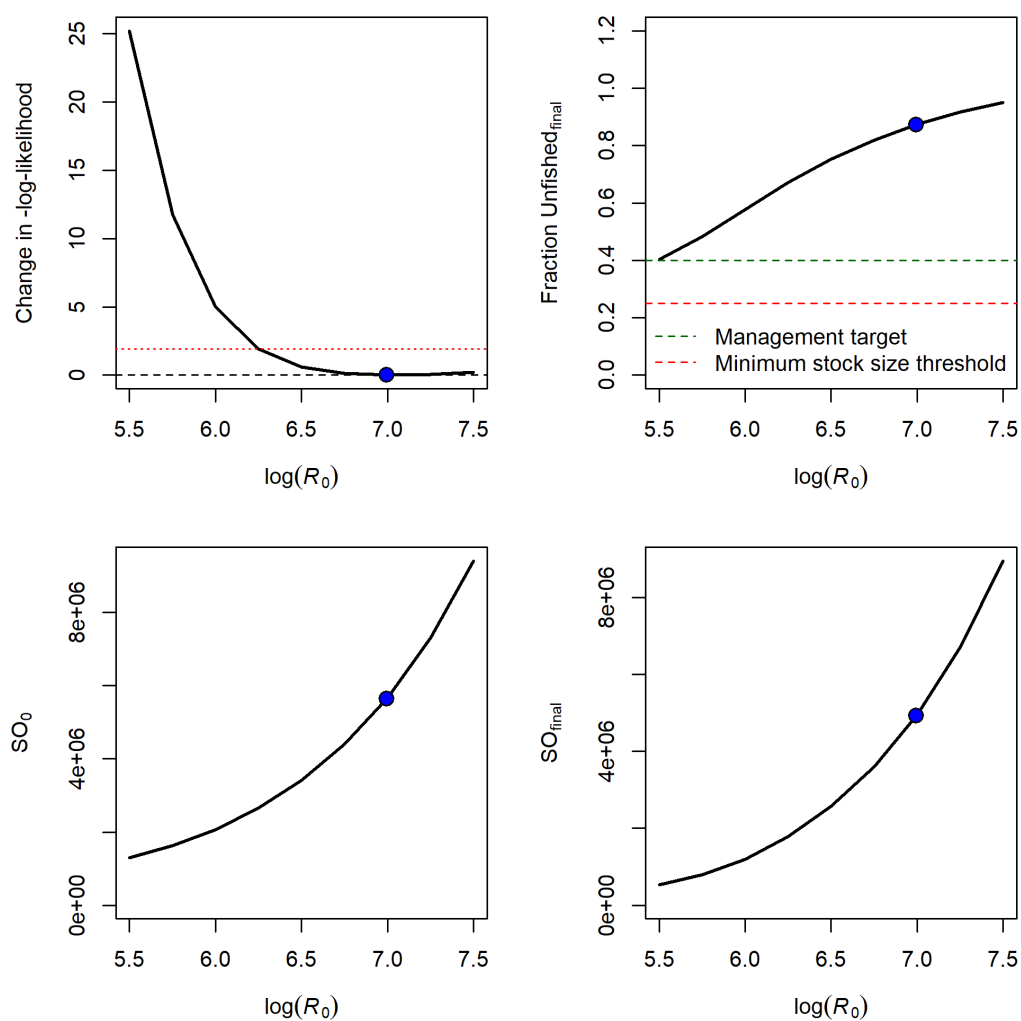


Figure 163: Initial recruitment ( $\ln R_0$ ) likelihood profile (change in the negative log-likelihood across a range of  $\ln(R_0)$  values) and derived quantities. Red line in the top left figure indicates the significance level in likelihood difference.

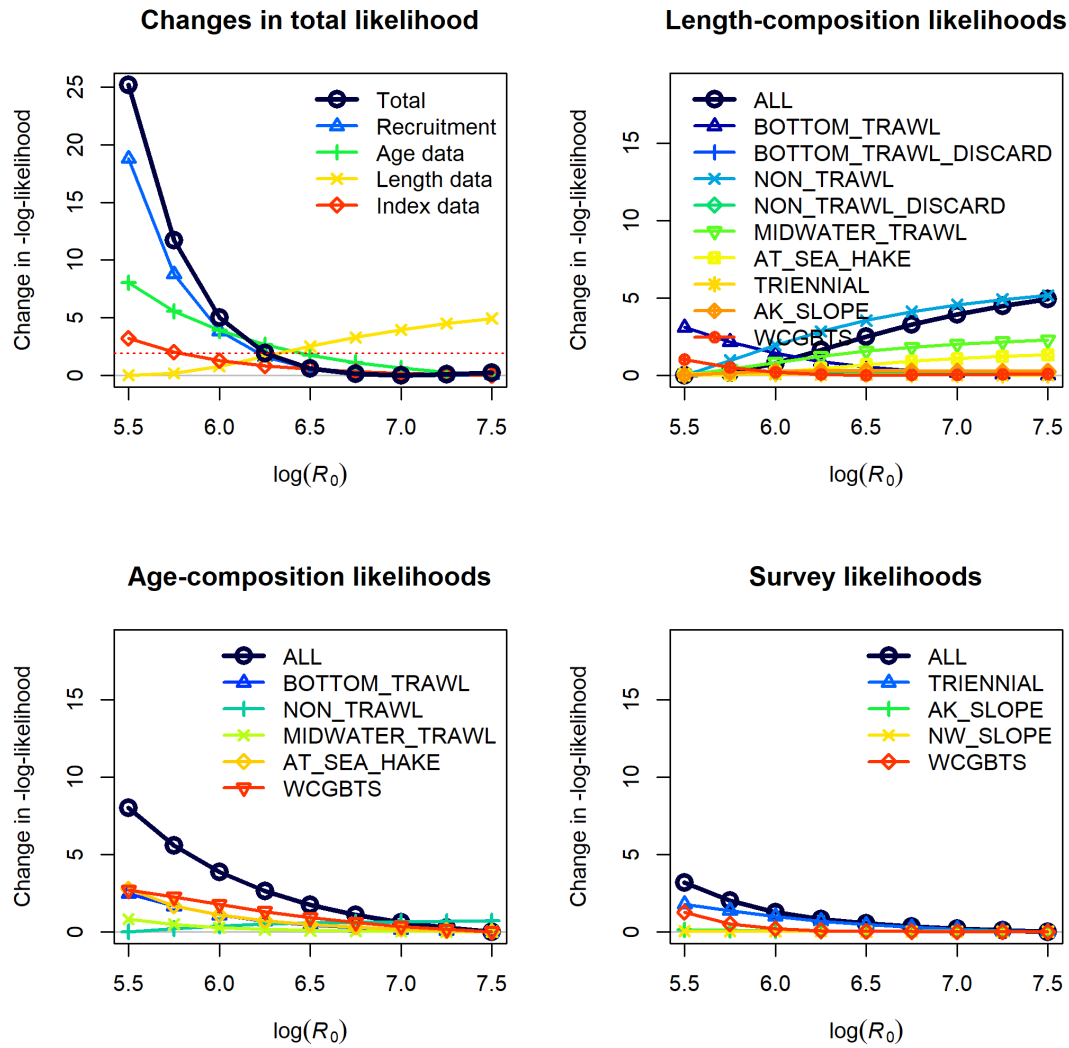


Figure 164: Initial recruitment ( $\ln(R_0)$ ) likelihood profile for each of the likelihood components.

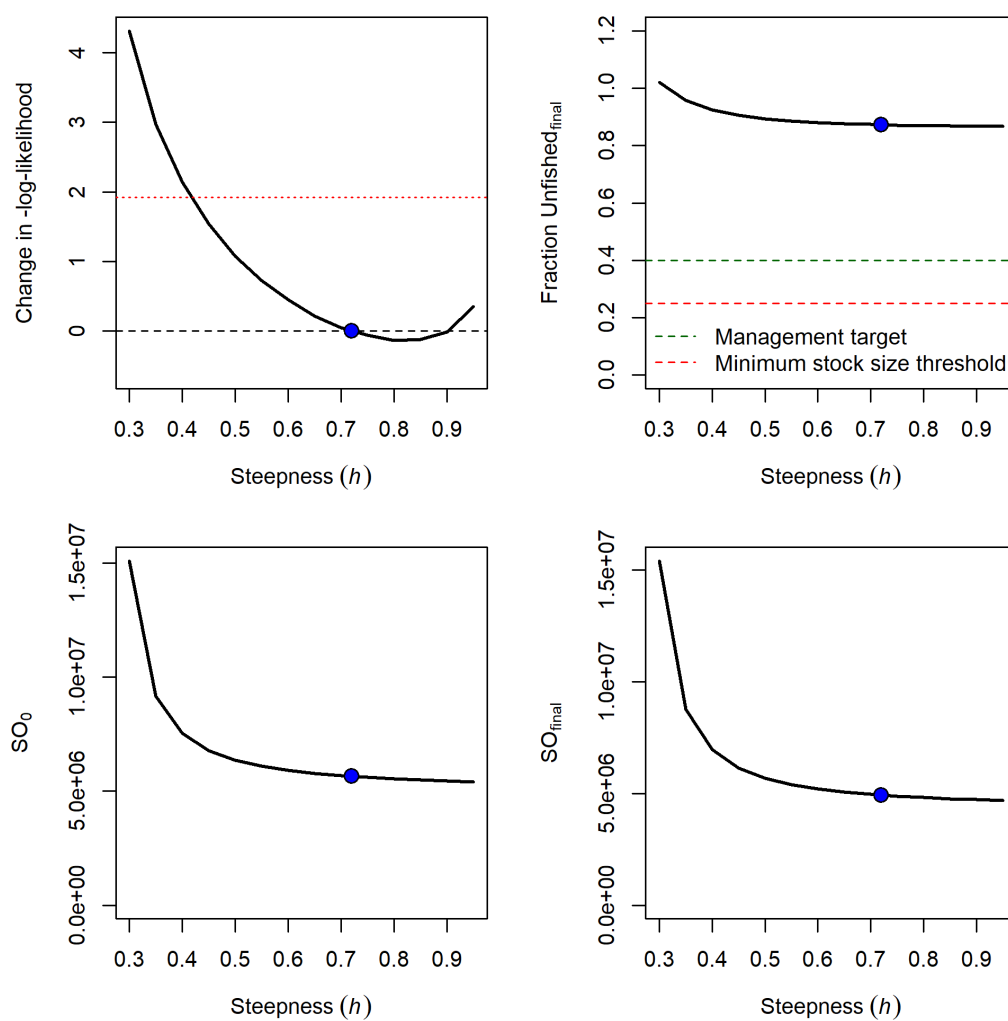
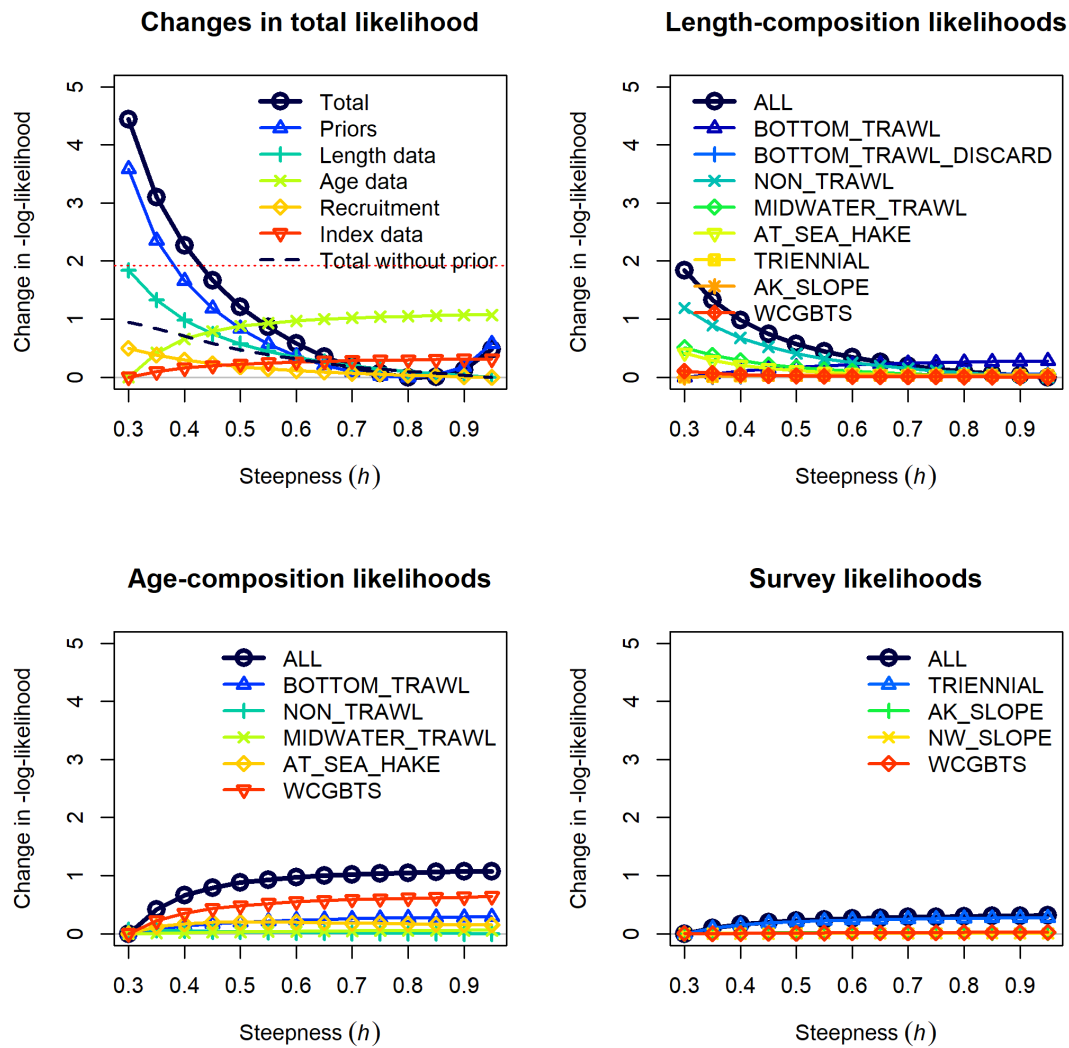


Figure 165: Steepness ( $h$ ) likelihood profile (change in the negative log-likelihood across a range of  $h$  values) and derived quantities. Red line in the top left figure indicates the significance level in likelihood difference.

Figure 166: Steepness ( $h$ ) likelihood profile for each of the likelihood components.

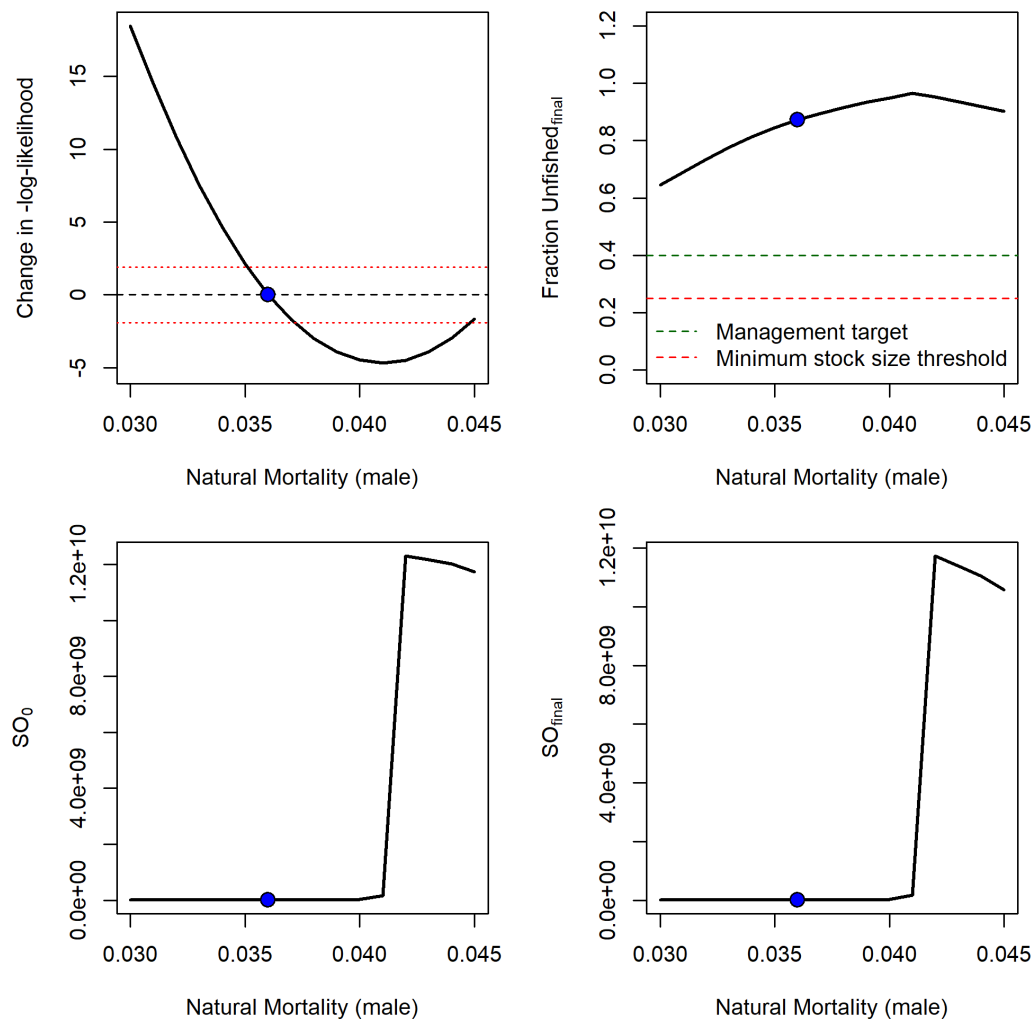


Figure 167: Male natural mortality ( $M$ ) likelihood profile (change in the negative log-likelihood across a range of male  $M$  values) and derived quantities. Red line in the top left figure indicates the significance level in likelihood difference.

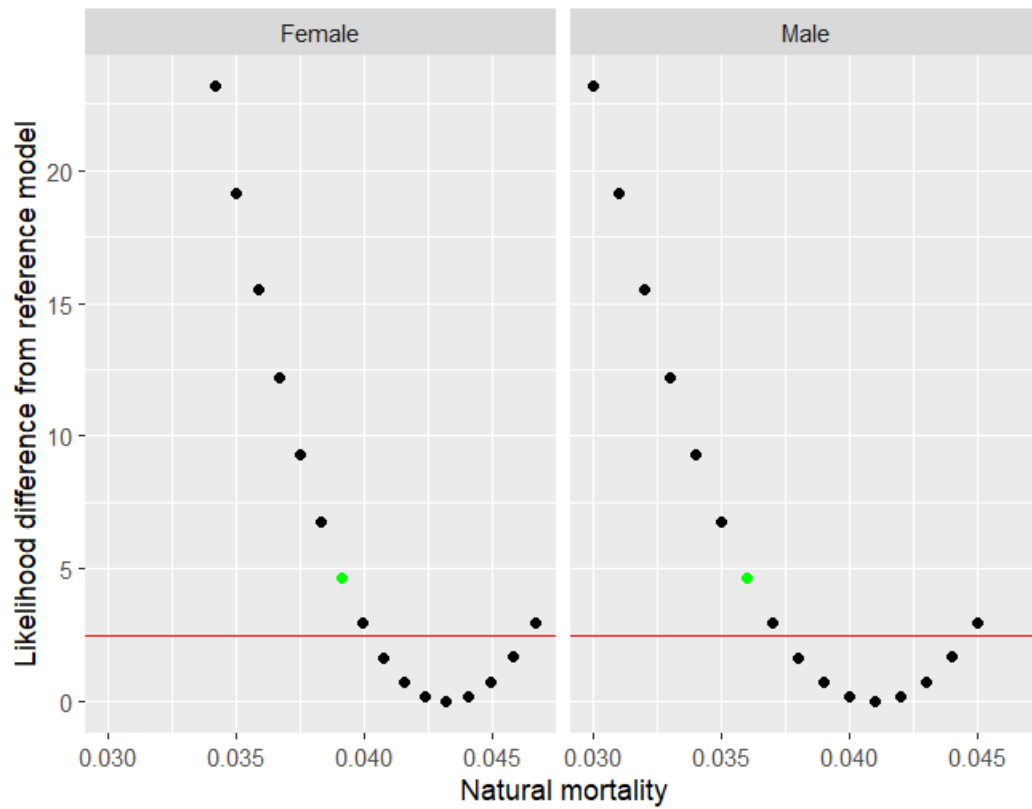


Figure 168: Likelihood differences by natural mortality ( $M$ ) by sex.



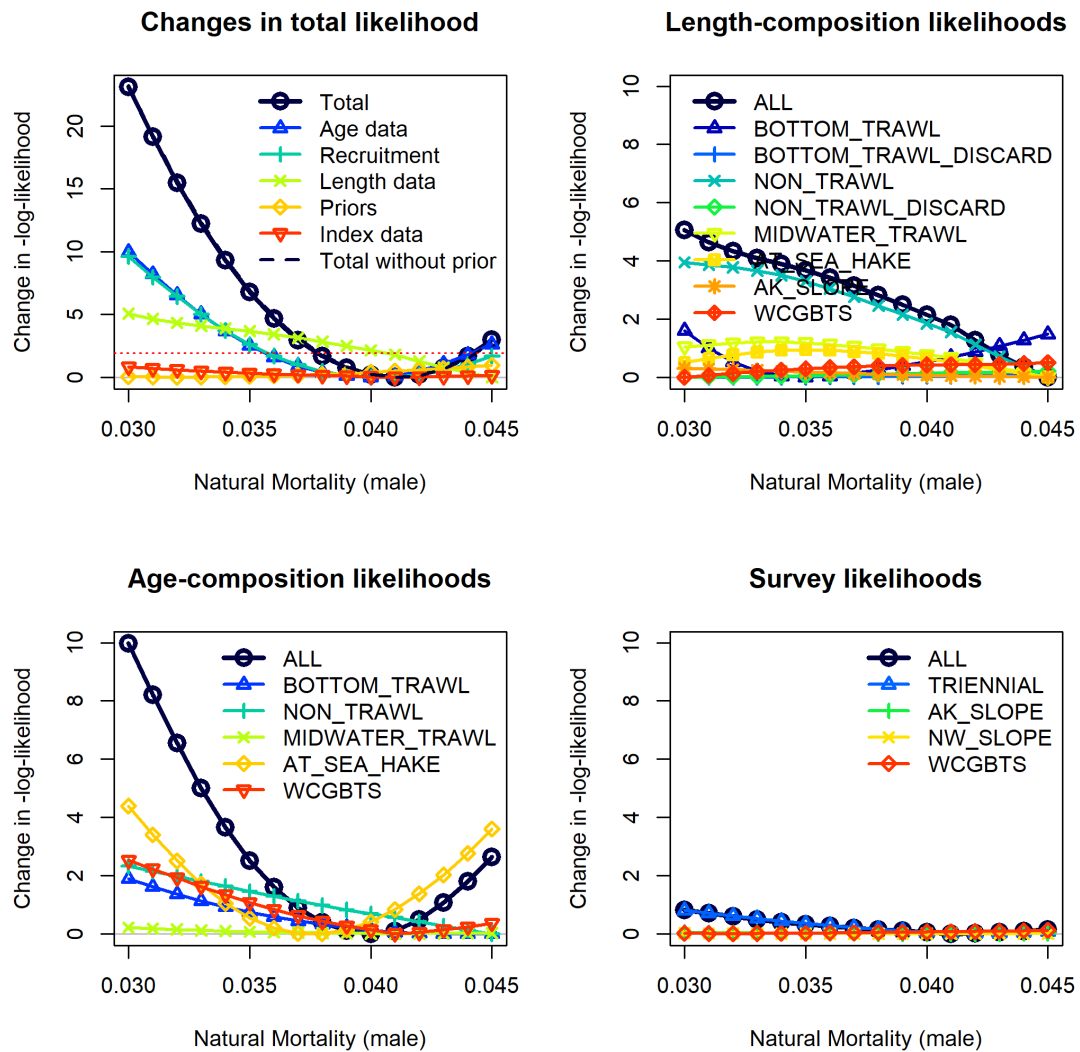


Figure 169: Male natural mortality ( $M$ ) likelihood profile for each of the likelihood components.

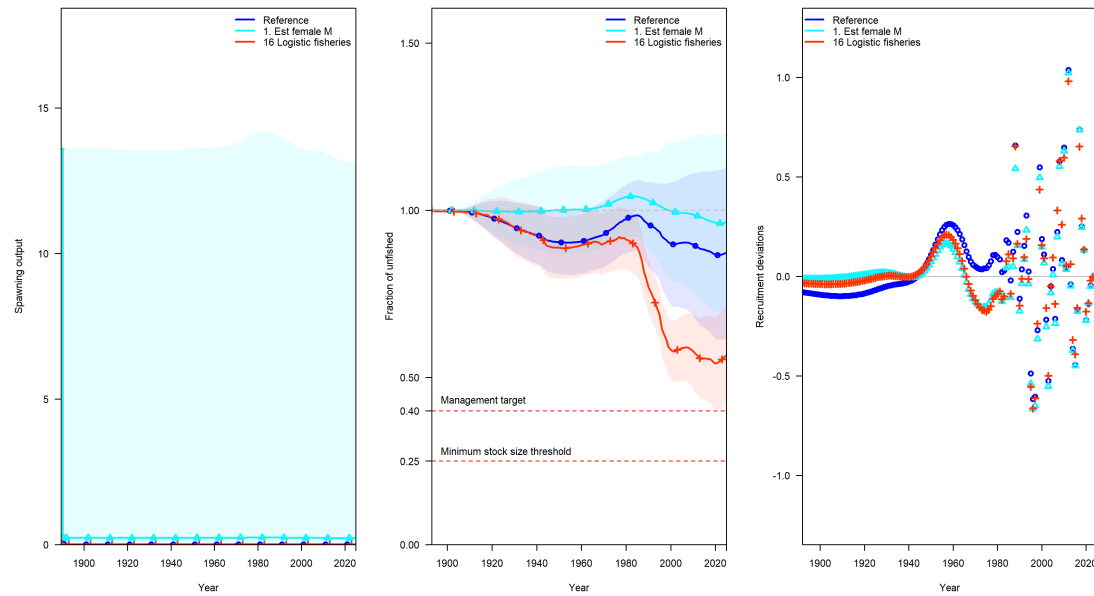


Figure 170: Comparison of spawning output (left panel), relative stock status (middle panel) and recruitment estimation (right panel) for the reference model and high and low states of nature.

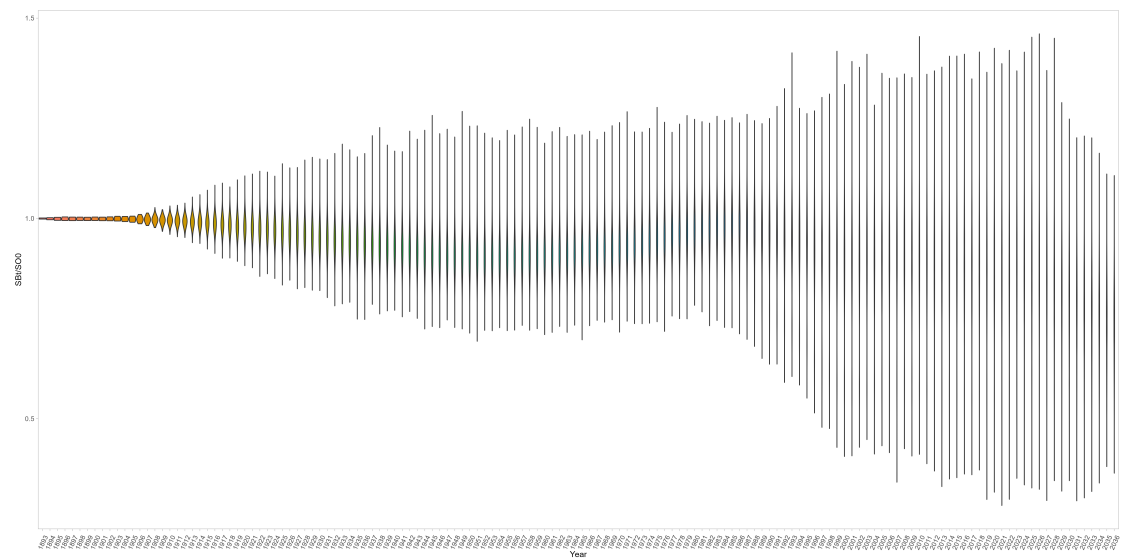


Figure 171: Ensemble relative stock status for the reference model and two states of nature.

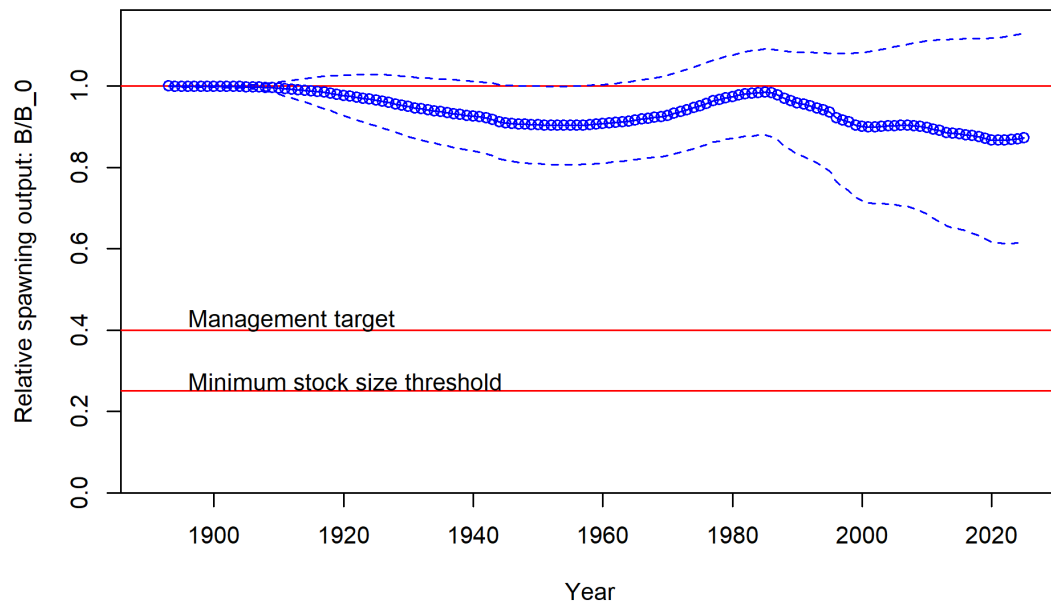


Figure 172: Placeholder.

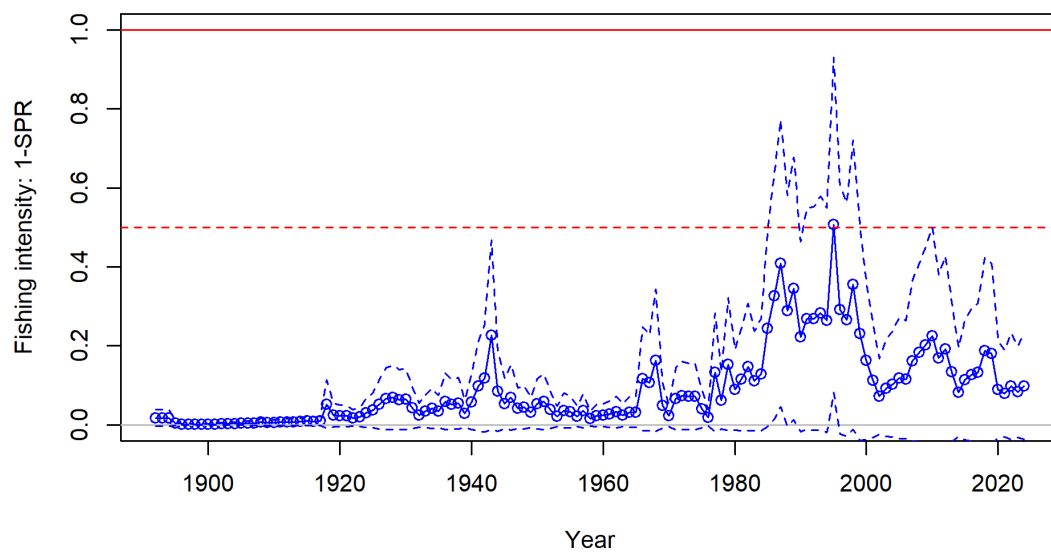


Figure 173: Placeholder.

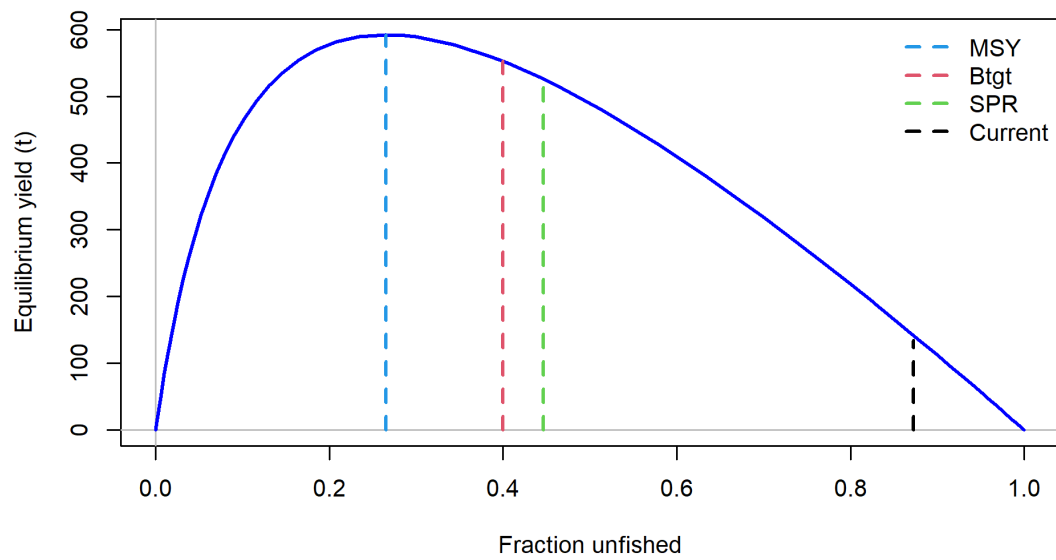


Figure 174: Placeholder.

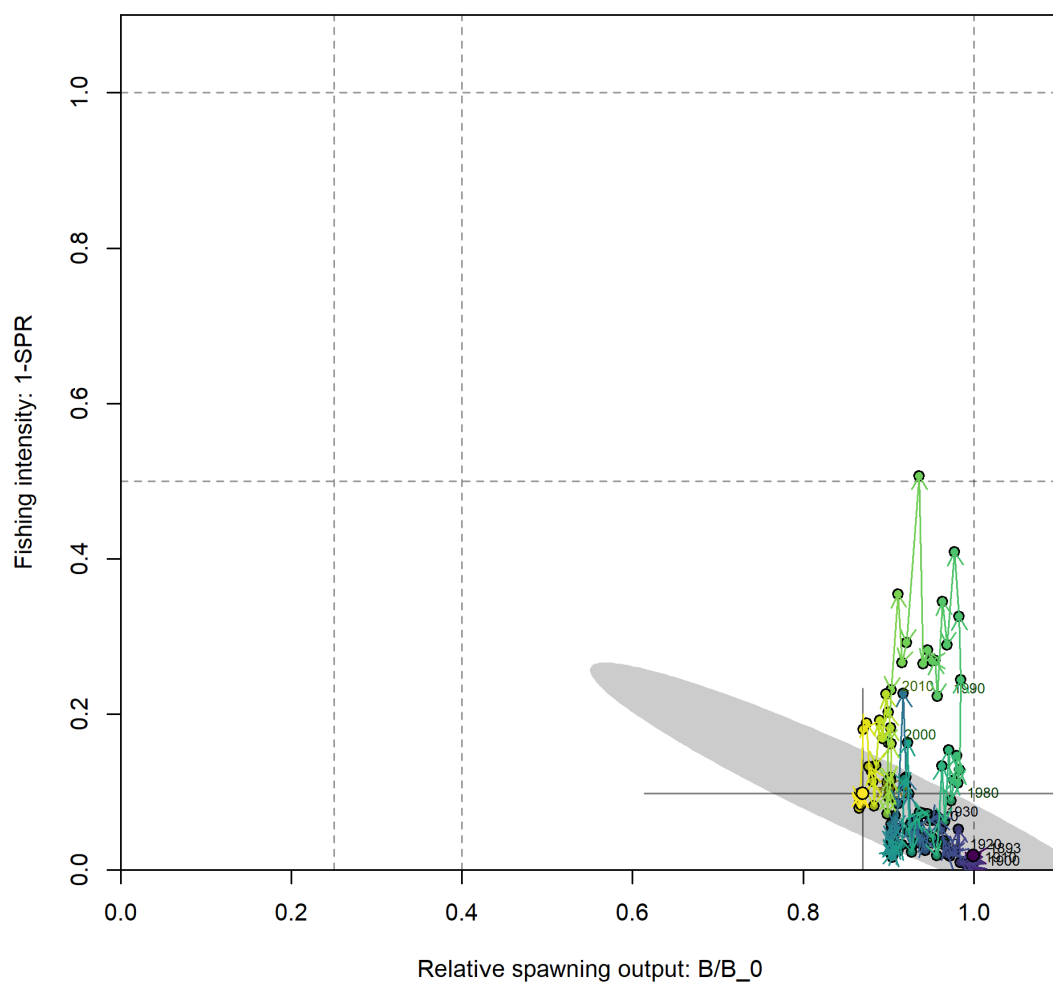


Figure 175: Placeholder.

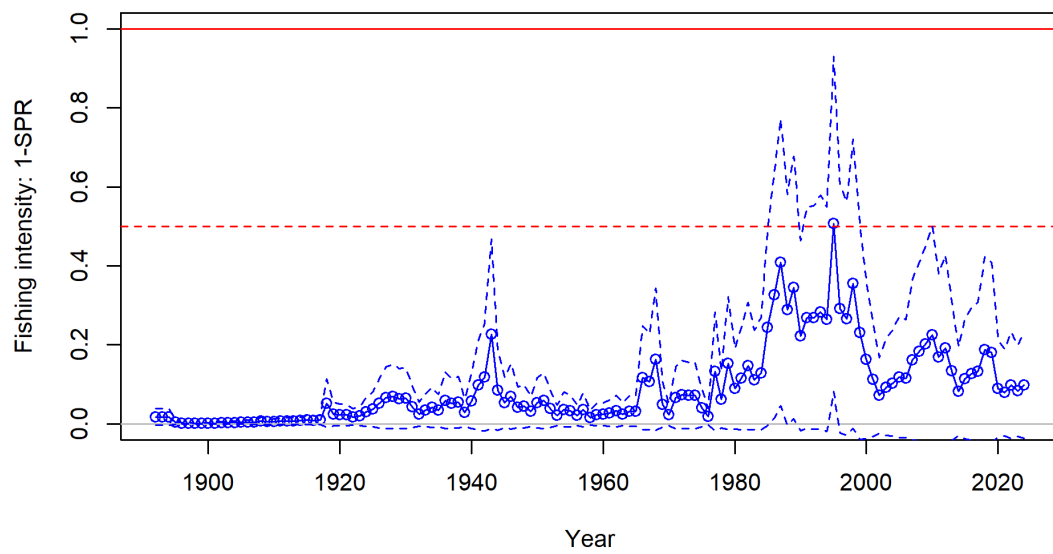


Figure 176: Estimated time series of fishing intensity for the base model.

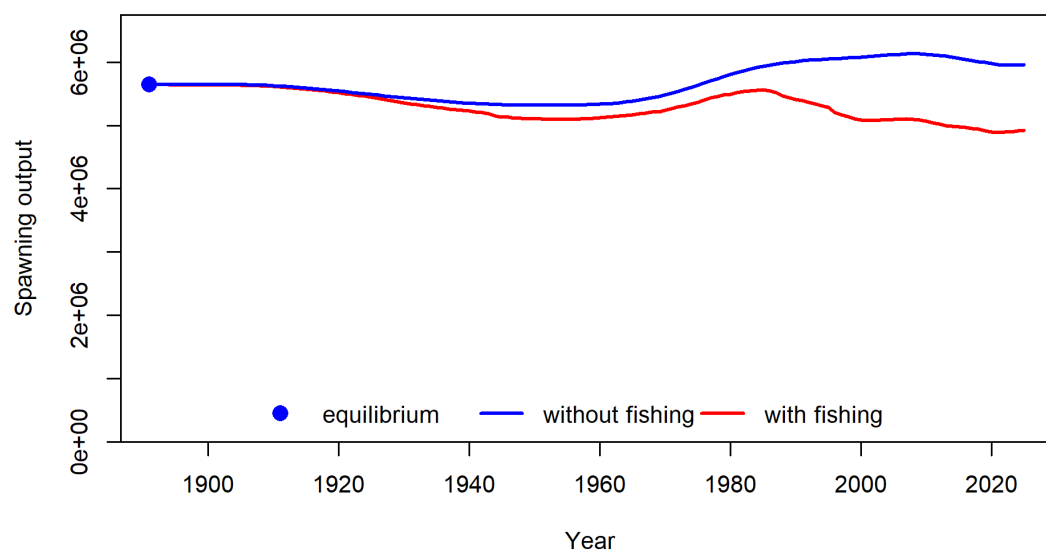


Figure 177: Dynamic B0 plot. The lower line shows the time series of estimated spawning output in the presence of fishing mortality. The upper line shows the time series that could occur under the same dynamics (including deviations in recruitment), but without fishing. The point at the left represents the unfished equilibrium.

## 9 Notes



## 10 Appendices